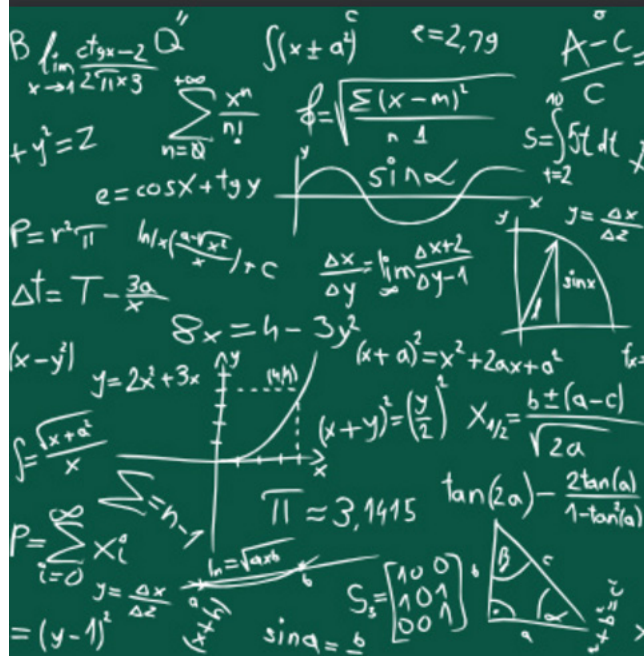


Sets



- Author: Fabio Furini

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1 Informal introduction to set theory

- Set theory is based on the following three key concepts:

1. Sets

The notion of set is generally taken as **primitive** (that is, not reducible to more elementary concepts). The words *collection*, *class*, *aggregate*, *family* are used as synonyms of set.

Example 1: sets

For example, we have the set of points of a plane, the set of students of a university, or the set of stars of a galaxy.

2. Elements

A set is determined by its *elements*, in the sense that a set is defined when we have a **criterion** to establish whether a given element is or is not an element of this set.

There are sets with a *finite number of elements* (for example the set of students of a university) and sets with an *infinite number of elements* (for example the set of points of the plane).

Sets are usually denoted by capital letters. The elements of a set are usually denoted by lowercase letters.

3. Membership

The concept of *membership* links elements to sets. When an object is an element of a set, we say that the element *belongs* to the set. To indicate that an element x belongs to a set A we write:

$$x \in A$$

1.1 Informal definition of sets

1. A first way to define sets is **definition by listing** (roster notation)

A set can be defined **by listing**, that is, by listing the elements that belong to it between curly brackets. This technique assumes that the set has a finite number of elements.

Example 2: definition by listing

The notation:

$$A = \{a, b, c\}$$

means that the set A has as elements the three letters a , b and c . For example, a belongs to A , that is, $a \in A$.

2. A second way to define sets is **definition by a property**:

A set can be defined **by a property** as follows:

$$A = \{ x \in U : p(x) \text{ is true} \}$$

where $p(x)$ is the property that the element x of the set U must have in order to belong to the set A . This technique can be used to define sets with a finite or even an infinite number of elements.

Example 3: definition by a property

Consider for example the set of letters of the Latin alphabet:

$$U = \{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z\}$$

Using for example the property $p(x)$ defined as “ x is a vowel” we can define the following set of vowels:

$$A = \underbrace{\{x \in U : x \text{ is a vowel}\}}_{\{a, e, i, o, u\}}$$

Note that to define a set A by a property we need a set U to which all the elements of the set A we want to define belong. The set U plays the role of the **universal set**.

It is important that the property $p(x)$ we use makes sense for every x of the set U (universal set), and therefore is true or false (without ambiguity of meaning) for every particular $x \in U$; the set A will then consist of exactly those x belonging to U for which the property $p(x)$ is true.

One must be careful when defining sets, since contradictions may arise. There is a formal definition of the concept of set, developed to avoid contradictions, but it is beyond the scope of this course.

- For example, the set of all sets that do not contain themselves is not a set in the formal definition of sets. Admitting this set would generate the contradiction: “the set of all sets that do not belong to themselves belongs to itself if and only if it does not belong to itself” (**Russell’s paradox** – section 5.1).

2 Number sets

Definition 1: numeral system

A **positional numeral system** is a way of encoding numbers using a **sequence of digits**, where each digit contributes differently to the number depending on its position. The number of distinct digits is the **base** of the system.

The **decimal expansion** of a number is a numerical representation in base 10; it expresses a number as a sum of powers of 10, and it can be finite or infinite depending on the type of number.

- Informal definition of the main number sets:

1. We denote by $\mathbb{N}$ the set of **natural numbers**, that is, the set of numbers that can be written as **decimal expansions without decimal point and without sign**. We will use the informal notation:

$$\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$$

2. We denote by $\mathbb{Z}$ the set of **integers**, that is, the set of numbers that can be written as **decimal expansions without decimal point and with a sign**. We will use the informal notation:

$$\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \pm 4, \dots\}$$

3. We denote by $\mathbb{Q}$ the set of **rational numbers**, that is, the set of numbers that can be written as **finite or infinite periodic decimal expansions**. In other words, it is the set of numbers that can be written as a fraction $\frac{p}{q}$ where p is an integer and q is a natural number different from zero.

Example 4: rational numbers

- For example, with $p = 2$ and $q = 5$ we have the fraction $\frac{2}{5}$, whose decimal expansion is 0.4.
- For example, with $p = 4$ and $q = 10$ we have the fraction $\frac{4}{10}$, whose decimal expansion is again 0.4. Note that a rational number can be written with more than one fraction.
- For example, with $p = 13$ and $q = 30$ we have the fraction $\frac{13}{30}$, whose decimal expansion is $0.4\bar{3} = 0.43333\dots$

However, we can represent every rational number different from 0 by a single fraction $\frac{p}{q}$ by choosing $p \in \mathbb{Z}$ and $q \in \mathbb{N}$ coprime (that is, relatively prime: p and q are not both divisible by the same integer greater than 1).

Remark 1

$$0,\overline{9} = 1$$

Proof. There are different proofs of this remark, based on different mathematical techniques.

- (a) A simple proof follows directly from the definition of 1 divided by 3; indeed, we have:

$$\begin{aligned}\frac{1}{3} &= 0,\overline{3} \\ \frac{1}{3} \cdot 3 &= 0,\overline{3} \cdot 3 \\ 1 &= 0,\overline{9}\end{aligned}$$

- (b) Using algebraic arguments we can write:

$$\begin{array}{ll}x = 0.999\dots & \\10x = 9.999\dots & \text{multiplying by 10} \\10x = 9 + 0.999\dots & \text{separating the integer part from the fractional part} \\10x = 9 + x & \text{by definition of } x \\9x = 9 & \text{subtracting } x \\x = 1 & \text{dividing by 9}\end{array}$$

- (c) A proof by contradiction is the following:

$$\begin{aligned}0,\overline{9} &\neq 1 \\ 0,\overline{9} \cdot 9 &\neq 1 \cdot 9 \\ 0,\overline{9} \cdot 9 + 0,\overline{9} &\neq 1 \cdot 9 + 0,\overline{9} \\ 0,\overline{9} \cdot 9 + 0,\overline{9} &\neq 9,\overline{9} \\ 0,\overline{9} \cdot (9 + 1) &\neq 9,\overline{9} \\ 0,\overline{9} \cdot (10) &\neq 9,\overline{9} \\ 9,\overline{9} &\neq 9,\overline{9} \quad \text{contradiction!}\end{aligned}$$

□

4. We denote by $\mathbb{R}$ the set of **real numbers**, that is, the set of numbers identified with finite or infinite decimal expansions, periodic or non-periodic.

Example 5: real numbers

- Consider for example the number

$$0,10110111011110\dots$$

obtained by putting after the decimal point one digit equal to 1, then 0, then two digits equal to 1, then 0, then three digits equal to 1 ...and so on. The string of nonzero digits after the decimal point is neither finite nor periodic: therefore this number is real but not rational.

- Other examples of real but not rational numbers are $\sqrt{2}$ and $\sqrt{3}$, or π and Euler's number e (Napier's constant), which have infinite non-periodic decimal expansions and are therefore real but not rational numbers.

- There are formal definitions of the sets of natural, integer and rational numbers that are beyond the scope of this course. Later we will give a formal definition of the real numbers.

3 Relations between sets

1. Equality

Two sets A and B are equal when they have the same elements. We write

$$A = B$$

and this means that every element that belongs to A also belongs to B and every element that belongs to B also belongs to A .

2. Inclusion

It may happen that only one of the two requirements expressed by the equality relation holds. For example, if we only know that every element of A is also an element of B , we can say that A is contained in B . We write:

$$A \subseteq B \text{ or } A \subset B$$

and we read “ A is contained in B ” or “ A is a **subset** of B ”. If we state that $A \subseteq B$, we do not exclude that $A = B$.

Saying that $A = B$ is equivalent to saying that $A \subseteq B$ and $B \subseteq A$.

Be careful not to confuse “belongs to” and “is contained in” (in symbols, $\in$ and $\subseteq$). In a sense, they are two ways of indicating that “something is inside something else”, but they have a fundamental logical difference:

- a set is contained in another set;
- an element belongs to a set.

Example 6: “belongs to” vs “is contained in”

For example:

$$3 \in \{1, 3, 4\};$$

$$\{3\} \subseteq \{1, 3, 4\};$$

$$\{1, 4\} \subseteq \{1, 3, 4\}.$$

In particular, do not confuse 3 (which is a number) with $\{3\}$, which is the set that contains the number 3 as its only element.

Sometimes we consider sets whose elements are other sets. In this case too, the symbols $\in$ and $\subseteq$ are not interchangeable, but must be used correctly.

Example 7: “belongs to” vs “is contained in”

For example, if we define the set

$$A = \{ \{1\}, \{2\}, \{1, 2\} \},$$

then it is correct to state that $\{2\} \in A$, because $\{2\}$ is a set that plays the role of an element of A , whereas it would be incorrect to say that $\{2\} \subseteq A$, because this would mean that $2 \in A$, while A has $\{2\}$ as an element but not 2. Instead, the subset of A containing only the element $\{2\}$ is denoted by $\{\{2\}\}$ and we have $\{\{2\}\} \subset A$.

3.1 Empty set, cardinality and power set

Definition 2: of empty set

The **empty set** is the set that contains no elements. It is denoted by $\emptyset$.

Remark 2

Given a set A , we have:

$$\emptyset \subseteq A$$

Proof. To prove it, we should show that every element that belongs to $\emptyset$ also belongs to A ; but no element belongs to $\emptyset$, so the thesis holds. $\square$

Definition 3: of cardinality of a set

The number of elements of a set A is the **cardinality** of the set. It is denoted by $|A|$.

Definition 4: of power set

Given a set A , the set whose elements are all the subsets of A is called the **power set** of A and is denoted by the symbol $\mathcal{P}(A)$.

- Every set A has two trivial subsets, namely A itself and the empty set $\emptyset$ (they could coincide, if A is empty).

Example 8: power set

For example, given

$$A = \{1, 2, 3\},$$

then

$$\mathcal{P}(A) = \left\{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\} \right\}$$

Remark 3

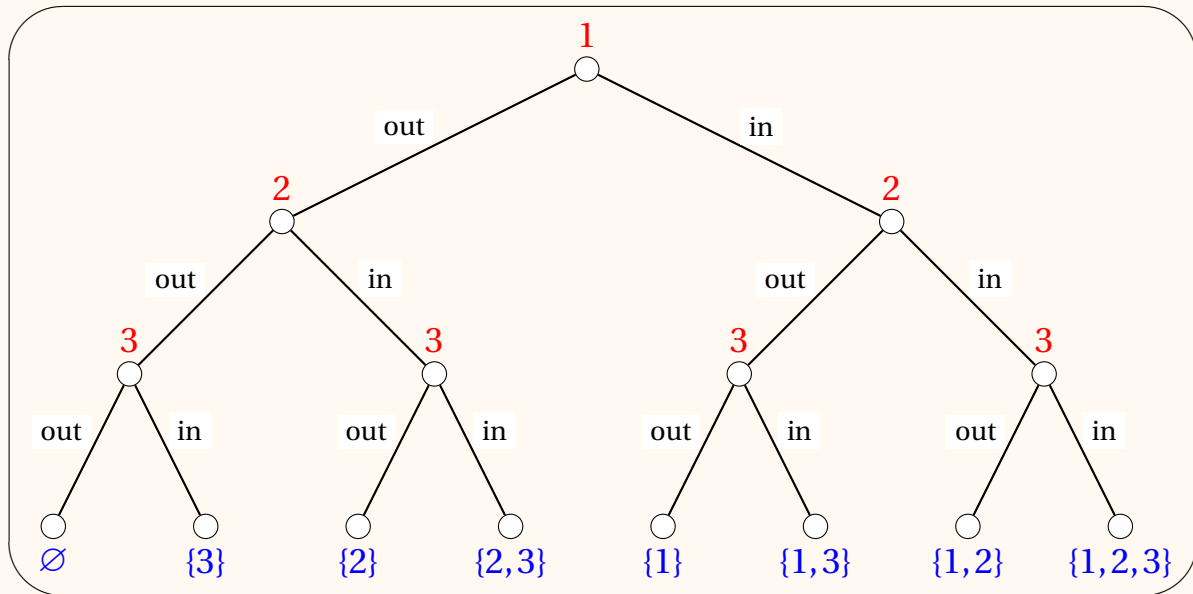
Given a set A with n elements, the power set $\mathcal{P}(A)$ has 2^n elements:

$$|\mathcal{P}(A)| = 2^n$$

Proof. For each of the elements of A , the subsets of A may or may not contain that element. Hence we must make a choice between two options n times. The total number of possible subsets is therefore 2^n . $\square$

Example 9: construction of the power set with a binary tree

Given the set $A = \{1, 2, 3\}$ with $n = 3$ elements, the cardinality of its power set is $2^3 = 8$. The construction of the power set $\mathcal{P}(A)$ can be visualized through the following **binary tree** (an undirected, connected and acyclic graph) in which at each level we decide whether or not to include the object in the subset:



4 Operations on sets

Definition 5: of intersection of sets

The *intersection* of two sets $A, B \subseteq U$ is the set defined by:

$$A \cap B = \{x \in U : x \in A \text{ and } x \in B\}$$

It is the set of elements that belong both to the first and to the second set.

Definition 6: of union of sets

The **union** of two sets $A, B \subseteq U$ is the set defined by:

$$A \cup B = \{x \in U : x \in A \text{ or } x \in B\}$$

It is the set of elements that belong to the first or to the second set, where “or” is meant in the non-exclusive sense (the set of elements that belong to A or to B or to both).

Definition 7: of difference of sets

The **difference** of two sets $A, B \subseteq U$ is the set defined by:

$$A \setminus B = \{x \in A : x \notin B\}$$

It is the set of elements that belong to the first but not to the second set. The symbol “ $\setminus$ ” can also be written “ $-$ ” by analogy with arithmetic subtraction.

4.1 Complementary sets and disjoint sets

Example 10: universal sets

For example, in questions of arithmetic we could have $U = \mathbb{N}$, while in questions of analysis we could have $U = \mathbb{R}$.

Definition 8: of set complementation and complementary sets

The **complementation** of a set $A \subseteq U$ is the set defined by:

$$\overline{A} = \{x \in U : x \notin A\}$$

This set is called the **complement** (complementary set) of A with respect to U

- For every set $A \subseteq U$, where U is the universal set, we have the following relations:

$$\begin{aligned}\overline{\overline{A}} &= A \\ A \cap \overline{A} &= \emptyset \\ A \cup \overline{A} &= U\end{aligned}$$

$$\begin{aligned}A \cap \emptyset &= \emptyset \\ A \cap U &= A \\ A \cup \emptyset &= A \\ A \cup U &= U\end{aligned}$$

$$\overline{U} = \emptyset$$

$$\overline{\emptyset} = U$$

Definition 1: of disjoint sets

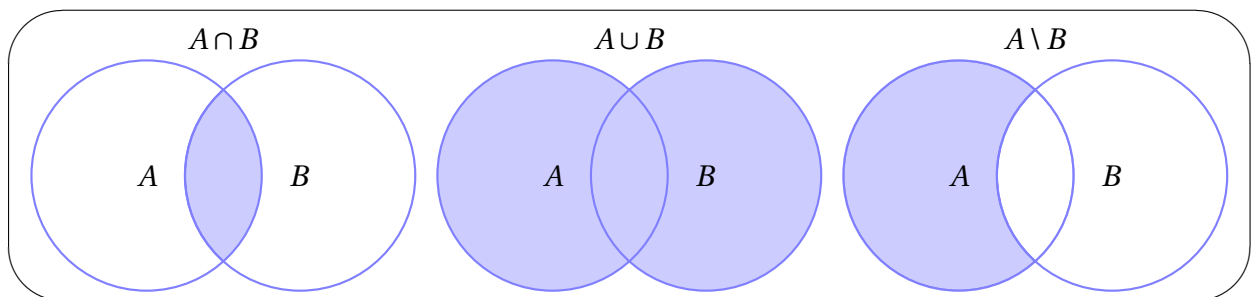
Two sets A and B are **disjoint** if they have no elements in common:

$$A \cap B = \emptyset$$

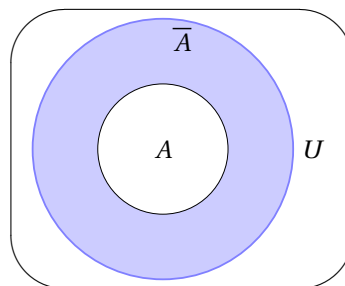
4.2 Venn diagrams

Venn diagrams are graphical representations in which sets are represented as regions of the plane

- Venn diagrams of intersection, union and difference:



- Venn diagram of the complement of a set:



4.3 Cartesian product

There is another operation on sets, which can be performed on any two sets (i.e., two sets not necessarily contained in the same universal set):

Definition 9: of Cartesian product

Given two (not necessarily distinct) sets A and B , the set consisting of all ordered pairs (a, b) , with $a \in A$ and $b \in B$, is called the **Cartesian product** of A and B and is denoted by the symbol $A \times B$.

When A and B are two finite sets, the cardinality of their Cartesian product is:

$$|A \times B| = |A| \cdot |B|.$$

Example 11: Cartesian product

$$\begin{aligned} \{a, b\} \times \{a, b, c\} &= \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c)\} \\ |\{a, b\}| &= 2, \quad |\{a, b, c\}| = 3, \quad |\{a, b\} \times \{a, b, c\}| = 2 \cdot 3 = 6 \end{aligned}$$

- A typical use of the Cartesian product is $\mathbb{R} \times \mathbb{R}$, which is abbreviated by the symbol $\mathbb{R}^2$ and denotes the set of ordered pairs of real numbers.
- Similarly, $\mathbb{R}^n$ (abbreviation of the Cartesian product of n sets equal to $\mathbb{R}$) is the set of *ordered n -tuples of real numbers*

$$\mathbb{R}^n = \{ (x_1, x_2, \dots, x_n) : x_i \in \mathbb{R}, i \in \{1, 2, \dots, n\} \}$$

4.4 Properties of operations on sets

Remark 4: properties of intersection

Given three sets A, B and C , intersection has the following properties:

- **Commutative:**

$$A \cap B = B \cap A$$

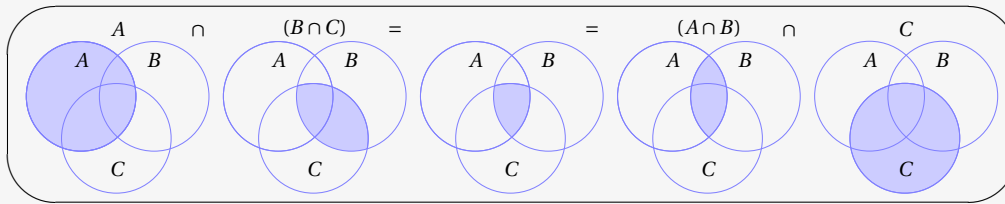
- **Associative:**

$$A \cap (B \cap C) = (A \cap B) \cap C$$

- **Idempotence:**

$$A \cap A = A$$

Proof. The graphical proof of the associative property of intersection is the following:



□

Remark 5: properties of union

Given three sets A , B and C , union has the following properties:

- **Commutative:**

$$A \cup B = B \cup A$$

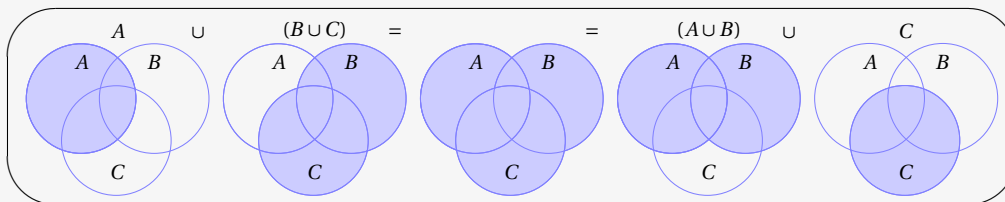
- **Associative:**

$$A \cup (B \cup C) = (A \cup B) \cup C$$

- **Idempotence:**

$$A \cup A = A$$

Proof. The graphical proof of the associative property of union is the following:



□

Remark 6: distributive properties (linking union and intersection)

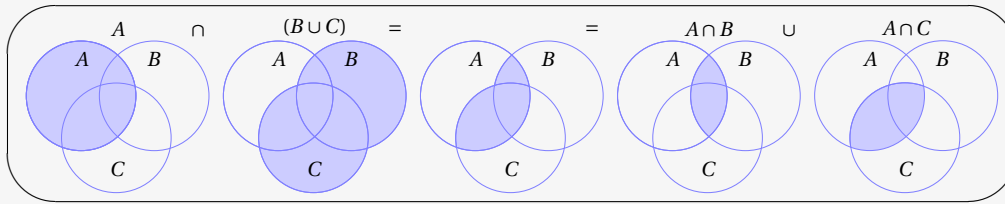
Given three sets A , B and C we have:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

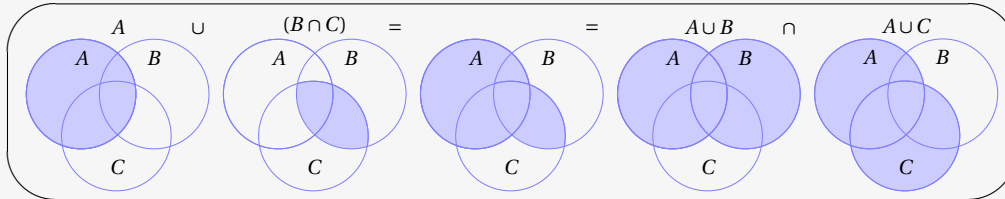
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

The distributive properties link union and intersection to each other.

Proof. The graphical proof of the first property is the following:



The graphical proof of the second property is the following:



□

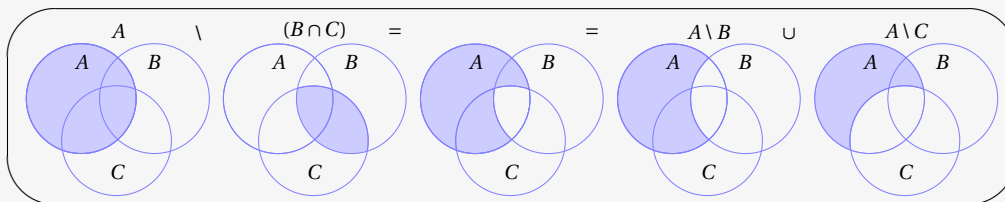
Proposition 1: De Morgan's laws (first version)

Given three sets A , B and C we have:

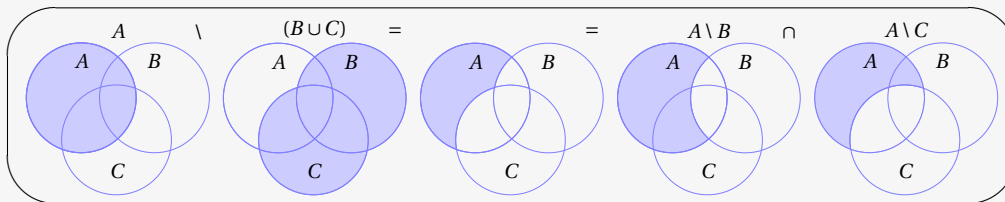
$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$$

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$

Proof. The graphical proof of the first property is the following:



The graphical proof of the second property is the following:



□

Proposition 2: De Morgan's laws (second version)

Given the sets $B, C \subseteq U$, we have

$$\overline{B \cap C} = \overline{B} \cup \overline{C}$$

$$\overline{B \cup C} = \overline{B} \cap \overline{C}$$

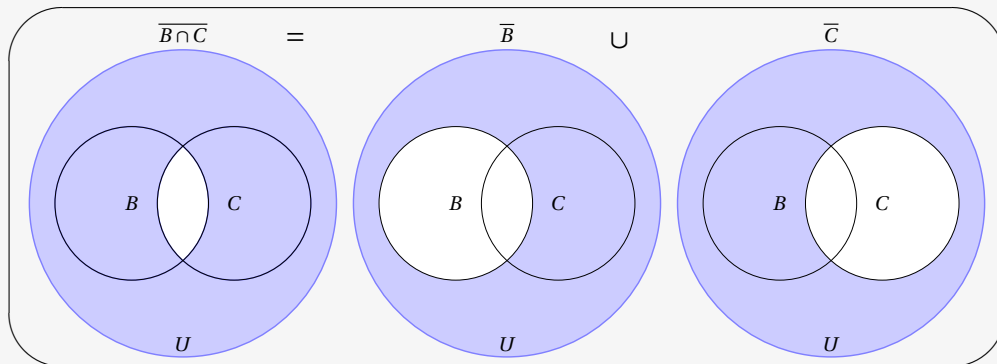
Proof. These laws can be derived by setting A equal to U in the previous De Morgan's laws, as follows:

$$\underbrace{U \setminus (B \cap C)}_{=\overline{B \cap C}} = \underbrace{(U \setminus B)}_{=\overline{B}} \cup \underbrace{(U \setminus C)}_{=\overline{C}}$$

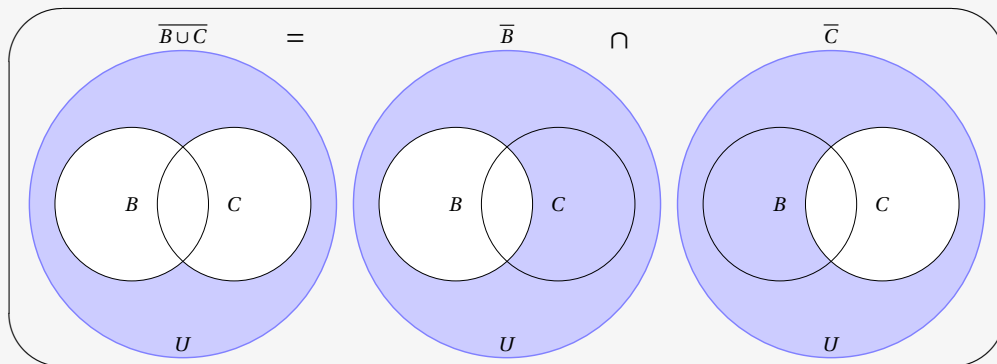
$$\underbrace{U \setminus (B \cup C)}_{=\overline{B \cup C}} = \underbrace{(U \setminus B)}_{=\overline{B}} \cap \underbrace{(U \setminus C)}_{=\overline{C}}$$

□

Proof. The graphical proof of the first property is the following:

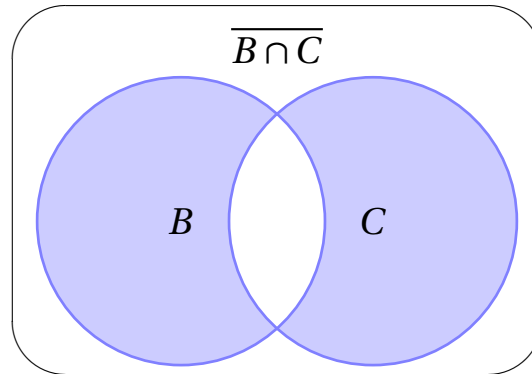


The graphical proof of the second property is the following:



□

- The set of elements that belong to a set B or to a set C but not to both (meaning “or” in the exclusive sense) is obtained by setting $U = B \cup C$ and taking the complement of the intersection:



Relation between inclusion and the operations of union and intersection:

$$A \subseteq B \text{ if and only if } A \cap B = A \text{ and } A \cup B = B.$$

4.5 Properties of the cardinality of sets

- For any two finite sets A and B , we have

$$|A \cup B| = |A| + |B| - |A \cap B|$$

from which we conclude that

$$|A \cup B| \leq |A| + |B|$$

- If A and B are disjoint then

$$|A \cap B| = 0 \text{ and hence } |A \cup B| = |A| + |B|$$

- If $A \subsetneq B$, then $|A| < |B|$

5 Further topics

5.1 Russell's paradox

“can a set be an element of itself or not?”

- For example, the set of all books in a library is not an element of itself (a set of books is not a book). On the other hand, the set of all sets with more than 20 elements is an element of itself.
- Following this reasoning, two categories of sets can be defined:
 1. sets that are not elements of themselves
 2. sets that are elements of themselves

If we consider the set of all sets that are not elements of themselves, is it an element of itself or not?

Let us call this set S ; two hypotheses can be made:

- 1 If we suppose $S \in S$, then S contains itself as an element and therefore does not belong to S (since by definition a set belongs to S only if it does not contain itself as an element). Hence $S \notin S$, and we have a contradiction. We conclude that the hypothesis must be wrong.
- 2 If we suppose $S \notin S$, then S does not contain itself as an element and therefore belongs to S (since by definition a set belongs to S if it does not contain itself as an element). Hence $S \in S$ and we have another contradiction. We conclude that this hypothesis must be wrong too!

Russell's paradox: The set of all sets that do not belong to themselves belongs to itself if and only if it does not belong to itself.

- The formal definition of the concept of set is based on the **Zermelo-Fraenkel system of axioms**, abbreviated as **ZF**. This system of axioms includes the standard axioms of axiomatic set theory on which, together with the *axiom of choice*, all of ordinary mathematics is based.

- The **axiom of regularity** states that “Every non-empty set A contains an element disjoint from A ”.
- The **axiom of pairing** states that “Given two objects, there exists a set whose elements are the two objects”

Remark 7

No set is an element of itself

Proof. Given a set A , we apply the axiom of regularity to $\{A\}$, which is a set by the axiom of pairing. We thus obtain the set $\{A, A\}$, which we abbreviate to $\{A\}$ since sets cannot contain repeated objects (it is a special case of a pair). By the axiom of regularity there must exist an element of $\{A\}$ disjoint from $\{A\}$. Since the only element of $\{A\}$ is A , A is disjoint from $\{A\}$. Hence, since $A \cap \{A\} = \emptyset$, we cannot have $A \in A$ (by the definition of disjoint). $\square$

- The other ZF axioms are not part of the basic course in Mathematical Analysis.

- A further proof of

$$0,\overline{9} = 1$$

starts from the assumption that two numbers are equal if and only if their difference is equal to zero, and it is based on computing the value of $1 - 0,\overline{9}$.

- This proof is based on the fact that 0 is the only non-negative number less than all the reciprocals of the positive integers, or equivalently that there is no number greater than every integer. This is the **Archimedean property**, which holds for the rational and the real numbers.

Proof. We write the number 0,999... with n digits after the decimal point as $0,(9)_n$, hence $0,(9)_1 = 0.9$, $0,(9)_2 = 0.99$, $0,(9)_3 = 0.999$, and so on.

Given $\frac{1}{10^n} = 0,0\dots 01$, with n digits after the decimal point, the addition rules for decimal numbers imply

$$0,(9)_n + \frac{1}{10^n} = 1 \text{ moreover } 0,(9)_n < 1, \forall n \in \mathbb{N}.$$

We must prove that 1 is the smallest number that is not less than all the $0,(9)_n$. For this it is enough to prove that, if a number x is not greater than 1 and not less than all the $0,(9)_n$, then $x = 1$.

So let x be such that

$$0,(9)_n \leq x \leq 1$$

for every positive integer n . Hence

$$1 - 1 \leq 1 - x \leq 1 - 0,(9)_n$$

which, using basic arithmetic and the first equality established above, simplifies to

$$0 \leq 1 - x \leq \frac{1}{10^n}$$

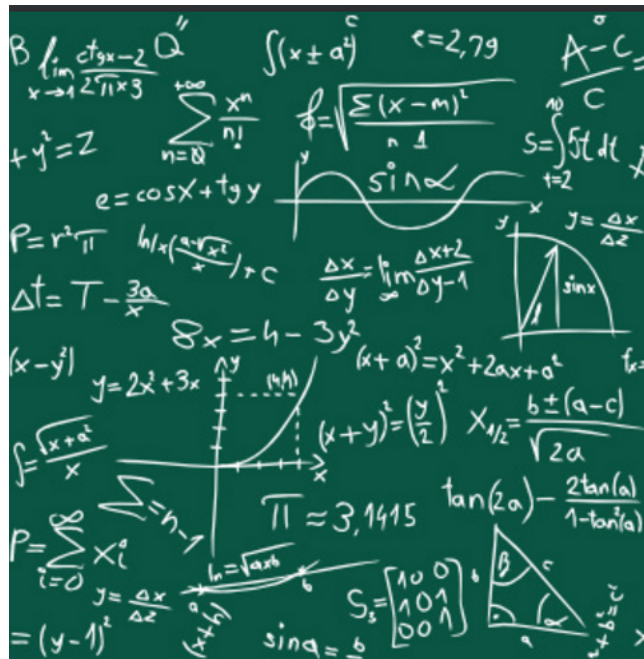
This implies that the difference between 1 and x is less than the reciprocal of any positive integer. Hence this difference must be zero, and therefore $x = 1$; which in turn implies

$$0.999\dots = 1$$

□



Basics of logic and proof techniques



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Logical symbols

- a) the symbol " $\forall$ " is called the *universal quantifier* and is read "for every", "for all", "for each"
- b) the symbol " $\exists$ " is called the *existential quantifier* and is read "there exists", "there exist"
- c) the symbol " $\Rightarrow$ " is called *logical implication* and is read "implies" or "if ... then"

- a) the symbol " $:$ " is read "such that"
- b) the symbol " $\in$ " is read "belongs to"
- c) the symbol " $\notin$ " is read "does not belong to"

- a) the symbol " $\vee$ " is called *logical disjunction* and is read "or"
- b) the symbol " $\wedge$ " is called *logical conjunction* and is read "and"
- c) the symbol " $\neg$ " is called *logical negation* and is read "not".

2 Universal implications and proofs

Predicates (or properties) and **propositions** (or statements)

- Consider the following assertion:

"the natural number n is odd" (1)

and let us ask whether it is true. Obviously the answer is: "it depends on n ". Indeed, in (1) the symbol n represents a variable that can take different values and make the assertion **true** or **false**.

A sentence of this kind is called a **predicate** (or *property*): its truth or falsity depends on the values of the **variable(s)** appearing in it.

- Now consider the following assertion:

“for every natural number n , if n is odd then n^2 is odd” (2)

which we can write more formally as follows:

$$\forall n \in \mathbb{N} (n \text{ odd} \Rightarrow n^2 \text{ odd}) \quad (3)$$

In this case we say that the variable n is not free, but bound by the *quantifier* $\forall$. As a consequence, (3) is true or false “once and for all” and is called a **proposition** (or *statement*).

In particular, the components of (3) are the set $\mathbb{N}$, two predicates defined on $\mathbb{N}$ given by

$$p(n): “n \text{ odd}” \text{ and } q(n): “n^2 \text{ odd}”$$

and the implication

$$p(n) \Rightarrow q(n).$$

Definition 1: of universal implication

In general, a statement involving a **set** A , **two predicates** $p(x)$ and $q(x)$ whose argument x ranges over A , and the **logical structure**:

$$\forall x \in A (p(x) \Rightarrow q(x)) \quad (4)$$

is called a **universal implication**.

Most **theorems** consist of **universal implications**, in which the predicate $p(x)$ plays the role of the **hypothesis** and the predicate $q(x)$ plays the role of the **thesis** (conclusion).

- In particular, (3) is a proposition (or a statement):

Proposition 1

$$\forall n \in \mathbb{N} (n \text{ odd} \Rightarrow n^2 \text{ odd}) \quad (5)$$

In this case one is easily convinced that the proposition is true, but how can we prove it rigorously?

- For example, is it enough to observe that 3 is odd and $3^2 = 9$ is odd to claim that the proposition is true? Certainly not, since the proposition requires the universal implication to hold for every natural number. However, there are infinitely many odd numbers: how can we prove a universal implication for infinitely many numbers?
- The key procedure is this: we consider a generic n satisfying the hypothesis (being odd) and we prove that n satisfies the thesis (its square is odd).

Let us see how to proceed to prove the previous proposition:

Proof. Let n be odd; we prove that then n^2 is odd.

Any odd number can be written in the form $2k + 1$, for a suitable $k \in \mathbb{N}$. We also observe that $2k$ is an even number for any $k \in \mathbb{N}$.

So let $n = 2k + 1$ be an odd number ($k \in \mathbb{N}$); we therefore need to write n^2 as an even integer plus one. We have:

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2(2k^2 + 2k)$ is an even integer, n^2 is odd. □

To prove the correctness of a universal implication such as (4), we consider a generic x satisfying the hypothesis $p(x)$ and we try to prove that the thesis $q(x)$ is true.

- Let us now prove a similar relation for even numbers:

Proposition 2

$$\forall n \in \mathbb{N} \ (n \text{ even} \Rightarrow n^2 \text{ even}) \quad (6)$$

Proof. Let n be even; we prove that then n^2 is even.

So let $n = 2k$ be an even number ($k \in \mathbb{N}$); we therefore need to write n^2 as an even integer. We have:

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

Since $2(2k^2)$ is an even integer, n^2 is even. □

2.1 Counterexamples

Counterexamples are an important technique for proving the **falsity** of a universal implication.

- Let us ask, for example, whether the following universal implication is true or false:

$$\forall n \in \mathbb{N} (n \text{ prime} \Rightarrow n \text{ odd}) \quad (7)$$

A moment's reflection shows that **this proposition is false**. Indeed, the number 2 is prime but it is even.

To claim that a universal implication is true, a proof is needed (an example is not enough), whereas to prove that a universal implication is false, one counterexample is enough.

- The universal implication requires that every x satisfying the hypothesis also satisfies the thesis: therefore, if we find even a single example of x that satisfies the hypothesis but not the thesis, this means that the universal implication is false. Not “false in one case”, but simply “false”, because the universal implication is true or false once and for all.

Definition 2: of counterexample

*In general, an example that satisfies the hypothesis but not the thesis of a universal implication, and that therefore proves its falsity, is called a **counterexample**.*

- The formal proof that the previous universal implication is false is the following:

Proof. The number 2 is a counterexample to the universal implication: “For every natural number n , if n is prime then n is odd”.

□

The negation of the proposition

“for every $x \in A$, if $p(x)$ holds then $q(x)$ holds”

$$\forall x \in A (p(x) \Rightarrow q(x))$$

is the proposition

“there exists $x \in A$ for which $p(x)$ holds and $q(x)$ does not hold”

$$\exists x \in A (p(x) \wedge \neg q(x))$$

This particular x is a **counterexample**.

3 Law of contraposition

- It is an indirect proof technique

The universal implication

$$\forall x \in A \ (p(x) \Rightarrow q(x)) \quad (8)$$

is logically equivalent to

$$\forall x \in A \ (\text{not } q(x) \Rightarrow \text{not } p(x)) \quad (9)$$

The second implication is called the **contrapositive** of the first.

- For example: since we know that the universal implication (proposition 5) holds

$$\forall n \in \mathbb{N} \ (n \text{ odd} \Rightarrow n^2 \text{ odd})$$

the following proposition holds:

Proposition 3

$$\forall n \in \mathbb{N} \ (n^2 \text{ even} \Rightarrow n \text{ even}) \quad (10)$$

Proof. Let n^2 be even; we prove that then n is even. We prove (10) relying on the truth of (5).

If n is not even then it is odd and therefore n^2 is odd by (5). This case contradicts the hypothesis that n^2 is even (hence it cannot occur).

Consequently n is even and (10) is proved. □

- The reasoning used in the previous proof has general validity, and shows precisely that if (8) is true then (9) is true; moreover, if the second is true then the first is true (because “not not $p(x)$ ” is logically equivalent to $p(x)$), so the two are logically equivalent.

The equivalence between (8) and (9) is called the **law of contraposition**. It is a method of *indirect proof* that consists in proving (9) to show that (8) is true (it requires proving that the negation of the thesis implies the negation of the hypothesis).

When using the law of contraposition, one must be able to construct the **correct negation** of a given proposition or property.

- Given any two predicates or properties $p(x)$ and $q(x)$, we list schematically some **rules** for constructing the **negation** of a proposition or property.

The negation of

“for every $x \in A$, $p(x)$ holds”

$$\forall x \in A (p(x))$$

is

“there exists $x \in A$ for which $p(x)$ does not hold”

$$\exists x \in A (\neg p(x))$$

The negation of

“there exists $x \in A$ for which $p(x)$ holds”

$$\exists x \in A (p(x))$$

is

“for every $x \in A$, $p(x)$ does not hold”

$$\forall x \in A (\neg p(x))$$

The negation of

“ $p(x)$ holds and $q(x)$ holds”

$$(p(x) \wedge q(x))$$

is

“ $p(x)$ does not hold or $q(x)$ does not hold”

$$(\neg p(x) \vee \neg q(x))$$

The negation of

“ $p(x)$ holds or $q(x)$ holds”

$$(p(x) \vee q(x))$$

is

“ $p(x)$ does not hold and $q(x)$ does not hold”

$$(\neg p(x) \wedge \neg q(x))$$

4 Sufficient conditions and necessary conditions

Definition 3: of sufficient condition

A **sufficient condition** is one that, if satisfied, guarantees the truth of the proposition.

Definition 4: of necessary condition

A **necessary condition** is one that must be satisfied for the proposition to be true.

Let $p(x)$ and $q(x)$ be any two predicates or properties. If $p(x)$ implies $q(x)$, formally:

$$p(x) \Rightarrow q(x)$$

then:

- $p(x)$ is a **sufficient condition** for $q(x)$
- $q(x)$ is a **necessary condition** for $p(x)$

- We have seen that the following universal implication is true (proposition (5)):

$$\forall n \in \mathbb{N} (n \text{ odd} \Rightarrow n^2 \text{ odd})$$

Hence “ n odd” is a sufficient condition for “ n^2 odd” and “ n^2 odd” is a necessary condition for “ n odd”.

- Let us now prove that the following proposition is also true:

Proposition 4

$$\forall n \in \mathbb{N} (n^2 \text{ odd} \Rightarrow n \text{ odd}) \quad (11)$$

Proof. Let n^2 be odd; we prove that then n is odd.

So let $n^2 = 2(2k^2 + 2k) + 1$ be an odd number ($k \in \mathbb{N}$), since $2(2k^2 + 2k)$ is an even number). We therefore need to write n as an even integer plus one. We have:

$$n = \sqrt{2(2k^2 + 2k) + 1} = \sqrt{4k^2 + 4k + 1} = \sqrt{(2k + 1)^2} = 2k + 1.$$

Since $2k + 1$ is an odd integer, n is odd.

□

In this way we have proved that “ n^2 odd” is a necessary and sufficient condition for “ n odd” and also that “ n odd” is a necessary and sufficient condition for “ n^2 odd”:

Proposition 5

$$\forall n \in \mathbb{N} (n \text{ odd} \iff n^2 \text{ odd}) \quad (12)$$

- We have seen that the following universal implication is true (proposition (6)):

$$\forall n \in \mathbb{N} (n \text{ even} \Rightarrow n^2 \text{ even})$$

Hence “ n even” is a sufficient condition for “ n^2 even” and “ n^2 even” is a necessary condition for “ n even”.

- Let us now prove that the following proposition is also true (direct proof, without using the law of contraposition as seen previously in (10)):

$$\forall n \in \mathbb{N} (n^2 \text{ even} \Rightarrow n \text{ even})$$

Proof. Let n^2 be even; we prove that then n is even.

So let $n^2 = 2 (2 k^2)$ be an even number ($k \in \mathbb{N}$); we therefore need to write n as an even integer. We have:

$$n = \sqrt{2 (2 k^2)} = \sqrt{4 k^2} = 2 k.$$

Since $2 k$ is an even integer, n is even. □

In this way we have proved that “ n^2 even” is a necessary and sufficient condition for “ n even” and also that “ n even” is a necessary and sufficient condition for “ n^2 even”:

Proposition 6

$$\forall n \in \mathbb{N} \ (n \text{ even} \iff n^2 \text{ even}) \quad (13)$$

Example 1: necessary and sufficient conditions

For example, for a square matrix of real numbers, the fact that its determinant is different from zero is a necessary and sufficient condition for it to be invertible.

Proposition 7

$$\forall n \in \mathbb{N} \ (n \text{ prime} > 2 \Rightarrow n \text{ odd}) \quad (14)$$

Proof. Let $n > 2$ be a prime number; we prove that then n is odd.

If n is not odd, it is even. But no even number greater than two is prime, a fact that contradicts the hypothesis that n is prime and greater than two (hence this case cannot occur).

Consequently n is odd. □

- Hence “ n odd” is a necessary condition for “ n prime > 2 ” and “ n prime > 2 ” is a sufficient condition for “ n odd”.

- However, “ n odd” does not imply “ n prime > 2 ”, since for example the number 9 is not prime (counterexample). That is:

$$\forall n \in \mathbb{N} (n \text{ odd} \not\Rightarrow n \text{ prime number greater than } 2)$$

Hence “ n odd” is a necessary but not sufficient condition for “ n prime > 2 ” and “ n prime > 2 ” is a sufficient but not necessary condition for “ n odd”.

Proposition 8

$$\forall n \in \mathbb{N} (n \text{ divisible by } 6 \Rightarrow n \text{ even}) \quad (15)$$

Proof. Let n be a number divisible by six; we prove that then n is even.

If n is not even, it is odd. But no odd number is divisible by six, a fact that contradicts the hypothesis that n is divisible by six (hence this case cannot occur).

Consequently n is even. □

- Hence “ n even” is a necessary condition for “ n divisible by 6” and “ n divisible by 6” is a sufficient condition for “ n even”.
- However, “ n even” does not imply “ n divisible by 6”, since for example the number 2 is even but not divisible by six (counterexample). That is:

$$\forall n \in \mathbb{N} (n \text{ even} \not\Rightarrow n \text{ divisible by } 6)$$

Hence “ n even” is a necessary but not sufficient condition for “ n divisible by 6” and “ n divisible by 6” is a sufficient but not necessary condition for “ n even”.

Example 2: necessary/sufficient but not sufficient/necessary conditions

Being a square implies being a rectangle:

$$\text{being a square} \Rightarrow \text{being a rectangle}$$

since all squares are rectangles.

Hence “being a rectangle” is a necessary condition for “being a square” and “being a square” is a sufficient condition for “being a rectangle”.

But being a rectangle does not imply being a square

$$\text{being a rectangle} \not\Rightarrow \text{being a square}$$

because there are rectangles that are not squares.

Hence “being a rectangle” is not a sufficient condition (but it is a necessary one) for “being a square” and “being a square” is not a necessary condition (but it is a sufficient one) for “being a rectangle”.

5 Proofs by contradiction

- It is an indirect proof technique

Definition 5: of proof by contradiction

*In general, a **proof by contradiction** consists in assuming that the hypothesis of the theorem and the negation of the thesis are true, and deducing from these facts a contradiction of any kind.*

- We illustrate **proof by contradiction** with the following theorem.

Theorem 1

There is no rational number whose square is 2.

Proof. Suppose, by contradiction, that there exists a number $r \in \mathbb{Q}$ such that $r^2 = 2$.

We can write $r = \frac{n}{m}$ with $n, m \in \mathbb{Z}$, $m \neq 0$.

Moreover, we can assume that the fraction $\frac{n}{m}$ is already reduced to lowest terms, that is, “simplified” (in other words: n , m have no common factors).

We then have the chain of implications:

$$\left(\frac{n}{m}\right)^2 = 2$$

$$n^2 = 2 m^2$$

so n^2 is even; but then by (13) n is also even and we can write $n = 2k$ for some $k \in \mathbb{Z}$.

Hence the relation $n^2 = 2 m^2$ can be rewritten as:

$$(2k)^2 = 2 m^2$$

$$4 k^2 = 2 m^2$$

$$m^2 = 2 k^2.$$

So m^2 is even. But then by (13) m is also even.

Therefore both n and m are even, and this is a **contradiction**, because we had assumed that the fraction $\frac{n}{m}$ had already been simplified.

The proof can be found in Euclid’s Elements (around 300 BC).

□

6 Logic and sets

The language of logic and the language of sets are two sides of the same coin.

6.1 Logical implication and set inclusion

- There is a parallel between the relation of set inclusion and logical implication. To explain it, consider the universal implication:

$$\forall n \in \mathbb{N} \ (n \text{ divisible by } 4 \Rightarrow n \text{ divisible by } 2) \quad (16)$$

If we denote by:

$$D_4 = \{ n \in \mathbb{N} : n \text{ is divisible by } 4 \} \text{ and } D_2 = \{ n \in \mathbb{N} : n \text{ is divisible by } 2 \}$$

we can observe that the universal implication written above is equivalent to the assertion:

$$D_4 \subseteq D_2$$

Indeed, this inclusion means that every element belonging to D_4 also belongs to D_2 , that is, that every natural number divisible by 4 is also divisible by 2.

The universal implication:

“for every $x \in A$, if $p(x)$ holds then $q(x)$ holds”

is equivalent to the set inclusion:

$$\{ x \in A : p(x) \text{ is true} \} \subseteq \{ x \in A : q(x) \text{ is true} \}.$$

6.2 Equality of sets and universal implications

- Proving the equality of two sets, i.e., $A = B$, requires proving two universal implications. Formally:

$$\forall x (x \in A \Rightarrow x \in B) \quad \text{and} \quad \forall x (x \in B \Rightarrow x \in A). \quad (17)$$

- Stating that $A \subsetneq B$ means stating that “Every element that belongs to A also belongs to B , and there exists an element of B that does not belong to A ”. Formally:

$$\forall x (x \in A \Rightarrow x \in B) \quad \text{and} \quad \exists x \in B : x \notin A. \quad (18)$$

6.3 Set operations and logical operations

There is a relation between operations on sets and logical operations. Precisely:

1. **Set intersection** is defined by means of “and” (**logical conjunction**).
2. **Set union** is defined by means of “or” (**logical disjunction**).
3. **Set difference** and the **complement operation** are defined by means of “not” (**logical negation**).

- The distributive properties of union and intersection of sets:

Remark 1: distributive properties (sets)

Given three sets A, B and C we have:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

- They can be rewritten in terms of predicates by observing that the intersection symbol $\cap$ corresponds to the conjunction $\wedge$ (“and”) and that the union symbol $\cup$ corresponds to the disjunction $\vee$ (“or”).

Remark 2: distributive properties (predicates)

Given three predicates $p(x), q(x)$ and $r(x)$ we have:

$$p(x) \wedge (q(x) \vee r(x)) = (p(x) \wedge q(x)) \vee (p(x) \wedge r(x))$$

$$p(x) \vee (q(x) \wedge r(x)) = (p(x) \vee q(x)) \wedge (p(x) \vee r(x)).$$

- De Morgan's laws:

Proposition 9: De Morgan's laws (sets)

Given the sets $B, C \subseteq U$, we have

$$\overline{B \cap C} = \overline{B} \cup \overline{C}$$

$$\overline{B \cup C} = \overline{B} \cap \overline{C}$$

- They can be rewritten in terms of predicates by observing that the complement operation corresponds to the negation $\neg$ (“not”).

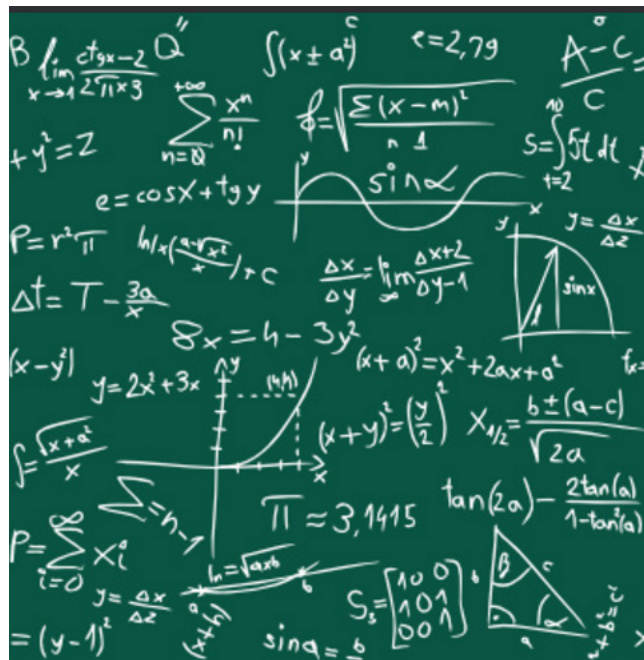
Proposition 10: De Morgan's laws (predicates)

Given two predicates $q(x)$ and $r(x)$ we have:

$$\neg(q(x) \wedge r(x)) = \neg q(x) \vee \neg r(x)$$

$$\neg(q(x) \vee r(x)) = \neg q(x) \wedge \neg r(x)$$

Number sets and intervals



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 The five main number sets

1.1 The natural numbers

1.2 The integers

$\mathbb{Z}$ is the set of **integers** (also called signed numbers), that is:

$$0, 1, -1, 2, -2, \dots$$

1.3 The rational numbers

$\mathbb{Q}$ is the set of **rational numbers**, that is, of the fractions:

$$\frac{n}{m}, \text{ where } n, m \in \mathbb{Z}, \text{ and } m \neq 0.$$

Rational numbers can also be written in decimal form. For example:

$$\frac{3}{4} = 0,75, \text{ while } \frac{1}{3} = 0,333\dots = 0,\bar{3}.$$

A rational number, written in decimal form, may have after the decimal point a finite number of digits (different from zero), or an infinite number of digits different from zero, which however repeat *periodically*.

Remark 1

$$0,\bar{9} = 1$$

There are different proofs of this remark, based on different mathematical techniques.

Proof. A simple proof follows directly from the definition of 1 divided by 3; indeed, we have:

$$\begin{aligned}\frac{1}{3} &= 0,\bar{3} \\ \frac{1}{3} \cdot 3 &= 0,\bar{3} \cdot 3 \\ 1 &= 0,\bar{9}\end{aligned}$$

□

Proof. Using algebraic arguments we can write:

$$\begin{array}{ll}
 x = 0.999\dots & \\
 10x = 9.999\dots & \text{multiplying by 10} \\
 10x = 9 + 0.999\dots & \text{separating the integer part from the fractional part} \\
 10x = 9 + x & \text{by definition of } x \\
 9x = 9 & \text{subtracting } x \\
 x = 1 & \text{dividing by 9}
 \end{array}$$

□

Proof. A proof by contradiction is the following:

$$\begin{array}{l}
 0.\overline{9} \neq 1 \\
 0.\overline{9} \cdot 9 \neq 1 \cdot 9 \\
 0.\overline{9} \cdot 9 + 0.\overline{9} \neq 1 \cdot 9 + 0.\overline{9} \\
 0.\overline{9} \cdot 9 + 0.\overline{9} \neq 9.\overline{9} \\
 0.\overline{9} \cdot (9 + 1) \neq 9.\overline{9} \\
 0.\overline{9} \cdot (10) \neq 9.\overline{9} \\
 9.\overline{9} \neq 9.\overline{9} \quad \text{contradiction!}
 \end{array}$$

□

Other proofs start from the assumption that two numbers are identical if and only if their difference is equal to zero, and they are based on computing the value of $1 - 0.\overline{9}$.

Proof. We write the number $0.999\dots$ with n digits after the decimal point as $0.(9)_n$, hence $0.(9)_1 = 0.9$, $0.(9)_2 = 0.99$, $0.(9)_3 = 0.999$, and so on.

Given $\frac{1}{10^n} = 0.0\dots 01$, with n digits after the decimal point, the addition rules for decimal numbers imply

$$0.(9)_n + \frac{1}{10^n} = 1 \text{ moreover } 0.(9)_n < 1, \forall n \in \mathbb{N}.$$

We must prove that 1 is the smallest number that is not less than all the $0.(9)_n$. For this it is enough to prove that, if a number x is not greater than 1 and not less than all the $0.(9)_n$, then $x = 1$.

So let x be such that

$$0.(9)_n \leq x \leq 1$$

for every positive integer n . Hence

$$1 - 1 \leq 1 - x \leq 1 - 0.(9)_n$$

which, using basic arithmetic and the first equality established above, simplifies to

$$0 \leq 1 - x \leq \frac{1}{10^n}$$

This implies that the difference between 1 and x is less than the reciprocal of any positive integer. Hence this difference must be zero, and therefore $x = 1$; which in turn implies

$$0.999\dots = 1$$

□

This proof is based on the fact that 0 is the only non-negative number less than all the reciprocals of the positive integers, or equivalently that there is no number greater than every integer.

This is the **Archimedean property**, which holds for the rational and the real numbers.

1.4 The real numbers

$\mathbb{R}$ is the set of **real numbers**, that is, those which, written in decimal form, have after the decimal point any sequence of digits different from zero, possibly even *infinite* and *non-periodic*.

That numbers of this last kind exist (that is, real but not rational), can be understood by reflecting on examples such as:

$$0,10110111011110\dots$$

After the decimal point, the previous number has: one digit equal to 1, then 0, then two digits equal to 1, then 0, then three digits equal to 1 ... and so on. It is clear that this rule defines a decimal number precisely. On the other hand, the string of nonzero digits after the decimal point is neither finite nor periodic: therefore this number is irrational.

1.5 The complex numbers

$\mathbb{C}$ is the set of **complex numbers**, that is, numbers of the form $a + i b$, where a, b are real numbers, and i is the *imaginary unit*, that is, a number whose square is -1 .

Among the number sets introduced, the following inclusions hold:

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}.$$

As indicated by the symbols, all the inclusions are strict:

- there exist integers that are not natural numbers (the negative numbers),
- there exist rational numbers that are not integers (the proper fractions),
- there exist real numbers that are not rational (the irrational numbers),
- there exist complex numbers that are not real (the imaginary numbers).

2 Intervals

Definition 1: of interval

Given two real numbers a, b , an **interval** with endpoints a and b is one of the following sets:

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}, \quad [a, b) = \{x \in \mathbb{R} : a \leq x < b\}$$

$$(a, b] = \{x \in \mathbb{R} : a < x \leq b\}, \quad (a, b) = \{x \in \mathbb{R} : a < x < b\}$$

- As can be seen, the square (round) bracket at one of the two endpoints indicates that this endpoint is included in (excluded from) the interval.

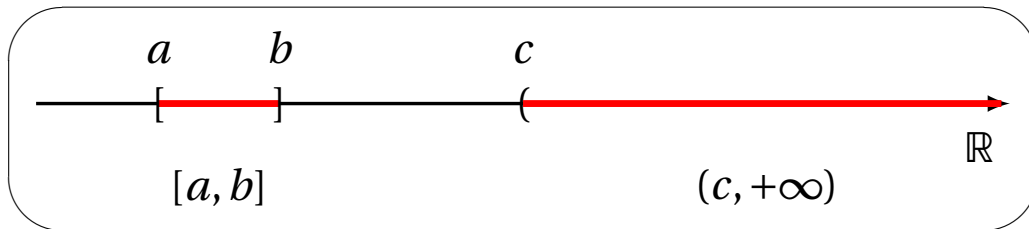
- The intervals $[a, b]$ are called **closed**; the intervals (a, b) are called **open**.
- All the intervals listed above are bounded; half-lines are also called (unbounded) intervals, for example:

$$(-\infty, b) = \{x \in \mathbb{R} : x < b\}$$

$$[a, +\infty) = \{x \in \mathbb{R} : x \geq a\}$$

or the whole line

$$\mathbb{R} = (-\infty, +\infty)$$



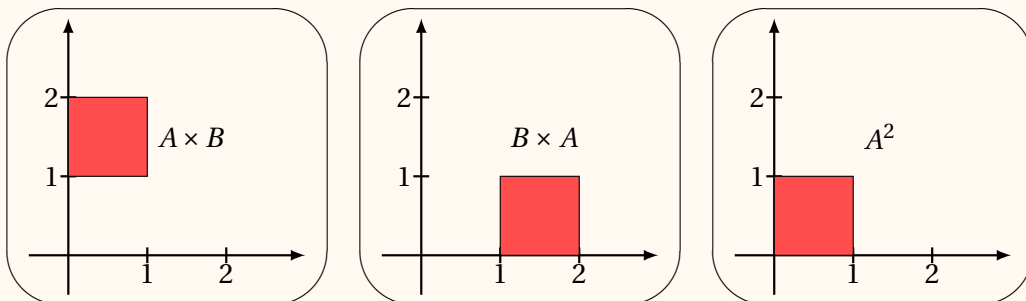
It can be proved that the intervals, bounded or unbounded, are exactly the subsets I of $\mathbb{R}$ that satisfy the following property (called **connectedness**):

$$x_1 < x_2 < x_3, \text{ if } x_1, x_3 \in I, \text{ then } x_2 \in I$$

- In what follows we will sometimes consider the Cartesian product of two (or more) intervals, which can be given the geometric meaning of a rectangle (in two dimensions) or of a parallelepiped (in three dimensions).

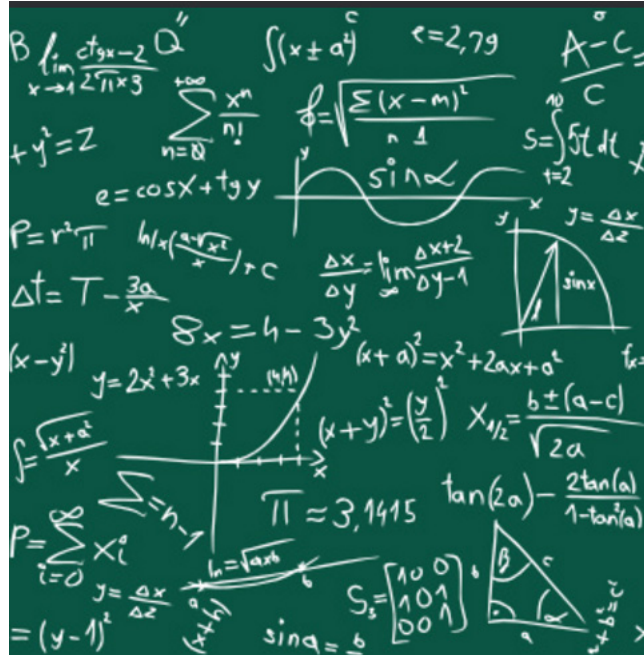
Example 1: Cartesian product of intervals

Let $A = [0, 1]$, $B = [1, 2]$; the figure



illustrates the sets $A \times B$, $B \times A$, $A \times A$, also denoted by A^2 . In general, $A \times B$ is different from $B \times A$.

Binary relations



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Binary relations

Definition 1: of binary relation

Given two sets A and B , a **binary relation** R is a subset of the Cartesian product $A \times B$

When we say that R is a binary relation on a single set A , we mean that R is a subset of $A \times A$.

- A binary relation will simply be called a relation (however, non-binary relations also exist).

Example 1:

- The relation “less than” (“ $<$ ”) on the natural numbers is the set:

$$R_{<} = \{(a, b) : a, b \in \mathbb{N} \text{ and } a < b\}.$$

- The relation “less than or equal to” (“ $\leq$ ”) on the natural numbers is the set:

$$R_{\leq} = \{(a, b) : a, b \in \mathbb{N} \text{ and } a \leq b\}.$$

- The relation “is a subset of” $R_{\subseteq}$ on the power set of the natural numbers (denoted by $2^{\mathbb{N}}$) is the set:

$$R_{\subseteq} = \{(A, B) : A, B \in 2^{\mathbb{N}} \text{ and } A \subseteq B\}.$$

- A relation $R \subseteq A \times A$ is **reflexive** if:

$$\forall a \in A, \quad (a, a) \in R$$

- A relation $R \subseteq A \times A$ is **symmetric** if:

$$\forall a, b \in A, \quad (a, b) \in R \Rightarrow (b, a) \in R$$

- A relation $R \subseteq A \times A$ is **transitive** if:

$$\forall a, b, c \in A, \quad (a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \in R$$

Example 2:

- The relation $R_{\leq}$ is reflexive, but $R_{<}$ is not.
- The relations $R_{<}$ and $R_{\leq}$ are not symmetric.
- The relations $R_{<}$ and $R_{\leq}$ are transitive, but for example the relation:

$$R_{ab}^1 = \{(a, b) : a, b \in \mathbb{N} \text{ and } a = b - 1\}$$

is not, e.g., $(3, 4) \in R_{ab}^1$ and $(4, 5) \in R_{ab}^1$ but $(3, 5) \notin R_{ab}^1$.

2 Partial order relations

- A relation $R \subseteq A \times A$ is **antisymmetric** if:

$$\forall a, b \in A, \quad (a, b) \in R \text{ and } (b, a) \in R \Rightarrow a = b$$

Example 3:

- The relation $R_{\leq}$ is antisymmetric, since $a \leq b$ and $b \leq a$ imply $a = b$.

Definition 2: of partial order relation

*A reflexive, antisymmetric and transitive relation is a **partial order relation**.*

Example 4:

- The relation $R_{\leq}$ is a partial order relation, but the relation $R_{<}$ is not, since it is not reflexive.

Remark 1

The relation $R_{\leq}$ is a partial order relation

Proof. We must prove that the relation is reflexive, antisymmetric and transitive.

- For it to be reflexive we must prove that $(S, S) \in R_{\subseteq}$, which is true since $S \subseteq S$.
- For it to be antisymmetric we must prove that if $S_1 \neq S_2$ then $S_1 \not\subseteq S_2$ or $S_2 \not\subseteq S_1$ or both, the negation of the desired property.

Since $S_1 \neq S_2$:

either there exists some element that is in S_1 but not in S_2 , hence $S_1 \not\subseteq S_2$

or there exists some element that is in S_2 but not in S_1 , hence $S_2 \not\subseteq S_1$

or both of the previous options hold

- For it to be transitive we must prove that $(S_1, S_2) \in R_{\subseteq}$ and $(S_2, S_3) \in R_{\subseteq}$ imply $(S_1, S_3) \in R_{\subseteq}$. Clearly, since $S_1 \subseteq S_2$ and $S_2 \subseteq S_3$, we have $S_1 \subseteq S_3$.

□

Definition 3: of partially ordered set

A **partially ordered set** is a pair consisting of a set and a partial order relation defined on it.

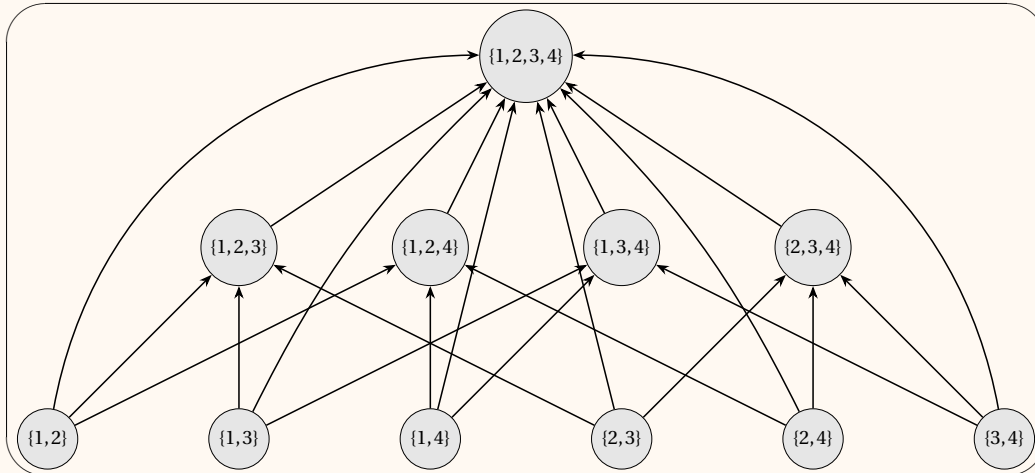
Example 5:

- The sets of natural, rational or real numbers with the relation $R_{\leq}$ are partially ordered sets.
- The relation “is a descendant of” defined on a subset of people is a partial order relation (if we consider individuals as descendants of themselves). Consequently, the subset of people considered with the relation “is a descendant of” is a partially ordered set.

Relations can be represented by a **directed graph**, where the **vertices** are the elements of the set on which the relation R is defined and an **arc** (a, b) means that $(a, b) \in R$. If the graph is acyclic, then the relation is a partial order relation.

Example 6:

The directed acyclic graph associated with the relation $R_{\subseteq}$ on the set $\{1, 2, 3, 4\} \subseteq \mathbb{N}$ is the following (the sets consisting of a single number and the empty set are not shown in the figure):



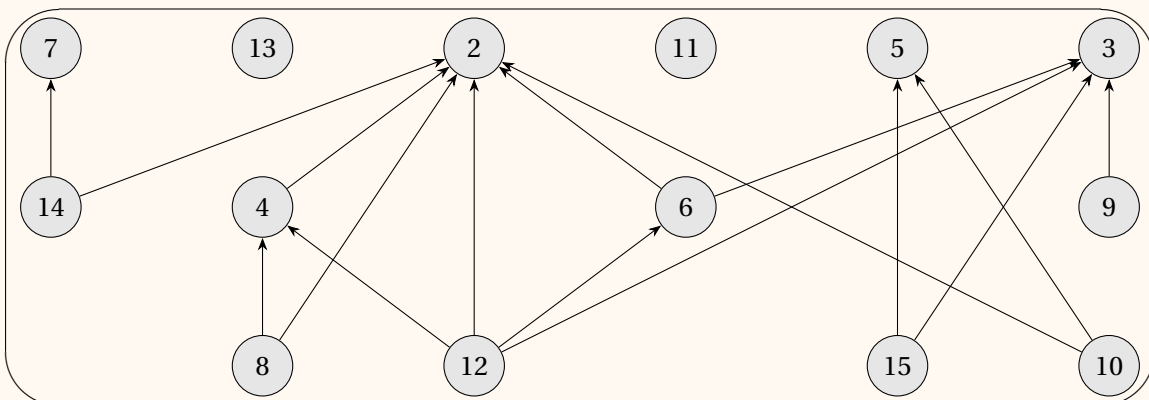
- In a partially ordered set there may be no single **maximum element**, that is, an element a such that:

$$\forall b \in A, \quad (b, a) \in R$$

A partially ordered set may therefore contain several maximal elements a such that, for no $b \in A$ with $b \neq a$, we have $(a, b) \in R$.

Example 7:

The directed acyclic graph associated with the partial order relation “is a divisor of” on the set $\{2, 3, \dots, 15\} \subseteq \mathbb{N}$ is the following:



Example 8:

Given a set of boxes of different sizes, the relation “one box is contained in the other” on the set of boxes may have several maximal boxes, that is, boxes that are not contained in any other box.

3 Total order relations

Definition 4: of total relation

A relation R on a set A is a **total relation** if:

$$\forall a, b \in A, \quad (a, b) \in R \text{ or } (b, a) \in R \text{ (or both)}$$

Example 9:

- The relation $R_{\leq}$ is a total relation.
- The relation $R_{\subseteq}$ is not a total relation since, taking for example $S_1 = \{1, 2\}$ and $S_2 = \{2, 3\}$, $(S_1, S_2) \notin R_{\subseteq}$ and $(S_2, S_1) \notin R_{\subseteq}$.
- The relation “is a descendant of” is not a total relation since there exist pairs of individuals (a, b) for which neither a descends from b nor b descends from a .

Definition 5: of total order relation

A partial order relation that is also a total relation is a **total order relation**.

Example 10:

- The relation $R_{\leq}$ is a total order relation.

Definition 6: of totally ordered set

A **totally ordered set** is a pair consisting of a set and a total order relation defined on it.

Example 11:

- The sets of natural, rational or real numbers with the relation $R_{\leq}$ are totally ordered sets.

4 Functions

Definition 7: of function

Given two sets A and B , a binary relation f on A and B is a **function** if, for each $a \in A$, there exists one and only one $b \in B$ such that $(a, b) \in f$.

Definition 8: of domain and codomain

The set A is called the **domain** of f , and the set B is called the **codomain** of f .

- We write:

$$f : A \rightarrow B$$

and if $(a, b) \in f$, we write:

$$b = f(a)$$

since b is uniquely determined by the choice of a .

- Intuitively, the function f assigns an element of B to each element of A . No element of A is associated with two different elements of B . The same element of B can however be assigned to different elements of A .

Example 12:

- The binary relation:

$$f = \{(a, b) : a, b \in \mathbb{N} \text{ and } b = a \bmod 2\}$$

is a function $f : \mathbb{N} \rightarrow \{0, 1\}$ since for all natural numbers a , there is exactly one value $b \in \{0, 1\}$ such that $b = a \bmod 2$. For example,

$$0 = f(0), \quad 1 = f(1), \quad 0 = f(2), \dots$$

Example 13:

- The binary relation

$$g = \{(a, b) : a, b \in \mathbb{N} \text{ and } a + b \text{ is even}\}$$

is not a function, since for example $(1, 3)$ and $(1, 5)$ are both in g . In other words, for $a = 1$, we do not have one and only one b such that $(a, b) \in g$.

Definition 9: of argument and value

Given a function $f : A \rightarrow B$, if $b = f(a)$, we say that $a \in A$ is the **argument** of f and that $b \in B$ is the **value** of f associated with a .

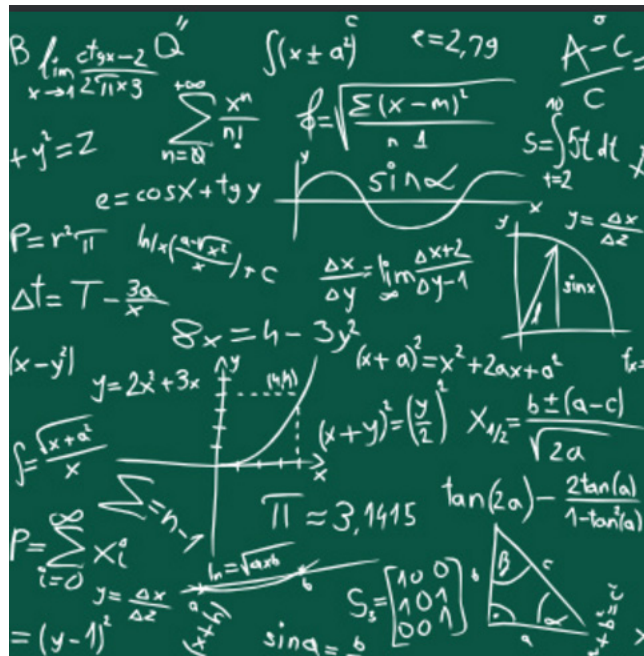
- We can **define a function** by *directly defining its value* for all the *elements* of its *domain*.

Example 14:

For example, we can define $f(n) = 2n$ for $n \in \mathbb{N}$, which means:

$$f = \{(n, 2n) : n \in \mathbb{N}\} \text{ and } f : \mathbb{N} \rightarrow \mathbb{N}$$

Ordered fields, supremum/infimum and the completeness axiom



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Properties R_1 and R_2

- We now begin to study more closely the **structure of the number sets** that we introduced earlier, and in particular: the set $\mathbb{Q}$ of **rational numbers** and the set $\mathbb{R}$ of **real numbers**.
- We denote by a, b, c three generic **rational** or **real** numbers.

$R_1 \rightarrow$ In $\mathbb{Q}$ and $\mathbb{R}$ an operation is defined (called **addition** or **sum**) which has the following four properties:

1 $\forall a, b, \quad a + b = b + a$ (**commutative property**)

2 $\forall a, b, c, \quad (a + b) + c = a + (b + c)$ (**associative property**)

3 there exists a **neutral element for the sum**, denoted by 0, such that:

$$\forall a, \quad a + 0 = a$$

4 for every a there exists an **inverse** element of a **with respect to the sum**, called the **opposite** of a and denoted by $-a$, such that:

$$a + (-a) = 0$$

$R_2 \rightarrow$ In $\mathbb{Q}$ and $\mathbb{R}$ an operation is defined (called **multiplication** or **product**) which has the following four properties:

1 $\forall a, b, \quad a \cdot b = b \cdot a$ (**commutative property**)

2 $\forall a, b, c, \quad (a \cdot b) \cdot c = a \cdot (b \cdot c)$ (**associative property**)

3 there exists a **neutral element for the product**, denoted by 1, such that:

$$\forall a, \quad a \cdot 1 = a$$

4 for every $a \neq 0$ there exists an **inverse** element of a **with respect to the product**, called the **reciprocal** of a and denoted by a^{-1} , such that:

$$a \cdot a^{-1} = 1$$

- The operations of sum and product are linked by the following property:

$$\forall a, b, c, \quad (a + b) \cdot c = a \cdot c + b \cdot c \quad (\text{distributive property})$$

- From properties R_1 and R_2 follows the **possibility of performing the four fundamental operations without restrictions**:

1. **addition** and **multiplication** (defined above)

2. **subtraction**, by setting:

$$a - b = a + (-b)$$

3. **division**, by setting:

$$a/b = a \cdot b^{-1} \quad \text{provided that } b \neq 0$$

2 Commensurable magnitudes

In mathematics, the word **magnitude** denotes a **property** of a **geometric figure** that can be **measured**. Examples of magnitudes are: the length of a segment, the measure of an angle, the area of a plane figure or the volume of a solid.

Definition 1: of homogeneous magnitudes

Two magnitudes are **homogeneous** if they have the same dimension, i.e., if they can be expressed using the same unit of measurement.

- The length of a segment and the height of a solid are two homogeneous magnitudes; indeed, both can be expressed with the same unit of length.
- The area of a square and the volume of a cone are not homogeneous magnitudes; indeed, area is expressed in square meters or in other units of surface, which however cannot be used to express volume.

Definition 2: of commensurable and incommensurable magnitudes

Two homogeneous magnitudes a and b are called **commensurable** if their ratio is a rational number, i.e., if $a/b \in \mathbb{Q}$. They are instead called **incommensurable** if their ratio is not a rational number, i.e., if $a/b \notin \mathbb{Q}$.

Example 1: of commensurable magnitudes

The volume of a sphere and the volume of the cylinder circumscribed about the sphere are commensurable. In formulas, the volume V_s of the sphere as a function of its radius r is given by the formula:

$$V_s = \frac{4}{3} \pi r^3$$

The cylinder circumscribed about it has base radius equal to the radius r of the sphere and height h equal to twice the radius. The volume V_c of this cylinder is therefore given by the formula:

$$V_c = \pi r^2 h = \pi r^2 2r = 2\pi r^3$$

Taking the ratio we obtain:

$$\frac{V_s}{V_c} = \frac{\frac{4}{3} \pi r^3}{2\pi r^3} = \frac{2}{3} \in \mathbb{Q}$$

Example 2: of incommensurable magnitudes

The length of a circle and the length of its diameter are incommensurable. In formulas, denoting by d the diameter of the circle and by c the length of the circle, we have:

$$c = \pi d$$

Now, taking the ratio between the length of the circle and its diameter, we obtain:

$$\frac{c}{d} = \frac{\pi d}{d} = \pi \notin \mathbb{Q}$$

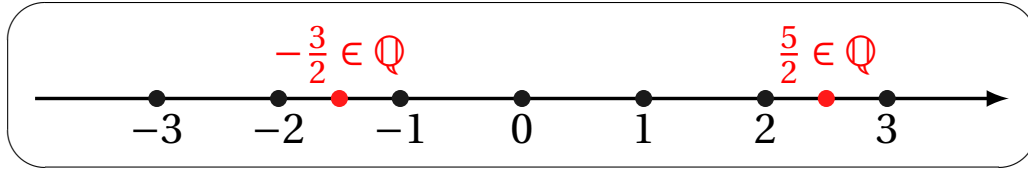
A proof that $\pi \notin \mathbb{Q}$ will be given later.

3 Geometric representation of $\mathbb{Q}$

A geometric representation of $\mathbb{Q}$ can be obtained by associating with each rational number a point of the **Euclidean line** (which can be imagined as a line in the plane of infinite length)

- With an arbitrarily chosen point of the Euclidean line we associate 0, and with another point, distinct from the first, we associate 1, thus identifying the oriented segment 01, which constitutes the **unit of measurement**
- At this point we have a **one-to-one correspondence** between the rational numbers and those points P of the line that are endpoints of oriented segments $0P$ **commensurable** with 01. They are clearly commensurable since:

$$\frac{P}{1} \in \mathbb{Q} \text{ with } P \in \mathbb{Q}$$



4 Total order relations

Remark 1

The sets of rational numbers $\mathbb{Q}$ and of real numbers $\mathbb{R}$ with the relation “less than or equal to” (“ $\leq$ ”) are *totally ordered sets*.

- The relations “less than or equal to” (“ $\leq$ ”) for the set of rational or real numbers:

$$R_{\leq} = \{(a, b) : a, b \in \mathbb{Q} \text{ and } a \leq b\}, \quad R_{\leq} = \{(a, b) : a, b \in \mathbb{R} \text{ and } a \leq b\}$$

are **partial order relations**. That is, they satisfy the following properties:

$$\forall a, \quad a \leq a \quad (\text{reflexive property})$$

$$\forall a, b, \quad a \leq b \text{ and } b \leq a \Rightarrow a = b \quad (\text{antisymmetric property})$$

$$\forall a, b, c, \quad a \leq b \text{ and } b \leq c \Rightarrow a \leq c \quad (\text{transitive property})$$

Moreover, the relations “less than or equal to” (“ $\leq$ ”) for $\mathbb{Q}$ and $\mathbb{R}$ are **total relations** since:

$$\forall a, b, \quad a \leq b \text{ or } b \leq a$$

- Consequently, the pair consisting of the set $\mathbb{Q}$ or $\mathbb{R}$ and the corresponding relation $R_{\leq}$ is a totally ordered set.

5 Property R_3

- We now highlight the properties of the ordering of rational or real numbers:

$R_3 \rightarrow$ In $\mathbb{Q}$ and $\mathbb{R}$ the total order relation “less than or equal to” (“ $\leq$ ”) is defined, compatible with the algebraic structure ^a, that is:

$$1. \forall a, b, c, \quad a \leq b \Rightarrow a + c \leq b + c$$

$$2. \forall a, b, c > 0, \quad a \leq b \Rightarrow a \cdot c \leq b \cdot c$$

^aan algebraic structure is a set, called the underlying set (of the structure), equipped with one or more operations

- We observe that all the rules of algebraic computation follow from the properties: R_1 , R_2 , R_3 .

Example 3:

For example, all the usual procedures used to solve inequalities are a consequence of the axioms and of the algebraic properties of the sum and the product, expressed by R_1 and R_2 .

6 Ordered fields

Definition 3: of (ordered) field

An **ordered field** is a set on which two operations (sum and product) and a total order relation are defined, satisfying the properties R_1 , R_2 , R_3 . A set with only the properties R_1 , R_2 is called a **field**.

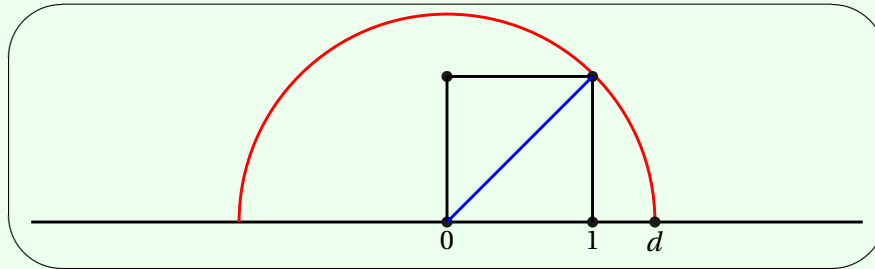
- Everything we have said so far about the operations of sum and product and about the total order relation “less than or equal to” (“ $\leq$ ”) holds both for the set of rational numbers and for the set of real numbers.

Remark 2

The sets of rational numbers $\mathbb{Q}$ and of real numbers $\mathbb{R}$ are ordered fields.

- From this point of view, therefore, $\mathbb{Q}$ and $\mathbb{R}$ appear to have similar properties. In what follows we will highlight the **property that substantially distinguishes** $\mathbb{R}$ from $\mathbb{Q}$ and that makes $\mathbb{R}$ the **right setting for developing mathematical analysis**.
- We begin by observing that the **set of rational numbers is inadequate for expressing the lengths of segments** (but also areas, volumes, times, speeds, etc.)

As we have seen, there exist magnitudes that are not **commensurable** with each other. The classic example is given by the diagonal and the side of a square: if the side measures 1, the abscissa d that measures the diagonal is not a rational number.



Indeed, by the Pythagorean Theorem we have:

$$d^2 = 1^2 + 1^2 = 2$$

but, as we have seen, there is no rational number whose square is 2. Hence the point d on the line does not represent any rational number. This means that, after “occupying” the points of the line with the rational numbers, there are still some **empty places** left on it.

7 Bounded sets, maxima and minima

Definition 4: of bounded set

Let E be a set contained in $\mathbb{Q}$ or in $\mathbb{R}$. The set E is called **bounded** if there exist two numbers m and M such that:

$$\forall x \in E, \quad m \leq x \leq M$$

- E is **bounded below** if there exists a number m such that:

$$\forall x \in E, \quad m \leq x$$

- E is **bounded above** if there exists a number M such that:

$$\forall x \in E, \quad x \leq M$$

- It makes no difference whether we require m and/or M to belong to $\mathbb{Q}$ or to $\mathbb{R}$, since the condition only concerns the existence of such numbers.

Definition 5: of maximum and minimum of a set

An element $x_M \in E$ is called the **maximum** of E if:

$$\forall x \in E, \quad x \leq x_M$$

An element $x_m \in E$ is called the **minimum** of E if:

$$\forall x \in E, \quad x_m \leq x$$

The existence of the maximum and of the minimum of a set implies that the set is bounded, that is:

$$\text{existence of maximum and minimum} \Rightarrow \text{bounded set} \quad (1)$$

but the converse is not true (we will give a counterexample later):

$$\text{bounded set} \not\Rightarrow \text{existence of maximum and minimum} \quad (2)$$

Hence the fact that a set is bounded is a necessary but not sufficient condition for the set to have a maximum and a minimum. Moreover, the existence of a maximum and a minimum is a sufficient but not necessary condition for a set to be bounded.

From the contrapositive of (1) we have:

$$\text{unbounded set} \Rightarrow \text{non-existence of maximum and minimum}$$

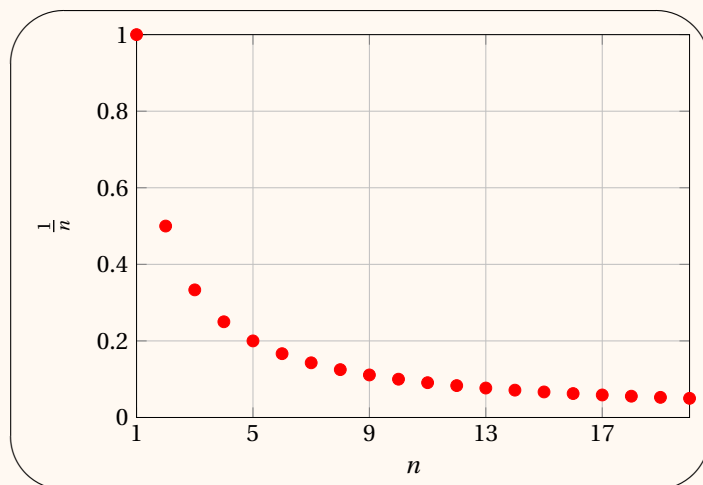
that is, if a set is not bounded, it has no maximum or it has no minimum.

Example 4: of maxima and minima

Example	set E	minimum	maximum
I)	$\mathbb{N}$	0	does not exist
II)	even integers	does not exist	does not exist

Example 5: of maxima and minima

Example	set E	minimum	maximum
III)	$\left\{ \frac{1}{n} : n \in \mathbb{N} \setminus \{0\} \right\}$	does not exist	1

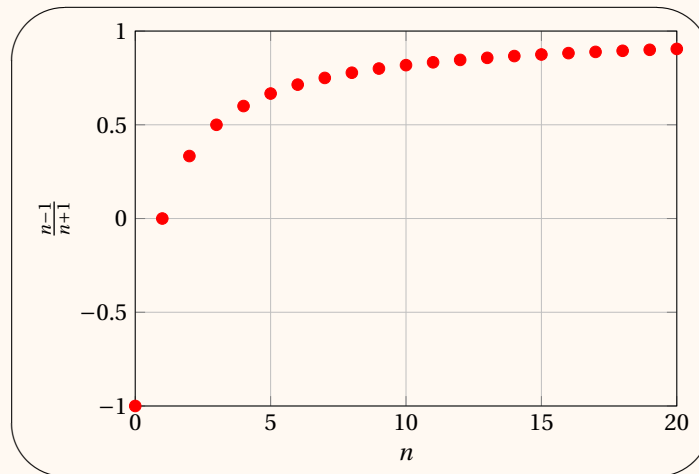


Example 6: of maxima and minima

Example	set E	minimum	maximum
IV)	$\left\{ \frac{n-1}{n+1} : n \in \mathbb{N} \right\}$	-1	does not exist

We have:

$$\frac{n-1}{n+1} = \frac{n+1-1-1}{n+1} = 1 - \frac{2}{n+1}$$



Note that sometimes, even though the set is bounded, it may have no maximum or no minimum.

Example 7: of maxima and minima

Example	set E	minimum	maximum
V)	$\{x \in \mathbb{R} : 27 \leq x^3\}$	3	does not exist

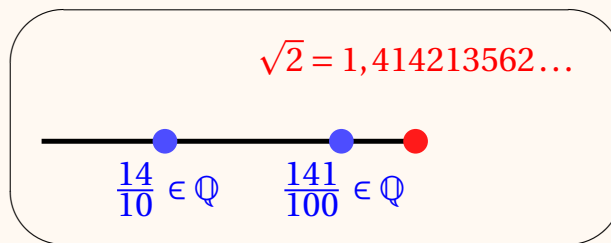
We have:

$$27 \leq x^3 \iff \sqrt[3]{27} = 3 \leq x \quad \text{and} \quad 3 \in E$$

Example 8: of maxima and minima

Example	set E	minimum	maximum
VI)	$\{x \in \mathbb{Q} : x \geq 0, x < \sqrt{2}\}$	0	does not exist

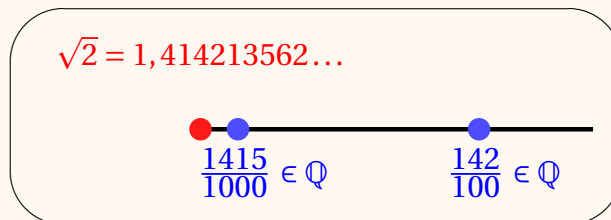
The value $\sqrt{2}$ is irrational, hence by truncating we obtain rational numbers that get closer and closer to $\sqrt{2}$, none of which has the property of being the maximum.



It is an example of a bounded set that has no maximum.

Example	set E	minimum	maximum
VII)	$\{x \in \mathbb{Q} : x > \sqrt{2}, x \leq 4\}$	does not exist	4

The value $\sqrt{2}$ is irrational, hence by truncating and rounding up we obtain rational numbers that get closer and closer to $\sqrt{2}$, none of which has the property of being the minimum.



It is an example of a bounded set that has no minimum.

8 Upper/lower bounds and suprema/infima

Definition 6: of upper bound of a set

Let E be a set contained in $\mathbb{Q}$ or in $\mathbb{R}$; a number k belonging to $\mathbb{Q}$ or to $\mathbb{R}$ respectively is called an **upper bound** of E if:

$$\forall x \in E, \quad x \leq k$$

Definition 7: of lower bound of a set

Let E be a set contained in $\mathbb{Q}$ or in $\mathbb{R}$; a number k belonging to $\mathbb{Q}$ or to $\mathbb{R}$ respectively is called a **lower bound** of E if:

$$\forall x \in E, \quad k \leq x$$

We observe that a set bounded above (below) has many upper (lower) bounds. Moreover, the upper (lower) bounds do not necessarily belong to the set itself.

Definition 8: of supremum

The **supremum** of E (denoted by $\sup E$) is the minimum of the upper bounds of E .

Definition 9: of infimum

The **infimum** of E (denoted by $\inf E$) is the maximum of the lower bounds of E .

We observe that the supremum (infimum) may not exist (unbounded sets). We also observe that if the set has a maximum (minimum), it coincides with the supremum (infimum).

Example 9: of suprema and infima

Example	set E	$\inf E$	$\sup E$	minimum	maximum
I)	$\{x \in \mathbb{Q} : x \geq 0, x < \sqrt{2}\}$	0	does not exist	0	does not exist

We have:

$$\sqrt{2} \notin E \quad \text{and} \quad \sqrt{2} \notin \mathbb{Q}$$

hence the maximum does not exist, and neither does the sup.

Example	set E	$\inf E$	$\sup E$	minimum	maximum
II)	$\{x \in \mathbb{R} : x \geq 0, x < \sqrt{2}\}$	0	$\sqrt{2}$	0	does not exist

We have:

$$\sqrt{2} \notin E \quad \text{and} \quad \sqrt{2} \in \mathbb{R}$$

that is, the maximum does not exist, but $\sqrt{2}$ is the minimum of the upper bounds and therefore it is the sup.

9 Axiomatic definition of the real numbers

A totally ordered number set X is said to have the **supremum property** (least upper bound property) if:

$R_4 \rightarrow$ Every non-empty set $E \subset X$ that is bounded above has a supremum in X

- It is easy to prove that, if this property holds, then it is also true that every non-empty subset of X that is bounded below has an infimum.
- It is not required that X itself has a supremum (for example, in the case $X = \mathbb{Q}$ or $\mathbb{R}$ this is certainly false!), but that every non-empty subset of X that is bounded above has one.

Example VI shows that $\mathbb{Q}$ certainly does not have the supremum property. On the other hand, $\mathbb{R}$ has this property. ^a

^aHowever, we will not present a formal proof.

- In the **axiomatic definition** of $\mathbb{R}$, this property is part of the very definition of $\mathbb{R}$

Definition 10: (axiomatic) of the real numbers

We call $\mathbb{R}$ a set that satisfies the properties R_1, R_2, R_3, R_4 , that is, an ordered field that has the supremum property

- The supremum property is also called the **Dedekind axiom**, or **axiom of continuity**, or **completeness axiom**, and it can also be stated in the following equivalent form.

Let $\{A, B\}$ be a partition of $\mathbb{R}$ (i.e., A and B are non-empty disjoint sets whose union is $\mathbb{R}$); the partition is called a **cut** (section) if:

$$\forall a \in A \text{ and } \forall b \in B \text{ we have } a < b.$$

Then one proves that:

$R'_4 \rightarrow$ For every cut $\{A, B\}$ of $\mathbb{R}$ there exists a unique real number s (called the **separating element**) such that:

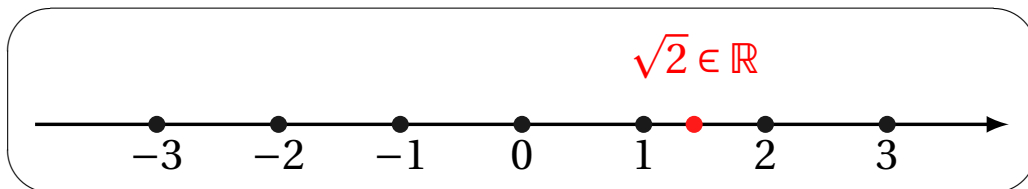
$$\forall a \in A \text{ and } \forall b \in B, \quad a \leq s \leq b$$

This separating element is $\sup A$ and $\inf B$, hence $\sup A = \inf B$

10 Geometric representation of $\mathbb{R}$

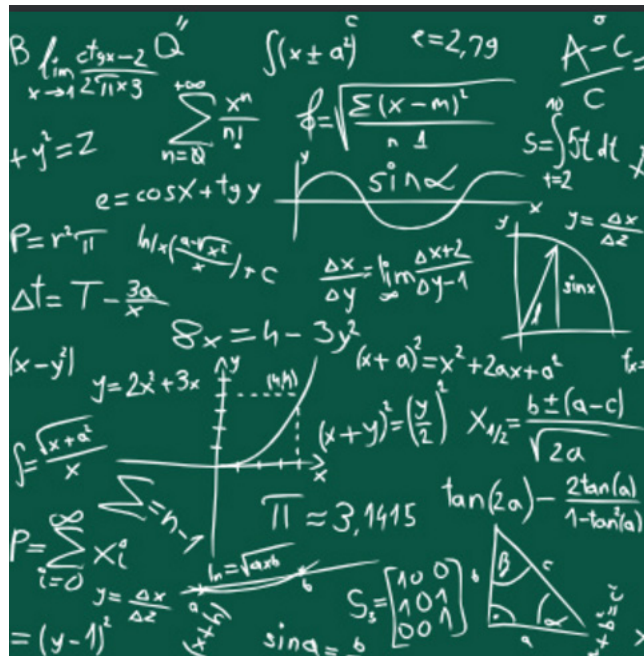
Can the set of real numbers $\mathbb{R}$ be put in one-to-one correspondence with the points of the Euclidean line?

- Let us now return to the problem of the incommensurability of the side and the diagonal of a square. When, with straightedge and compass, having fixed the unit side of the square, we construct its diagonal and transfer it onto the starting line, the arc drawn by the compass "splits the line in two", generating what we have called a cut; the separating element, which exists by axiom R_4 , geometrically represents the point of intersection, and therefore that number measures the diagonal.



Hence the geometric interpretation of property R_4 shows us how the set $\mathbb{R}$ is an adequate representation of our intuitive idea of a line, so adequate that **in mathematics the expression “the real line” is often used to denote** $\mathbb{R}$, identifying the number set with its natural geometric representation.

Roots, powers, logarithms and modular arithmetic



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Roots, powers, logarithms

- As a consequence of property R_4 we can perform, in the real field, operations that are only occasionally possible in the rational field, such as extracting roots or raising to a power.

1.1 Arithmetic n -th roots

Theorem 1

For every $y \in \mathbb{R}$, $y > 0$ and $n \in \mathbb{N}$, $n \geq 1$, there exists one and only one $x \in \mathbb{R}$, $x > 0$, such that $x^n = y$.

- This number is called the arithmetic n -th root of y and is denoted by one of the symbols

$$\sqrt[n]{y} \text{ or } y^{\frac{1}{n}}.$$

Proof. The proof of this Theorem will be given using the properties of continuous functions. □

The arithmetic n -th root is non-negative.

Example 1: arithmetic n -th root

For example:

$$\sqrt{4} = 2, \quad \sqrt{9} = 3.$$

Moreover, we have:

$$\sqrt{x^2} = |x| \tag{1}$$

Construction of the decimal representation of the n -th root.

- We look for the decimal expansion of

$$\sqrt{2} \approx 1.41421356237$$

this number, not being rational, will be represented by an infinite (non-periodic) expansion.

- We proceed as follows: we build a class of rational numbers of the form:

$$\begin{aligned}
 0 &< a_0 \\
 &< a_0, a_1 \\
 &< a_0, a_1 a_2 \\
 &< a_0, a_1 a_2 a_3 \\
 &< \dots\dots\dots \\
 &< a_0, a_1 a_2 a_3 \dots a_n \\
 &< \dots\dots\dots
 \end{aligned}$$

The rule is: each of these numbers is the largest among those with the same number of digits after the decimal point whose square is less than 2. The first of these numbers are:

$$\begin{array}{lll}
 1 & 1^2 & = 1 \\
 1,4 & (1,4)^2 & = 1,96 \\
 1,41 & (1,41)^2 & = 1,9881 \\
 1,414 & (1,414)^2 & = 1,999396 \\
 1,4142 & (1,4142)^2 & = 1,99996164 \\
 \dots & &
 \end{array}$$

- This set of numbers, which we call E_- , is bounded above (each one is < 2); by property R_4 it has a supremum, and this is used to define $\sqrt{2}$.

The number $\sqrt{2}$ is defined precisely as $\sup E_-$

- We could also have built a class of numbers E_+ like the previous one, with the rule that each of them is the smallest among those with the same number of digits after the decimal point whose square is greater than 2; we would have obtained:

$$\begin{array}{lll}
 2 & 2^2 & = 4 \\
 1,5 & (1,5)^2 & = 2,25 \\
 1,42 & (1,42)^2 & = 2,0164 \\
 1,415 & (1,415)^2 & = 2,002225 \\
 1,4143 & (1,4143)^2 & = 2,00024449 \\
 \dots & &
 \end{array}$$

This set E_+ is bounded below (every element is > 1), therefore it has an infimum; one proves that:

$$\sqrt{2} = \inf E_+ = \sup E_-$$

The numbers of the class E_- approximate $\sqrt{2}$ from below, those of the class E_+ from above.

1.2 Powers with real exponent

Extracting the n -th root is the inverse operation of raising to an integer power.

Rational exponent

- The operation of raising to a power can be extended to every rational exponent if the base is positive (using the previous theorem).

$$\text{If } r = \frac{m}{n} \text{ and } a > 0 \text{ then } a^r = (a^m)^{\frac{1}{n}} = \sqrt[n]{a^m}$$

(we assume $m \in \mathbb{Z}$ and $n \in \mathbb{Z}$ positive)

Real exponent

- If the exponent is real, $b = b_0, b_1 b_2 b_3 \dots b_n \dots$, the number a^b ($a > 0$) will be determined by the class of numbers

$$a^{b_0} \quad a^{b_0, b_1} \quad a^{b_0, b_1 b_2} \quad \dots$$

in a way similar to the case of the root.

If the base a is negative, the operation of raising to a power a^b is defined only in certain cases:

- if the exponent b is an integer, or
- if the exponent $b = \frac{n}{m}$ is rational, provided that it is not the case that n is odd and m is even.

If $c < 0$ and m is odd, we define

$$\sqrt[m]{c} = -\sqrt[m]{-c}$$

Example 2: negative base

For example:

$$(-2)^{\frac{3}{5}} = \sqrt[5]{(-2)^3} = \sqrt[5]{-8} = -\sqrt[5]{8}$$

$$(-2)^{\frac{2}{5}} = \sqrt[5]{(-2)^2} = \sqrt[5]{4}$$

- When we say “does not exist in $\mathbb{R}$ ” we mean that it is not possible to define this operation in such a way that the usual rules of computation remain valid.

Example 3: non-existence of a power with real exponent

For example:

$$(-2)^{\frac{3}{4}}$$

does not exist in the real field!

Let

$$a, b \in \mathbb{R}, \quad a > 0 \text{ and } b > 0 \quad (\text{positive reals})$$

and

$$c, d \in \mathbb{R} \quad (\text{arbitrary reals}),$$

the main properties of raising to a power are:

$$a^0 = 1 \quad \forall a \neq 0 \quad (2a)$$

$$1^c = 1 \quad \forall c \quad (2b)$$

$$a^c > 0 \quad \forall c \quad (2c)$$

$$a^c > 1 \quad \forall a > 1 \text{ and } c > 0 \quad (2d)$$

$$a^c < 1 \quad \forall a < 1 \text{ and } c > 0 \quad (2e)$$

$$a^{c+d} = a^c \cdot a^d \quad (3a)$$

$$(a \cdot b)^c = a^c \cdot b^c \quad (3b)$$

$$(a^b)^c = a^{b \cdot c} \quad (3c)$$

$$c < d \Rightarrow a^c < a^d \text{ if } a > 1 \quad (4a)$$

$$c < d \Rightarrow a^c > a^d \text{ if } a < 1 \quad (4b)$$

$$0 < a \leq b \Rightarrow a^c \leq b^c \quad \forall c > 0 \quad (4c)$$

1.3 Logarithms

- Consider the equation

$$a^x = y, \quad a > 0$$

with y given and x unknown. First of all, if $a = 1$, it is solvable only if $y = 1$ (and in that case every real number x is a solution). So let $a \neq 1$.

If $y \leq 0$ it has no solution, see property (2c).

The following theorem tells us that it has exactly one solution for every $y > 0$:

Theorem 2

Let $a > 0$, $a \neq 1$, $y > 0$. There exists a unique real number x such that $a^x = y$.

- This number is called the logarithm to base a of y and is denoted by the symbol:

$$\log_a y$$

- The logarithm is the inverse operation of raising to a power.
- the logarithm of a number to a given base is the exponent to which the base must be raised to obtain the number itself.

Let

$$x, y, a \in \mathbb{R}, \quad x > 0, y > 0, a > 0 \text{ (positive reals) and } a \neq 1,$$

the main properties of logarithms are (they follow from those of exponentials):

$$\log_a (x \cdot y) = \log_a x + \log_a y \quad (5a)$$

$$\log_a \left(\frac{x}{y} \right) = \log_a x - \log_a y \quad (5b)$$

$$\log_a \left(\frac{1}{y} \right) = -\log_a y \quad (5c)$$

$$\log_a x^\alpha = \alpha \log_a x \quad (\forall \alpha \in \mathbb{R}) \quad (6a)$$

$$\log_b x = \frac{\log_a x}{\log_a b} \quad (\forall b > 0, b \neq 1) \quad (6b)$$

$$\log_a x = \frac{1}{\log_x a} = -\log_{\frac{1}{a}} x \quad (x \neq 1) \quad (6c)$$

Notation:

- $\lg n = \log_2 n \rightarrow$ **binary logarithm**
- $\ln n = \log_e n \rightarrow$ **natural logarithm** (where $e \approx 2.718$ is **Euler's number** (Napier's constant)) (sometimes the natural logarithm $\ln$ is also denoted by $\log$)
- $\lg^k n = (\lg n)^k \rightarrow$ **power**
- $\lg \lg n = \lg(\lg n) \rightarrow$ **composition**

1.4 Approximations

- a rational number can always be expressed with absolute precision, using either the fractional or the decimal notation (possibly with repeating digits).
- On the other hand, it is of course not possible to write all the decimal digits of an irrational number, since there are infinitely many of them and they follow one another without periodicity.
- What does it mean, then, to “know” or “specify” an irrational number? It means knowing some algorithm that allows us (at least in theory) to write as many exact decimal digits as we wish.

Example 4: irrational numbers

- In the (already considered) example of the irrational number:

$$0,101001000100001\dots$$

built according to the rule: write one digit 1, one digit 0, one digit 1, two digits 0, one digit 1, three digits 0, and so on) it is clear that we could write as many digits as we wish.

- In other cases, such as

$$\sqrt{2} \text{ or } \log_2 3,$$

things are more laborious and require iterative computations to determine each successive decimal digit; nevertheless, these computations can actually be carried out.

- At other times, in Mathematical Analysis, a number is specified in a non-constructive way, by denoting it as the unique number that solves a given problem (once it has been proved that such a solution exists and is unique). This is a less satisfactory approach from an operational point of view, but theoretically flawless.

Practical aspects of computing irrational numbers

- Whenever we perform computations with a **pocket calculator**, it will only display numbers with a fixed number of decimal digits after the decimal point (typically 9). This means that we are working only with rational numbers, indeed with a finite subset of the set $\mathbb{Q}$.
- Working with a **computer** things improve somewhat, but we still remain within finite subsets of $\mathbb{Q}$. Of course, one needs to be aware of this.
- Besides the approximation produced by computing tools, sometimes we ourselves are not interested in too many decimal digits; we then deliberately introduce approximations, writing for example

$$\sqrt{2} \approx 1,414$$

Rounding rule: the last digit written is rounded down (up) if the first digit discarded is from 0 to 4 (respectively, from 5 to 9).

Example 5: Rounding

For example:

$$2,4138 \approx 2,41$$

$$2,4152 \approx 2,42$$

2 Modular arithmetic

Definition 1: of integer part

Given a real number $a \in \mathbb{R}$, we denote by $[a]$ or $\lfloor a \rfloor$ the **integer part** (or “**floor**”) of a :

$$[a] = \text{integer } n \text{ such that } n \leq a < n + 1$$

Example 6: Integer part

$$[2,38] = 2; \quad [3] = 3; \quad [-1,8] = -2.$$

- While for positive numbers the integer part is obtained simply by “throwing away the digits after the decimal point”, for negative numbers one must take the largest integer $\leq a$, which is different from what is obtained by throwing away the digits after the decimal point (except when a is already an integer)

Definition 2: of fractional part

Given a real number $a \in \mathbb{R}$, we denote by (a) the **fractional part** (or decimal part, in Italian “**mantissa**”) of a :

$$(a) = a - [a]$$

Example 7: Fractional part

$$(2,38) = 0.38; \quad (3) = 0; \quad (-1,8) = 0.2.$$

- The fractional part is therefore not an integer but a real number, lying in $[0, 1)$.
- For positive numbers, it is obtained simply by “throwing away the digits before the decimal point”; for negative numbers the fractional part is the **one’s complement** of the number obtained by throwing away the digits before the decimal point.

Definition 3: of upper integer part

Given a real number $a \in \mathbb{R}$, we denote by $\lceil a \rceil$ the **upper integer part** (or “**ceil**”) of a :

$$\lceil a \rceil = \text{integer } n \text{ such that } n - 1 < a \leq n$$

Example 8: of ceil

$$\lceil 2,38 \rceil = 3; \quad \lceil 3 \rceil = 3; \quad \lceil -1,8 \rceil = -1.$$

Some properties:

Given a real number $a \in \mathbb{R}$, we have

$$a - 1 < \lfloor a \rfloor \leq a \leq \lceil a \rceil < a + 1$$

Given an integer $n \in \mathbb{Z}$, we have

$$\left\lfloor \frac{n}{2} \right\rfloor + \left\lceil \frac{n}{2} \right\rceil = n$$

Given two positive integers $r, s \in \mathbb{Z}$, $r, s > 0$, we have

$$\left\lceil \frac{r}{s} \right\rceil \leq \frac{r + (s - 1)}{s} \quad (7a)$$

$$\left\lfloor \frac{r}{s} \right\rfloor \geq \frac{r - (s - 1)}{s} \quad (7b)$$

Given, in addition, a non-negative real number $p \in \mathbb{R}$, $p \geq 0$, we have

$$\left\lceil \frac{\left\lceil \frac{p}{r} \right\rceil}{s} \right\rceil = \left\lceil \frac{p}{r s} \right\rceil \quad (8a)$$

$$\left\lfloor \frac{\left\lfloor \frac{p}{r} \right\rfloor}{s} \right\rfloor = \left\lfloor \frac{p}{r s} \right\rfloor \quad (8b)$$

Definition 4: Divisor

A **divisor** of an integer $n \in \mathbb{Z}$, also called a **factor** of n , is an integer $m \in \mathbb{Z}$ that can be multiplied by some integer $q \in \mathbb{Z}$ to obtain n , i.e., if $n = q \cdot m$.

- If m is a divisor of n , n is a **multiple** of m .
- An integer n is **divisible** by another integer m if m is a *divisor* of n .
- For an integer $a \in \mathbb{Z}$ and a positive integer $n \in \mathbb{Z}, n > 0$, the value $a \bmod n$ is the remainder of the division $\frac{a}{n}$.

Definition 5: of modulo (remainder)

Given $a \in \mathbb{Z}$ and $n \in \mathbb{Z}, n > 0$,

$$a \bmod n = a - n \left\lfloor \frac{a}{n} \right\rfloor$$

- It follows that $0 \leq a \bmod n < n$

If $(a \bmod n) = (b \bmod n)$, we write

$$a \equiv b \pmod{n}$$

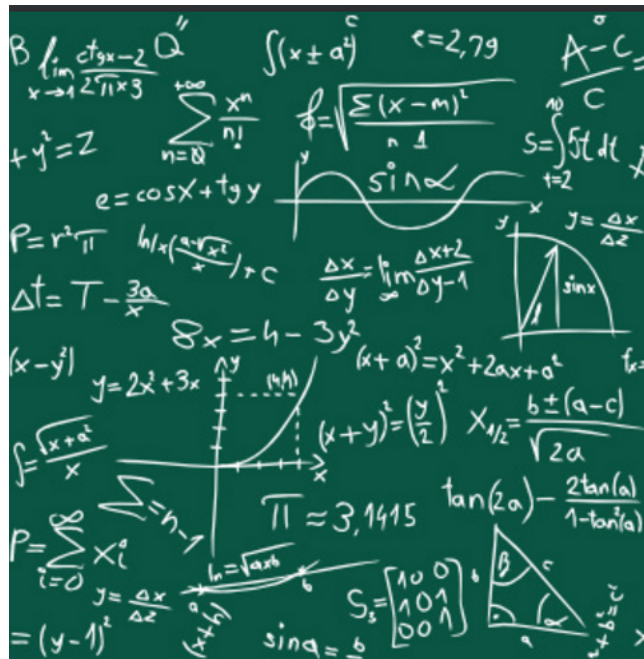
and we say that a is **equivalent** to b , modulo n .

Example 9:

For example, 23 and 13 are equivalent modulo 5 and we write $23 \equiv 13 \pmod{5}$

- Equivalently, $a \equiv b \pmod{n}$ if a and b have the same remainder when divided by n .
- Equivalently, $a \equiv b \pmod{n}$ if and only if n is a divisor of $|b - a|$.

Summations and geometric progressions



- **Author:** Fabio Furini.
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1 Summations

Definition 1: summation

Let $a_1, a_2, \dots, a_n$ be n real numbers. Their sum

$$a_1 + a_2 + \dots + a_n$$

can be written in compact form with the summation symbol:

$$\sum_{k=1}^n a_k$$

which is read: “sum for k from 1 to n of a_k ”. The symbol k is called the summation index.

- The summation symbol is therefore just a shorthand, which is nevertheless very useful when the terms a_k are defined explicitly as a function of the index k .

Example 1: summations

$$\sum_{k=1}^{10} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10}$$

$$\sum_{k=3}^n k^2 = 3^2 + 4^2 + 5^2 + \dots + n^2$$

- The summation index is a **dummy index**. This means that if we replace k with i , j or any other index (in all its occurrences), the value of the summation does not change.

Example 2: dummy index

We have:

$$\sum_{k=1}^n k^2 = \sum_{i=1}^n i^2$$

since both symbols denote the sum of the squares of the first n natural numbers (without zero).

On the other hand, we have:

$$\sum_{k=1}^n k^2 \neq \sum_{k=1}^m k^2$$

since the two symbols denote the sum of, respectively, the first n or the first m squares (if $n \neq m$ the result will be different).

1.1 Main properties of summations

Remark 1

Given $c \in \mathbb{R}$, we have:

$$\sum_{k=1}^n (c \cdot a_k) = c \sum_{k=1}^n a_k \quad (\text{product by a constant}) \quad (1)$$

$$\sum_{k=1}^n c = c \cdot n \quad (\text{summation with a constant term}) \quad (2)$$

Proof. First property (1). By the distributive property we have:

$$\underbrace{c a_1 + c a_2 + \cdots + c a_n}_{=\sum_{k=1}^n (c \cdot a_k)} = \underbrace{c (a_1 + a_2 + \cdots + a_n)}_{=c \sum_{k=1}^n a_k}$$

Second property (2):

$$\underbrace{c + c + \cdots + c}_{=\sum_{k=1}^n c \quad \text{that is, } c \text{ added } n \text{ times}} = c n$$

□

Remark 2

Given two summations $\sum_{k=1}^n a_k$ and $\sum_{k=1}^n b_k$, we have:

$$\sum_{k=1}^n a_k + \sum_{k=1}^n b_k = \sum_{k=1}^n (a_k + b_k) \quad (3)$$

Proof. We have:

$$\underbrace{a_1 + a_2 + \cdots + a_n + b_1 + b_2 + \cdots + b_n}_{=\sum_{k=1}^n a_k + \sum_{k=1}^n b_k} = \underbrace{a_1 + b_1 + a_2 + b_2 + \cdots + a_n + b_n}_{=\sum_{k=1}^n (a_k + b_k)}$$

□

Remark 3

Given two natural numbers $n, m \in \mathbb{N}$, we have:

$$\sum_{k=1}^{n+m} a_k = \sum_{k=1}^n a_k + \sum_{k=n+1}^{n+m} a_k \quad (\text{splitting}) \quad (4)$$

$$\sum_{k=1}^n a_k = \sum_{k=1+m}^{n+m} a_{k-m} = \sum_{k=1-m}^{n-m} a_{k+m} \quad (\text{index shift}) \quad (5)$$

$$\sum_{k=1}^n a_k = \sum_{k=1}^n a_{n-k+1} = \sum_{k=0}^{n-1} a_{n-k} \quad (\text{index reflection}) \quad (6)$$

Proof. The three properties are simply different ways of writing and/or ordering the terms of the summations. □

1.2 Some important summations**Remark 4: sum of the first n natural numbers (without zero)**

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Proof.

$$\begin{aligned} \sum_{k=1}^n k &= \frac{1}{2} \left(\sum_{k=1}^n k + \sum_{k=1}^n k \right) = \frac{1}{2} \left(\sum_{k=1}^n k + \sum_{k=1}^n (n - k + 1) \right) \\ &= \frac{1}{2} \sum_{k=1}^n (k + n - k + 1) = \frac{1}{2} \sum_{k=1}^n (n + 1) = \frac{n(n+1)}{2} \end{aligned}$$

□

Remark 5: sum of the first n odd numbers

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^n (2k-1) = n^2 \quad \text{or equivalently} \quad \sum_{k=0}^{n-1} (2k+1) = n^2$$

Proof. Using the properties of summations we have:

$$\begin{aligned} \sum_{k=1}^n (2k-1) &= 2 \sum_{k=1}^n k - \sum_{k=1}^n 1 \\ &= 2 \left(\frac{n(n+1)}{2} \right) - n = n^2 + n - n = n^2 \\ \\ \sum_{k=0}^{n-1} (2k+1) &= 2 \sum_{k=0}^{n-1} k + \sum_{k=0}^{n-1} 1 \\ &= 2 \sum_{k=1}^n (k-1) + n = 2 \left(\sum_{k=1}^n k - \sum_{k=1}^n 1 \right) + n \\ &= 2 \left(\frac{n(n+1)}{2} - n \right) + n = n^2 + n - 2n + n = n^2 \end{aligned}$$

□

Remark 6: sum of the first n even numbers (without zero)

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^n 2k = n(n+1)$$

Proof.

$$\sum_{k=1}^n 2k = 2 \sum_{k=1}^n k = 2 \left(\frac{n(n+1)}{2} \right) = n(n+1)$$

□

2 Geometric progressions

Definition 2: geometric progression

A sequence of real numbers is a **geometric progression** if the ratio between each term (starting from the second one) and the previous one is constant. This constant is called the common ratio of the progression.

Given the first term $a \in \mathbb{R}$ and the common ratio $q \in \mathbb{R}$, the associated geometric progression is:

$$a, a q, a q^2, a q^3, a q^4, \dots$$

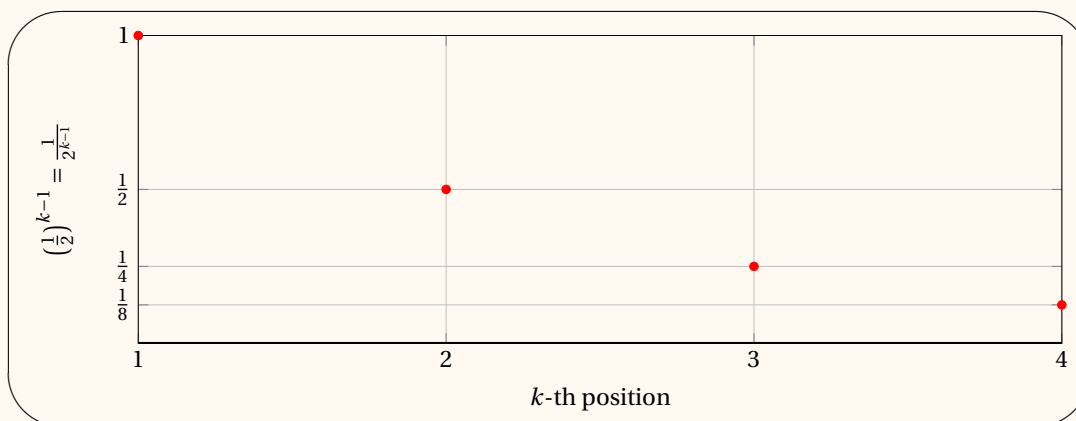
Each term (starting from the second one) is obtained from the previous one by multiplying it by q . The k -th term ($k \in \mathbb{N}$, $k \geq 1$) can be written as $a q^{k-1}$ and we have:

$a q^{1-1} = a q^0 = a$	first term, in position $k = 1$
$a q^{2-1} = a q^1 = a q$	second term, in position $k = 2$
$a q^{3-1} = a q^2$	third term, in position $k = 3$
$a q^{4-1} = a q^3$	fourth term, in position $k = 4$
$\dots$	$\dots$

Example 3: geometric progressions

- With $a = 1$ and $q = \frac{1}{2}$, the first 4 terms are:

$$1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}$$



- With $a = 1$ and $q = 2$, the first 4 terms are:

$$1, 2, 4, 8$$

2.1 Sums of the terms of geometric progressions

Remark 7: sum of the first n terms of the geometric progression ($a = 1$)

Given $q \in \mathbb{R}_+$, for every natural number $n \geq 1$ we have:

$$\sum_{k=1}^n q^{k-1} = \begin{cases} \frac{q^n - 1}{q - 1} & \text{if } q \neq 1 \\ n & \text{otherwise} \end{cases} \quad (7)$$

Proof. If $q \neq 1$, we prove the equation in the following equivalent form:

$$(q - 1) \sum_{k=1}^n q^{k-1} = q^n - 1$$

Applying the properties of summations, we get:

$$\begin{aligned} (q - 1) \sum_{k=1}^n q^{k-1} &= q \sum_{k=1}^n q^{k-1} - \sum_{k=1}^n q^{k-1} = \\ &= \sum_{k=1}^n q^k - \sum_{k=1}^n q^{k-1} = \sum_{k=1}^n q^k - \sum_{k=0}^{n-1} q^k = \\ &= \sum_{k=1}^{n-1} q^k + q^n - \left(1 + \sum_{k=1}^{n-1} q^k \right) = q^n - 1 \end{aligned}$$

If $q = 1$, we have instead:

$$\sum_{k=1}^n q^{k-1} = \sum_{k=1}^n 1 = n$$

□

- Given $q \in \mathbb{R}_+$, $n \in \mathbb{N}$, $n \geq 1$ and $a \in \mathbb{R}$, formula (7) extends as follows:

$$\sum_{k=1}^n a q^{k-1} = \begin{cases} a \left(\frac{q^n - 1}{q - 1} \right) & \text{if } q \neq 1 \\ a n & \text{otherwise} \end{cases} \quad \text{since } \sum_{k=1}^n a q^{k-1} = a \sum_{k=1}^n q^{k-1} \quad (8)$$

clearly:

$$\frac{q^n - 1}{q - 1} = \frac{1 - q^n}{1 - q} \quad \text{and hence we also have } \sum_{k=1}^n a q^{k-1} = \begin{cases} a \left(\frac{1 - q^n}{1 - q} \right) & \text{if } q \neq 1 \\ a n & \text{otherwise} \end{cases}$$

Example 4: sum of the first n terms of geometric progressions

- With $a = 1$ and $q = \frac{1}{2}$, the first 4 terms are:

$$1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8} \quad \text{and} \quad 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{8+4+2+1}{8} = \frac{15}{8}$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^4 \left(\frac{1}{2}\right)^{k-1} = \sum_{k=1}^4 \frac{1}{2^{k-1}} = \frac{1 - \frac{1}{2^4}}{1 - \frac{1}{2}} = \frac{1 - \frac{1}{16}}{\frac{1}{2}} = \frac{15}{16} \cdot 2 = \frac{15}{8}$$

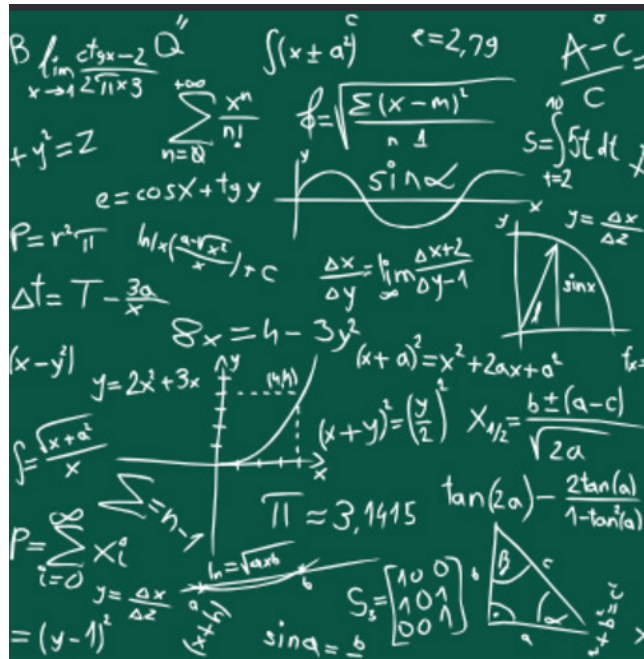
- With $a = 1$ and $q = 2$, the first 4 terms are:

$$1, 2, 4, 8 \quad \text{and} \quad 1 + 2 + 4 + 8 = 15$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^4 2^{k-1} = \frac{2^4 - 1}{2 - 1} = 16 - 1 = 15$$

Principle of mathematical induction



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 The principle of mathematical induction

- We now present a proof technique called proof by induction. This procedure can be applied to theorems with the following structure:

“ for every $n \in \mathbb{N}$, $n \geq n_0$, property $p(n)$ holds ”

- The number n_0 is the smallest integer for which we want the property to be true; if $n_0 = 0$ the theorem simply states that the property is true for every $n \in \mathbb{N}$.

A proof by induction consists of the following two steps:

1. We prove that $p(n)$ is true for $n = n_0$ (**base case of the induction**).
2. We prove that, if n is an arbitrary natural number $\geq n_0$, the fact that $p(n)$ is true implies that $p(n + 1)$ is true (**inductive step**).

We can then conclude that for every $n \geq n_0$, $p(n)$ is true.

- The validity of this proof technique, *intuitively*, is based on the following fact:
 1. By point 1, we know that $p(n_0)$ is true. Suppose for example $n_0 = 1$: we therefore know that $p(1)$ is true (this must be proved explicitly).
 2. By point 2, since $p(1)$ is true, $p(2)$ will be true: indeed, we have proved that for any n , if $p(n)$ is true then $p(n + 1)$ is also true. But then, since $p(2)$ is true, $p(3)$ will be true; but then $p(4)$ is true, ... and so on, hence $p(n)$ is true for every $n \geq 1$.
- In practice, the proof consists of two phases.
 1. Prove $p(n_0)$ directly;
 2. Assume $p(n)$ as a hypothesis (**inductive hypothesis**) and prove $p(n + 1)$.
- This is the delicate point, often subject to misunderstandings. “Assuming as a hypothesis” $p(n)$ does not mean assuming the thesis as a hypothesis. What must be proved is that:

“ for every $n \geq n_0$, if $p(n)$ is true then $p(n + 1)$ is also true ”

and not

“ if $p(n)$ is true for every n , then $p(n + 1)$ is also true ”

2 Proofs based on the principle of mathematical induction

2.1 Bernoulli's inequality

Remark 1: Bernoulli's inequality

For every integer $n \geq 0$, $x \in \mathbb{R}$, $x \geq -1$, we have:

$$(1 + x)^n \geq 1 + n x \quad (1)$$

Proof. By induction on n .

- **Base case of the induction**

Let $n = 0$. Then the statement becomes:

$$(1 + x)^0 \geq 1 + 0x \quad \text{i.e.} \quad 1 \geq 1$$

which is clearly true.

- **Inductive step**

Suppose it is true for n , and let us prove it for $(n + 1)$. By the inductive hypothesis, we have:

$$(1 + x)^n \geq 1 + nx$$

and moreover we have

$$(1 + x) \geq 0 \quad \text{since} \quad x \geq -1.$$

Then we can write:

$$\begin{aligned}(1 + x)^{n+1} &= (1 + x) \cdot (1 + x)^n \\ &\geq (1 + x) \cdot (1 + nx) \\ &= 1 + (n + 1)x + nx^2 \\ &\geq 1 + (n + 1)x\end{aligned}$$

where in the last inequality we used the fact that $nx^2 \geq 0$.

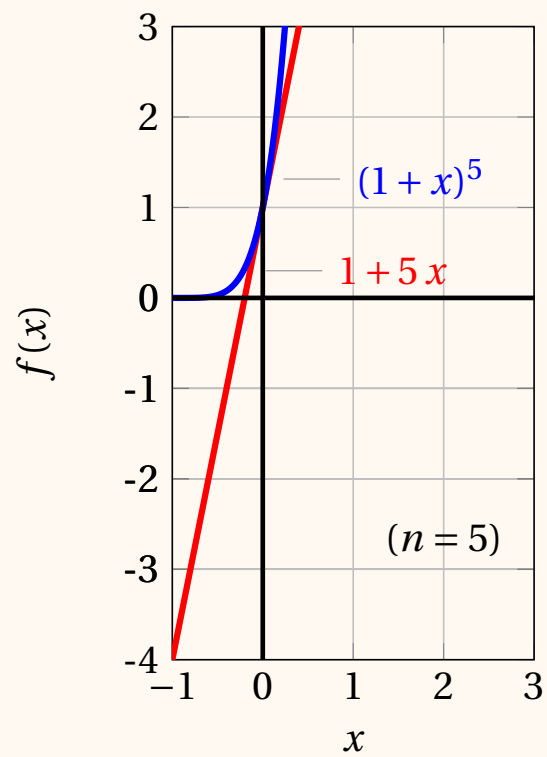
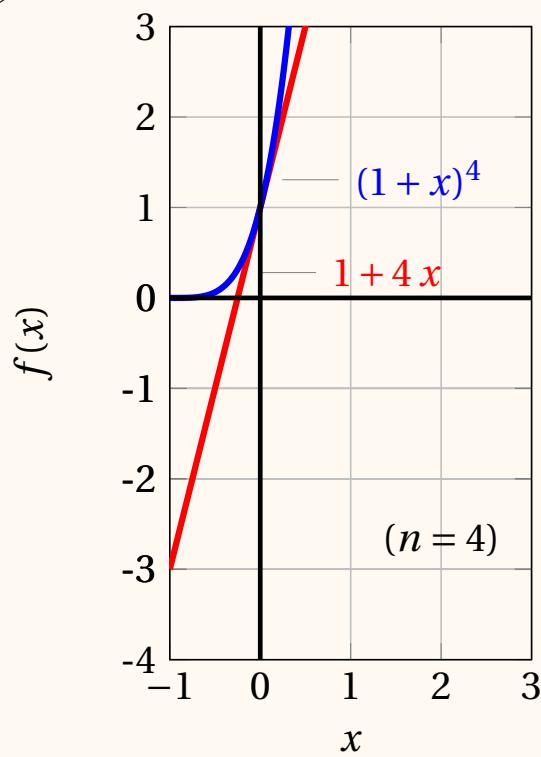
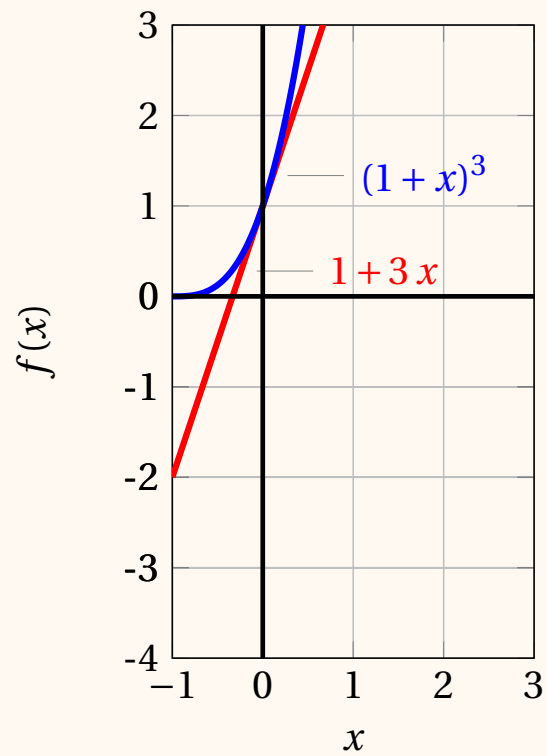
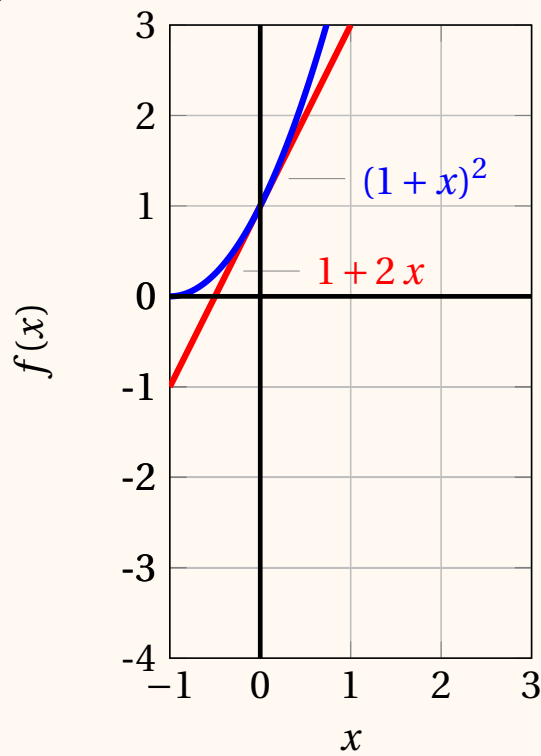
The chain of inequalities shows that, for $n + 1$, we have

$$(1 + x)^{n+1} \geq 1 + (n + 1)x$$

which is exactly the desired statement, for $n + 1$.

□

Example 1: Bernoulli's inequality



2.2 Some important summations

Remark 2: sum of the first n natural numbers

For every integer $n \geq 1$, we have:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Proof. By induction on n .

- **Base case of the induction**

Let $n = 1$. Then the statement becomes:

$$\sum_{k=1}^1 k = \frac{1(1+1)}{2} \quad \text{i.e.} \quad 1 = 1$$

which is clearly true.

- **Inductive step**

Suppose it is true for n , and let us prove it for $(n+1)$. By the inductive hypothesis, we have:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

Hence we can write:

$$\begin{aligned} \sum_{k=1}^{n+1} k &= \sum_{k=1}^n k + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{n(n+1) + 2(n+1)}{2} \\ &= \frac{(n+1)(n+2)}{2} = \frac{(n+1)((n+1)+1)}{2} \end{aligned}$$

which is exactly the desired statement, for $n+1$.

□

2.3 Sum of the terms of the geometric progression

Remark 3: sum of the first n terms of the geometric progression ($a = 1$)

Given $q \in \mathbb{R}_+$, for every integer $n \geq 1$ we have:

$$\sum_{k=1}^n q^{k-1} = \begin{cases} \frac{q^n - 1}{q - 1} & \text{if } q \neq 1 \\ n & \text{otherwise} \end{cases} \quad (2)$$

Proof. By induction on n .

- **Base case of the induction**

Let $n = 1$. Then the statement becomes:

$$\sum_{k=1}^1 q^{k-1} = \frac{q^1 - 1}{q - 1} \quad \text{i.e. } 1 = 1$$

which is clearly true.

- **Inductive step**

Suppose it is true for n , and let us prove it for $(n + 1)$. By the inductive hypothesis, we have:

$$\sum_{k=1}^n q^{k-1} = \frac{q^n - 1}{q - 1}$$

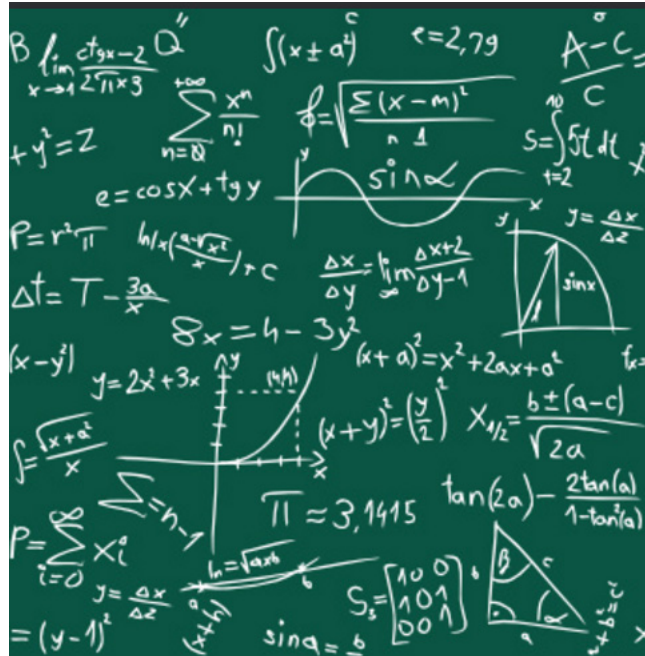
Hence we can write

$$\begin{aligned} \sum_{k=1}^{n+1} q^{k-1} &= \sum_{k=1}^n q^{k-1} + q^n = \frac{q^n - 1}{q - 1} + q^n \\ &= \frac{q^n - 1 + q^{n+1} - q^n}{q - 1} = \frac{q^{n+1} - 1}{q - 1} \end{aligned}$$

which is exactly the desired statement, for $n + 1$.

□

Factorials, binomial coefficients and triangle inequality



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Factorials

Definition 1: factorial of n

The factorial of n is the product of the first n positive integers. It is denoted by $n!$ and is read “ n factorial”. In formulas:

$$n! = \prod_{k=1}^n k = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (n-1) \cdot n$$

By definition, we set $0! = 1$.

- The number $n!$ grows very rapidly as n increases. The first values are:

n	0	1	2	3	4	5	6	7	8	9	10
$n!$	1	1	2	6	24	120	720	5.040	40.320	362.880	3.628.800

- Some properties of the factorial, which are immediate to verify, are:

$$n! = n \cdot (n-1)!$$

$$\frac{n!}{(n-k)!} = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-k+1), \text{ with } k \geq 1 \quad (1)$$

With $k \geq 1$, it becomes the product of k factors, starting from n and decreasing by one unit at a time.

Example 1: Computing the factorial

$$\frac{100!}{95!} = \frac{100!}{(100-5)!} = 100 \cdot 99 \cdot 98 \cdot 97 \cdot 96 = 9.034.502.400$$

It is always convenient to simplify expressions containing the factorial as much as possible, before computing them!

2 Binomial coefficients

Definition 2: binomial coefficient

The **binomial coefficient** is defined as the number:

$$c_{n,k} = \frac{n!}{k!(n-k)!} \quad \text{with } 0 \leq k \leq n. \quad (2)$$

The binomial coefficient $c_{n,k}$ is usually denoted by the symbol: $\binom{n}{k}$ which is read “ n choose k ”.

- Given rule (1), we have:

$$c_{n,k} = \frac{n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-k+1)}{k!} \quad (3)$$

with $k \geq 1$, an expression that is more convenient for computing the binomial coefficient.

We have:

$$\binom{n}{n-k} = \binom{n}{k} \quad (4)$$

Since:

$$\binom{n}{n-k} = \frac{n!}{(n-k)!(n-(n-k))!} = \frac{n!}{k!(n-k)!} = \binom{n}{k}$$

We have:

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k} \quad (5)$$

Since:

$$\begin{aligned} \binom{n-1}{k-1} + \binom{n-1}{k} &= \frac{(n-1)!}{(k-1)! \underbrace{(n-1-(k-1))!}_{=(n-k)! = (n-k)(n-k-1)!}} + \frac{(n-1)!}{k!(n-k-1)!} \\ &= \frac{(n-1)!}{(k-1)!(n-k)(n-k-1)!} + \frac{(n-1)!}{k(k-1)!(n-k-1)!} \\ &= \frac{k(n-1)! + (n-k)(n-1)!}{k(k-1)!(n-k)(n-k-1)!} = \frac{\overbrace{(n-1)!}^{=n!} n}{\underbrace{k(k-1)!}_{k!} \underbrace{(n-k)(n-k-1)!}_{(n-k)!}} \\ &= \frac{n!}{k!(n-k)!} = \binom{n}{k} \end{aligned}$$

It also follows that:

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$$

2.1 Newton's formula

The n -th power of a binomial $(a + b)$ can be computed with the following formula (from which the name binomial coefficient derives):

Remark 1: Newton's formula

For every integer $n \geq 0$, with $a, b \in \mathbb{R}$, we have:

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \quad (6)$$

Proof. By induction on n .

- **Base case of the induction**

Let $n = 0$. Then the statement becomes: $(a + b)^0 = \binom{0}{0} a^0 b^0$ i.e. $1 = 1$ which is clearly true.

- **Inductive step**

Suppose it is true for n , and let us prove it for $(n + 1)$. By the inductive hypothesis, we have: $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$. Then:

$$\begin{aligned}
 (a + b)^{n+1} &= (a + b) \cdot (a + b)^n = (a + b) \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \\
 &= \sum_{k=0}^n \binom{n}{k} a^{n-k+1} b^k + \underbrace{\sum_{k=0}^n \binom{n}{k} a^{n-k} b^{k+1}}_{= \sum_{k=1}^{n+1} \binom{n}{k-1} a^{n-k+1} b^k} \\
 &= a^{n+1} + \sum_{k=1}^n \binom{n}{k} a^{n-k+1} b^k + b^{n+1} + \sum_{k=1}^n \binom{n}{k-1} a^{n-k+1} b^k \\
 &= a^{n+1} + b^{n+1} + \sum_{k=1}^n \underbrace{\left(\binom{n}{k} + \binom{n}{k-1} \right)}_{= \binom{n+1}{k}} a^{n-k+1} b^k \\
 &= \sum_{k=0}^{n+1} \binom{n+1}{k} a^{n+1-k} b^k
 \end{aligned}$$

which is exactly the desired statement, for $n + 1$.

□

Remark 2

For every integer $n \geq 0$ and integer k such that $0 \leq k \leq n$, we have:

$$\sum_{k=0}^n \binom{n}{k} = 2^n \tag{7}$$

Proof. We write:

$$2^n = (1 + 1)^n$$

We now apply Newton's formula (6):

$$(1 + 1)^n = \sum_{k=0}^n \binom{n}{k} 1^{n-k} 1^k = \sum_{k=0}^n \binom{n}{k}$$

□

2.2 Recursive computation of binomial coefficients

Relation (5) allows us to compute the binomial coefficients $\binom{n}{k}$ by means of the so-called **Pascal's triangle** (or Tartaglia's triangle).

The rules for building the triangle are:

1. At the top of the triangle we place the number $\binom{0}{0} = 1$ (by definition).
2. On the sides we place the numbers $\binom{n}{0} = \binom{n}{n} = 1$ for every $n \geq 1$.
3. For $0 < k < n$, the number $\binom{n}{k}$ is written at the intersection of the n -th row and the k -th column.
4. The number $\binom{n}{k}$ is the sum of the two numbers located in the previous row, the one in the same column and the one in the previous column.

Example 2: Pascal's triangle (or Tartaglia's triangle)

Pascal's triangle with $n \leq 10$ and $k \leq 10$ is:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$
$n=0$	1										
$n=1$	1	1									
$n=2$	1	2	1								
$n=3$	1	3	3	1							
$n=4$	1	4	6	4	1						
$n=5$	1	5	10	10	5	1					
$n=6$	1	6	15	20	15	6	1				
$n=7$	1	7	21	35	35	21	7	1			
$n=8$	1	8	28	56	70	56	28	8	1		
$n=9$	1	9	36	84	126	126	84	36	9	1	
$n=10$	1	10	45	120	210	252	210	120	45	10	1

To compute the binomial coefficient $\binom{5}{3}$, corresponding to the blue cell, we can use relation (5) and add the two binomial coefficients $\binom{4}{2}$ and $\binom{4}{3}$, corresponding to the red cells:

$$\binom{5}{3} = \binom{4}{2} + \binom{4}{3} = 6 + 4 = 10.$$

Example 3: computing the power of a binomial using Pascal's triangle

Thanks to the previous Pascal's triangle and to Newton's formula (6) we can compute:

$$(a+b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$$

3 Absolute value

Definition 3: absolute value

The **absolute value** of a real number $a \in \mathbb{R}$ (or **modulus** of a) is the non-negative number defined as follows:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases} \quad (8)$$

From the definition of absolute value it immediately follows that:

$$\forall \varepsilon \geq 0, a \in \mathbb{R}, \quad |a| \leq \varepsilon \iff -\varepsilon \leq a \leq \varepsilon \quad (9)$$

3.1 Triangle inequality in $\mathbb{R}$

Remark 3: triangle inequality in $\mathbb{R}$

$$|b + c| \leq |b| + |c| \quad \forall b, c \in \mathbb{R} \quad (10)$$

Proof. We write the two relations:

$$-|b| \leq b \leq |b|, \quad -|c| \leq c \leq |c|$$

and add them side by side:

$$-(|b| + |c|) \leq b + c \leq |b| + |c|$$

Hence, by (9), with $\varepsilon = |b| + |c| \geq 0$ and $a = b + c$, (10) follows. $\square$

- The triangle inequality is also used in the following form:

$$|d - e| \leq |d - f| + |e - f| \quad \forall d, e, f \in \mathbb{R} \quad (11)$$

To obtain it, it suffices to set in (10):

$$b = d - f, \quad c = f - e$$

we obtain:

$$|d - f + f - e| = |d - e| \leq |d - f| + |f - e| = |d - f| + |e - f|$$

since:

$$|f - e| = |e - f| \quad \forall e, f \in \mathbb{R}$$

- Moreover, the triangle inequality can also be written in the following form:

$$|g| \leq |g - h| + |h| \quad \text{i.e.} \quad |g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R} \quad (12)$$

To obtain it, it suffices to set in (10):

$$b = g - h, \quad c = h$$

Remark 4: reverse triangle inequality in $\mathbb{R}$

$$||g| - |h|| \leq |g - h|, \quad \forall g, h \in \mathbb{R}. \quad (13)$$

Proof. From (12) we have

$$|g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R}$$

Similarly, swapping g and h in (12) we obtain:

$$|h| - |g| \leq |h - g| = |g - h| \quad \text{that is} \quad |g| - |h| \geq -|g - h|$$

hence we have

$$-(|g - h|) \leq |g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R}$$

Hence, by (9), with $\varepsilon = |g - h| \geq 0$ and $a = |g| - |h|$, the reverse triangle inequality follows. $\square$

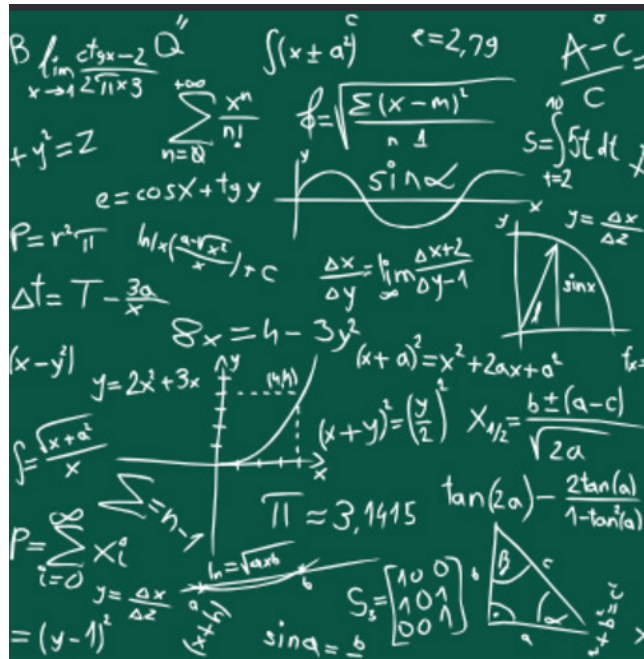
- Inequality (10) can easily be extended to the case of k terms:

$$\left| \sum_{i=1}^k b_i \right| \leq \sum_{i=1}^k |b_i|. \quad (14)$$

- The following immediate properties also hold:

$$|bc| = |b||c|, \quad \left| \frac{b}{c} \right| = \frac{|b|}{|c|}, \quad |-b| = |b| \quad \forall b, c \in \mathbb{R}. \quad (15)$$

The Fibonacci sequence



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 The Fibonacci sequence

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1 The Fibonacci sequence

Definition 1: Fibonacci sequence

$$\begin{aligned}F_0 &= 0, \\F_1 &= 1, \\F_i &= F_{i-1} + F_{i-2}, \quad \text{with } i \geq 2, i \in \mathbb{N}\end{aligned}\tag{1}$$

- Therefore, each Fibonacci number is the sum of the two previous ones, which gives the sequence

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \dots$$

- The Fibonacci sequence is related to the **golden ratio** ϕ and to the **conjugate of the golden ratio** $\hat{\phi}$,

Definition 2: golden ratio

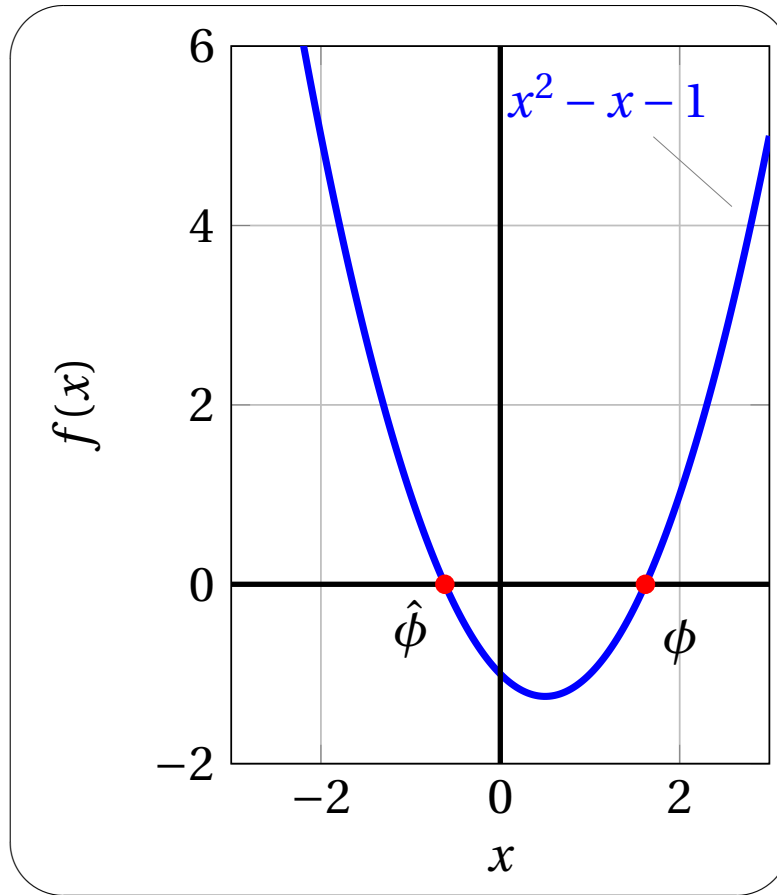
The **golden ratio** ϕ and the **conjugate of the golden ratio** $\hat{\phi}$ are the two roots of the equation:

$$x^2 = x + 1$$

and are given by the formulas:

$$\begin{aligned}\phi &= \frac{1 + \sqrt{5}}{2} & \hat{\phi} &= \frac{1 - \sqrt{5}}{2} \\ &\approx 1.61803 & &\approx -0.61803\end{aligned}$$

The following figure shows the graph of the function $f(x) = x^2 - x - 1$ on the interval $[-3, 3]$; the red dots correspond to the roots of the equation $x^2 = x + 1$.



The golden ratio ϕ and the conjugate of the golden ratio $\hat{\phi}$ clearly satisfy the equation $x^2 = x + 1$ since:

$$\phi^2 = \left(\frac{1 + \sqrt{5}}{2} \right)^2 = \frac{1 + 2\sqrt{5} + 5}{4} = \frac{3 + \sqrt{5}}{2} = \frac{1 + \sqrt{5}}{2} + 1 = \phi + 1$$

$$\hat{\phi}^2 = \left(\frac{1 - \sqrt{5}}{2} \right)^2 = \frac{1 - 2\sqrt{5} + 5}{4} = \frac{3 - \sqrt{5}}{2} = \frac{1 - \sqrt{5}}{2} + 1 = \hat{\phi} + 1$$

Remark 1

$$F_i = \frac{\phi^i - \hat{\phi}^i}{\sqrt{5}}, \quad i = 0, 1, 2, \dots$$

Proof. By induction on i .

- **Base case of the induction**

We prove that the formula holds for $i = 0$ and $i = 1$:

$$F_0 = \frac{\phi^0 - \hat{\phi}^0}{\sqrt{5}} = \frac{1 - 1}{\sqrt{5}} = 0, \quad F_1 = \frac{\phi^1 - \hat{\phi}^1}{\sqrt{5}} = \frac{\sqrt{5}}{\sqrt{5}} = 1.$$

- **Inductive step**

Suppose it is true for $i = k$ and $i = k - 1$ with $k \geq 1$, and let us prove it for $i = k + 1$. By the inductive hypothesis, we have: $F_{k+1} = F_k + F_{k-1}$, hence we can write

$$\begin{aligned} F_{k+1} &= F_k + F_{k-1} \\ &= \frac{\phi^k - \hat{\phi}^k}{\sqrt{5}} + \frac{\phi^{k-1} - \hat{\phi}^{k-1}}{\sqrt{5}} = \frac{(\phi^k - \hat{\phi}^k) + (\phi^{k-1} - \hat{\phi}^{k-1})}{\sqrt{5}} \\ &= \frac{(\phi^k + \phi^{k-1}) - (\hat{\phi}^k + \hat{\phi}^{k-1})}{\sqrt{5}} = \frac{\phi^{k-1}(\phi + 1) - \hat{\phi}^{k-1}(\hat{\phi} + 1)}{\sqrt{5}} \\ &= \frac{\phi^{k-1}(\phi^2) - \hat{\phi}^{k-1}(\hat{\phi}^2)}{\sqrt{5}} = \frac{\phi^{k+1} - \hat{\phi}^{k+1}}{\sqrt{5}} \end{aligned}$$

which is exactly the desired statement, for $i = k + 1$.

□

Remark 2

$$F_i = \left\lfloor \frac{\phi^i}{\sqrt{5}} + \frac{1}{2} \right\rfloor, \quad i = 0, 1, 2, \dots$$

Proof. Since $|\hat{\phi}| < 1$, we have

$$\frac{|\hat{\phi}^i|}{\sqrt{5}} \leq \frac{1}{\sqrt{5}} < \frac{1}{2} \quad \text{and} \quad -\frac{1}{2} < \frac{\hat{\phi}^i}{\sqrt{5}} < \frac{1}{2}, \quad i = 0, 1, 2, \dots$$

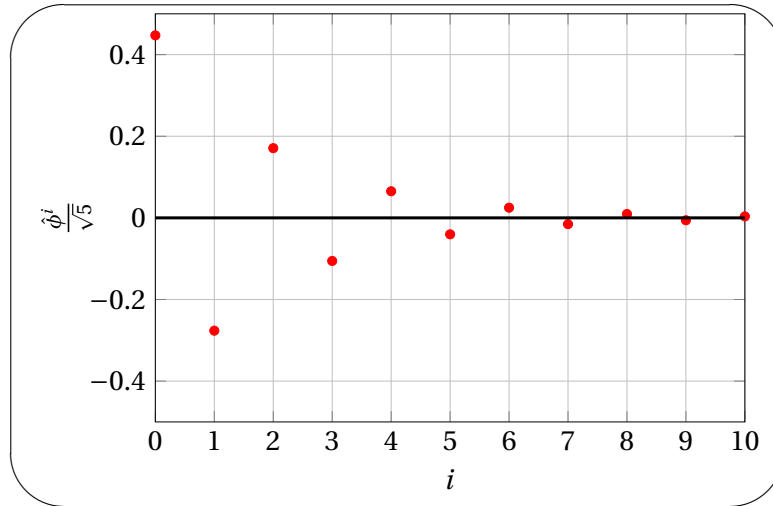
Since $F_i \in \mathbb{N}$ and $F_i = \frac{\phi^i}{\sqrt{5}} - \frac{\hat{\phi}^i}{\sqrt{5}}$, the i -th Fibonacci number F_i is equal to $\frac{\phi^i}{\sqrt{5}}$ rounded to the nearest integer:

$$F_i = \left\lfloor \frac{\phi^i}{\sqrt{5}} + \frac{1}{2} \right\rfloor, \quad i = 0, 1, 2, \dots$$

□

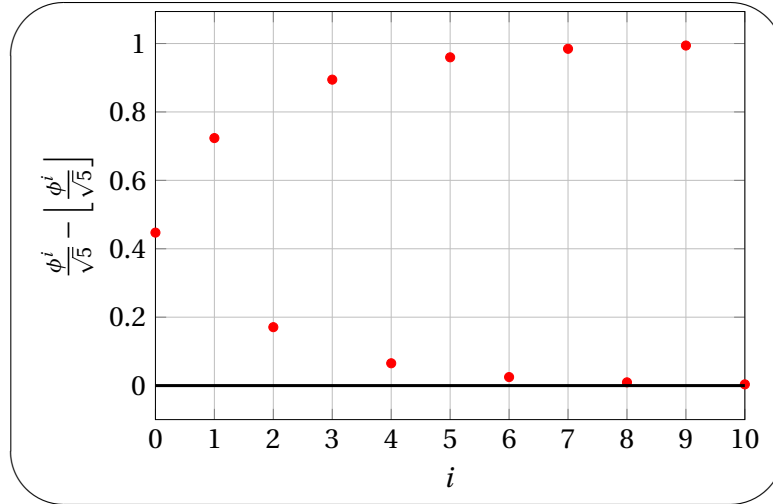
- The values of

$\frac{\hat{\phi}^i}{\sqrt{5}}$ with $i = 0, 1, \dots, 10$ are :



- The values of

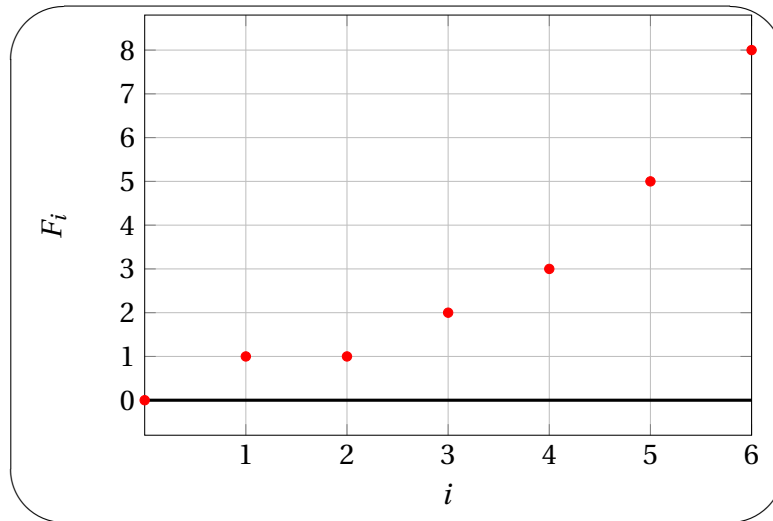
$\frac{\phi^i}{\sqrt{5}} - \left\lfloor \frac{\phi^i}{\sqrt{5}} \right\rfloor$ with $i = 0, 1, \dots, 10$ are :



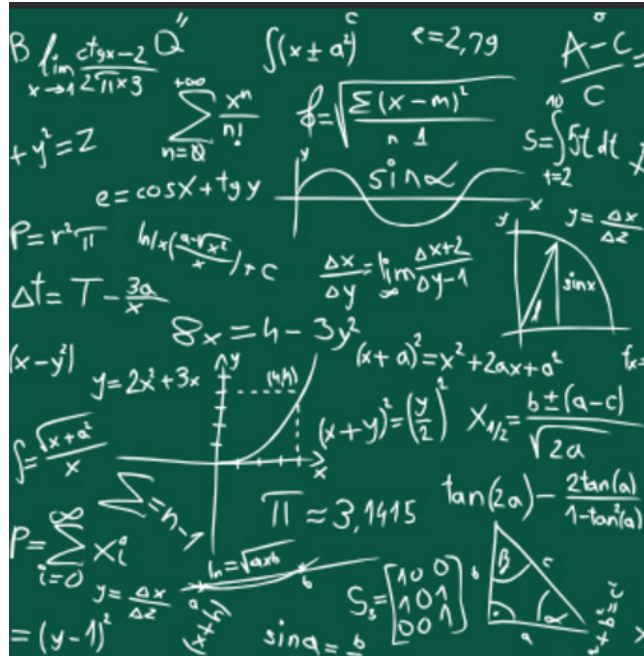
- Since $F_i \in \mathbb{N}$, we moreover have:

$$\left(\frac{\phi^i}{\sqrt{5}} - \left\lfloor \frac{\phi^i}{\sqrt{5}} \right\rfloor \right) - \frac{\hat{\phi}^i}{\sqrt{5}} \in \{0, 1\}, \quad i = 0, 1, 2, \dots$$

- The values of F_i , with $i = 0, 1, \dots, 10$, are:



Cardinality of infinite sets



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Countable cardinality

Is it possible to compare the “size” of infinite sets?

- Let us approach the question starting from the important number sets that we have introduced: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and ask ourselves: how many elements does each of these sets have?
- Intuitively, the answer seems obvious: each of these sets has infinitely many elements; however, the rational numbers are more numerous than the integers, since $\mathbb{Z} \subset \mathbb{Q}$, and for the same reason the real numbers are more numerous than the rational numbers, since $\mathbb{Q} \subset \mathbb{R}$.
- How can we state, at the same time, that two sets are both infinite, but one is more numerous than the other?

How is it possible, in general, to compare the size of infinite sets?

- To give meaning to these questions, even before being able to answer them, we must define what we mean by two sets having the same size.
- Abstracting from the experience of counting the elements of a finite set, the idea of **same size** has been identified with that of **one-to-one correspondence** (bijection):

Definition 1: equal cardinality of two sets

Two sets A , B are said to have **equal cardinality** (or power) if they can be put in one-to-one correspondence with each other, i.e., if there exists a rule that associates with each element of A one and only one element of B , and vice versa.

- The cardinality (or power) of a set expresses the intuitive idea of size

Remark 1

The set of integers $\mathbb{Z}$ and the set of natural numbers $\mathbb{N}$ have the same cardinality

Proof. The following rule puts the integers and the natural numbers in one-to-one correspondence:

$\mathbb{Z}$	0	1	-1	2	-2	...	n	$-n$	...
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$		$\downarrow$	$\downarrow$	
$\mathbb{N}$	0	1	2	3	4	...	$2n-1$	$2n$	...

□

- Therefore, even though from the point of view of inclusion $\mathbb{Z}$ has “more elements” than $\mathbb{N}$ (in the sense that it has all the elements of $\mathbb{N}$ plus others), the two sets have the same cardinality.

Two sets that have the same cardinality are also called **equipotent** and should be thought of as equally numerous.

Definition 2: countable set

A set that has the same cardinality as $\mathbb{N}$ is called **countable**.

- The term countable indicates that the elements of the set can be enumerated, that is, arranged in a numbered list (place 1, place 2, place 3, ...)

The cardinality of $\mathbb{N}$ is called the **cardinality of the countable** (countable infinity).

Theorem 1

The set $\mathbb{Q}$ is countable

Proof. • We begin by proving that the set of positive rational numbers is countable.

- To do this, we represent the positive rational numbers as fractions n/m , with n, m positive integers, and we arrange these fractions in an **infinite triangular table**, in the following way:

$\frac{n}{m}$, with $n + m = \dots$				
2	$\frac{1}{1}$			
3	$\frac{1}{2}$	$\frac{2}{1}$		
4	$\frac{1}{3}$	$\frac{2}{2}$	$\frac{3}{1}$	
5	$\frac{1}{4}$	$\frac{2}{3}$	$\frac{3}{2}$	$\frac{4}{1}$
...	...			

Note that: all the fractions representing positive rational numbers appear in this table at least once; some numbers are repeated several times (e.g., $\frac{1}{1} = \frac{2}{2}$). Each row has finite length.

- Hence we can put $\mathbb{N}$ in one-to-one correspondence with the set of positive rationals, by going through the table row by row (and skipping an element when it is equal to one already encountered). For example:

1	2	3	4	5	6	7	8	9	...
$\frac{1}{1}$	$\frac{1}{2}$	$\frac{2}{1}$	$\frac{1}{3}$	$\frac{3}{1}$	$\frac{1}{4}$	$\frac{2}{3}$	$\frac{3}{2}$	$\frac{4}{1}$	...

(note that we skipped $\frac{2}{2}$ because it is equal to $\frac{1}{1}$, already encountered).

- This proves that the set of positive rationals is countable.
- Then $\mathbb{Q}$ is also countable, and this can be proved with an argument similar to the one we used to prove that $\mathbb{Z}$ is countable: denoting by $q_1, q_2, q_3, \dots$ the positive rationals, we set:

$\mathbb{N}$	0	1	2	3	4	5	6	...
	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	
$\mathbb{Q}$	0	q_1	$-q_1$	q_2	$-q_2$	q_3	$-q_3$	...

which gives a one-to-one correspondence between $\mathbb{N}$ and $\mathbb{Q}$.

□

- Having discovered that several infinite sets, each properly contained in the next ($\mathbb{N}, \mathbb{Z}, \mathbb{Q}$), have the same cardinality, might suggest that this is true for all infinite sets. This is not true, as shown in the next section.

2 Cardinality of the continuum

Theorem 2

The set of real numbers $\mathbb{R}$ is not countable

- The proof technique is called **Cantor's diagonal argument**

Proof. • We will prove that the interval $[0, 1]$ has uncountable cardinality, from which the uncountability of $\mathbb{R}$ obviously follows

- Suppose therefore, **by contradiction**, that $[0, 1]$ is countable, and arrange **all** the real numbers of the interval $[0, 1]$ in a list $r_1, r_2, r_3, \dots$
- We write each number r_i in decimal form:

$$0.a_1 a_2 a_3 \dots$$

where the a_i are digits from 0 to 9; if the digits are all zero we have $r_i = 0$ and if they are all nine we have $r_i = 0.\bar{9} = 1$):

$$r_1 = 0.a_{11} a_{12} a_{13} \dots$$

$$r_2 = 0.a_{21} a_{22} a_{23} \dots$$

$$r_3 = 0.a_{31} a_{32} a_{33} \dots$$

...

- We now define the following decimal number:

$$r = 0.b_1 b_2 b_3 \dots$$

where the digits b_i are defined by the following rule:

$$b_i = \begin{cases} 5 & \text{if } a_{ii} \text{ is one of the digits } 0, 1, 2, 3 \text{ or } 4 \\ 4 & \text{if } a_{ii} \text{ is one of the digits } 5, 6, 7, 8 \text{ or } 9 \end{cases}$$

With this definition we have $b_i \neq a_{ii}$ for every i . Note that the number r has been constructed by reasoning on the diagonal of the infinite table whose rows are the numbers r_i , hence the name of the argument.

□

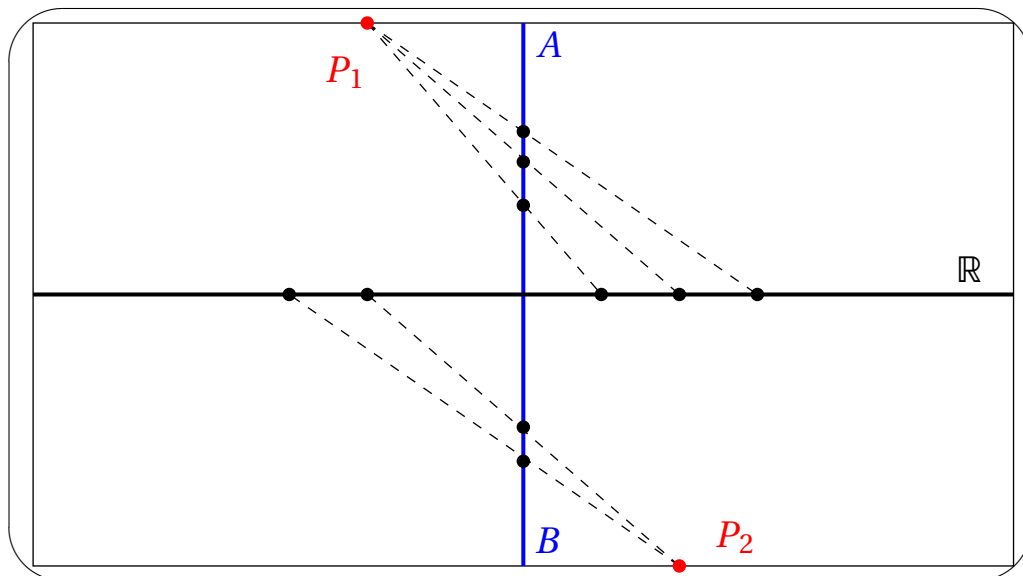
Proof. • We now observe that r is a real number belonging to the interval $[0, 1]$ (because it is of the form $0.b_1b_2b_3\dots$ with all digits equal to 4 or 5) and on the other hand it is not equal to any of the numbers r_i in the list, since:

- $r \neq r_1$: the first digit of r is different from the first digit of r_1 ($b_1 \neq a_{11}$);
- $r \neq r_2$: the second digit of r is different from the second digit of r_2 ($b_2 \neq a_{22}$);
- ...

- But this leads to a **contradiction**, since we had assumed that the r_i completely exhausted the set of real numbers of the interval $[0, 1]$. Therefore $[0, 1]$ is not countable. □

Since $\mathbb{N}$ cannot be put in one-to-one correspondence with $\mathbb{R}$, but it can be put in one-to-one correspondence with a proper subset of $\mathbb{R}$ ($\mathbb{N}$ itself!), we say that $\mathbb{N}$ has **smaller cardinality** than $\mathbb{R}$.

- We also note that the interval $[0, 1]$, and any interval of $\mathbb{R}$ (open or closed, bounded or unbounded), has the same cardinality as $\mathbb{R}$.
- This fact can be proved, for example, with an elementary geometric construction that gives a one-to-one correspondence between the points of the line and those of a segment $[A, B]$; as shown in the following figure, by drawing suitable segments from the two fixed points P_1, P_2 to the line itself:

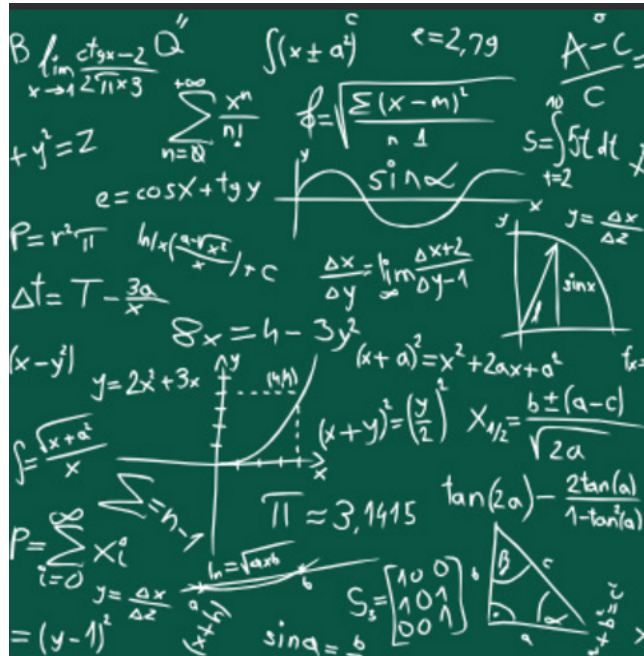


- This implies that, for example, the points of a line have the same cardinality as the points of a segment.

The cardinality of $\mathbb{R}$ is called the **cardinality of the continuum**.

- It is not only every interval of $\mathbb{R}$ that has this same cardinality. It can be proved that the same holds for the plane, for three-dimensional space and for their “continuous” subsets: for example, a plane and a sphere both have the cardinality of the continuum.
- So far we have encountered only two hierarchical levels of infinity: the cardinality of the countable and that of the continuum. One should not believe that only these two exist! For example, the set of all subsets of $\mathbb{R}$ (its power set) is even more numerous than $\mathbb{R}$; and with this procedure we can always construct a set more numerous than a given set.
- Therefore the hierarchical levels of infinite cardinalities are themselves infinite.

Complex numbers



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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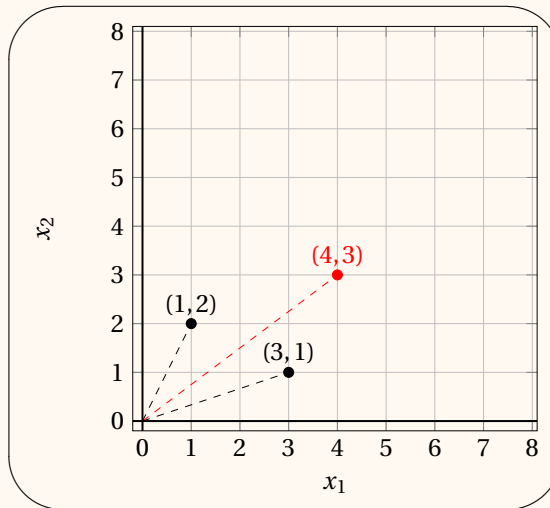
1 Definition of $\mathbb{C}$ and field structure

- We have denoted by $\mathbb{R}^2$ (short for $\mathbb{R} \times \mathbb{R}$) the set of ordered pairs (a, b) of real numbers.
- On these pairs we directly define the operations of sum and product with the following rules:

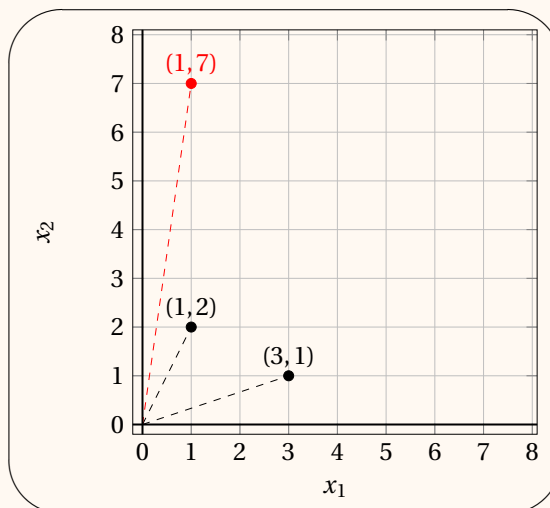
$$(a, b) + (c, d) = (a + c, b + d) \quad (1)$$

$$(a, b) \cdot (c, d) = (ac - bd, ad + bc) \quad (2)$$

Example 1: sum



Example 2: product



- This “sum” and this “product” satisfy the **commutative**, **associative** and **distributive** properties
- We also observe that, $\forall (a, b) \in \mathbb{R}^2$:

$$(a, b) + (0, 0) = (0, 0) + (a, b) = (a, b)$$

hence the pair $(0, 0)$ is the **identity element for the sum**. Moreover:

$$(a, b) \cdot (1, 0) = (1, 0) \cdot (a, b) = (a, b)$$

hence the pair $(1, 0)$ is the **identity element for the product**. We have:

$$(a, b) + (-a, -b) = (0, 0)$$

hence $(-a, -b)$ is the **opposite** of (a, b) . Moreover, if $(a, b) \neq (0, 0)$ then:

$$(a, b) \cdot \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right) = (1, 0)$$

hence the pair $(a/(a^2 + b^2), -b/(a^2 + b^2))$ is the **reciprocal** of (a, b) .

Definition 1: field of complex numbers

*Properties R_1, R_2 are satisfied by the sum and the product defined above, and therefore the set $\mathbb{R}^2$ with this structure is a field, which we will call the **field of complex numbers** and denote by $\mathbb{C}$*

- We now observe that $\mathbb{C}$ contains the subset $\mathbb{C}_0$ of pairs of the form $(a, 0)$; it is a subfield of $\mathbb{C}$, since the sum and product of pairs of this form are again pairs of the same form; indeed we have:

$$(a, 0) + (b, 0) = (a + b, 0) \quad \text{and} \quad (a, 0) \cdot (b, 0) = (a \cdot b, 0)$$

Moreover, $\mathbb{C}_0$ can be ordered by setting $(a, 0) < (b, 0)$ if $a < b$.

- If we then put the set of real numbers $\mathbb{R}$ in one-to-one correspondence with $\mathbb{C}_0$, by setting

$$(a, 0) \longleftrightarrow a$$

we can identify the real numbers $a \in \mathbb{R}$ with the complex numbers of the form $(a, 0) \in \mathbb{R}^2$. In this sense the field of complex numbers $\mathbb{C}$ is an **extension** of the field of real numbers $\mathbb{R}$.

Let us now consider the number $(0, 1) \in \mathbb{C}$. It has the remarkable property that:

$$(0, 1) \cdot (0, 1) = (-1, 0)$$

i.e., its square coincides with the real number -1

Definition 2: imaginary unit

*The pair $(0, 1) \in \mathbb{C}$ is denoted by the letter “ i ” and is called the **imaginary unit***

2 Algebraic form of complex numbers

We observe that, if we write any complex number $(c, 0)$ simply as c , we have:

$$(a, b) = (a, 0) + \underbrace{(0, 1)}_{=i} \cdot (b, 0) = a + ib$$

With this notation, rules (1) and (2) are the ordinary rules of algebraic calculation, keeping in mind that $i^2 = -1$:

$$(a + ib) + (c + id) = (a + c) + i(b + d) \quad (3)$$

$$(a + ib) \cdot (c + id) = (ac - bd) + i(ad + bc) \quad (4)$$

Definition 3: algebraic form, real part and imaginary part

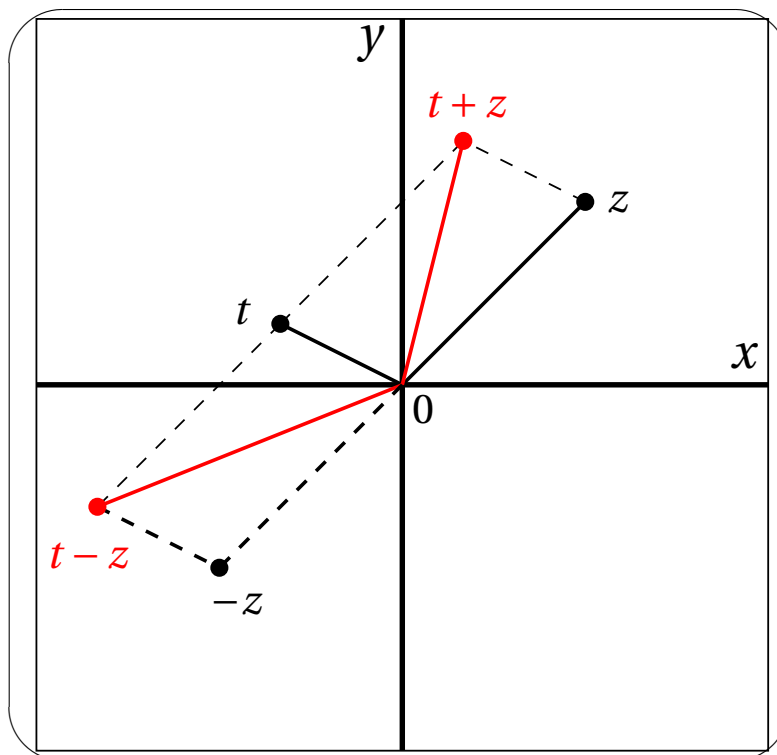
The expression:

$$z = a + ib \quad (5)$$

is called the **algebraic form of complex numbers**; a is called the **real part** of z and is denoted by $\Re(z)$ (or $\text{Re}(z)$) while b is called the **imaginary part** and is denoted by $\Im(z)$ (or $\text{Im}(z)$).

Complex plane

- In a Cartesian plane, the complex numbers $a + ib$ can be represented as points with coordinates (a, b) . In this context:
 - the plane is called the **complex plane** or **Gauss plane**
 - the x, y axes are called the **real axis** and the **imaginary axis**
 - the points on the real axis are the **real numbers**
 - the points on the imaginary axis are the **purely imaginary numbers** (i.e., of the form ib)
- The **sum of two complex numbers** is the complex number whose coordinates are the sums of the coordinates: the geometric meaning of this fact is that the point $z + t$ is constructed from the points z, t according to the “**parallelogram rule**”, illustrated in the following figure:



Remark 1

The set of complex numbers $\mathbb{C}$ is not an ordered field

Proof. • We have seen that $\mathbb{C}$ satisfies the field axioms; however, it does not satisfy those of an ordered field, that is, it is not possible to define a relation $\leq$ between complex numbers in such a way that properties $R3$ hold

- It can be proved that properties $R3$ imply that the square of any number is never negative, and on the other hand, if a number is positive its opposite is negative. Now, in $\mathbb{C}$ we have:

$$1^2 = 1 \quad \text{and} \quad i^2 = -1$$

We therefore have two squares, each the opposite of the other. Neither of them, however, can be negative (because they are squares), and this is absurd (because between a and $-a$ one must be negative, if $a \neq 0$). We conclude that $\mathbb{C}$ is not an ordered field.

□

Definition 4: conjugate

The complex number $a - ib$ is called the **complex conjugate** of $z = a + ib$ and is denoted by $\bar{z}$.

We have:

$$z + \bar{z} = 2a = (a + ib) + (a - ib) = 2a = 2\Re(z) \quad (6)$$

$$z - \bar{z} = 2ib = (a + ib) - (a - ib) = 2ib = 2i\Im(z) \quad (7)$$

$$z \cdot \bar{z} = (a + ib) \cdot (a - ib) = a^2 - \underbrace{aib + iba}_{=-1} b^2 = a^2 + b^2 \geq 0 \quad (8)$$

- The conjugation operation has the following elementary properties with respect to sum and product:

$$\overline{(z_1 + z_2)} = \bar{z}_1 + \bar{z}_2 \quad (9)$$

$$\overline{(z_1 \cdot z_2)} = \bar{z}_1 \cdot \bar{z}_2 \quad (10)$$

$$\overline{\left(\frac{1}{z}\right)} = \frac{1}{\bar{z}} \quad (11)$$

Definition 5: modulus

The **modulus** of $z = a + ib$ is the non-negative real number $\sqrt{a^2 + b^2}$; it is denoted by $|z|$.

- If $z = a$ is real, its modulus is called the absolute value and is still denoted by $|a|$. The following properties hold:
 - a) $|z| = 0 \iff z = 0$, moreover $|z| \geq 0$
 - b) $|z| = |\bar{z}|$
 - c) $\Re(z) \leq |z|$ $\Im(z) \leq |z|$ $|z| \leq |\Re(z)| + |\Im(z)|$
 - d) $|z_1 + z_2| \leq |z_1| + |z_2|$ **triangle inequality**
 - e) $|z_1 + z_2| \geq ||z_1| - |z_2||$
- Properties a), b), c) are immediately verified.

Proof. • Let us prove properties d) and e). They are equivalent to the following:

$$(|z_1| - |z_2|)^2 \leq |z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2$$

• Setting $z_1 = a + ib$, $z_2 = c + id$ we obtain:

$$\begin{aligned} |z_1 + z_2|^2 &= |(a + ib) + (c + id)|^2 \\ &= |(a + c) + i(b + d)|^2 \\ &= \left(\sqrt{(a + c)^2 + (b + d)^2} \right)^2 = (a + c)^2 + (b + d)^2 \end{aligned}$$

and hence we have:

$$\begin{aligned} \underbrace{\left(\sqrt{a^2 + b^2} - \sqrt{c^2 + d^2} \right)^2}_{=(a^2 + b^2) + (c^2 + d^2) - 2\sqrt{a^2 + b^2} \cdot \sqrt{c^2 + d^2}} &\leq \underbrace{(a + c)^2 + (b + d)^2}_{=a^2 + 2ac + c^2 + b^2 + 2bd + d^2} \leq \underbrace{\left(\sqrt{a^2 + b^2} + \sqrt{c^2 + d^2} \right)^2}_{=(a^2 + b^2) + (c^2 + d^2) + 2\sqrt{a^2 + b^2} \cdot \sqrt{c^2 + d^2}} \end{aligned}$$

After simplification, this double inequality reduces to:

$$-\sqrt{a^2 + b^2} \cdot \sqrt{c^2 + d^2} \leq ac + bd \leq \sqrt{a^2 + b^2} \cdot \sqrt{c^2 + d^2}$$

which is equivalent to the following:

$$|ac + db| \leq \sqrt{a^2 + b^2} \cdot \sqrt{c^2 + d^2}$$

Squaring both sides we arrive at:

$$\underbrace{(ac + db)^2}_{=a^2c^2 + d^2b^2 + 2acbd} \leq \underbrace{(a^2 + b^2) \cdot (c^2 + d^2)}_{=a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2}$$

that is,

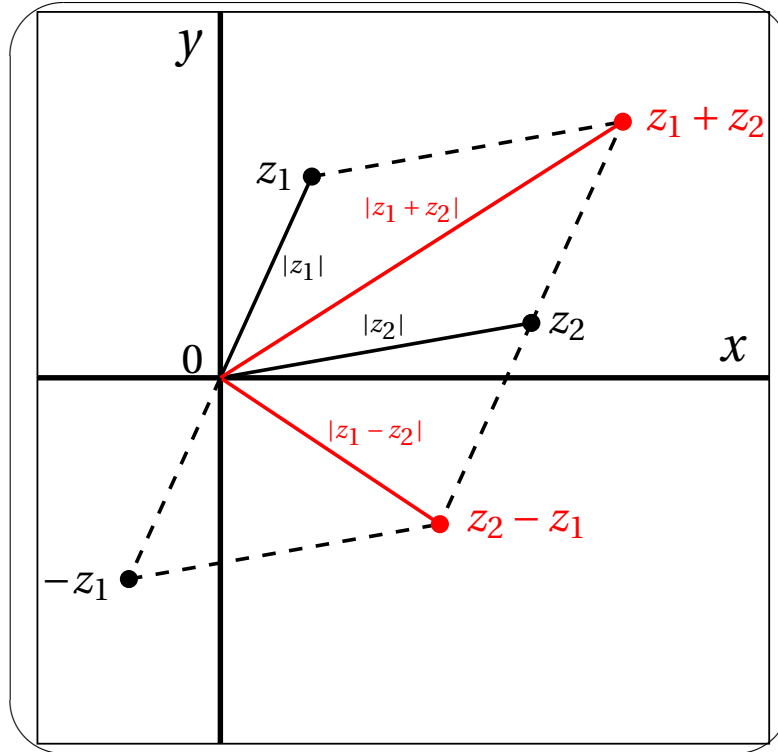
$$0 \leq -2acbd + a^2d^2 + b^2c^2 = (ad - bc)^2$$

which is true for every $a, b, c, d \in \mathbb{R}$.

□

- Geometrically, $|z|$ represents the distance of the point (or complex number) z from the origin; $|z_1 - z_2|$ represents the distance between the two points z_1 and z_2 ; inequalities d) and e) express the well-known theorem on the **lengths of the sides of a triangle**¹:

¹In a non-degenerate triangle, the sum of the lengths of two sides is greater than the length of the third one.



- Using the concepts just introduced, we can write the quotient of two complex numbers in algebraic form:

$$\frac{a + ib}{c + id}$$

it suffices to multiply the numerator and the denominator by $c - id$, and we obtain:

$$\frac{a + ib}{c + id} = \frac{(a + ib)(c - id)}{\underbrace{(c + id)(c - id)}_{=c^2+d^2}} = \frac{ac - aid + ibc - i^2bd}{c^2 + d^2} = \frac{(ac + bd)}{c^2 + d^2} + i \frac{(bc - ad)}{c^2 + d^2}$$

2.1 Equations in the complex field

- Let us see how to solve an equation in the complex field, when it involves the unknown $z = x + iy$ also through $\Re(z)$, $\Im(z)$, z , $|z|$.
- We illustrate, with the following example, the procedure of transforming the equation in one complex unknown into a system of two equations in two real unknowns. The method consists in taking the real and imaginary parts of the equation.

Example 3: equations in the complex field (algebraic method)

- We want to solve:

$$z^2 + i \Im z + 2 \bar{z} = 0$$

We set $z = x + i y$, with x, y real unknowns, and rewrite the equation:

$$\begin{cases} z^2 = (x + i y)^2 = x^2 - y^2 + 2 i x y \\ i \Im z = i y \\ 2 \bar{z} = 2(x - i y) = 2x - 2 i y \end{cases}$$

Substituting, we therefore obtain:

$$(x^2 - y^2 + 2 i x y) + (i y) + (2x - 2 i y) = 0$$

- Now, a complex number is zero if and only if its real part and imaginary part are zero. Therefore we separate the real part and the imaginary part of the left-hand side:

$$(x^2 - y^2 + 2x) + i(2xy + y - 2y) = 0$$

and set both equal to zero:

$$\begin{cases} x^2 - y^2 + 2x = 0 \\ 2xy - y = 0 \end{cases}$$

We have thus transformed the equation in one complex unknown into a system of two equations in two real unknowns.

- We solve the system. The second equation gives:

$$y = 0 \quad \text{or} \quad x = \frac{1}{2}$$

1. For $y = 0$ the first equation becomes

$$x^2 + 2x = 0 \quad \text{which gives} \quad x = 0 \quad \text{or} \quad x = -2$$

2. For $x = \frac{1}{2}$ the first equation becomes

$$-y^2 + \frac{5}{4} = 0 \quad \text{which gives} \quad y = \pm \frac{\sqrt{5}}{2}$$

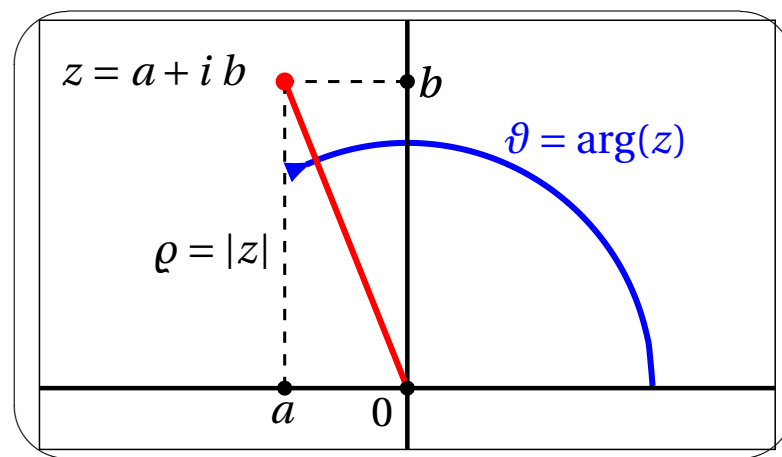
- Hence the equation has the following 4 solutions:

$$z = 0, \quad z = -2, \quad z = \frac{1}{2} + i \frac{\sqrt{5}}{2}, \quad z = \frac{1}{2} - i \frac{\sqrt{5}}{2}$$

The method seen in this example can in principle be applied to any equation in $\mathbb{C}$, but a generic system of two equations in two unknowns is almost always impossible to solve algebraically.

3 Trigonometric form of complex numbers

- As is known from Geometry, the points of the plane can be identified not only by their Cartesian coordinates, but also by their **polar coordinates**:
 - $\rho \rightarrow$ **polar radius**, i.e., the distance of the point from the origin
 - $\vartheta \rightarrow$ **polar angle**, i.e., the angle that the line joining the point to the origin forms with the positive x -axis, measured counterclockwise.
- Clearly, a pair ρ, ϑ , with $\rho > 0$, identifies a well-defined point of the plane; conversely, a point of the plane uniquely determines the coordinate ρ , but the angle ϑ , measured in radians, is determined only up to multiples of 2π .



- Given a complex number z , its **modulus** $|z|$ coincides with the **polar radius** ρ of the point representing it in the complex plane.
- We call **argument** of z , and denote by $\arg(z)$, any of the angles ϑ associated with the point z . In this way the argument of z is not uniquely determined. Often this indeterminacy does not cause any problem. At other times, however, it is preferable to assign a well-defined argument to a complex number. This can be done in infinitely many ways, by fixing any interval of length 2π within which the angle ϑ is allowed to vary
- The intervals most commonly used for this purpose are $[0, 2\pi)$ and $(-\pi, \pi]$; the argument of z is then called the **principal argument**.

Example 4: arguments of complex numbers

- the number $-i$ has argument $-\pi/2$ or $3\pi/2$ or any other value of the form $-\pi/2 + 2k\pi$ with $k \in \mathbb{Z}$. Its principal argument will be $3\pi/2$ if we adopt the convention $\vartheta \in [0, 2\pi)$, and $-\pi/2$ with the convention $\vartheta \in (-\pi, \pi]$.
- Positive real numbers have principal argument 0 and negative ones π , with both conventions.

Given the number $z = a + ib$, from trigonometry we immediately obtain the relations between the Cartesian coordinates a, b and the polar ones ρ, ϑ :

$$a = \rho \cos \vartheta \quad \text{and} \quad b = \rho \sin \vartheta \quad (12)$$

The inverse relations are:

$$\rho = \sqrt{a^2 + b^2}, \quad \cos \vartheta = \frac{a}{\sqrt{a^2 + b^2}} \quad \text{and} \quad \sin \vartheta = \frac{b}{\sqrt{a^2 + b^2}} \quad (13)$$

Definition 6: trigonometric form

A complex number $z = a + ib$ can also be written in the form

$$z = \rho (\cos \vartheta + i \sin \vartheta) \quad (14)$$

which is called the **trigonometric form** of complex numbers.

Example 5: trigonometric form

Let us write the following complex number in trigonometric form:

$$z = \sqrt{3} + i$$

We have $\rho = \sqrt{a^2 + b^2} = \sqrt{3 + 1} = 2$, hence:

$$\sqrt{3} + i = 2 \left(\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

3.1 De Moivre's formulas

- The trigonometric form is convenient for expressing products and quotients of complex numbers. Indeed, if we have:

$$z_1 = \rho_1 (\cos \vartheta_1 + i \sin \vartheta_1) \quad z_2 = \rho_2 (\cos \vartheta_2 + i \sin \vartheta_2)$$

for the **product** we obtain:

$$\begin{aligned} z_1 z_2 &= \rho_1 \rho_2 \cdot \left\{ \cos \vartheta_1 \cos \vartheta_2 - \sin \vartheta_1 \sin \vartheta_2 + i (\sin \vartheta_1 \cos \vartheta_2 + \cos \vartheta_1 \sin \vartheta_2) \right\} \\ &= \rho_1 \rho_2 \cdot \left\{ \cos (\vartheta_1 + \vartheta_2) + i \sin (\vartheta_1 + \vartheta_2) \right\} \end{aligned} \quad (15)$$

for the **quotient**, if $z_2 \neq 0$, we have:

$$\frac{z_1}{z_2} = \frac{\rho_1}{\rho_2} \cdot \frac{\cos \vartheta_1 + i \sin \vartheta_1}{\cos \vartheta_2 + i \sin \vartheta_2},$$

multiplying numerator and denominator by $(\cos \vartheta_2 - i \sin \vartheta_2)$ and, taking into account that $(\cos \vartheta_2)^2 + (\sin \vartheta_2)^2 = 1$, we obtain:

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{\rho_1}{\rho_2} \cdot \left\{ (\cos \vartheta_1 + i \sin \vartheta_1) \cdot (\cos \vartheta_2 - i \sin \vartheta_2) \right\} \\ &= \frac{\rho_1}{\rho_2} \cdot \left\{ \cos(\vartheta_1 - \vartheta_2) + i \sin(\vartheta_1 - \vartheta_2) \right\} \end{aligned} \quad (16)$$

- Therefore the modulus of the product and of the quotient of two complex numbers is, respectively, the product and the quotient of the moduli; the argument is, respectively, the sum and the difference of the arguments:

$$|z_1 \cdot z_2| = |z_1| \cdot |z_2| \quad \text{and} \quad \arg(z_1 \cdot z_2) = \arg(z_1) + \arg(z_2) \quad (17)$$

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} \quad \text{and} \quad \arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2) \quad (18)$$

- Formula (15) generalizes to the case of any number of factors $z_1, z_2, \dots, z_n$:

$$z_1 z_2 \dots z_n = \rho_1 \rho_2 \dots \rho_n \cdot \left\{ \cos(\vartheta_1 + \vartheta_2 + \dots + \vartheta_n) + i \sin(\vartheta_1 + \vartheta_2 + \dots + \vartheta_n) \right\} \quad (19)$$

If, moreover, all the factors are equal, we obtain:

$$z^n = \rho^n \cdot \left\{ \cos(n \vartheta) + i \sin(n \vartheta) \right\} \quad (20)$$

- These relations on products and quotients of complex numbers are known as De Moivre's formulas.

Example 6: powers of complex numbers with De Moivre's formulas

Write in algebraic form:

$$z = (1 + i)^7$$

We determine the modulus and argument of $(1 + i)$ and then apply De Moivre's formula.

$$|1 + i| = \sqrt{2} \quad \text{and} \quad \arg(1 + i) = \frac{\pi}{4} \quad \left(\cos \vartheta = \frac{1}{\sqrt{2}}, \quad \sin \vartheta = \frac{1}{\sqrt{2}} \right)$$

Hence we have:

$$|(1 + i)^7| = \left(\sqrt{2}\right)^7 = 2^{\frac{7}{2}} = 2^{3+\frac{1}{2}} = 8\sqrt{2} \quad \text{and} \quad \arg(1 + i)^7 = \frac{7}{4}\pi$$

Consequently:

$$(1 + i)^7 = 8\sqrt{2} \left(\underbrace{\cos \frac{7}{4}\pi}_{=\frac{1}{\sqrt{2}}} + i \underbrace{\sin \frac{7}{4}\pi}_{=-\frac{1}{\sqrt{2}}} \right) = 8\sqrt{2} \left(\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} \right) = 8 - 8i$$

- De Moivre's formulas allow us to give a geometric interpretation of the product of complex numbers.
- Let z , to begin with, be a complex number with modulus 1, hence of the form $(\cos \vartheta + i \sin \vartheta)$. Then, multiplying a number by z means adding ϑ to its argument, i.e., performing a **rotation by the angle ϑ** .
- If z has modulus ρ instead of 1, in addition to a rotation we perform a **dilation by the factor ρ** .

Example 7: geometric interpretation of the product of complex numbers

- multiplying by i means performing a rotation by $\frac{\pi}{2}$;
- multiplying by -1 means performing a rotation by π ;
- multiplying by $(1 + i)$ means performing a dilation by the factor $\sqrt{2}$ and a rotation by $\frac{\pi}{4}$

Example 8: equations in the complex field (trigonometric method)

- We want to solve:

$$z^3 - |z| = 0$$

We rewrite the equation in the form

$$z^3 = |z|$$

and we set $z = \rho (\cos \vartheta + i \sin \vartheta)$ (trigonometric form) and rewrite the equation:

$$\begin{cases} z^3 = \rho^3 (\cos 3\vartheta + i \sin 3\vartheta) \\ |z| = \rho \end{cases}$$

Substituting, we therefore obtain:

$$\rho^3 (\cos 3\vartheta + i \sin 3\vartheta) = \rho$$

The equation is satisfied if and only if the two sides have equal moduli and arguments that differ by multiples of 2π (the right-hand side has argument 0), that is:

$$\begin{cases} \rho^3 = \rho \\ 3\vartheta = 2k\pi \quad \text{with } k \in \mathbb{Z} \end{cases}$$

The first equation gives $\rho = 0$ and $\rho = 1$ (careful: ρ must be ≥ 0 because it is the modulus of the complex number; therefore $\rho = -1$ is not acceptable); the second one gives $\vartheta = \frac{2k\pi}{3}, k \in \mathbb{Z}$. Hence:

$$z = 0, \quad z = \cos \frac{2k\pi}{3} + i \sin \frac{2k\pi}{3} \quad \text{with } k \in \mathbb{Z}$$

Explicitly:

$$z = 0, \quad z = 1, \quad z = -\frac{1}{2} + i \frac{\sqrt{3}}{2}, \quad z = -\frac{1}{2} - i \frac{\sqrt{3}}{2}$$

3.2 n -th roots of complex numbers

Definition 7: n -th root of a complex number

Given a complex number w , we say that z is an n -th (complex) root of w if $z^n = w$.

Theorem 1

Let $w \in \mathbb{C}$, $w \neq 0$, and n an integer ≥ 1 . There exist exactly n complex n -th roots $z_0, z_1, \dots, z_{n-1}$ of w ; setting

$$w = r (\cos \varphi + i \sin \varphi) \quad \text{and} \quad z_k = \varrho_k (\cos \vartheta_k + i \sin \vartheta_k)$$

we have

$$\begin{cases} \varrho_k = r^{1/n} \\ \vartheta_k = \frac{\varphi + 2k\pi}{n} \end{cases} \quad k = 0, 1, 2, \dots, n-1 \quad (21)$$

Proof. • The numbers z_k are clearly roots of w , as can be seen by computing z_k^n with De Moivre's formula. Let us show that there are no others.

- If a number $R(\cos \psi + i \sin \psi)$ is an n -th root of w , we must have

$$R^n = r \quad \text{and} \quad n\psi = \varphi + 2h\pi \quad \text{with} \quad h \in \mathbb{Z}$$

or equivalently:

$$R = r^{1/n} \quad \text{and} \quad \psi = \varphi/n + 2h\pi/n \quad \text{with} \quad h \in \mathbb{Z}$$

- Giving h the values $0, 1, \dots, n-1$ we find exactly the numbers z_k .
- Giving h any other value $\bar{h}$ different from the previous ones, it can be written in the form $\bar{h} = k + mn$ ($m \in \mathbb{Z}$ is the quotient and k is the remainder of the division of $\bar{h}$ by n), so that we would have

$$\psi = \frac{\varphi}{n} + \frac{2k\pi}{n} + 2m\pi = \vartheta_k + 2m\pi$$

and we would find again the same z_k as before.

□

Example 9: fifth root of a complex number

- Let us compute $\sqrt[5]{1+i}$: the complex number $(1+i)$ has modulus $\sqrt{2}$ and argument $\pi/4$. The fifth roots will therefore be:

$$\begin{aligned}\sqrt[5]{1+i} &= \sqrt[5]{\sqrt{2}} \left[\cos\left(\frac{\frac{\pi}{4} + 2k\pi}{5}\right) + i \sin\left(\frac{\frac{\pi}{4} + 2k\pi}{5}\right) \right] = \\ &= \sqrt[10]{2} \left[\cos\left(\frac{\pi}{20} + \frac{2}{5}k\pi\right) + i \sin\left(\frac{\pi}{20} + \frac{2}{5}k\pi\right) \right] \quad \text{with } k = 0, 1, 2, 3, 4\end{aligned}$$

- The angles found are not standard angles, but if desired, their sines and cosines can be computed approximately using a calculator.

- Unfortunately, a somewhat ambiguous notation is used for complex roots, the same one used to denote the arithmetic root; that is, $\sqrt[n]{z}$ or $z^{1/n}$ denotes the set of the n complex roots of z .
- This can create confusion when z is real. Indeed, the symbol $\sqrt{4}$, understood as the arithmetic root of 4, is 2; understood as the complex root of 4, it is the set of the two numbers +2 and -2.

Example 10: cube root of a complex number

- Let us compute $\sqrt[3]{-1}$: the complex number (-1) has modulus 1 and argument π . The cube roots will therefore be:

$$\sqrt[3]{-1} = \underbrace{\sqrt[3]{1}}_{=1} \left[\cos\left(\frac{\pi}{3} + \frac{2}{3}k\pi\right) + i \sin\left(\frac{\pi}{3} + \frac{2}{3}k\pi\right) \right] \quad \text{with } k = 0, 1, 2$$

that is, the numbers z_k of the form:

$$\cos \vartheta_k + i \sin \vartheta_k \quad \text{with } \vartheta_k = \frac{\pi}{3} + \frac{2}{3}k\pi \quad \text{and } k = 0, 1, 2$$

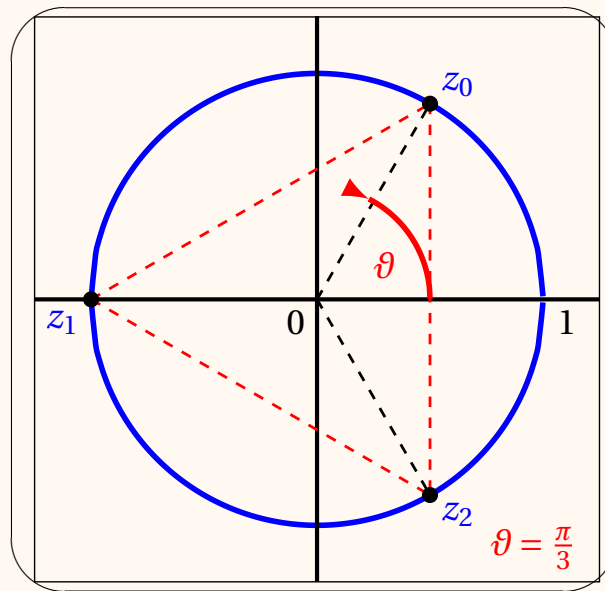
- Explicitly we have:

$$\begin{cases} z_0 = \cos\left(\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{3}\right) = \frac{1}{2} + i \frac{\sqrt{3}}{2} = \frac{1}{2}(1 + i\sqrt{3}) \\ z_1 = \cos\left(\frac{\pi}{3} + \frac{2\pi}{3}\right) + i \sin\left(\frac{\pi}{3} + \frac{2\pi}{3}\right) = \cos(\pi) + i \sin(\pi) = -1 \\ z_2 = \cos\left(\frac{\pi}{3} + \frac{4\pi}{3}\right) + i \sin\left(\frac{\pi}{3} + \frac{4\pi}{3}\right) = \cos\left(\frac{5\pi}{3}\right) + i \sin\left(\frac{5\pi}{3}\right) = \frac{1}{2} - i \frac{\sqrt{3}}{2} = \frac{1}{2}(1 - i\sqrt{3}) \end{cases}$$

- The arrangement of the roots of complex numbers in the Gauss plane is not random.
- Indeed, if $w = r(\cos \varphi + i \sin \varphi)$, the n -th roots $z_0, z_1, \dots, z_{n-1}$ of w lie at the vertices of the regular polygon with n sides inscribed in the circle with center 0 and radius $r^{1/n}$, with the vertex z_0 located at the point with argument $\vartheta = \varphi/n$.

Example 11: roots in the Gauss plane

- The figure shows the cube roots of -1 : z_0, z_1, z_2 from the previous exercise:

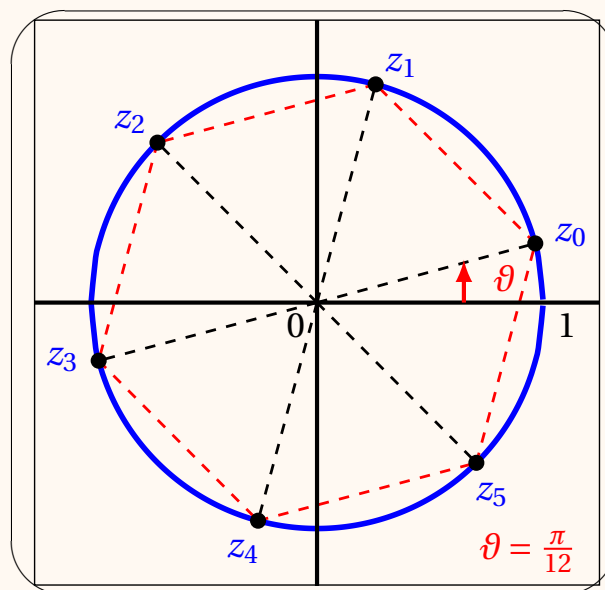


- The figure shows the sixth roots of i : $z_0, z_1, z_2, z_3, z_4, z_5$,

$$\sqrt[6]{i} = \underbrace{\sqrt[6]{1}}_{=1} \left[\cos \left(\frac{\pi}{2 \cdot 6} + \frac{2}{6} k \pi \right) + i \sin \left(\frac{\pi}{2 \cdot 6} + \frac{2}{6} k \pi \right) \right] \quad \text{with } k = 0, 1, 2, 3, 4, 5$$

that is, the numbers z_k of the form:

$$\cos \vartheta_k + i \sin \vartheta_k \quad \text{with } \vartheta_k = \frac{\pi}{12} + \frac{1}{3} k \pi \quad \text{and } k = 0, 1, 2, 3, 4, 5$$

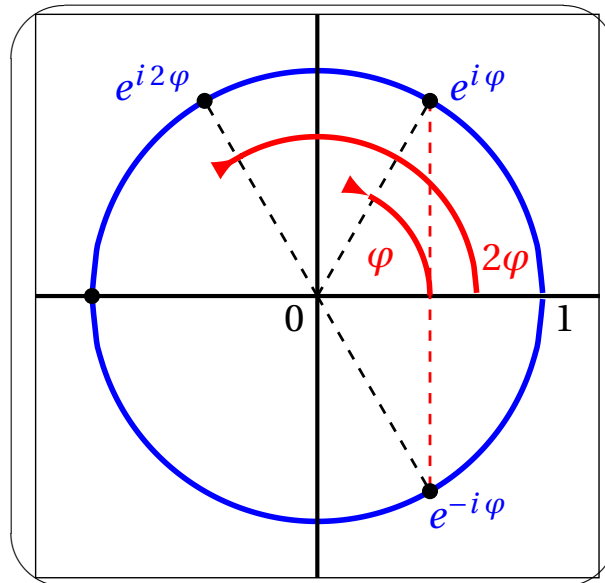


3.3 Exponential form of complex numbers

- The following notation is useful:

$$e^{i\varphi} = \cos \varphi + i \sin \varphi \quad \text{with } \varphi \in \mathbb{R}$$

where e is Euler's number (Napier's constant).



With this notation, the *trigonometric form* can be equivalently rewritten as:

$$z = |z| (\cos \varphi + i \sin \varphi) = |z| e^{i\varphi}, \quad \varphi = \arg(z) \quad \text{with } z \neq 0$$

This notation is called the **exponential form** of complex numbers.

- This form is useful since we have:

$$e^{i\varphi_1} \cdot e^{i\varphi_2} = e^{i(\varphi_1 + \varphi_2)}$$

that is, it is very easy to compute products of complex numbers.

De Moivre's formulas become:

$$(e^{i\varphi})^n = e^{in\varphi}$$

The exponential notation is widely used in applications because it greatly simplifies manipulations involving trigonometric quantities.

We have:

$$e^{2k\pi i} = 1 \quad \text{and} \quad e^{i(\varphi+2k\pi)} = e^{i\varphi} \quad \forall k \in \mathbb{Z}, \quad \text{moreover} \quad |e^{i\varphi}| = 1$$

We also have:

$$e^{i\pi} = -1, \quad e^{i\frac{\pi}{2}} = i, \quad e^{i\frac{3\pi}{2}} = -i, \quad e^{i\frac{3\pi}{4}} = \frac{-1+i}{\sqrt{2}}$$

Rewriting, we obtain **Euler's formula**:

$$e^{i\pi} + 1 = 0$$

which links in a simple way five of the most important constants: 0, 1, i , π , e .

The exponential notation can be generalized consistently with the properties of powers, by defining the **complex exponential**:

$$e^z = e^{x+iy} = e^x e^{iy} = e^x (\cos y + i \sin y), \quad \forall z = x + iy \in \mathbb{C}$$

- From the definition of exponential form it follows that

$$e^{-i\varphi} = \cos(-\varphi) + i \sin(-\varphi) = \cos \varphi - i \sin \varphi$$

and we easily obtain:

$$\cos \varphi = \frac{e^{i\varphi} + e^{-i\varphi}}{2}, \quad \sin \varphi = \frac{e^{i\varphi} - e^{-i\varphi}}{2i} \quad \text{with } \varphi \in \mathbb{R}$$

These formulas allow us to express sine and cosine as combinations of complex exponentials.

4 Quadratic equations

- A **second-degree equation** or **quadratic equation** in one **real** unknown x is an algebraic equation in which the highest power of the unknown is 2, and it can always be written in the form:

$$ax^2 + bx + c = 0 \quad (a \neq 0) \tag{22}$$

where a, b and c are real numbers.

The solutions are also called **roots** or **zeros** of the equation.

Remark 2: quadratic formula

Given a quadratic equation $a x^2 + b x + c = 0$ ($a \neq 0$), its zeros or roots are:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (23)$$

- A quadratic equation is called a “complete quadratic equation” when all its coefficients are different from 0. It is solved with the so-called **method of completing the square**, so called because the equation is modified until its left-hand side becomes the square of a binomial.

Proof. Isolating the constant term, we obtain:

$$a x^2 + b x = -c$$

$$\underbrace{4 a^2 x^2}_{=(2 a x)^2} + \underbrace{4 a b x}_{=2 (2 a x) b} = -4 a c$$

$$(2 a x)^2 + 2 (2 a x) b + b^2 = b^2 - 4 a c$$

$$(2 a x + b)^2 = \underbrace{b^2 - 4 a c}_{:=\Delta \text{ (discriminant)}}$$

$$2 a x + b = \pm \sqrt{b^2 - 4 a c}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4 a c}}{2 a}$$

□

- If the discriminant Δ is negative there are no real solutions.
- If $\Delta = 0$, the quadratic formula becomes:

$$x = -\frac{b}{2a}$$

therefore there is only one root, with *multiplicity two*.

- Finally, if $\Delta < 0$, the equation has no real solutions. In particular, there are always two solutions, but they belong to the field of complex numbers: they are **two complex conjugate numbers** and are computed with the two formulas:

$$x_{\pm} = \frac{-b}{2a} \pm i \left(\frac{\sqrt{4ac - b^2}}{2a} \right)$$

$$x_- = \frac{-b}{2a} - i \left(\frac{\sqrt{4ac - b^2}}{2a} \right)$$

where i is the **imaginary unit** ($i^2 = -1$).

A quadratic equation in one **complex** unknown z has the following form

$$a z^2 + b z + c = 0 \quad (a \neq 0) \quad (24)$$

where a, b and c are complex numbers. It is solved with the same formula as in the case of a real unknown:

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (25)$$

where the square root is understood in the complex sense (the $\pm$ sign is superfluous, because in the complex field the root denotes two numbers, each the opposite of the other)

Example 12: quadratic equations in the complex field

- We want to solve:

$$z^2 + 2iz - \sqrt{3}i = 0 \quad a = 1, b = 2i, c = -\sqrt{3}i$$

we have (switching to the trigonometric form to compute the root):

$$\begin{aligned} z &= \frac{-2i \pm \sqrt{2^2 i^2 + 2^2 \sqrt{3}i}}{2} = \frac{-2i \pm 2\sqrt{-1 + \sqrt{3}i}}{2} = -i \pm \sqrt{-1 + \sqrt{3}i} = -i \pm \sqrt{2 \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)} \\ &= -i \pm \sqrt{2 \left(\cos \frac{2}{3}\pi + i \sin \frac{2}{3}\pi \right)} = -i \pm \sqrt{2} \left(\underbrace{\cos \frac{\pi}{3}}_{=\frac{1}{2}} + i \underbrace{\sin \frac{\pi}{3}}_{=\frac{\sqrt{3}}{2}} \right) = \pm \frac{\sqrt{2}}{2} + i \left(-1 \pm \frac{\sqrt{6}}{2} \right) \end{aligned}$$

- The previous theorem tells us that a polynomial of the form $z^n + a$ (with a complex) has exactly n roots in $\mathbb{C}$; in the real field, instead, the equation $x^n + a = 0$ may have two, one, or no roots (examples: $x^2 - 1 = 0$, $x^3 - 1 = 0$, $x^2 + 1 = 0$); we now know that such an equation always has n roots in $\mathbb{C}$, but only occasionally one or two of them lie in $\mathbb{R}$.
- The result is much more general, as stated by the following theorem, whose proof we do not report.

Theorem 2: fundamental theorem of algebra

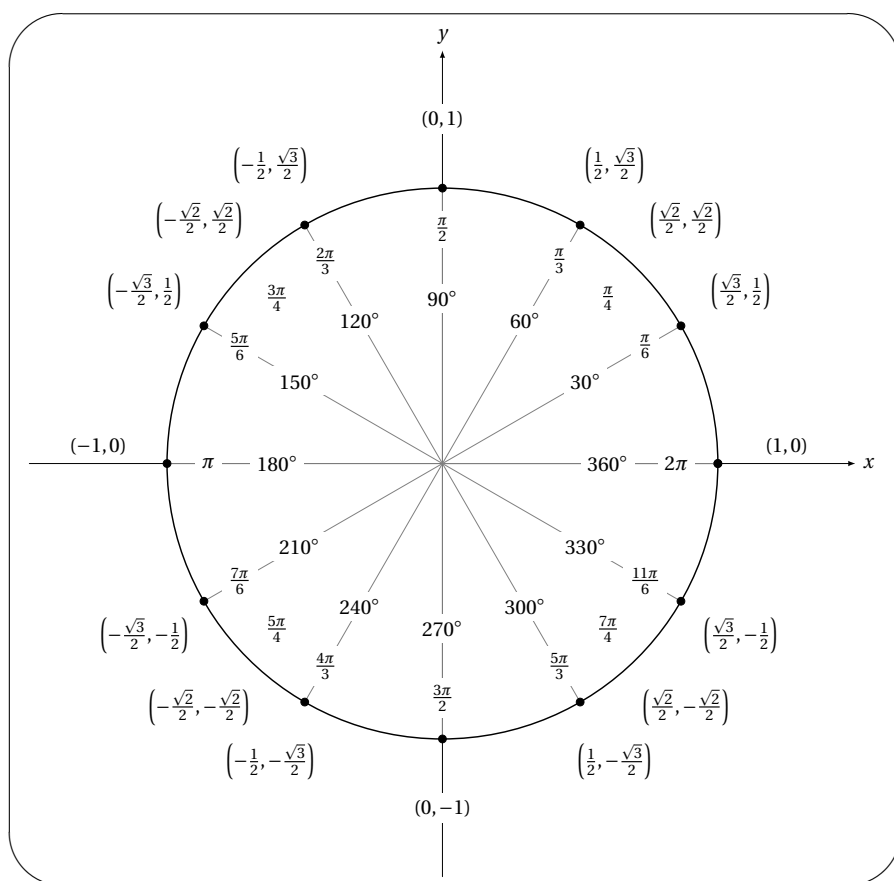
A polynomial equation of the form

$$a_0 + a_1 z + \cdots + a_n z^n = 0 \quad (a_n \neq 0)$$

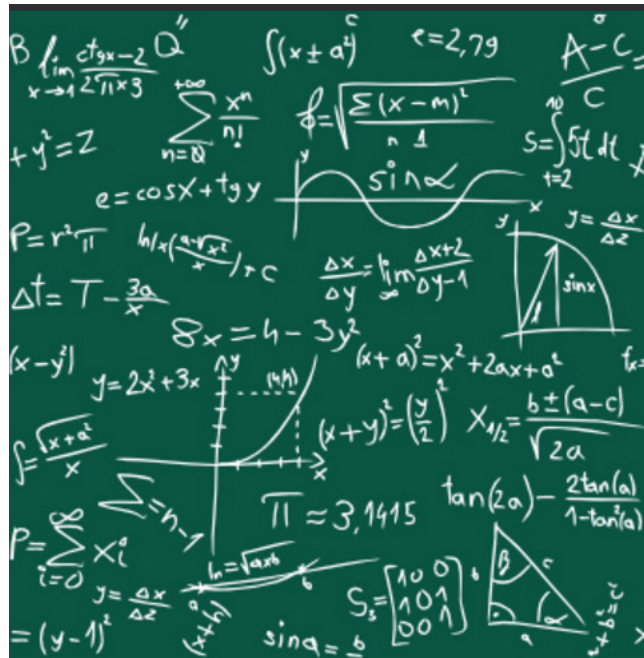
with arbitrary complex coefficients has exactly n roots in $\mathbb{C}$, if each of them is counted with its multiplicity^a.

^aIf $P(z)$ is a polynomial in z of degree n and z_0 is one of its roots, we say that z_0 has multiplicity k (k integer, ≥ 1) if the formula $P(z) = (z - z_0)^k Q(z)$ holds, where Q is a polynomial such that $Q(z_0) \neq 0$.

Cosines and sines of the main angles



Functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

Contents

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2	Definition of function, domain, codomain and image	4
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1 The concept of function

The existence of a **variable quantity** implies the existence of a **relation** between **two quantities**, that is, the dependence of one quantity on another.

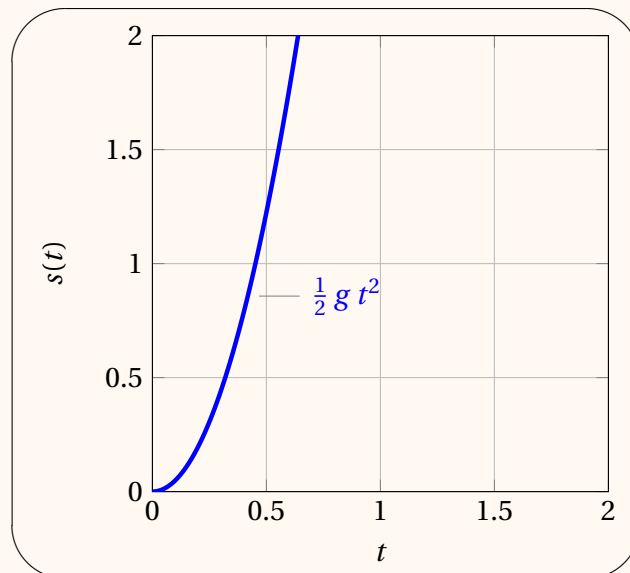
- In each case, this relation follows a certain **law** or **formula**.

Example 1: Laws/formulas $\rightarrow$ functions

If a heavy object is dropped from a certain height, the distance traveled by the object varies with time t according to the formula:

$$s(t) = \frac{1}{2} g t^2 \quad \text{with } t \geq 0$$

where $g \approx 9,8$ is the gravitational acceleration constant.



Hence, the time t is associated with the distance traveled $s(t)$: $t \mapsto s(t)$.

Example 2: Laws/formulas → functions

If a unit capital is invested for one year at the annual rate i , with interest paid monthly at a rate $\frac{i}{12}$, we have:

- after the first month, a capital equal to:

$$1 + \frac{i}{12} \cdot 1 = 1 + \underbrace{\frac{i}{12}}_{\alpha}$$

- after the second month, a capital equal to:

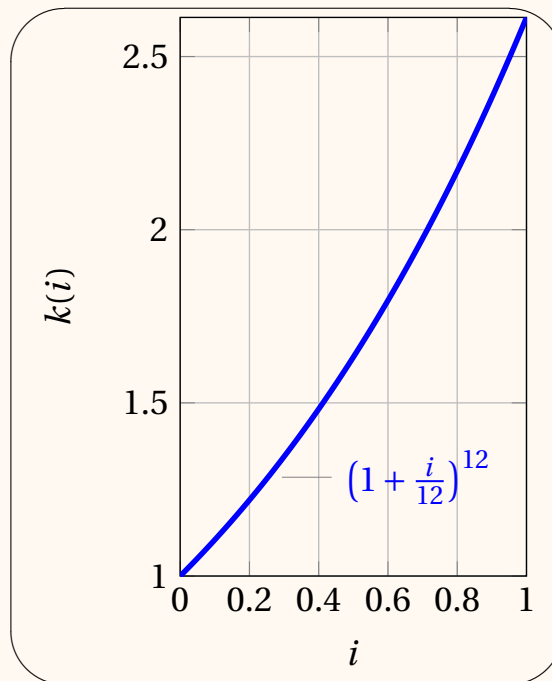
$$\underbrace{1 + \frac{i}{12}}_{\alpha} + \frac{i}{12} \underbrace{\left(1 + \frac{i}{12}\right)}_{\alpha} = \left(1 + \frac{i}{12}\right) \left(1 + \frac{i}{12}\right) = \underbrace{\left(1 + \frac{i}{12}\right)^2}_{\beta}$$

- after the third month, a capital equal to:

$$\underbrace{\left(1 + \frac{i}{12}\right)^2}_{\beta} + \frac{i}{12} \underbrace{\left(1 + \frac{i}{12}\right)^2}_{\beta} = \left(1 + \frac{i}{12}\right)^2 \left(1 + \frac{i}{12}\right) = \left(1 + \frac{i}{12}\right)^3$$

Repeating the argument, the capital at the end of the year therefore depends on the rate i according to the formula:

$$k(i) = \left(1 + \frac{i}{12}\right)^{12} \quad \text{with } i \in [0, 1]$$



Hence, the rate i is associated with the capital $k(i)$ at the end of the year: $i \mapsto k(i)$

- In each of the previous examples, a *real number* (**input**) is associated *uniquely* with another *real number* (**output**). It is precisely the **uniqueness of the relation** that characterizes a function.
- In general, the **admissible inputs** for a given relation (function) are subject to natural restrictions, related to the very nature of the relation.

Example 3: Admissible inputs

- In the first example the “starting” real number has the meaning of *time*; imagining that the object is dropped at an initial time $t = 0$, it is clear that we must restrict ourselves to times $t \geq 0$.
- In the second example, clearly we must have $0 \leq i \leq 1$, where 1 corresponds to a rate of 100%.

2 Definition of function, domain, codomain and image

Definition 1: of domain

The set of admissible inputs for a given function is called the **domain**.

- The expressions **independent variable** (for a generic *input*) and **dependent variable** (for the *output*) are often used.

Example 4: Other types of relations

Let us consider the set of students of a given course.

- If A is the set of students and B is the set of their names, the relation that associates each student with his or her name is *unique* (single-valued).
- In this case neither A nor B is a set of numbers, but nevertheless a unique (single-valued) relation between these two sets is well defined.

On the other hand, the converse is not necessarily true: there could be two students with the same name.

Definition 2: of function

Given any two sets A , B , a **function** f with **domain** A and values in B (or also “with **codomain** B ”) is any law that associates with each element of A one and only one element of B .

- The notation:

$$f : A \rightarrow B$$

(which reads “ f defined from A to B ”), indicates the **domain** and the **codomain** of the function f .

- The notation

$$f : x \mapsto f(x)$$

(which reads “ f maps x to $f(x)$ ”) indicates how the function f acts on the elements.

- The **symbol** $f(x)$ **denotes the output or the value** that the function f associates with x , and it must not be confused with the symbol f , which denotes the **function itself**.

Example 5: Notation for functions

For the function

$$k(i) = \left(1 + \frac{i}{12}\right)^{12}$$

of the previous example we would have:

$$k : [0, 1] \rightarrow \mathbb{R}$$

$$k : i \mapsto \left(1 + \frac{i}{12}\right)^{12}$$

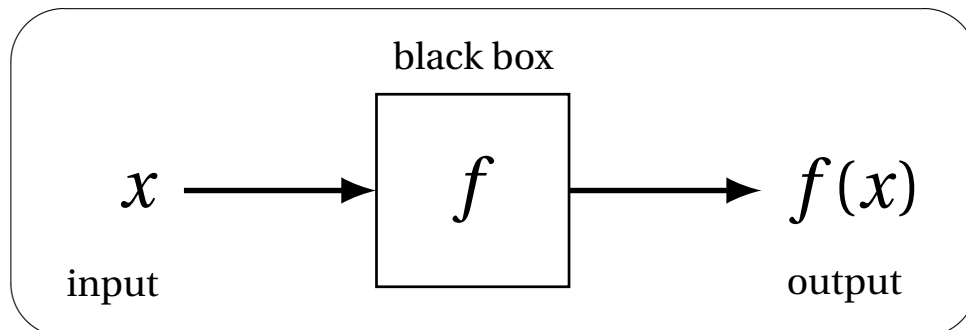
- The notation

$$y = f(x)$$

is also used to indicate that the variable y is a function of x . In this case y is the output or the value that f associates with the input x .

For brevity, sometimes the value of $f(x)$ is defined directly in terms of x , as for example $f(x) = x^2 + 10$.

- In general, we can think of a function as a **black box** that associates with each admissible input x (**input**) a unique output $f(x)$ (**output**):



Definition 3: of image and image of the domain

The output corresponding to x is called the **image** of x ; the set of possible outputs is called the **image of the domain A under f** and is denoted by the symbol $f(A)$ or $\text{Im } f$.

- Note that in the notation $f : A \rightarrow B$, the codomain B can be larger than the image $f(A)$, that is, in general we have

$$f(A) \subseteq B.$$

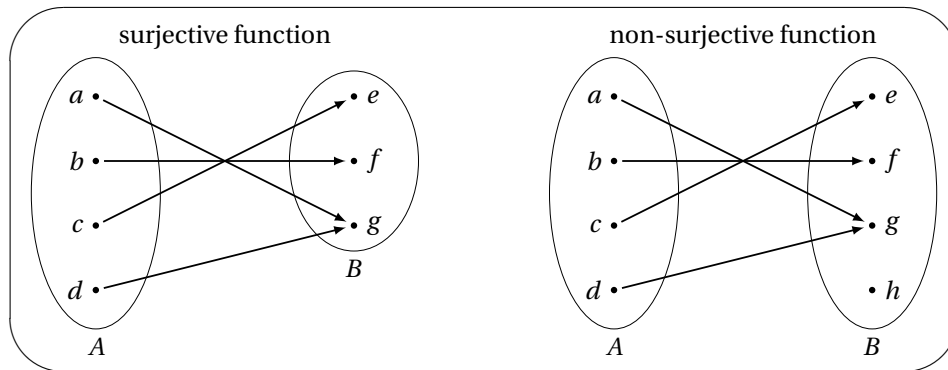
- If f is real-valued, we usually write $f : A \rightarrow \mathbb{R}$ without specifying the actual image of f .

3 Surjections, injections and bijections

Definition 4: of surjection (surjective function)

A function f is a **surjection** if the image of its domain coincides with its codomain.

- A surjection $f : A \rightarrow B$ is sometimes also called a mapping *onto* B (from A).
- Set representation:



Example 6: surjective and non-surjective functions

- The function $f(n) = \lfloor \frac{n}{2} \rfloor$ is a surjective function from $\mathbb{N}$ to $\mathbb{N}$, since every element of the codomain $\mathbb{N}$ is the image of some value of the domain.
- The function $f(n) = 2n$ is not a surjective function from $\mathbb{N}$ to $\mathbb{N}$, since no argument of f produces 3 as a value.
- The function $f(n) = 2n$ is, however, a surjective function from the natural numbers to the even numbers.

Remark 1

Given two sets A and B , and a function $f : A \rightarrow B$, if f is surjective then $|A| \geq |B|$.

Proof. We proceed by induction on the number n of elements in the codomain of the function.

- **Base case**

If there is only one element in the codomain ($|B| = 1$), since the function is surjective, we have $|A| \geq 1$ (every element of B is the image of at least one element of A). Hence $1 \geq 1$, which is clearly true.

- **Inductive step**

Assume then that all functions with codomain of size n satisfy $|A_n| \geq |B_n|$.

Considering codomains of size $n + 1$ we have $|B_{n+1}| = |B_n| + 1$. Since the codomain has one more element and the function is surjective, we have $|A_{n+1}| \geq |A_n| + 1$. Substituting we obtain:

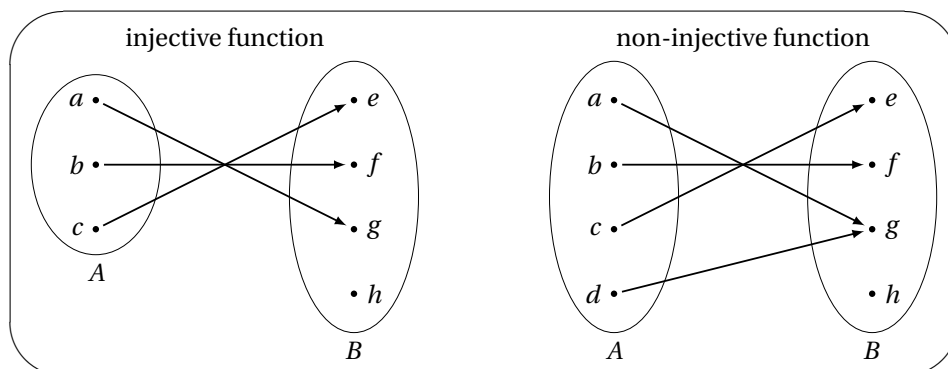
$$\underbrace{|A_n|}_{\leq |A_{n+1}| - 1} \geq \underbrace{|B_n|}_{= |B_{n+1}| - 1} \quad \text{hence } |A_{n+1}| \geq |B_{n+1}|.$$

□

Definition 5: of injection (injective function)

A function f is an **injection** if distinct arguments of f produce distinct values, that is, if $a \neq b$ implies $f(a) \neq f(b)$.

- set representation:



Example 7:

- The function $f(n) = 2n$ is an injective function from $\mathbb{N}$ to $\mathbb{N}$, since each even number b is the image under f of exactly one element of the domain, namely $n = \frac{b}{2}$
- The function $f(n) = \lfloor \frac{n}{2} \rfloor$ is not an injective function, since the value 1 is obtained from two arguments: 2 and 3.

- An injection is also called a **one-to-one** function.

Remark 2

Given two sets A and B , and a function $f : A \rightarrow B$, if f is injective then $|A| \leq |B|$.

Proof. We proceed by induction on the number n of elements in the domain of the function.

- **Base case**

If there is only one element in the domain ($|A| = 1$), since each element of the domain is associated with one and only one element of the codomain, we have $|B| \geq 1$. Hence $1 \leq 1$, which is clearly true.

- **Inductive step**

Assume then that all functions with domain of size n satisfy $|A_n| \leq |B_n|$.

Considering domains of size $n + 1$ we have $|A_{n+1}| = |A_n| + 1$ and, since the function is injective, $|B_{n+1}| \geq |B_n| + 1$. Substituting we obtain

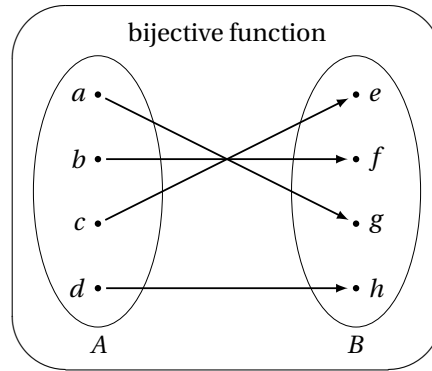
$$\underbrace{|A_n|}_{= |A_{n+1}| - 1} \leq \underbrace{|B_n|}_{\leq |B_{n+1}| - 1} \quad \text{hence} \quad |A_{n+1}| \leq |B_{n+1}|.$$

□

Definition 6: of bijection (bijective function or one-to-one correspondence)

A function f is a **bijection** if: (i) it is injective and (ii) it is surjective.

- set representation:



Example 8:

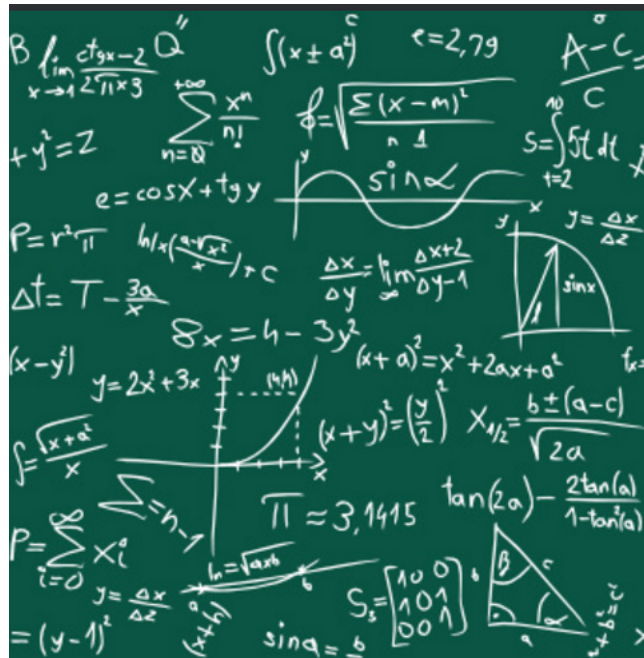
- The function $f(n) = (-1)^n \lceil \frac{n}{2} \rceil$ is a bijection from $\mathbb{N}$ to $\mathbb{Z}$. The values of the function are:

$$f(0) = 0, \quad f(1) = -1, \quad f(2) = 1, \quad f(3) = -2, \quad f(4) = 2 \dots$$

The function is injective, since no element of $\mathbb{Z}$ is the image of more than one element of $\mathbb{N}$. The function is surjective, since all elements of $\mathbb{Z}$ are images of some element of $\mathbb{N}$.

- A bijection is also called a **one-to-one** correspondence, since it pairs elements of the domain with elements of the codomain.
- A bijection from a set A to itself is also called a **permutation**.

Real functions of a real variable



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1	Real function of a real variable	2
2	Bounded functions	4
3	Symmetric functions	7
4	Monotonic functions	8
5	Periodic functions	10

1 Real function of a real variable

Definition 1: real function of a real variable

A function whose domain D is a subset of $\mathbb{R}$ and whose codomain is $\mathbb{R}$:

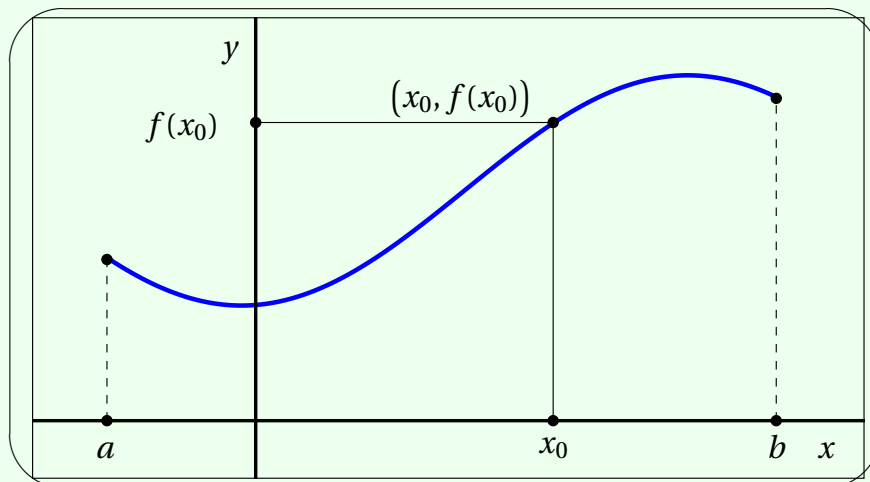
$$f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto f(x)$$

is called a **real function of a real variable**.

- These are functions in which the “input” variable x and the “output” variable $f(x)$ are real numbers.
- The most common real functions of a real variable have as domain D and as image $f(D)$ an **interval** (possibly the whole of $\mathbb{R}$) or the union of a **finite number of intervals**.

The dependence of the output $f(x)$ on the input x is effectively visualized by drawing the **graph** of f , that is, the set of points of the plane with coordinates (x, y) such that $y = f(x)$ and x in the domain D .

Example of the graph of a real function of a real variable with domain $D = [a, b]$ (a closed and bounded interval):



Every line parallel to the y -axis that intersects the x -axis at a point x_0 of the domain D intersects the graph of f at one and only one point. ^a **Therefore, not all curves are graphs of functions.**

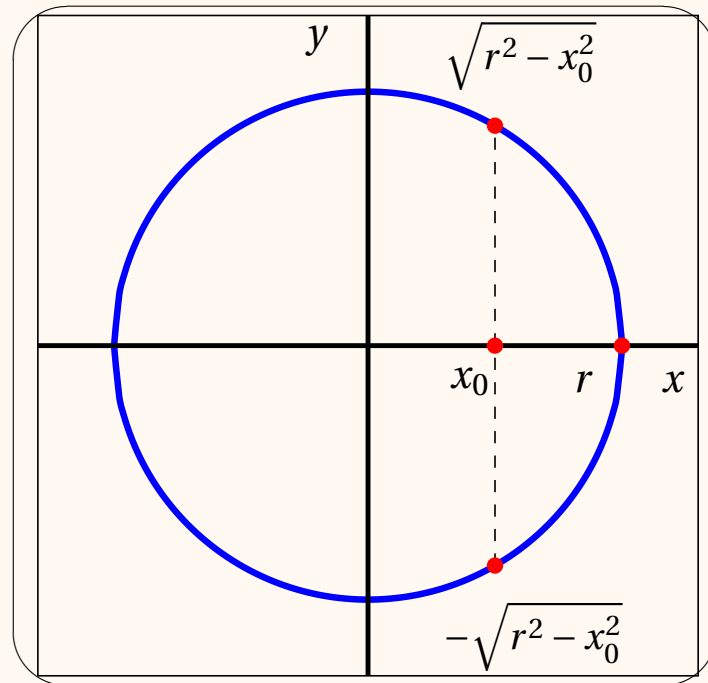
Note, instead, that nothing prevents a line parallel to the x -axis from intersecting the graph of f at several points or at no point at all.

^aIf the line did not intersect the graph, this would mean that no output corresponds to the input x . If the line intersected the graph at more than one point, this would mean that several distinct outputs correspond to the input x , and hence the function would no longer be single-valued.

Example 1: curve that does not correspond to the graph of a function

Consider, for example, the curve made of the points of the circle of radius r :

$$x^2 + y^2 = r^2 \iff y = \pm \sqrt{r^2 - x^2}$$



It is not possible to associate a unique output with the input $x_0 \in (-r, r)$, hence this curve does not correspond to the graph of a function!

2 Bounded functions

Definition 2: bounded functions

Given a function $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$, the function is said to be

$$\left\{ \begin{array}{ll} \text{bounded above} & \text{if } \exists M \in \mathbb{R} : f(x) \leq M, \forall x \in D \\ \text{bounded below} & \text{if } \exists M \in \mathbb{R} : f(x) \geq M, \forall x \in D \\ \text{bounded} & \text{if } \exists M \in \mathbb{R}_{\geq 0} : |f(x)| \leq M, \forall x \in D \end{array} \right.$$

- Graphically, we have:

1. a function is bounded above if its graph is contained in the lower half-plane bounded by a line parallel to the x -axis

2. a function is bounded below if its graph is contained in the upper half-plane bounded by a line parallel to the x -axis
3. a function is bounded if its graph is contained in a horizontal strip

Example 2: bounded function

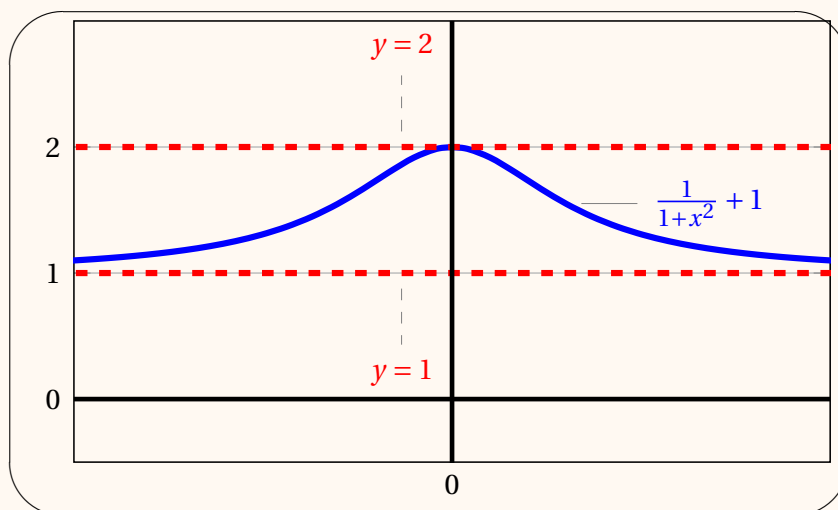
Consider the function:

$$f: \mathbb{R} \rightarrow \mathbb{R}, f: x \mapsto \frac{1}{1+x^2} + 1$$

we have

$$\frac{1}{1+x^2} + 1 = \frac{1+x^2-x^2}{1+x^2} + 1 = 2 - \frac{x^2}{1+x^2} \quad \text{and} \quad \frac{x^2}{1+x^2} \geq 0, \quad \forall x \in \mathbb{R}$$

$$\text{hence } 1 < \frac{1}{1+x^2} + 1 \leq 2, \quad \forall x \in \mathbb{R}$$

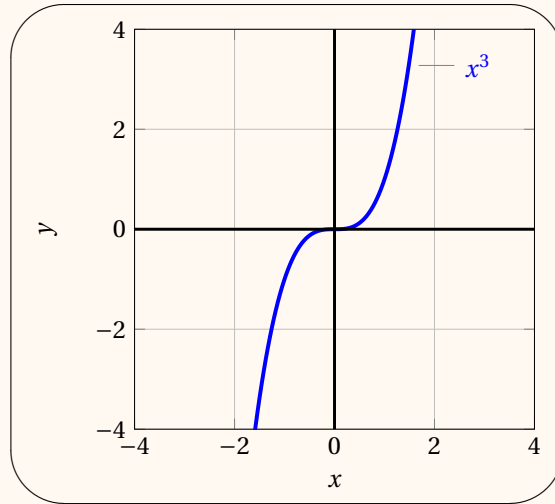


Example 3: Unbounded function

The function

$$f: \mathbb{R} \rightarrow \mathbb{R}, f: x \mapsto x^3 \quad (y = x^3)$$

is bounded neither above nor below.

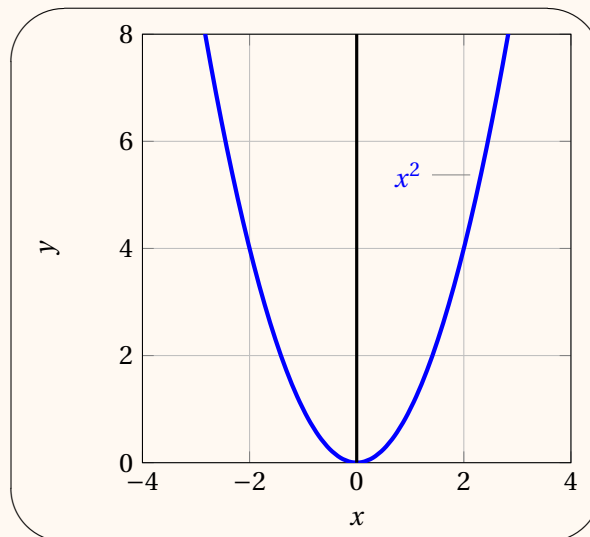


Example 4: Function bounded below

The function

$$f: \mathbb{R} \rightarrow \mathbb{R}, f: x \mapsto x^2 \quad (y = x^2)$$

is bounded below; indeed, $x^2 \geq 0, \forall x \in \mathbb{R}$



- Equivalently, we can say that a function is *bounded above* (*bounded below*, *bounded*) if, respectively, its **image** is a subset of $\mathbb{R}$ that is *bounded above* (*bounded below*, *bounded*).

3 Symmetric functions

Definition 3: even function

Functions whose graph is symmetric with respect to the y -axis are called **even**.

- They are characterized by the relation

$$f(-x) = f(x)$$

which expresses the equality of the ordinates corresponding to the points x and $-x$, which are symmetric with respect to $x = 0$.

Definition 4: Odd function

Functions whose graph is symmetric with respect to the origin are called **odd**.

- They are characterized by the relation

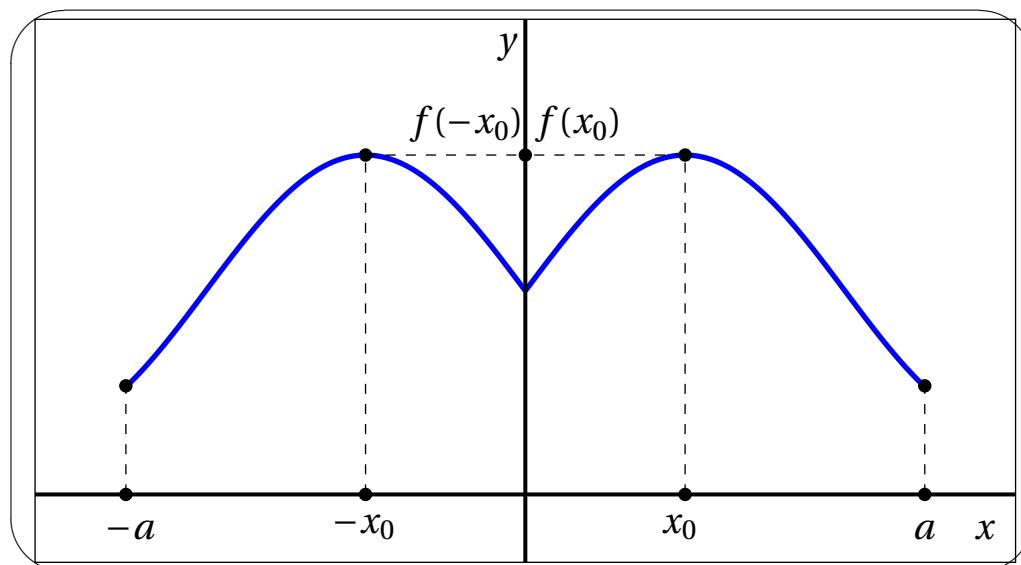
$$f(-x) = -f(x)$$

which expresses the fact that the ordinates corresponding to the points x and $-x$, which are symmetric with respect to $x = 0$, are the opposite of each other.

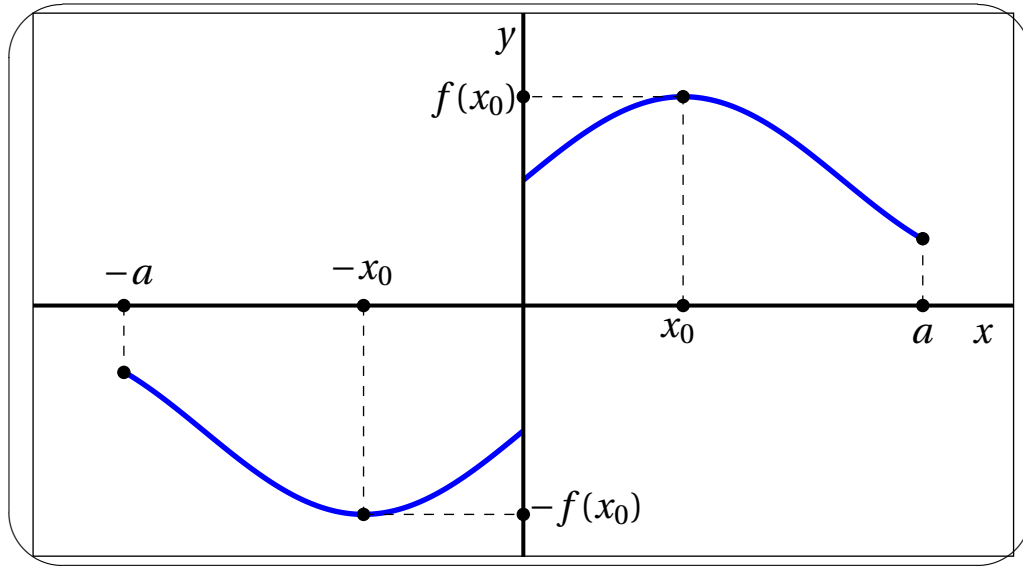
Example 5: Even and odd functions

For example, the function $x \mapsto x^2$ is even, while $x \mapsto x^3$ is odd. More generally, powers with integer exponent are even functions if the exponent is even and odd functions if the exponent is odd.

1. Example of the graph of an even function:



2. Example of the graph of an odd function:



A function *cannot* have a graph that is symmetric with respect to the x -axis, since the correspondence would no longer be single-valued.

4 Monotonic functions

Definition 5: Increasing function

A function is called **non-decreasing** if for every pair of points x_1, x_2 in the domain of f we have:

$$x_1 > x_2 \implies f(x_1) \geq f(x_2) \quad (1)$$

Definition 6: Strictly increasing function

A function is called **increasing** if for every pair of points x_1, x_2 in the domain of f we have:

$$x_1 > x_2 \implies f(x_1) > f(x_2) \quad (2)$$

Definition 7: Decreasing function

A function is called **non-increasing** if for every pair of points x_1, x_2 in the domain of f we have:

$$x_1 > x_2 \implies f(x_1) \leq f(x_2) \quad (3)$$

Definition 8: Strictly decreasing function

A function is called **decreasing** if for every pair of points x_1, x_2 in the domain of f we have:

$$x_1 > x_2 \implies f(x_1) < f(x_2) \quad (4)$$

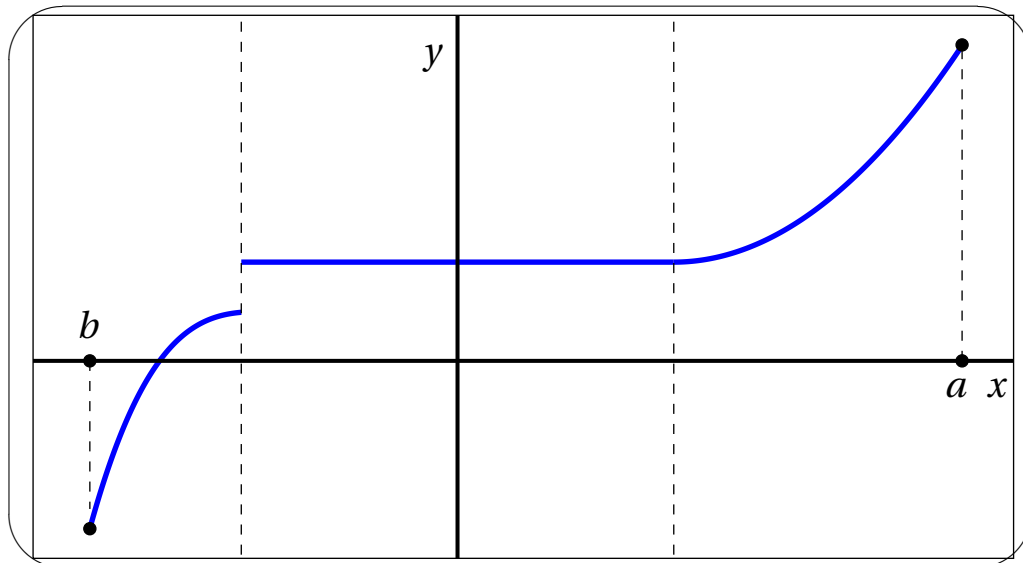
- A function f is *non-decreasing* if, as x increases, the corresponding ordinate on the graph of the function does not decrease (hence it either stays the same or increases);
- A function f is *non-increasing* if, as x increases, the corresponding ordinate on the graph of the function does not increase (hence it either stays the same or decreases).

Increasing or decreasing functions are called **monotonic**. Strictly increasing or strictly decreasing functions are called **strictly monotonic**.

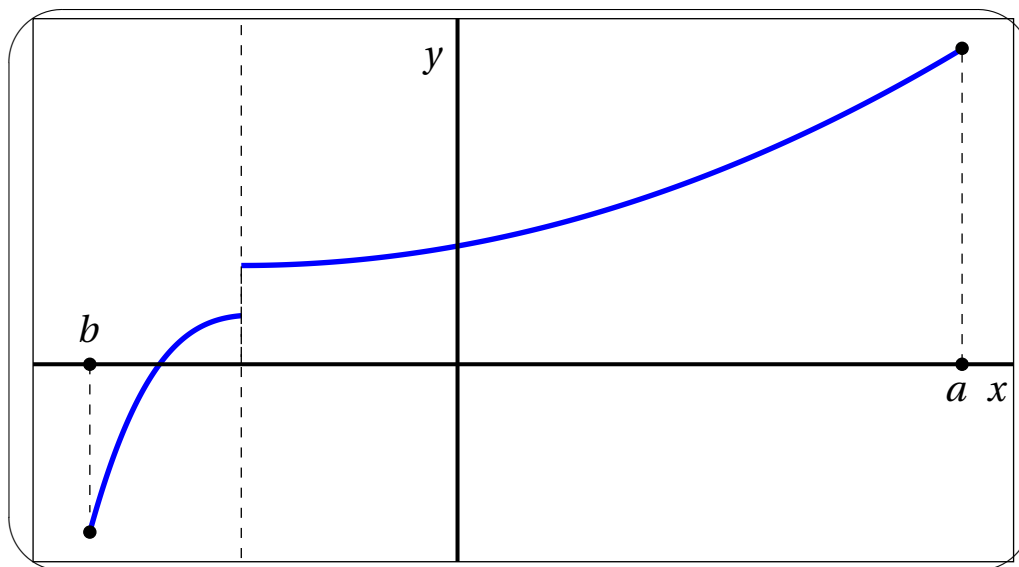
Example 6: Monotonic functions

For example, the function $x \mapsto x^3$ is strictly monotonic increasing; the constant function $x \mapsto k$ (whose graph is the line with equation $y = k$) is both non-decreasing and non-increasing.

1. Example of the graph of a non-decreasing function (horizontal segment):



2. Example of the graph of an increasing function:



- Increasing or decreasing functions (strictly or not) are called **monotonic**.

5 Periodic functions

Definition 9: Periodic function

A (non-constant) function $f : D \rightarrow \mathbb{R}$ is **periodic** with period T , $T > 0$, if T is the smallest positive real number such that

$$f(x + T) = f(x) \text{ for every } x \in D$$

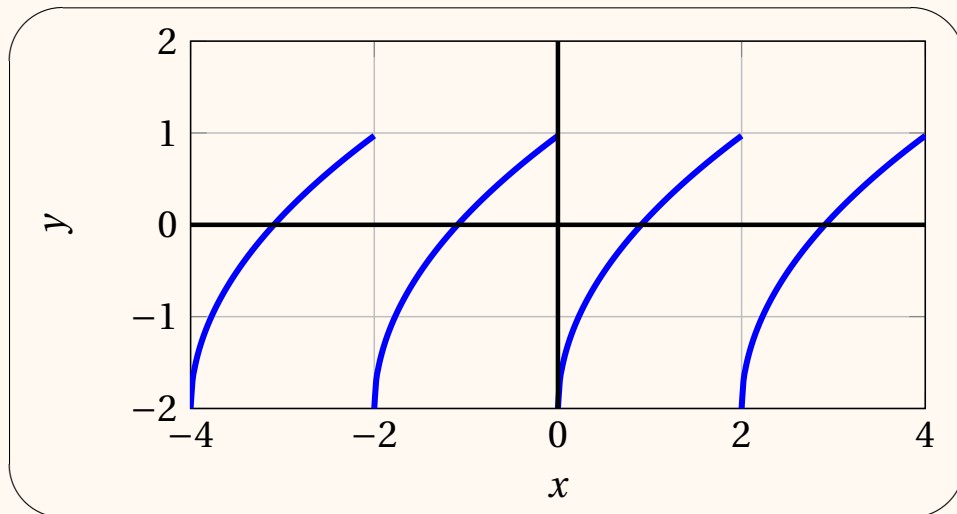
- Every interval of length T contained in D is called a **periodicity interval**.

Example 7: Periodic functions

Typical examples of periodic functions are the *trigonometric functions* $x \mapsto \sin(x)$ ($T = 2\pi$), $x \mapsto \cos(x)$ ($T = 2\pi$) and $x \mapsto \tan(x)$ ($T = \pi$).

Example 8: Graph of periodic functions

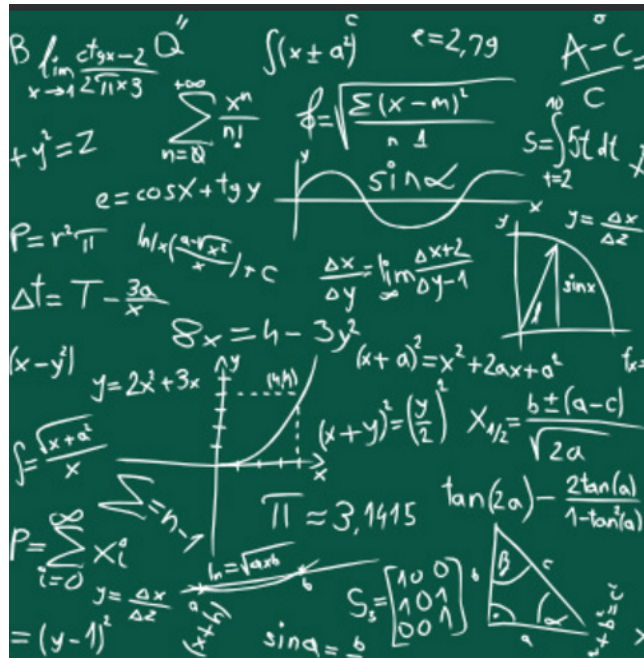
Graph of a periodic function with period $T = 2$:



- Lines parallel to the x -axis have equation

$$y = k \quad k \in \mathbb{R}$$

Power functions



- Author: Fabio Furini.

Contents

1	Power functions	2
1.1	Rational exponent	2
1.2	Real exponent	6
1.3	Polynomials	7
2	Direct and inverse proportionality functions	8

1 Power functions

Definition 1: power functions

Given $\alpha \in \mathbb{R}, \alpha \neq 0$, the function:

$$f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto x^\alpha \quad (1)$$

is called the **power function** with exponent α . The domain D depends on the value of α :

- with rational exponent $\alpha = \frac{m}{n} \in \mathbb{Q}$, $m \in \mathbb{Z}$ and $n \in \mathbb{N}_+$ coprime, we have four cases:

$$D = \begin{cases} \mathbb{R} & \text{if } n \text{ odd and } \frac{m}{n} > 0 \\ \mathbb{R} \setminus \{0\} & \text{if } n \text{ odd and } \frac{m}{n} < 0 \\ [0, +\infty) & \text{if } n \text{ even and } \frac{m}{n} > 0 \\ (0, +\infty) & \text{if } n \text{ even and } \frac{m}{n} < 0 \end{cases}$$

- with real, non-rational exponent $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, we have two cases:

$$D = \begin{cases} [0, +\infty) & \text{if } \alpha > 0 \\ (0, +\infty) & \text{if } \alpha < 0 \end{cases}$$

1.1 Rational exponent

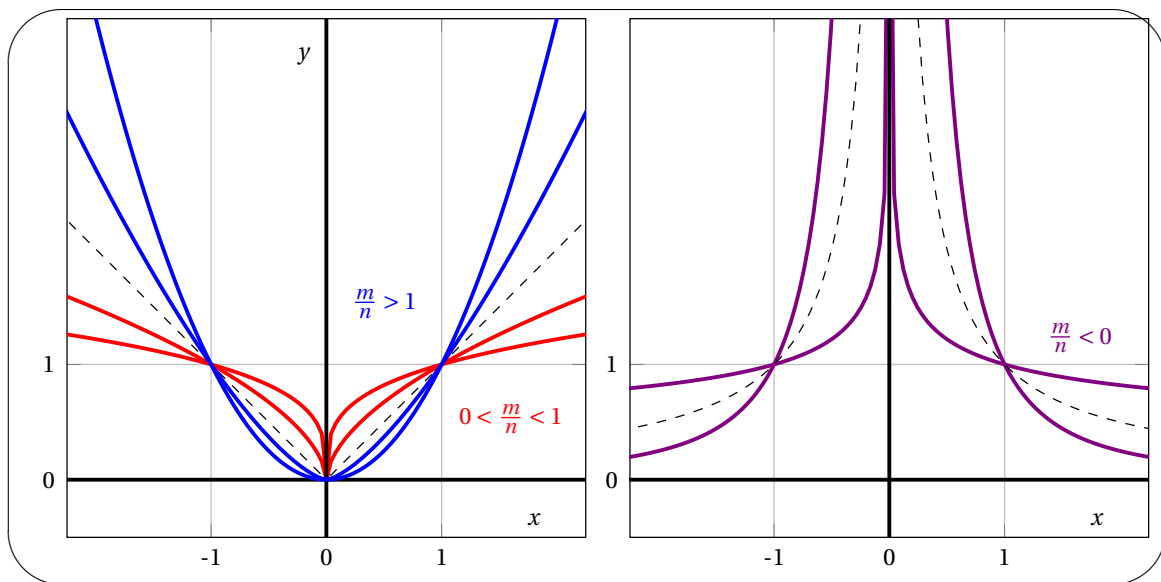
With $\alpha = \frac{m}{n} \in \mathbb{Q}$, $m \in \mathbb{Z}$ and $n \in \mathbb{N}_+$ coprime, we have a power function with rational exponent:

$$f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto x^{\frac{m}{n}} = \sqrt[n]{x^m}$$

We have the following three cases, depending on whether m and n are even or odd.

1. With n odd and m even, it is an **even function**, i.e., $(-x)^{\frac{m}{n}} = x^{\frac{m}{n}}$. For the sign and the monotonicity we have:

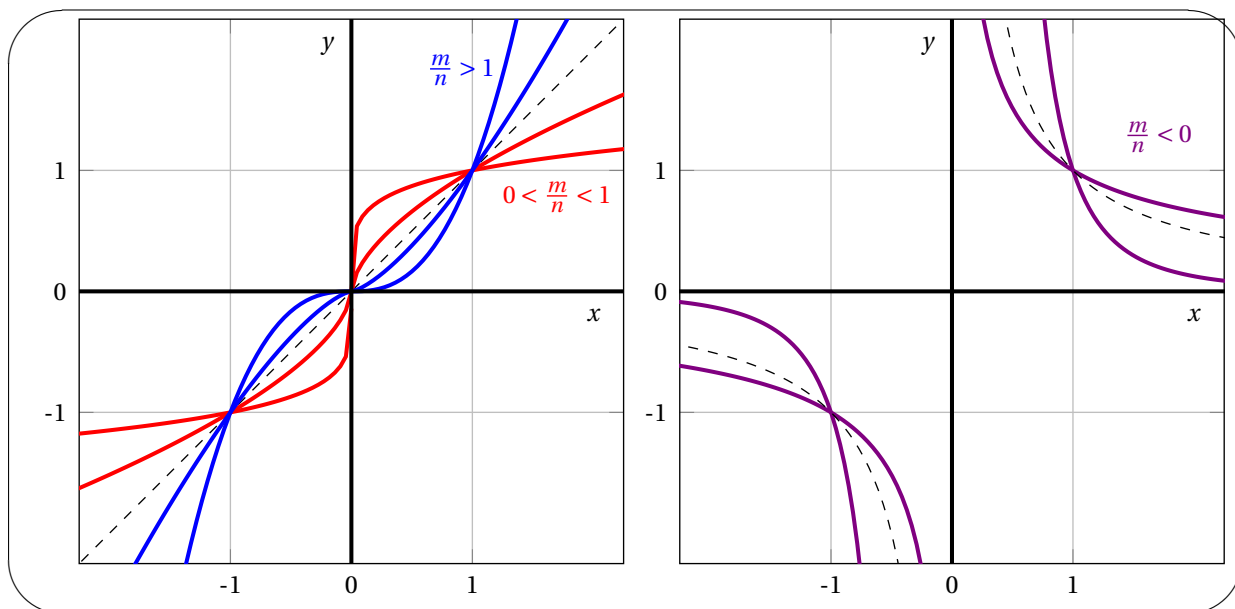
$$\begin{aligned} & x^{\frac{m}{n}} > 0, \quad \forall x \in \mathbb{R} \setminus \{0\}; \quad x^{\frac{m}{n}} = 0 \iff x = 0 \text{ and } \frac{m}{n} > 0 \\ & \begin{cases} \text{if } \frac{m}{n} > 0 \text{ then } f \text{ is decreasing in } (-\infty, 0] \text{ and increasing in } [0, +\infty) \\ \text{if } \frac{m}{n} < 0 \text{ then } f \text{ is increasing in } (-\infty, 0) \text{ and decreasing in } (0, +\infty) \end{cases} \end{aligned}$$



2. With n odd and m odd, it is an **odd function**, i.e., $(-x)^{\frac{m}{n}} = -x^{\frac{m}{n}}$. For the sign and the monotonicity we have:

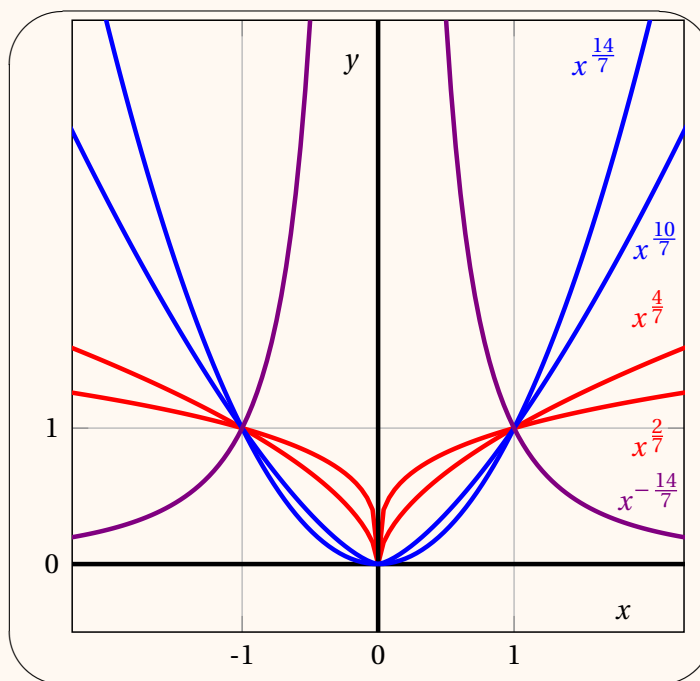
$$x^{\frac{m}{n}} < 0, \forall x < 0, \quad x^{\frac{m}{n}} > 0, \forall x > 0; \quad x^{\frac{m}{n}} = 0 \iff x = 0 \text{ and } \frac{m}{n} > 0$$

$$\begin{cases} \text{if } \frac{m}{n} > 0 \text{ then } f \text{ is increasing in } \mathbb{R} \\ \text{if } \frac{m}{n} < 0 \text{ then } f \text{ is decreasing in } (-\infty, 0) \text{ and in } (0, +\infty) \end{cases}$$

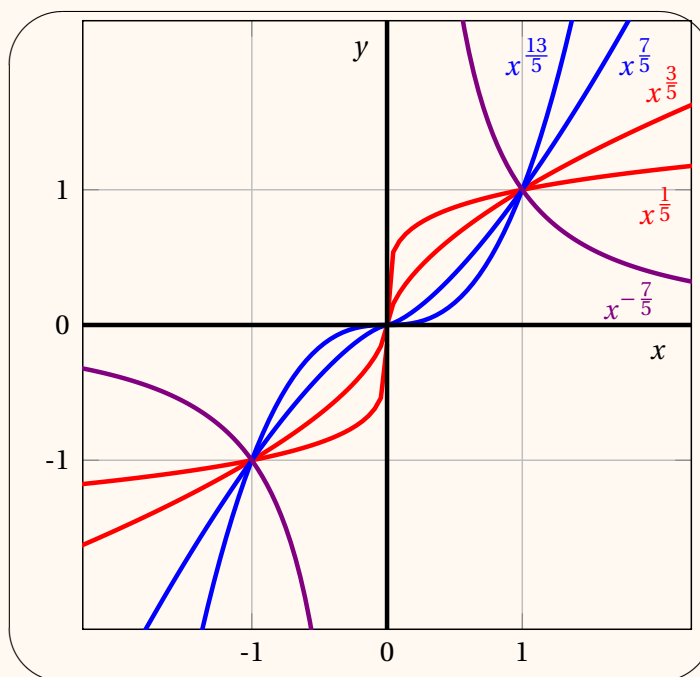


Example 1: graphs of power functions with rational exponent $\frac{m}{n}$ and n odd

With m even:



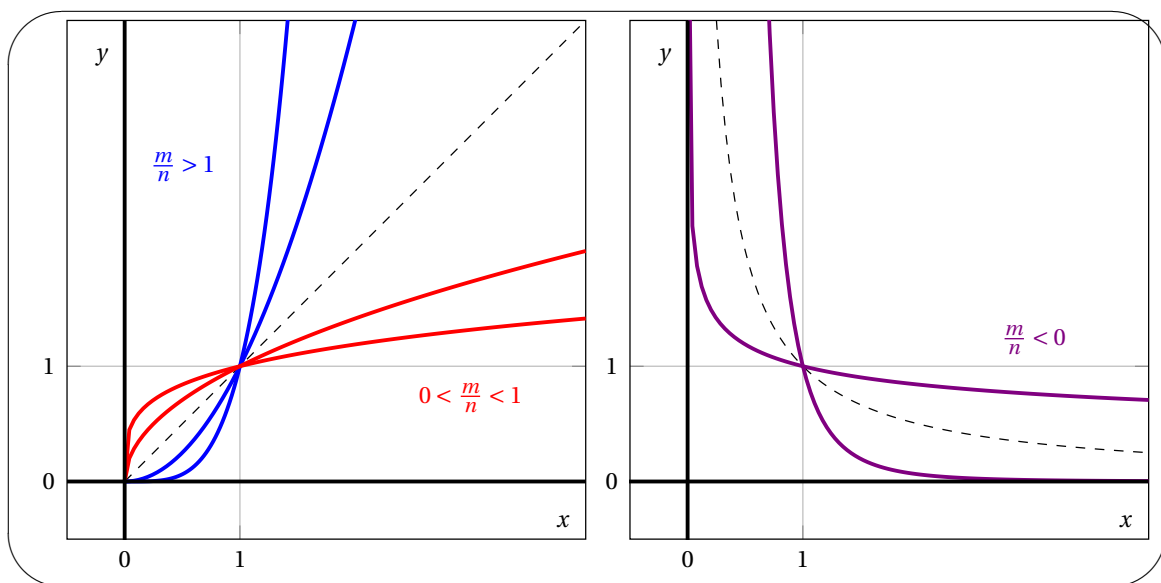
With m odd:



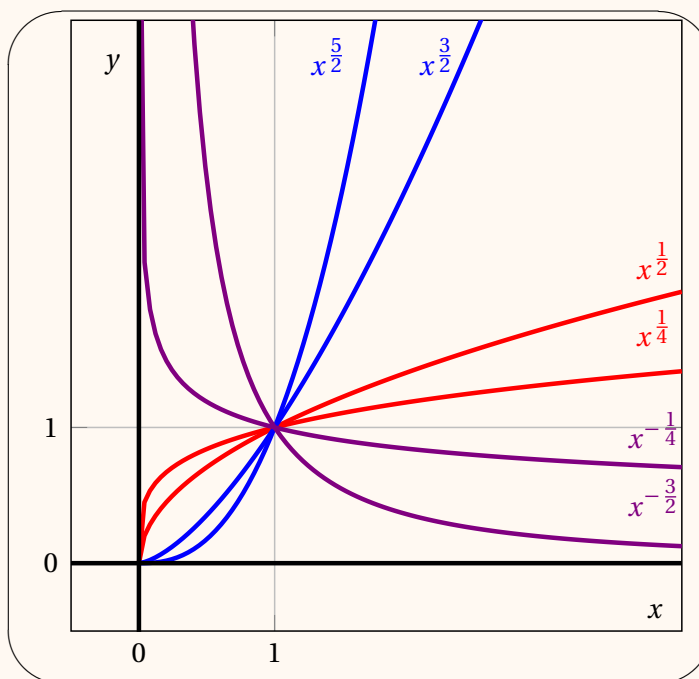
3. With n even and m odd (m cannot be even since they are coprime), for the sign and the monotonicity we have:

$$x^{\frac{m}{n}} > 0, \quad \forall x > 0; \quad x^{\frac{m}{n}} = 0 \iff x = 0 \text{ and } \frac{m}{n} > 0$$

$$\begin{cases} \text{if } \frac{m}{n} > 0 \text{ then } f \text{ is increasing in } [0, +\infty) \\ \text{if } \frac{m}{n} < 0 \text{ then } f \text{ is decreasing in } (0, +\infty) \end{cases}$$



Example 2: graphs of power functions with rational exponent $\frac{m}{n}$ and n even

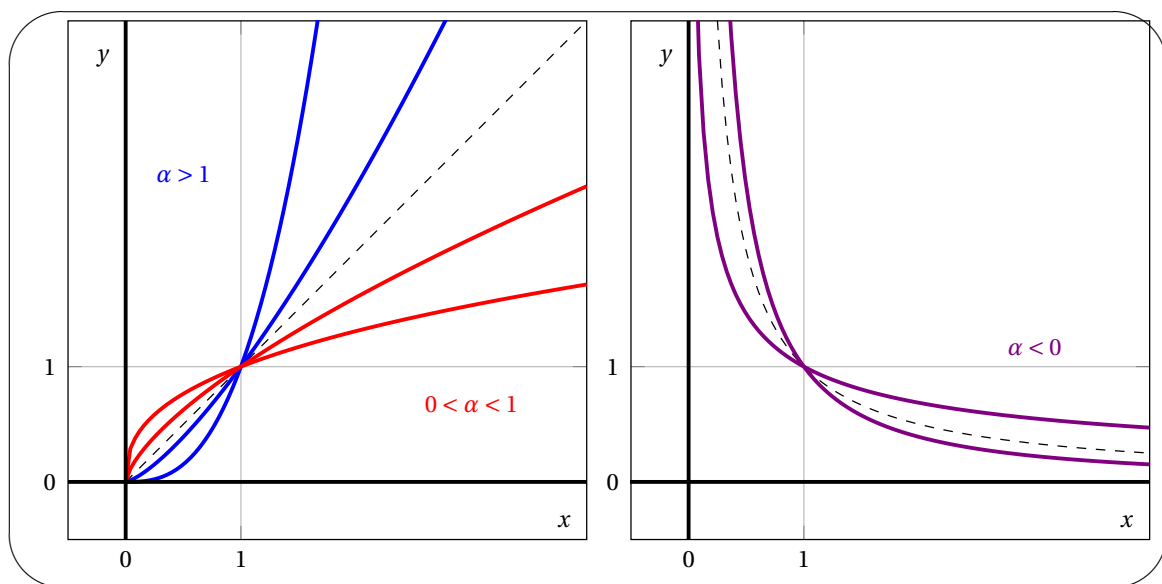


1.2 Real exponent

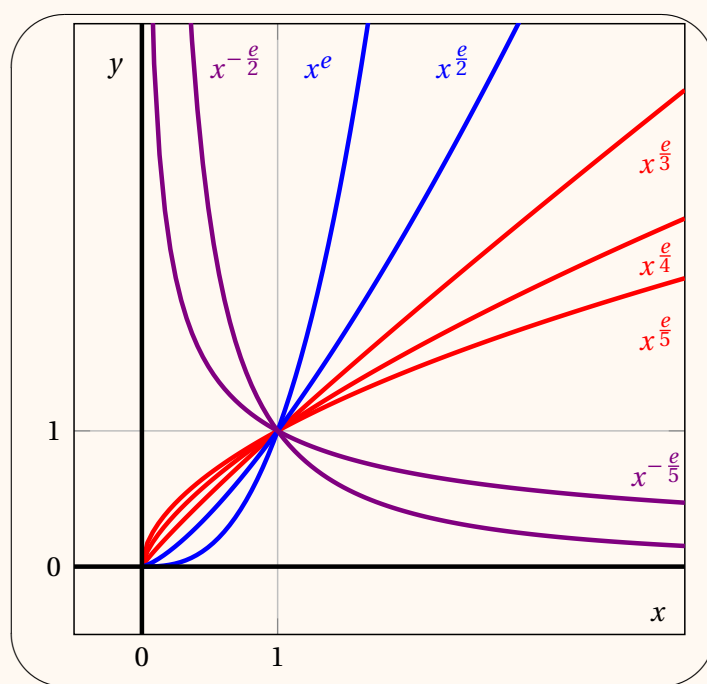
- With $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, for the sign and the monotonicity we have:

$$x^\alpha > 0, \quad \forall x > 0; \quad x^\alpha = 0 \iff x = 0 \text{ and } \alpha > 0$$

$$\begin{cases} \text{if } \alpha > 0 \text{ then } f \text{ is increasing in } [0, +\infty) \\ \text{if } \alpha < 0 \text{ then } f \text{ is decreasing in } (0, +\infty) \end{cases}$$



Example 3: graphs of power functions with real exponent



1.3 Polynomials

Definition 2: polynomial

Given $n + 1$ values $a_i \in \mathbb{R}$ with $i \in \{0, 1, \dots, n\}$ and $a_n \neq 0$, the function:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \underbrace{\sum_{i=0}^n a_i x^i}_{=P_n(x)} \quad (2)$$

is called a **polynomial** of degree n .

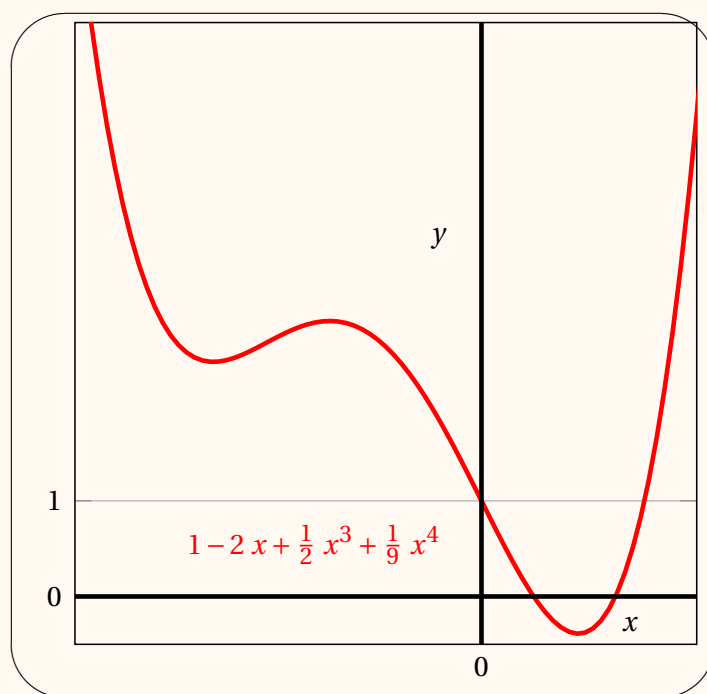
- For each **monomial** $i \in \{0, 1, \dots, n\}$:
 - the value a_i is the **coefficient of the monomial**
 - the power function x^i with integer exponent i is the **literal part** of the monomial

The value a_0 is the **constant term** of the polynomial since $x^0 = 1$.

Example 4: polynomial

For example, with $n = 4$, $a_0 = 1$, $a_1 = -2$, $a_2 = 0$, $a_3 = \frac{1}{2}$ and $a_4 = \frac{1}{9}$ we have the following polynomial of degree four:

$$P_4(x) = \sum_{i=0}^4 a_i x^i = 1 - 2x + \frac{1}{2}x^3 + \frac{1}{9}x^4$$

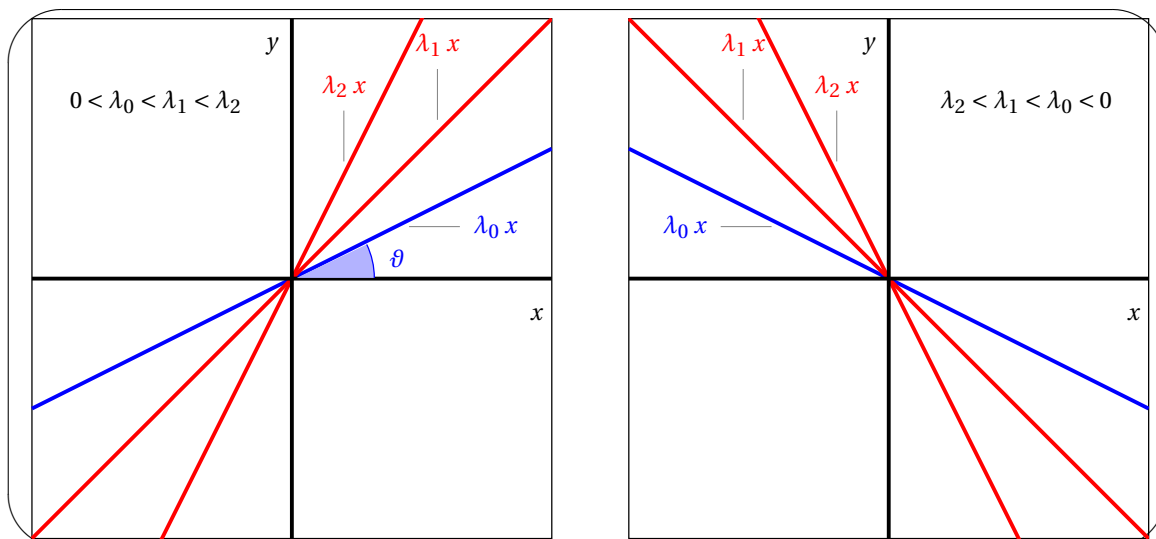


2 Direct and inverse proportionality functions

With $\alpha = 1$ and given $\lambda \in \mathbb{R}$, we have the family of **direct proportionality (or linear) functions**:

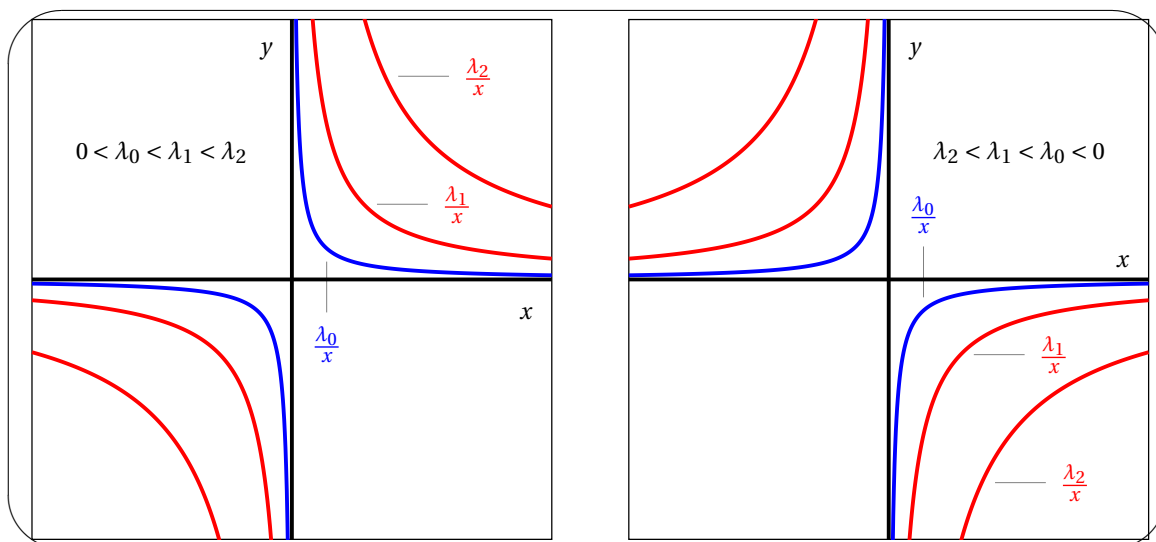
$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \lambda x \quad \text{where } \lambda \text{ is the constant of direct proportionality}$$

Moreover, $\lambda = \tan \vartheta$ and ϑ is the angle between the line $y = \lambda x$ and the x -axis.



With $\alpha = -1$ and given $\lambda \in \mathbb{R}$, we have the family of **inverse proportionality functions (or rectangular hyperbolas)**:

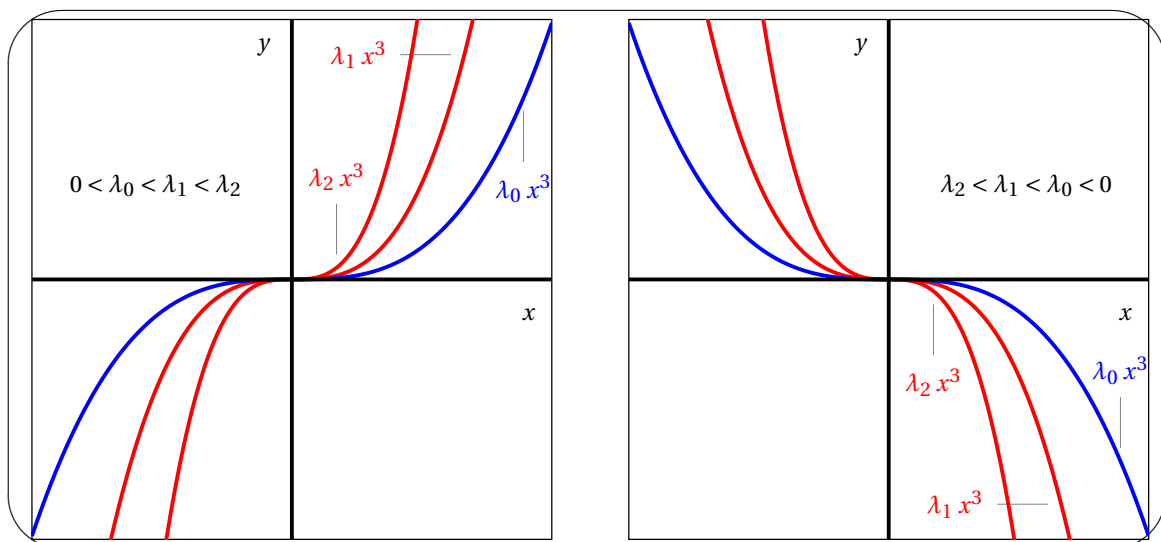
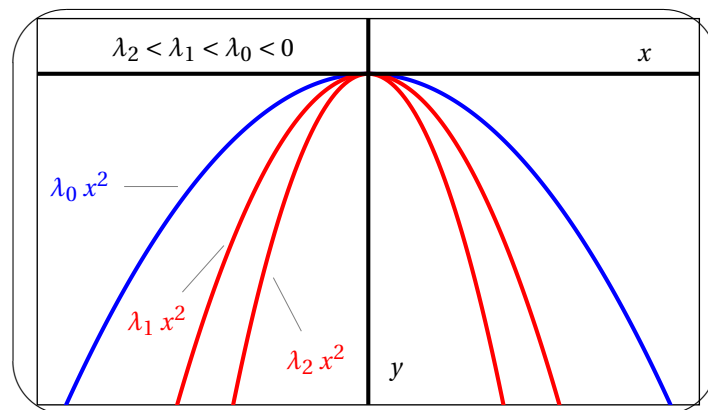
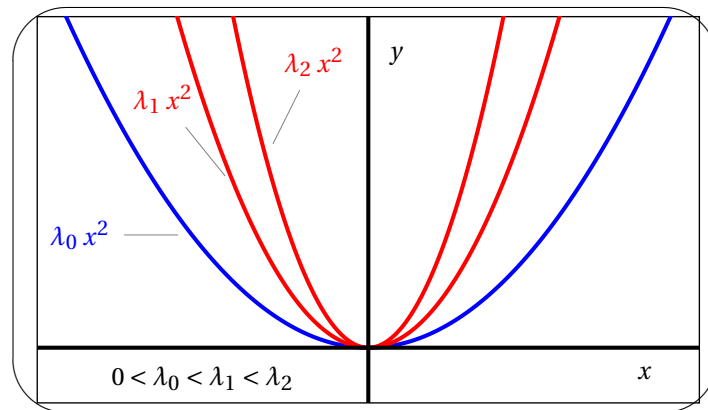
$$f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}, \quad f: x \mapsto \frac{\lambda}{x} \quad \text{where } \lambda \text{ is the constant of inverse proportionality}$$



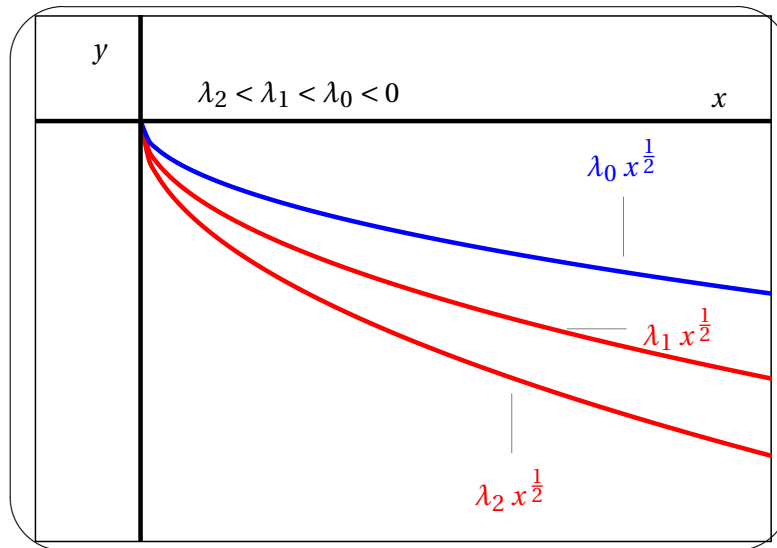
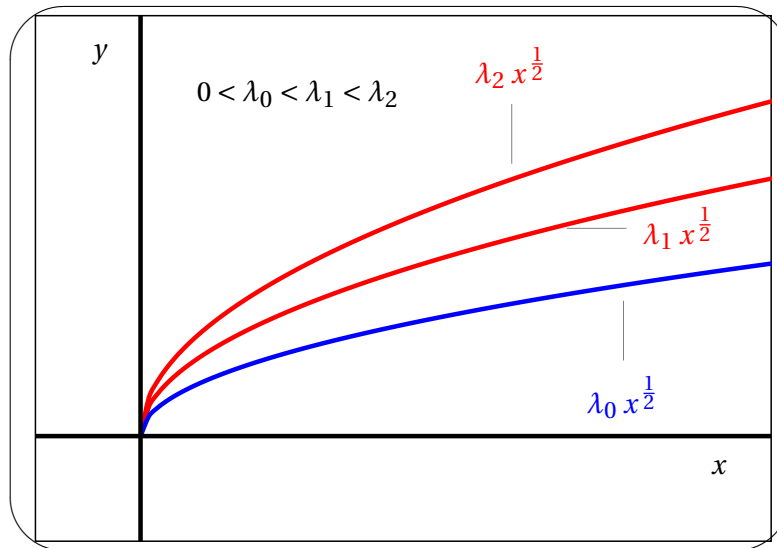
Given $\alpha \in \mathbb{R}$ and $\lambda \in \mathbb{R}$, we have the family of **power functions multiplied by a constant**:

$$f: D \subseteq \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \lambda x^\alpha \quad \text{where } \lambda \text{ is the multiplicative constant}$$

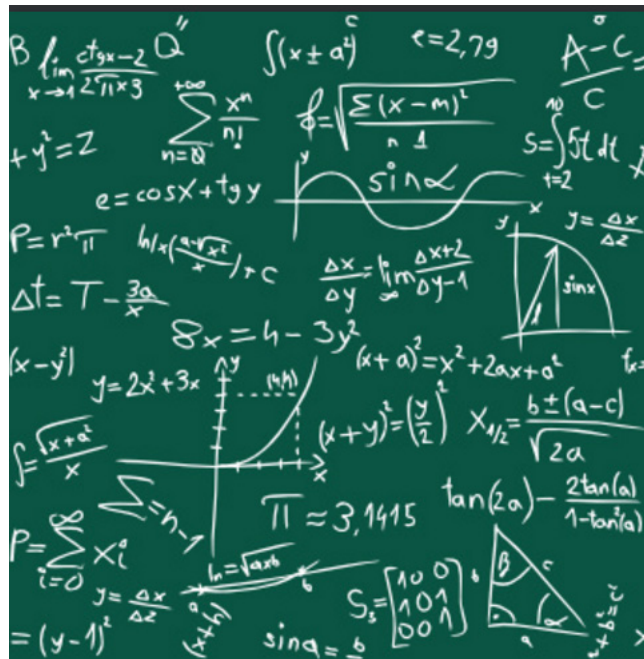
- For example, with α equal to 2 or equal to 3 and $\lambda \in \mathbb{R}$ we have:



- For example, with $\alpha = \frac{1}{2}$ and $\lambda \in \mathbb{R}$ we have:



Exponential and logarithmic functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1	Exponential functions	2
2	Logarithmic functions	2
3	Change of base	3

1 Exponential functions

Definition 1: exponential functions

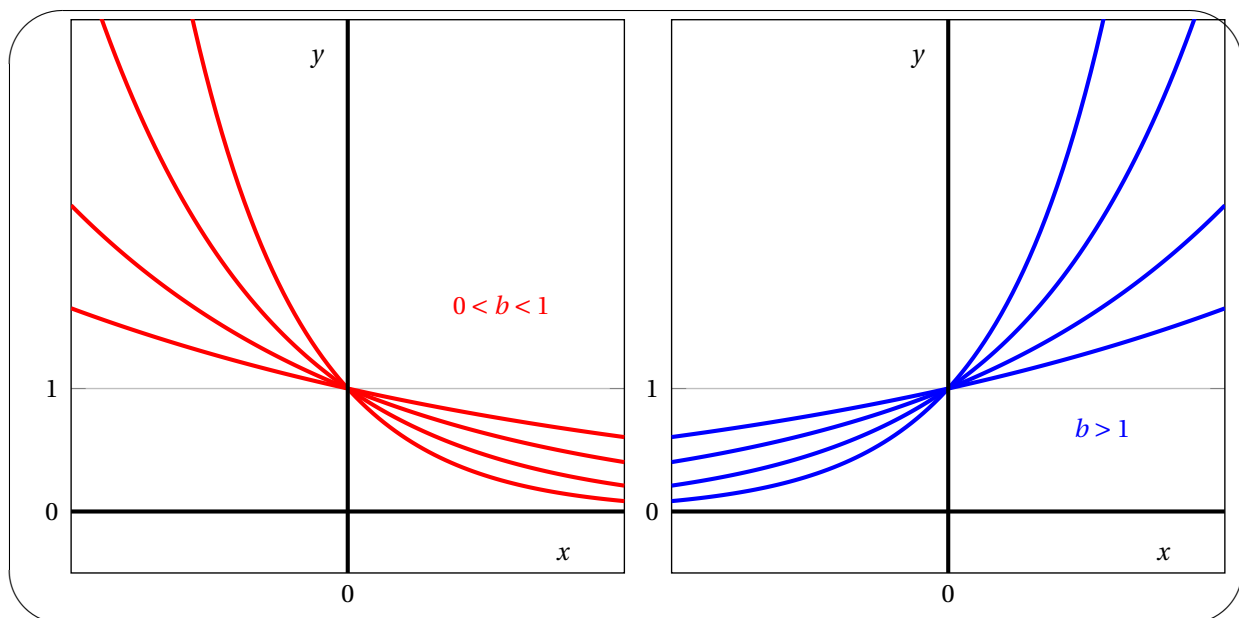
Given $b \in \mathbb{R}_+ \setminus \{1\}$, the function:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto b^x \quad (1)$$

is called the **exponential function** with base b .

- Regarding positivity and monotonicity, we have:

$$b^x > 0, \quad \forall x \in \mathbb{R} \quad \text{and} \quad \begin{cases} \text{if } 0 < b < 1 \text{ then } f \text{ is decreasing on } \mathbb{R} \\ \text{if } b > 1 \text{ then } f \text{ is increasing on } \mathbb{R} \end{cases}$$



2 Logarithmic functions

Definition 2: logarithmic functions

Given $a \in \mathbb{R}_{>0} \setminus \{1\}$, the function:

$$f: (0, +\infty) \rightarrow \mathbb{R}, \quad f: x \mapsto \log_a x \quad (2)$$

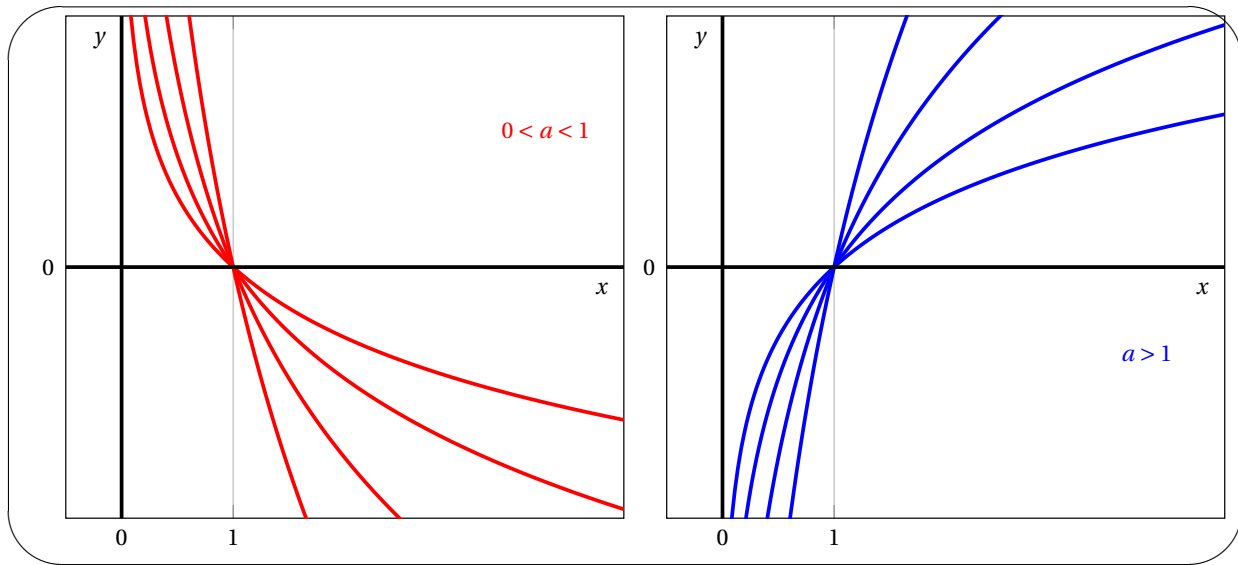
is called the **logarithmic function** with base a .

- Regarding positivity and monotonicity, we have:

$$\begin{cases} \text{if } 0 < a < 1 & \log_a x > 0, \forall x \in (0, 1), \log_a x < 0, \forall x \in (1, +\infty) \text{ and } f \text{ is decreasing on } \mathbb{R}_{>0} \\ \text{if } a > 1 & \log_a x < 0, \forall x \in (0, 1), \log_a x > 0, \forall x \in (1, +\infty) \text{ and } f \text{ is increasing on } \mathbb{R}_{>0} \end{cases}$$

Moreover, we have:

$$\log_a x = 0 \iff x = 1, \forall a \in \mathbb{R}_{>0} \setminus \{1\}$$



3 Change of base

Given any base $c \in \mathbb{R}_{>0} \setminus \{1\}$, all exponential and logarithmic functions can be rewritten in terms of another base $d \in \mathbb{R}_{>0} \setminus \{1\}$ as follows:

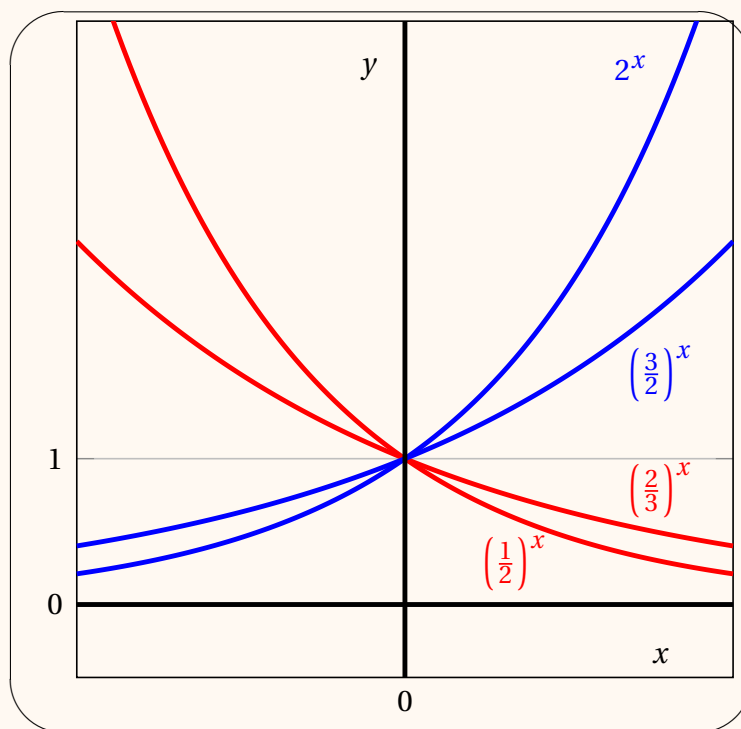
$$c^x = d^{\log_d c^x} = d^{x \log_d c} \quad \text{hence with } \lambda = \log_d c \quad \text{we get} \quad c^x = d^{\lambda x}$$

$$\log_c x = \frac{\log_d x}{\log_d c} \quad \text{hence with } \lambda = \frac{1}{\log_d c} \quad \text{we get} \quad \log_c x = \lambda \log_d x$$

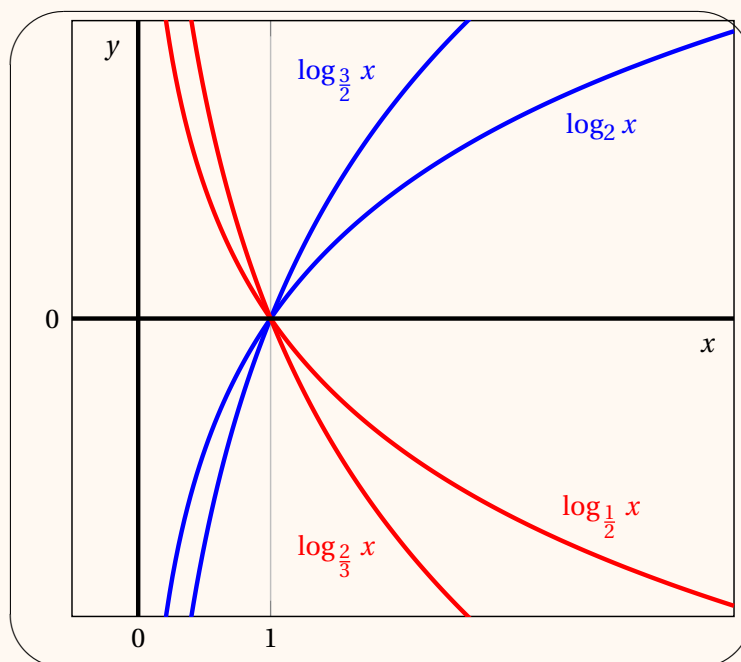
The logarithmic function with base e is also written $\log x$ or $\ln x$, while the one with base 2 is also written $\lg x$.

Example 1: graphs of exponential and logarithmic functions

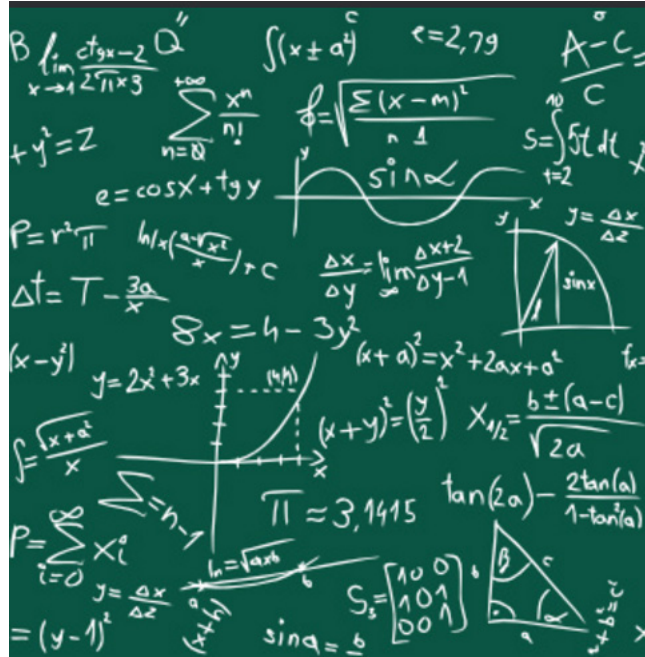
Exponential functions:



Logarithmic functions:



Trigonometric functions



- Author: Fabio Furini.

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1 Trigonometric functions

Definition 1: sine function and cosine function

The trigonometric functions:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \sin x \quad (1)$$

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \cos x \quad (2)$$

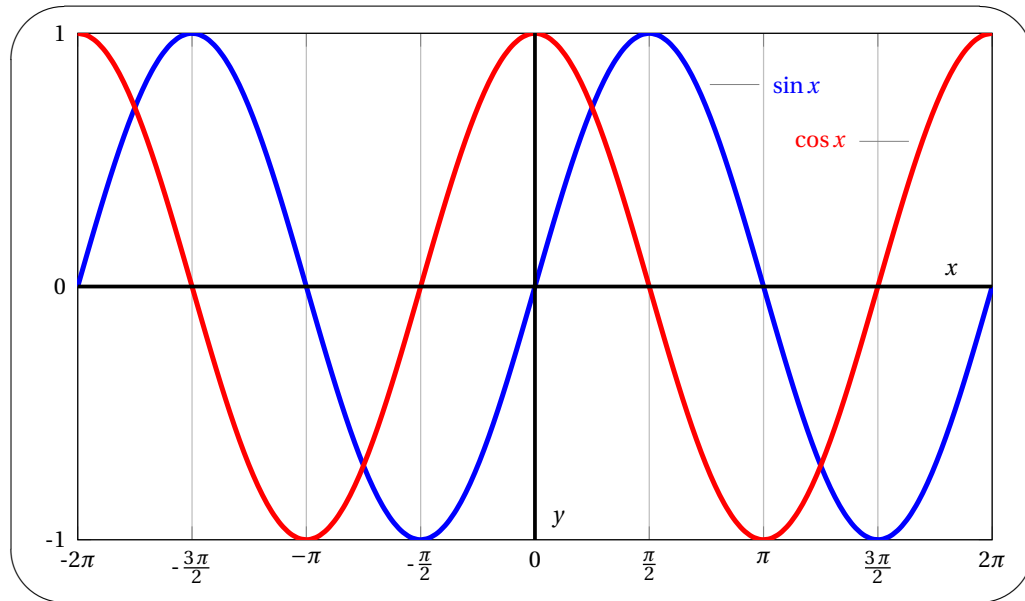
are called **sine** and **cosine**.

- The sine is an **odd** function, while the cosine is an **even** function. Hence we have:

$$\sin(-x) = -\sin x, \quad \forall x \in \mathbb{R} \quad \text{and} \quad \cos(-x) = \cos x, \quad \forall x \in \mathbb{R}$$

- The sine and the cosine are periodic functions with period:

$$T = 2\pi \quad \text{and we have the fundamental identity} \quad \sin^2 x + \cos^2 x = 1, \quad \forall x \in \mathbb{R}$$



We have

$$\cos\left(x - \frac{\pi}{2}\right) = \sin x, \quad \forall x \in \mathbb{R}$$

- From the sign and the monotonicity of the sine function we have:

$$\sin x = 0 \iff x = k\pi, \quad \forall k \in \mathbb{Z} \quad \begin{cases} \sin x > 0 & \text{if } x \in (2k\pi, 2k\pi + \pi), \quad \forall k \in \mathbb{Z} \\ \sin x < 0 & \text{if } x \in (2k\pi - \pi, 2k\pi), \quad \forall k \in \mathbb{Z} \end{cases}$$

$$\begin{cases} \text{if } x \in (2k\pi - \frac{\pi}{2}, 2k\pi + \frac{\pi}{2}), \forall k \in \mathbb{Z} & f \text{ is increasing} \\ \text{if } x \in (2k\pi + \frac{\pi}{2}, 2k\pi + \frac{3\pi}{2}), \forall k \in \mathbb{Z} & f \text{ is decreasing} \end{cases}$$

- From the sign and the monotonicity of the cosine function we have:

$$\cos x = 0 \iff x = k\pi - \frac{\pi}{2}, \forall k \in \mathbb{Z} \quad \begin{cases} \cos x > 0 & \text{if } x \in (2k\pi - \frac{\pi}{2}, 2k\pi + \frac{\pi}{2}), \forall k \in \mathbb{Z} \\ \cos x < 0 & \text{if } x \in (2k\pi + \frac{\pi}{2}, 2k\pi + \frac{3\pi}{2}), \forall k \in \mathbb{Z} \end{cases}$$

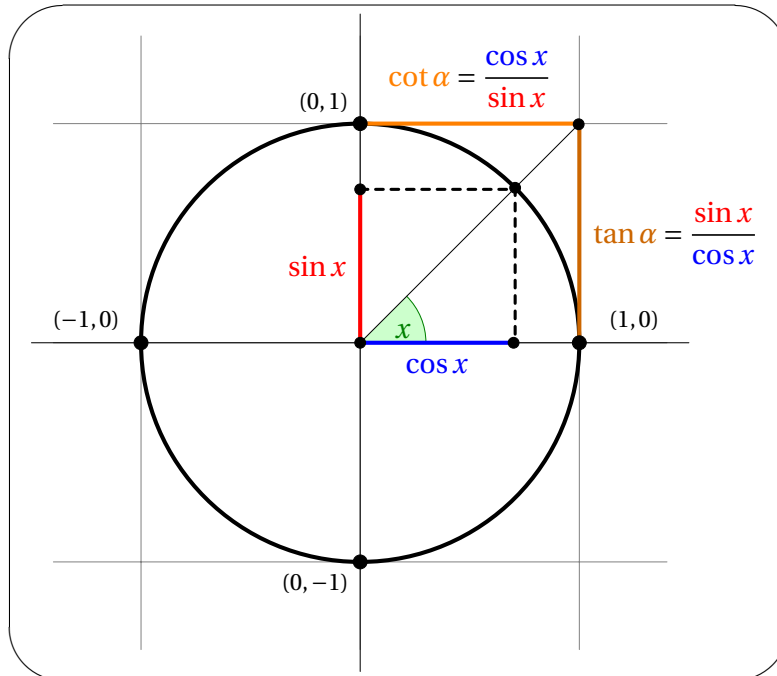
$$\begin{cases} \text{if } x \in (2k\pi + \pi, 2k\pi + 2\pi), \forall k \in \mathbb{Z} & f \text{ is increasing} \\ \text{if } x \in (2k\pi, 2k\pi + \pi), \forall k \in \mathbb{Z} & f \text{ is decreasing} \end{cases}$$

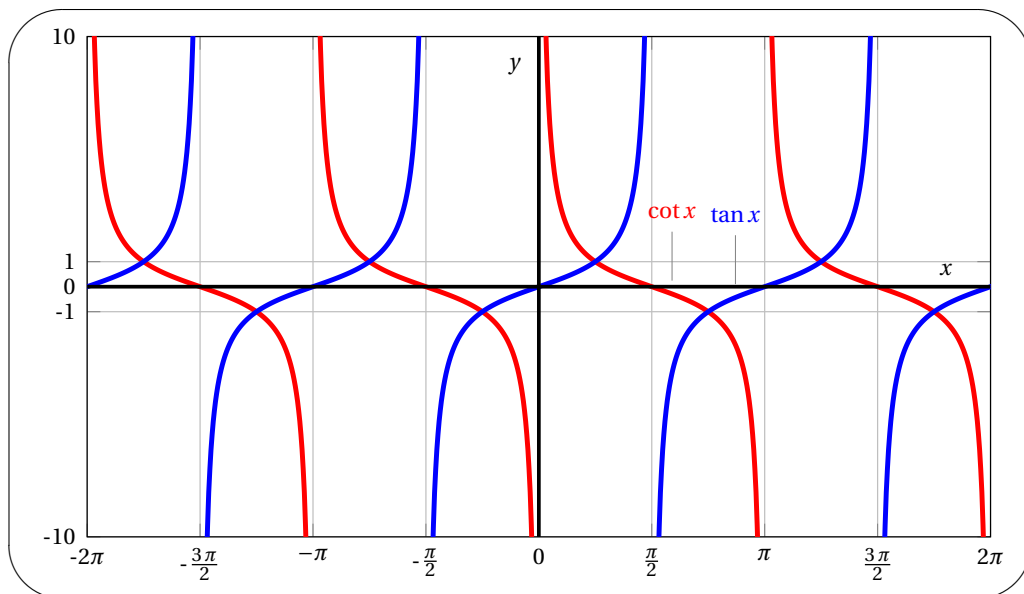
The trigonometric functions:

$$f: \mathbb{R} \setminus \{k\pi - \frac{\pi}{2}, \forall k \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad f: x \mapsto \tan x = \frac{\sin x}{\cos x} \quad (3)$$

$$f: \mathbb{R} \setminus \{k\pi, \forall k \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad f: x \mapsto \cot x = \frac{\cos x}{\sin x} \quad (4)$$

are called **tangent** and **cotangent**; they are both odd and periodic with period $T = \pi$.



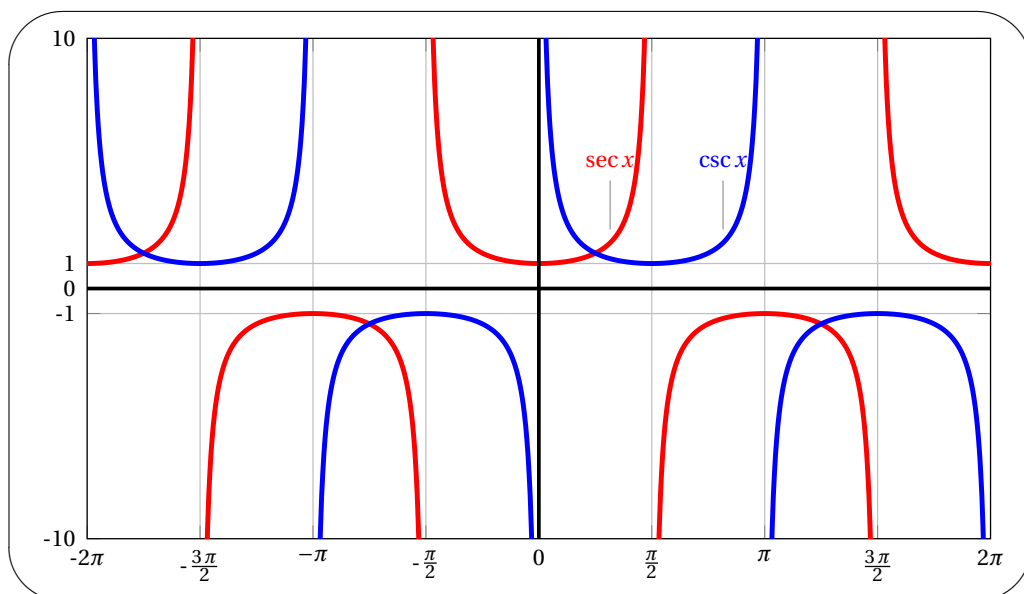


The trigonometric functions:

$$f: \mathbb{R} \setminus \{k\pi - \frac{\pi}{2}, \forall k \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad f: x \mapsto \sec x = \frac{1}{\cos x} \quad (5)$$

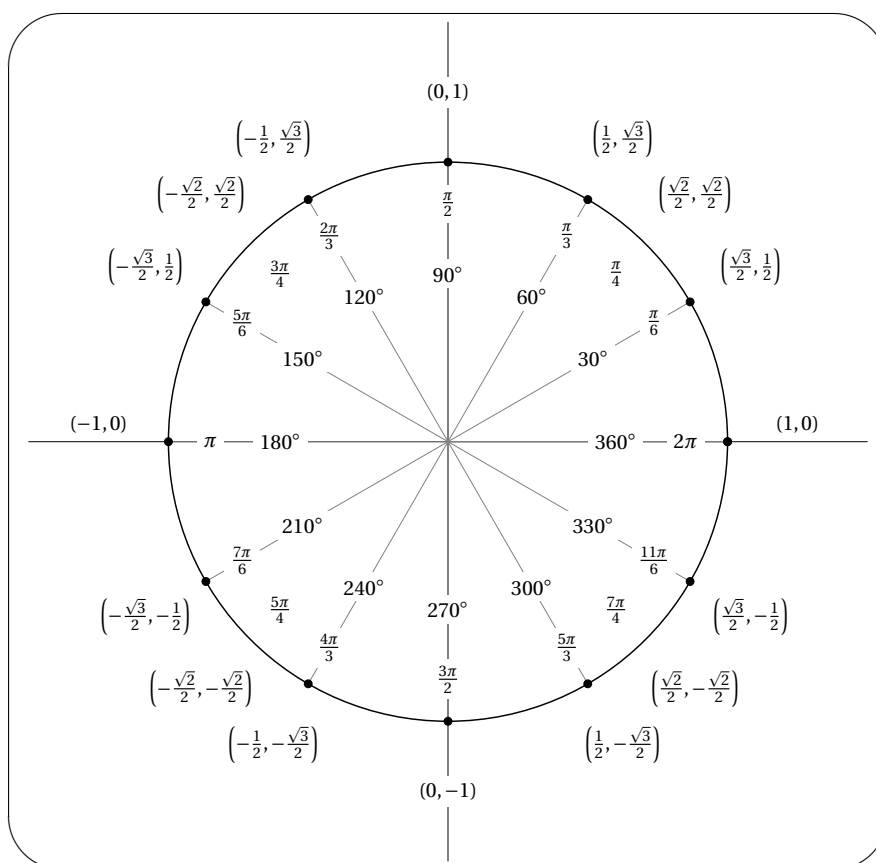
$$f: \mathbb{R} \setminus \{k\pi, \forall k \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad f: x \mapsto \csc x = \frac{1}{\sin x} \quad (6)$$

are called **secant** and **cosecant** and they are periodic with period $T = 2\pi$. The secant function is even, while the cosecant function is odd.



2 Values of the trigonometric functions

	cos	sin	tan	cot	sec	csc
0	1	0	0	$\pm\infty$	1	$\pm\infty$
$x = \frac{\pi}{6}$ (30°)	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{3}$	$\sqrt{3}$	$\frac{2}{3}\sqrt{3}$	2
$x = \frac{\pi}{4}$ (45°)	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	1	$\sqrt{2}$	$\sqrt{2}$
$x = \frac{\pi}{3}$ (60°)	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\sqrt{3}$	$\frac{\sqrt{3}}{3}$	2	$\frac{2}{3}\sqrt{3}$
$x = \frac{\pi}{2}$ (90°)	0	1	$\pm\infty$	0	$\pm\infty$	1
$x = \pi$ (180°)	-1	0	0	$\pm\infty$	-1	$\pm\infty$



	cos	sin	tan	cot
x	$\cos x$	$\sin x$	$\tan x$	$\cot x$
$-x$	$\cos x$	$-\sin x$	$-\tan x$	$-\cot x$
$\frac{\pi}{2} + x$	$-\sin x$	$\cos x$	$-\cot x$	$-\tan x$
$\frac{\pi}{2} - x$	$\sin x$	$\cos x$	$\cot x$	$\tan x$
$\pi + x$	$-\cos x$	$-\sin x$	$\tan x$	$\cot x$
$\pi - x$	$-\cos x$	$\sin x$	$-\tan x$	$-\cot x$

3 Main trigonometric formulas

Relations among the trigonometric functions

	$\sin x$	$\cos x$	$\tan x$	$\cot x$
$\sin x$	-	$\pm \sqrt{1 - \cos^2 x}$	$\pm \sqrt{\frac{\tan^2 x}{1 + \tan^2 x}}$	$\pm \sqrt{\frac{1}{1 + \cot^2 x}}$
$\cos x$	$\pm \sqrt{1 - \sin^2 x}$	-	$\pm \sqrt{\frac{1}{1 + \tan^2 x}}$	$\pm \sqrt{\frac{\cot^2 x}{1 + \cot^2 x}}$
$\tan x$	$\pm \sqrt{\frac{\sin^2 x}{1 - \sin^2 x}}$	$\pm \sqrt{\frac{1 - \cos^2 x}{\cos^2 x}}$	-	$\frac{1}{\cot x}$
$\cot x$	$\pm \sqrt{\frac{1 - \sin^2 x}{\sin^2 x}}$	$\pm \sqrt{\frac{\cos^2 x}{1 - \cos^2 x}}$	$\frac{1}{\tan x}$	-

where the symbol $\pm$ means that the sign depends on the quadrant in which x lies

Addition and subtraction

$$\sin(x_1 \pm x_2) = \sin x_1 \cos x_2 \pm \sin x_2 \cos x_1 \quad (7a)$$

$$\cos(x_1 \pm x_2) = \cos x_1 \cos x_2 \mp \sin x_1 \sin x_2 \quad (7b)$$

$$\tan(x_1 \pm x_2) = \frac{\tan x_1 \pm \tan x_2}{1 \mp \tan x_1 \tan x_2} \quad (8a)$$

$$\cot(x_1 \pm x_2) = \frac{\cot x_1 \cot x_2 \mp 1}{\cot x_2 \pm \cot x_1} \quad (8b)$$

Double-angle formulas

$$\sin(2x) = 2 \sin x \cos x \quad (9a)$$

$$\begin{aligned} \cos(2x) &= \cos^2 x - \sin^2 x \\ &= 2 \cos^2 x - 1 \\ &= 1 - 2 \sin^2 x \end{aligned} \quad (9b)$$

$$\tan(2x) = \frac{2 \tan x}{1 - \tan^2 x} \quad (10a)$$

$$\cot(2x) = \frac{\cot^2 x - 1}{2 \cot x} \quad (10b)$$

Triple-angle formulas

$$\sin(3x) = 3 \sin x - 4 \sin^3 x \quad (11a)$$

$$\cos(3x) = 4 \cos^3 x - 3 \cos x \quad (11b)$$

Half-angle formulas

$$\sin \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{2}} \quad (12a)$$

$$\cos \frac{x}{2} = \pm \sqrt{\frac{1 + \cos x}{2}} \quad (12b)$$

$$\begin{aligned} \tan \frac{x}{2} &= \pm \sqrt{\frac{1 - \cos x}{1 + \cos x}} \\ &= \frac{\sin x}{1 + \cos x} \\ &= \frac{1 - \cos x}{\sin x} \end{aligned} \quad (13a)$$

$$\begin{aligned} \cot \frac{x}{2} &= \pm \sqrt{\frac{1 + \cos x}{1 - \cos x}} \\ &= \frac{\sin x}{1 - \cos x} \\ &= \frac{1 + \cos x}{\sin x} \end{aligned} \quad (14a)$$

Parametric formulas (tangent half-angle substitution)

$$\sin x = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \quad (15a)$$

$$\cos x = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \quad (15b)$$

Sum-to-product formulas (prosthaphaeresis)

$$\sin x_1 + \sin x_2 = 2 \sin \frac{x_1 + x_2}{2} \cos \frac{x_1 - x_2}{2} \quad (16a)$$

$$\sin x_1 - \sin x_2 = 2 \sin \frac{x_1 - x_2}{2} \cos \frac{x_1 + x_2}{2} \quad (16b)$$

$$\cos x_1 + \cos x_2 = 2 \cos \frac{x_1 + x_2}{2} \cos \frac{x_1 - x_2}{2} \quad (16c)$$

$$\cos x_1 - \cos x_2 = -2 \sin \frac{x_1 + x_2}{2} \sin \frac{x_1 - x_2}{2} \quad (16d)$$

Product-to-sum formulas (Werner's formulas)

$$\sin x_1 \sin x_2 = \frac{1}{2} \left(\cos(x_1 - x_2) - \cos(x_1 + x_2) \right) \quad (17a)$$

$$\sin x_1 \cos x_2 = \frac{1}{2} \left(\sin(x_1 - x_2) + \sin(x_1 + x_2) \right) \quad (17b)$$

$$\cos x_1 \cos x_2 = \frac{1}{2} \left(\cos(x_1 - x_2) + \cos(x_1 + x_2) \right) \quad (17c)$$

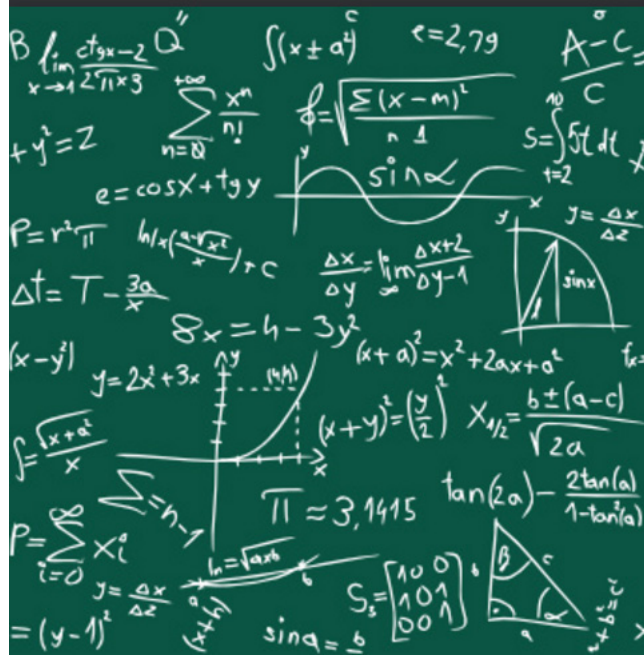
Additional formulas

$$\sin^2 x = \frac{1 - \cos(2x)}{2} \quad (18a)$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2} \quad (18b)$$

$$\sin x \cos x = \frac{\sin(2x)}{2} \quad (18c)$$

Vibration phenomena



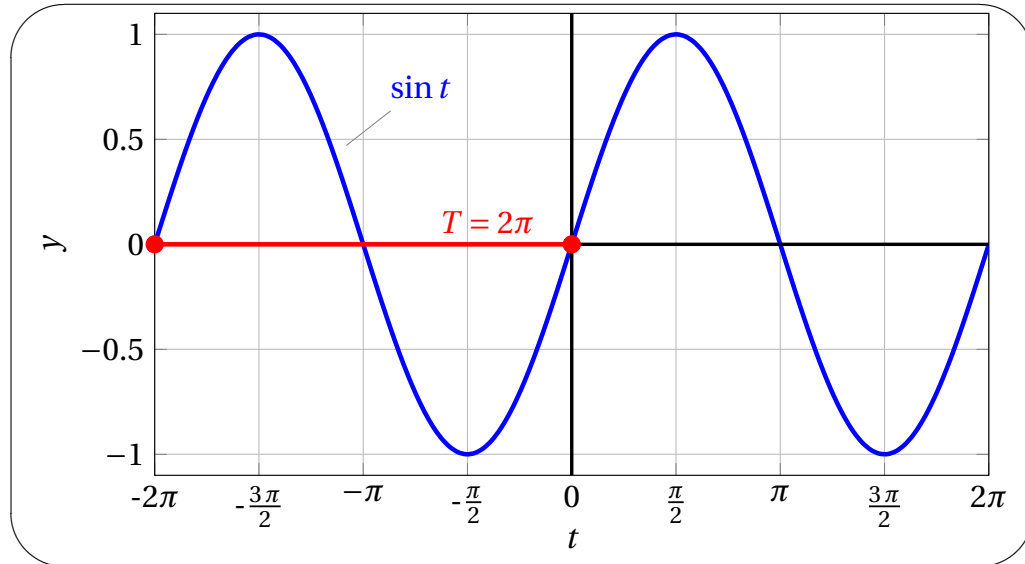
- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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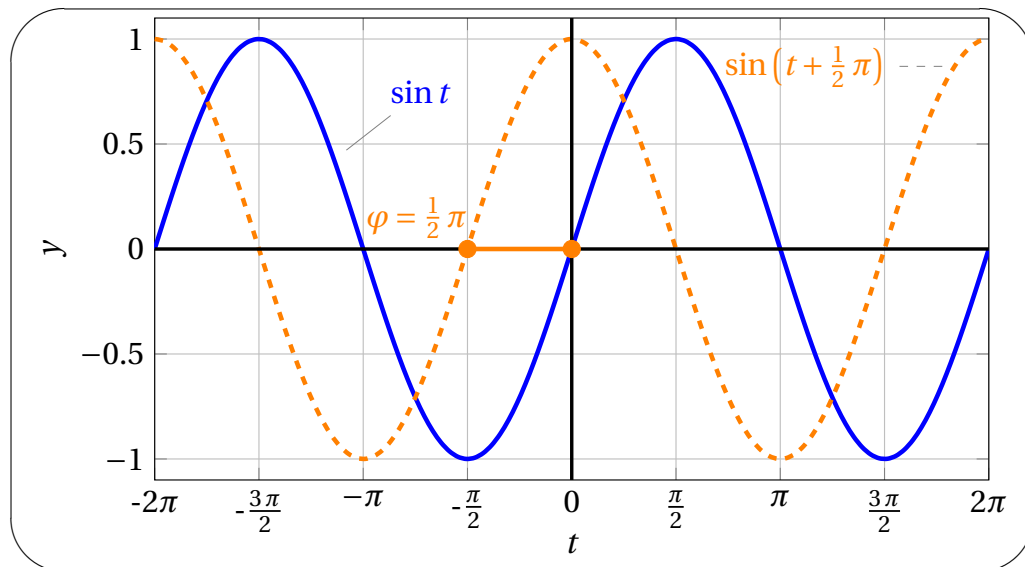
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1 Vibration phenomena

- We have seen that the sine and cosine functions are periodic with **period** 2π . We will use the variable t to denote time.



- Recall that $\sin(t + \varphi)$ corresponds to a **phase shift** φ (horizontal translation).



The functions:

$$t \mapsto a \sin(\omega t), \quad t \mapsto b \cos(\omega t) \quad (1)$$

where a , b and ω are positive real numbers, are called **elementary vibrations**. They are periodic functions with

$$\text{period } T = \frac{2\pi}{\omega}$$

Example 1: period

For example, considering $t' = t + \frac{2\pi}{\omega}$:

$$\sin \left[\omega \underbrace{\left(t + \frac{2\pi}{\omega} \right)}_{t'} \right] = \sin(\omega t + 2\pi) = \sin \omega t$$

- Clearly we have:

$$|a \sin(\omega t)| \leq a \quad \text{and} \quad |b \cos(\omega t)| \leq b$$

The functions (1) describe elementary vibrations characterized by

amplitude a and b (respectively)

$$\text{angular frequency } \omega = \frac{2\pi}{T}$$

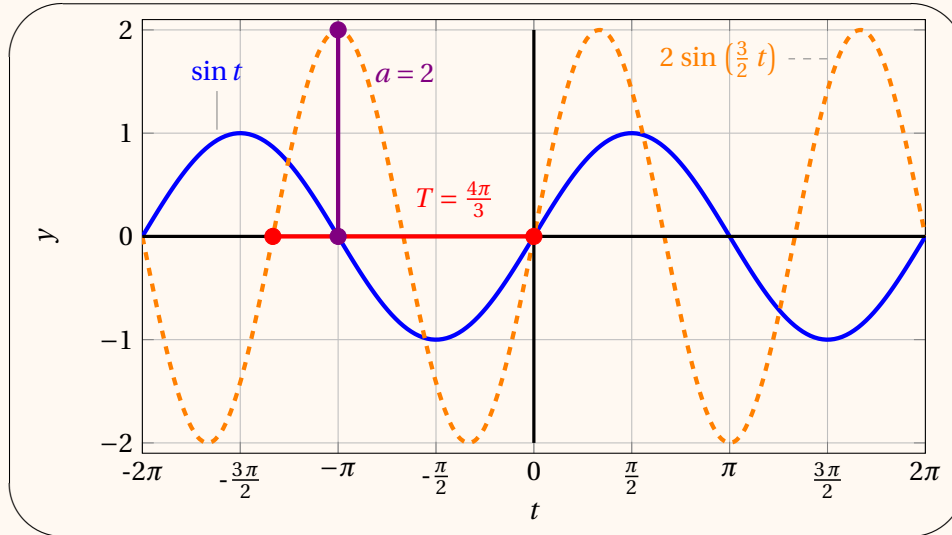
also called *angular velocity*, which indicates how many periods there are in an interval of length 2π . Moreover, they are characterized by the

$$\text{frequency } \nu = \frac{\omega}{2\pi}$$

which indicates how many times the function repeats itself in an interval of length 1

Example 2: of amplitude, angular frequency and frequency

$$t \mapsto 2 \sin\left(\frac{3}{2}t\right) \quad a = 2, \omega = \frac{3}{2}, T = \frac{4\pi}{3}$$



The angular frequency $\omega = \frac{3}{2}$ means that there are 1.5 periods in an interval of length 2π . The frequency ν is $\frac{3}{4\pi}$.

- The function

$$h(t) = a \sin(\omega t) + b \cos(\omega t) \quad (2)$$

describes the **superposition** of the two elementary vibrations with the **same angular frequency** ω . The latter is again an elementary vibration, **phase-shifted** with respect to the previous ones.

- Indeed, setting

$$A = \sqrt{a^2 + b^2}$$

we can write

$$h(t) = A \underbrace{\frac{a}{\sqrt{a^2 + b^2}}}_{\alpha} \sin(\omega t) + A \underbrace{\frac{b}{\sqrt{a^2 + b^2}}}_{\beta} \cos(\omega t) \quad (3)$$

- Now observe that the numbers

$$\alpha = \frac{a}{\sqrt{a^2 + b^2}} \quad \text{and} \quad \beta = \frac{b}{\sqrt{a^2 + b^2}}$$

satisfy the conditions

$$-1 \leq \alpha \leq 1 \quad -1 \leq \beta \leq 1 \quad \alpha^2 + \beta^2 = 1$$

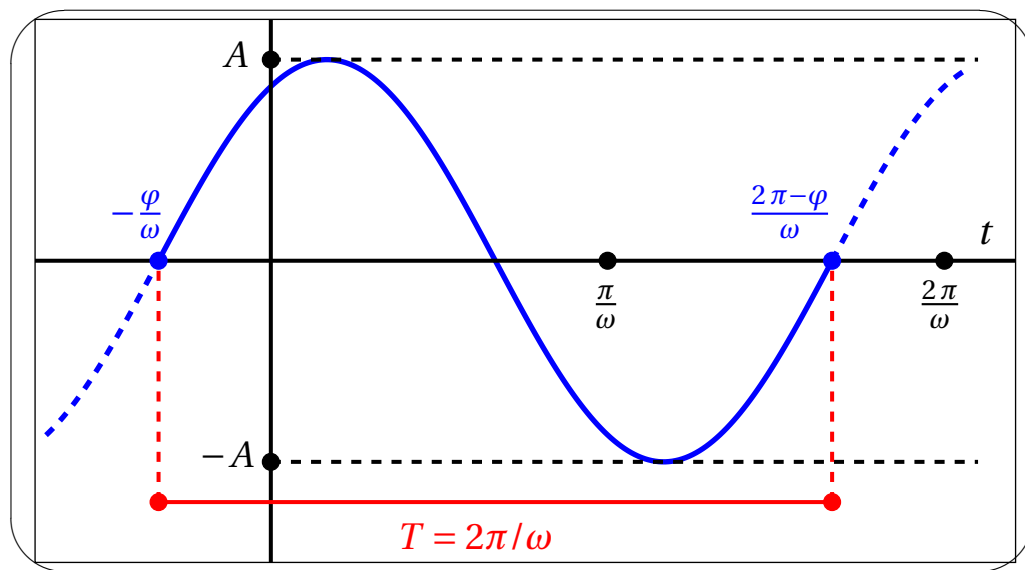
Hence there exists a **unique angle** φ such that

$$\cos \varphi = \alpha \quad \sin \varphi = \beta$$

so that (3) can be rewritten in the following form

$$\begin{aligned} h(t) &= A \cos \varphi \sin(\omega t) + A \sin \varphi \cos(\omega t) \\ &= A \sin(\omega t + \varphi) \end{aligned} \tag{4}$$

Therefore $h(t)$ represents an elementary vibration with amplitude A , angular frequency ω , phase-shifted by an angle φ .



In summary:

$$a \sin(\omega t) + b \cos(\omega t) = A \sin(\omega t + \varphi)$$

$$A = \sqrt{a^2 + b^2} \quad \begin{cases} a = A \cos \varphi \\ b = A \sin \varphi \end{cases}$$

- Under fairly general conditions, a periodic natural phenomenon can be written as the superposition of a finite or infinite number of elementary vibrations with different frequencies (**Fourier series**).

2 Damping and amplification effects

- By multiplying an elementary vibration by **powers** or **exponentials**, we can model **damping** or **amplification** effects.

First example

- For example, consider the function

$$h(t) = t \sin(\omega t)$$

which models an **amplified vibration**.

- Since $-1 \leq \sin(\omega t) \leq 1$, we have

$$-t \leq t \sin(\omega t) \leq t$$

and hence the graph of $h(t)$ lies between the graphs of the lines with equations $y = -t$, $y = t$.

- At the points where

$$\sin(\omega t) = 1$$

i.e.,

$$t = \frac{\pi}{2\omega} + k \frac{2\pi}{\omega} \quad (k = 0, 1, 2, \dots)$$

the graph of $h(t)$ touches that of $y = t$.

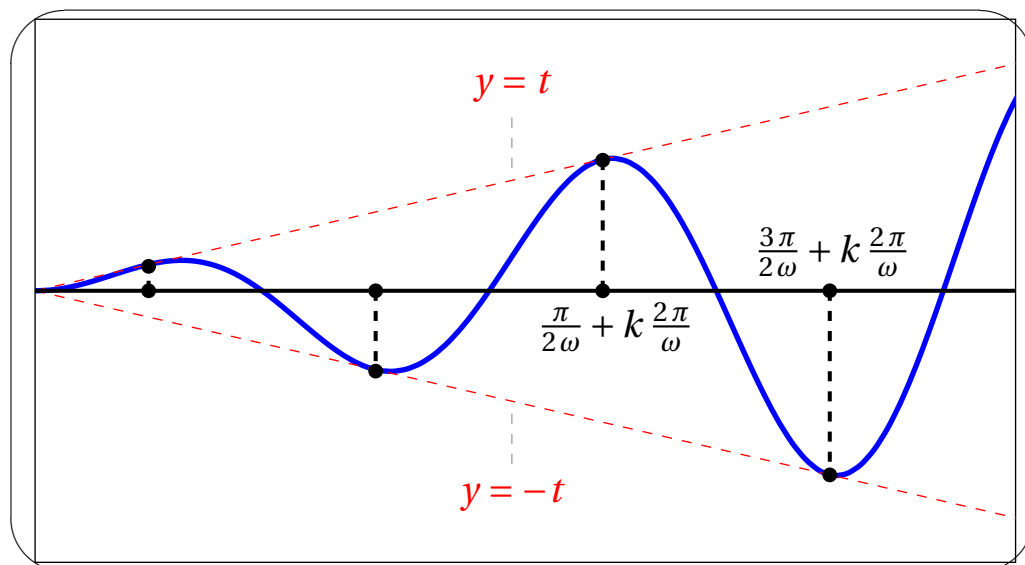
- At the points where

$$\sin(\omega t) = -1$$

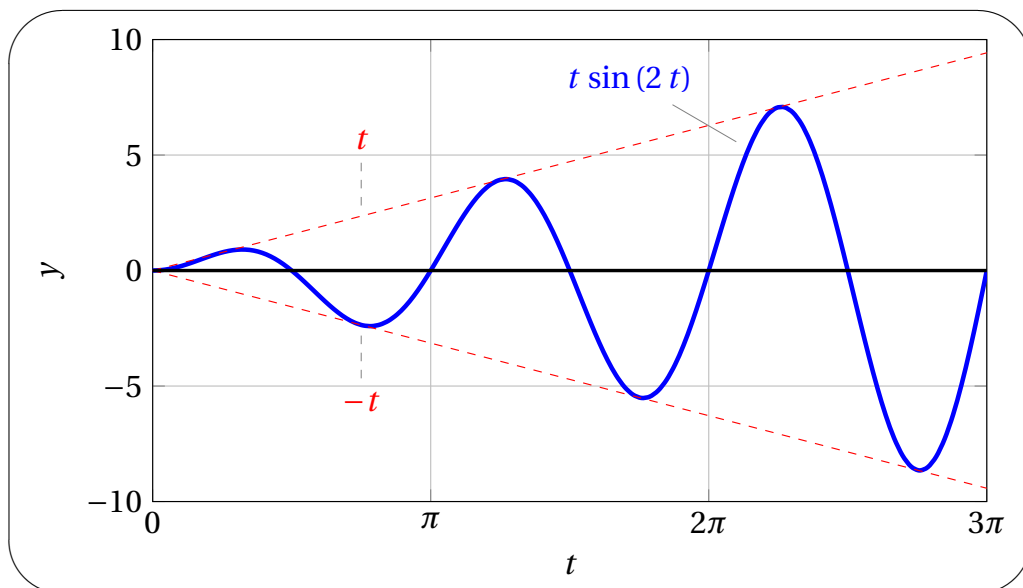
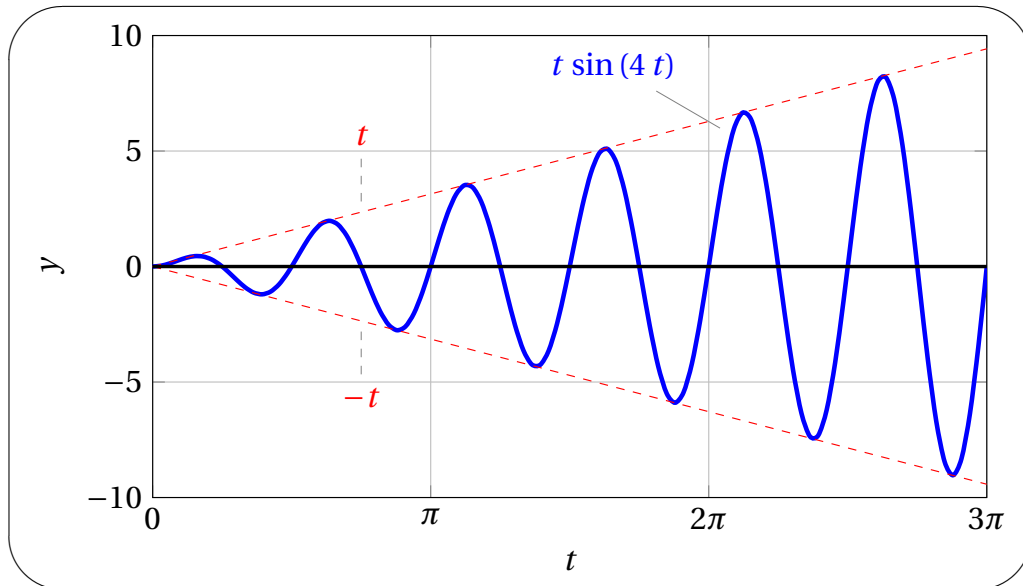
i.e.,

$$t = \frac{3\pi}{2\omega} + k \frac{2\pi}{\omega} \quad (k = 0, 1, 2, \dots)$$

the graph of $h(t)$ touches that of $y = -t$.



- From the graph we can see that multiplication by t has the effect of amplifying the vibration as t increases.



Second example

- For example, consider the function

$$k(t) = e^{-\alpha t} \sin(\omega t) \quad (\alpha > 0)$$

which models a **damped vibration**.

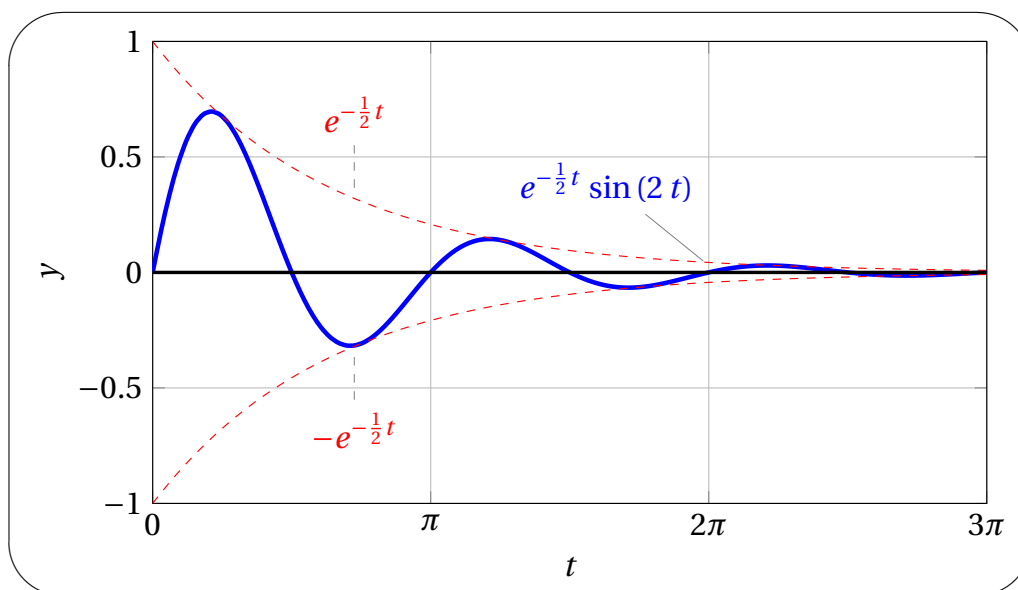
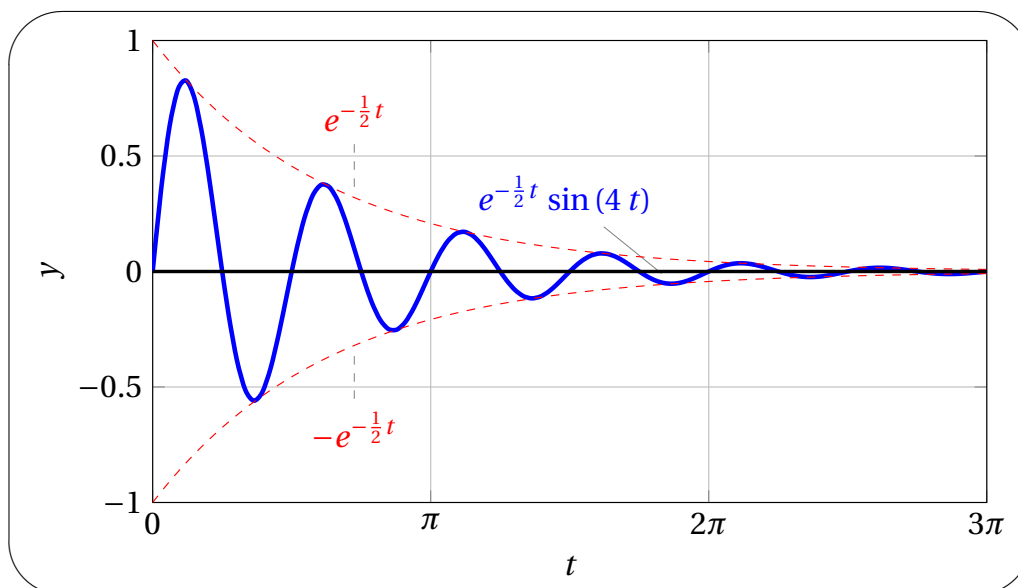
- Recalling that

$$e^{-\alpha t} = \left(\frac{1}{e^\alpha}\right)^t$$

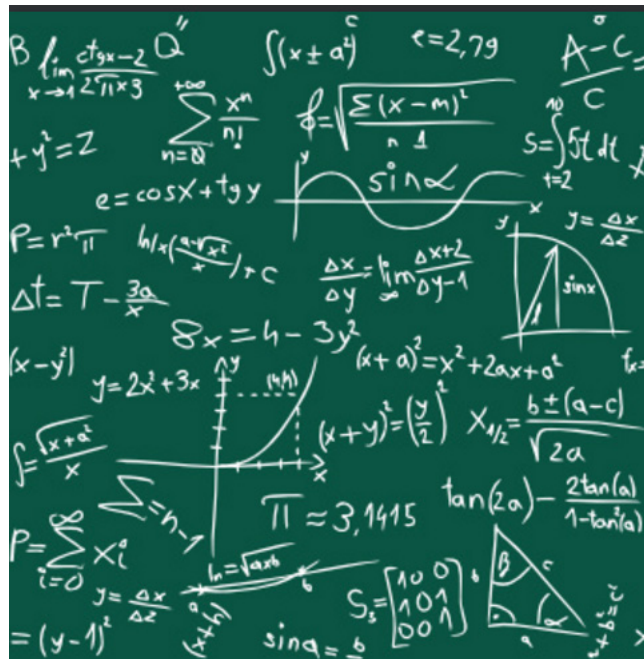
is an exponential with base less than 1, considerations analogous to those made for the function $h(t)$ show that the graph of k lies between the graphs of the functions

$$y_1 = e^{-\alpha t} \quad \text{and} \quad y_2 = -e^{-\alpha t}$$

as shown in the figure:



Integer part and fractional part functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1	Integer part and fractional part functions	2
2	Piecewise-defined functions	4

1 Integer part and fractional part functions

- Two functions that are typically encountered when writing *algorithms* are the integer part (floor) function and the fractional part (or decimal part) function.

Definition 1: of the integer part function

The **integer part (floor) function** is:

$$f : \mathbb{R} \rightarrow \mathbb{Z}, x \mapsto \lfloor x \rfloor \quad (\text{or } x \mapsto \lfloor x \rfloor) \quad (1)$$

Definition 2: of the upper integer part function

The **upper integer part (ceiling) function** is:

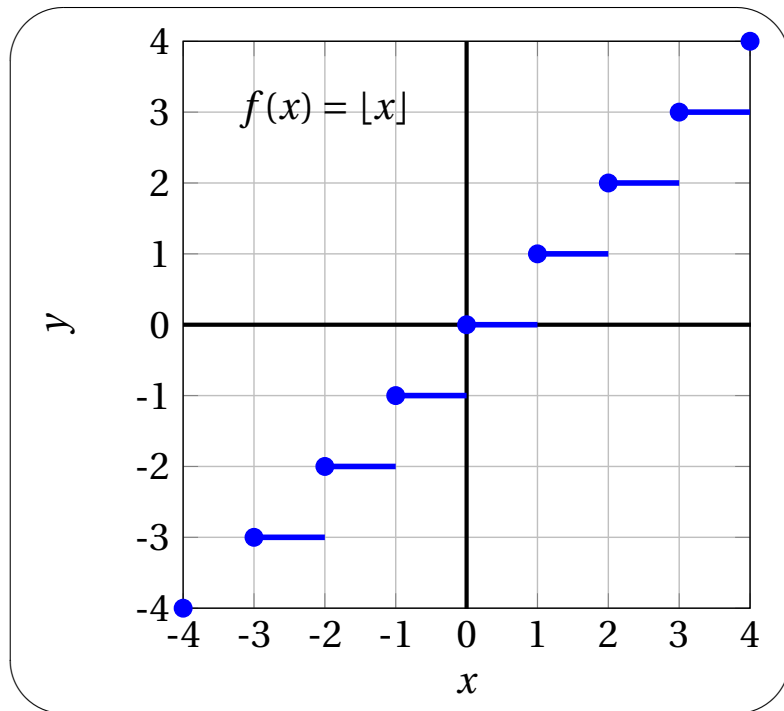
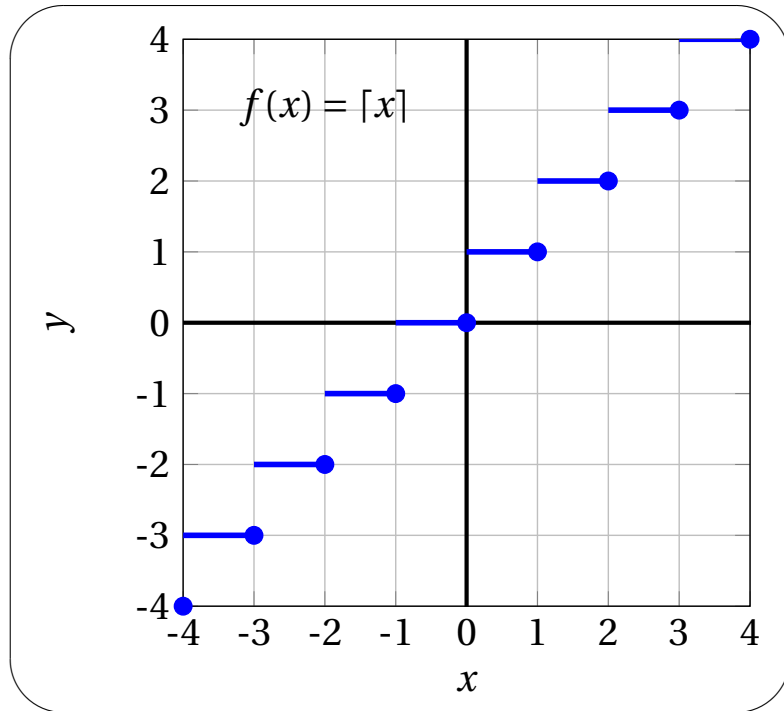
$$f : \mathbb{R} \rightarrow \mathbb{Z}, x \mapsto \lceil x \rceil \quad (2)$$

Definition 3: of the fractional part function

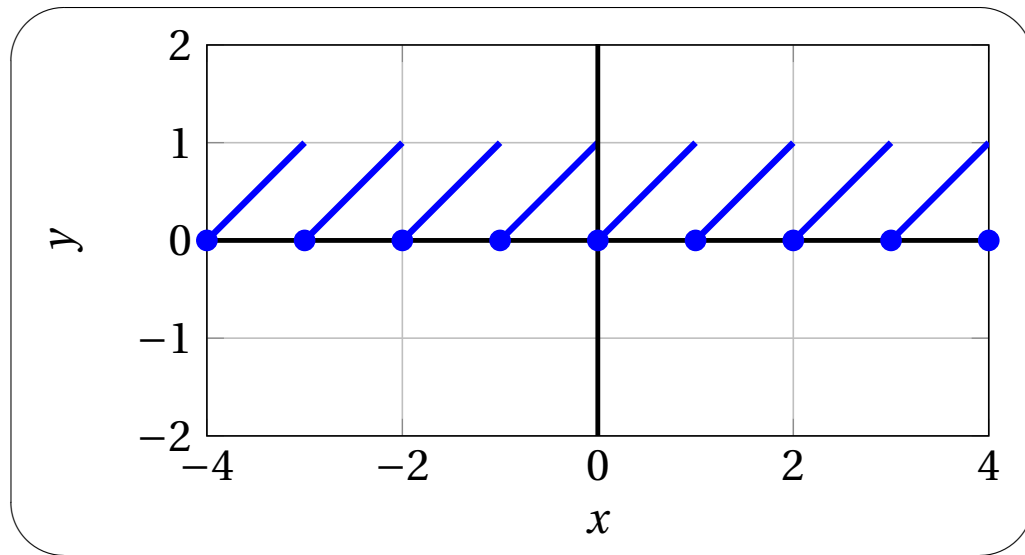
The **fractional part function** is:

$$f : \mathbb{R} \rightarrow [0, 1), x \mapsto (x) \quad (3)$$

- The integer part function $f(x) = \lfloor x \rfloor$ is *monotone increasing*, like the upper integer part function $f(x) = \lceil x \rceil$.
- Graphs of the functions $f(x) = \lceil x \rceil$ and $f(x) = \lfloor x \rfloor$ on the interval $[-4, 4]$.



- Note that the fractional part is a periodic function with period 1:



2 Piecewise-defined functions

- Starting from the elementary functions, new ones can be built by using **different analytic definitions on different intervals**.

Definition 4: of piecewise-defined function

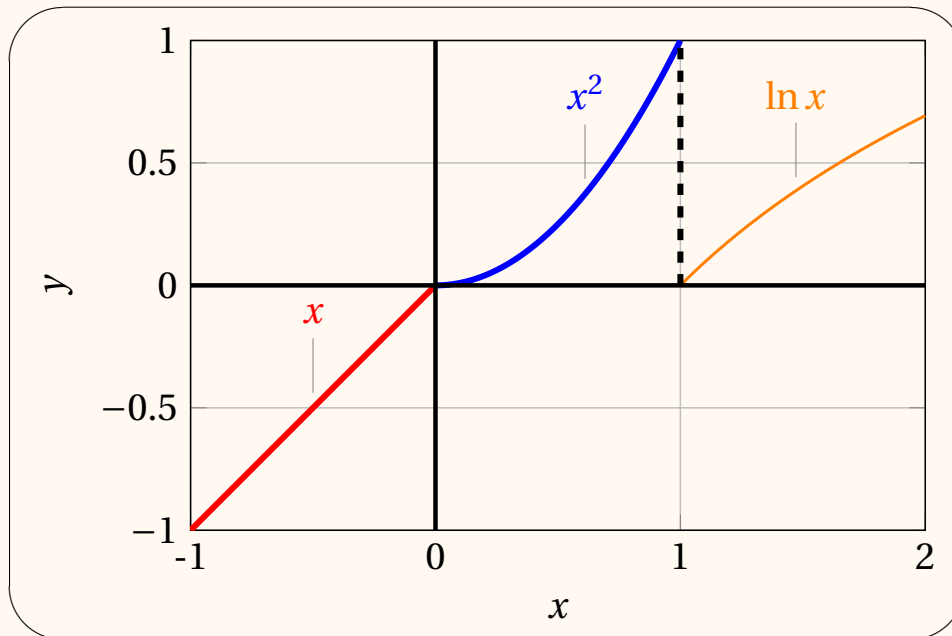
A function f whose value $f(x)$ is computed through different “instructions” depending on the interval in which x lies is called a **piecewise-defined function**.

Example 1: Piecewise-defined functions

Consider for example

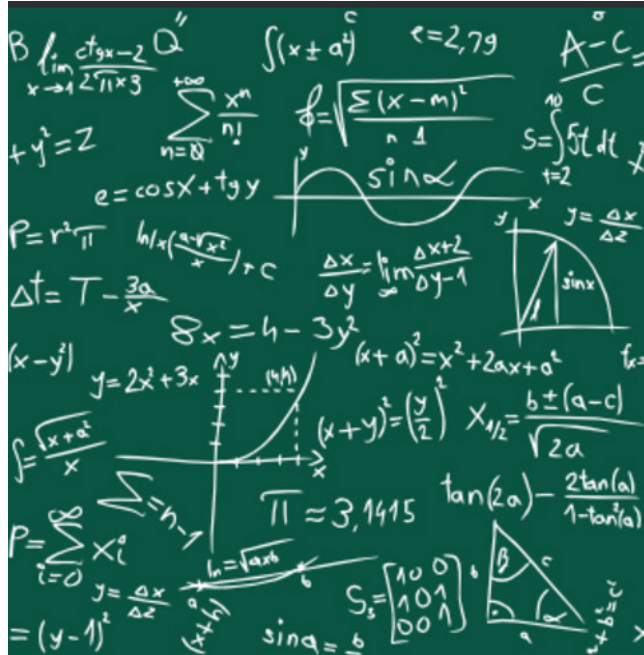
$$f(x) = \begin{cases} \ln x & \text{if } x > 1, \\ x^2 & \text{if } 0 < x \leq 1, \\ x & \text{if } x \leq 0 \end{cases}$$

its graph is:



- From the *logical / algorithmic* point of view, one can say that a function of this kind has the peculiarity of being built using not only mathematical functions but also the **logical function** “if ... then”.

Hyperbolic functions



- Author: Fabio Furini.

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1	Hyperbolic functions	2
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1 Hyperbolic functions

Definition 1: hyperbolic sine function and hyperbolic cosine function

The hyperbolic functions:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \text{Sh } x = \frac{e^x - e^{-x}}{2} \quad (1)$$

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \text{Ch } x = \frac{e^x + e^{-x}}{2} \quad (2)$$

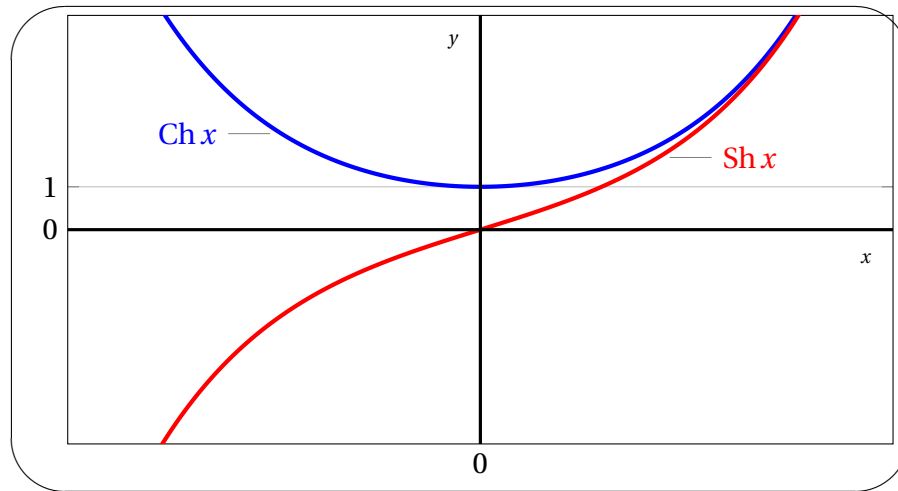
are called **hyperbolic sine** and **hyperbolic cosine**.

- The hyperbolic sine is an **odd** function, while the hyperbolic cosine is an **even** function. Hence we have:

$$\text{Sh}(-x) = -\text{Sh } x, \quad \forall x \in \mathbb{R} \quad \text{and} \quad \text{Ch}(-x) = \text{Ch } x, \quad \forall x \in \mathbb{R}$$

- We have the fundamental relations:

$$\text{Ch}^2 x - \text{Sh}^2 x = 1, \quad \forall x \in \mathbb{R} \quad \text{and} \quad \text{Sh } x \leq \frac{e^x}{2} \leq \text{Ch } x, \quad \forall x \in \mathbb{R} \quad (3)$$



- For the sign and monotonicity of the hyperbolic sine function we have:

$$\text{Sh } x = 0 \iff x = 0 \quad \begin{cases} \text{Sh } x > 0 & \text{if } x > 0 \\ \text{Sh } x < 0 & \text{if } x < 0 \end{cases}$$

f is increasing $\forall x \in \mathbb{R}$

- For the sign and monotonicity of the hyperbolic cosine function we have:

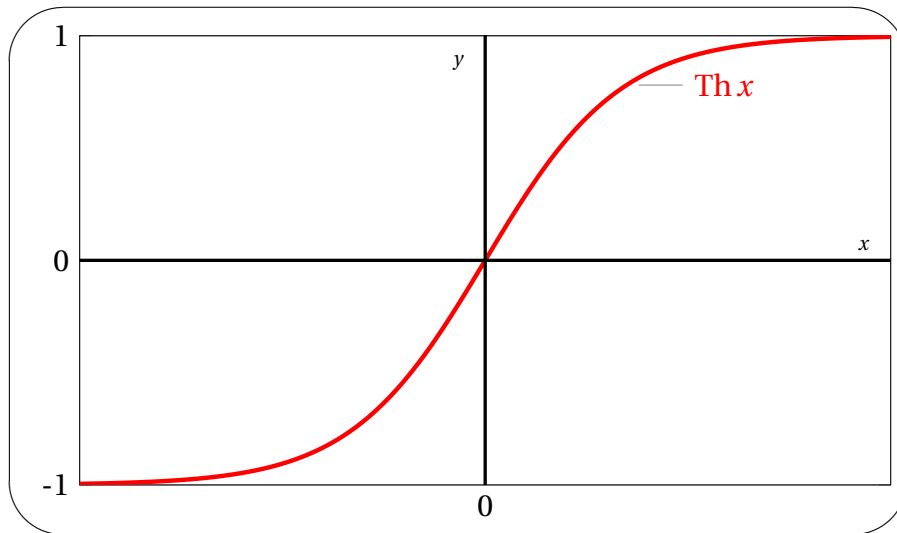
$$\text{Ch } x > 0, \quad \forall x \in \mathbb{R}$$

$$\begin{cases} \text{if } x < 0 & f \text{ is decreasing} \\ \text{if } x > 0 & f \text{ is increasing} \end{cases}$$

The hyperbolic function:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \text{Th } x = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (4)$$

is called the **hyperbolic tangent** and it is an odd function.



2 Main hyperbolic formulas

Addition

$$\text{Sh}(x_1 + x_2) = \text{Sh } x_1 \text{ Ch } x_2 + \text{Sh } x_2 \text{ Ch } x_1 \quad (5a)$$

$$\text{Ch}(x_1 + x_2) = \text{Ch } x_1 \text{ Ch } x_2 + \text{Sh } x_1 \text{ Sh } x_2 \quad (5b)$$

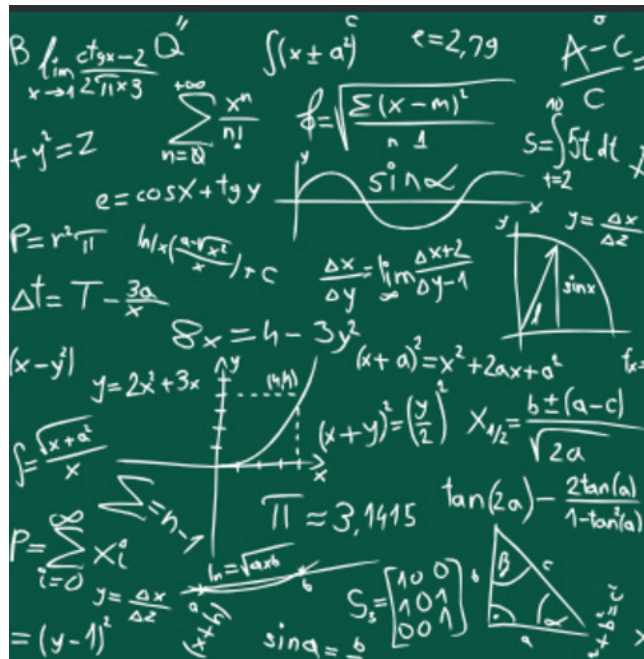
$$\text{Th}(x_1 + x_2) = \frac{\text{Th } x_1 + \text{Th } x_2}{1 + \text{Th } x_1 \text{ Th } x_2} \quad (5c)$$

Double-argument formulas

$$\operatorname{Sh}(2x) = 2 \operatorname{Sh} x \operatorname{Ch} x \quad (6a)$$

$$\operatorname{Ch}(2x) = \operatorname{Ch}^2 x + \operatorname{Sh}^2 x \quad (6b)$$

Operations on graphs



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Operations on graphs

- Knowing the graph of a function $y = f(x)$, by means of simple geometric transformations it is possible to draw the graph of the following functions:

$$y_1 = f(x) + a, \quad a \in \mathbb{R} \quad (1a)$$

$$y_2 = f(x + a), \quad a \in \mathbb{R} \quad (1b)$$

$$y_3 = k f(x), \quad k \in \mathbb{R} \quad (1c)$$

$$y_4 = f(kx), \quad k \in \mathbb{R} \quad (1d)$$

$$y_5 = |f(x)| \quad (1e)$$

$$y_6 = f(|x|) \quad (1f)$$

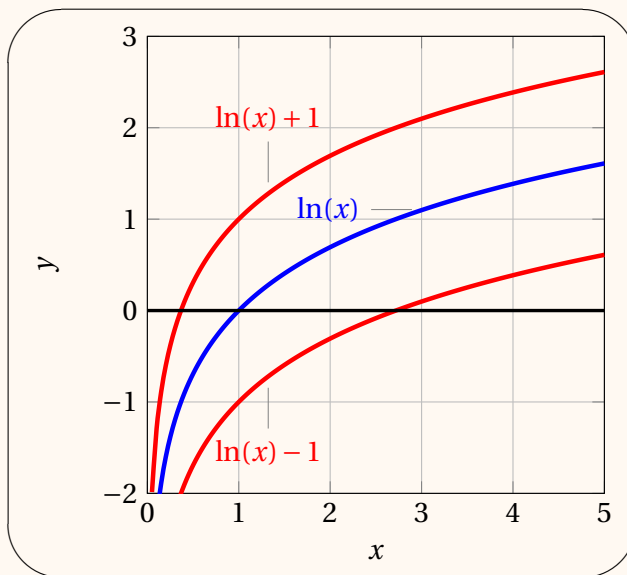
- Therefore, starting from the elementary functions, it is possible to build, by means of these operations, a large variety of new functions. These operations create **translations**, **dilations** and **reflections** of the original function.

1.1 Operations related to $y_1 = f(x) + a$

Example 1: Operations on graphs related to $y_1 = f(x) + a$

Consider $y = \ln x$. Then

$$y_1 = \ln(x) + a \quad (x > 0).$$



- The graph of y_1 is obtained from that of y by a **translation** of a units **upward** if $a > 0$, **downward** if $a < 0$.

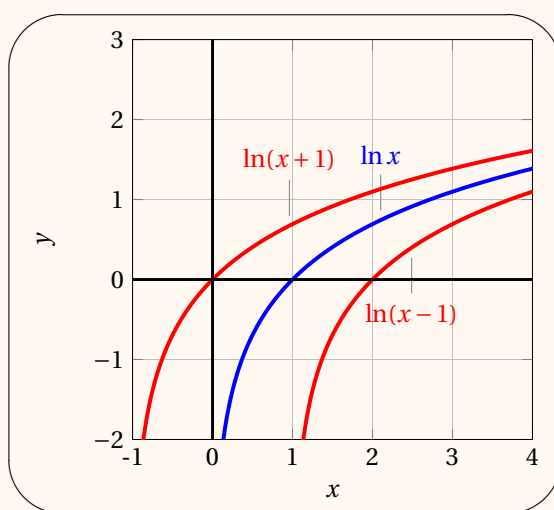
1.2 Operations related to $y_2 = f(x + a)$

Example 2: Operations on graphs related to $y_2 = f(x + a)$

Consider again $y = \ln x$. Then

$$y_2 = \ln(x + a).$$

Since $\ln x$ is defined for $x > 0$, $\ln(x + a)$ will be defined for $x + a > 0$, i.e., for $x > -a$. Since $\ln x = 0$ if $x = 1$, we have $\ln(x + a) = 0$ for $x + a = 1$, i.e., $x = 1 - a$.



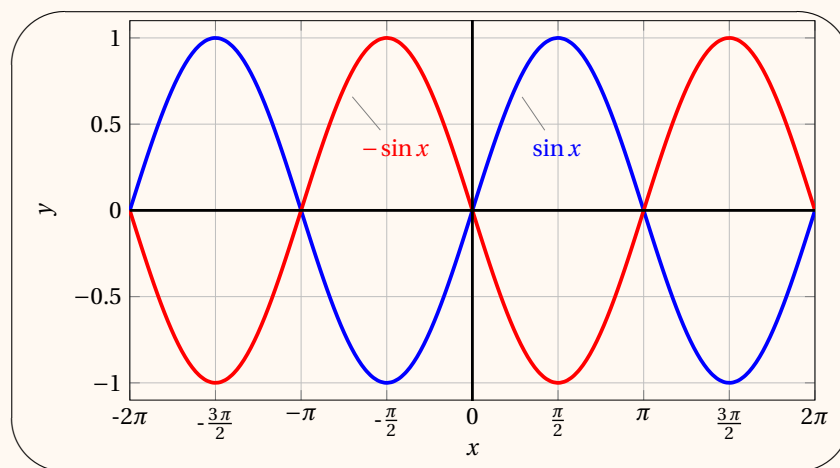
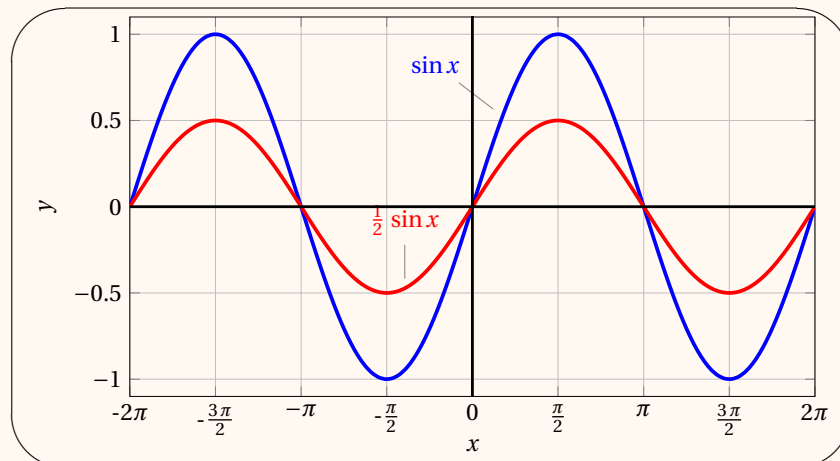
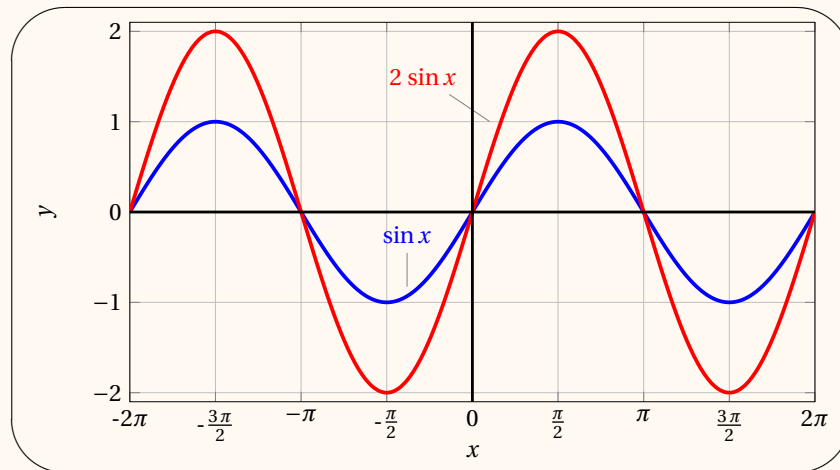
- The graph of y_2 is obtained from that of y by a **translation** of a units to the **left** if $a > 0$, to the **right** if $a < 0$.

1.3 Operations related to $y_3 = k f(x)$

- The graph of $y_3 = k f(x)$ is obtained from that of f by multiplying all the ordinates $f(x)$ by k .
- In particular, if $k = -1$ the ordinates simply change sign, so that the graph of y_3 is symmetric to that of f with respect to the x -axis.
- Note that if $k > 1$, the graph is “**stretched**” in the **vertical direction**, dilating the **positive** ordinates **upward** and the **negative** ones **downward**.
- Conversely, if $0 < k < 1$ the graph “**shrinks**”, again in the vertical direction.
- Therefore, the operation of multiplying $f(x)$ by k has the geometric meaning of a **dilation** (if $|k| > 1$) or a **contraction** (if $|k| < 1$) along the y -axis, possibly accompanied by a **reflection with respect to the x -axis**, if $k < 0$.

Example 3: Operations on graphs related to $y_3 = k f(x)$

Consider $y = \sin x$, then $y_3 = k \sin(x)$.

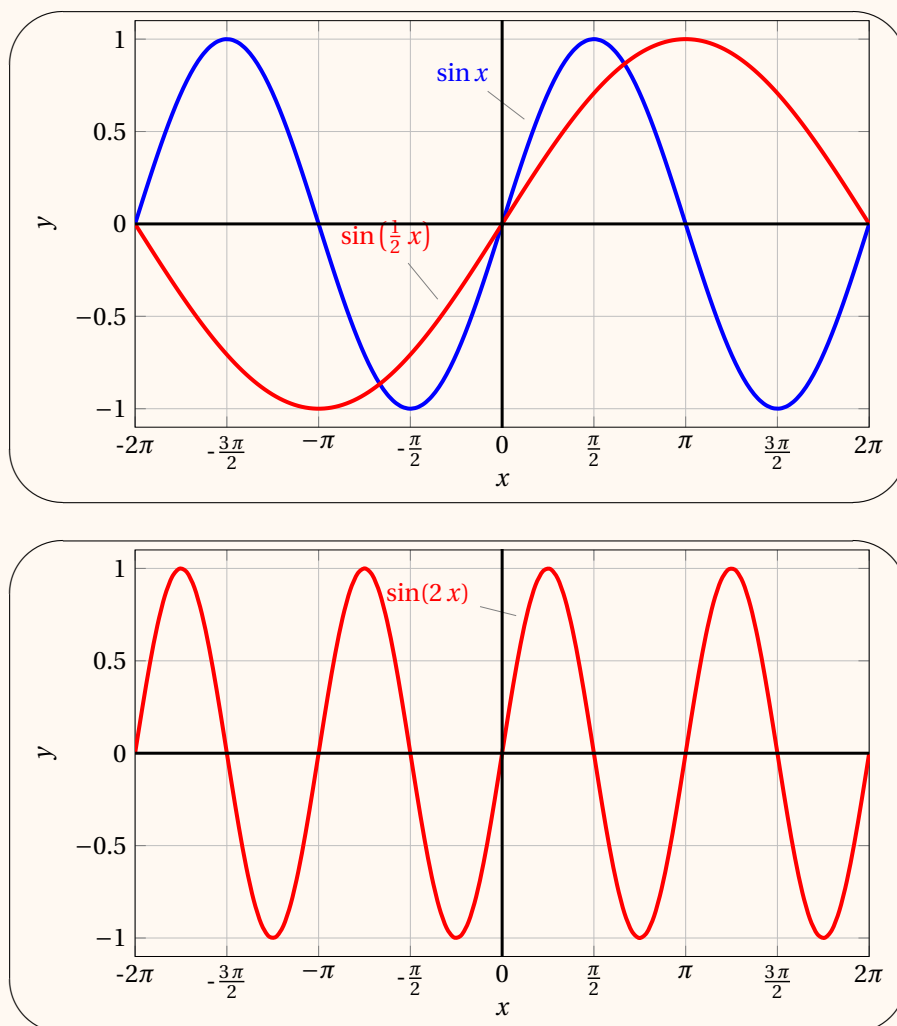


1.4 Operations related to $y_4 = f(kx)$

- The graph of $y_4 = f(kx)$ is obtained from that of $f(x)$ by a **change of scale** along the x -axis.
- If $k > 1$, kx grows faster than x , and therefore the graph of y_4 will be similar to that of f but with faster oscillations, that is, it will be “**compressed**” in the **horizontal direction** by a factor $\frac{1}{k}$.
- Similarly, if $0 < k < 1$ the graph will appear “**dilated**” in the **horizontal direction**, with gentler oscillations.
- If $k < 0$, in addition to a compression (if $|k| > 1$) or a dilation (if $|k| < 1$) along the x -axis, there will be a **reflection with respect to the y -axis**.

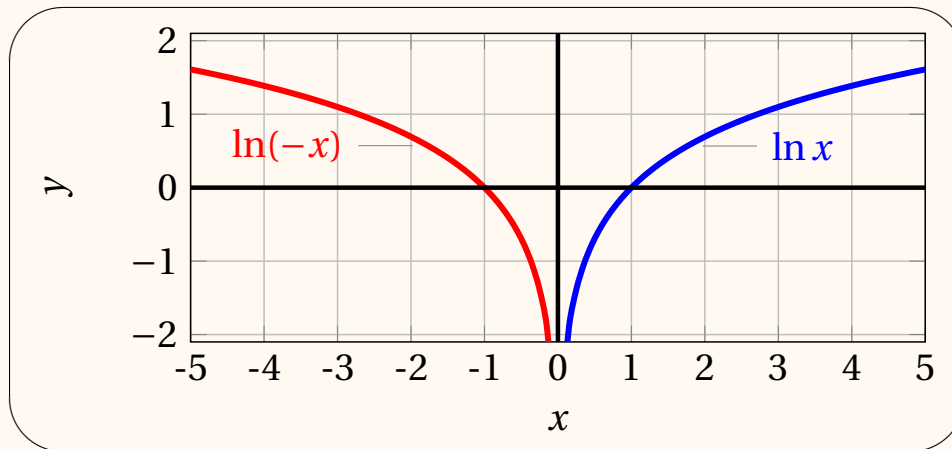
Example 4: Operations on graphs related to $y_4 = f(kx)$

Consider $y = \sin x$, then $y_4 = \sin(kx)$. With $k = \frac{1}{2}$ and $k = 2$ we have:



Example 5: Operations on graphs related to $y_4 = f(kx)$

Consider $y = \ln x$, then $y_4 = \ln(kx)$. With $k = -1$ we have:



1.5 Operations related to $y_5 = |f(x)|$

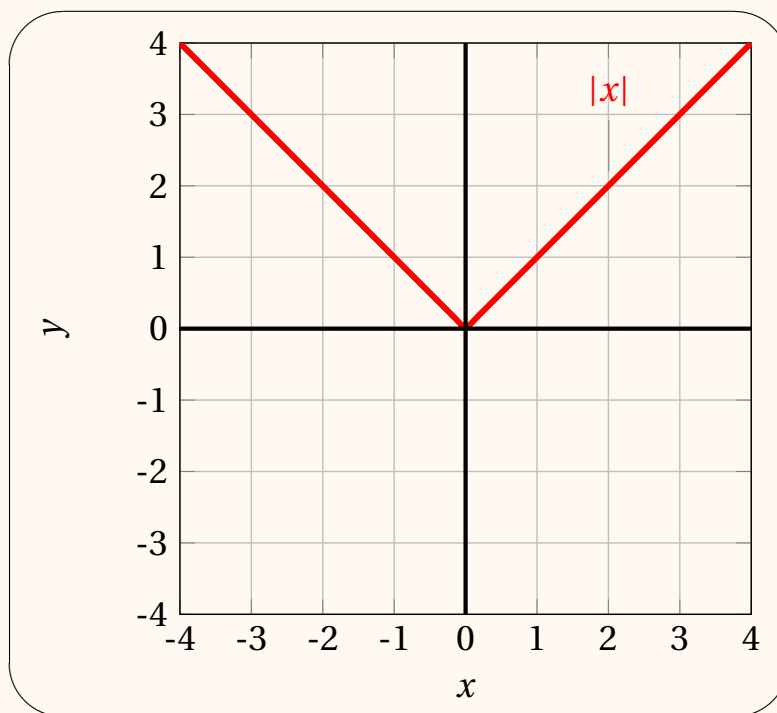
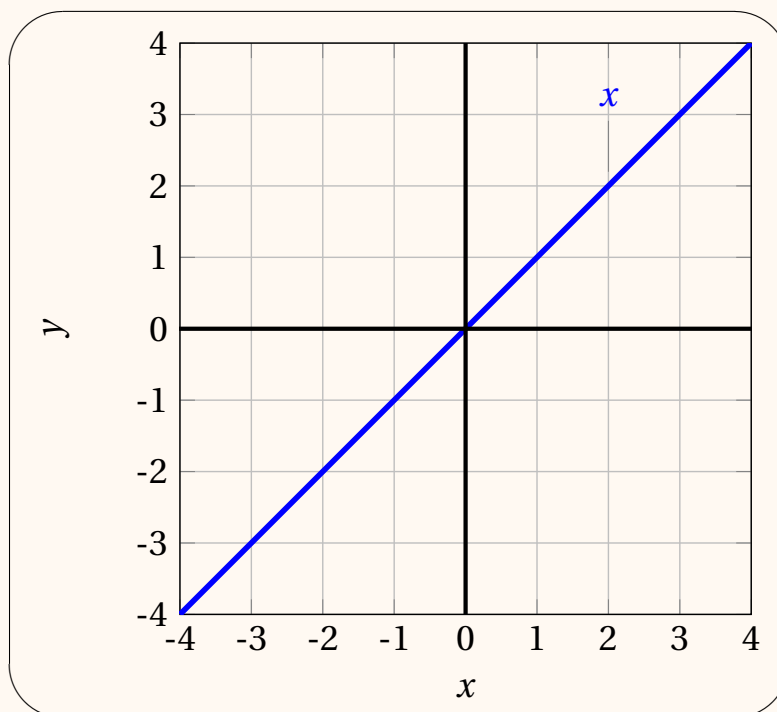
Absolute value of $f(x)$:

$$|f(x)| = \begin{cases} f(x) & \text{if } f(x) \geq 0, \\ -f(x) & \text{if } f(x) < 0 \end{cases}$$

- When passing from the graph of $y = f(x)$ to that of $y_5 = |f(x)|$, the points with **non-negative ordinate remain unchanged**, while those with **negative ordinate are transformed into their symmetric points with respect to the x -axis**.
- The graph of $y_5 = |f(x)|$ is obtained from that of f by “flipping” symmetrically with respect to the x -axis the part of the graph of f that lies in the lower half-plane, and leaving the rest unchanged.

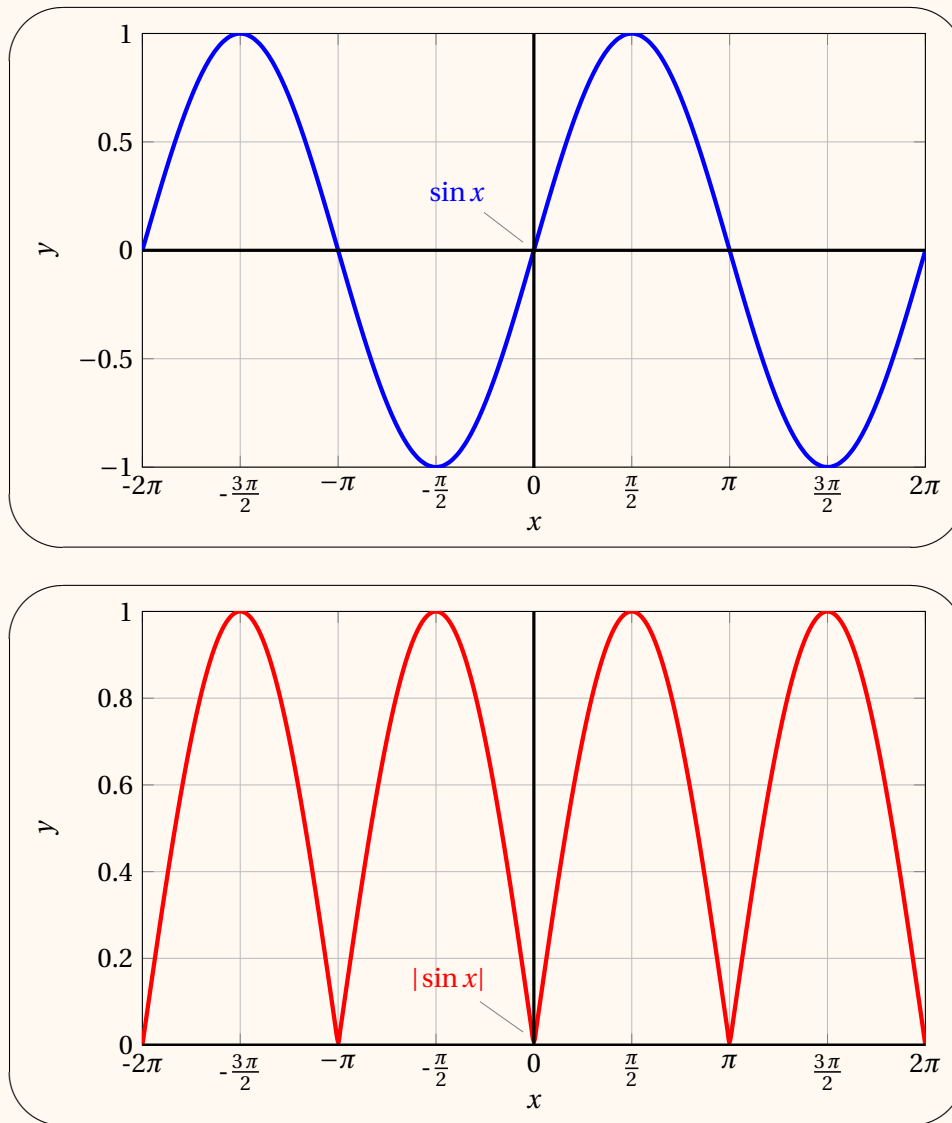
Example 6: Operations on graphs related to $y_5 = |f(x)|$

Consider $y = x$, then $y_5 = |x|$



Example 7: Operations on graphs related to $y_5 = |f(x)|$

Consider $y = \sin x$, then $y_5 = |\sin x|$

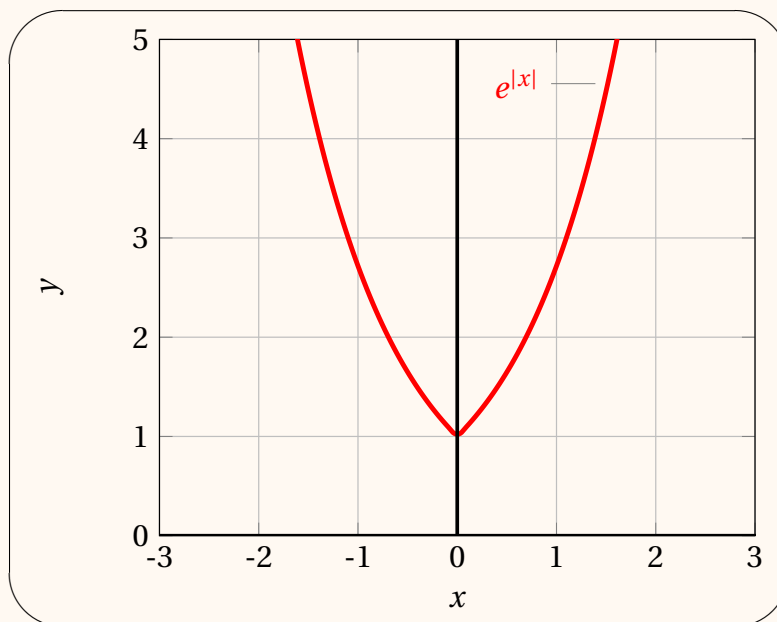
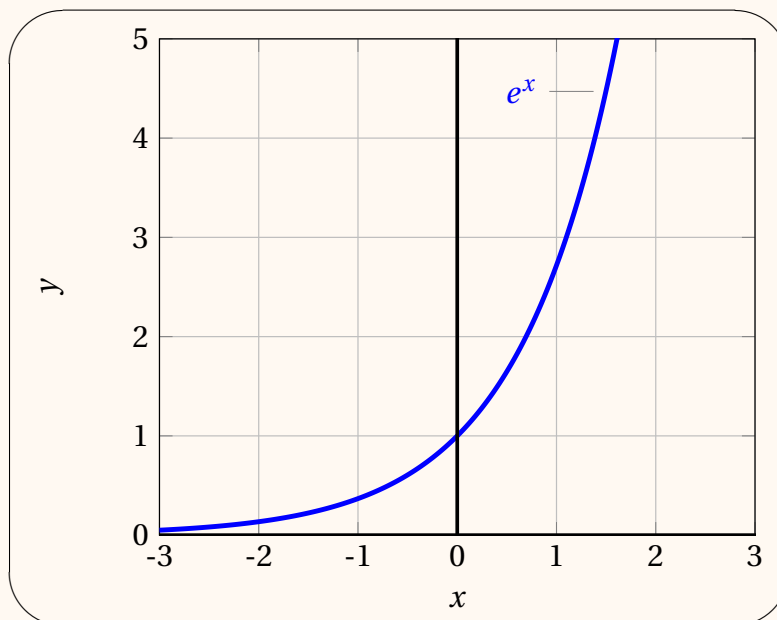


1.6 Operations related to $y_6 = f(|x|)$

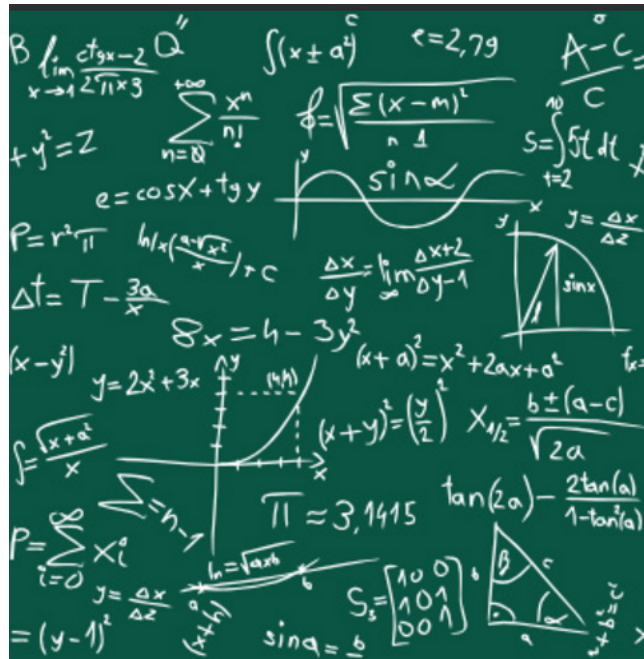
- Finally, to draw the graph of $y_6 = f(|x|)$, we observe that $|x| = x$ for $x \geq 0$, and hence in the right half-plane the two graphs coincide.
- We have $|-x| = |x|$, and hence y_6 is an even function, therefore symmetric with respect to the y -axis.
- Consequently, the graph of $y_6 = f(|x|)$ is drawn by **leaving the graph of f unchanged in the right half-plane and flipping it symmetrically with respect to the y -axis.**

Example 8: Operations on graphs related to $y_6 = f(|x|)$

Consider $y = e^x$, then $y_6 = e^{|x|}$



Composite functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Composite functions

2

1 Composite functions

Definition 1: of composite function

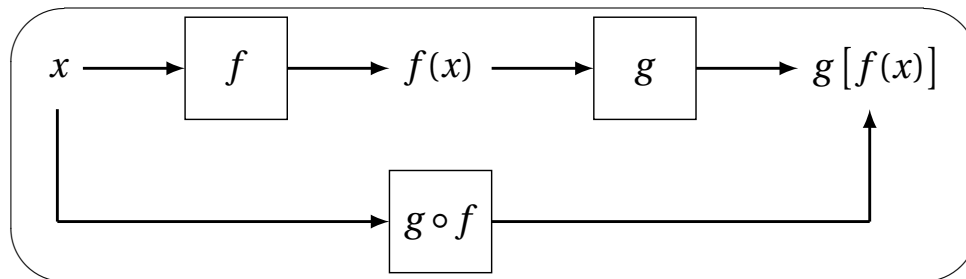
Given two functions:

$$f: E \rightarrow \mathbb{R} \quad \text{and} \quad g: F \rightarrow \mathbb{R},$$

if $f(E) \subseteq F$ (i.e., if for every $x \in E$ we have $f(x) \in F$), one can define the function $h: E \rightarrow \mathbb{R}$, the **composite** of f and g (in this order), denoted by the symbol $g \circ f$, through the formula

$$h(x) = (g \circ f)(x) = g[f(x)]$$

- The composition scheme is the following:

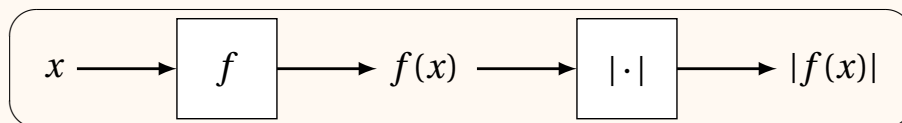


Example 1: of composite function

The function $x \mapsto |f(x)|$ is actually “composed” of two functions:

- given x , we compute $f(x)$
- once $f(x)$ is computed, we compute $|f(x)|$

This amounts to operating in series with two black boxes, the first corresponding to f , the second to its absolute value, according to the following scheme:



- It may happen that both compositions $(g \circ f)$ and $(f \circ g)$ are well defined, but in general

$$(f \circ g) \neq (g \circ f)$$

In other words, **the commutative property does not hold.**

Example 2: of composition of functions

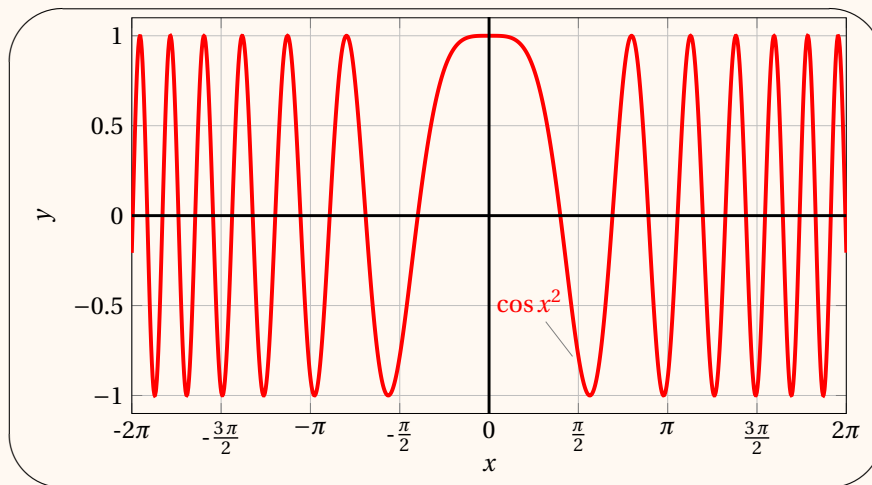
Consider the functions:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto x^2 \quad \text{and} \quad g: \mathbb{R} \rightarrow \mathbb{R}, \quad g: x \mapsto \cos x$$

that is, $f(x) = x^2$ and $g(x) = \cos x$.

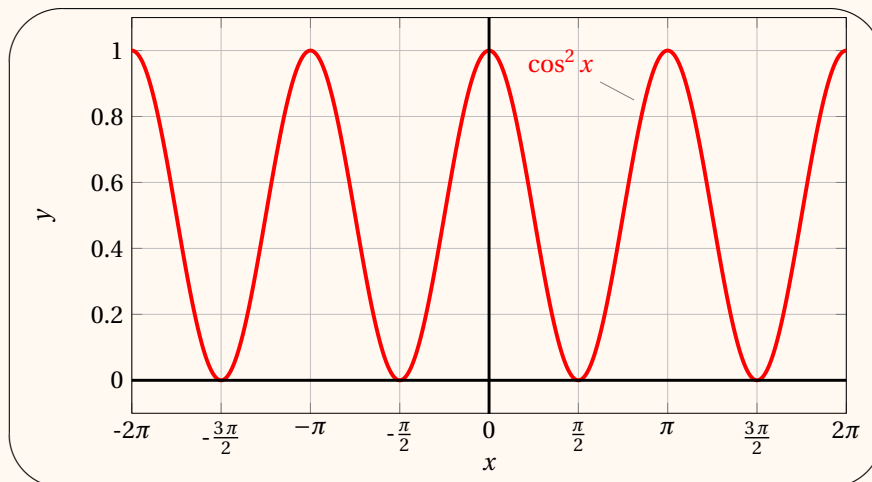
1. Since g is defined on all of $\mathbb{R}$, $h = g \circ f$ is well defined on $\mathbb{R}$ and the following formula holds

$$h(x) = (g \circ f)(x) = g[f(x)] = \cos x^2$$



2. Since f is defined on all of $\mathbb{R}$, $k = f \circ g$ is well defined on $\mathbb{R}$ and the following formula holds

$$k(x) = (f \circ g)(x) = f[g(x)] = \cos^2 x$$



Example 3: of composition of functions

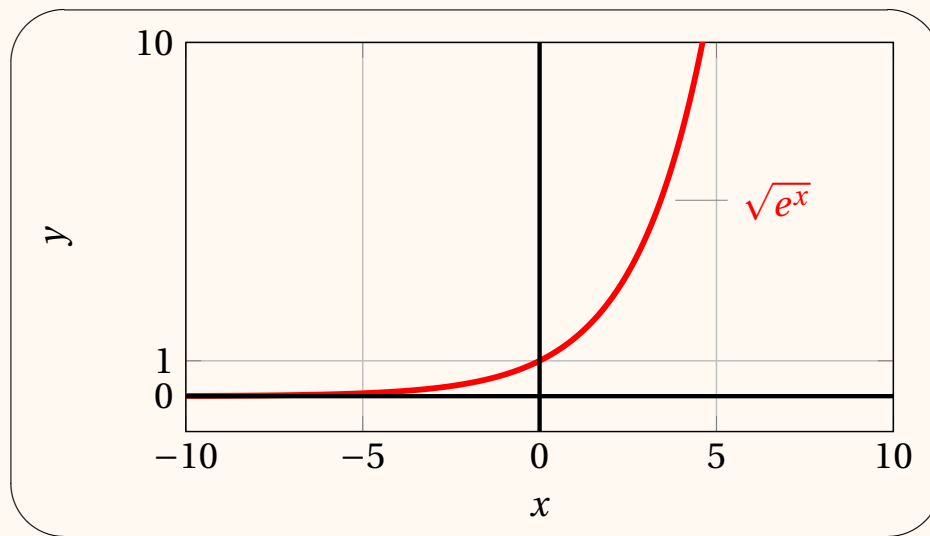
Consider the functions:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto e^x \quad g: [0, +\infty) \rightarrow \mathbb{R}, \quad g: x \mapsto \sqrt{x}$$

that is, $f(x) = e^x$ and $g(x) = \sqrt{x}$.

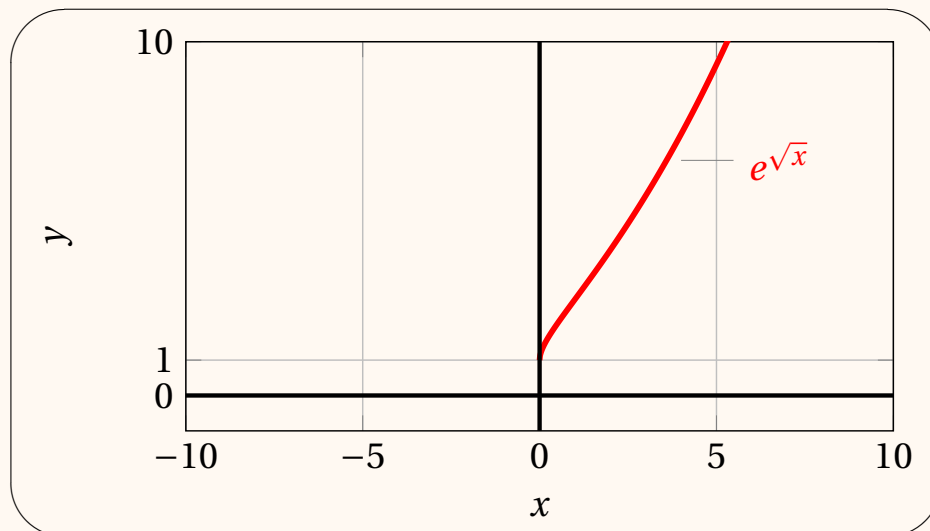
1. Since g is defined on $[0, +\infty)$ and $f(x) > 0$ for every $x \in \mathbb{R}$, $h = g \circ f$ is well defined on $\mathbb{R}$ and the following formula holds

$$h(x) = (g \circ f)(x) = g[f(x)] = \sqrt{e^x}$$



2. Since g is defined only on $[0, +\infty)$, $k = f \circ g$ is well defined only on $\mathbb{R}_+$ and the following formula holds

$$k(x) = (f \circ g)(x) = f[g(x)] = e^{\sqrt{x}}$$



- The composition operation can be extended to three or more factors. One can verify that if the composition $(f \circ g) \circ r$ exists, then $f \circ (g \circ r)$ also exists and they are equal

$$(f \circ g) \circ r = f \circ (g \circ r)$$

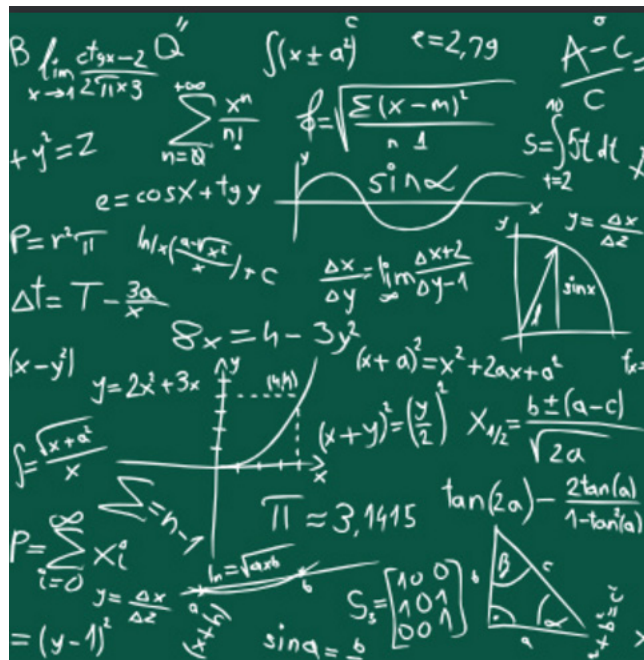
In other words, **the associative property holds**.

- If a function $f : D \rightarrow \mathbb{R}$ is such that $f(D) \subseteq D$, then it can be composed with itself

$$f^2 = f \circ f \text{ that is } f^2(x) = f[f(x)]$$

f^2 is called the **second iterate** of f . Similarly, f^n is called the **n -th iterate function** of f .

Inverse functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1.4	Inverse hyperbolic functions	17

1 Invertible functions and inverse functions

- Given a real function of a real variable $f : D \rightarrow \mathbb{R}$, for every input x in the domain D there exists a unique output $y = f(x)$ in the image of the domain $f(D)$.
- If, in addition, for every output $y = f(x) \in f(D)$ there exists a unique input $x \in D$, then f is called **invertible**, and it establishes a **one-to-one correspondence** between the domain D of f and the image of the domain $f(D)$.

Definition 1: of invertible function

A function $f : D \rightarrow \mathbb{R}$ is **invertible** on the domain D if one of the following equivalent conditions holds:

$$\forall x_1, x_2 \in D, \quad x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$$

$$\forall x_1, x_2 \in D, \quad f(x_1) = f(x_2) \implies x_1 = x_2$$

$$\forall y \in f(D), \quad \exists! x \in D : y = f(x)$$

that is, if f is **injective**.

WARNING: the definition of invertible function given here for real functions of a real variable is different from the usual one given for functions between arbitrary sets, which also requires surjectivity. In this case *injectivity is enough* because we define the inverse on the image of f and not on its whole codomain.

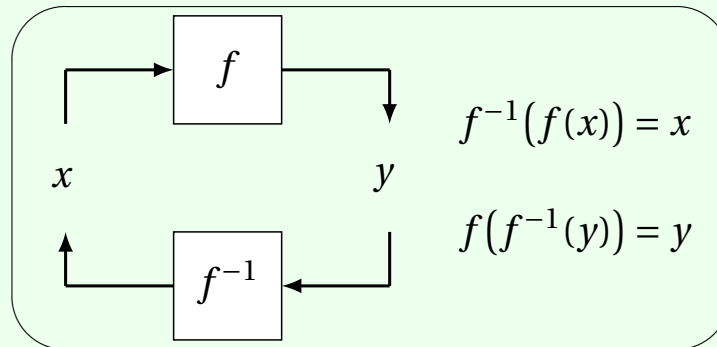
Definition 2: of inverse function

Given an invertible function $f : D \rightarrow \mathbb{R}$, the function that associates to each output $y \in f(D)$ the unique input $x \in D$ such that $f(x) = y$ is called the **inverse function** of f and is denoted by the symbol f^{-1} .

The pair f and f^{-1} is written as:

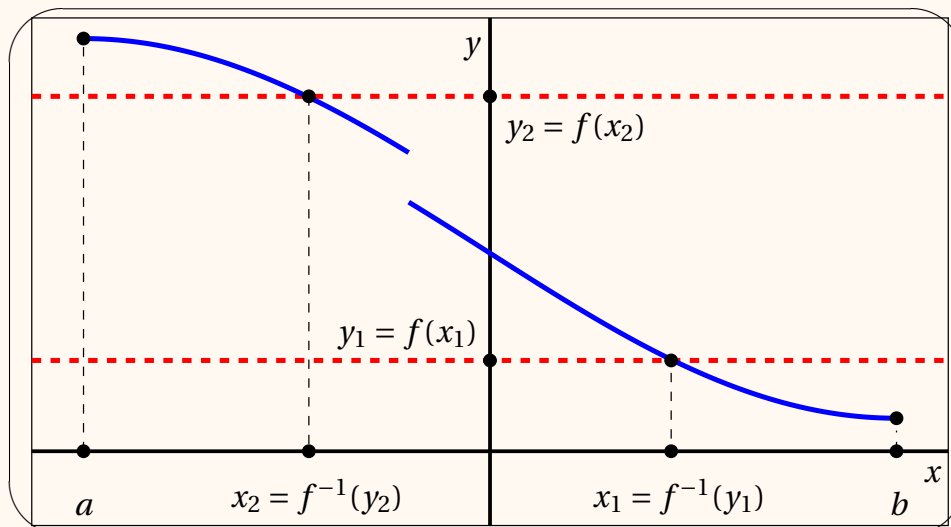
$$f: \begin{cases} y = f(x) \\ x \in D \end{cases} \quad f^{-1}: \begin{cases} x = f^{-1}(y) \\ y \in f(D) \end{cases} \quad (1)$$

The black box of f^{-1} works backwards with respect to that of f , according to the following scheme:



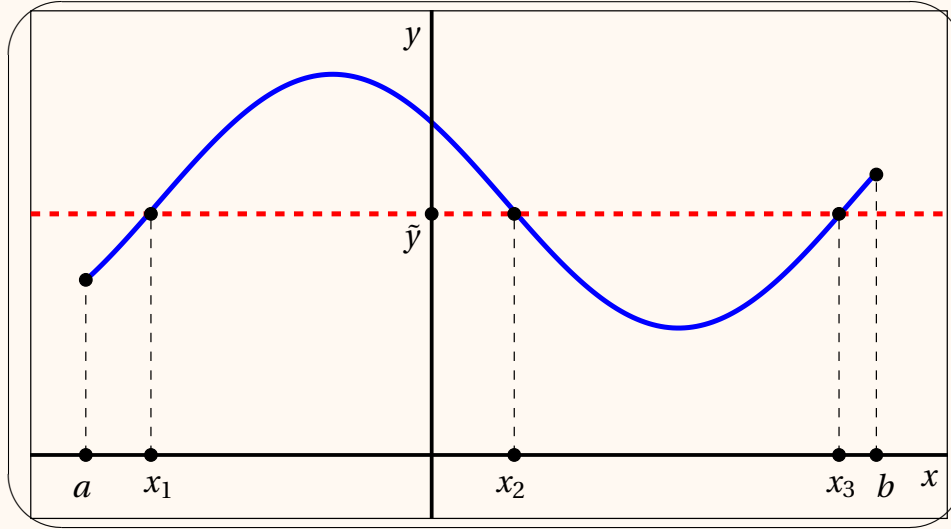
The invertibility condition is equivalent to requiring that the graph of f be intersected in at most one point by every line parallel to the x -axis.

Example 1: graph of an invertible function



Graph of a function that is **invertible** on $[a, b]$, since every line parallel to the x -axis either does not intersect the graph of f or intersects it in exactly one point.

Example 2: graph of a non-invertible function

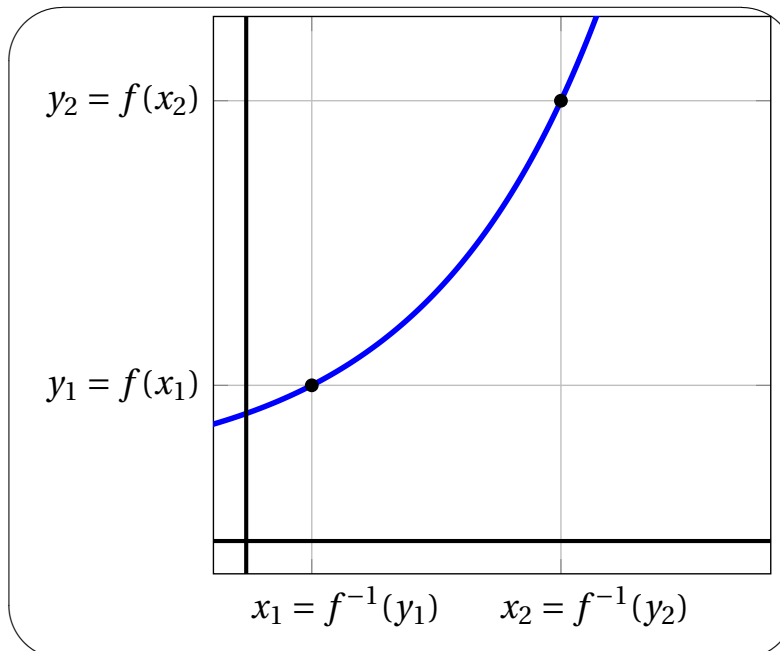


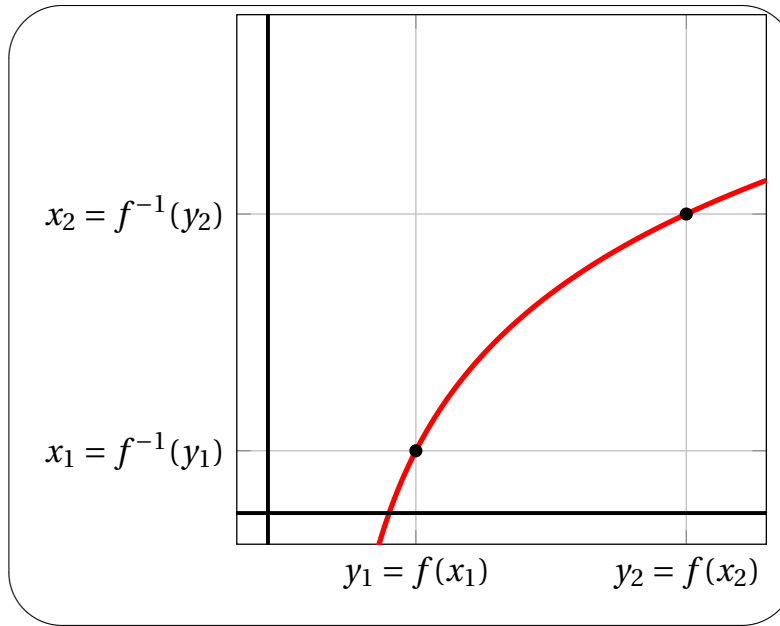
Graph of a function that is **not invertible** on $[a, b]$, since for the indicated value $\tilde{y}$ there are several values x (precisely x_1 , x_2 and x_3) whose image is $\tilde{y}$.

Theorem 1

If a function $f : D \rightarrow \mathbb{R}$ is strictly increasing (decreasing) on D , then it is invertible on D . Moreover, its inverse function is strictly increasing (decreasing).

For the strictly increasing case, graphically we have:





Proof. Consider the case of a function that is strictly increasing on D and two arbitrary values $x_1, x_2 \in D$ (the strictly decreasing case is analogous).

If $x_1 \neq x_2$, then either $x_1 < x_2$ or $x_1 > x_2$. Since f is strictly increasing, we have:

$$\text{if } x_1 < x_2, \text{ then } f(x_1) < f(x_2)$$

$$\text{if } x_1 > x_2, \text{ then } f(x_1) > f(x_2)$$

In both cases $f(x_1) \neq f(x_2)$, therefore f is invertible.

Consider the inverse function f^{-1} , that is, $x = f^{-1}(y)$, and let us prove that it is strictly increasing. Consider two arbitrary values $y_1, y_2 \in f(D)$, with $y_1 < y_2$.

If we had $f^{-1}(y_1) = x_1 \geq x_2 = f^{-1}(y_2)$, since f is strictly increasing we would have $y_1 = f(x_1) \geq f(x_2) = y_2$, that is, $y_1 \geq y_2$. This cannot happen, since it contradicts $y_1 < y_2$ (**contradiction**).

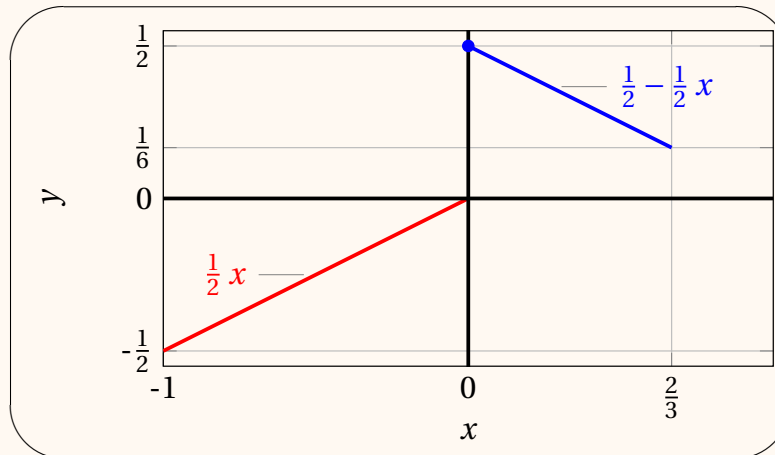
Consequently, we have $f^{-1}(y_1) = x_1 < x_2 = f^{-1}(y_2)$, that is, $f^{-1}(y_1) < f^{-1}(y_2)$, and hence f^{-1} is strictly increasing. □

However, a function can be invertible even without being strictly increasing or decreasing.

Example 3: invertible function that is neither strictly increasing nor strictly decreasing

For example, the following piecewise-defined function is invertible:

$$f(x) = \begin{cases} \frac{1}{2}x & \text{if } -1 \leq x < 0, \\ \frac{1}{2} - \frac{1}{2}x & \text{if } 0 \leq x \leq \frac{2}{3} \end{cases}$$



- Two classes of functions that are certainly not invertible are:

- even (symmetric) functions, since:

$$f(-x) = f(x)$$

- periodic functions, since:

$$f(x + T) = f(x)$$

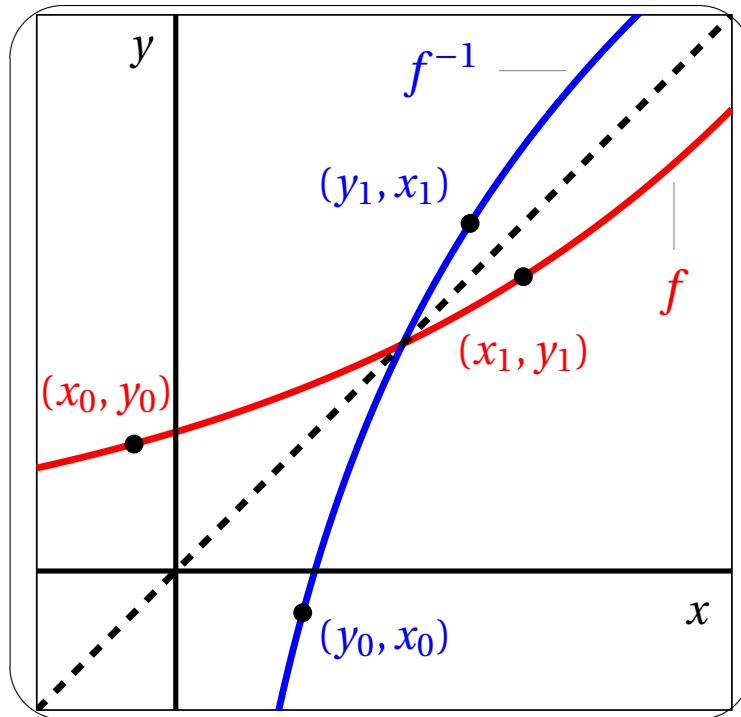
1.1 Graph of the inverse function

- The relations between an invertible function f and its inverse function f^{-1} :

$$f: \begin{cases} y = f(x) \\ x \in D \end{cases} \quad f^{-1}: \begin{cases} x = f^{-1}(y) \\ y \in f(D) \end{cases}$$

indicate that if the point (x_0, y_0) lies on the graph of f , then the point (y_0, x_0) lies on the graph of f^{-1} .

- Since the points (x_0, y_0) and (y_0, x_0) are **symmetric with respect to the bisector** with equation $y = x$, the graph of f^{-1} is obtained from that of f by symmetry with respect to the bisector.



If the analytic expression of f is known and f is invertible, the **analytic expression** of f^{-1} is found by trying to solve for x the equation:

$f(x) = y$ that is, find the input x that produces the output y

Note that it is not always possible to find it, even in cases where the inverse function exists!

Given $a, b \in \mathbb{R}, a \neq 0$, we have:

$$f: \begin{cases} y = ax + b \\ x \in \mathbb{R} \end{cases} \quad f^{-1}: \begin{cases} x = \frac{y-b}{a} \\ y \in \mathbb{R} \end{cases}$$

Example 4: inverse function of an affine function

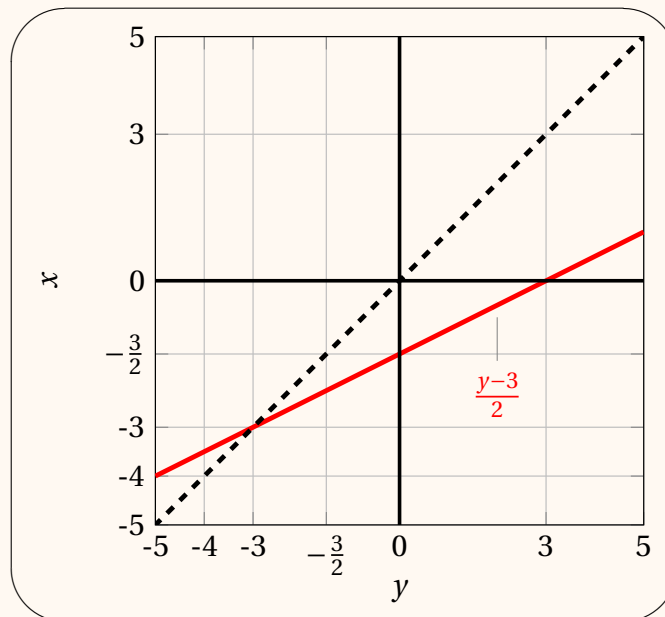
The function $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto 2x + 3$ is strictly increasing, hence invertible on $\mathbb{R}$, and the equation

$$2x + 3 = y \text{ solved for } x, \text{ that is } x = \frac{y-3}{2}$$

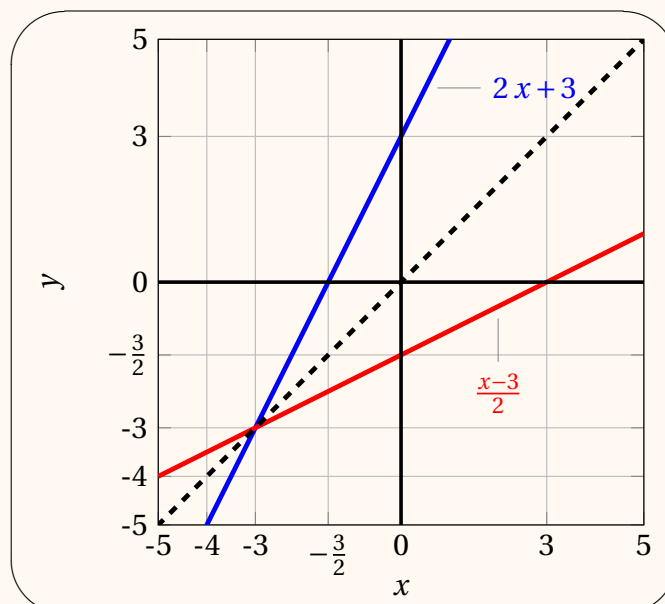
gives for f^{-1} the analytic expression:

$$f^{-1}(y) = \frac{y-3}{2}$$

that is, the function: $f^{-1} : \mathbb{R} \rightarrow \mathbb{R}, y \mapsto \frac{y-3}{2}$.



The graphs $y = 2x + 3$ and $y = \frac{x-3}{2}$ are symmetric with respect to the bisector $y = x$



We have:

$$f: \begin{cases} y = x^2 \\ x \geq 0 \end{cases} \quad f^{-1}: \begin{cases} x = \sqrt{y} \\ y \geq 0 \end{cases}$$

Example 5: inverse function of $f(x) = x^2$

The function $f: \mathbb{R} \rightarrow \mathbb{R}, x \mapsto x^2$ is strictly increasing on the interval $[0, +\infty)$, hence invertible there; the equation

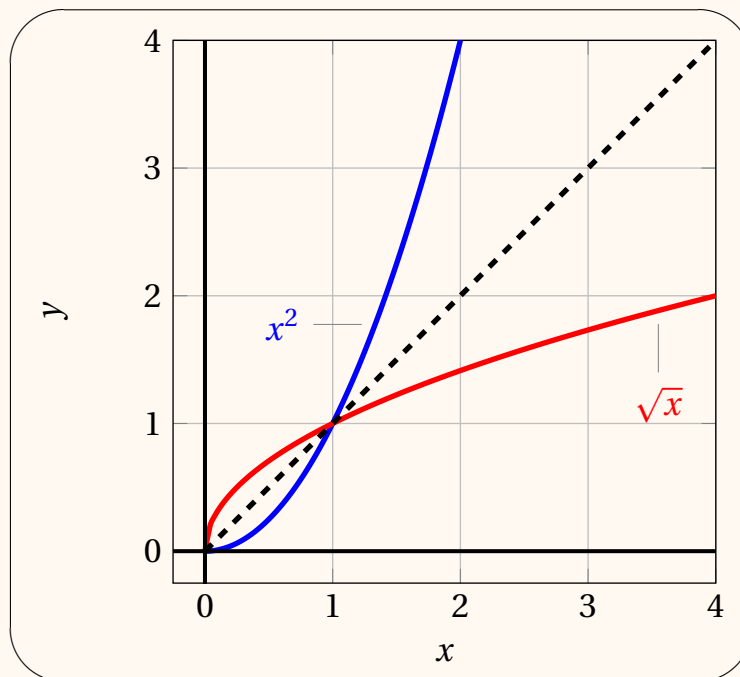
$$x^2 = y \text{ solved for } x, \text{ that is } x = \sqrt{y}$$

gives for f^{-1} the analytic expression

$$f^{-1}(y) = \sqrt{y}$$

that is, the function: $f^{-1}: [0, +\infty) \rightarrow \mathbb{R}, y \mapsto \sqrt{y}$.

The graphs $y = x^2$ and $y = \sqrt{x}$ are symmetric with respect to the bisector $y = x$



Given $a \in \mathbb{R}_+, a \neq 1$, we have:

$$f: \begin{cases} y = a^x \\ x \in \mathbb{R} \end{cases} \quad f^{-1}: \begin{cases} x = \log_a y \\ y > 0 \end{cases}$$

Example 6: inverse function of $f(x) = e^x$

The function $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto e^x$ is strictly increasing on all of $\mathbb{R}$ and hence invertible; the equation

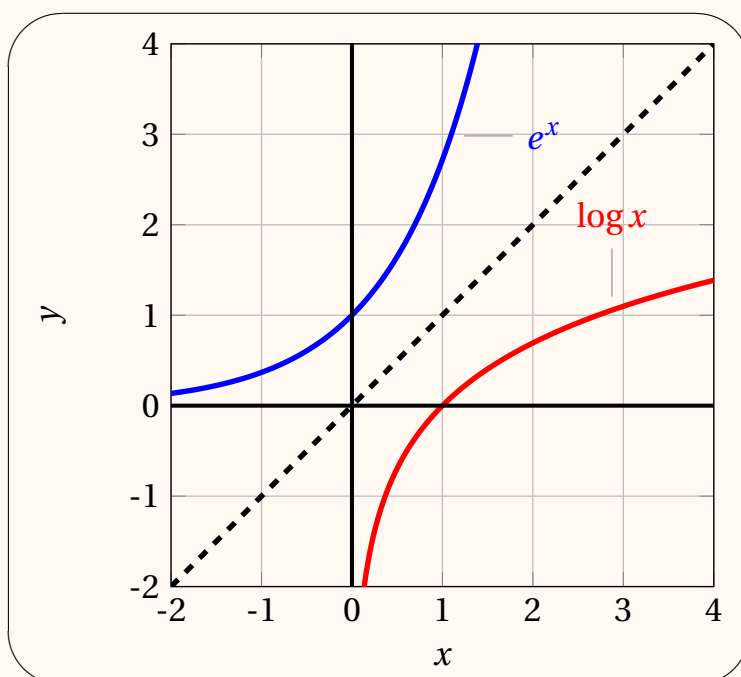
$$e^x = y \text{ solved for } x, \text{ that is } x = \log y$$

gives for f^{-1} the analytic expression

$$f^{-1}(y) = \log y$$

that is, the function: $f^{-1} : (0, +\infty) \rightarrow \mathbb{R}, y \mapsto \log y$.

The graphs $y = e^x$ and $y = \log x$ are symmetric with respect to the bisector $y = x$



- Now consider the function:

$$f(x) = x + e^x$$

Being the sum of two functions that are strictly increasing on all of $\mathbb{R}$, $f(x)$ is strictly increasing and hence invertible on all of $\mathbb{R}$ (see Theorem (??)). However, we would try in vain to solve the equation $x + e^x = y$ for x . In other words, f^{-1} exists, but we cannot write it explicitly.

1.2 Inverse power functions

Given $\alpha \in \mathbb{R}, \alpha \neq 0$, we have:

$$f: \begin{cases} y = x^\alpha \\ x > 0 \end{cases} \quad f^{-1}: \begin{cases} x = y^{\frac{1}{\alpha}} \\ y > 0 \end{cases}$$

If $\alpha > 0$ we have:

$$f: \begin{cases} y = x^\alpha \\ x \geq 0 \end{cases} \quad f^{-1}: \begin{cases} x = y^{\frac{1}{\alpha}} \\ y \geq 0 \end{cases}$$

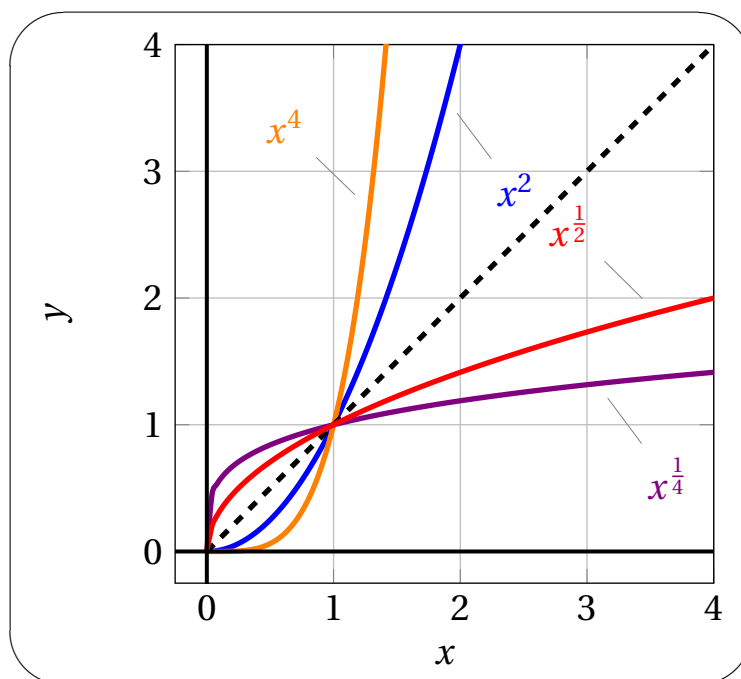
If $\alpha = \frac{m}{n} \in \mathbb{Q}$, with $m \in \mathbb{Z}, n \in \mathbb{N}_+$ odd and coprime, we have:

$$f: \begin{cases} y = x^{\frac{m}{n}} \\ x \in \mathbb{R} \end{cases} \quad f^{-1}: \begin{cases} x = y^{\frac{n}{m}} \\ y \in \mathbb{R} \end{cases}$$

- The **even powers**:

$$x^{2n} \quad \text{with } n = 1, 2, \dots$$

are not invertible on the whole real line, but only on the half-line $x \geq 0$.



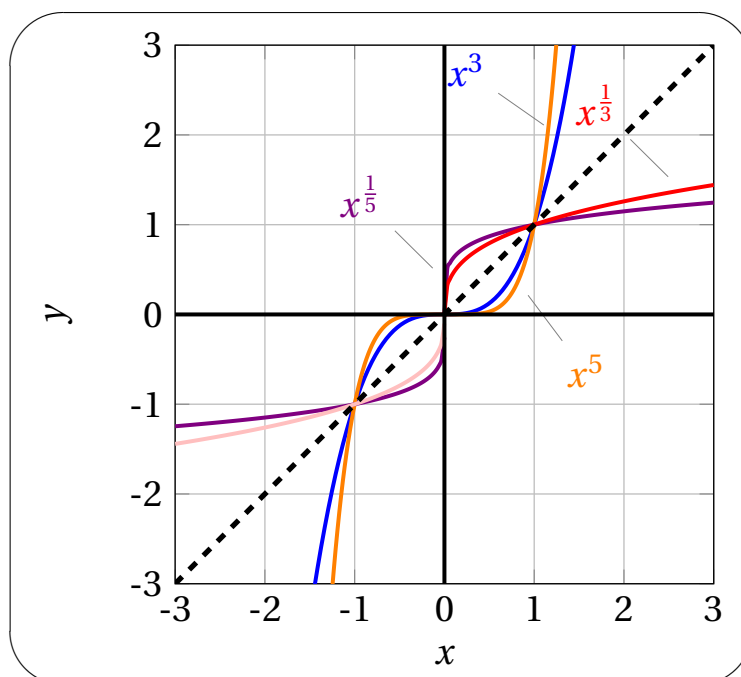
- The **odd powers**:

$$x^{2n+1} \quad \text{with } n = 1, 2, \dots$$

and the **powers with rational exponent**

$$x^{\frac{m}{n}} \quad \text{with } n, m \text{ positive odd integers}$$

being strictly increasing monotone functions, are invertible from $-\infty$ to $+\infty$.



1.3 Inverse trigonometric functions

- Being periodic, the trigonometric functions cannot be invertible. Indeed, for example, the equation

$$y = \sin x$$

has infinitely many solutions if $-1 \leq y \leq 1$ (the output y corresponds to infinitely many inputs), or has no real solutions if $|y| > 1$.

- To speak of inverse functions of sine, cosine and tangent, we will need to restrict ourselves to intervals on which these functions are strictly monotone and therefore invertible.
- An interval on which the sine function is invertible is $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

Definition 3: of arcsine

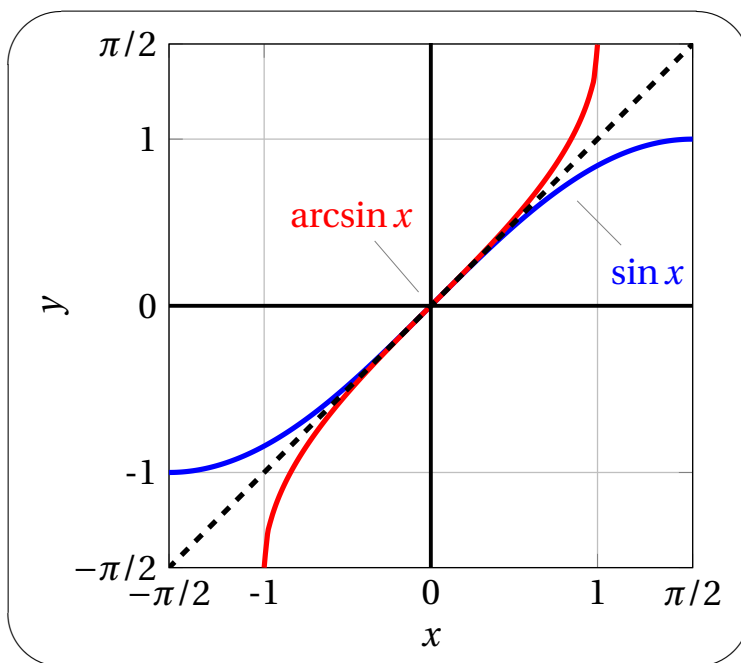
The inverse function of the sine on the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is the **arcsine**:

$$f : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad f : y \mapsto \arcsin y$$

We have:

$$f : \begin{cases} y = \sin x \\ x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \end{cases} \quad f^{-1} : \begin{cases} x = \arcsin y \\ y \in [-1, 1] \end{cases} \quad (2)$$

- The graph of the arcsine is obtained from the arc of the sine curve restricted to the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (strictly monotone), by symmetry with respect to the bisector $y = x$,



- An interval on which the cosine function is invertible is $[0, \pi]$.

Definition 4: of arccosine

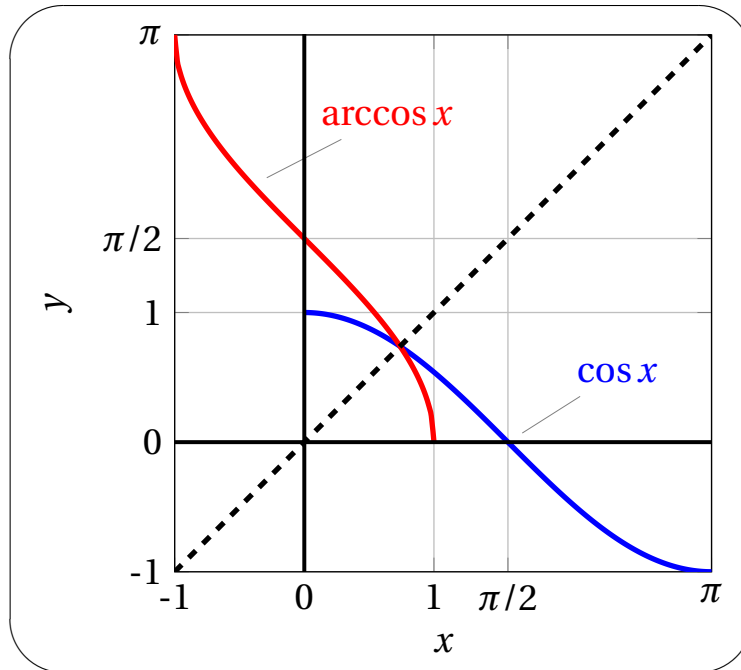
The inverse function of the cosine on the interval $[0, \pi]$ is the **arccosine**:

$$f : [-1, 1] \rightarrow [0, \pi], \quad f : y \mapsto \arccos y$$

We have:

$$f : \begin{cases} y = \cos x \\ x \in [0, \pi] \end{cases} \quad f^{-1} : \begin{cases} x = \arccos y \\ y \in [-1, 1] \end{cases} \quad (3)$$

- The graph of the arccosine is obtained from the arc of the cosine curve restricted to the interval $[0, \pi]$ (strictly monotone), by symmetry with respect to the bisector $y = x$,



- Note that on the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ the tangent is strictly monotone and hence invertible. Its inverse function is called the arctangent (arctan), and it is defined on $\mathbb{R}$.

Definition 5: of arctangent

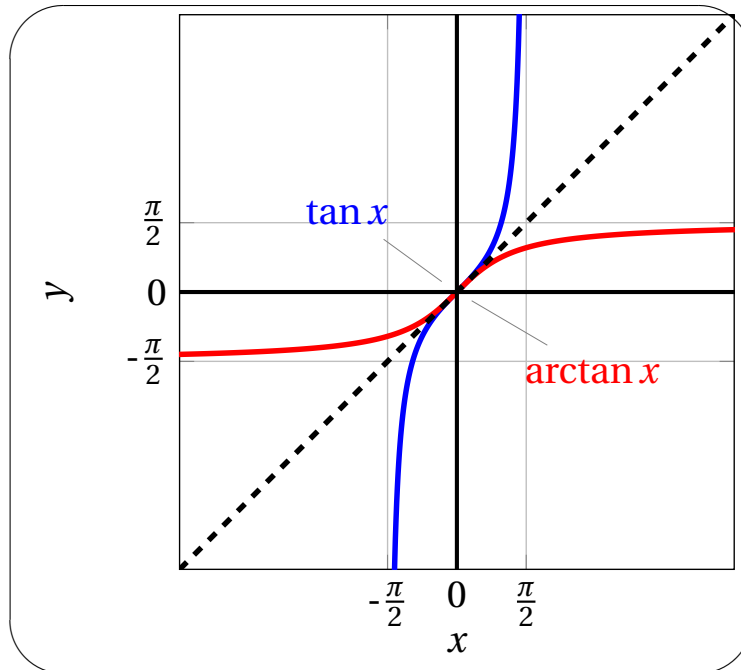
The inverse function of the tangent on the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ is the **arctangent**:

$$f: \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad f: y \mapsto \arctan y$$

We have:

$$f: \begin{cases} y = \tan x \\ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \end{cases} \quad f^{-1}: \begin{cases} x = \arctan y \\ y \in \mathbb{R} \end{cases} \quad (4)$$

- The graph of the arctangent is obtained from that of the tangent restricted to the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ (strictly monotone), by symmetry with respect to the bisector $y = x$,



- By means of the inverse trigonometric functions, we can express the solutions of a **trigonometric equation or inequality** when it involves angles that are not standard angles.

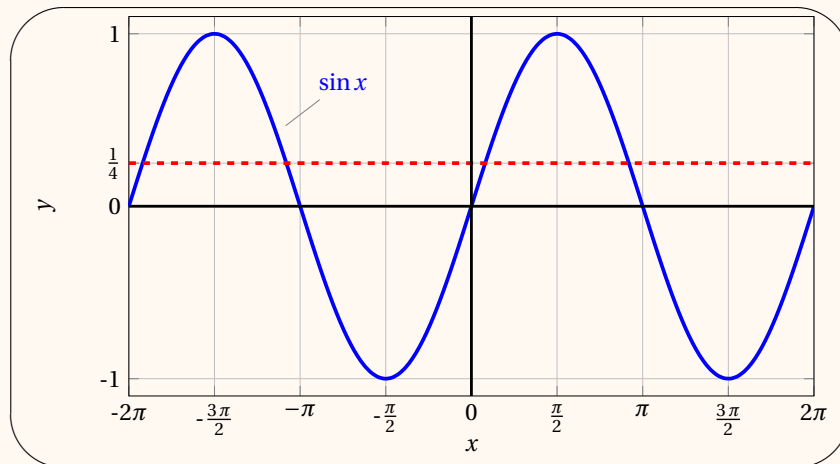
Example 7: Trigonometric equations/inequalities

- The solutions of:

$$\sin x = \frac{1}{4}$$

are:

$$(i) \quad x = \arcsin \frac{1}{4} + 2k\pi; \quad (ii) \quad x = \pi - \arcsin \frac{1}{4} + 2k\pi \quad (k \in \mathbb{Z})$$

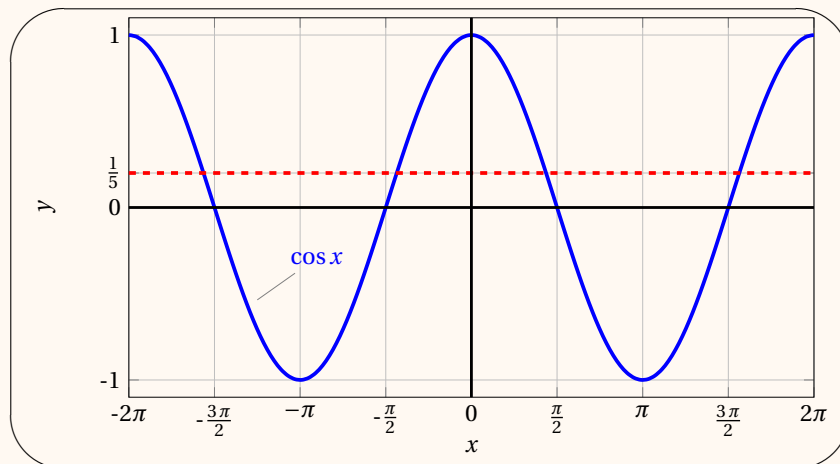


- The solutions of:

$$\cos x < \frac{1}{5}$$

are:

$$\arccos \frac{1}{5} + 2k\pi < x < 2\pi - \arccos \frac{1}{5} + 2k\pi \quad (k \in \mathbb{Z})$$



- The solutions of:

$$\tan x \geq 3$$

are:

$$\arctan 3 + k\pi \leq x < \frac{\pi}{2} + k\pi \quad (k \in \mathbb{Z})$$

1.4 Inverse hyperbolic functions

- Consider the **hyperbolic sine** function:

$$y = \text{Sh } x = \frac{e^x - e^{-x}}{2}$$

It is defined and strictly increasing on all of $\mathbb{R}$, therefore it is invertible.

- To solve the equation for x , multiplying both sides by e^x we obtain:

$$\begin{aligned} 0 &= e^x y - e^x \frac{e^x - e^{-x}}{2} \\ &= 2 e^x y - (e^x e^x - \underbrace{e^x e^{-x}}_{=e^{x-x}=e^0}) \\ &= 2 y e^x - e^{2x} + 1 \\ &= e^{2x} - 2 y e^x - 1 \end{aligned}$$

which is a quadratic equation in the unknown e^x . Setting $t = e^x$ we obtain

$$0 = t^2 - 2 y t - 1$$

We obtain:

$$t = y \pm \sqrt{y^2 + 1} \quad \text{and substituting back} \quad e^x = y \pm \sqrt{y^2 + 1}$$

since $e^x > 0$, the solution with the minus sign must be discarded. Hence we are left with:

$$e^x = \underbrace{y + \sqrt{y^2 + 1}}_{>0, \forall y \in \mathbb{R}} \quad \text{and hence} \quad x = \log \left(y + \sqrt{y^2 + 1} \right)$$

- This is the analytic expression of the inverse function of $\text{Sh } x$, which is called the **inverse hyperbolic sine** (area hyperbolic sine), and is also denoted by $\text{SettSh } y$. It is defined for every real y .

Definition 6: of inverse hyperbolic sine

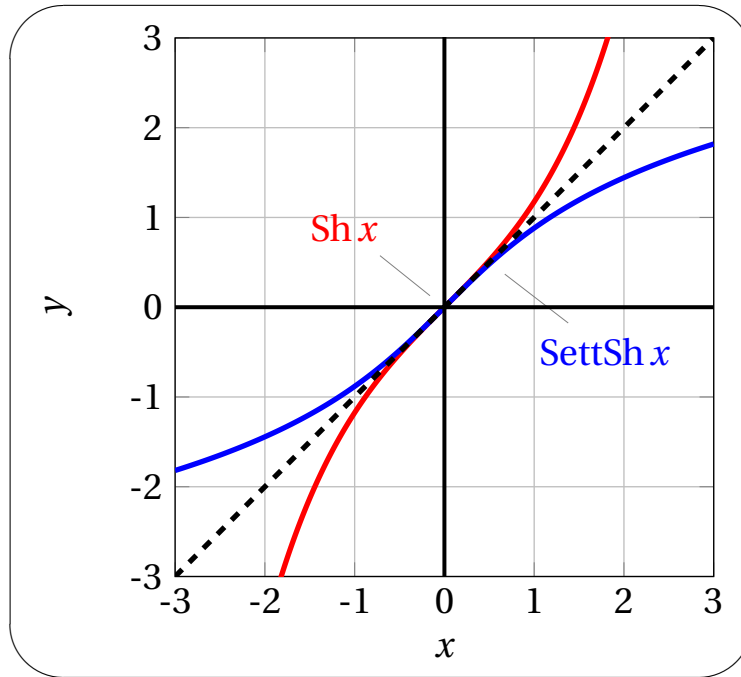
The inverse function of the hyperbolic sine is the **inverse hyperbolic sine**:

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : y \mapsto \text{SettSh } y$$

We have:

$$f : \begin{cases} y = \text{Sh } x \\ x \in \mathbb{R} \end{cases} \quad f^{-1} : \begin{cases} x = \text{SettSh } y \\ y \in \mathbb{R} \end{cases} \quad (5)$$

- The graph of the inverse hyperbolic sine is obtained from that of the hyperbolic sine on $\mathbb{R}$ (strictly monotone), by symmetry with respect to the bisector $y = x$



- Consider the **hyperbolic cosine** function:

$$y = \text{Ch } x = \frac{e^x + e^{-x}}{2}$$

It is defined on all of $\mathbb{R}$, strictly increasing for $x \geq 0$, decreasing for $x \leq 0$. Therefore it is not invertible on all of $\mathbb{R}$.

- Its restriction to $x \geq 0$, however, is invertible. We want to determine the analytic expression of the inverse function of this restriction.
- Proceeding as before, we obtain:

$$e^x = y \pm \sqrt{y^2 - 1}$$

- This time both numbers $y \pm \sqrt{y^2 - 1}$ are positive; recall, however, that we are reasoning only for $x \geq 0$, which is equivalent to choosing the plus sign. Therefore:

$$x = \log\left(y + \sqrt{y^2 - 1}\right)$$

This is the analytic expression of the inverse function of $\text{Ch } x$, which is called the inverse hyperbolic cosine (area hyperbolic cosine), and is also denoted by SettCh . Note that it is defined for $y \geq 1$.

Definition 7: of inverse hyperbolic cosine

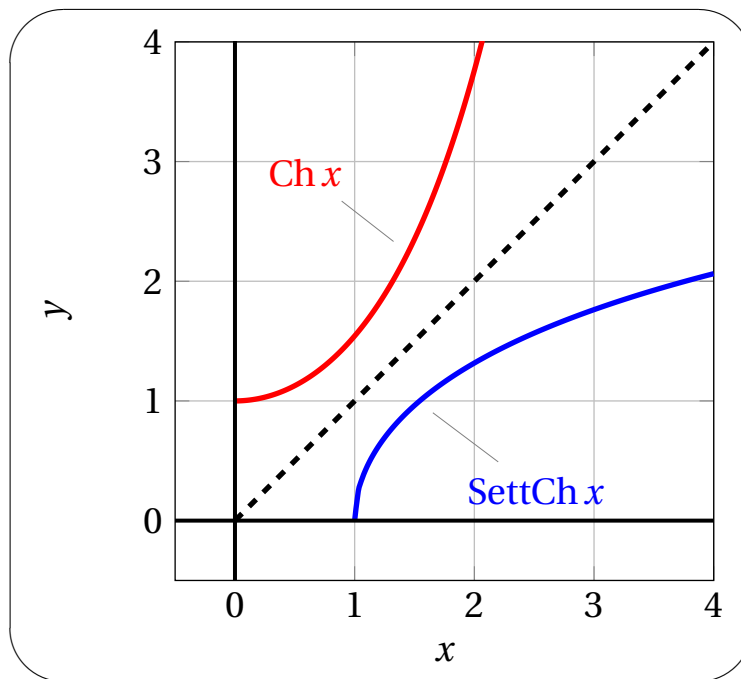
The inverse function of the hyperbolic cosine on the interval $[0, +\infty)$ is the **inverse hyperbolic cosine**:

$$f: [1, +\infty) \rightarrow [0, +\infty), \quad f: y \mapsto \text{SettCh } y$$

We have:

$$f: \begin{cases} y = \text{Ch } x \\ x \in [0, +\infty) \end{cases} \quad f^{-1}: \begin{cases} x = \text{SettCh } y \\ y \in [1, +\infty) \end{cases}$$

- The graph of the inverse hyperbolic cosine is obtained from that of the hyperbolic cosine on $[0, +\infty)$ (strictly monotone), by symmetry with respect to the bisector $y = x$.



Example 8: Hyperbolic equations

- The equation:

$$\operatorname{Sh} x = 2$$

has the unique solution:

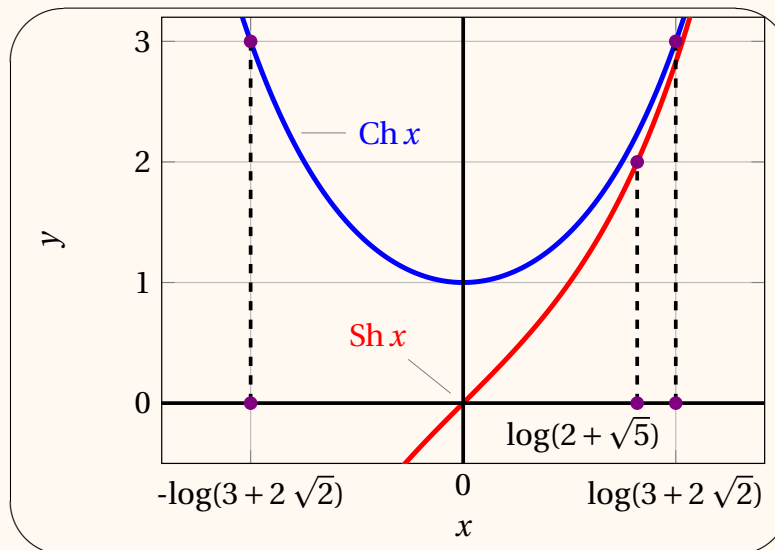
$$x = \operatorname{SettSh} 2 = \log(2 + \sqrt{5})$$

- The equation:

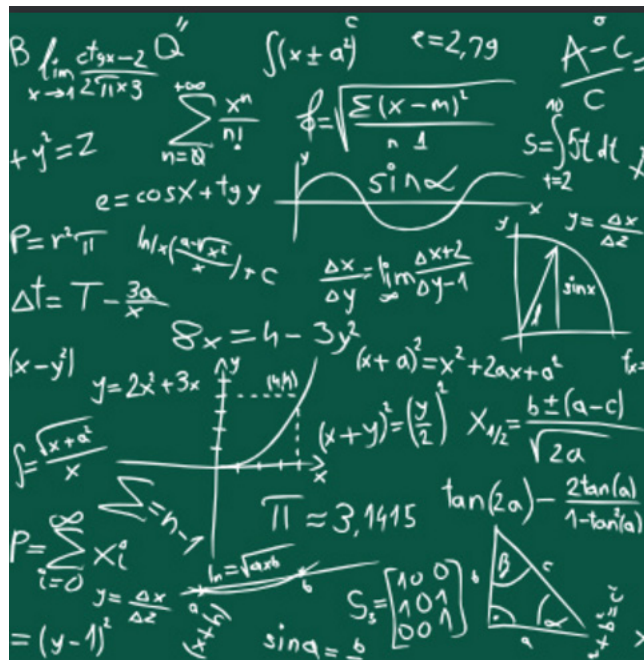
$$\operatorname{Ch} x = 3$$

has two solutions:

$$x = \pm \operatorname{SettCh} 3 = \pm \log(3 + 2\sqrt{2})$$



Sequences and limits of sequences



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagni and Sandro Salsa, Zanichelli.

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1 Definition of sequence and properties

- Consider the set $\mathbb{N}$ of non-negative integers, ordered according to the natural order

$$\mathbb{N}: 0, 1, 2, 3, \dots, n, \dots$$

Definition 1: Sequence

A **sequence** is a relation that associates with each natural number $n \in \mathbb{N}$ (or from a certain natural number n_0 onward) a real number $a_n \in \mathbb{R}$.

- A sequence is therefore a **function**:

$$f: \mathbb{N} \rightarrow \mathbb{R}$$

$$f: n \mapsto a_n$$

or possibly

$$f: \{n \in \mathbb{N} : n \geq n_0\} \rightarrow \mathbb{R}$$

for some fixed integer $n_0 \in \mathbb{N}$.

The sequence associates with the **input variable** n the **output value** a_n .

- The fact that the domain of the function f is the set of natural numbers makes it possible to write the sequence by **listing its values**, in the order in which they follow one another as n increases:

$$a_0, a_1, a_2, \dots, a_n, \dots$$

- The dots after a_n indicate that we are not considering only the first n terms of the sequence (i.e., a **finite set** of numbers), but the whole sequence of *infinitely many terms* (i.e., an **infinite set** of numbers).

Example 1: Sequences

$$n \in \mathbb{N}$$

$$n \mapsto n^2$$

$$0, 1, 4, 9, 16, \dots$$

$$n \in \mathbb{N}$$

$$n \mapsto (-1)^n$$

$$1, -1, 1, -1, 1, \dots$$

$$n \in \mathbb{N}$$

$$n \mapsto a \in \mathbb{R}$$

$$a, a, a, a, a, \dots$$

The last sequence is called a **constant sequence**.

Example 2: Sequences

$$n \in \mathbb{N}, n \geq 1$$

$$n \mapsto \frac{1}{n}$$

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots$$

$$n \in \mathbb{N}, n \geq 2$$

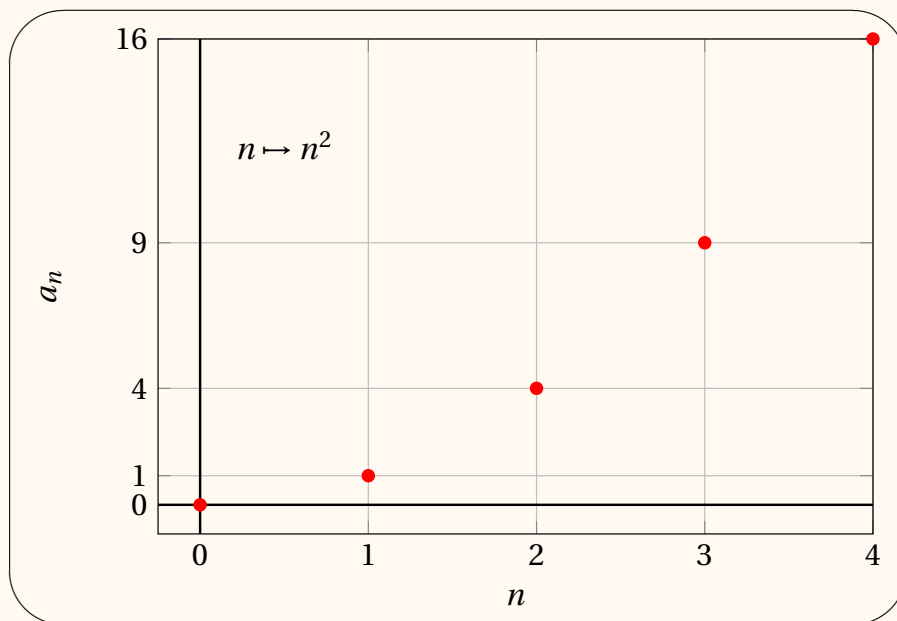
$$n \mapsto \frac{n+1}{n-1}$$

$$3, 2, \frac{5}{3}, \frac{6}{4}, \frac{7}{5}, \dots$$

We can represent sequences graphically by the **points** of the Cartesian plane with coordinates (n, a_n) .

Example 3: Graph of a sequence

The graph of the sequence $n \mapsto n^2$ with $n \in \{0, 1, 2, 3, 4\}$ is:



To denote a sequence we will use the notation:

$$\{a_n\} \quad \text{or} \quad n \mapsto a_n$$

possibly specifying the set in which the input variable n varies (the whole set $\mathbb{N}$ or from a certain value $n_0 \in \mathbb{N}$ onward).

A sequence $\{a_n\}$ is **bounded** if there exist two numbers $m \in \mathbb{R}$ and $M \in \mathbb{R}$ such that:

$$m \leq a_n \leq M, \quad \forall n \in \mathbb{N}$$

It is **bounded below** if m exists. It is **bounded above** if M exists.

Example 4: Bounded sequences

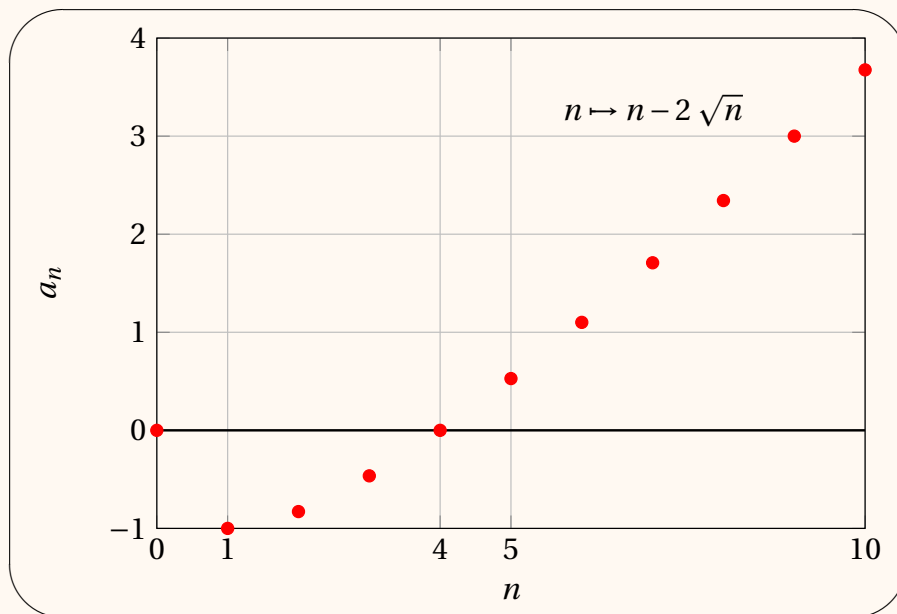
- the sequence $\{(-1)^n\}$ is bounded
- the sequence $\{n^2\}$ is only bounded below
- the sequence $\{(-2)^n\}$ is not bounded (neither below nor above).

Definition 2: Property that holds eventually

We say that a sequence $\{a_n\}$ has (or acquires) a certain property **eventually** if there exists $\tilde{n} \in \mathbb{N}$ such that a_n satisfies that property for every $n \geq \tilde{n}$.

Example 5: Properties that hold eventually

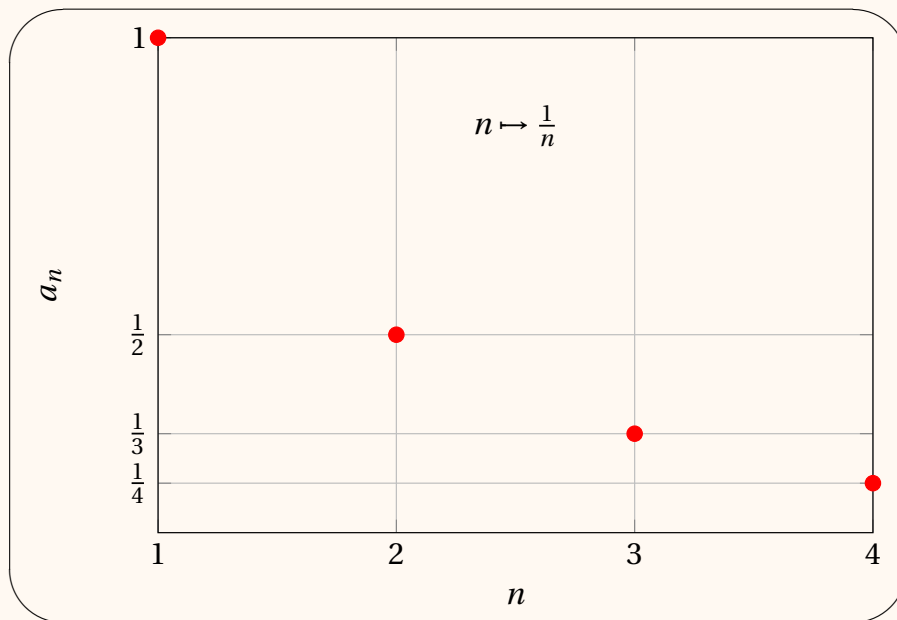
Consider the sequence $\{n - 2\sqrt{n}\}$. The graph of the sequence with $n \in \{0, 1, 2, \dots, 10\}$ is:



This sequence is eventually positive. With $n = 4$ we have $a_n = 0$, hence taking $\tilde{n} = 5$, we have $a_n > 0$ for $n \geq \tilde{n}$.

Example 6: Properties that hold eventually

Now consider the sequence $\{\frac{1}{n}\}$. The graph of the sequence with $n \in \{1, 2, \dots, 4\}$ is:



This sequence is eventually less than 10^{-100} . With $n = 10^{100}$ we have $a_n = 10^{-100}$, hence taking for example $\tilde{n} = 10^{100} + 1$ we have $a_n < 10^{-100}$ for $n \geq \tilde{n}$.

1.1 Convergent sequences and definition of the limit of a sequence

Definition 3: Convergent sequence

A sequence $\{a_n\}$ is called **convergent** if there exists a number $\ell \in \mathbb{R}$ such that:

$$|a_n - \ell| < \varepsilon, \quad \text{eventually} \quad (1)$$

for every $\varepsilon > 0$.

A sequence $\{a_n\}$ is therefore called **convergent** if for every $\varepsilon > 0$ (as small as we like) there exists a number $n(\varepsilon) \in \mathbb{N}$ such that:

$$|a_n - \ell| < \varepsilon \quad \text{for every } n \geq n(\varepsilon)$$

The number $n(\varepsilon)$ depends (in general) on the value of ε . If the sequence $\{a_n\}$ is convergent, then the number $\ell \in \mathbb{R}$ is associated with it.

Definition 4: Limit of a sequence

The number $\ell \in \mathbb{R}$ appearing in inequality (1) is called the **limit of the sequence** $\{a_n\}$, and we write:

$$\lim_{n \rightarrow +\infty} a_n = \ell \quad \text{or, equivalently} \quad a_n \rightarrow \ell \quad \text{for } n \rightarrow +\infty$$

- read, respectively:

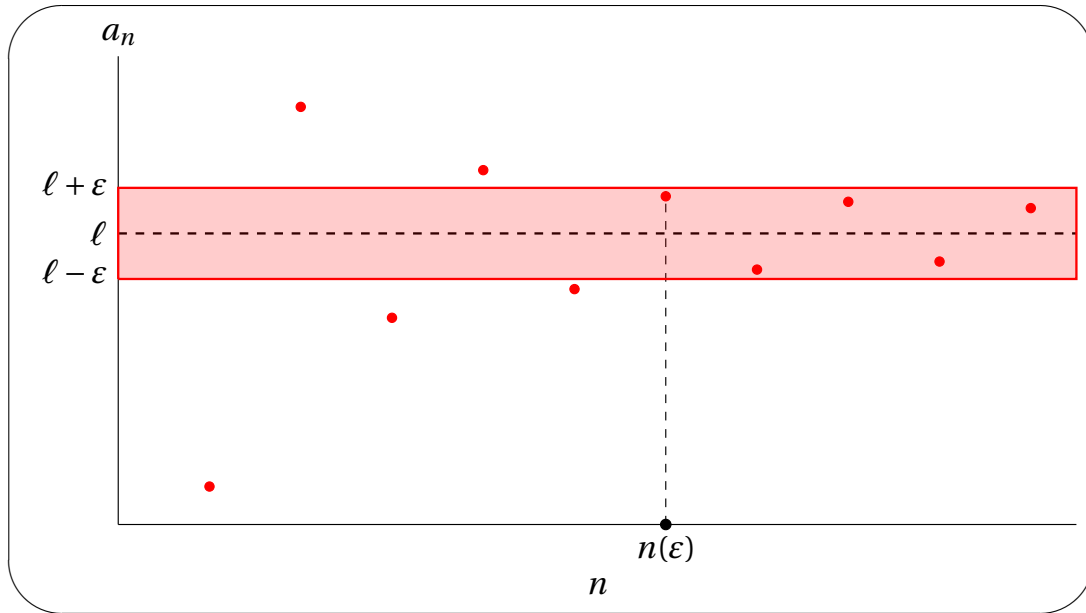
the limit of a_n , as n tends to infinity, is ℓ

a_n tends to ℓ as n tends to infinity

Inequality (1) corresponds to the following two:

$$\ell - \varepsilon < a_n < \ell + \varepsilon \quad (2)$$

- Representing the points of a sequence graphically, we have:



The convergence condition means that, having fixed a horizontal strip “as narrow as we like”:

$$[\ell - \varepsilon, \ell + \varepsilon]$$

from a certain value of n onward, called $n(\varepsilon)$, the points a_n of the sequence no longer leave this strip. In the previous graph, having fixed the width of the strip, the values a_n lie inside the strip for $n \geq n(\varepsilon)$.

Theorem 1: Uniqueness of the limit of a sequence

If a sequence $\{a_n\}$ converges to the limit $\ell \in \mathbb{R}$, then this limit is unique.

Proof. Suppose, by contradiction, that there exist two different limits, ℓ_1 and ℓ_2 , associated with the same sequence $\{a_n\}$. Then, eventually, for every $\varepsilon > 0$ we would have:

$$|\ell_1 - \ell_2| = |\ell_1 - a_n + a_n - \ell_2| \leq \underbrace{|\ell_1 - a_n|}_{=|a_n - \ell_1| < \varepsilon} + \underbrace{|a_n - \ell_2|}_{< \varepsilon} < 2\varepsilon \quad (3)$$

We used the triangle inequality. Since $\varepsilon > 0$ can be chosen arbitrarily small, (3) can be satisfied if and only if:

$$\ell_1 = \ell_2$$

Hence two different values ℓ_1 and ℓ_2 cannot exist and consequently the limit (if it exists) is unique. $\square$

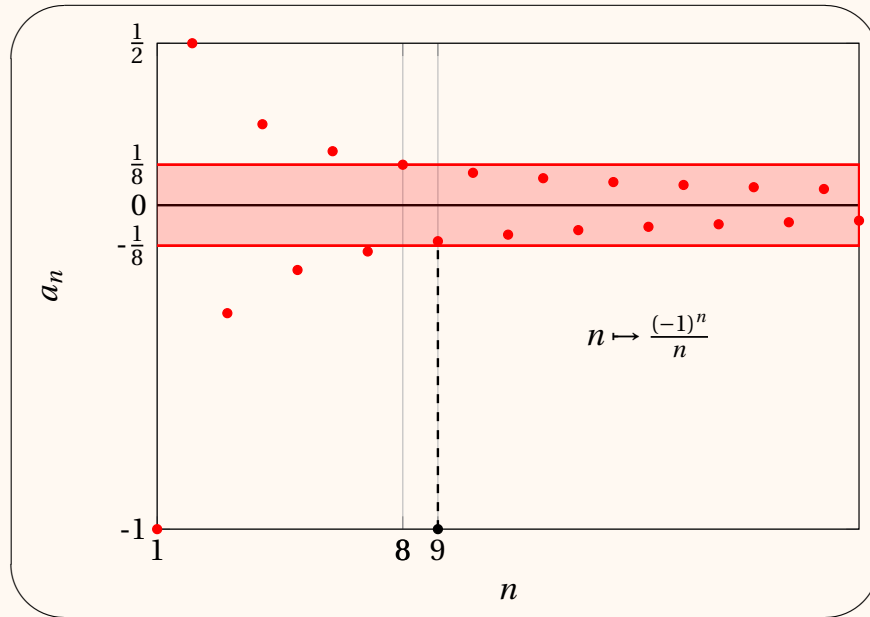
Example 7: Verifying the limit of a sequence

The graph of the sequence $n \mapsto \frac{(-1)^n}{n}$, for example starting from $n = 9$, lies within the horizontal strip:

$$\left[-\frac{1}{8}, \frac{1}{8} \right]$$

given by $\ell = 0$ and $\varepsilon = \frac{1}{8}$. With $n = 8$ we have $a_n = \frac{1}{8}$, with $n = 9$ we have $a_n = -\frac{1}{9}$, hence

$$|a_n| < \frac{1}{8} \quad \text{for every } n \geq 9$$



To prove that:

$$a_n \rightarrow 0 \quad \text{for } n \rightarrow +\infty$$

we must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow |a_n| < \varepsilon$$

The inequality is equivalent to

$$\left| \frac{(-1)^n}{n} \right| < \varepsilon \quad \text{which is satisfied for } n > \frac{1}{\varepsilon}$$

Having fixed $\varepsilon > 0$, it will suffice to choose the first integer

$$n(\varepsilon) > \frac{1}{\varepsilon}$$

to satisfy the condition required by the definition of limit.

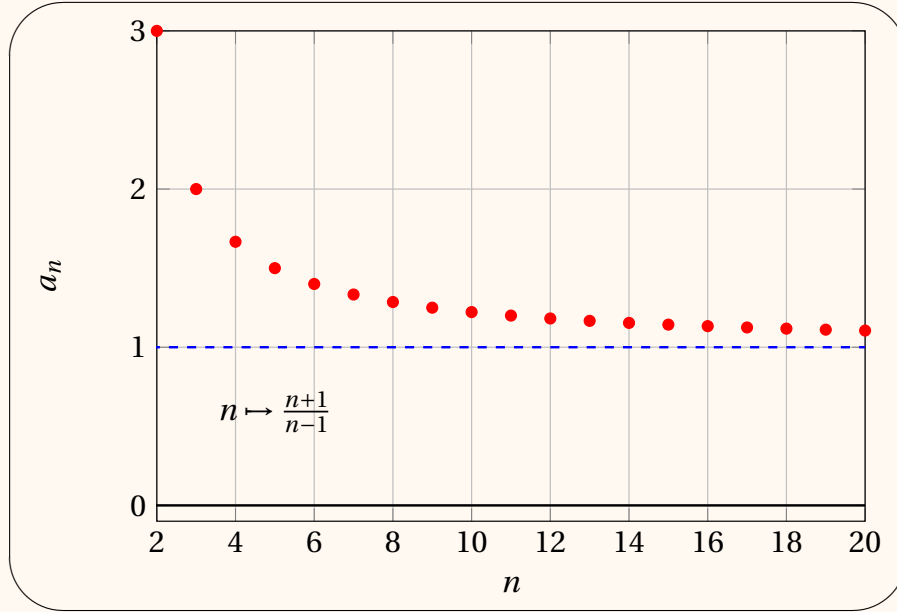
Convergent sequences are (eventually) bounded.

Example 8: Verifying the limit of a sequence

Consider the sequence:

$$n \mapsto \frac{n+1}{n-1} \quad \left(\frac{n+1}{n-1} = \frac{n-1+1+1}{n-1} = 1 + \frac{2}{n-1} \right)$$

This is the graph of the sequence with $n \in \{2, 3, \dots, 20\}$:



We see that the values a_n get closer to 1, hence we try to prove, using the definition of limit, that:

$$\lim_{n \rightarrow +\infty} \frac{n+1}{n-1} = 1$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that:

$$n > n(\varepsilon) \Rightarrow 1 - \varepsilon < \frac{n+1}{n-1} < 1 + \varepsilon$$

The left inequality is always satisfied (the numerator of the fraction is always larger than the denominator). We take the right one:

$$\frac{n+1}{n-1} - 1 < \varepsilon, \quad \frac{n+1 - (n-1)}{n-1} < \varepsilon, \quad \frac{2}{n-1} < \varepsilon, \quad n-1 > \frac{2}{\varepsilon}$$

$$\text{hence it is satisfied if } n > \frac{2+\varepsilon}{\varepsilon}$$

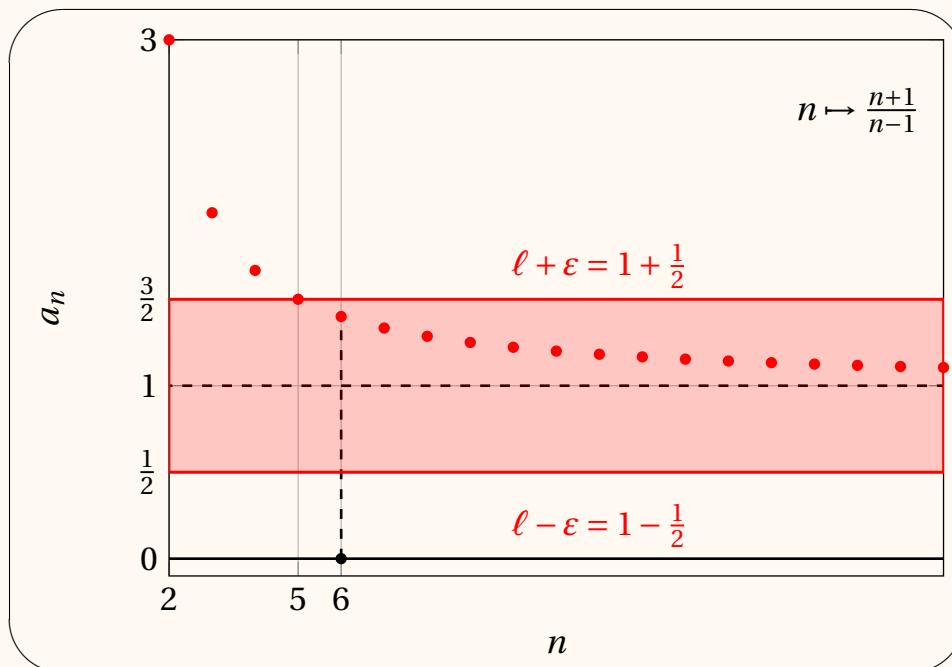
Having fixed $\varepsilon > 0$, it will suffice to choose the first integer

$$n(\varepsilon) > \frac{2+\varepsilon}{\varepsilon}$$

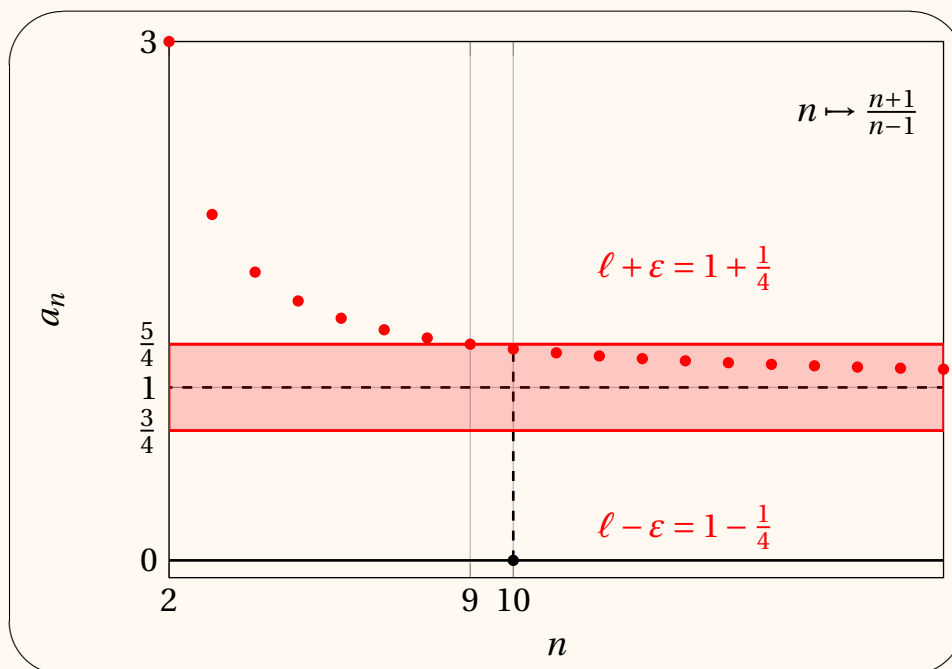
to satisfy the condition required by the definition of limit.

Example 9: Verifying the limit of a sequence

For the previous example, we proved that the limit equals 1. Let us now check what happens when we fix $\varepsilon = \frac{1}{2}$. In this case $n(\frac{1}{2}) > \frac{2+1/2}{1/2} = 5$.



If instead we fix $\varepsilon = \frac{1}{4}$, in this case we have $n(\frac{1}{4}) > \frac{2+1/4}{1/4} = 9$.

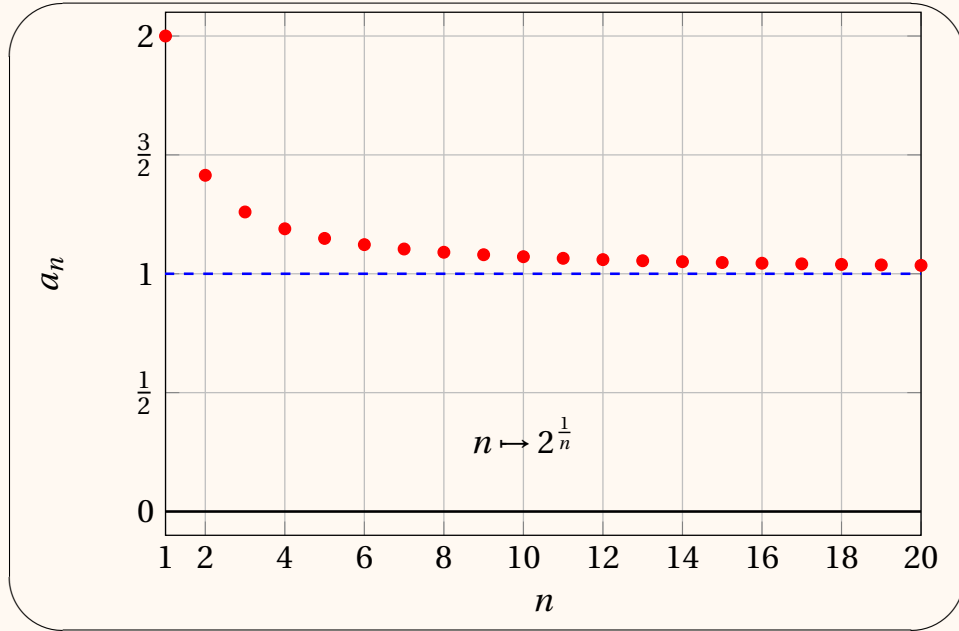


Example 10: Verifying the limit of a sequence

Consider the sequence:

$$n \mapsto 2^{\frac{1}{n}}$$

This is the graph of the sequence with $n \in \{1, 2, \dots, 20\}$:



We see that the values a_n get closer to 1, hence we try to prove, using the definition of limit, that:

$$\lim_{n \rightarrow +\infty} 2^{\frac{1}{n}} = 1$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that:

$$n > n(\varepsilon) \Rightarrow 1 - \varepsilon < 2^{\frac{1}{n}} < 1 + \varepsilon.$$

The left inequality is always satisfied (2 raised to a positive rational number), while for the right one, taking the logarithm to base 2, we obtain:

$$\frac{1}{n} < \log_2(1 + \varepsilon)$$

Hence it is satisfied if:

$$n > \frac{1}{\log_2(1 + \varepsilon)}$$

Having fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{1}{\log_2(1 + \varepsilon)}$$

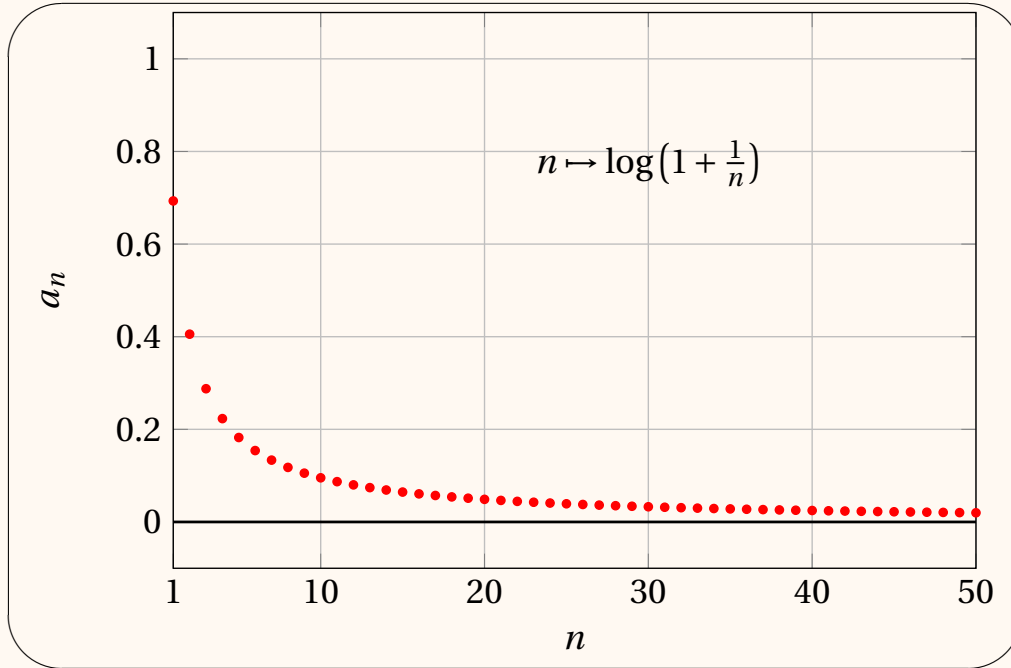
to satisfy the condition required by the definition of limit.

Example 11: Limit of sequences

Consider the sequence:

$$n \mapsto \log\left(1 + \frac{1}{n}\right).$$

This is the graph of the sequence with $n \in \{1, 2, \dots, 50\}$:



We see that the values a_n get closer to 0, hence we try to prove, using the definition of limit, that:

$$\lim_{n \rightarrow +\infty} \log\left(1 + \frac{1}{n}\right) = 0$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that:

$$n > n(\varepsilon) \Rightarrow -\varepsilon < \log\left(1 + \frac{1}{n}\right) < \varepsilon$$

The left inequality is always satisfied (the logarithm to base e of a number greater than 1), while for the right one, exponentiating, we obtain:

$$\frac{1}{n} + 1 < e^\varepsilon \quad \text{hence it is satisfied if} \quad n > \frac{1}{e^\varepsilon - 1} \quad (e^\varepsilon - 1 > 0 \text{ with } \varepsilon > 0)$$

Having fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{1}{e^\varepsilon - 1}$$

to satisfy the condition required by the definition of limit.

1.2 Divergent sequences and irregular sequences

Definition 5: Sequence divergent to $+\infty$

A sequence $\{a_n\}$ is called **divergent** to $+\infty$ if for every $M > 0$ there exists a number $n(M) \in \mathbb{N}$ such that:

$$a_n > M \text{ for every } n \geq n(M)$$

Definition 6: Sequence divergent to $-\infty$

A sequence $\{a_n\}$ is called **divergent** to $-\infty$ if for every $M > 0$ there exists a number $n(M) \in \mathbb{N}$ such that:

$$a_n < -M \text{ for every } n \geq n(M)$$

The number $n(M)$ depends (in general) on the value of M .

- In the two cases we will say, respectively, that $+\infty$ and $-\infty$ are the limits of the sequence and we will write, respectively:

$$\lim_{n \rightarrow +\infty} a_n = +\infty \quad \text{or} \quad \lim_{n \rightarrow +\infty} a_n = -\infty$$

The values of a sequence divergent to $+\infty$ eventually exceed any fixed real number. The values of a sequence divergent to $-\infty$ eventually fall below any fixed real number.

The symbols $+\infty$ and $-\infty$ are not numbers.

- If we represent the real numbers on the **Euclidean line**, each number corresponds to a point and each point to a number.
- With the symbols $+\infty$ and $-\infty$ we agree to denote two “points”:

$+\infty$ lies to the *right* of every point of $\mathbb{R}$

$-\infty$ lies to the *left* of every point of $\mathbb{R}$

however, no number corresponds to these two points.

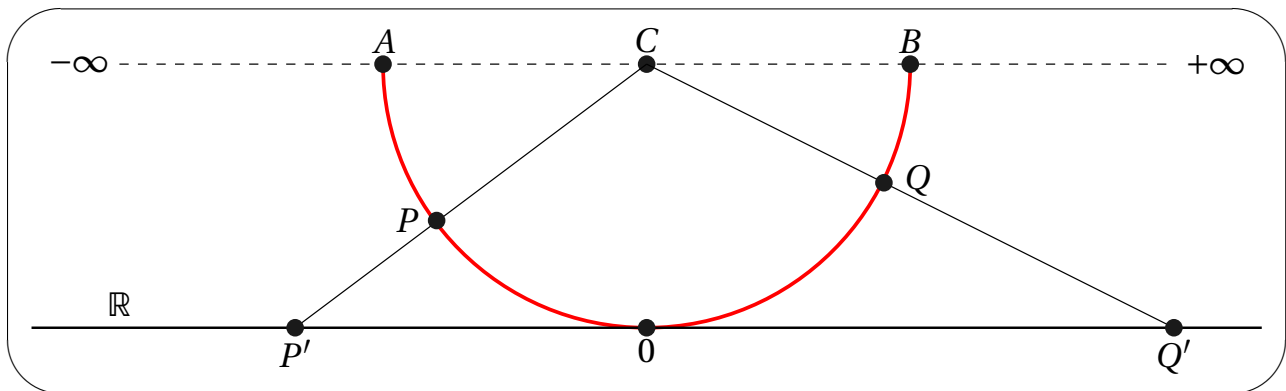
- On the symbols $+\infty$ and $-\infty$ the operations of sum and product with the properties stated in R_1 and R_2 are not defined, even though we will be able to perform these operations “partially” (as we will see later).

Definition 7: The set $\mathbb{R}^*$

The set of real numbers $\mathbb{R}$ with the addition of the two elements $+\infty$ and $-\infty$ will be denoted by:

$$\mathbb{R}^* = \mathbb{R} \cup \{+\infty\} \cup \{-\infty\}$$

- We can represent the set $\mathbb{R}^*$ “visually” by putting the points of the **line** in one-to-one correspondence with those of a **semicircle** (projecting them from the center of the semicircle onto the line $\mathbb{R}$):



- No point on $\mathbb{R}$ corresponds to the points A and B ; we will say that $-\infty$ is the “counterpart” of the point A and $+\infty$ the “counterpart” of B .
- The limit operation becomes *fully meaningful* if set in $\mathbb{R}^*$ instead of $\mathbb{R}$, i.e., the limit of a sequence can be a real number ℓ or $+\infty$ or $-\infty$.

Sequences whose limit is a real number are **convergent**; those whose limit is $+\infty$ or $-\infty$ are **divergent**.

Example 12: Convergent and divergent sequences

- the canonical sequence $\{n\}$ is divergent to $+\infty$;
- the sequence $\{2^n\}$ is divergent to $+\infty$;
- the sequence $\{-2^n\}$ is divergent to $-\infty$;
- the sequence $\{2^{\frac{1}{n}}\}$ is convergent to 1.

Remark 1

$$\lim_{n \rightarrow +\infty} n^\alpha = \begin{cases} +\infty & \text{if } \alpha > 0 \\ 1 & \text{if } \alpha = 0 \\ 0 & \text{if } \alpha < 0 \end{cases}$$

Proof. If $\alpha > 0$, we must prove that the sequence diverges to $+\infty$. We must therefore verify that for every $M > 0$ there exists $n(M) \in \mathbb{N}$ such that

$$n > n(M) \Rightarrow a_n > M$$

The inequality

$$n^\alpha > M \text{ is satisfied for } n > M^{\frac{1}{\alpha}}$$

Hence, having fixed $M > 0$, it will suffice to choose the first integer

$$n(M) > M^{\frac{1}{\alpha}}$$

to satisfy the required divergence condition.

Hence there is no $M > 0$ such that $n^\alpha \leq M$ for $n \in \mathbb{N}$, and the sequence is not bounded. Hence $n^\alpha \rightarrow +\infty$ for $n \rightarrow +\infty$.

If $\alpha < 0$, we must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow -\varepsilon < n^\alpha < \varepsilon$$

The left inequality is always satisfied. For the right inequality, since $n^\alpha = \frac{1}{n^{|\alpha|}}$, we obtain

$$\frac{1}{n^{|\alpha|}} < \varepsilon$$

which is satisfied if

$$n^{|\alpha|} > \frac{1}{\varepsilon} \text{ i.e., for } n > \frac{1}{\varepsilon^{1/|\alpha|}}$$

Having fixed $\varepsilon > 0$, it will suffice to choose the first integer

$$n(\varepsilon) > \frac{1}{\varepsilon^{1/|\alpha|}}$$

to satisfy the condition required by the definition of limit.

□

- Finally, we observe that there are sequences that are neither convergent nor divergent

Definition 8: Irregular (indeterminate) sequence

A sequence that is neither convergent nor divergent is called **irregular** (oscillating) or **indeterminate**.

Example 13: Irregular sequences

- the sequence $\{(-1)^n\}$ is neither convergent nor divergent (but it is bounded)
 - the sequence $\{(-2)^n\}$ is neither convergent nor divergent (and not even bounded).
-
- For **irregular** sequences, the limit operation is not defined, i.e., their **limit does not exist**.

Summarizing, the operation of computing the limit allows us to answer rigorously the question: **how do the numbers a_n behave as n becomes larger and larger?**

2 Unbounded sets and their suprema/infima

- It is convenient to adopt the convention introduced for limits also for the sup and the inf, extending the definition of these quantities as follows

Definition 9: Supremum and infimum sup and inf (unbounded sets)

If a set $E \subseteq \mathbb{R}$ is not bounded above (below), we will say that

$$\sup E = +\infty \quad (\inf E = -\infty)$$

- In this way, property R_4 of the real numbers can be stated as follows:

$R_4 \rightarrow$ every non-empty set $E \subseteq \mathbb{R}$ has a supremum and an infimum; $\sup E$ ($\inf E$) is a number if E is bounded above (below), otherwise it is $+\infty$ ($-\infty$).

3 Infinitesimal and infinite sequences

Definition 10: Infinitesimal sequence

A sequence $\{a_n\}$ tending to zero is called **infinitesimal**

Example 14: Infinitesimal sequences

- the sequence $\{\frac{1}{n}\}$ is infinitesimal
- the sequence $\{\frac{1}{n^2}\}$ is infinitesimal

- The concept of *infinitesimal* plays a central role and is also fundamental for having a correct and effective **intuitive picture** of the concepts of *infinitesimal calculus*.

an “**infinitesimal**” is not an “infinitely small number” (a meaningless concept) but a **variable quantity** (a sequence or, as we will see, a function) that **becomes indefinitely small**.

Definition 11: Infinite sequence

A sequence $\{a_n\}$ tending to $\pm\infty$ is called **infinite** (an infinity)

Example 15: Infinite sequences

- the sequence $\{n^2\}$ is infinite
- the sequence $\{n!\}$ is infinite

- Sometimes it is possible to specify whether a convergent sequence **approaches its limit from above or from below**

Definition 12: Limit from above

We say that the sequence $\{a_n\}$ tends to $\ell \in \mathbb{R}$ **from above** and we write

$$\lim_{n \rightarrow +\infty} a_n = \ell^+$$

if for every $\varepsilon > 0$ we have that

$$0 \leq a_n - \ell < \varepsilon, \quad \text{eventually.}$$

Definition 13: Limit from below

We say that the sequence $\{a_n\}$ tends to $\ell \in \mathbb{R}$ **from below** and we write

$$\lim_{n \rightarrow +\infty} a_n = \ell^-$$

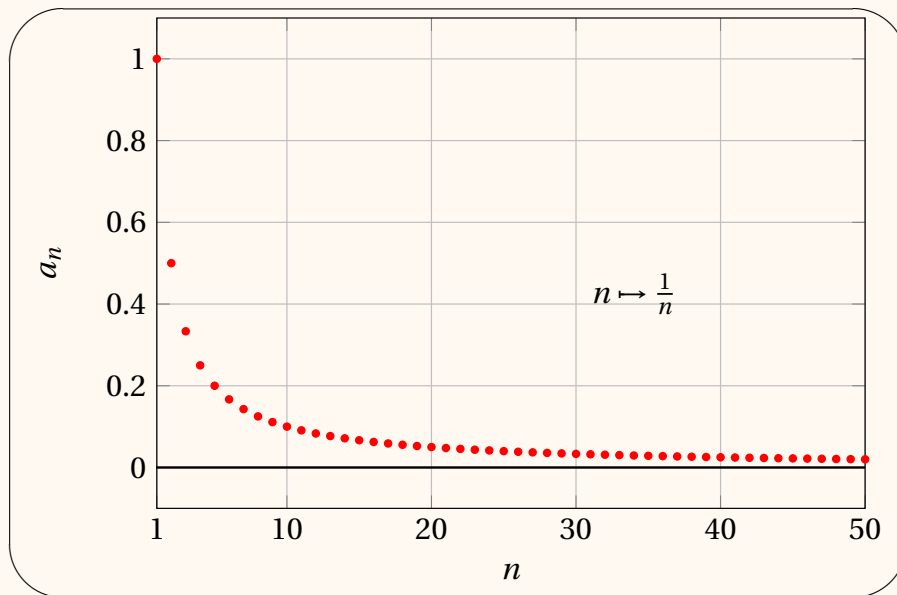
if for every $\varepsilon > 0$ we have that

$$0 \leq \ell - a_n < \varepsilon, \quad \text{eventually.}$$

- Saying that $a_n \rightarrow \ell^+$ for $n \rightarrow +\infty$ means stating that $a_n \rightarrow \ell$ and moreover $a_n \geq \ell$ eventually; hence a_n approaches ℓ *from above*, i.e., it approximates ℓ from above.
- Saying that $a_n \rightarrow \ell^-$ for $n \rightarrow +\infty$ means stating that $a_n \rightarrow \ell$ and moreover $a_n \leq \ell$ eventually; hence a_n approaches ℓ *from below*, i.e., it approximates ℓ from below.

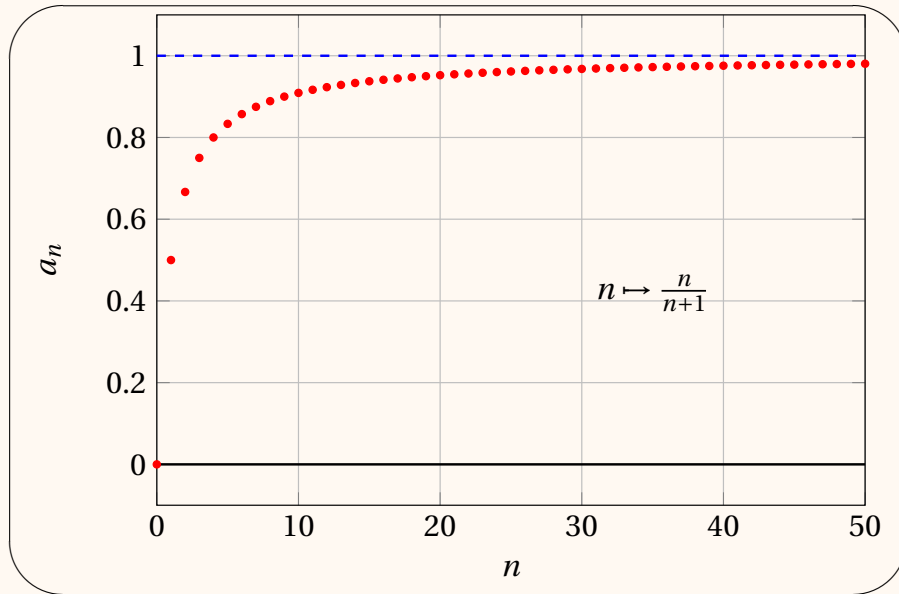
Example 16: Sequence with limit from above

$$\lim_{n \rightarrow +\infty} \frac{1}{n} = 0^+$$



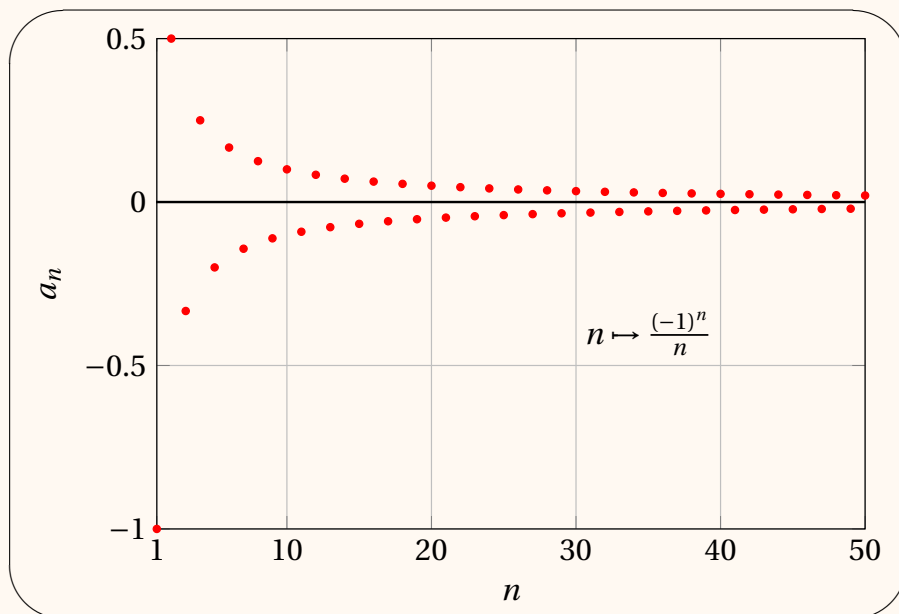
Example 17: Sequence with limit from below

$$\lim_{n \rightarrow +\infty} \frac{n}{n+1} = 1^- \quad \left(\frac{n}{n+1} = \frac{n+1-1}{n+1} = 1 - \frac{1}{n+1} \right)$$



Example 18: Sequence with a limit, but neither from below nor from above

$$\lim_{n \rightarrow +\infty} \frac{(-1)^n}{n} = 0$$



In this case we can state neither that $a_n \rightarrow 0^+$ nor that $a_n \rightarrow 0^-$ for $n \rightarrow +\infty$.

4 Monotone sequences

Definition 14: Monotone sequences

A sequence $\{a_n\}$ is called **monotone increasing** if:

$$a_n \leq a_{n+1}, \forall n$$

A sequence $\{a_n\}$ is called **strictly monotone increasing** if:

$$a_n < a_{n+1}, \forall n$$

A sequence $\{a_n\}$ is called **monotone decreasing** if:

$$a_n \geq a_{n+1}, \forall n$$

A sequence $\{a_n\}$ is called **strictly monotone decreasing** if:

$$a_n > a_{n+1}, \forall n$$

Example 19: Monotone increasing/decreasing sequences

- The sequence $\{n^2\}$ is strictly monotone increasing
 - The sequence $\{\frac{1}{n}\}$ is strictly monotone decreasing
 - The sequence $\{(-1)^n\}$ is not monotone
 - every constant sequence is monotone (increasing or decreasing, not strictly)
- With regard to the limit operation, these sequences are of particular importance; indeed, they are **never irregular**, but are **convergent or divergent** depending on whether they are **bounded or not**.

Theorem 2: Monotone sequence theorem

- Let $\{a_n\}$ be a **monotone increasing sequence that is bounded above**. Then $\{a_n\}$ is convergent, and its limit equals

$$\sup\{a_n : n \in \mathbb{N}\}.$$

- Let $\{a_n\}$ be a **monotone decreasing sequence that is bounded below**. Then $\{a_n\}$ is convergent, and its limit equals

$$\inf\{a_n : n \in \mathbb{N}\}.$$

Proof. We consider the case of monotone increasing sequences that are bounded above.

Since the sequence is bounded above, the set of values taken by the sequence $\{a_n : n \in \mathbb{N}\}$ is bounded above.

By the *supremum property* enjoyed by $\mathbb{R}$, there therefore exists a finite supremum, which we denote by ℓ :

$$\ell = \sup\{a_n : n \in \mathbb{N}\} \quad \text{and} \quad \ell \in \mathbb{R}.$$

We now need to prove that

$$\lim_{n \rightarrow +\infty} a_n = \ell$$

and hence that for every $\varepsilon > 0$ we have

$$\ell - \varepsilon < a_n < \ell + \varepsilon, \quad \text{eventually.}$$

The second inequality is obvious. For every $n \in \mathbb{N}$ we have

$$a_n \leq \ell \quad (\text{and hence } a_n < \ell + \varepsilon)$$

since ℓ is the supremum of $\{a_n : n \in \mathbb{N}\}$ and hence an upper bound.

By definition of supremum, ℓ is the least of the upper bounds of the set $\{a_n : n \in \mathbb{N}\}$. Therefore, since

$$\ell - \varepsilon < \ell,$$

certainly $\ell - \varepsilon$ is not an upper bound of the set $\{a_n : n \in \mathbb{N}\}$. This means that there exists an $n(\varepsilon) \in \mathbb{N}$ for which

$$a_{n(\varepsilon)} > \ell - \varepsilon.$$

On the other hand, the sequence is monotone increasing, therefore for every $n \geq n(\varepsilon)$ we have $a_n \geq a_{n(\varepsilon)}$. We have thus proved that

$$a_n \geq a_{n(\varepsilon)} > \ell - \varepsilon \quad \text{for every } n \geq n(\varepsilon)$$

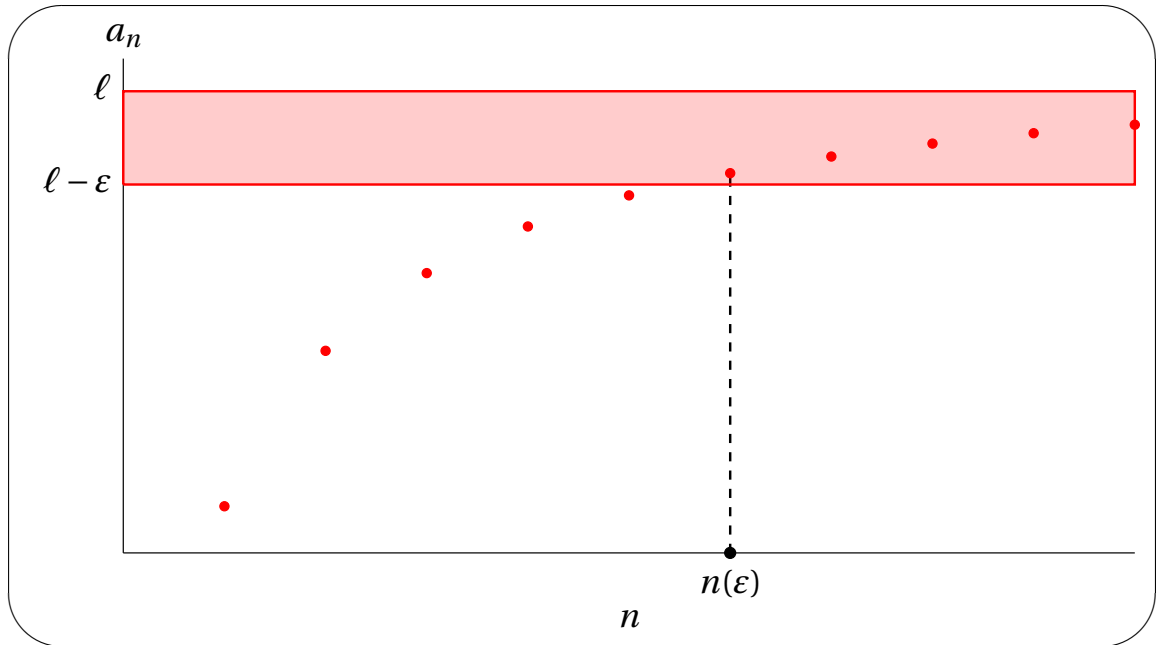
Consequently, we have the claim:

$$\lim_{n \rightarrow +\infty} a_n = \ell$$

The case of monotone decreasing sequences that are bounded below is proved analogously.

□

- The idea of the proof is conveyed by the following figure:



- To express symbolically as well that the limit is the sup (or the inf) of an increasing (or decreasing) sequence, we use the notation

$$a_n \uparrow \ell \text{ or } a_n \downarrow \ell$$

This in particular implies that $a_n \rightarrow \ell^-$ (respectively, ℓ^+), but it contains a further piece of information: the monotonicity of the sequence.

- This theorem is a consequence of the completeness axiom (axiom of continuity) R_4 of the real numbers and therefore holds if the setting we consider is $\mathbb{R}$. For example, it is not true that an increasing and bounded sequence of rational numbers always has a rational limit, i.e., in $\mathbb{Q}$.

Example 20: Increasing and bounded sequence in $\mathbb{Q}$

Let $\{a_n\}$ be the sequence defined as follows:

$$a_0 = 0, a_1 = 0, 1, a_2 = 0, 1011, a_3 = 0, 10110111, a_4 = 0, 1011011101111 \dots$$

At step n we append to the decimal number obtained at the previous step a digit zero followed by n digits equal to 1. The sequence $\{a_n\}$ is evidently increasing, and bounded above (for example, $a_n \leq 1$).

- In $\mathbb{R}$ the sequence converges to the number $\sup\{a_n : n \in \mathbb{N}\}$, which after the decimal point has an infinite, non-periodic decimal expansion (one digit 1, one digit 0, two digits 1, one digit 0, three digits 1, one digit 0, and so on forever)
- Hence the limit of the sequence is an irrational number. This example shows that in the set $\mathbb{Q}$ the monotone sequence theorem is false.

- The monotone sequence theorem can be completed with the next corollary, which considers bounded or unbounded sequences.

Corollary 1: Of the monotone sequence theorem

Let $\{a_n\}$ be a monotone increasing sequence. Then there exists

$$\lim_{n \rightarrow +\infty} a_n = \sup\{a_n : n \in \mathbb{N}\}.$$

Proof. If $\{a_n\}$ is bounded above, the statement is contained in Theorem (??) (monotone sequence theorem).

If instead $\{a_n\}$ is unbounded above, this means that, having fixed $M > 0$, there exists an $n(M) \in \mathbb{N}$ such that

$$a_{n(M)} > M$$

On the other hand, the sequence is increasing, therefore for every $n \geq n(M)$ we have

$$a_n \geq a_{n(M)} > M$$

We have thus proved that for every $M > 0$ we have $a_n > M$, eventually. This means that $a_n \rightarrow +\infty$ for $n \rightarrow +\infty$.

□

Summarizing, we have:

- if $\{a_n\}$ is bounded above, then it converges (and its limit equals the supremum of its values, which in this case is a real number)
- if instead $\{a_n\}$ is unbounded above, then a_n tends to $+\infty$ (which in this case equals the supremum of its values).

A monotone sequence either converges or diverges (it cannot be irregular).

5 Limits of geometric progressions

- Consider the **geometric progression** with common ratio $a \in \mathbb{R}$:

$$1, a, a^2, a^3, \dots, a^n, \dots$$

it is equivalent to the sequence $\{a^n\}$:

$$n \mapsto a^n$$

- If $a > 1$, the sequence is monotone increasing and unbounded above.
- If $a = 1$, the sequence is constant.
- If $0 < a < 1$, the sequence is monotone decreasing and tends to zero.
- If a is negative, the sequence is not monotone.

Remark 2

$$\lim_{n \rightarrow +\infty} a^n = \begin{cases} +\infty & \text{if } a > 1 \\ 1 & \text{if } a = 1 \\ 0 & \text{if } |a| < 1 \\ \text{does not exist} & \text{if } a \leq -1 \end{cases}$$

Proof. If $|a| < 1$, we must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow -\varepsilon < a_n < +\varepsilon$$

Hence we must verify that

$$n > n(\varepsilon) \Rightarrow -\varepsilon < |a|^n < \varepsilon$$

The first inequality is always true, while the second is satisfied if

$$n > \log_{|a|} \varepsilon$$

Having fixed $\varepsilon > 0$, it will suffice to choose the first integer

$$n(\varepsilon) > \log_{|a|} \varepsilon$$

to satisfy the condition required by the definition of limit.

If $a > 1$, we must verify that for every $M > 0$ there exists $n(M) \in \mathbb{N}$ such that

$$n > n(M) \Rightarrow a_n > M$$

The inequality

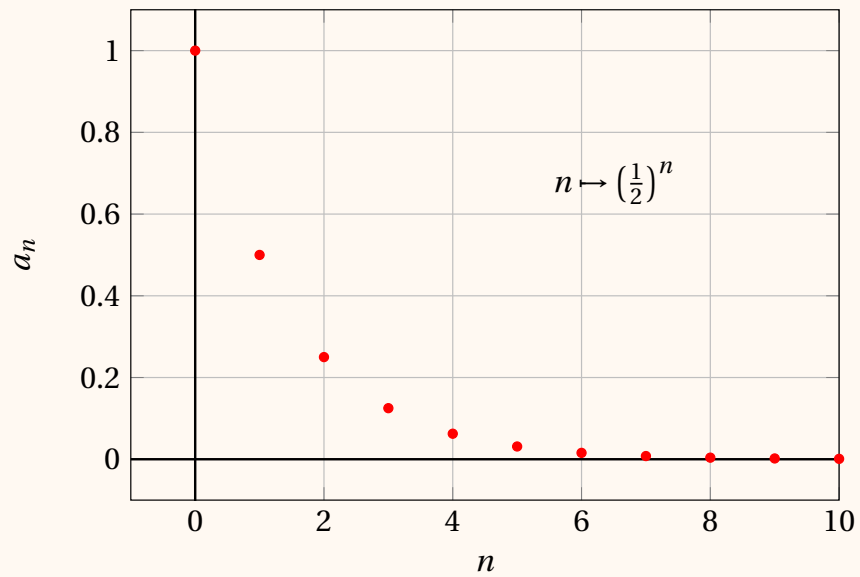
$$a^n > M \text{ is satisfied for } n > \log_a M$$

Hence, having fixed $M > 0$, it will suffice to choose the first integer

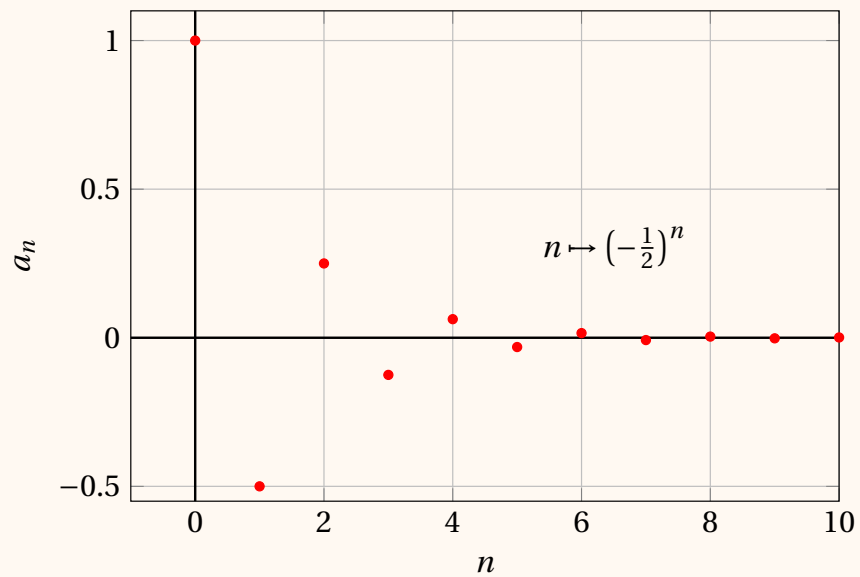
$$n(M) > \log_a M$$

to satisfy the required divergence condition. □

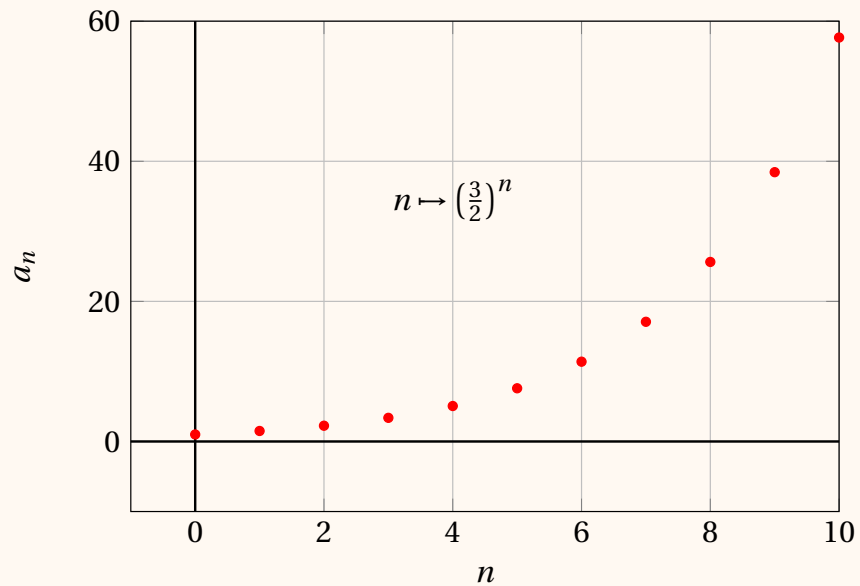
Example 21: Infinitesimal and monotone decreasing geometric progression



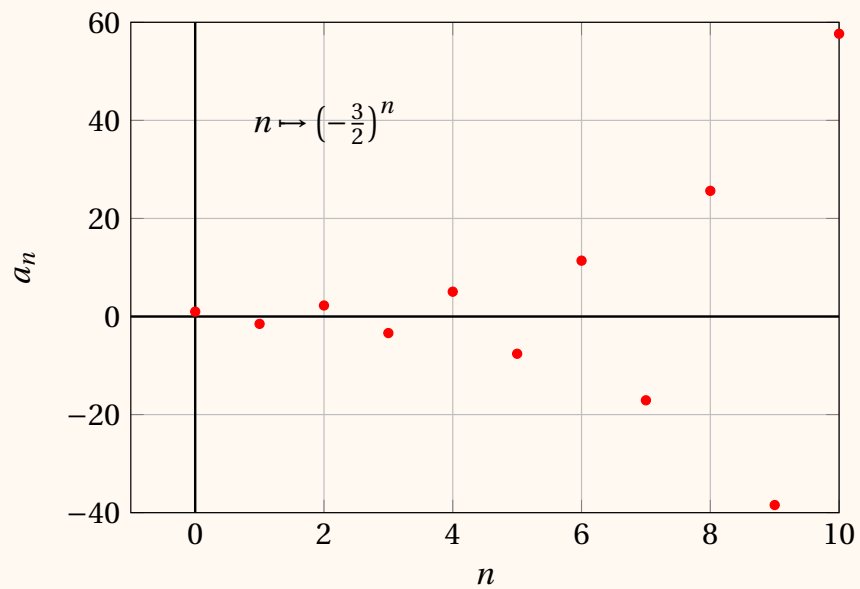
Example 22: Infinitesimal (but not monotone) geometric progression



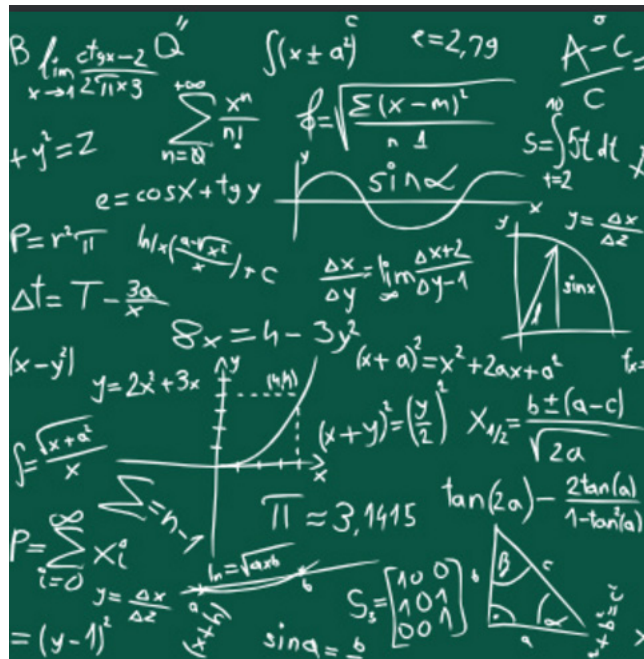
Example 23: Divergent geometric progression



Example 24: Geometric progression that is neither convergent nor divergent



Computing limits of sequences



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

Contents

1 Computing limits of sequences

2

1 Computing limits of sequences

- The proofs of the **basic theorems** on computing limits are based on the **definition of limit**, on the **use of inequalities**, and on the use of **properties that are eventually true**.
- In particular, these theorems illustrate the relation between the limit operation and the algebraic structures¹ and the order structures² present in $\mathbb{R}$.

We will simply write $a_n \rightarrow \ell$ to mean $a_n \rightarrow \ell$ for $n \rightarrow +\infty$

Properties of the limit operation with respect to algebraic operations.

Theorem 1: Algebra of limits, case of finite limits

Hypotheses:

$$1. a_n \rightarrow \ell_a \in \mathbb{R} \quad 2. b_n \rightarrow \ell_b \in \mathbb{R}.$$

Claim:

$$\begin{aligned} 1. a_n \pm b_n &\rightarrow \ell_a \pm \ell_b & 2. \frac{a_n}{b_n} &\rightarrow \frac{\ell_a}{\ell_b} \quad (b_n, \ell_b \neq 0, \text{ eventually}) \\ 3. a_n b_n &\rightarrow \ell_a \ell_b & 4. a_n^{b_n} &\rightarrow \ell_a^{\ell_b} \quad (a_n, \ell_a > 0, \text{ eventually}). \end{aligned}$$

Proof. We prove that:

$$a_n \rightarrow \ell_a \in \mathbb{R}, b_n \rightarrow \ell_b \in \mathbb{R} \Rightarrow a_n + b_n \rightarrow \ell_a + \ell_b$$

We consider

$$|(a_n + b_n) - (\ell_a + \ell_b)| = |(a_n - \ell_a) + (b_n - \ell_b)| \leq |a_n - \ell_a| + |b_n - \ell_b| \quad (1)$$

by the triangle inequality. Since by hypothesis $a_n \rightarrow \ell_a$ and $b_n \rightarrow \ell_b$, we have that

$$|a_n - \ell_a| < \varepsilon_a \quad \text{and} \quad |b_n - \ell_b| < \varepsilon_b \quad (\text{eventually})$$

for every $\varepsilon_a > 0$ and $\varepsilon_b > 0$. Bounding from above the terms on the right-hand side of (1) we conclude that

$$|(a_n + b_n) - (\ell_a + \ell_b)| < \underbrace{2 \max\{\varepsilon_a, \varepsilon_b\}}_{=\tilde{\varepsilon} \text{ and } >0}$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

¹A set on which an operation is defined is called an algebraic structure

²A set is endowed with an order structure if an order relation is defined on it

Proof. We prove that:

$$a_n \rightarrow \ell_a \in \mathbb{R}, b_n \rightarrow \ell_b \in \mathbb{R} \Rightarrow a_n - b_n \rightarrow \ell_a - \ell_b$$

For every $\varepsilon_a > 0$ and $\varepsilon_b > 0$, we have:

$$|(a_n - b_n) - (\ell_a - \ell_b)| = |(a_n - \ell_a) + (\ell_b - b_n)| \leq \underbrace{|a_n - \ell_a|}_{< \varepsilon_a} + \underbrace{|\ell_b - b_n|}_{= |b_n - \ell_b| < \varepsilon_b}$$

Hence

$$|(a_n - b_n) - (\ell_a - \ell_b)| < \underbrace{2 \max\{\varepsilon_a, \varepsilon_b\}}_{=\tilde{\varepsilon} \text{ and } >0}$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

Proof. We prove that:

$$a_n \rightarrow \ell_a \in \mathbb{R}, b_n \rightarrow \ell_b \in \mathbb{R} \Rightarrow a_n b_n \rightarrow \ell_a \ell_b$$

We consider

$$\begin{aligned} |(a_n b_n) - (\ell_a \ell_b)| &= |a_n b_n - a_n \ell_b + a_n \ell_b - \ell_a \ell_b| \\ &= |a_n (b_n - \ell_b) + \ell_b (a_n - \ell_a)| \leq |a_n (b_n - \ell_b)| + |\ell_b (a_n - \ell_a)| \\ &= |a_n| |b_n - \ell_b| + |\ell_b| |a_n - \ell_a| \end{aligned}$$

by the triangle inequality and the properties of the absolute value. Hence

$$|(a_n b_n) - (\ell_a \ell_b)| \leq |a_n| |b_n - \ell_b| + |\ell_b| |a_n - \ell_a| \quad (2)$$

Since by hypothesis $a_n \rightarrow \ell_a$ and $b_n \rightarrow \ell_b$, for every $\varepsilon_a > 0$ and $\varepsilon_b > 0$ we have

$$|a_n - \ell_a| < \varepsilon_a \quad \text{and} \quad |b_n - \ell_b| < \varepsilon_b \quad (\text{eventually}).$$

Moreover, since

$$|a_n - \ell_a| \geq |a_n| - |\ell_a| \quad \text{we have} \quad |a_n| < |\ell_a| + \varepsilon_a, \quad \text{eventually.}$$

Therefore, bounding from above the terms on the right-hand side of (2), we conclude that

$$|(a_n b_n) - (\ell_a \ell_b)| < (|\ell_a| + \varepsilon_a) \varepsilon_b + |\ell_b| \varepsilon_a = \underbrace{|\ell_a| \varepsilon_b + |\ell_b| \varepsilon_a + \varepsilon_a \cdot \varepsilon_b}_{= \tilde{\varepsilon} \text{ and } > 0}$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

- Moreover, the **limit operation preserves the ordering**

Theorem 2: Sign-preservation, first form

Hypotheses:

$$1. a_n \rightarrow \ell_a \quad 2. \ell_a \leq 0.$$

Claim:

$$a_n \leq 0, \text{ eventually.}$$

Proof. We consider the case $\ell_a > 0$. By definition of limit we have that

$$|a_n - \ell_a| < \varepsilon, \quad \text{eventually,}$$

for every $\varepsilon > 0$, which we rewrite in the form:

$$\ell_a - \varepsilon < a_n < \ell_a + \varepsilon, \quad \text{eventually.}$$

Since $\ell_a > 0$, we can choose $\varepsilon > 0$ such that $\ell_a - \varepsilon > 0$; then the inequality

$$0 < \ell_a - \varepsilon < a_n$$

shows that $a_n > 0$, eventually.

The case $\ell_a < 0$ is proved analogously. □

Theorem 3: Sign-preservation, second form (part I)

Hypotheses:

$$1. \ a_n \rightarrow \ell_a \in \mathbb{R} \quad 2. \ a_n \geq 0, \text{ eventually.}$$

Claim:

$$\ell_a \geq 0.$$

Proof. It follows from the previous theorem. Indeed, if by contradiction we had $\ell_a < 0$, from the previous theorem we would have $a_n < 0$ eventually, which is incompatible with the hypothesis that $a_n \geq 0$ eventually.

This case cannot occur, i.e., the opposite holds, which is the claim of the theorem. □

Theorem 4: Sign-preservation, second form (part II)

Hypotheses:

$$1. \ a_n \rightarrow \ell_a \in \mathbb{R} \quad 2. \ b_n \rightarrow \ell_b \in \mathbb{R} \quad 3. \ a_n \geq b_n, \text{ eventually.}$$

Claim:

$$\ell_a \geq \ell_b.$$

Proof. We consider the sequence $a_n - b_n$; by the theorem on the algebra of limits we have

$$a_n - b_n \rightarrow \ell_a - \ell_b.$$

Since by hypothesis

$$a_n \geq b_n \text{ and hence } a_n - b_n \geq 0,$$

by the sign-preservation theorem, second form (part I), applied to the sequence $a_n - b_n$ we have

$$\ell_a - \ell_b \geq 0 \text{ and hence } \ell_a \geq \ell_b.$$

□

- This theorem tells us that in an inequality between two sequences we can pass to the limit on both sides, keeping the “ $\leq$ ” or the “ $\geq$ ”.
- Note, instead, that in general strict inequalities “ $<$ ” and “ $>$ ” are not preserved when passing to the limit.

Example 1: Passing to the limit with strict inequalities

For example, even if the a_n are strictly positive, their limit ℓ_a is positive or zero, as shown by the simple example $\frac{1}{n} \rightarrow 0$.

Theorem 5: Comparison (squeeze) theorem

Hypotheses:

$$1. \ a_n \rightarrow \ell, c_n \rightarrow \ell \text{ and } \ell \in \mathbb{R}, \quad 2. \ a_n \leq b_n \leq c_n, \text{ eventually.}$$

Claim:

$$b_n \rightarrow \ell.$$

Proof. By definition of limit we have, eventually, that

$$\ell - \varepsilon_a < a_n < \ell + \varepsilon_a \quad \text{and} \quad \ell - \varepsilon_c < c_n < \ell + \varepsilon_c$$

for every $\varepsilon_a > 0$ and $\varepsilon_c > 0$. Hence:

$$\ell - \max\{\varepsilon_a, \varepsilon_c\} < a_n < \ell + \max\{\varepsilon_a, \varepsilon_c\} \quad \text{and} \quad \ell - \max\{\varepsilon_a, \varepsilon_c\} < c_n < \ell + \max\{\varepsilon_a, \varepsilon_c\}$$

From the hypotheses of the theorem we therefore have, eventually, that

$$\ell - \max\{\varepsilon_a, \varepsilon_c\} < a_n \leq b_n \leq c_n < \ell + \max\{\varepsilon_a, \varepsilon_c\}$$

But then, eventually, we have

$$\ell - \underbrace{\max\{\varepsilon_a, \varepsilon_c\}}_{=\tilde{\varepsilon} \text{ and } >0} < b_n < \ell + \underbrace{\max\{\varepsilon_a, \varepsilon_c\}}_{=\tilde{\varepsilon} \text{ and } >0}.$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

- Frequently used special cases of this theorem are expressed by the following corollaries, which are very useful when studying the product of an oscillating (but bounded) sequence and one that tends to zero

Corollary 1: Of the comparison theorem (part I)

Hypotheses:

1. $c_n \rightarrow 0$
2. $|b_n| \leq c_n$, eventually.

Claim:

$$b_n \rightarrow 0.$$

Proof. We know that eventually we have $-c_n \leq b_n \leq c_n$. Obviously

$$\text{if } c_n \rightarrow 0 \text{ then } -c_n \rightarrow 0.$$

Hence by the comparison theorem (with $a_n = -c_n$ and $\ell = 0$) we have that $b_n \rightarrow 0$. □

Corollary 2: Of the comparison theorem (part II)

Hypotheses:

1. $c_n \rightarrow 0$
2. $\{b_n\}$ is bounded (but not necessarily convergent).

Claim:

$$c_n b_n \rightarrow 0.$$

Proof. If $\{b_n\}$ is bounded, then $|b_n| \leq M$ for some $M > 0$ and for every $n \in \mathbb{N}$. We can therefore write

$$|b_n c_n| \leq M |c_n|.$$

Since

$$c_n \rightarrow 0 \text{ also } M |c_n| \rightarrow 0,$$

by Corollary (??) we conclude that $b_n c_n \rightarrow 0$. □

The product of an infinitesimal sequence and a bounded one is infinitesimal.

Example 2: Application of the corollary

Consider the sequence given by a ratio of two expressions, each consisting of a sum of powers of n , such as:

$$n \mapsto \frac{n^{5/2} - 3n + 7}{n^3 + \sqrt{n} - 3n^2}$$

Factoring out the highest power both in the numerator and in the denominator we obtain:

$$\frac{n^{5/2} \left(1 - \frac{3}{n^{3/2}} + \frac{7}{n^{5/2}}\right)}{n^3 \left(1 + \frac{1}{n^{5/2}} - \frac{3}{n}\right)} = \frac{1}{\sqrt{n}} \frac{1 - \frac{3}{n^{3/2}} + \frac{7}{n^{5/2}}}{1 + \frac{1}{n^{5/2}} - \frac{3}{n}}$$

Now, by Theorem (??) on the algebra of limits and knowing that negative powers of n tend to zero, we can state that:

$$1 - \underbrace{\frac{3}{n^{3/2}}}_{\rightarrow 0} + \underbrace{\frac{7}{n^{5/2}}}_{\rightarrow 0} \rightarrow 1, \quad 1 + \underbrace{\frac{1}{n^{5/2}}}_{\rightarrow 0} - \underbrace{\frac{3}{n}}_{\rightarrow 0} \rightarrow 1 \quad \text{and} \quad \left(\frac{1 - \frac{3}{n^{3/2}} + \frac{7}{n^{5/2}}}{1 + \frac{1}{n^{5/2}} - \frac{3}{n}} \right) \rightarrow 1$$

hence the last sequence is convergent and consequently bounded. Now, by Corollary (??) and since

$$\frac{1}{\sqrt{n}} \rightarrow 0 \text{ we have } \frac{n^{5/2} - 3n + 7}{n^3 + \sqrt{n} - 3n^2} \rightarrow 0$$

Example 3: Application of the corollary

The sequence

$$n \mapsto \frac{\sin n}{n}$$

is the product of two sequences

$$n \mapsto \frac{1}{n} \text{ (convergent, infinitesimal) and } n \mapsto \sin n \text{ (irregular),}$$

hence Theorem (??) on the algebra of limits cannot be applied (the second limit does not exist).

- However, Corollary (??) can be applied. The sequence $\{\frac{1}{n}\}$ is infinitesimal and, since $|\sin n| \leq 1$, the sequence $\{\sin n\}$ is bounded; therefore we have

$$\lim_{n \rightarrow +\infty} \frac{\sin n}{n} = 0$$

- So far we have seen theorems that work on pairs of sequences that are both convergent or at least bounded.

Sequences with limits $+\infty$ and $-\infty$

- Suppose, for example, that

$$a_n \rightarrow \ell_a \text{ and } b_n \rightarrow +\infty$$

then it is easy (and intuitive) to see that

$$a_n + b_n \rightarrow +\infty$$

We will abbreviate this as follows:

$$\ell_a + \infty = +\infty$$

- Reasoning in a similar way we can summarize the rules for the limit of the sum (or difference) of two sequences, one or both of which are divergent.

Rules of partial arithmetization of the infinity symbol

Theorem 6: Partial arithmetization of the infinity symbol (addition)

Hypotheses:

$$1. a_n \rightarrow \ell_a \in \mathbb{R} \quad 2. b_n \rightarrow +\infty \quad 3. c_n \rightarrow +\infty.$$

Claim:

$$\begin{aligned} 1. a_n + b_n &\rightarrow \ell_a + \infty = +\infty & 2. a_n - b_n &\rightarrow \ell_a - \infty = -\infty \\ 3. b_n + c_n &\rightarrow +\infty + \infty = +\infty & 4. -b_n - c_n &\rightarrow -\infty - \infty = -\infty. \end{aligned}$$

Theorem 7: Partial arithmetization of the infinity symbol (product)

Hypotheses:

$$1. a_n \rightarrow \ell_a \in \mathbb{R} \quad 2. b_n \rightarrow 0 \quad 3. c_n \rightarrow \infty.$$

Claim:

$$1. a_n c_n \rightarrow \ell_a \infty = \infty \quad (\ell_a \neq 0) \quad 2. \frac{a_n}{b_n} \rightarrow \frac{\ell_a}{0} = \infty \quad (\ell_a \neq 0) \quad 3. \frac{a_n}{c_n} \rightarrow \frac{\ell_a}{\infty} = 0.$$

- the **sign** of ∞ must be determined with the **usual rule of signs**.

Example 4: Rule of signs

We have:

- if $a_n \rightarrow \ell_a \in \mathbb{R}, \ell_a > 0$ and $b_n \rightarrow 0^+$ then $\frac{a_n}{b_n} \rightarrow +\infty$
- if $a_n \rightarrow \ell_a \in \mathbb{R}, \ell_a < 0$ and $b_n \rightarrow 0^-$ then $\frac{a_n}{b_n} \rightarrow +\infty$
- if $a_n \rightarrow \ell_a \in \mathbb{R}, \ell_a > 0$ and $b_n \rightarrow 0^-$ then $\frac{a_n}{b_n} \rightarrow -\infty$
- if $a_n \rightarrow \ell_a \in \mathbb{R}, \ell_a < 0$ and $b_n \rightarrow 0^+$ then $\frac{a_n}{b_n} \rightarrow -\infty$

Therefore, in order to apply the rules of partial arithmetization of the infinity symbol, it is necessary to determine whether b_n tends to zero from above or from below.

Proof. We prove that:

$$a_n \rightarrow \ell_a \in \mathbb{R}, c_n \rightarrow +\infty \Rightarrow \frac{a_n}{c_n} \rightarrow 0.$$

For every $\varepsilon > 0$, since $a_n \rightarrow \ell_a$, eventually we have

$$\underbrace{|a_n - \ell_a|}_{\geq |a_n| - |\ell_a|} < \varepsilon \quad \text{hence} \quad |a_n| < |\ell_a| + \varepsilon.$$

Moreover, since $c_n \rightarrow +\infty$, eventually we have

$$c_n > \frac{1}{\varepsilon}$$

It follows that, eventually, we have

$$\frac{|a_n|}{|c_n|} < \frac{|\ell_a| + \varepsilon}{\frac{1}{\varepsilon}} \quad \text{and hence} \quad \left| \frac{a_n}{c_n} \right| < \varepsilon (|\ell_a| + \varepsilon) = \underbrace{|\ell_a| \varepsilon + \varepsilon^2}_{=\tilde{\varepsilon} \text{ and } > 0}$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

The four missing operations:

$$+\infty - \infty, \quad 0 \cdot \infty, \quad \frac{0}{0} \quad \text{and} \quad \frac{\infty}{\infty}$$

are called **indeterminate forms**, since no rule can be established a priori to determine their result.

Example 5: Resolving indeterminate forms $+\infty - \infty$

Consider the sequence

$$n \mapsto \sqrt{n+1} - \sqrt{n-1}$$

We have the difference of two sequences:

$$n \mapsto \sqrt{n+1} \quad \text{with} \quad \lim_{n \rightarrow +\infty} \sqrt{n+1} = +\infty$$

$$n \mapsto \sqrt{n-1} \quad \text{with} \quad \lim_{n \rightarrow +\infty} \sqrt{n-1} = +\infty$$

Hence we fall into the indeterminate form $+\infty - \infty$.

Multiplying and dividing by $\sqrt{n+1} + \sqrt{n-1}$ we obtain

$$\frac{(\sqrt{n+1} - \sqrt{n-1})(\sqrt{n+1} + \sqrt{n-1})}{\sqrt{n+1} + \sqrt{n-1}} = \frac{(\sqrt{n+1})^2 - (\sqrt{n-1})^2}{\sqrt{n+1} + \sqrt{n-1}} = \frac{2}{\sqrt{n+1} + \sqrt{n-1}}$$

recalling that $a^2 - b^2 = (a - b)(a + b)$. Now, considering the sequence in the denominator, we have

$$\lim_{n \rightarrow +\infty} \sqrt{n+1} + \sqrt{n-1} = +\infty + \infty = +\infty$$

Using the rule

$$a_n \rightarrow \ell_a \in \mathbb{R}, \quad c_n \rightarrow \infty \Rightarrow \frac{a_n}{c_n} \rightarrow 0$$

of the theorem on the partial arithmetization of the infinity symbol (product), we have:

$$\lim_{n \rightarrow +\infty} \sqrt{n+1} - \sqrt{n-1} = \lim_{n \rightarrow +\infty} \frac{2}{\sqrt{n+1} + \sqrt{n-1}} = \frac{2}{+\infty} = 0$$

- Limits of sequences of the form:

$$\{a_n^{b_n}\}$$

can be handled by considering the sequence of their logarithms, taking for simplicity the base e .

Given a sequence $\{a_n^{b_n}\}$, we have that:

$$\text{if } b_n \log a_n \rightarrow \ell \text{ then } a_n^{b_n} \rightarrow e^\ell \quad (3)$$

$$\text{if } b_n \log a_n \rightarrow +\infty \text{ then } a_n^{b_n} \rightarrow +\infty \quad (4)$$

$$\text{if } b_n \log a_n \rightarrow 0 \text{ then } a_n^{b_n} \rightarrow 1 \quad (5)$$

$$\text{if } b_n \log a_n \rightarrow -\infty \text{ then } a_n^{b_n} \rightarrow 0 \quad (6)$$

If the sequence $\{b_n \log a_n\}$ is indeterminate (has no limit), then $\{a_n^{b_n}\}$ is indeterminate as well.

Example 6: Computing limits by taking logarithms

$$\lim_{n \rightarrow +\infty} (3n)^{(-3n^2+7)}$$

taking logarithms we have

$$\underbrace{(-3n^2+7)}_{b_n} \log \underbrace{3n}_{a_n}$$

since

$$\lim_{n \rightarrow +\infty} (-3n^2+7) \log(3n) = -\infty \cdot +\infty = -\infty$$

then

$$\lim_{n \rightarrow +\infty} (3n)^{(-3n^2+7)} = 0$$

Example 7: Computing limits by taking logarithms (alternative method)

$$\lim_{n \rightarrow +\infty} (3n)^{(-3n^2+7)} = \lim_{n \rightarrow +\infty} e^{\log((3n)^{(-3n^2+7)})} = \lim_{n \rightarrow +\infty} e^{(-3n^2+7) \log(3n)}$$

since

$$\lim_{n \rightarrow +\infty} (-3n^2+7) \log(3n) = -\infty \cdot +\infty = -\infty$$

then

$$\lim_{n \rightarrow +\infty} (3n)^{(-3n^2+7)} = e^{-\infty} = 0$$

We also have the following **indeterminate forms**:

$$1^\infty, \quad 0^0, \quad \infty^0$$

Taking logarithms, they correspond to the indeterminate form

$$0 \cdot \infty$$

since:

$$\log(1^\infty) = \infty \log 1 = \infty \cdot 0$$

$$\log(0^0) = 0 \log 0 = 0 \cdot -\infty$$

$$\log(+\infty^0) = 0 \log(+\infty) = 0 \cdot +\infty$$

Finally, since $-\infty^0 = -1 \cdot (+\infty^0)$, we have

$$-1 \cdot (\log(+\infty^0)) = -1 \cdot (0 \log +\infty) = -1 \cdot (0 \cdot +\infty)$$

Limits of the form

$$0^{+\infty} \text{ and } 0^{-\infty}$$

are not indeterminate forms. We have:

$$0^{+\infty} = 0 \text{ and } 0^{-\infty} = +\infty.$$

since, taking logarithms, we have

$$\log(0^{+\infty}) = +\infty \cdot \log 0 = +\infty \cdot -\infty = -\infty \quad \text{and} \quad e^{-\infty} = 0$$

$$\log(0^{-\infty}) = -\infty \cdot \log 0 = -\infty \cdot -\infty = +\infty \quad \text{and} \quad e^{+\infty} = +\infty$$

Example 8: Limits of the forms $0^{+\infty}$ and $0^{-\infty}$

- Example of the case $0^{+\infty} = 0$:

$$\lim_{n \rightarrow +\infty} \left(\frac{1}{n}\right)^{\log n} = 0^{+\infty} = 0.$$

Indeed we have

$$\left(\frac{1}{n}\right)^{\log n} = e^{\log\left(\frac{1}{n}\right)^{\log n}} = e^{\log n \log \frac{1}{n}}$$

hence

$$\lim_{n \rightarrow +\infty} \left(\frac{1}{n}\right)^{\log n} = \lim_{n \rightarrow +\infty} e^{\overbrace{\log n}^{\rightarrow +\infty} \overbrace{\log \frac{1}{n}}^{\rightarrow -\infty}} = e^{-\infty} = 0.$$

- Example of the case $0^{-\infty} = +\infty$:

$$\lim_{n \rightarrow +\infty} \left(\frac{1}{n}\right)^{-\log n} = 0^{-\infty} = +\infty.$$

Indeed we have

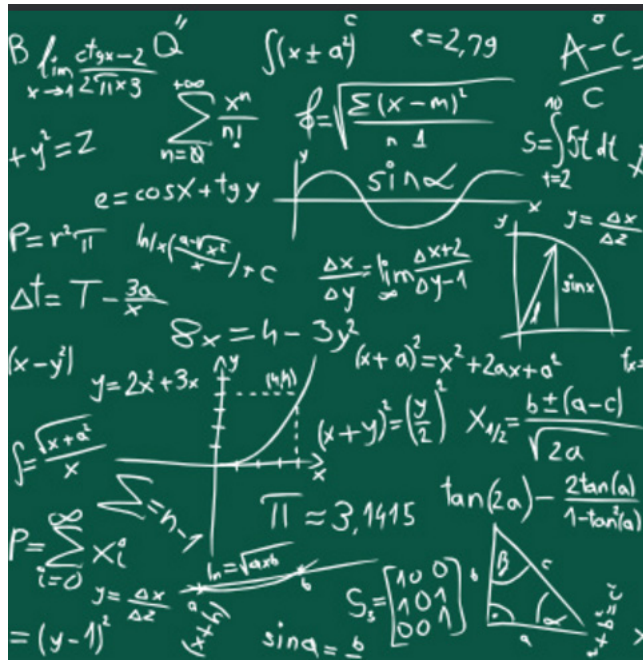
$$\left(\frac{1}{n}\right)^{-\log n} = e^{\log\left(\frac{1}{n}\right)^{-\log n}} = e^{-\log n \log \frac{1}{n}}$$

hence

$$\lim_{n \rightarrow +\infty} \left(\frac{1}{n}\right)^{-\log n} = \lim_{n \rightarrow +\infty} e^{\overbrace{-\log n}^{\rightarrow -\infty} \overbrace{\log \frac{1}{n}}^{\rightarrow -\infty}} = e^{+\infty} = +\infty.$$

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Euler's number



- **Author:** Fabio Furini.
- **Book:** “*Analisi matematica 1*”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Euler's number e (Napier's constant)

Theorem 1

The sequence

$$a_n = \left(1 + \frac{1}{n}\right)^n \quad \text{with } n \geq 1$$

is convergent.

Proof. We will prove that the sequence $\{a_n\}$ is monotone increasing ($a_n \geq a_{n-1}, \forall n \in \mathbb{N}, n \geq 2$) and bounded ($m \leq a_n \leq M, \forall n \in \mathbb{N}, n \geq 1$); hence it is convergent by the monotone sequence theorem.

To prove that $\{a_n\}$ is monotone increasing, we study, for $n \geq 2$, the ratio:

$$\begin{aligned} \frac{a_n}{a_{n-1}} &= \frac{\left(1 + \frac{1}{n}\right)^n}{\left(1 + \frac{1}{n-1}\right)^{n-1}} = \frac{\left(\frac{n+1}{n}\right)^n}{\left(\frac{n}{n-1}\right)^{n-1}} \\ &= \left(\frac{n+1}{n}\right)^n \frac{1}{\left(\frac{n}{n-1}\right)^{n-1}} = \left(\frac{n^2-1}{n^2}\right)^n \frac{1}{\left(\frac{n-1}{n}\right)} \\ &= \frac{\left(1 - \frac{1}{n^2}\right)^n}{1 - \frac{1}{n}} \geq \frac{1 - n \cdot \frac{1}{n^2}}{1 - \frac{1}{n}} = 1 \end{aligned}$$

where for the “ $\geq$ ” we applied Bernoulli's inequality:

$$(1+x)^n \geq 1 + nx \quad \text{with } x = -\frac{1}{n^2} \geq -1 \quad \text{and } n \geq 2$$

Hence we have

$$\frac{a_n}{a_{n-1}} \geq 1$$

i.e., $a_n \geq a_{n-1}$ and the sequence is monotone increasing.

To prove that $\{a_n\}$ is bounded, we observe that, since $a_1 = 2$, it follows that $a_n \geq 2, \forall n \geq 1$.

Now consider the sequence

$$b_n = \left(1 + \frac{1}{n}\right)^{n+1} \quad \text{note that } b_n = a_n \left(1 + \frac{1}{n}\right) \quad \text{therefore } b_n > a_n, \forall n \in \mathbb{N}, n \geq 1$$

□

Proof. To prove that $\{b_n\}$ is monotone decreasing, we study, for $n \geq 2$, the ratio:

$$\begin{aligned}
 \frac{b_n}{b_{n-1}} &= \frac{\left(1 + \frac{1}{n}\right)^{n+1}}{\left(1 + \frac{1}{n-1}\right)^n} = \frac{\left(\frac{n+1}{n}\right)^{n+1}}{\left(\frac{n}{n-1}\right)^n} \\
 &= \left(\frac{\frac{n+1}{n}}{\frac{n}{n-1}}\right)^n \left(\frac{n+1}{n}\right) = \frac{1}{\left(\frac{n^2}{n^2-1}\right)^n} \left(\frac{n+1}{n}\right) \\
 &= \frac{1}{\left(1 + \frac{1}{n^2-1}\right)^n} \left(\frac{n+1}{n}\right) \leq \frac{1}{\left(1 + \frac{n}{n^2-1}\right)} \left(\frac{n+1}{n}\right) \\
 &< \frac{1}{\left(1 + \frac{1}{n}\right)} \left(1 + \frac{1}{n}\right) = 1
 \end{aligned}$$

where for the “ $\leq$ ” we applied Bernoulli’s inequality:

$$(1+x)^n \geq 1+n x \text{ with } x = \frac{1}{n^2-1} \geq -1 \text{ and } n \geq 2$$

and for the “ $<$ ” we applied the inequality

$$\frac{n}{n^2-1} > \frac{1}{n} \text{ since } n^2 > n^2-1 \text{ for } n \geq 2$$

Hence we have proved that

$$\frac{b_n}{b_{n-1}} < 1 \text{ hence } b_n < b_{n-1}$$

and the sequence $\{b_n\}$ is (strictly) monotone decreasing.

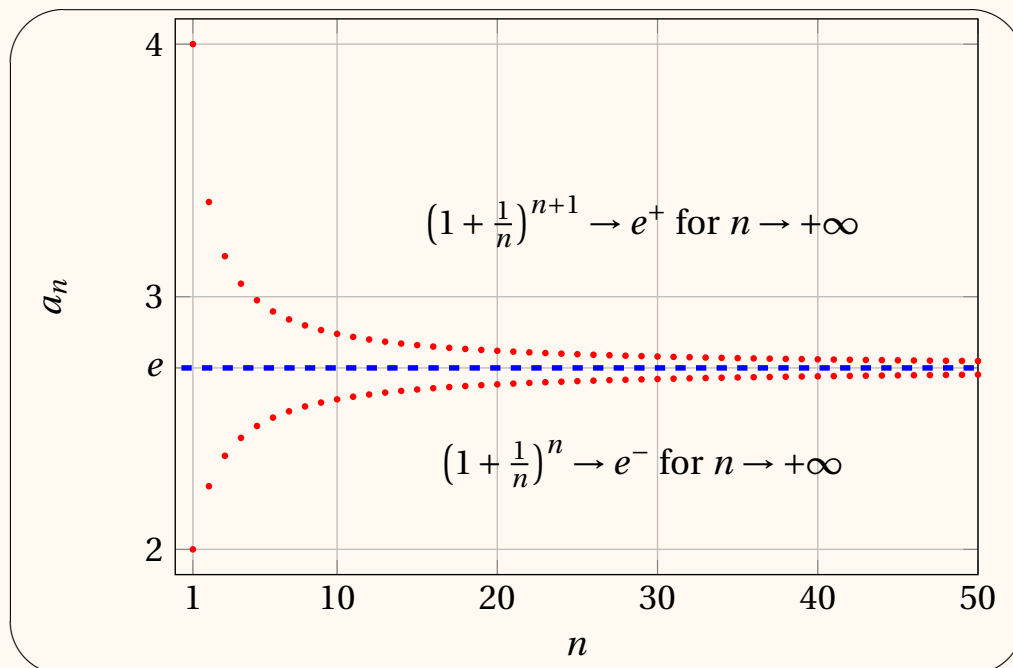
Since $b_1 = 4$, we therefore obtain

$$a_n < b_n \leq b_1 = 4, \forall n \geq 1$$

and $\{a_n\}$ is bounded.

□

Example 1: Graphs of the sequences $\{a_n\}$ and $\{b_n\}$



- The limit of the sequence a_n just studied is an **irrational** number that is very important in mathematics. This limit is denoted by the letter e (**Euler's number**, or Napier's constant) and its decimal representation begins as follows:

2.7182818284...

By definition, we have:

$$e = \lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n \quad (1)$$

- This number is very often used as the base of logarithms, which, when this base is used, are called natural or Napierian logarithms (after the mathematician **John Napier**) and are denoted simply by the symbol $\log$ (or $\ln$) without indicating the base.
- Consequently, we have:

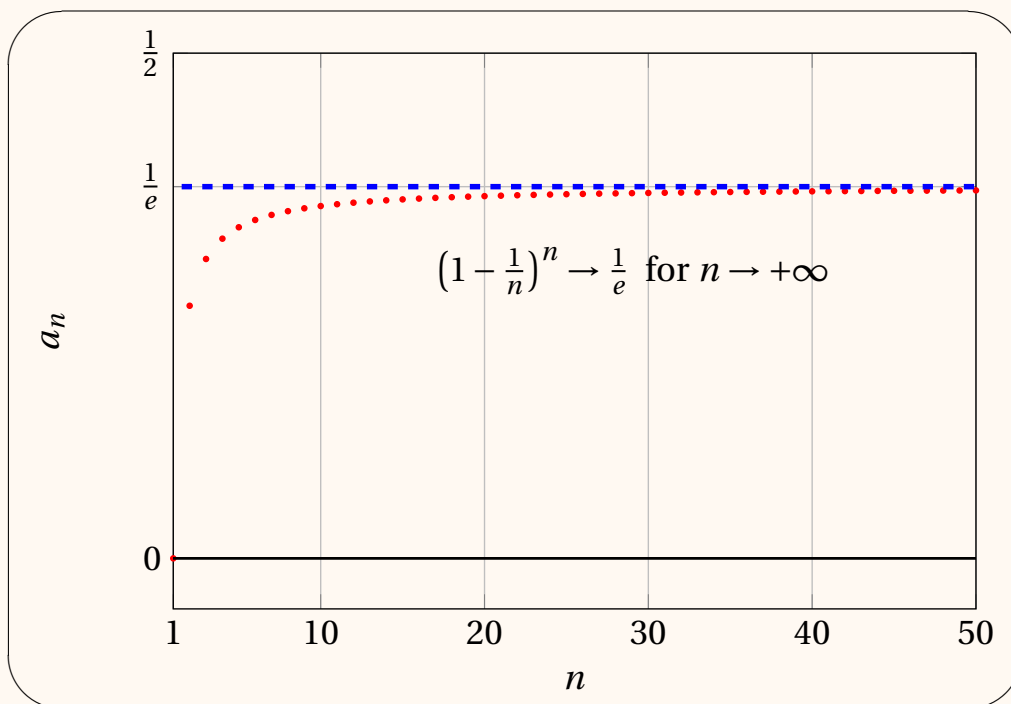
$$\lim_{n \rightarrow +\infty} n \log \left(1 + \frac{1}{n}\right) = \lim_{n \rightarrow +\infty} \log \left(1 + \frac{1}{n}\right)^n = \log e = 1$$

Example 2: Computing limits with the sequence tending to e

$$\lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n}\right)^n = \frac{1}{e}$$

Since:

$$\left(\frac{n-1}{n}\right)^n = \frac{1}{\left(\frac{n}{n-1}\right)^n} = \frac{1}{\left(1 + \frac{1}{n-1}\right)^n} = \underbrace{\frac{1}{\left(1 + \frac{1}{n-1}\right)^{n-1}}}_{\rightarrow e} \cdot \underbrace{\frac{1}{\left(1 + \frac{1}{n-1}\right)}}_{\rightarrow 1}$$



Theorem 2

Let $\{c_n\}$ be any divergent sequence (to $+\infty$ or $-\infty$); then

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{c_n}\right)^{c_n} = e, \quad \lim_{n \rightarrow +\infty} \left(1 - \frac{1}{c_n}\right)^{c_n} = \frac{1}{e} \quad (2)$$

Proof. Recall that, given $x \in \mathbb{R}$, its integer part, denoted by $\lfloor x \rfloor$, is the largest integer not exceeding x .

If $c_n \rightarrow +\infty$, since $\lfloor c_n \rfloor > c_n - 1$, by comparison we also have $\lfloor c_n \rfloor \rightarrow +\infty$. Hence, by (1) and by the definition of limit we have

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{\lfloor c_n \rfloor + 1}\right)^{\lfloor c_n \rfloor + 1} = \lim_{n \rightarrow +\infty} \left(1 + \frac{1}{\lfloor c_n \rfloor}\right)^{\lfloor c_n \rfloor} = e$$

Using:

$$\lfloor c_n \rfloor \leq c_n < \lfloor c_n \rfloor + 1$$

we obtain

$$\left(1 + \frac{1}{c_n}\right)^{c_n} < \left(1 + \frac{1}{\lfloor c_n \rfloor}\right)^{\lfloor c_n \rfloor + 1} = \underbrace{\left(1 + \frac{1}{\lfloor c_n \rfloor}\right)^{\lfloor c_n \rfloor}}_{\rightarrow e} \cdot \underbrace{\left(1 + \frac{1}{\lfloor c_n \rfloor}\right)}_{\rightarrow 1}$$

and also

$$\left(1 + \frac{1}{c_n}\right)^{c_n} > \left(1 + \frac{1}{\lfloor c_n \rfloor + 1}\right)^{\lfloor c_n \rfloor} = \underbrace{\left(1 + \frac{1}{\lfloor c_n \rfloor + 1}\right)^{\lfloor c_n \rfloor + 1}}_{\rightarrow e} \cdot \underbrace{\left(1 + \frac{1}{\lfloor c_n \rfloor + 1}\right)^{-1}}_{\rightarrow 1}$$

hence the first limit of the theorem follows from the comparison theorem.

If instead $c_n \rightarrow -\infty$, then the sequence $d_n = -c_n \rightarrow +\infty$, and we have

$$\left(1 + \frac{1}{c_n}\right)^{c_n} = \left(1 - \frac{1}{d_n}\right)^{-d_n} = \left(\frac{d_n}{d_n - 1}\right)^{d_n} = \left(1 + \frac{1}{d_n - 1}\right)^{d_n - 1} \cdot \left(1 + \frac{1}{d_n - 1}\right)$$

Since $d_n - 1 \rightarrow +\infty$, the claim follows from the previous case.

The second limit of the theorem is proved analogously.

□

This theorem is useful for computing limits involving the indeterminate form 1^∞

Example 3: Computing limits with the sequence tending to e

We compute

$$\lim_{n \rightarrow +\infty} \left(\frac{n}{3+n} \right)^{5n+1} = [1^\infty]$$

since:

$$\lim_{n \rightarrow +\infty} \frac{n}{3+n} = \lim_{n \rightarrow +\infty} 1 - \frac{3}{3+n} = 1, \quad \lim_{n \rightarrow +\infty} 5n+1 = +\infty$$

We rewrite the sequence as follows:

$$\begin{aligned} \left(\frac{n}{3+n} \right)^{5n+1} &= \left(\frac{3+n}{n} \right)^{-(5n+1)} = \frac{1}{\left(1 + \frac{3}{n} \right)^{5n+1}} \\ &= \frac{1}{\left(\left(1 + \frac{3}{n} \right)^{\frac{n}{3}} \right)^{\frac{3(5n+1)}{n}}} = \frac{1}{\left(\left(1 + \frac{1}{\frac{n}{3}} \right)^{\frac{n}{3}} \right)^{\frac{3(5n+1)}{n}}} \end{aligned}$$

and we consider the denominator

$$\underbrace{\left(1 + \frac{1}{\frac{n}{3}} \right)^{\frac{n}{3}}}_{\rightarrow e}^{\frac{3(5n+1)}{n}} \rightarrow e^{15} \quad \text{since } \frac{n}{3} \rightarrow +\infty \text{ for } n \rightarrow +\infty$$

Summarizing, we have:

$$\lim_{n \rightarrow +\infty} \left(\frac{n}{3+n} \right)^{5n+1} = \frac{1}{e^{15}}$$

Alternative method:

$$\lim_{n \rightarrow +\infty} \left(\frac{n}{3+n} \right)^{5n+1} = \lim_{n \rightarrow +\infty} \left(1 - \frac{3}{3+n} \right)^{5n+1} = \lim_{n \rightarrow +\infty} \left(\underbrace{\left(1 - \frac{1}{\frac{3+n}{3}} \right)^{\frac{3+n}{3}}}_{\rightarrow \frac{1}{e}}^{\frac{3(5n+1)}{3+n}} \right) = \frac{1}{e^{15}}$$

since

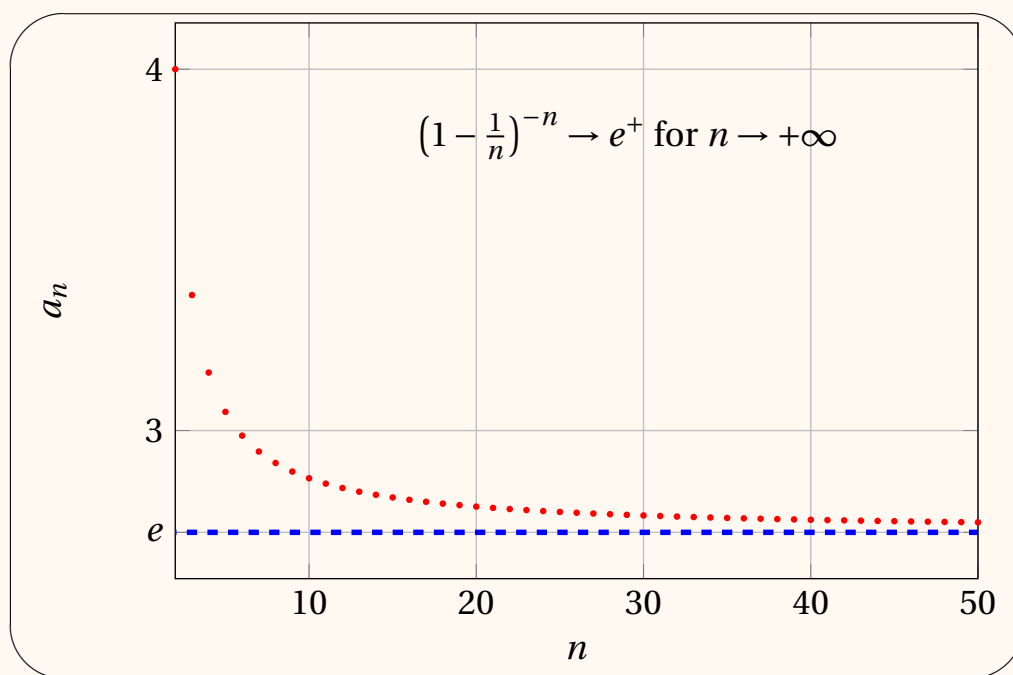
$$\frac{3+n}{3} \rightarrow +\infty \text{ for } n \rightarrow +\infty$$

Example 4: Computing limits with the sequence tending to e

$$\lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n}\right)^{-n} = e$$

Since:

$$\left(\frac{n-1}{n}\right)^{-n} = \left(\frac{n}{n-1}\right)^n = \left(1 + \frac{1}{n-1}\right)^n = \underbrace{\left(1 + \frac{1}{n-1}\right)^{n-1}}_{\rightarrow e} \cdot \underbrace{\left(1 + \frac{1}{n-1}\right)}_{\rightarrow 1}$$



Example 5: Computing limits with the sequence tending to e

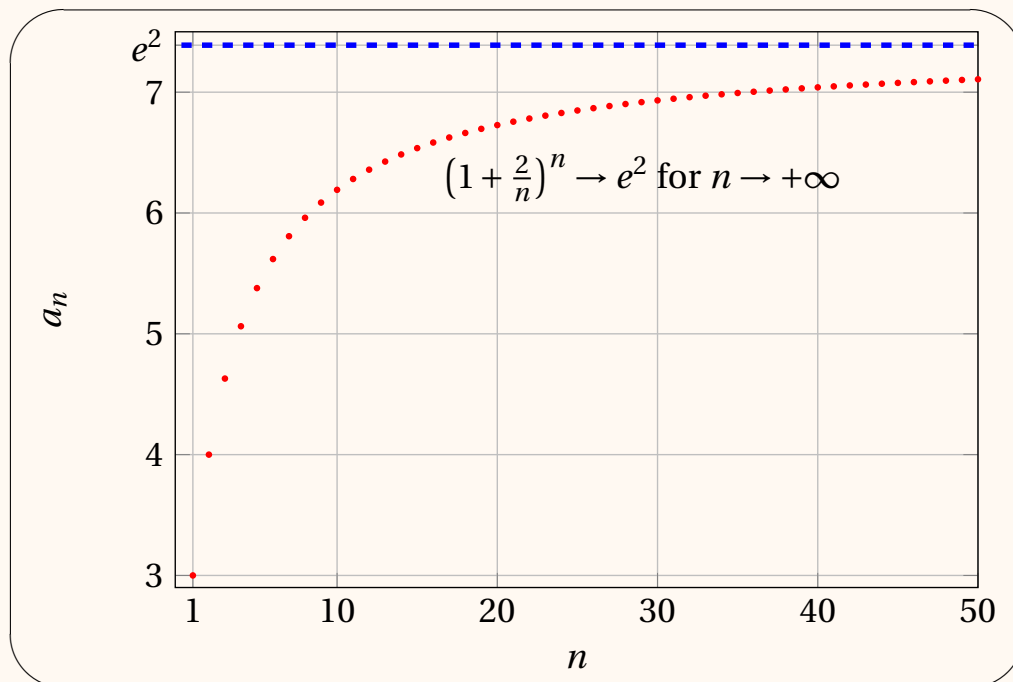
$$\lim_{n \rightarrow +\infty} \left(1 + \frac{\alpha}{n}\right)^n = e^\alpha \quad \text{with } \alpha \in \mathbb{R}$$

Since:

$$\left(1 + \frac{\alpha}{n}\right)^n = \left(1 + \frac{1}{\frac{n}{\alpha}}\right)^n = \left(\underbrace{\left(1 + \frac{1}{\frac{n}{\alpha}}\right)^{\frac{n}{\alpha}}}_{\rightarrow e}\right)^\alpha$$

For example, with $\alpha = 2$ we have:

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{2}{n}\right)^n = e^2$$



2 Euler's number in finance

- Suppose we own a capital of unit value and an annual interest rate t . If the interest is paid annually, after one year the capital owned will be

$$1 + t \cdot 1 = 1 + t$$

- If instead the interest is paid monthly, we will have

- after the first month, a capital equal to

$$1 + \frac{t}{12} \cdot 1 = \underbrace{1 + \frac{t}{12}}_{\alpha}$$

- after the second month, a capital equal to

$$\underbrace{1 + \frac{t}{12}}_{\alpha} + \frac{t}{12} \underbrace{\left(1 + \frac{t}{12}\right)}_{\alpha} = \underbrace{\left(1 + \frac{t}{12}\right)^2}_{\beta}$$

- after the third month, a capital equal to

$$\underbrace{\left(1 + \frac{t}{12}\right)^2}_{\beta} + \frac{t}{12} \underbrace{\left(1 + \frac{t}{12}\right)^2}_{\beta} = \left(1 + \frac{t}{12}\right)^3$$

- at the end of the year we will have a capital equal to

$$\left(1 + \frac{t}{12}\right)^{12}$$

- If the interest is computed every n -th of a year, at the end we will have a capital equal to

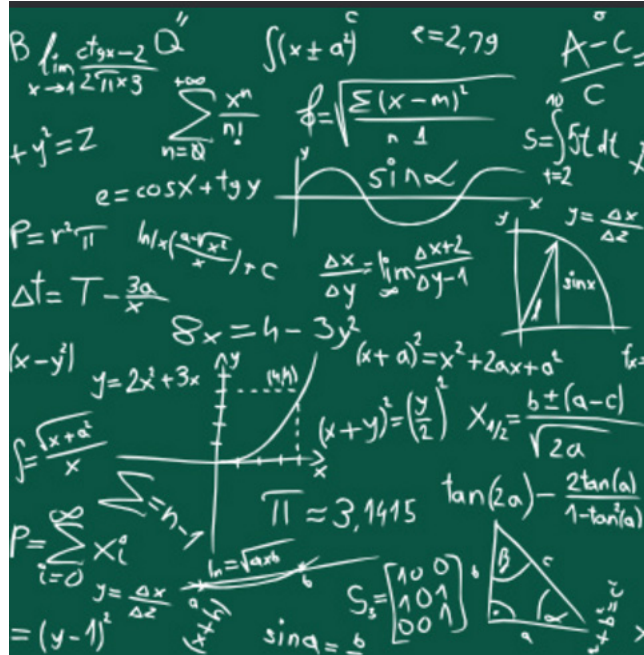
$$\left(1 + \frac{t}{n}\right)^n$$

- For $t = 1$ (a 100% return) we obtain exactly the sequence that defines e

Hence, even if the interest is paid an infinite number of times per year, the capital does not grow to infinity but tends to e , since:

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n = e$$

Comparisons and asymptotic estimates



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

Contents

1 Comparisons and asymptotic estimates

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1 Comparisons and asymptotic estimates

- We have seen that a sequence tending to 0 is an **infinitesimal**; a sequence that diverges (to $+\infty$, to $-\infty$) is called an **infinity** (infinite quantity).
- When two sequences are both infinitesimals or both infinities, it is useful to be able to compare them, to understand which of the two tends “**more rapidly**” to 0 or to infinity.

Example 1: Infinities

Examples of infinities are the following sequences:

$$\{\log n\}, \quad \{\sqrt{n}\}, \quad \{n^2\}, \quad \{2^n\}$$

Example 2: Infinitesimals

Examples of infinitesimals are the following sequences:

$$\left\{\frac{1}{\log n}\right\}, \quad \left\{\frac{1}{\sqrt{n}}\right\}, \quad \left\{\frac{1}{n^2}\right\}, \quad \left\{\frac{1}{2^n}\right\}$$

- Let $\{a_n\}$ and $\{b_n\}$ be **two infinities**; considering the limit of their ratio we have 4 cases:

$$\lim_{n \rightarrow +\infty} \frac{a_n}{b_n} = \begin{cases} 0 & \text{case 1): } \{a_n\} \text{ is an infinity of } \mathbf{lower\ order} \text{ than } \{b_n\} \\ l \in \mathbb{R}, l \neq 0 & \text{case 2): } \{a_n\} \text{ and } \{b_n\} \text{ are infinities of the } \mathbf{same\ order} \\ \pm\infty & \text{case 3): } \{a_n\} \text{ is an infinity of } \mathbf{higher\ order} \text{ than } \{b_n\} \\ \text{nonexistent} & \text{case 4): } \{a_n\} \text{ and } \{b_n\} \text{ are not comparable} \end{cases}$$

- Let $\{a_n\}$ and $\{b_n\}$ be **two infinitesimals** (with b_n eventually different from zero); considering the limit of their ratio we have 4 cases:

$$\lim_{n \rightarrow +\infty} \frac{a_n}{b_n} = \begin{cases} 0 & \text{case 1): } \{a_n\} \text{ is an infinitesimal of } \mathbf{higher\ order} \text{ than } \{b_n\} \\ l \in \mathbb{R}, l \neq 0 & \text{case 2): } \{a_n\} \text{ and } \{b_n\} \text{ are infinitesimals of the } \mathbf{same\ order} \\ \pm\infty & \text{case 3): } \{a_n\} \text{ is an infinitesimal of } \mathbf{lower\ order} \text{ than } \{b_n\} \\ \text{nonexistent} & \text{case 4): } \{a_n\} \text{ and } \{b_n\} \text{ are not comparable} \end{cases}$$

- The case

$$\frac{a_n}{b_n} \rightarrow 1$$

is particularly important: in this case we say that the two sequences $\{a_n\}$ and $\{b_n\}$ are **asymptotic** (asymptotically equivalent).

- To indicate this, we write

$$a_n \sim b_n$$

(read: a_n is asymptotic to b_n)

- The asymptotic symbol is very useful in computing limits, thanks to the following **properties**:

Proposition 1: Asymptotic behavior

1. If $a_n \sim b_n$, the two sequences have the same behavior:

- either they converge to the same limit,
- or they both diverge to $\pm\infty$,
- or neither of them has a limit.

2. We can write chains of asymptotic relations, i.e.:

$$\text{if } a_n \sim b_n \sim \dots \sim c_n \text{ then } a_n \sim c_n$$

3. An expression made of a product or quotient of several factors can be estimated factor by factor:

$$\text{if } a_n \sim a'_n, b_n \sim b'_n, c_n \sim c'_n \text{ then } \frac{a_n b_n}{c_n} \sim \frac{a'_n b'_n}{c'_n}$$

- **Warning:** the same does not hold for sums or for exponentials.

Proof. 1. We prove the first statement

if $a_n \sim b_n$ then $\{a_n\}$ and $\{b_n\}$ have the same behavior

- If $a_n \rightarrow l \in \mathbb{R}$, since

$$b_n = \frac{b_n}{a_n} \cdot a_n \text{ and } \frac{b_n}{a_n} \rightarrow 1 \text{ (by definition of asymptotic),}$$

then by the theorem on the algebra of limits we have

$$b_n \rightarrow 1 \cdot l = l.$$

- With the same steps, the theorem on the partial arithmetization of the infinity symbol allows us to conclude that

$$\text{if } a_n \rightarrow \pm\infty \text{ and } a_n \sim b_n \text{ then } b_n \rightarrow \pm\infty.$$

Observing that the asymptotic relation is symmetric, what we have just proved also shows that if $\{b_n\}$ converges (diverges), then $\{a_n\}$ converges (diverges) as well.

- We conclude that if $\{a_n\}$ is irregular, then $\{b_n\}$ is irregular as well, because if, by contradiction, it were not, then by what we have just proved $\{a_n\}$ would also be convergent or divergent.

2. We prove the transitivity of the asymptotic relation:

$$\text{if } a_n \sim b_n \sim c_n \text{ then } a_n \sim c_n.$$

The hypotheses mean that

$$\frac{a_n}{b_n} \rightarrow 1 \text{ and } \frac{b_n}{c_n} \rightarrow 1$$

Then by the theorem on the algebra of limits we have

$$\frac{a_n}{c_n} = \frac{a_n}{b_n} \cdot \frac{b_n}{c_n} \rightarrow 1.$$

3. The third property is proved analogously.

□

Example 3: Asymptotic sequences by applying the definition

We prove that:

$$\log n \sim \log(n+1)$$

Factoring out n in the second sequence we obtain

$$\log(n+1) = \log\left(n\left(1 + \frac{1}{n}\right)\right) = \log n + \log\left(1 + \frac{1}{n}\right)$$

Hence

$$\lim_{n \rightarrow +\infty} \frac{\log n}{\log(n+1)} = \lim_{n \rightarrow +\infty} \frac{\log n}{\underbrace{\log n + \log\left(1 + \frac{1}{n}\right)}_{\rightarrow 0}} = 1$$

- A typical way to show that

$$a_n \sim b_n$$

consists in writing

$$a_n = b_n c_n \text{ with } c_n \rightarrow 1.$$

That is, decomposing $\{a_n\}$ into the product of a sequence $\{b_n\}$ and a sequence $\{c_n\}$ that tends to 1.

Example 4: Asymptotic sequences via the decomposition method

Example:

$$\underbrace{2n^2 + 3n + 1}_{a_n} = \underbrace{2n^2}_{b_n} \underbrace{\left(1 + \frac{3}{2n} + \frac{1}{2n^2}\right)}_{c_n} \sim 2n^2 \quad \text{hence } a_n \sim b_n$$

since

$$\underbrace{\left(1 + \frac{3}{2n} + \frac{1}{2n^2}\right)}_{c_n} \rightarrow 1$$

Example 5: Computing limits with asymptotic estimates

We compute the limit

$$\lim_{n \rightarrow +\infty} \frac{2n^3 + 4n + 1}{5(n+1)^3} = \left[\frac{\infty}{\infty} \right] \quad (\text{indeterminate form})$$

Proceeding as before we can write

$$\begin{aligned} \underbrace{2n^3 + 4n + 1}_{a_n} &= \underbrace{2n^3}_{a'_n} \underbrace{\left(1 + \frac{2}{n^2} + \frac{1}{2n^3}\right)}_{\rightarrow 1} \sim 2n^3 \\ 5(n+1)^3 &= \underbrace{5(n^3 + 3n^2 + 3n + 1)}_{c_n} = \underbrace{5n^3}_{c'_n} \underbrace{\left(1 + \frac{3}{n} + \frac{3}{n^2} + \frac{1}{n^3}\right)}_{\rightarrow 1} \sim 5n^3 \end{aligned}$$

Using point 3 of Proposition (??) on asymptotic behavior we can write:

$$\text{if } a_n \sim a'_n, c_n \sim c'_n \quad \text{then} \quad \frac{a_n}{c_n} \sim \frac{a'_n}{c'_n}$$

and obtain

$$\frac{2n^3 + 4n + 1}{5(n+1)^3} \sim \frac{2n^3}{5n^3}$$

that is, the two sequences have the same behavior. Hence

$$\lim_{n \rightarrow +\infty} \frac{2n^3 + 4n + 1}{5(n+1)^3} = \lim_{n \rightarrow +\infty} \frac{2n^3}{5n^3} = \frac{2}{5}$$

- Another way to show that

$$a_n \sim b_n$$

is to use the **substitution principle**.

- For example, knowing that

$$\lim_{n \rightarrow +\infty} \underbrace{\log n}_{a_n} \rightarrow +\infty$$

we can state

$$\lim_{n \rightarrow +\infty} \log c_n \rightarrow +\infty$$

where $\{c_n\}$ is any sequence diverging to $+\infty$, hence

$$\underbrace{\log n}_{a_n} \sim \underbrace{\log c_n}_{b_n}$$

Example 6: Computing limits with the substitution principle and asymptotic estimates

We compute the limit

$$\lim_{n \rightarrow +\infty} \log_3(n^2 + 4n + 1)$$

using the substitution principle and defining

$$c_n = n^2 + 4n + 1 \quad \text{we have} \quad \lim_{n \rightarrow +\infty} c_n = +\infty$$

then

$$\log_3 n \sim \log_3(n^2 + 4n + 1)$$

and hence

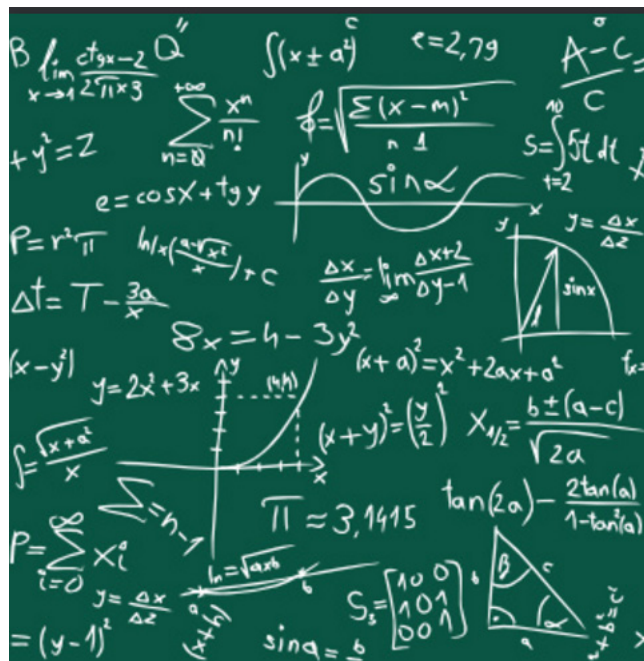
$$\lim_{n \rightarrow +\infty} \log_3(n^2 + 4n + 1) = \lim_{n \rightarrow +\infty} \log_3 n = +\infty$$

The fact that the asymptotic relation satisfies the 3 properties:

1. *Reflexive*: $a_n \sim a_n$
2. *Symmetric*: if $a_n \sim b_n$ then $b_n \sim a_n$
3. *Transitive*: if $a_n \sim b_n$ and $b_n \sim c_n$ then $a_n \sim c_n$

means that “**asymptotic**” is an **equivalence relation**.

Hierarchies of infinities and the ratio test



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1	Hierarchies of infinities for sequences, part 1 and part 2	2
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1 Hierarchies of infinities for sequences, part 1 and part 2

Theorem 1: Hierarchy of infinities (part I)

$$\lim_{n \rightarrow +\infty} \frac{\log_a n}{n^\alpha} = 0 \quad (1)$$

for every $a > 1$ and $\alpha > 0$.

Proof. We begin by establishing a useful inequality between any positive real number and its logarithm.

For $x \in \mathbb{R}$, $x > 0$, let k be the integer part of x , i.e., the integer k for which we have

$$k \leq x < k + 1, \text{ moreover we have } 2^x \geq 2^k = (1 + 1)^k \geq 1 + k > x,$$

the first inequality follows from the monotonicity of the exponential function, the second from the binomial expansion (or from Bernoulli's inequality). Taking logarithms to base $a > 1$, we obtain

$$\log_a x < x \log_a 2 \quad (\text{for every } x \in \mathbb{R}, x > 0).$$

We now apply this inequality to the number $x = n^{\alpha/2}$, and we have

$$\frac{\alpha}{2} \log_a n < n^{\alpha/2} \log_a 2 \quad \text{and} \quad \frac{\log_a n}{n^{\alpha/2}} < \frac{2}{\alpha} \log_a 2,$$

hence

$$\frac{\log_a n}{n^\alpha} = \frac{\log_a n}{n^{\alpha/2}} \frac{1}{n^{\alpha/2}} \leq \underbrace{\frac{2}{\alpha} \log_a 2}_{\text{constant}} \frac{1}{n^{\alpha/2}} \quad \text{and} \quad \frac{1}{n^{\alpha/2}} \rightarrow 0.$$

By the corollary of the comparison theorem, the claim follows. □

Theorem 2: Hierarchy of infinities (part II)

$$\lim_{n \rightarrow +\infty} \frac{n^\alpha}{a^n} = 0 \quad (2)$$

for every $a > 1$ and $\alpha > 0$.

Proof. We use the hierarchy of infinities theorem (part I), replacing the integer n with the integer 2^n :

$$0 = \lim_{n \rightarrow +\infty} \frac{\log_a(2^n)}{(2^n)^\alpha} = \lim_{n \rightarrow +\infty} \frac{n \overbrace{\log_a 2}^{\text{constant}}}{(2^\alpha)^n} \text{ hence } \frac{n}{(2^\alpha)^n} \rightarrow 0.$$

If now $a > 1$ is fixed, choosing $\alpha > 0$ such that $2^\alpha = a$ we obtain that $\frac{n}{a^n} \rightarrow 0$, that is, the second relation in the special case where n is raised to the exponent 1. The general case follows from the identity:

$$\frac{n^\alpha}{a^n} = \left(\frac{n}{a^{n/\alpha}} \right)^\alpha = \left(\frac{n}{(a^{1/\alpha})^n} \right)^\alpha$$

Indeed, by the previous result $\frac{n}{(a^{1/\alpha})^n} \rightarrow 0$ (the base $a^{1/\alpha}$ is still a number > 1), hence it follows that

$$\left(\frac{n}{(a^{1/\alpha})^n} \right)^\alpha \rightarrow 0 \text{ and from this the claim.}$$

□

- These limits describe the “speed” at which logarithms (with base > 1), powers (with exponent > 0) and exponentials (with base > 1) go to infinity. Logarithms with base > 1 go more slowly than any power with exponent > 0 ; powers with exponent > 0 go more slowly than any exponential with base > 1 .
- Exponentials with base > 1 are infinities of higher order than powers with exponent > 0 and than logarithms with base > 1 . Powers with exponent > 0 are infinities of higher order than logarithms with base > 1 .

Example 1: Computing limits using the hierarchy of infinities

We compute the limit

$$\lim_{n \rightarrow +\infty} \sqrt[n]{n} = [\infty^0] \quad (\text{indeterminate form})$$

Now we write

$$\sqrt[n]{n} = n^{\frac{1}{n}} = e^{\log n^{1/n}} = e^{\frac{\log n}{n}}$$

and studying the sequence in the exponent

$$a_n = \frac{\log n}{n} \text{ we have } a_n \rightarrow 0$$

thanks to the hierarchy of infinities theorem. Hence we have:

$$\lim_{n \rightarrow +\infty} \sqrt[n]{n} = \lim_{n \rightarrow +\infty} e^{\frac{\log n}{n}} = 1$$

Example 2: Computing limits using the hierarchy of infinities

We compute the limit

$$\lim_{n \rightarrow +\infty} \frac{2^n + n}{2^{n+1}} = \left[\frac{\infty}{\infty} \right] \quad (\text{indeterminate form})$$

We can write:

$$\underbrace{2^n + n}_{a_n} = \underbrace{2^n}_{a'_n} \underbrace{\left(1 + \frac{n}{2^n}\right)}_{\rightarrow 1} \sim 2^n$$

since, thanks to the hierarchy of infinities theorem, we have

$$\frac{n}{2^n} \rightarrow 0,$$

i.e., 2^n is an infinity of higher order than n . Hence we can write:

$$\frac{2^n + n}{2^{n+1}} \sim \frac{2^n}{2^{n+1}}$$

Now, since

$$\frac{2^n}{2^{n+1}} = \frac{2^n}{2^n \cdot 2} = \frac{1}{2}$$

then

$$\lim_{n \rightarrow +\infty} \frac{2^n + n}{2^{n+1}} = \frac{1}{2}$$

2 The ratio test theorem

Theorem 3: Ratio test

Let $\{a_n\}$ be a positive sequence (i.e., $a_n > 0$ for every n).

$$\text{If there exists } \lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} = l \text{ and } \begin{cases} l < 1, \text{ then } a_n \rightarrow 0 \\ l > 1 \text{ (or } l = +\infty), \text{ then } a_n \rightarrow +\infty \end{cases}$$

- The previous theorem reduces the study of the limit of a positive sequence $\{a_n\}$ to the computation of the limit of another sequence, the sequence (of ratios)

$$n \mapsto \frac{a_{n+1}}{a_n}$$

In some cases the latter is easier to study than the original one, as we will see in the examples.

Note that in the case $l = 1$ the theorem does not allow us to conclude anything.

Proof.

1. Suppose that:

$$\frac{a_{n+1}}{a_n} \rightarrow l < 1$$

Then, for every $\varepsilon > 0$, we have, for $n \geq n(\varepsilon)$,

$$\frac{a_{n+1}}{a_n} < l + \varepsilon$$

We can then write the chain of inequalities:

$$a_{n(\varepsilon)+1} < (l + \varepsilon) a_{n(\varepsilon)}, \quad a_{n(\varepsilon)+2} < (l + \varepsilon) \underbrace{a_{n(\varepsilon)+1}}_{< (l + \varepsilon) a_{n(\varepsilon)}} < (l + \varepsilon)^2 a_{n(\varepsilon)}, \quad \dots$$

hence

$$a_{n(\varepsilon)+k} < (l + \varepsilon)^k a_{n(\varepsilon)}$$

Choosing ε small enough to have $l + \varepsilon < 1$, we have

$$(l + \varepsilon)^k \rightarrow 0 \text{ for } k \rightarrow +\infty$$

On the other hand, $n(\varepsilon)$ is fixed and consequently $a_{n(\varepsilon)}$ is fixed as well; hence for k large enough the right-hand side (and therefore the left-hand side) is as small as we like. This proves the first claim, that is: $a_n \rightarrow 0$.

2. Suppose that:

$$\frac{a_{n+1}}{a_n} \rightarrow l > 1$$

Then, for every $\varepsilon > 0$, we have, for $n \geq n(\varepsilon)$,

$$\frac{a_{n+1}}{a_n} > l - \varepsilon$$

We choose ε small enough to have $l - \varepsilon > 1$; with steps similar to before we can write

$$a_{n(\varepsilon)+k} > (l - \varepsilon)^k a_{n(\varepsilon)} \quad \text{and} \quad (l - \varepsilon)^k \rightarrow +\infty \text{ for } k \rightarrow +\infty.$$

On the other hand, $n(\varepsilon)$ is fixed and consequently $a_{n(\varepsilon)}$ is fixed as well; hence for k large enough the right-hand side (and therefore the left-hand side) is as large as we like. This proves the second claim, that is: $a_n \rightarrow +\infty$.

□

Example 3: Using the ratio test theorem

Let us try to compute, with the ratio test, the limit

$$\lim_{n \rightarrow +\infty} \frac{\log n}{n} = \left[\frac{\infty}{\infty} \right] \quad (\text{indeterminate form})$$

we have

$$\frac{a_{n+1}}{a_n} = \frac{\log(n+1)}{n+1} \cdot \frac{n}{\log n} = \frac{n}{n+1} \cdot \frac{\log(n+1)}{\log n}$$

we have

$$\frac{n}{n+1} = \frac{n+1-1}{n+1} = 1 - \underbrace{\frac{1}{n+1}}_{\rightarrow 0} \quad \text{hence} \quad \lim_{n \rightarrow +\infty} \frac{n}{n+1} = 1$$

moreover we have

$$\frac{\log(n+1)}{\log n} \sim \frac{\log n}{\log n} = 1 \quad \text{hence} \quad \lim_{n \rightarrow +\infty} \frac{\log(n+1)}{\log n} = 1$$

where $\log(n+1) \sim \log(n)$ by the substitution principle. Now, using the theorem on the algebra of limits, we have

$$\lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} = 1 \cdot 1 = 1$$

hence Theorem (??) (ratio test) does not allow us, in this case, to conclude anything.

3 Hierarchies of infinities for sequences, part 3 and part 4

Theorem 4: Hierarchy of infinities (part III)

$$\lim_{n \rightarrow +\infty} \frac{a^n}{n!} = 0 \quad (3)$$

for every $a > 0$.

- hence exponentials with base > 0 go more slowly than the factorial

Proof. We apply the ratio test to the sequence

$$b_n = \frac{a^n}{n!} \quad \text{we have} \quad \frac{b_{n+1}}{b_n} = \frac{a^{n+1}}{(n+1)!} \cdot \frac{n!}{a^n} = \frac{a}{n+1} \quad \text{and} \quad \frac{a}{n+1} \rightarrow 0$$

Using the ratio test theorem, we obtain the claim. $\square$

Theorem 5: Hierarchy of infinities (part IV)

$$\lim_{n \rightarrow +\infty} \frac{n!}{n^n} = 0 \quad (4)$$

- hence the factorial goes more slowly than n^n

Proof. We apply the ratio test to the sequence

$$b_n = \frac{n!}{n^n}$$

we have

$$\begin{aligned} \frac{b_{n+1}}{b_n} &= \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{(n+1) \cdot n!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} \\ &= \frac{(n+1) \cdot n^n}{(n+1)^n \cdot (n+1)} = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1 + \frac{1}{n}\right)^n} \end{aligned}$$

and since

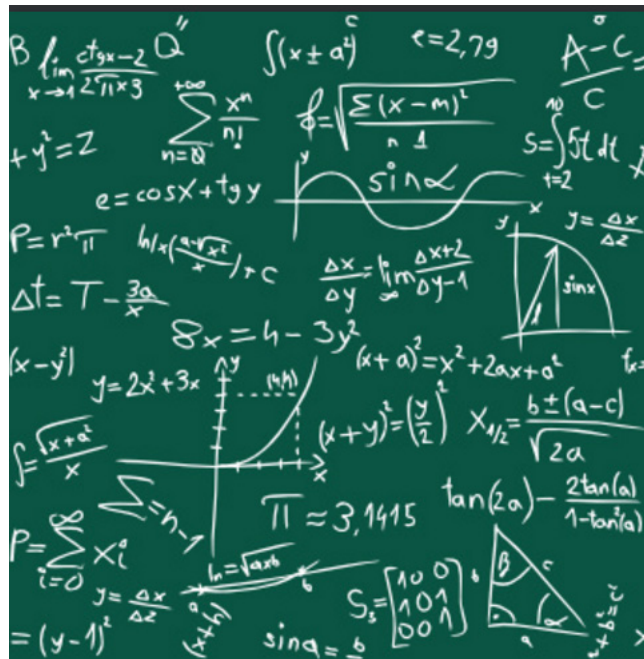
$$\left(1 + \frac{1}{n}\right)^n \rightarrow e$$

then

$$\frac{b_{n+1}}{b_n} \rightarrow \frac{1}{e} < 1$$

Using the ratio test theorem, we obtain the claim. □

Recursively defined sequences



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Recursively defined sequences

- We begin with the following problem:

The bank lends us a_0 euros, with a management fee of b euros per year, at a fixed annual interest rate r ($r \in (0, 1)$). How much will we have to pay back after n years?

- After one year, the amount to be paid back will be

$$a_1 = a_0 + r a_0 + b = (1 + r) a_0 + b$$

after two years

$$a_2 = a_1 + r a_1 + b = (1 + r) a_1 + b$$

after three years

$$a_3 = a_2 + r a_2 + b = (1 + r) a_2 + b$$

and so on... Each year the amount is computed using the amount of the previous year. We therefore have the sequence:

$$\underbrace{a_0 \text{ given,}}_{\text{given value}} \quad \underbrace{a_{n+1} = (1 + r) a_n + b}_{\text{recursive relation}} \text{ with } n \in \mathbb{N}$$

This is a very simple case of a **recursively defined sequence**, in which each term is defined starting from the previous one (or the previous ones).

- In this case it is easy to convince ourselves that the sequence can be rewritten in **closed form** (i.e., explicitly).

After one year:

$$a_1 = (1 + r) a_0 + b$$

after two years:

$$\begin{aligned} a_2 &= (1 + r) a_1 + b = (1 + r) ((1 + r) a_0 + b) + b \\ &= (1 + r)^2 a_0 + b (1 + (1 + r)) \end{aligned}$$

after three years:

$$\begin{aligned} a_3 &= (1 + r) a_2 + b = (1 + r) \left((1 + r)^2 a_0 + b (1 + (1 + r)) \right) + b \\ &= (1 + r)^3 a_0 + b (1 + (1 + r) + (1 + r)^2) \end{aligned}$$

and so on ... hence

$$a_n = s^n a_0 + b(1 + s + s^2 + \cdots + s^{n-1}), \text{ where } s = (1 + r) > 1$$

We can therefore easily compute the limit as follows:

$$\lim_{n \rightarrow +\infty} a_n = \lim_{n \rightarrow +\infty} s^n a_0 + b(1 + s + s^2 + \cdots + s^{n-1}) = +\infty$$

However, it is not always possible to make explicit a sequence given in iterative form. In that case, we are left with the **problem of understanding its behavior and its limit**.

Example 1: Behavior of a recursively defined sequence

Given $b > 0$, we determine the behavior of the sequence defined by

$$a_0 = b, \quad a_{n+1} = \frac{a_n}{1 + a_n} \text{ with } n \in \mathbb{N}$$

All the terms of the sequence are positive, hence the denominator of the fraction appearing in the definition never vanishes.

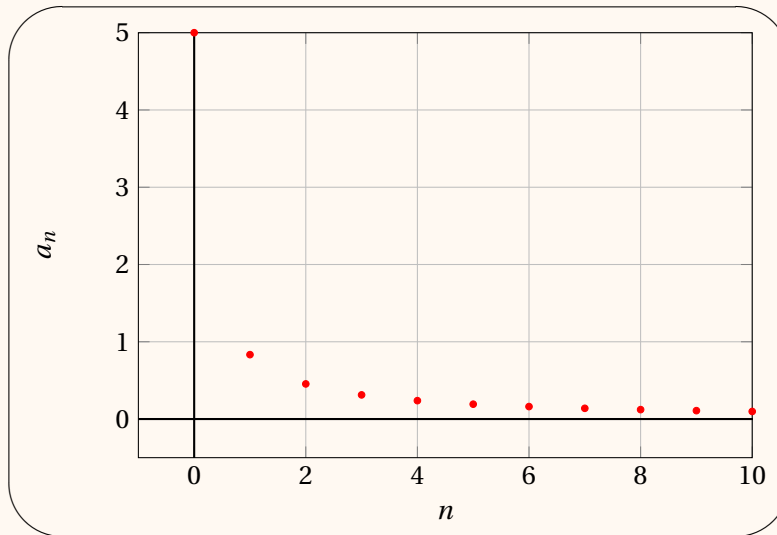
To verify this we proceed by induction:

$$a_0 = b > 0, \text{ and if } a_n > 0 \text{ then } a_{n+1} = \frac{a_n}{1 + a_n} > 0 \text{ (quotient of two positive numbers)}$$

Consequently, all the terms of the sequence after the initial one are less than 1, since $a_n < 1 + a_n$.

For example, with $b = 5$, we have:

$$a_0 = 5, \quad a_1 = \frac{5}{6}, \quad a_2 = \frac{\frac{5}{6}}{1 + \frac{5}{6}} = \frac{5}{11}, \quad a_3 = \frac{\frac{5}{11}}{1 + \frac{5}{11}} = \frac{5}{16} \dots$$



Example 2: Behavior of a recursively defined sequence

Let us try to understand whether or not the sequence is monotone. We know that $a_n > 0, \forall n \in \mathbb{N}$; then the inequality $a_n \leq a_{n+1}$ becomes

$$a_n \leq a_{n+1} \iff a_n \leq \frac{a_n}{1 + a_n} \iff 1 + a_n \leq 1 \iff a_n \leq 0$$

which we know to be false; hence the opposite inequality holds, i.e., the sequence is strictly decreasing ($a_n > a_{n+1}, \forall n \in \mathbb{N}$).

Since it is also bounded below (by zero), a_n converges to a finite non-negative limit, which we call $\ell \geq 0$.

We pass to the limit in the recursive relation.

$$\underbrace{\lim_{n \rightarrow +\infty} a_{n+1}}_{=\ell} = \underbrace{\lim_{n \rightarrow +\infty} \frac{a_n}{1 + a_n}}_{=\frac{\ell}{1+\ell}} \quad \text{hence} \quad \ell = \frac{\ell}{1 + \ell} \quad \text{and consequently} \quad \ell = 0$$

since

$$\ell - \frac{\ell}{1 + \ell} = 0, \quad \frac{\ell(1 + \ell) - \ell}{1 + \ell} = 0, \quad \frac{\ell^2}{1 + \ell} = 0 \iff \ell = 0$$

Example 3: Closed formula of the recurrence

Given $b > 0$, we look for the closed formula of the sequence:

$$a_0 = b, \quad a_{n+1} = \frac{a_n}{1 + a_n} \quad \text{for } n \in \mathbb{N}$$

With $n = 1$, we have

$$a_1 = \frac{a_0}{1 + a_0}$$

With $n = 2$, we have

$$a_2 = \frac{a_1}{1 + a_1} = \frac{\frac{a_0}{1 + a_0}}{1 + \frac{a_0}{1 + a_0}} = \frac{\frac{a_0}{1 + a_0}}{\frac{1 + 2a_0}{1 + a_0}} = \frac{a_0}{1 + a_0} \frac{1 + a_0}{1 + 2a_0} = \frac{a_0}{1 + 2a_0}$$

With $n = 3$, we have

$$a_3 = \frac{a_2}{1 + a_2} = \frac{\frac{a_0}{1 + 2a_0}}{1 + \frac{a_0}{1 + 2a_0}} = \frac{\frac{a_0}{1 + 2a_0}}{\frac{1 + 3a_0}{1 + 2a_0}} = \frac{a_0}{1 + 2a_0} \frac{1 + 2a_0}{1 + 3a_0} = \frac{a_0}{1 + 3a_0}$$

and so on ... hence

$$a_n = \frac{a_0}{1 + n a_0} \quad \text{with } n \in \mathbb{N} \quad \text{and we have} \quad \lim_{n \rightarrow +\infty} \frac{a_0}{1 + n a_0} = 0$$

The previous example illustrates a good two-step strategy for studying the behavior of recursively defined sequences:

1. we try to understand whether the sequence satisfies some monotonicity property;
2. if so, we examine the possible limits, taking into account the recursive relation.

2 Heron's algorithm

- Given $b > 0$ and $c > 0$, we study the following recursively defined sequence:

$$a_0 = b, \quad a_{n+1} = \frac{1}{2} \left(a_n + \frac{c}{a_n} \right) \text{ with } n \in \mathbb{N}$$

It is evident that the sequence consists of positive numbers. To verify this we proceed by induction:

$$a_0 = b > 0, \text{ and if } a_n > 0 \text{ then } a_{n+1} = \frac{1}{2} \left(a_n + \frac{c}{a_n} \right) = \frac{1}{2} \left(\frac{a_n^2 + c}{a_n} \right) > 0$$

since we have the quotient of two positive numbers.

Let us try to understand whether or not the sequence is monotone. We know that $a_n > 0, \forall n \in \mathbb{N}$; the inequality $a_{n+1} < a_n$ becomes

$$a_{n+1} < a_n \iff \frac{1}{2} \left(a_n + \frac{c}{a_n} \right) < a_n \iff c < a_n^2 \iff a_n > \sqrt{c}$$

since

$$\frac{1}{2} \left(a_n + \frac{c}{a_n} \right) < a_n, \quad \frac{a_n^2 + c}{a_n} < 2 a_n, \quad a_n^2 + c < 2 a_n^2, \quad c < a_n^2$$

hence the sequence is strictly decreasing if and only if:

$$a_n > \sqrt{c}$$

We now need to understand whether or not this condition is true. We have

$$a_{n+1} = \frac{1}{2} \left(a_n + \frac{c}{a_n} \right) > \sqrt{c} \iff (a_n - \sqrt{c})^2 > 0 \iff a_n \neq \sqrt{c}$$

since

$$\frac{1}{2} \left(a_n + \frac{c}{a_n} \right) - \sqrt{c} > 0, \quad \frac{a_n^2 + c}{2 a_n} - \sqrt{c} > 0, \quad \frac{a_n^2 + c - 2 a_n \sqrt{c}}{2 a_n} > 0, \quad \frac{(a_n - \sqrt{c})^2}{2 a_n} > 0$$

hence $a_n > \sqrt{c}$ for every $n \geq 1$ if $\underbrace{b}_{=a_0} \neq \sqrt{c}$. And hence we always have $a_1 > \sqrt{c}, \forall b > 0$.

1. If $b > \sqrt{c}$ the sequence is strictly decreasing, therefore it has a limit (finite and positive), which we denote by $\ell \geq 0$.

To identify it, we pass to the limit in the recursive relation, obtaining

$$\underbrace{\lim_{n \rightarrow +\infty} a_{n+1}}_{=\ell} = \underbrace{\lim_{n \rightarrow +\infty} \frac{1}{2} \left(a_n + \frac{c}{a_n} \right)}_{=\frac{1}{2}(\ell + \frac{c}{\ell})} \text{ hence } \ell = \frac{1}{2} \left(\ell + \frac{c}{\ell} \right) \text{ and consequently } \ell = \sqrt{c}$$

since

$$\ell - \frac{1}{2} \left(\ell + \frac{c}{\ell} \right) = 0, \quad \ell - \frac{1}{2} \left(\frac{\ell^2 + c}{\ell} \right) = 0, \quad \left(\frac{2\ell^2 - \ell^2 - c}{2\ell} \right) = 0, \quad \left(\frac{\ell^2 - c}{2\ell} \right) = 0$$

Therefore, if $b > \sqrt{c}$, the whole sequence converges to $\sqrt{c}$ while decreasing.

2. If instead $0 < b < \sqrt{c}$, then by what we have seen before we will have $a_1 > \sqrt{c}$, and from that point on the sequence will start decreasing, again tending to $\sqrt{c}$.
3. Finally, if $b = \sqrt{c}$, the sequence remains constant, since

$$a_1 = \frac{1}{2} \left(\sqrt{c} + \frac{c}{\sqrt{c}} \right) = \frac{1}{2} \frac{\sqrt{c}\sqrt{c} + c}{\sqrt{c}} = \frac{1}{2} \frac{2c}{\sqrt{c}} = \frac{\sqrt{c}\sqrt{c}}{\sqrt{c}} = \sqrt{c}$$

and all the other terms a_n , with $n > 1$, will consequently have the same value $\sqrt{c}$.

- Moreover, $\forall b > 0$ and $n \geq 1$ we have:

$$\frac{c}{a_n} < \sqrt{c} \text{ since } a_n > \sqrt{c}, \quad c = \sqrt{c} \cdot \underbrace{\sqrt{c}}_{< a_n} \text{ and } c < \sqrt{c} \cdot a_n$$

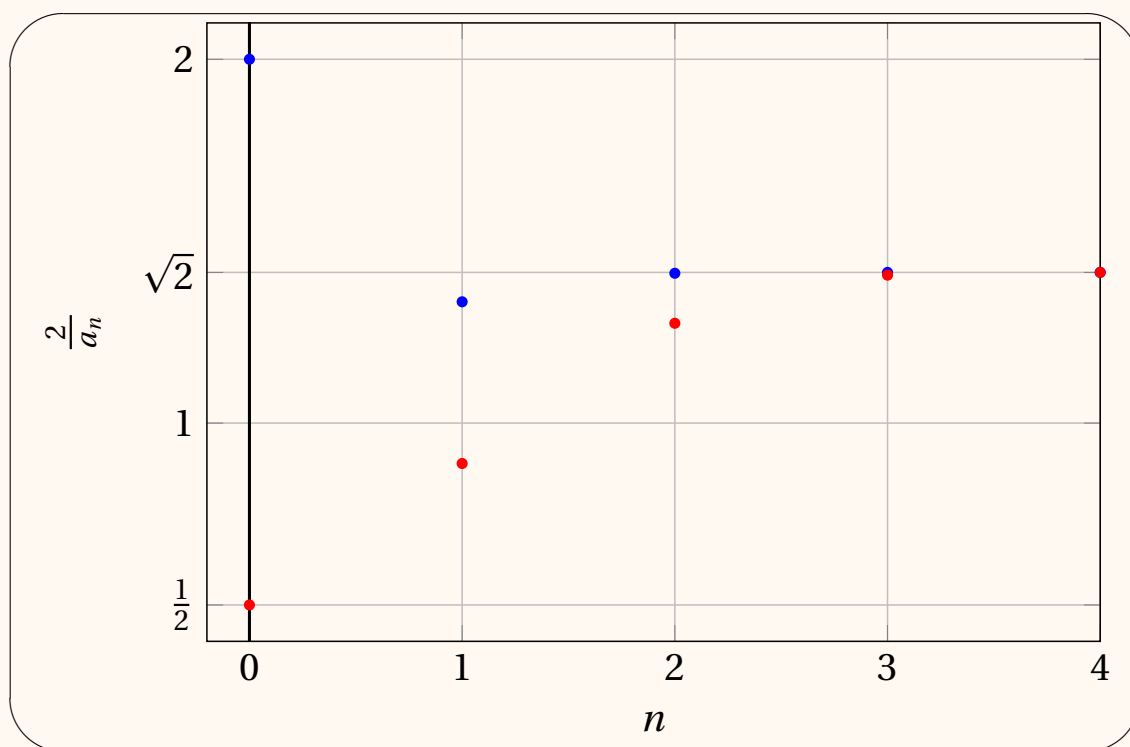
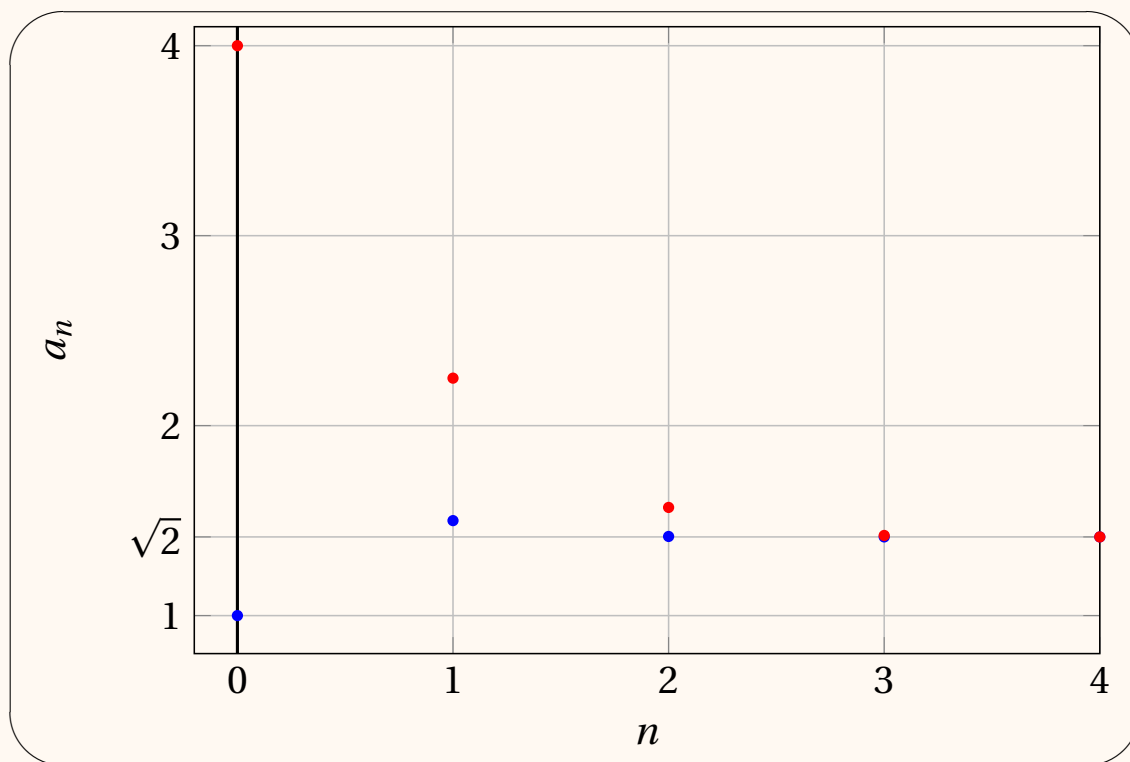
Summarizing, $\forall b > 0$ and $n \geq 1$ we have:

$$\frac{c}{a_n} < \sqrt{c} < a_n, \quad a_n \rightarrow \sqrt{c} \text{ and also } \frac{c}{a_n} \rightarrow \sqrt{c} \text{ since } c = \sqrt{c}\sqrt{c}$$

For every choice of $b > 0$ the sequence a_n tends to $\sqrt{c}$ and can therefore be used to approximate the square root.

Example 4: Behavior of the sequence

For example, with $c = 2$ and $b = 4 > \sqrt{2}$ (red points) or $b = 1 < \sqrt{2}$ (blue points), we have:



The recursive computation of the values of the sequence just seen is called Heron's **algorithm**, originally proposed to compute **the side of a square of area c** .

- The steps of the algorithm are:

1. Having fixed the base equal to a_0 (the initial value b), we construct the associated rectangle of area c . Its height will be $\frac{c}{a_0}$, since the area of the rectangle is fixed equal to that of the square, that is:

$$c = \left(a_0 \cdot \frac{c}{a_0} \right)$$

2. Since the side of the square we are looking for lies between a_0 and $\frac{c}{a_0}$, we take as the new approximation a_1 the average of a_0 and $\frac{c}{a_0}$:

$$a_1 = \frac{1}{2} \left(a_0 + \frac{c}{a_0} \right)$$

which becomes the new base of the rectangle. Then we take a_1 in place of a_0 and repeat, and so on.

For example, let us try to compute the side of a square of area $c = 2$; in other words, we try to estimate the value $\sqrt{2}$.

At the **first iteration** (with $n=0$) we consider, for example, a rectangle with base $a_0 = b = 4$ (the first upper **approximation** of $\sqrt{2}$); consequently, the height is $\frac{c}{a_0} = \frac{1}{2}$ (the first lower approx. of $\sqrt{2}$). Hence: $\frac{1}{2} < \sqrt{2} < 4$.

We compute the value a_1 :

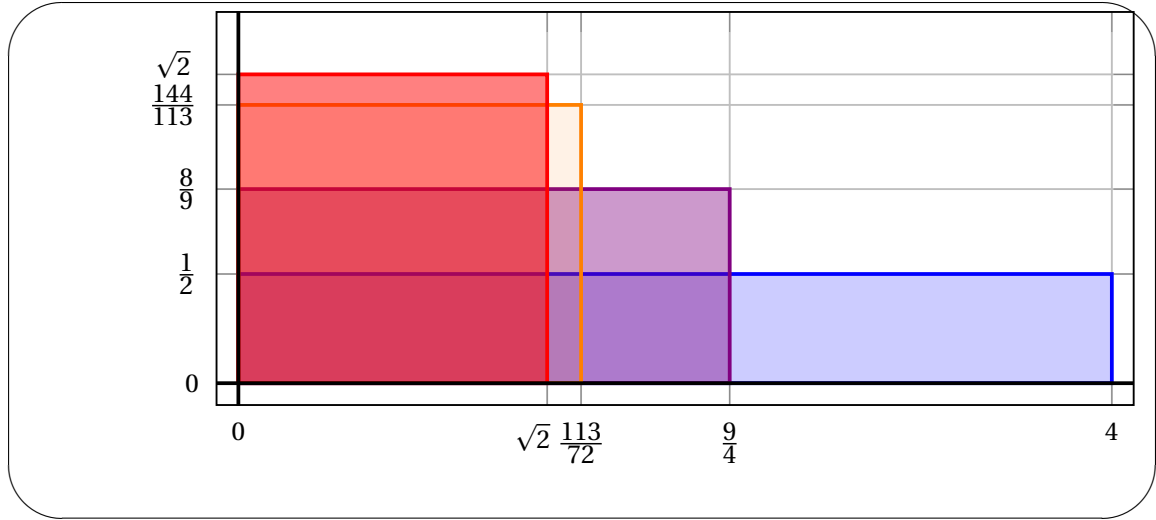
$$a_1 = \frac{1}{2} \left(4 + \frac{1}{2} \right) = \frac{9}{4}$$

At the **second iteration** (with $n=1$) we have a rectangle with base $a_1 = \frac{9}{4}$ (the second upper approx. of $\sqrt{2}$); consequently, the height is $\frac{c}{a_1} = \frac{2}{9/4} = \frac{8}{9}$ (the second lower approx. of $\sqrt{2}$). Hence: $\frac{8}{9} < \sqrt{2} < \frac{9}{4}$.

We compute the value a_2 :

$$a_2 = \frac{1}{2} \left(\frac{9}{4} + \frac{8}{9} \right) = \frac{113}{72}$$

At the **third iteration** (with $n=2$) we have a rectangle with base $a_2 = \frac{113}{72}$ (the third upper approx. of $\sqrt{2}$); consequently, the height is $\frac{c}{a_2} = \frac{2}{113/72} = \frac{144}{113}$ (the third lower approx. of $\sqrt{2}$). Hence: $\frac{144}{113} < \sqrt{2} < \frac{113}{72}$.



We compute the value a_3 :

$$a_3 = \frac{1}{2} \left(\frac{113}{72} + \frac{144}{113} \right) = \frac{23,137}{16,272}$$

At the **fourth iteration** (with $n=3$) we have a rectangle with base $a_3 = \frac{23,137}{16,272}$ (the fourth upper approx. of $\sqrt{2}$); consequently, the height is $\frac{c}{a_3} = \frac{2}{23,137/16,272} = \frac{32,544}{23,137}$ (the fourth lower approx. of $\sqrt{2}$). Hence: $\frac{32,544}{23,137} < \sqrt{2} < \frac{23,137}{16,272}$.

Already at the fourth iteration we have a fairly good estimate of $\sqrt{2}$:

$$\frac{23137}{16272} = 1.421890363... \quad \text{and} \quad \sqrt{2} = 1.414213562...$$

that is, an estimate that is correct up to the first decimal digit.

Finally, we compute the value a_4 :

$$a_4 = \frac{1}{2} \left(\frac{23,137}{16,272} + \frac{32,544}{23,137} \right) = \frac{1,064,876,737}{752,970,528}$$

With the new estimate we have:

$$\frac{1,064,876,737}{752,970,528} = 1.414234285... \quad \text{and} \quad \sqrt{2} = 1.414213562...$$

which is correct up to the fourth decimal digit, i.e., with just one additional iteration we have fixed 3 more decimal digits.

2.1 Error estimates for Heron's algorithm

- The **relative error** $\tilde{\varepsilon}_n$ at iteration n is:

$$\tilde{\varepsilon}_n = \frac{|a_n - \sqrt{c}|}{\sqrt{c}}$$

we have $\sqrt{c} < a_n, \forall b > 0$ and $n \geq 1$, hence we can simply consider:

$$\tilde{\varepsilon}_n = \frac{a_n - \sqrt{c}}{\sqrt{c}}$$

The value of $\sqrt{c}$ is unknown, hence we need a way to analyze the values of the relative errors that is independent of $\sqrt{c}$.

We first consider the **absolute error**:

$$\varepsilon_n = a_n - \sqrt{c}$$

we know that

$$\frac{c}{a_n} < \sqrt{c} < a_n, \quad \forall b > 0, n \geq 1$$

hence we can obtain an estimate of the absolute error ε_n made at iteration n as follows:

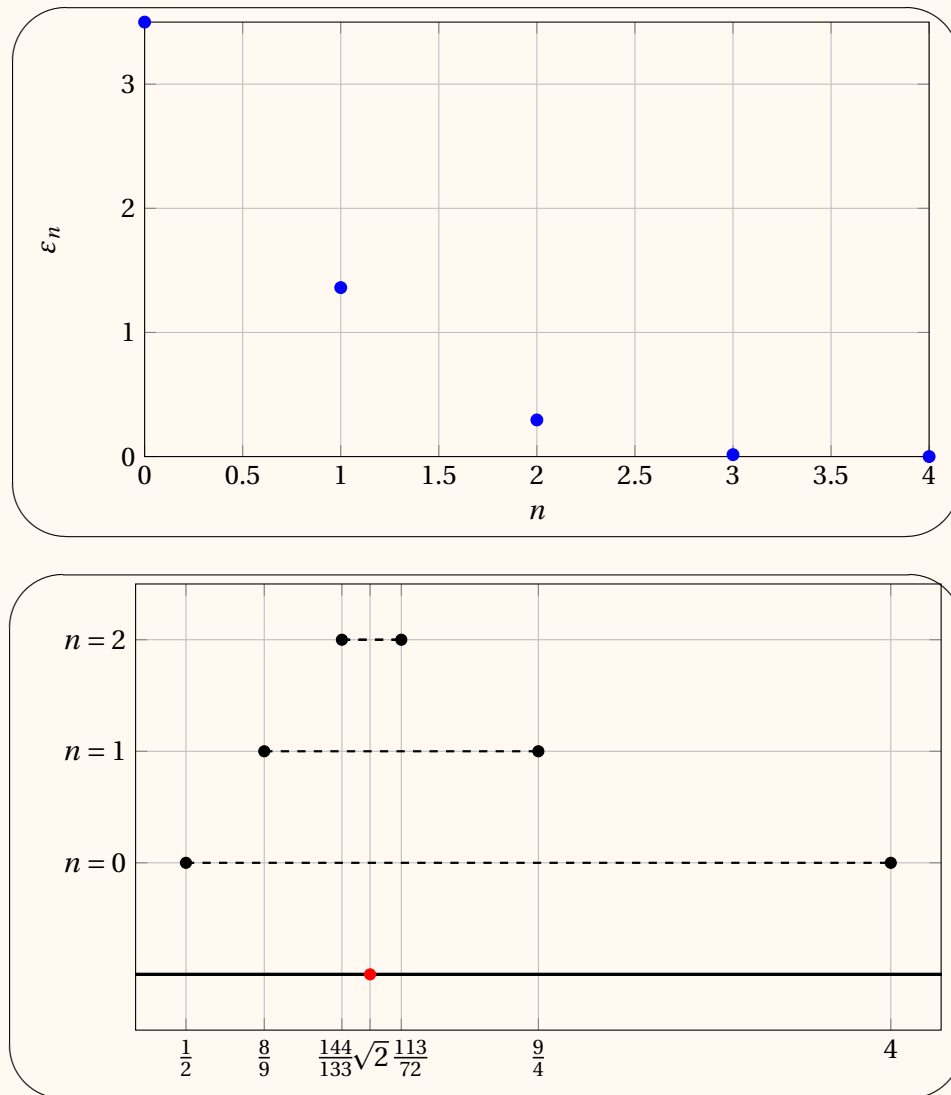
$$\varepsilon_n = a_n - \sqrt{c} < a_n - \frac{c}{a_n} \quad \text{and} \quad \varepsilon_n \rightarrow 0 \text{ for } n \rightarrow +\infty \text{ since } a_n \rightarrow \sqrt{c}$$

Example 5: Error estimates for Heron's algorithm ($\sqrt{2} = 1.414213562\dots$)

We go back to the previous example with $c = 2$ and $b = 4$, and we have:

iter.	$\frac{2}{a_n}$	a_n	estimate of $\sqrt{2}$	ε_n
$n = 0$	$\frac{1}{2}$	4	4	$< \frac{7}{2} = 3.5$
$n = 1$	$\frac{8}{9}$	$\frac{9}{4}$	2.25	$< \frac{49}{36} = 1.361\dots$
$n = 2$	$\frac{144}{113}$	$\frac{113}{72}$	1.569444444...	$< \frac{2,401}{8,136} = 0.295\dots$
$n = 3$	$\frac{32,544}{23,137}$	$\frac{23,137}{16,272}$	1.421890363...	$< \frac{5,764,801}{376,485,264} = 0.0153\dots$
$n = 4$	$\frac{1,505,941,056}{1,064,876,737}$	$\frac{1,064,876,737}{752,970,528}$	1.414234285...	$< \frac{33,232,930,569,601}{801,820,798,913,807,136} = 0.000041\dots$

Example 6: Errors and intervals considered by Heron's algorithm



We now want to understand how “fast” the error decreases by comparing the relative errors of two consecutive iterations. This estimate gives us information about the quality of the algorithm, i.e., we want to determine the “**rate of convergence**” of the algorithm.

We begin by observing that the value $\sqrt{c}$ can be written as follows:

$$\sqrt{c} = \frac{1}{2} \left(\sqrt{c} + \frac{c}{\sqrt{c}} \right) \text{ since } \frac{1}{2} \left(\sqrt{c} + \frac{c}{\sqrt{c}} \right) = \frac{1}{2} \frac{\sqrt{c}\sqrt{c} + c}{\sqrt{c}} = \frac{1}{2} \frac{2c}{\sqrt{c}} = \frac{\sqrt{c}\sqrt{c}}{\sqrt{c}} = \sqrt{c}$$

We now establish the relation between ε_{n+1} and ε_n ; we have:

$$\begin{aligned}
\varepsilon_{n+1} &= a_{n+1} - \sqrt{c} = \frac{1}{2} \left(a_n + \frac{c}{a_n} \right) - \frac{1}{2} \left(\sqrt{c} + \frac{c}{\sqrt{c}} \right) \\
&= \frac{1}{2} \left(a_n - \sqrt{c} + \frac{c}{a_n} - \frac{c}{\sqrt{c}} \right) = \frac{1}{2} \left(\varepsilon_n + \frac{c}{a_n} - \frac{c}{\sqrt{c}} \right) \\
&= \frac{1}{2} \left(\varepsilon_n + \frac{c(\sqrt{c} - a_n)}{\sqrt{c} a_n} \right) = \frac{1}{2} \left(\varepsilon_n - \frac{c \varepsilon_n}{\sqrt{c} a_n} \right) = \frac{1}{2} \left(\varepsilon_n - \frac{\sqrt{c} \varepsilon_n}{a_n} \right) \\
&= \frac{1}{2} \varepsilon_n \left(1 - \frac{\sqrt{c}}{a_n} \right) = \frac{1}{2} \varepsilon_n \left(\frac{a_n - \sqrt{c}}{a_n} \right) \\
&= \frac{1}{2} \frac{\varepsilon_n^2}{a_n}
\end{aligned}$$

We now look for the relation between $\tilde{\varepsilon}_{n+1}$ and $\tilde{\varepsilon}_n$, and we substitute $\varepsilon_n = \sqrt{c} \tilde{\varepsilon}_n$ into the previous formula:

$$\sqrt{c} \tilde{\varepsilon}_{n+1} = \frac{1}{2} \frac{(\sqrt{c} \tilde{\varepsilon}_n)^2}{a_n} = \frac{1}{2} \frac{\sqrt{c} \sqrt{c} \tilde{\varepsilon}_n^2}{a_n} \quad \text{hence} \quad \tilde{\varepsilon}_{n+1} = \frac{1}{2} \frac{\sqrt{c} \tilde{\varepsilon}_n^2}{a_n}$$

We have:

$$\tilde{\varepsilon}_n = \frac{\varepsilon_n}{\sqrt{c}} = \frac{a_n - \sqrt{c}}{\sqrt{c}} = \frac{a_n}{\sqrt{c}} - 1 \quad \text{hence} \quad \frac{\sqrt{c}}{a_n} = \frac{1}{\tilde{\varepsilon}_n + 1}$$

and hence we obtain:

$$\tilde{\varepsilon}_{n+1} = \frac{1}{2} \frac{\sqrt{c} \tilde{\varepsilon}_n^2}{a_n} = \frac{1}{2} \frac{\tilde{\varepsilon}_n^2}{1 + \tilde{\varepsilon}_n}$$

We have obtained the relation between $\tilde{\varepsilon}_{n+1}$ and $\tilde{\varepsilon}_n$, independent of both a_n and $\sqrt{c}$.

From the formula just obtained we can deduce the following simplified and more intuitive one:

$$0 \leq \tilde{\varepsilon}_{n+1} < \frac{1}{2} \min \{ \tilde{\varepsilon}_n, \tilde{\varepsilon}_n^2 \}$$

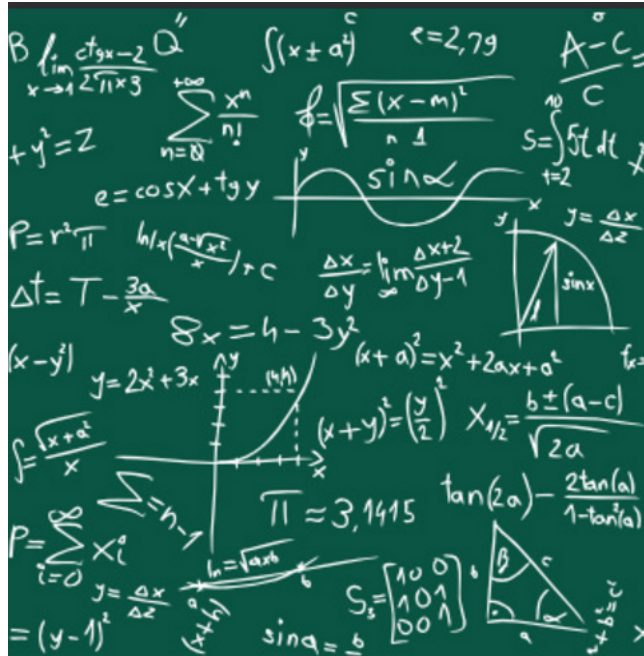
Hence:

$$\text{if } \tilde{\varepsilon}_n < 1, \quad \text{we have } \tilde{\varepsilon}_{n+1} < \frac{1}{2} \tilde{\varepsilon}_n^2$$

For example, if the relative error at a certain iteration equals 10^{-3} , at the next step it will be less than $5 \cdot 10^{-7}$. In other words, the number of zeros after the decimal point (which equals the number of correctly estimated digits after the decimal point) doubles at each iteration.

The relative error of Heron's algorithm at each step is proportional to the square of the relative error at the previous step. **The rate of convergence is quadratic.**

Limits of functions, asymptotes and continuity



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Sequential definition of limit

The limit operation can be extended from sequences to real functions of a real variable.

In this way we can describe the behavior of the function when the independent variable **moves close to a given point** or **becomes very large** (in absolute value).

- Consider an **interval** I , a **point** $c \in I$ and a **function** f defined in I , except possibly at the point c .
- The **interval** I can be **bounded** or **unbounded**, **closed** or **open**.
- The **point** c can be **interior** to the interval or **one of its endpoints** (possibly $+\infty$ or $-\infty$).

Definition 1: Limit (sequential definition)

With $\ell, c \in \mathbb{R}^*$, we write

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c$$

if for every sequence $\{x_n\}$ of points of I such that $x_n \rightarrow c$ as $n \rightarrow +\infty$ and $x_n \neq c, \forall n$, we have

$$f(x_n) \rightarrow \ell \text{ as } n \rightarrow +\infty$$

In these cases we can also write:

$$\lim_{x \rightarrow c} f(x) = \ell$$

which is read: the limit of $f(x)$, as x tends to c , is ℓ .

In other words, if **for any sequence** $\{x_n\}$ **of inputs** the **sequence of outputs** $\{f(x_n)\}$ tends to the limit ℓ (finite or infinite), then we say that the limit of $f(x)$, as x tends to c , is ℓ .

The sequential definition of limit reduces the concept of limit of a function to that of limit of a sequence.

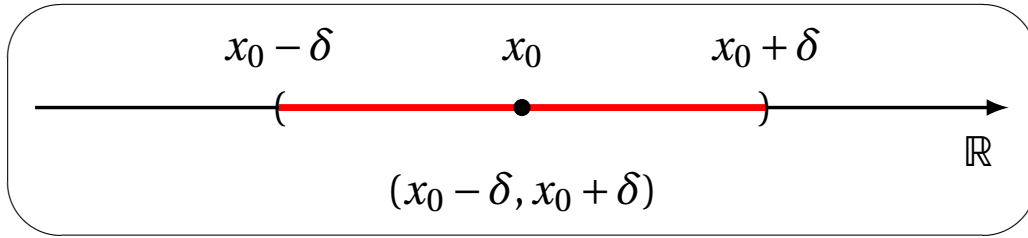
It is not the only possible definition. Another definition is the *topological definition of limit*. The two definitions are perfectly equivalent.

1.1 Neighborhoods and topological definition of limit

Definition 2: Centered neighborhood of a point

Given $\delta > 0$ and $x_0 \in \mathbb{R}$, the **neighborhood** centered at x_0 associated with δ is the open interval:

$$(x_0 - \delta, x_0 + \delta) = \{x \in \mathbb{R} : x_0 - \delta < x < x_0 + \delta\}$$



Saying that “ x moves in a neighborhood of x_0 ” means stating that

$$x \in (x_0 - \delta, x_0 + \delta), \text{ where we think of } 0 < \delta \ll 1.$$

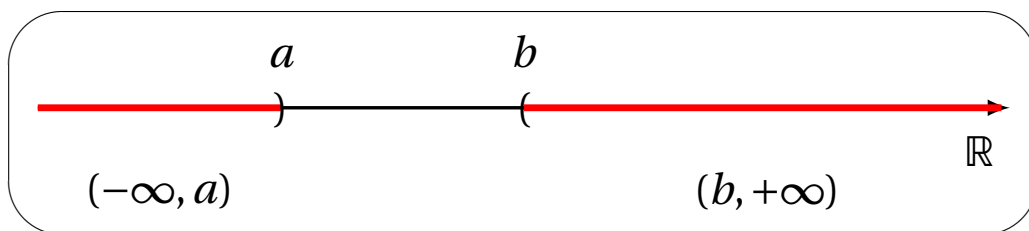
Definition 3: Neighborhood of $\pm\infty$

Given $a, b \in \mathbb{R}$, the neighborhood of $-\infty$ associated with a is the interval:

$$(-\infty, a) = \{x \in \mathbb{R} : x < a\}$$

and the neighborhood of $+\infty$ associated with b is the interval:

$$(b, +\infty) = \{x \in \mathbb{R} : x > b\}$$



- We now introduce the expression “eventually as $x \rightarrow c$ ”, analogously to what we did in the case of sequences

Definition 4: Property eventually satisfied by functions

We say that a function f **eventually** has a certain property as $x \rightarrow c \in \mathbb{R}^*$ if there exists a neighborhood U of c such that the property holds for $f(x)$ for every $x \in U, x \neq c$.

- We now give the topological definition of limit which, as already mentioned, is equivalent to the sequential one seen above, but turns out to be useful.

Definition 5: Limit (topological definition)

Let $c, \ell \in \mathbb{R}^*$, and let f be a function defined at least eventually as $x \rightarrow c$. We write

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c$$

if for every neighborhood U_ℓ of ℓ there exists a neighborhood U_c of c such that

$$\forall x \in U_c, x \neq c, \text{ we have } f(x) \in U_\ell$$

- This definition can be specialized to the specific cases:

Finite limit at a finite point: $f(x) \rightarrow \ell \in \mathbb{R}$ as $x \rightarrow c \in \mathbb{R}$

$$\forall \varepsilon > 0, \exists \delta > 0: \quad \forall x \neq c, |x - c| < \delta \implies |f(x) - \ell| < \varepsilon$$

Infinite limit at a finite point: $f(x) \rightarrow \pm\infty$ as $x \rightarrow c \in \mathbb{R}$

$$\forall M > 0, \exists \delta > 0: \quad \forall x \neq c, |x - c| < \delta \implies |f(x)| > M$$

Finite limit at infinity: $f(x) \rightarrow \ell \in \mathbb{R}$ as $x \rightarrow \pm\infty$

$$\forall \varepsilon > 0, \exists b > 0: \quad \forall |x| > b \implies |f(x) - \ell| < \varepsilon$$

Infinite limit at infinity: $f(x) \rightarrow \pm\infty$ as $x \rightarrow \pm\infty$

$$\forall M > 0, \exists b > 0: \quad \forall |x| > b \implies |f(x)| > M$$

The fact that we have already developed the basics of *limit calculus for sequences* will make it very convenient to use the sequential definition of limit to prove the theorems on limits of functions starting from those seen for sequences.

Theorem 1: Uniqueness of the limit of functions

Given a function f , if $f(x) \rightarrow \ell$ as $x \rightarrow c$, then this limit ℓ is unique.

Proof. If there were two limits ℓ_1 and ℓ_2 , different from each other, taking any sequence $\{x_n\}$ such that $x_n \rightarrow c$ and $x_n \neq c, \forall n$, we would have:

$$f(x_n) \rightarrow \ell_1 \quad \text{and} \quad f(x_n) \rightarrow \ell_2 \quad \text{as} \quad n \rightarrow +\infty$$

Hence the sequence $\{f(x_n)\}$ would have two distinct limits: a contradiction. $\square$

Definition 6: Infinitesimal function

A function f such that $f(x) \rightarrow 0$ as $x \rightarrow c$ is called **infinitesimal** as $x \rightarrow c$.

Definition 7: Infinite function

A function f such that $f(x) \rightarrow \pm\infty$ as $x \rightarrow c$ is called **infinite** as $x \rightarrow c$.

- When we write:

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c \text{ or } \lim_{x \rightarrow c} f(x) = \ell$$

we have

$$\text{a } \begin{cases} \text{finite limit} \\ \text{infinite limit} \end{cases} \quad \text{if} \quad \begin{cases} \ell \in \mathbb{R} \\ \ell = \pm\infty \end{cases}$$

we have

$$\text{a limit } \begin{cases} \text{at a finite point} \\ \text{at infinity} \end{cases} \quad \text{if} \quad \begin{cases} c \in \mathbb{R} \\ c = \pm\infty \end{cases}$$

1.2 Finite limit at infinity

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c \text{ with } \ell \in \mathbb{R} \text{ and } c = \pm\infty$$

Example 1: Finite limit at infinity (sequential definition of limit)

Let us prove that:

$$e^x \rightarrow 0 \text{ as } x \rightarrow -\infty$$

By the definition, we have to prove that for every sequence $\{x_n\}$ such that $x_n \rightarrow -\infty$ as $n \rightarrow +\infty$, we have:

$$e^{x_n} \rightarrow 0 \text{ as } n \rightarrow +\infty$$

By the definition of limit of a sequence, this means proving that, for every $\varepsilon > 0$, we have:

$$|e^{x_n}| < \varepsilon, \text{ eventually, that is } e^{x_n} < \varepsilon, \text{ eventually,}$$

since the exponential is always positive. The last inequality is equivalent to:

$$x_n < \log \varepsilon$$

If $\varepsilon > 0$ is a small number (< 1), $\log \varepsilon$ is a negative number (and large in absolute value). We set

$$M(\varepsilon) = -\log \varepsilon > 0$$

We therefore have to prove that, for every $M(\varepsilon) > 0$, we have:

$$x_n < -M(\varepsilon), \text{ eventually.}$$

But this is exactly what holds by hypothesis, because $x_n \rightarrow -\infty$. Hence $e^x \rightarrow 0$ as $x \rightarrow -\infty$.

1.2.1 Horizontal asymptote

Definition 8: Horizontal asymptote

We say that f has a **horizontal asymptote** with equation $y = \ell \in \mathbb{R}$ as $x \rightarrow +\infty$ ($x \rightarrow -\infty$) if:

$$f(x) \rightarrow \ell \in \mathbb{R} \text{ as } x \rightarrow +\infty \text{ (} x \rightarrow -\infty \text{)}$$

- Every situation of *finite limit at infinity*, therefore, corresponds graphically to the **presence of a horizontal asymptote**, that is, a horizontal line which the graph of the function gets closer and closer to.

1.2.2 Limit from above or from below

- When a function has a finite limit at infinity, it is **sometimes** possible to specify whether this limit is **from above** (ℓ^+) or **from below** (ℓ^-). Graphically, this means that the graph of the function approaches the level $y = \ell$ from **above** or from **below**.

Definition 9: Limit from above or from below

With $\ell, c \in \mathbb{R}^*$, we write

$$f(x) \rightarrow \ell^+ \text{ (} \ell^- \text{) as } x \rightarrow c$$

if for every sequence $\{x_n\}$ of points of I such that $x_n \rightarrow c$ as $n \rightarrow +\infty$ and $x_n \neq c, \forall n$, we have

$$f(x_n) \rightarrow \ell^+ \text{ (} \ell^- \text{) as } n \rightarrow +\infty$$

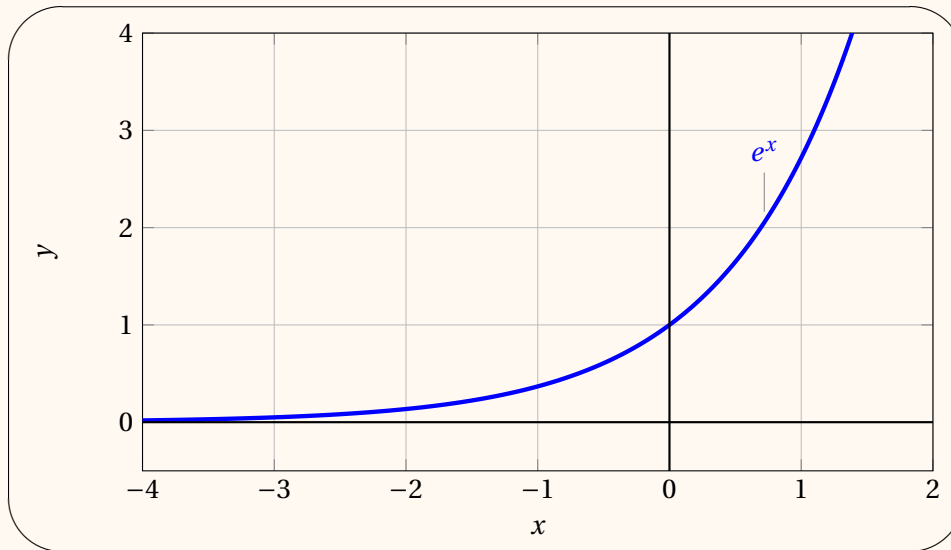
- Stating that $f(x) \rightarrow \ell^+$ means that $f(x) \rightarrow \ell$ and moreover $f(x) \geq \ell$, eventually.
- Stating that $f(x) \rightarrow \ell^-$ means that $f(x) \rightarrow \ell$ and moreover $f(x) \leq \ell$, eventually.
- In these cases we say that $f(x)$ tends to ℓ from above (from below) as x tends to c .

Example 2: Limits from above/below

For example:

$$\lim_{x \rightarrow -\infty} e^x = 0^+$$

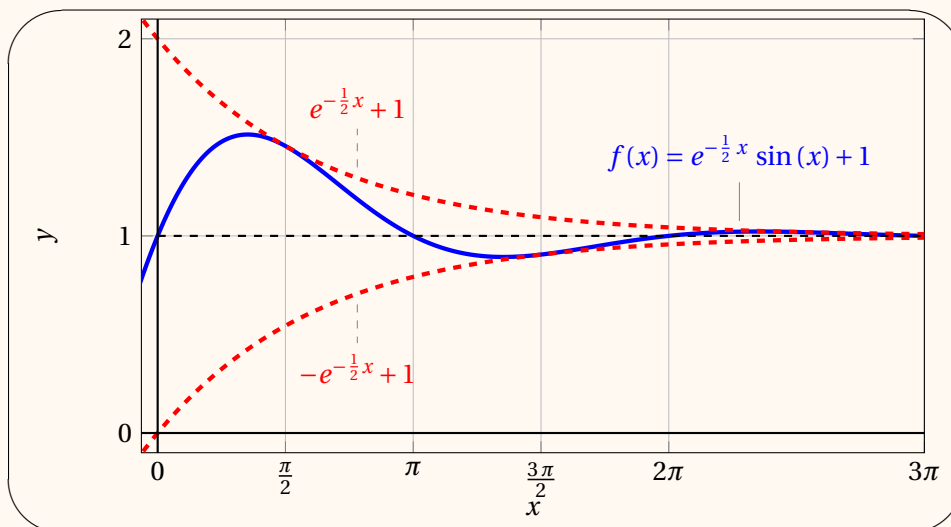
the notation 0^+ means that the function tends to 0 from above, that is, the values of $f(x)$ tend to zero while remaining non-negative.



Example 3: Limit from above/below

Note that not every finite limit is necessarily from above or from below, as shown for example by:

$$\lim_{x \rightarrow +\infty} e^{-\frac{1}{2}x} \sin(x) + 1 = 1$$



The function $f(x)$ tends to 1 as $x \rightarrow +\infty$, but we can state neither $f(x) \rightarrow 1^+$ nor $f(x) \rightarrow 1^-$.

1.3 Infinite limit at infinity

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c \text{ with } \ell = \pm\infty \text{ and } c = \pm\infty$$

Example 4: Infinite limit at infinity (sequential definition of limit)

Let us prove that:

$$\log_{1/2} x \rightarrow -\infty \text{ as } x \rightarrow +\infty$$

By the definition, we have to prove that for every sequence $\{x_n\}$ such that $x_n \rightarrow +\infty$ as $n \rightarrow +\infty$, we have:

$$\log_{1/2} x_n \rightarrow -\infty \text{ as } n \rightarrow +\infty$$

This means proving that for every $M > 0$ we have:

$$\log_{1/2} x_n < -M, \text{ eventually, that is } x_n > \left(\frac{1}{2}\right)^{-M} = 2^M, \text{ eventually}$$

But by hypothesis $x_n \rightarrow +\infty$, so, having fixed the positive quantity 2^M , we certainly have $x_n > 2^M$, eventually. Hence $\log_{1/2} x \rightarrow -\infty$ as $x \rightarrow +\infty$.

1.3.1 Oblique asymptote

- When a function has an infinite limit at infinity, it may happen (but it does not always happen) that there exists an oblique line which the graph of the function gets closer and closer to.

Definition 10: Oblique asymptote

We say that f has an **oblique asymptote** with equation $y = mx + q$ ($q, m \in \mathbb{R}, m \neq 0$) as $x \rightarrow +\infty$ ($-\infty$) if:

$$(f(x) - (mx + q)) \rightarrow 0 \text{ as } x \rightarrow +\infty \text{ } (-\infty)$$

Example 5: Oblique asymptote

Consider

$$f(x) = 2x + 1 + e^x$$

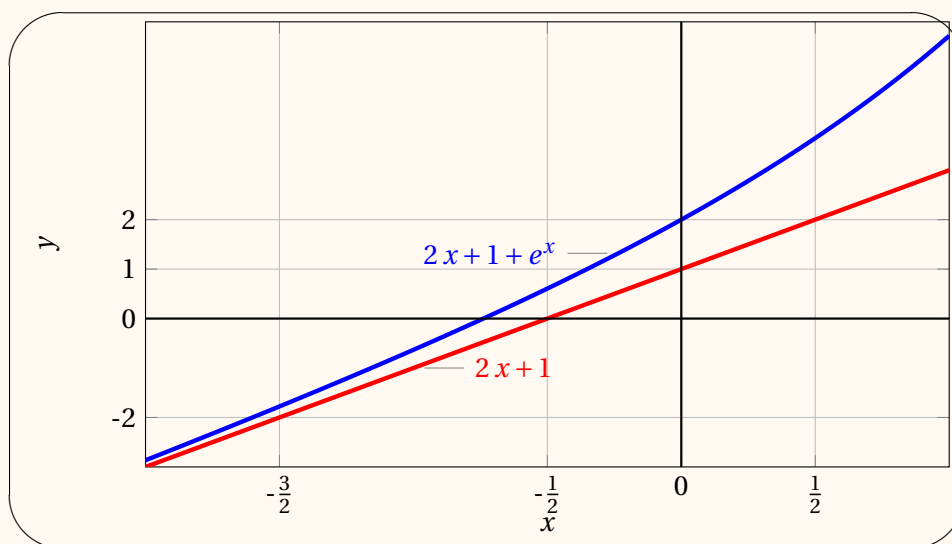
and we want to test whether or not $2x + 1$ is an oblique asymptote as $x \rightarrow -\infty$. We have

$$\lim_{x \rightarrow -\infty} (f(x) - (2x + 1)) = \lim_{x \rightarrow -\infty} e^x = 0$$

Therefore, by the definition of oblique asymptote, the line

$$y = 2x + 1$$

is an oblique asymptote of f as $x \rightarrow -\infty$.



- In less elementary cases, instead of having to “guess” the oblique asymptote, it is useful to have an operational criterion to find it.

Proposition 1: Existence of an oblique asymptote

A function f has an oblique asymptote with equation $y = mx + q$ as $x \rightarrow +\infty$ if and only if the following two conditions hold:

$$\lim_{x \rightarrow +\infty} \frac{f(x)}{x} = m \in \mathbb{R}, m \neq 0 \quad \text{and} \quad \lim_{x \rightarrow +\infty} (f(x) - mx) = q \in \mathbb{R}$$

An analogous criterion holds for $x \rightarrow -\infty$.

- Note that the first condition requires $f(x)$ to have the same order of infinity as $y = x$ as $x \rightarrow +\infty$, since:

$$\lim_{x \rightarrow -\infty} x = -\infty \quad \text{and} \quad \lim_{x \rightarrow +\infty} x = +\infty$$

Moreover, if $m > 0$ we have:

$$\lim_{x \rightarrow -\infty} f(x) = -\infty \quad \text{and} \quad \lim_{x \rightarrow +\infty} f(x) = +\infty$$

and if $m < 0$ we have:

$$\lim_{x \rightarrow -\infty} f(x) = +\infty \quad \text{and} \quad \lim_{x \rightarrow +\infty} f(x) = -\infty$$

The oblique asymptote $y = mx + q$ is a (non-horizontal) line that approximates the behavior of a function that diverges as $x \rightarrow \pm\infty$.

Proof. The first condition rules out the possibility that the asymptote is horizontal and requires $f(x)$ to have the same order of infinity as $y = x$. From the second condition:

$$\text{if } \lim_{x \rightarrow +\infty} (f(x) - mx) = q \quad \text{then} \quad \lim_{x \rightarrow +\infty} (f(x) - (mx + q)) = 0$$

Analogous reasoning for $x \rightarrow -\infty$. □

Example 6: Oblique asymptote

Consider

$$f(x) = 3x + \sqrt{x}$$

and compute

$$\lim_{x \rightarrow +\infty} \frac{3x + \sqrt{x}}{x} = \lim_{x \rightarrow +\infty} \left(3 + \frac{1}{\sqrt{x}} \right) = 3, \quad \lim_{x \rightarrow +\infty} (3x + \sqrt{x} - 3x) = \lim_{x \rightarrow +\infty} \sqrt{x} = +\infty$$

since the second limit is infinite, the second condition of the proposition does not hold and consequently the function has no oblique asymptote as $x \rightarrow +\infty$.

1.4 Infinite limit at a finite point

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c \text{ with } \ell = \pm\infty \text{ and } c \in \mathbb{R}$$

Example 7: Infinite limit at a finite point (sequential definition of limit)

Let us prove that:

$$\frac{1}{x^2} \rightarrow +\infty \text{ as } x \rightarrow 0$$

By the definition, we have to prove that for every sequence $\{x_n\}$ such that $x_n \rightarrow 0$ as $n \rightarrow +\infty$ and $x_n \neq 0, \forall n$, we have:

$$\frac{1}{x_n^2} \rightarrow +\infty \text{ as } n \rightarrow +\infty$$

Note that the condition $x_n \neq 0, \forall n$, contained in the definition of limit, was irrelevant in the previous examples where we had $x \rightarrow \pm\infty$, but now it clearly becomes relevant.

We therefore have to prove that for every $M > 0$ we have:

$$\frac{1}{x_n^2} > M, \text{ eventually, that is } |x_n| < \frac{1}{\sqrt{M}}, \text{ eventually}$$

But by hypothesis $x_n \rightarrow 0$, so, having fixed the positive quantity $\frac{1}{\sqrt{M}}$, we certainly have $|x_n| < \frac{1}{\sqrt{M}}$ eventually. Hence $1/x^2 \rightarrow +\infty$ as $x \rightarrow 0$.

1.4.1 Right and left limits

- Sometimes a function behaves differently, as far as its limit is concerned, depending on whether x approaches c from the **right** or from the **left**.

Definition 11: Right and left limits

With $\ell \in \mathbb{R}^*$ and $c \in \mathbb{R}$, we write

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c^+ \text{ (} c^- \text{)}$$

if for every sequence $\{x_n\}$ of points of I such that $x_n \rightarrow c^+ \text{ (} c^- \text{)}$ as $n \rightarrow +\infty$ and $x_n \neq c^+ \text{ (} c^- \text{)}, \forall n$, we have

$$f(x_n) \rightarrow \ell \text{ as } n \rightarrow +\infty$$

The limit $\lim_{x \rightarrow c} f(x)$ exists if and only if the right limit and the left limit exist and are both equal to ℓ . That is, if and only if:

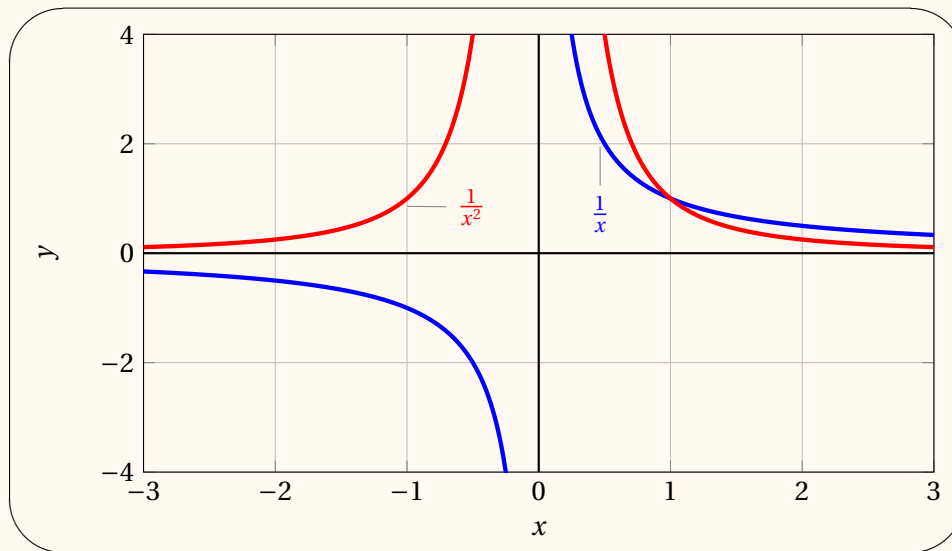
$$\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) = \ell$$

However, it may happen that the right limit and the left limit exist but are different from each other, or that only one of the two exists. In these cases:

$$\lim_{x \rightarrow c} f(x) \text{ does not exist}$$

Example 8: Right and left limits

Consider the functions $y = \frac{1}{x}$ and $y = \frac{1}{x^2}$:



$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty, \quad \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty \quad \text{while} \quad \lim_{x \rightarrow 0} \frac{1}{x} \text{ does not exist}$$

$$\lim_{x \rightarrow 0^+} \frac{1}{x^2} = +\infty, \quad \lim_{x \rightarrow 0^-} \frac{1}{x^2} = +\infty \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty$$

If f is defined in (a, b) , the limit as $x \rightarrow a$ (respectively $x \rightarrow b$) is automatically a right limit (respectively a left limit).

1.4.2 Vertical asymptote

Definition 12: Vertical asymptote

We say that f has a **vertical asymptote** with equation $x = c \in \mathbb{R}$ as $x \rightarrow c$ (c^+ or c^-) if:

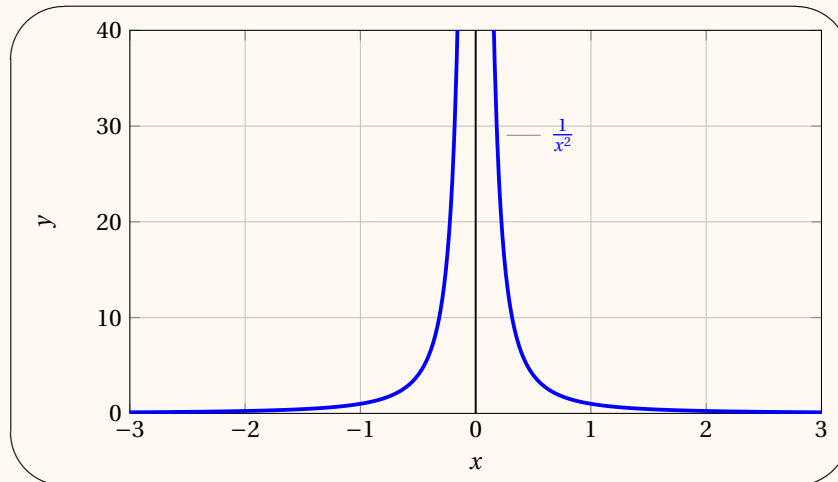
$$f(x) \rightarrow \pm\infty \text{ as } x \rightarrow c \text{ (} c^+ \text{ or } c^- \text{)}$$

- Every situation of **infinite limit at a finite point**, therefore, corresponds graphically to the presence of a **vertical asymptote**, that is, a **vertical line** which the graph of the function gets closer and closer to.

Example 9: Vertical asymptote

- $x = 0$ is a vertical asymptote of

$$\frac{1}{x^2} \text{ as } x \rightarrow 0$$



- $x = 0$ is a vertical asymptote of

$$\frac{1}{x} \text{ as } x \rightarrow 0^+ \text{ and as } x \rightarrow 0^-$$

- $x = 0$ is a vertical asymptote of

$$\log x \text{ as } x \rightarrow 0^+$$

1.5 Finite limit at a finite point

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c \text{ with } \ell \in \mathbb{R} \text{ and } c \in \mathbb{R}$$

Example 10: Finite limit at a finite point (sequential definition of limit)

Consider the function $f(x) = \sin x$ and let us prove that:

$$\sin x \rightarrow 0 \text{ as } x \rightarrow 0$$

By the definition, we have to prove that for every sequence $\{x_n\}$ such that $x_n \rightarrow 0$ as $n \rightarrow +\infty$ and $x_n \neq 0, \forall n$, we have:

$$\sin x_n \rightarrow 0 \text{ as } n \rightarrow +\infty$$

We start from the following elementary inequality:

$$|\sin x| \leq |x|, \quad \forall x \in \mathbb{R} \text{ hence } |\sin x_n| \leq |x_n|$$

By the comparison theorem:

$$\text{if } x_n \rightarrow 0 \text{ as } n \rightarrow +\infty \text{ then also } \sin x_n \rightarrow 0 \text{ as } n \rightarrow +\infty$$

Hence $\sin x \rightarrow 0$ as $x \rightarrow 0$ and we have:

$$\lim_{x \rightarrow 0} \sin x = f(0)$$

Example 11: Finite limit at a finite point (sequential definition of limit)

Consider the function:

$$f(x) = \begin{cases} 1 & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

and let us prove that:

$$f(x) \rightarrow 1 \text{ as } x \rightarrow 0$$

By the definition, we have to prove that for every sequence $\{x_n\}$ such that $x_n \rightarrow 0$ as $n \rightarrow +\infty$ and $x_n \neq 0, \forall n$, we have:

$$f(x_n) \rightarrow 1 \text{ as } n \rightarrow +\infty$$

We have

$$f(x_n) = 1, \quad \forall n, \text{ hence } f(x_n) \rightarrow 1 \text{ as } n \rightarrow +\infty$$

Consequently $f(x) \rightarrow 1$ as $x \rightarrow 0$. In this case, therefore (unlike the previous example), we have:

$$\lim_{x \rightarrow 0} f(x) \neq f(0)$$

- Think about the two examples just seen. In both cases the limit at a finite point of a certain function exists and is finite.
- In the first case, this limit coincides with the value of the function at the point considered; in the second case it does not.
- We therefore introduce the concept of continuity to distinguish these two cases.

1.6 Continuity

Definition 13: Continuity

If $f: I \rightarrow \mathbb{R}$ and $c \in I$, we say that f is **continuous** at c if

$$\lim_{x \rightarrow c} f(x) = f(c)$$

We say that f is continuous in I if it is continuous at each point of I .

- The continuity property on an interval has a simple geometric interpretation:

“the graph of a function continuous on an interval can be drawn, on that interval, without lifting the pen from the paper”

- A function that is **not continuous** at a point c is called **discontinuous** at c .

Example 12: Continuity

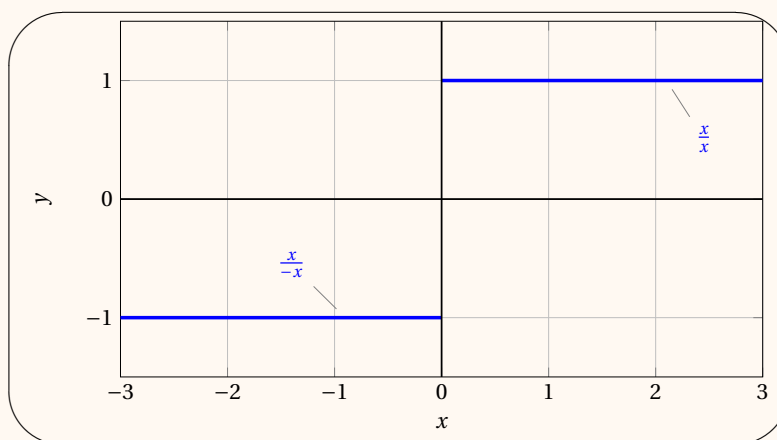
The function:

$$f(x) = \begin{cases} 1 & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is discontinuous at 0. The function $\sin x$ is continuous at 0.

Example 13: Discontinuity

Consider the function: $f(x) = x/|x|$



In this case it is not possible to compute $f(0)$ and the function $f(x)$ is discontinuous at 0 since:

$$\lim_{x \rightarrow 0^+} f(x) = 1 \text{ and } \lim_{x \rightarrow 0^-} f(x) = -1$$

Definition 14: Jump discontinuity point

We say that c is a **jump discontinuity point** of f when the right and left limits at c exist and are finite, but are different from each other. The jump is given by the difference of the limits:

$$\text{jump at } c = \lim_{x \rightarrow c^+} f(x) - \lim_{x \rightarrow c^-} f(x)$$

Example 14: Jump discontinuity

The function

$$f(x) = \frac{x}{|x|}$$

has a jump discontinuity point at $x = 0$, with jump equal to 2.

If one of the two limits, the right limit or the left limit (as $x \rightarrow c$), coincides with $f(c)$, we say that f is **continuous from the right or from the left**, respectively.

- Functions with jump discontinuities are well suited to model phenomena that undergo sudden changes.
- Continuous functions, on the other hand, owe their importance to the fact that

$$\lim_{x \rightarrow c} f(x) = f(c)$$

can be interpreted by saying that “if x is close to c ” then “ $f(x)$ is close to $f(c)$ ”, that is, small variations of x imply small variations of $f(x)$.

1.7 Non-existence of the limit

- The limit of a function may also not exist

$$\lim_{x \rightarrow c} f(x) \text{ does not exist}$$

Example 15: Non-existence of the limit (sequential definition of limit)

Let us prove that:

$$\lim_{x \rightarrow +\infty} \sin x \text{ does not exist}$$

To prove it, it is enough to find two sequences $\{x_n\}$ and $\{y_n\}$ such that $x_n \rightarrow +\infty$ and $y_n \rightarrow +\infty$ as $n \rightarrow +\infty$, with $\{\sin x_n\}$ and $\{\sin y_n\}$ tending to two different limits.

We take:

$$x_n = n\pi \quad \text{and} \quad y_n = \frac{\pi}{2} + 2n\pi, \quad \forall n$$

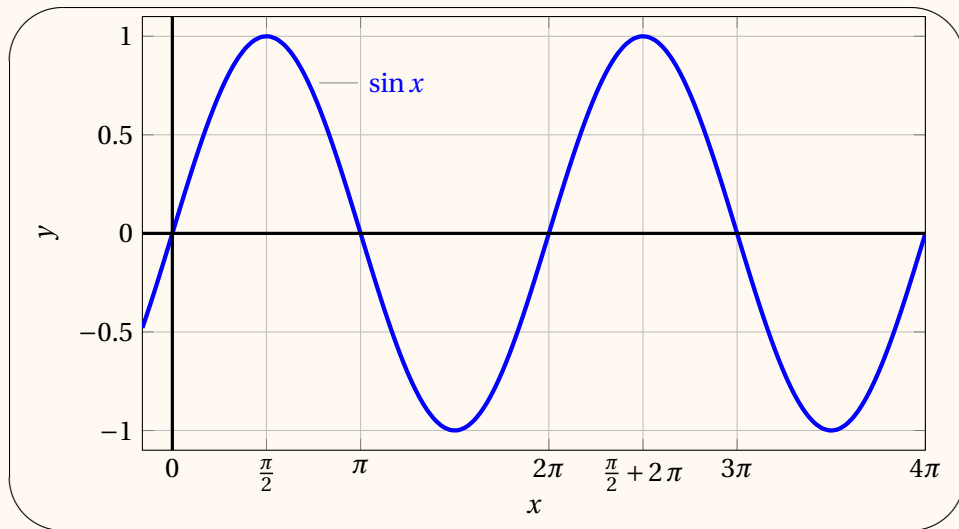
consequently we have:

$$\{n\pi\} \rightarrow +\infty \quad \text{and} \quad \left\{\frac{\pi}{2} + 2n\pi\right\} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty$$

For these sequences we have

$$\sin x_n = 0 \quad \text{and} \quad \sin y_n = 1, \quad \forall n$$

and consequently the two sequences tend to two different limits. Hence the definition of limit is not satisfied and the limit does not exist.



Example 16: Non-existence of the limit (sequential definition of limit)

Let us prove that:

$$\lim_{x \rightarrow 0} \sin \frac{1}{x} \text{ does not exist}$$

To prove it, it is enough to find two sequences $\{x_n\}$ and $\{y_n\}$ such that $x_n \rightarrow 0$, $y_n \rightarrow 0$ as $n \rightarrow +\infty$ and $x_n, y_n \neq 0, \forall n$, with $\left\{\sin \frac{1}{x_n}\right\}$ and $\left\{\sin \frac{1}{y_n}\right\}$ tending to two different limits.

We take:

$$x_n = \frac{1}{n\pi} \quad \text{and} \quad y_n = \frac{1}{\frac{\pi}{2} + 2n\pi}, \quad \forall n > 1$$

consequently we have:

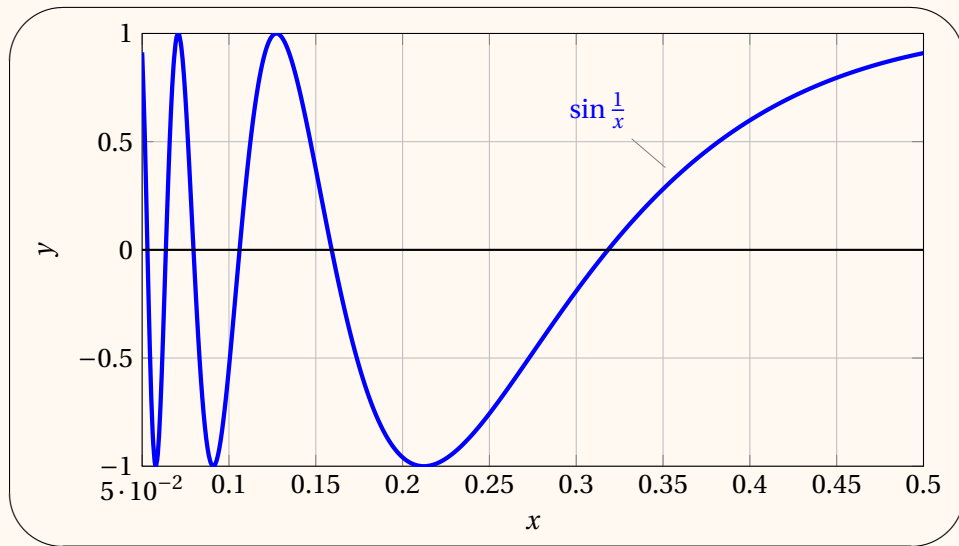
$$\left\{\frac{1}{n\pi}\right\} \rightarrow 0 \quad \text{and} \quad \left\{\frac{1}{\frac{\pi}{2} + 2n\pi}\right\} \rightarrow 0 \quad \text{as } n \rightarrow +\infty$$

For these sequences we have

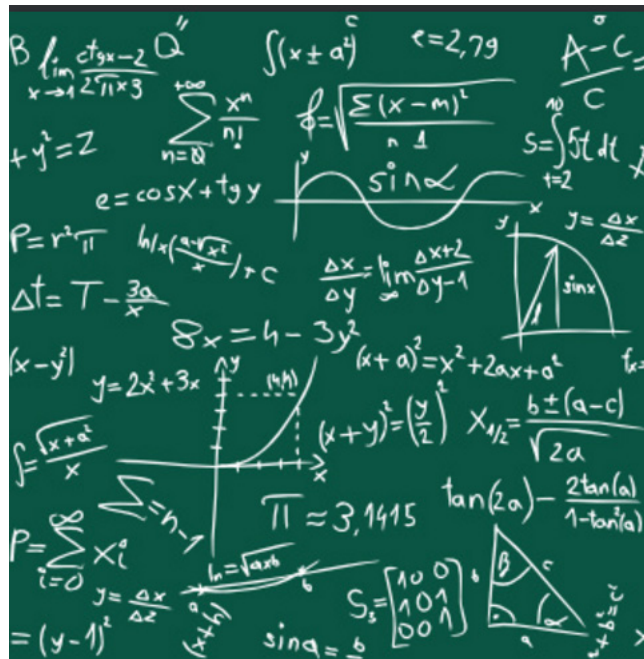
$$\sin \frac{1}{x_n} = 0 \quad \text{and} \quad \sin \frac{1}{y_n} = 1, \quad \forall n > 1$$

and consequently the two sequences tend to two different limits. Hence the definition of limit is not satisfied and the limit does not exist.

The function has infinitely many oscillations in a finite space, so it is not possible to draw its graph on an interval $(0, b]$ with $b \in \mathbb{R}_+$.



Computing limits of functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Computing limits of functions

- We state the theorems on limits of functions that follow immediately from the corresponding theorems on limits of sequences and from the sequential definition of limit

1.1 Algebra of limits theorem

Theorem 1: Algebra of limits, case of finite limits

Hypotheses as $x \rightarrow c$:

$$1. f(x) \rightarrow \ell_1 \in \mathbb{R}, \quad 2. g(x) \rightarrow \ell_2 \in \mathbb{R}.$$

Thesis as $x \rightarrow c$:

$$1. f(x) \pm g(x) \rightarrow \ell_1 \pm \ell_2, \quad 2. f(x) g(x) \rightarrow \ell_1 \ell_2,$$

$$3. \frac{f(x)}{g(x)} \rightarrow \frac{\ell_1}{\ell_2} \quad (\ell_2 \neq 0, g(x) \neq 0, \text{ eventually as } x \rightarrow c).$$

Proof. Let $\{x_n\}$ be any sequence such that

$$x_n \neq c, \forall n, \text{ and } x_n \rightarrow c \text{ as } n \rightarrow +\infty.$$

By hypothesis we have

$$f(x_n) \rightarrow \ell_1 \text{ and } g(x_n) \rightarrow \ell_2 \quad (\ell_1, \ell_2 \in \mathbb{R}).$$

From the theorem on the algebra of limits for sequences we therefore conclude that

$$f(x_n) \pm g(x_n) \rightarrow \ell_1 \pm \ell_2,$$

and hence

$$f(x) \pm g(x) \rightarrow \ell_1 \pm \ell_2.$$

Claims 2 and 3 of the theorem are proved in a perfectly analogous way.

□

1.2 Sign-preservation theorems

In the following statements ℓ and c will be points of $\mathbb{R}^*$ unless otherwise stated.

Theorem 2: Sign preservation, 1st form

Hypotheses:

$$1. f(x) \rightarrow \ell \text{ as } x \rightarrow c, \quad 2. \ell \leq 0.$$

Thesis:

$$f(x) \leq 0 \text{ eventually as } x \rightarrow c.$$

Proof. Let $\{x_n\}$ be any sequence such that

$$x_n \neq c, \forall n, \text{ and } x_n \rightarrow c \text{ as } n \rightarrow +\infty$$

By the hypothesis we have

$$f(x_n) \rightarrow \ell > 0$$

hence, by the sign-preservation theorem for sequences, applied to the sequence $\{f(x_n)\}$, we conclude that

$$f(x_n) > 0, \text{ eventually.}$$

Since this holds for every sequence such that $x_n \rightarrow c$, we conclude that

$$f(x) > 0, \text{ eventually, as } x \rightarrow c$$

□

Theorem 3: Sign preservation for functions, 2nd form

Hypotheses:

$$1. f(x) \rightarrow \ell \in \mathbb{R} \text{ as } x \rightarrow c, \\ 2. f(x) \geq 0 \text{ eventually as } x \rightarrow c.$$

Thesis:

$$\ell \geq 0.$$

Proof. It follows from the corresponding theorem for sequences.

□

- For functions we also have the following sign-preservation theorem.

Theorem 4: Sign preservation for continuous functions

Hypotheses:

1. f is continuous at $c \in \mathbb{R}$,
2. $f(c) > 0$.

Thesis:

$$f(x) > 0 \text{ eventually as } x \rightarrow c.$$

Proof. If f is continuous at c , then

$$f(c) = \lim_{x \rightarrow c} f(x)$$

hence the hypothesis $f(c) > 0$ means that

$$f(c) = \lim_{x \rightarrow c} f(x) > 0$$

and this, by the sign-preservation theorem (1st form), implies that

$$f(x) > 0 \text{ eventually as } x \rightarrow c.$$

□

1.3 Comparison theorem

Theorem 5: Comparison theorem

Hypotheses:

1. $f(x) \rightarrow \ell$ and $g(x) \rightarrow \ell$ as $x \rightarrow c$,
2. $f(x) \leq h(x) \leq g(x)$ eventually as $x \rightarrow c$.

Thesis:

$$h(x) \rightarrow \ell \text{ as } x \rightarrow c.$$

Proof. Let $\{x_n\}$ be any sequence such that

$$x_n \neq c, \forall n, \text{ and } x_n \rightarrow c \text{ as } n \rightarrow +\infty$$

We want to prove that

$$h(x_n) \rightarrow \ell \text{ as } n \rightarrow +\infty$$

By hypothesis we know that:

$$f(x_n) \leq h(x_n) \leq g(x_n), \text{ eventually,}$$

$$f(x_n) \rightarrow \ell \text{ and } g(x_n) \rightarrow \ell, \text{ as } x_n \rightarrow c$$

Hence, by the comparison theorem for sequences applied to:

$$\{f(x_n)\}, \{h(x_n)\} \text{ and } \{g(x_n)\}$$

we conclude that

$$h(x_n) \rightarrow \ell \text{ as } n \rightarrow +\infty$$

□

Corollary 1: Of the comparison theorem (part I)

Hypotheses:

1. $g(x) \rightarrow 0$ as $x \rightarrow c$,
2. $|h(x)| \leq g(x)$ eventually as $x \rightarrow c$.

Thesis:

$$h(x) \rightarrow 0 \text{ as } x \rightarrow c.$$

Proof. It follows from the corresponding corollary for sequences.

□

Example 1: Corollary of the comparison theorem

Let us prove that:

$$\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$$

We have

$$x \rightarrow 0 \text{ as } x \rightarrow 0 \text{ and } \lim_{x \rightarrow 0} \sin \frac{1}{x} \text{ does not exist}$$

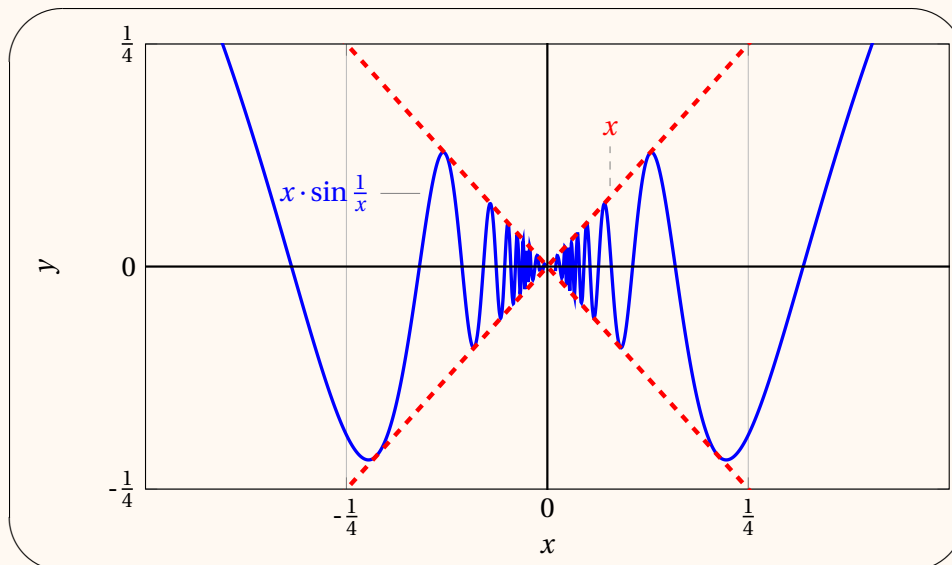
hence we cannot apply Theorem (??) on the algebra of limits for functions.

We have

$$\left| \sin \frac{1}{x} \right| \leq 1 \text{ hence } \left| x \sin \frac{1}{x} \right| \leq |x| \text{ and } |x| \rightarrow 0 \text{ as } x \rightarrow 0.$$

Hence, by Corollary (??) of the comparison theorem for functions, we have:

$$x \sin \frac{1}{x} \rightarrow 0 \text{ as } x \rightarrow 0.$$



Corollary 2: Of the comparison theorem (part II)

Hypotheses:

1. $f(x) \rightarrow 0$ as $x \rightarrow c$,
2. $g(x)$ is bounded, eventually, as $x \rightarrow c$.

Thesis:

$$f(x) g(x) \rightarrow 0 \text{ as } x \rightarrow c.$$

Proof. It follows from the corresponding corollary for sequences. □

Example 2: Corollary of the comparison theorem

Let us prove that:

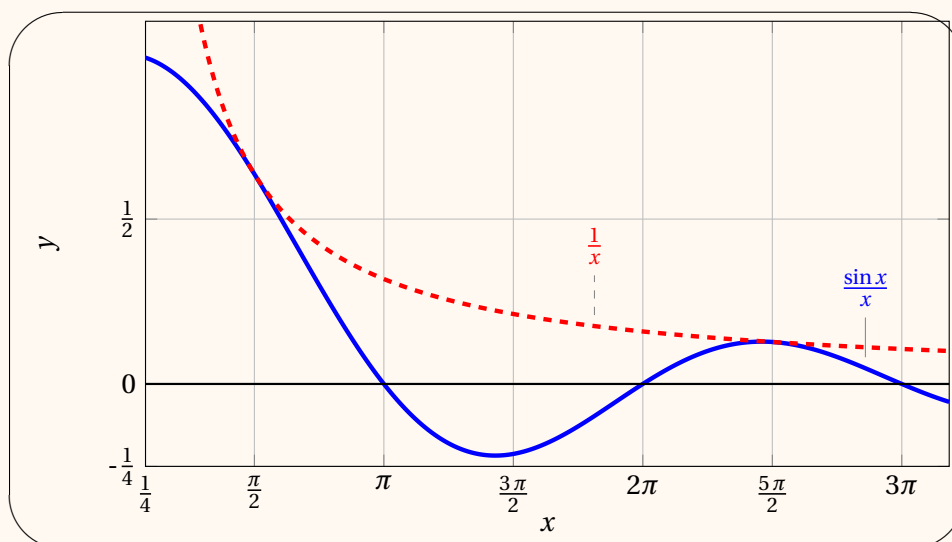
$$\lim_{x \rightarrow +\infty} \frac{x + \sin x}{2x + \cos x} = \frac{1}{2}$$

Factoring, we obtain

$$\frac{x + \sin x}{2x + \cos x} = \frac{x \left(1 + \frac{\sin x}{x}\right)}{2x \left(1 + \frac{\cos x}{2x}\right)} = \frac{1}{2} \left(\frac{1 + \frac{\sin x}{x}}{1 + \frac{\cos x}{2x}} \right)$$

By Corollary (??) of the comparison theorem for functions, we have:

$$\frac{\sin x}{x} \rightarrow 0 \text{ and } \frac{\cos x}{2x} \rightarrow 0 \text{ as } x \rightarrow +\infty.$$



Then, by Theorem (??) on the algebra of limits, we therefore have:

$$\frac{x + \sin x}{2x + \cos x} \rightarrow \frac{1}{2} \text{ as } x \rightarrow +\infty.$$

1.4 Theorems of partial arithmetization of the infinity symbol

Theorem 6: Partial arithmetization of the infinity symbol (addition)

Hypotheses as $x \rightarrow c$:

$$1. f(x) \rightarrow \ell \in \mathbb{R}, \quad 2. g(x) \rightarrow +\infty, \quad 3. h(x) \rightarrow +\infty.$$

Thesis as $x \rightarrow c$:

$$\begin{aligned} 1. f(x) + g(x) &\rightarrow \ell + \infty = +\infty, & 2. f(x) - g(x) &\rightarrow \ell - \infty = -\infty, \\ 3. g(x) + h(x) &\rightarrow +\infty + \infty = +\infty, & 4. -g(x) - h(x) &\rightarrow -\infty - \infty = -\infty. \end{aligned}$$

Proof. It follows from the corresponding theorem for sequences. □

Theorem 7: Partial arithmetization of the infinity symbol (product)

Hypotheses as $x \rightarrow c$:

$$1. f(x) \rightarrow \ell \in \mathbb{R}, \quad 2. g(x) \rightarrow 0, \quad 3. h(x) \rightarrow \infty.$$

Thesis as $x \rightarrow c$:

$$\begin{aligned} 1. f(x) h(x) &\rightarrow \ell \cdot \infty = \infty \quad (\ell \neq 0), & 2. \frac{f(x)}{g(x)} &\rightarrow \frac{\ell}{0} = \infty \quad (\ell \neq 0), \\ 3. \frac{f(x)}{h(x)} &\rightarrow \frac{\ell}{\infty} = 0. \end{aligned}$$

Proof. It follows from the corresponding theorem for sequences. □

- As for sequences, the **sign** of ∞ must be determined with the **rule of signs**.

Example 3: Partial arithmetization of the infinity symbol

$$\lim_{x \rightarrow -\infty} \left(\frac{1}{x} - 2 \right) x^3 = +\infty$$

Since $\left(\frac{1}{x} - 2 \right) \rightarrow -2$ and $x^3 \rightarrow -\infty$ as $x \rightarrow -\infty$.

2 Change of variable theorem for limits

Theorem 8: Change of variable in the limit

Let f and g be two functions for which the composition $f \circ g$ is defined, at least eventually as $x \rightarrow x_0 \in \mathbb{R}^*$. If:

1. $\lim_{x \rightarrow x_0} g(x) = t_0 \in \mathbb{R}^*$, 2. $\lim_{t \rightarrow t_0} f(t) = \ell \in \mathbb{R}^*$
3. $g(x) \neq t_0$, eventually, as $x \rightarrow x_0$

Then:

$$\lim_{x \rightarrow x_0} f(g(x)) = \lim_{t \rightarrow t_0} f(t).$$

Hypothesis 3. is not necessary when f is continuous at t_0 , or when $t_0 = \pm\infty$.

Proof. Let $\{x_n\}$ be any sequence such that

$$x_n \neq x_0, \forall n, \text{ and } x_n \rightarrow x_0 \text{ as } n \rightarrow +\infty.$$

We have

$$g(x_n) \rightarrow t_0 \text{ as } n \rightarrow +\infty \text{ (by hypothesis 1)}$$

and

$$g(x_n) \neq t_0, \text{ eventually (by hypothesis 3).}$$

Therefore

$$f(g(x_n)) \rightarrow \ell \text{ (by hypothesis 2).}$$

If $t_0 = \pm\infty$ the condition $g(x) \neq \pm\infty$ is obviously satisfied, while if f is continuous at t_0 , $\ell = f(t_0)$, so if $g(x_n) = t_0$ for some n we would have

$$f(g(x_n)) = f(t_0) = \ell$$

and hence the convergence

$$f(g(x_n)) \rightarrow \ell \text{ as } n \rightarrow +\infty$$

would be guaranteed anyway.

□

Example 4: Computing a limit with the change of variable theorem

Let us compute the limit

$$\lim_{x \rightarrow +\infty} \log \left(\frac{2x^3 + 4x + 1}{5(x+1)^3} \right)$$

The functions f and g are:

$$g(x) = \left(\frac{2x^3 + 4x + 1}{5(x+1)^3} \right) \quad \text{and} \quad f(t) = \log t.$$

We have $t = g(x)$ and

$$(f \circ g)(x) = f(g(x)) = \log \left(\frac{2x^3 + 4x + 1}{5(x+1)^3} \right) \quad \text{and} \quad x_0 = +\infty.$$

We compute

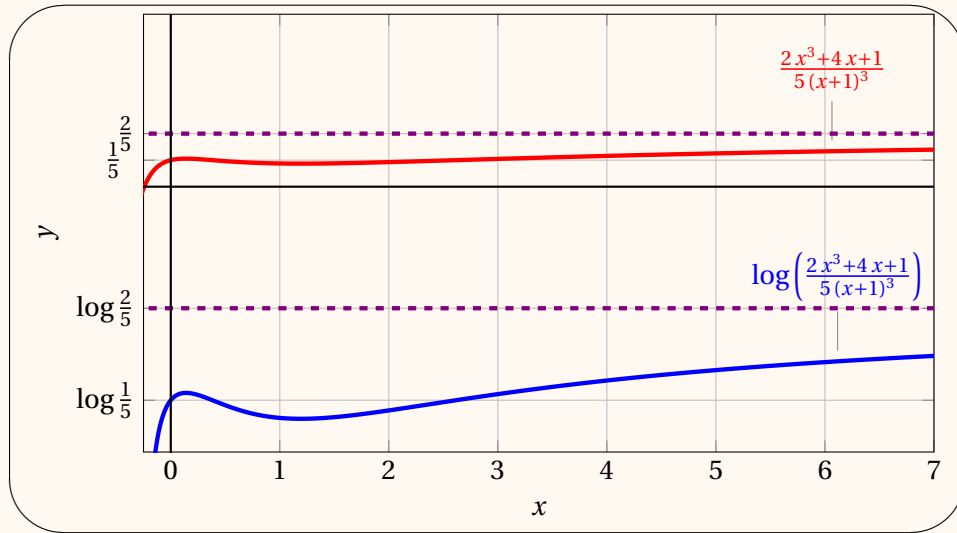
$$\lim_{x \rightarrow +\infty} g(x) = \frac{2}{5} \quad \text{and consequently} \quad t_0 = \frac{2}{5}.$$

Now we compute

$$\lim_{t \rightarrow 2/5} \log t = \log \frac{2}{5} \quad (\text{as we will see, the logarithm is continuous in } \mathbb{R}_+)$$

Hence

$$\lim_{x \rightarrow +\infty} f(g(x)) = \log \frac{2}{5}$$



Moreover:

$$\lim_{x \rightarrow 0} f(g(x)) = \log \frac{1}{5}$$

using the theorem we have:

$$\lim_{x \rightarrow 0} g(x) = \frac{1}{5} \quad \text{and} \quad \lim_{t \rightarrow 1/5} \log t = \log \frac{1}{5}$$

since

$$\lim_{x \rightarrow 0} 2x^3 + 4x + 1 = 1 \quad \text{and} \quad \lim_{x \rightarrow 0} 5(x+1)^3 = 5$$

Example 5: Computing a limit with the change of variable theorem

Let us compute the limits

$$\lim_{x \rightarrow 0^+} e^{\frac{1}{x}} \quad \text{and} \quad \lim_{x \rightarrow 0^-} e^{\frac{1}{x}}.$$

The functions f and g are:

$$g(x) = \frac{1}{x} \quad \text{and} \quad f(t) = e^t.$$

We have $t = g(x)$ and

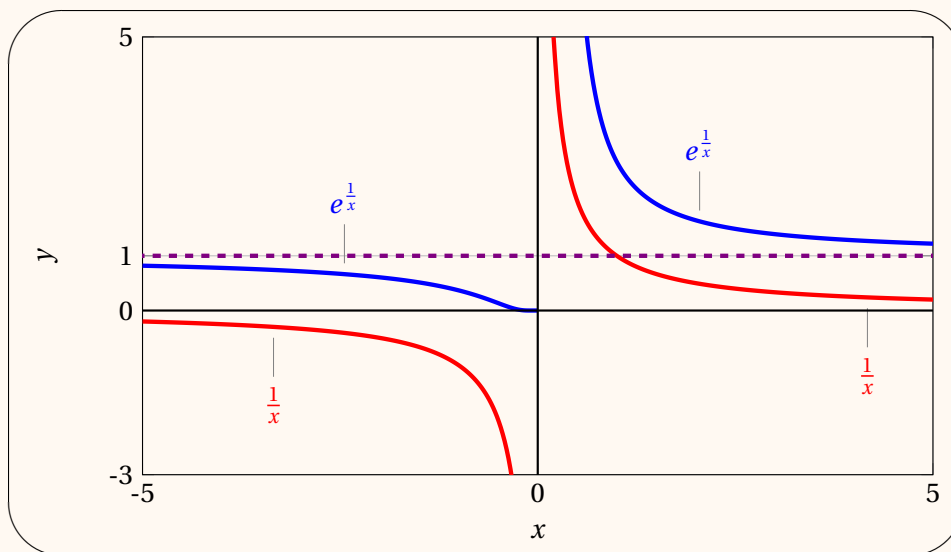
$$(f \circ g)(x) = f(g(x)) = e^{\frac{1}{x}} \quad \text{moreover} \quad x_0 = 0^+ \quad \text{and} \quad x_0 = 0^-.$$

1. $\frac{1}{x} \rightarrow +\infty$ as $x \rightarrow 0^+$ and hence $t_0 = +\infty$. Now we compute:

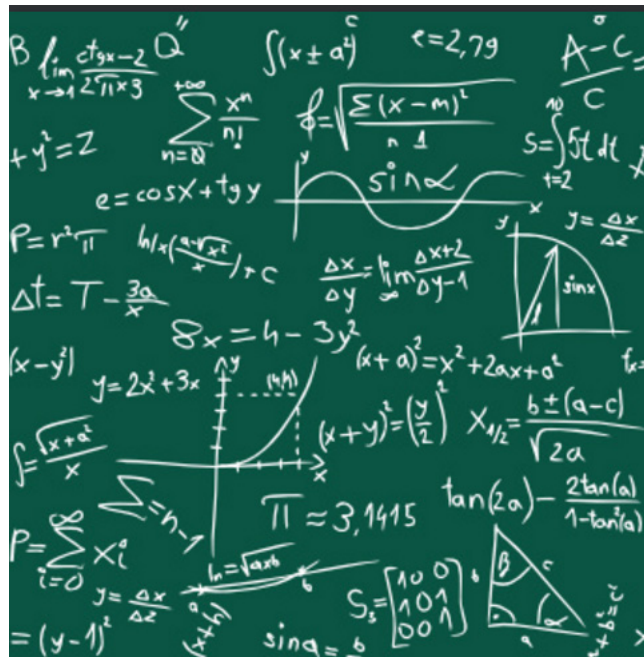
$$\lim_{t \rightarrow +\infty} e^t = +\infty \quad \text{hence} \quad \lim_{x \rightarrow 0^+} e^{\frac{1}{x}} = +\infty.$$

2. $\frac{1}{x} \rightarrow -\infty$ as $x \rightarrow 0^-$ and hence $t_0 = -\infty$. Now we compute:

$$\lim_{t \rightarrow -\infty} e^t = 0 \quad \text{hence} \quad \lim_{x \rightarrow 0^-} e^{\frac{1}{x}} = 0.$$



Limits of polynomials and rational functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

Contents

1	Limits of polynomials at $\pm\infty$	2
2	Limits of rational functions at $\pm\infty$	2
3	Limits of quotients of sums of powers with rational exponents	4

1 Limits of polynomials at $\pm\infty$

- A polynomial of (maximum) degree n can be written as:

$$P_n(x) = \sum_{i=0}^n a_i x^i \quad \text{with } a_i \in \mathbb{R}, \text{ for } i \in \{0, 1, \dots, n\}, \text{ and } a_n \neq 0.$$

The value a_i is the **coefficient of the monomial** i , with $i = 0, 1, \dots, n$, while x^i is the **literal part** of the monomial. In this notation, without loss of generality, the monomials are ordered by increasing values of the exponents.¹

- Factoring out $a_n x^n$, that is, the last monomial, which is the one of maximum degree, we have:

$$P_n(x) = a_n x^n \left(\overbrace{\frac{a_0}{a_n x^n} + \frac{a_1}{a_n x^{n-1}} + \dots + \frac{a_{n-1}}{a_n x}}^{\rightarrow 0 \text{ as } x \rightarrow \pm\infty} + 1 \right)$$

$$\lim_{x \rightarrow \pm\infty} P_n(x) = \lim_{x \rightarrow \pm\infty} a_n x^n$$

To compute the limit of a polynomial as $x \rightarrow \pm\infty$, it is enough to compute the limit of the monomial of maximum degree.

Example 1: Limits of polynomials as $x \rightarrow \pm\infty$

For example:

$$\lim_{x \rightarrow +\infty} (8 - x^2 + 3x^3) = \lim_{x \rightarrow +\infty} 3x^3 \left(\underbrace{\frac{8}{3x^3} - \frac{1}{3x}}_{\rightarrow 0 \text{ as } x \rightarrow +\infty} + 1 \right) = \lim_{x \rightarrow +\infty} 3x^3 = +\infty$$

$$\lim_{x \rightarrow -\infty} (120 + 4x - 2x^3) = \lim_{x \rightarrow -\infty} -2x^3 \left(\underbrace{\frac{120}{-2x^3} + \frac{4}{-2x^2}}_{\rightarrow 0 \text{ as } x \rightarrow -\infty} + 1 \right) = \lim_{x \rightarrow -\infty} -2x^3 = +\infty$$

2 Limits of rational functions at $\pm\infty$

- Let f be a rational function (ratio of polynomials):

$$f(x) = \frac{P_n(x)}{P_m(x)}$$

¹If a monomial of degree i is missing we have $a_i = 0$; a_0 is the constant term since $x^0 = 1$.

where $P_n(x)$ and $P_m(x)$ are polynomials of degree n and m , respectively:

$$P_n(x) = \sum_{i=0}^n a_i x^i \quad \text{with } a_i \in \mathbb{R}, \text{ for } i = 0, 1, \dots, n, \text{ and } a_n \neq 0.$$

$$P_m(x) = \sum_{i=0}^m b_i x^i \quad \text{with } b_i \in \mathbb{R}, \text{ for } i = 0, 1, \dots, m, \text{ and } b_m \neq 0.$$

- Factoring out $a_n x^n$ in the numerator and $b_m x^m$ in the denominator, that is, the monomials of maximum degree, we have:

$$\frac{P_n(x)}{P_m(x)} = \frac{a_n x^n}{b_m x^m} \frac{\left\{ \overbrace{\frac{a_0}{a_n x^n} + \frac{a_1}{a_n x^{n-1}} + \dots + \frac{a_{n-1}}{a_n x} + 1}^{\rightarrow 0 \text{ as } x \rightarrow \pm\infty} \right\}}{\left\{ \underbrace{\frac{b_0}{b_m x^m} + \frac{b_1}{b_m x^{m-1}} + \dots + \frac{b_{m-1}}{b_m x} + 1}_{\rightarrow 0 \text{ as } x \rightarrow \pm\infty} \right\}}.$$

$$\lim_{x \rightarrow \pm\infty} \frac{P_n(x)}{P_m(x)} = \lim_{x \rightarrow \pm\infty} \frac{a_n x^n}{b_m x^m} = \lim_{x \rightarrow \pm\infty} \frac{a_n}{b_m} x^{n-m}$$

To compute the limit of a ratio of polynomials as $x \rightarrow \pm\infty$, it is enough to compute the limit of the ratio of the monomials of maximum degree.

Example 2: Limits of rational functions as $x \rightarrow \pm\infty$

For example:

$$\lim_{x \rightarrow -\infty} \frac{1 + 3x^3}{2 + x^2} = \lim_{x \rightarrow -\infty} \frac{3x^3}{x^2} = \lim_{x \rightarrow -\infty} 3x = -\infty$$

$$\lim_{x \rightarrow +\infty} \frac{1 + 3x^3}{1 - 10x + x^4} = \lim_{x \rightarrow +\infty} \frac{3x^3}{x^4} = \lim_{x \rightarrow +\infty} 3 \frac{1}{x} = 0$$

$$\lim_{x \rightarrow -\infty} \frac{7x^2 + 6x^5}{-4 + 7x^5} = \lim_{x \rightarrow -\infty} \frac{6x^5}{7x^5} = \lim_{x \rightarrow -\infty} \frac{6}{7} = \frac{6}{7}$$

To compute the limit of rational functions (without the constant term) as $x \rightarrow 0$ we must factor out, in the numerator and in the denominator, the powers of minimum degree.

Example 3: Limits of rational functions as $x \rightarrow 0$

For example:

$$\lim_{x \rightarrow 0} \frac{-2x^2 + 3x^3 - 6x^5}{x + x^2 - x^7} = \lim_{x \rightarrow 0} \frac{-2x^2 \left(\overbrace{1 - \frac{3}{2}x + 3x^3}^{\rightarrow 0 \text{ as } x \rightarrow 0} \right)}{x \left(\overbrace{1 + x - x^6}^{\rightarrow 0 \text{ as } x \rightarrow 0} \right)} = \lim_{x \rightarrow 0} -2x = 0$$

3 Limits of quotients of sums of powers with rational exponents

- The same rules hold for quotients of sums of powers with rational exponents:

as $x \rightarrow \infty$ we must factor out the power with the maximum exponent

as $x \rightarrow 0^+$ we must factor out the power with the minimum exponent

Raising to a power can also be defined with a negative base if the exponent is a rational number (fraction) with an odd denominator. Hence in this case we can compute the limit as $x \rightarrow 0$ (otherwise only as $x \rightarrow 0^+$).

Example 4: Limits of quotients of sums of powers with rational exponents

For example:

$$\lim_{x \rightarrow 0} \frac{x^{2/3}}{3x^{1/3} + x + x^2}$$

Factoring out in the denominator the power with the minimum exponent, we have:

$$\frac{x^{2/3}}{3x^{1/3} + x + x^2} = \frac{x^{2/3}}{3x^{1/3} \left(1 + \frac{x}{3x^{1/3}} + \frac{x^2}{3x^{1/3}} \right)} = \frac{x^{2/3}}{3x^{1/3} \left(1 + \frac{x^{2/3}}{3} + \frac{x^{5/3}}{3} \right)} = \frac{1}{3} x^{1/3} \frac{1}{\left(1 + \frac{x^{2/3}}{3} + \frac{x^{5/3}}{3} \right)}$$

hence

$$\lim_{x \rightarrow 0} \frac{x^{2/3}}{3x^{1/3} + x + x^2} = \frac{1}{3} \lim_{x \rightarrow 0} x^{1/3} \frac{1}{\left(1 + \underbrace{\frac{x^{2/3}}{3} + \frac{x^{5/3}}{3}}_{\rightarrow 0 \text{ as } x \rightarrow 0} \right)} = 0$$

Another option to compute limits as $x \rightarrow 0$ is to make a change of variable:

$$x = \frac{1}{y}, \text{ if } x \rightarrow 0^+ \text{ then } y \rightarrow +\infty, \text{ if } x \rightarrow 0^- \text{ then } y \rightarrow -\infty$$

and hence we reduce to the case of limits at $\pm\infty$.

Example 5: Computing the limit of rational functions as $x \rightarrow 0^+$ with a change of variable

We want to compute the following limit:

$$\lim_{x \rightarrow 0} \frac{x^{2/3}}{3x^{1/3} + x + x^2}, \quad f(x) = \frac{x^{2/3}}{3x^{1/3} + x + x^2}$$

Making the change of variable $x = \frac{1}{y}$, we therefore have to compute the following two limits:

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{y \rightarrow +\infty} \frac{\left(\frac{1}{y}\right)^{2/3}}{3\left(\frac{1}{y}\right)^{1/3} + \frac{1}{y} + \left(\frac{1}{y}\right)^2} \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = \lim_{y \rightarrow -\infty} \frac{\left(\frac{1}{y}\right)^{2/3}}{3\left(\frac{1}{y}\right)^{1/3} + \frac{1}{y} + \left(\frac{1}{y}\right)^2}$$

We have:

$$\frac{\left(\frac{1}{y}\right)^{2/3}}{3\left(\frac{1}{y}\right)^{1/3} + \frac{1}{y} + \left(\frac{1}{y}\right)^2} = \frac{y^{-2/3}}{3y^{-1/3} + y^{-1} + y^{-2}}$$

Factoring out in the denominator the power with the maximum exponent, we have:

$$\frac{y^{-2/3}}{3y^{-1/3} + y^{-1} + y^{-2}} = \frac{y^{-2/3}}{3y^{-1/3}\left(1 + \frac{y^{-1}}{3y^{-1/3}} + \frac{y^{-2}}{3y^{-1/3}}\right)} = \frac{1}{3} \frac{1}{y^{1/3}} \frac{1}{\left(1 + \frac{1}{3y^{2/3}} + \frac{1}{3y^{5/3}}\right)}$$

Hence:

$$\frac{1}{3} \lim_{y \rightarrow -\infty} \frac{1}{y^{1/3}} \frac{1}{\left(1 + \frac{1}{3y^{2/3}} + \frac{1}{3y^{5/3}}\right)} = \frac{1}{3} \lim_{y \rightarrow +\infty} \frac{1}{y^{1/3}} \frac{1}{\left(1 + \frac{1}{3y^{2/3}} + \frac{1}{3y^{5/3}}\right)} = 0$$

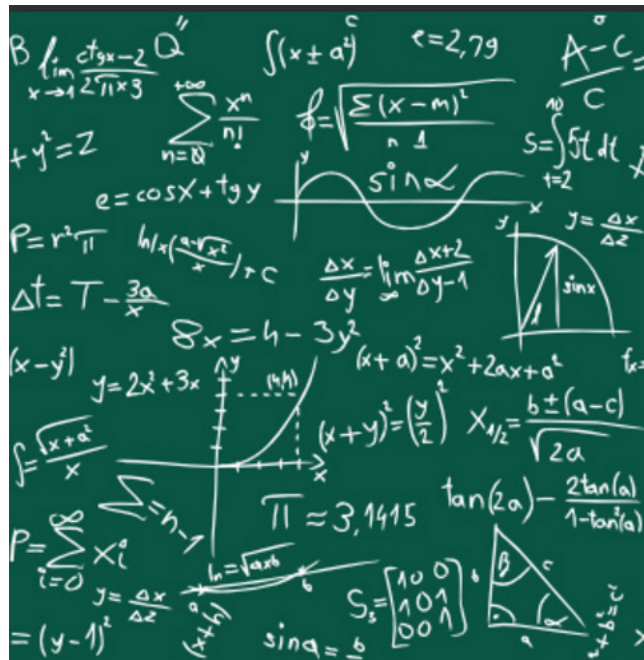
since, as $y \rightarrow \pm\infty$, we have

$$\frac{1}{y^{1/3}} \rightarrow 0 \quad \text{and} \quad \left(1 + \frac{1}{3y^{2/3}} + \frac{1}{3y^{5/3}}\right) \rightarrow 1 \quad \text{since} \quad \frac{1}{3y^{2/3}} \rightarrow 0, \quad \frac{1}{3y^{5/3}} \rightarrow 0.$$

Since the right limit and the left limit of $f(x)$ as $x \rightarrow 0$ exist and are equal to 0, then:

$$\lim_{x \rightarrow 0} f(x) = 0$$

Continuous functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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2	Continuity of elementary functions theorem	2
3	Continuity of composite functions theorem	5

1 Algebra of continuous functions theorem

Theorem 1: Algebra of continuous functions

Let f and g be two functions defined at least in a neighborhood of $x_0 \in \mathbb{R}$ and continuous at x_0 . Then:

1. $f(x) \pm g(x)$ is continuous at x_0 ; 2. $f(x) \cdot g(x)$ is continuous at x_0 ;

3. $\frac{f(x)}{g(x)}$ is continuous at x_0 provided that $g(x_0) \neq 0$.

Proof. Let us prove, for example, claim 3., the others being analogous. By hypothesis we know that f and g are continuous at x_0 , that is:

$$f(x) \rightarrow f(x_0) \text{ and } g(x) \rightarrow g(x_0) \text{ as } x \rightarrow x_0.$$

Moreover $g(x_0) \neq 0$ and hence, by the sign-preservation theorem for continuous functions, $g(x) \neq 0$ eventually as $x \rightarrow x_0$.

Then, by the theorem on the algebra of limits, we conclude that

$$\frac{f(x)}{g(x)} \rightarrow \frac{f(x_0)}{g(x_0)} \text{ as } x \rightarrow x_0$$

that is, $f(x)/g(x)$ is continuous at x_0 . □

2 Continuity of elementary functions theorem

Theorem 2: Continuity of elementary functions

The following elementary functions are continuous at all points of their domain:

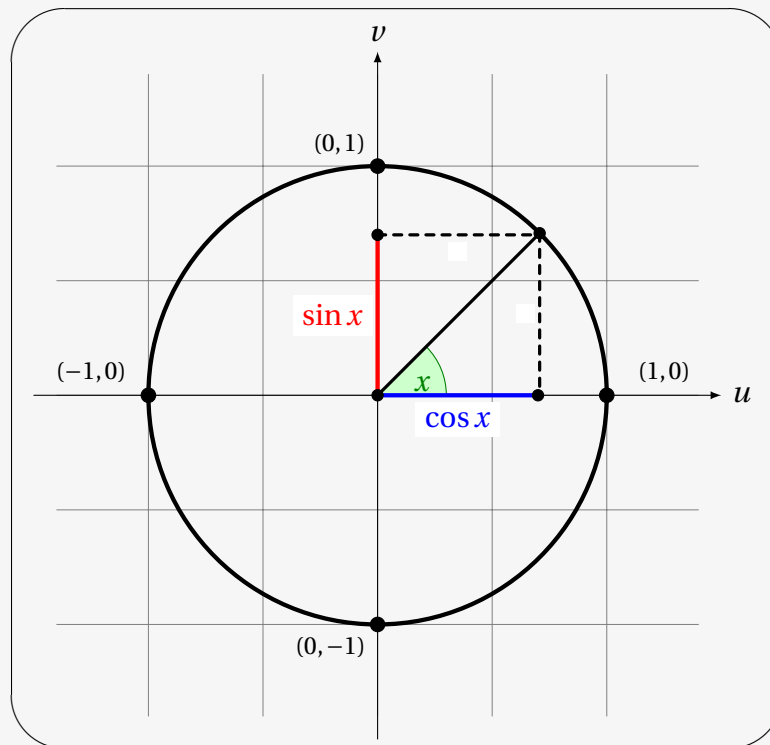
1. Powers with integer, rational or real exponent;
2. Exponential functions;
3. Logarithmic functions;
4. Elementary trigonometric functions ($\sin x$, $\cos x$)

Proof. For example, let us prove the continuity on the whole of $\mathbb{R}$ of the functions $\sin x$ and $\cos x$.

- We have seen that $\sin x$ is continuous at $x = 0$; let us show that $\cos x$ is also continuous at $x = 0$.
- The unit (trigonometric) circle shows that, if x is an angle in the first quadrant,

$$\sin^2 x + \cos^2 x = 1$$

since 1 is the hypotenuse of a right triangle with legs $\sin x$, $\cos x$.



- It follows that

$$0 \leq 1 - \cos x \leq \sin x \quad \text{for } x \in \left[0, \frac{\pi}{2}\right], \text{ and hence}$$

$$0 \leq 1 - \cos x \leq |\sin x| \quad \text{for } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

- Then, by the comparison theorem,

$$1 - \cos x \rightarrow 0 \text{ as } x \rightarrow 0 \text{ hence } \cos x \rightarrow 1 \text{ as } x \rightarrow 0$$

and therefore $\cos x$ is continuous at 0.

□

Proof. • To now prove the continuity of $\sin x$ at a generic point $x_0 \in \mathbb{R}$ we consider the chain of relations:

$$\begin{aligned} |\sin(x_0 + h) - \sin x_0| &= |\sin x_0 \cos h + \sin h \cos x_0 - \sin x_0| \\ &= |\sin x_0 (\cos h - 1) + \cos x_0 \sin h| \\ &\leq |\sin x_0 (\cos h - 1)| + |\cos x_0 \sin h| \\ &= |\sin x_0| |\cos h - 1| + |\cos x_0| |\sin h| \end{aligned}$$

Now $|\sin h|$ and $|\cos h - 1|$ tend to zero as $h \rightarrow 0$, by what we have just proved, while $|\sin x_0|$ and $|\cos x_0|$ are constants, so

$$\sin(x_0 + h) - \sin x_0 \rightarrow 0 \text{ as } h \rightarrow 0$$

that is

$$\sin(x_0 + h) \rightarrow \sin x_0 \text{ as } h \rightarrow 0$$

and $\sin x$ is continuous at x_0 . An analogous argument shows the continuity of $\cos x$.

□

- Having proved that $\sin x$, $\cos x$ are continuous on the whole of $\mathbb{R}$, we deduce that the functions $\tan x$, $\cot x$ are continuous on their domain (by the theorem on the algebra of continuous functions)
- Power functions with integer exponent are continuous (on the whole of $\mathbb{R}$), since $f(x) = x$ is obviously continuous, and $f(x) = x^n$ is the product of n continuous functions.
- Polynomials are continuous functions, since they are obtained by adding functions of the type $c x^n$, which are continuous by the previous point.
- Rational functions (i.e., quotients of polynomials) are continuous functions, except at the points where the denominator vanishes (the denominator is a polynomial, hence it vanishes at a finite number of points)

3 Continuity of composite functions theorem

Theorem 3: Continuity of the composite function

Let:

- g be a function defined at least in a neighborhood of x_0 and continuous at x_0 ,
- f be a function defined at least in a neighborhood of $t_0 = g(x_0)$ and continuous at t_0 ,

then $f \circ g$ is defined at least in a neighborhood of x_0 and is continuous at x_0 .

Proof. Since g is continuous at x_0 ,

$$\lim_{x \rightarrow x_0} g(x) = g(x_0) = t_0$$

then, by the change of variable theorem for limits, we have:

$$\lim_{x \rightarrow x_0} f(g(x)) = \lim_{t \rightarrow t_0} f(t)$$

and since f is continuous at t_0 , we have

$$\lim_{t \rightarrow t_0} f(t) = f(t_0)$$

and the thesis is proved. □

- It follows that all the functions that can be obtained from elementary functions by sums, products, quotients and compositions are continuous on their domain. **Hence, by combining functions in this way, we still obtain continuous functions.**

To summarize:

- sums, products and quotients of continuous functions give continuous functions (where the denominator does not vanish);
- the elementary functions of mathematical analysis are continuous on their domain;
- the composition of continuous functions gives a continuous function.

- It is therefore possible to know a priori that a function is continuous on its domain, without applying the definition of continuity case by case.

For continuous functions, if x_0 is a point of the domain, the limit as $x \rightarrow x_0$ is computed simply by substituting x_0 into the analytic expression of the function, that is:

$$f(x) \rightarrow f(x_0) \quad \text{as } x \rightarrow x_0$$

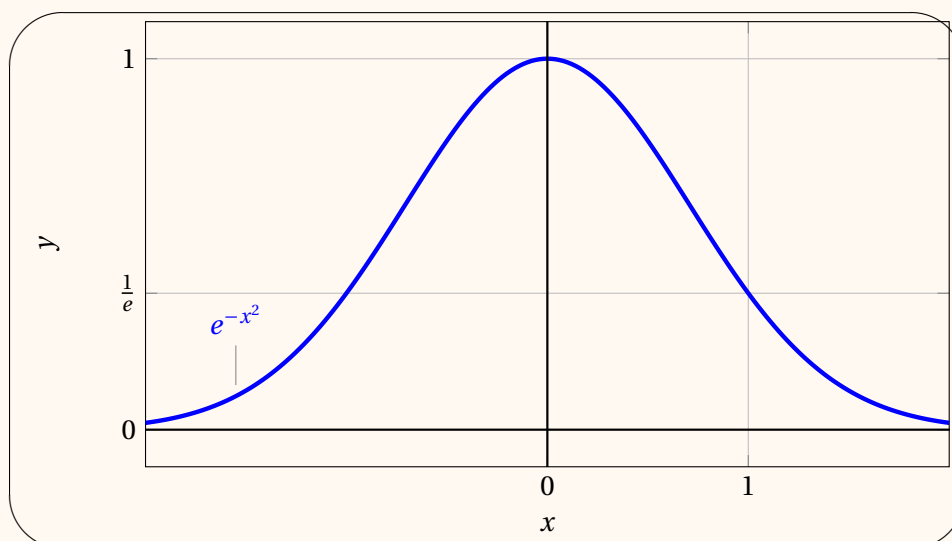
Example 1: Continuous functions and finite limits at a finite point

The following function is continuous for $x \in \mathbb{R}$:

$$f(x) = e^{-x^2}$$

hence:

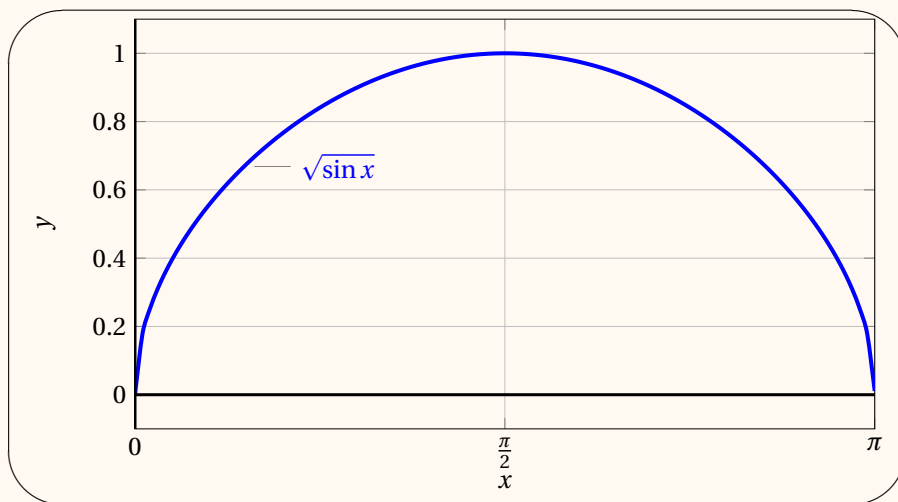
$$\lim_{x \rightarrow 1} e^{-x^2} = e^{-1} = \frac{1}{e}$$



Example 2: Continuous functions and finite limits at a finite point

The following function is continuous for $x \in \mathbb{R}$ with $2k\pi \leq x \leq 2k\pi + \pi, \forall k \in \mathbb{Z}$:

$$f(x) = \sqrt{\sin x} \quad \text{hence} \quad \lim_{x \rightarrow \pi^-} \sqrt{\sin x} = \sqrt{\sin \pi} = 0$$



Example 3: Continuous functions and finite limits at a finite point

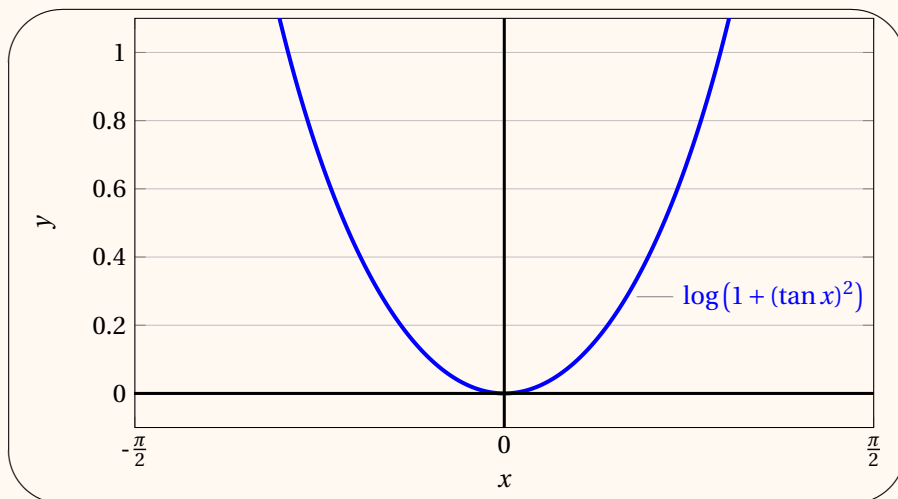
The following function is continuous for $x \in \mathbb{R}$ with $x \neq (2k+1)\frac{\pi}{2}, \forall k \in \mathbb{Z}$:

$$f(x) = \log_a(1 + (\tan x)^2), \quad \forall a > 0, a \neq 1$$

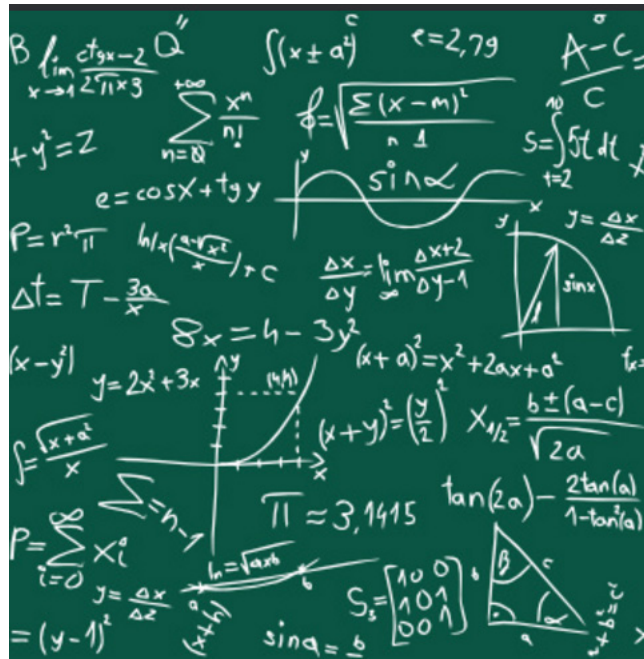
hence:

$$\lim_{x \rightarrow 0} \log_a(1 + (\tan x)^2) = \log_a(1 + (\tan 0)^2) = \log_a 1 = 0$$

For example, with $a = e$ we have:



Comparison of infinities



- **Author:** Fabio Furini.

Contents

1 Comparison of infinities for functions

1 Comparison of infinities for functions

Theorem 1: Comparison of infinities

Given $\alpha, \beta, \lambda > 0$, $a, b > 1$, we have

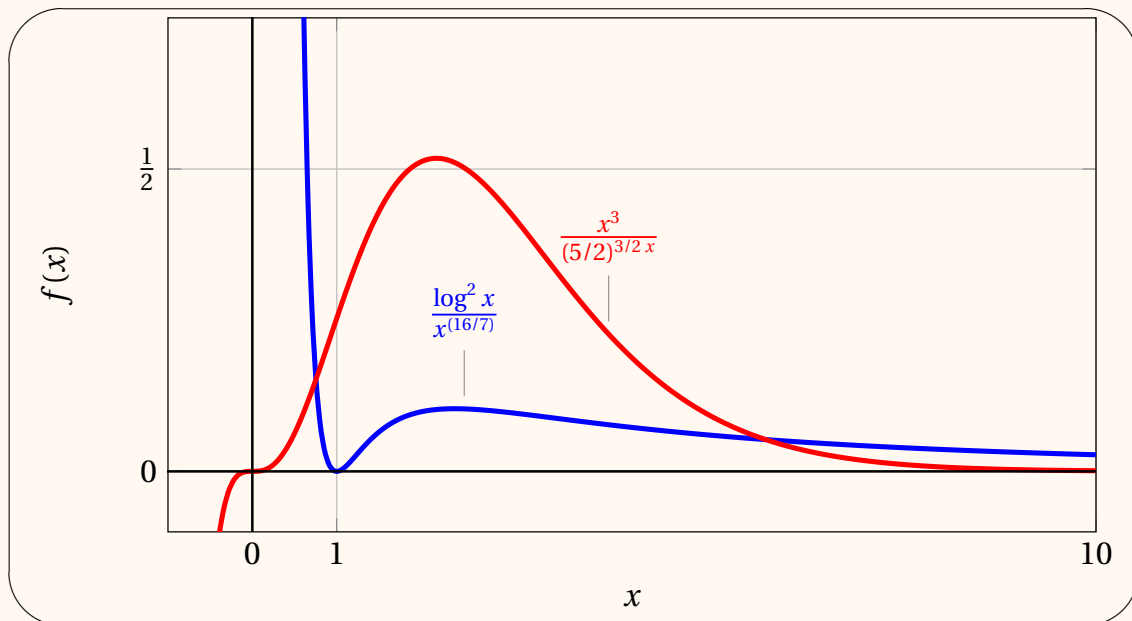
$$\lim_{x \rightarrow +\infty} \frac{\log_a^\beta x}{x^\alpha} = 0 \quad \text{and} \quad \lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = 0$$

Proof. The theorem will be proved with the tools of differential calculus. □

- Any power with positive exponent is an infinity of higher order than any power of logarithms with base > 1 , and
- Any exponential with base > 1 is an infinity of higher order than any power with positive exponent.

Example 1: Comparison of infinities

$$\lim_{x \rightarrow +\infty} \frac{\log^2 x}{x^{(16/7)}} = 0, \quad \lim_{x \rightarrow +\infty} \frac{x^3}{(5/2)^{3/2 x}} = 0$$



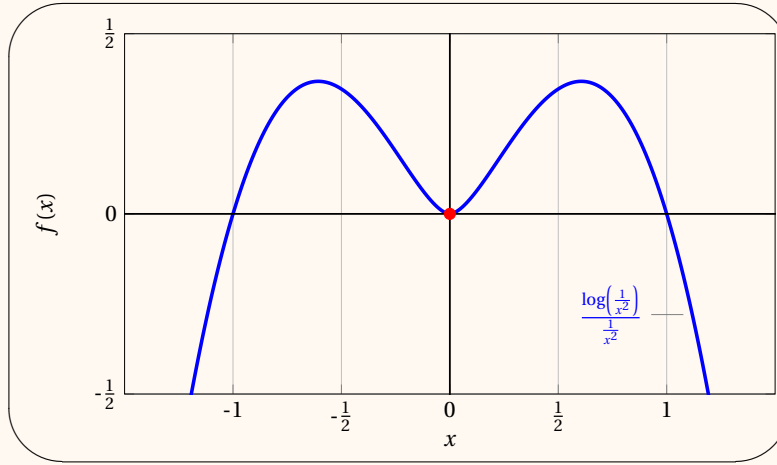
If a function $\eta(x) \rightarrow +\infty$ as $x \rightarrow c \in \mathbb{R}^*$, we have, from the comparison of infinities theorem:

$$\lim_{x \rightarrow c} \frac{(\log_a(\eta(x)))^\beta}{(\eta(x))^\alpha} = 0 \quad \text{and} \quad \lim_{x \rightarrow c} \frac{(\eta(x))^\alpha}{b^\lambda \eta(x)} = 0 \quad \text{with } \alpha, \beta, \lambda > 0, a, b > 1$$

Example 2: Comparison of infinities

$$\lim_{x \rightarrow 0} \frac{\log\left(\frac{1}{x^2}\right)}{\frac{1}{x^2}} = 0$$

since, with $\eta(x) = \frac{1}{x^2}$, we have $\eta(x) \rightarrow +\infty$ as $x \rightarrow 0$.



Remark 1

$$\lim_{x \rightarrow 0^+} x^\alpha \log_a^\beta x = 0 \quad (\alpha > 0, \beta = \frac{m}{n}, n, m \in \mathbb{N}, n \text{ odd}, a > 1)$$

Proof. Given $\alpha, \beta > 0, a > 1$ we have:

$$\lim_{x \rightarrow +\infty} \frac{\log_a^\beta x}{x^\alpha} = 0 \quad \text{and} \quad \frac{\log_a^\beta x}{x^\alpha} = \left(\frac{1}{x}\right)^\alpha \left(-\log_a\left(\frac{1}{x}\right)\right)^\beta$$

Setting $t = \frac{1}{x}$, $x \rightarrow +\infty$ is equivalent to $t \rightarrow 0^+$, and substituting we have:

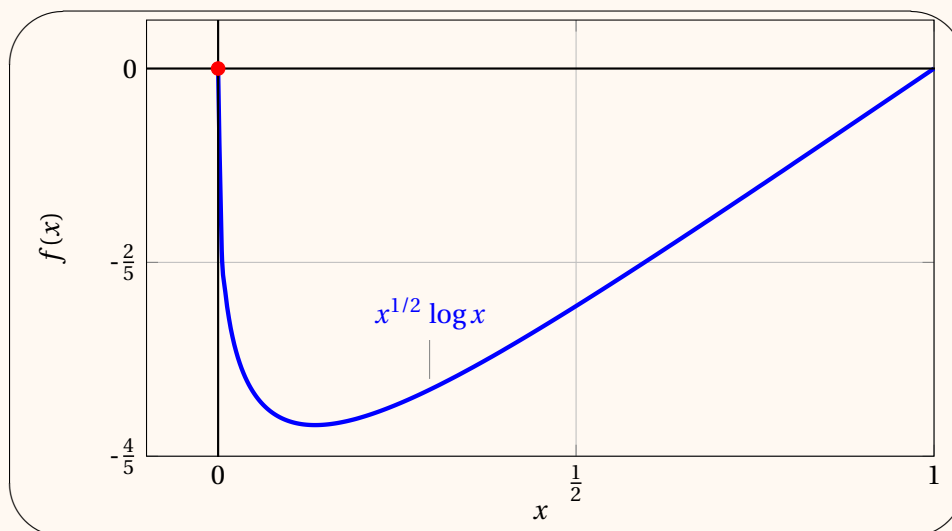
$$\lim_{x \rightarrow +\infty} \frac{\log_a^\beta x}{x^\alpha} = (-1)^\beta \lim_{t \rightarrow 0^+} t^\alpha \log_a^\beta t = 0$$

□

Example 3: Comparison of infinities

$$\lim_{x \rightarrow 0^+} x^{1/2} \log x = 0^-$$

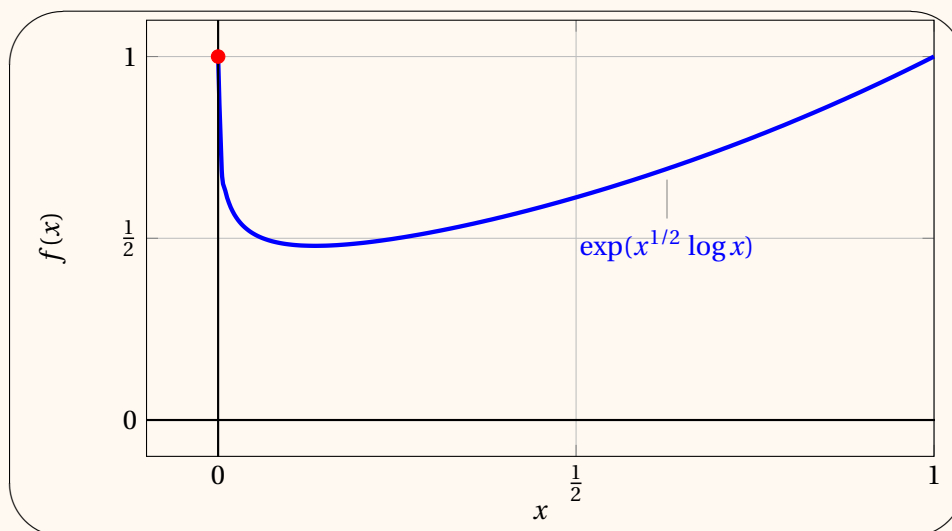
since we have $\alpha = \frac{1}{2} > 0, \beta = 1 > 0$ and $a = e > 1$.



$$\lim_{x \rightarrow 0^+} x^{\sqrt{x}} = 1^-$$

since

$$\lim_{x \rightarrow 0^+} x^{\sqrt{x}} = \lim_{x \rightarrow 0^+} \underbrace{e^{x^{1/2} \log x}}_{=\exp(x^{1/2} \log x)} = e^{0^-} = 1^-$$



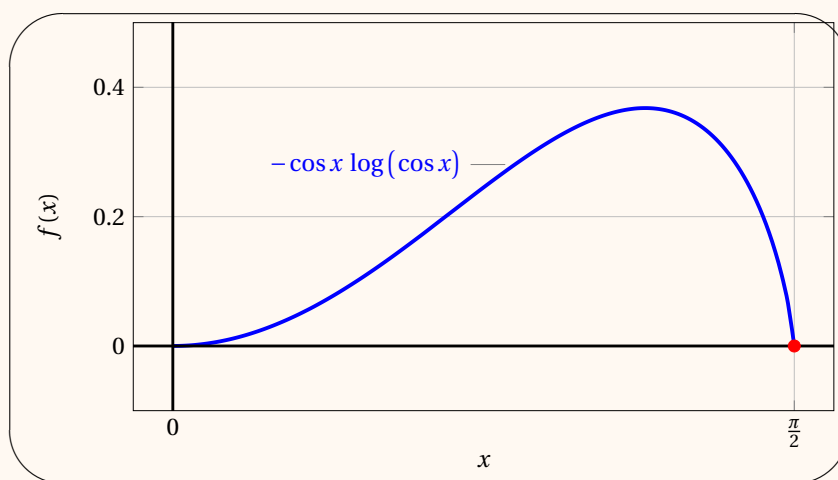
If a function $\varepsilon(x)$ tends to 0^+ as $x \rightarrow c \in \mathbb{R}^*$, it follows from the previous remark that:

$$\lim_{x \rightarrow c} (\varepsilon(x))^\alpha (\log_a(\varepsilon(x)))^\beta = 0 \quad (\alpha > 0, \beta = \frac{m}{n}, n, m \in \mathbb{N}, n \text{ odd}, a > 1)$$

Example 4: Comparison of infinities

$$\lim_{x \rightarrow \frac{\pi}{2}^-} -\cos x \log(\cos x) = 0^+$$

since, with $\varepsilon(x) = \cos x$, we have $\varepsilon(x) \rightarrow 0^+$ as $x \rightarrow \frac{\pi}{2}^-$ and $\alpha = \beta = 1$.



From the comparison of infinities theorem it follows that:

$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{\log_a^\beta x} = +\infty \quad \text{and} \quad \lim_{x \rightarrow +\infty} \frac{b^{\lambda x}}{x^\alpha} = +\infty \quad \text{with} \quad \alpha, \beta, \lambda > 0, a, b > 1$$

Remark 2

$$\lim_{x \rightarrow 0^+} x^\alpha b^{\lambda/x} = +\infty \quad (\alpha, \lambda > 0, b > 1)$$

Proof. Given $\alpha, \lambda > 0, b > 1$, we have:

$$\lim_{x \rightarrow +\infty} \frac{b^{\lambda x}}{x^\alpha} = +\infty$$

Setting $t = \frac{1}{x}$, $x \rightarrow +\infty$ is equivalent to $t \rightarrow 0^+$, and substituting we have:

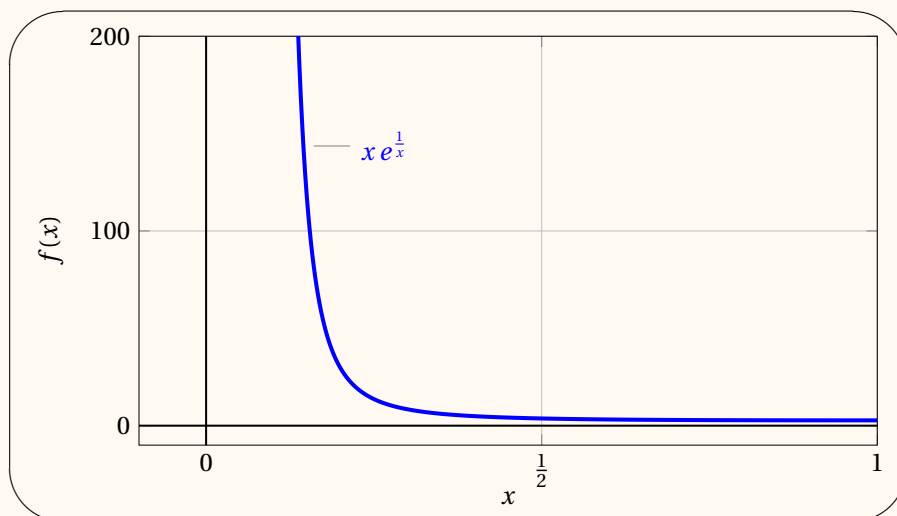
$$\lim_{x \rightarrow +\infty} \frac{b^{\lambda x}}{x^\alpha} = \lim_{x \rightarrow +\infty} \left(\frac{1}{x}\right)^\alpha b^{\lambda x} = \lim_{t \rightarrow 0^+} t^\alpha b^{\lambda/t} = +\infty$$

□

Example 5: Comparison of infinities

$$\lim_{x \rightarrow 0^+} x e^{\frac{1}{x}} = +\infty$$

since we have $\alpha = \lambda = 1$ and $b = e > 1$



Remark 3

$$\lim_{x \rightarrow 0^+} x^{-\alpha} b^{-\lambda/x} = 0 \quad (\alpha, \lambda > 0, b > 1)$$

Proof. Given $\alpha, \lambda > 0, b > 1$ we have:

$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = 0$$

Setting $t = \frac{1}{x}$, $x \rightarrow +\infty$ is equivalent to $t \rightarrow 0^+$, and substituting we have:

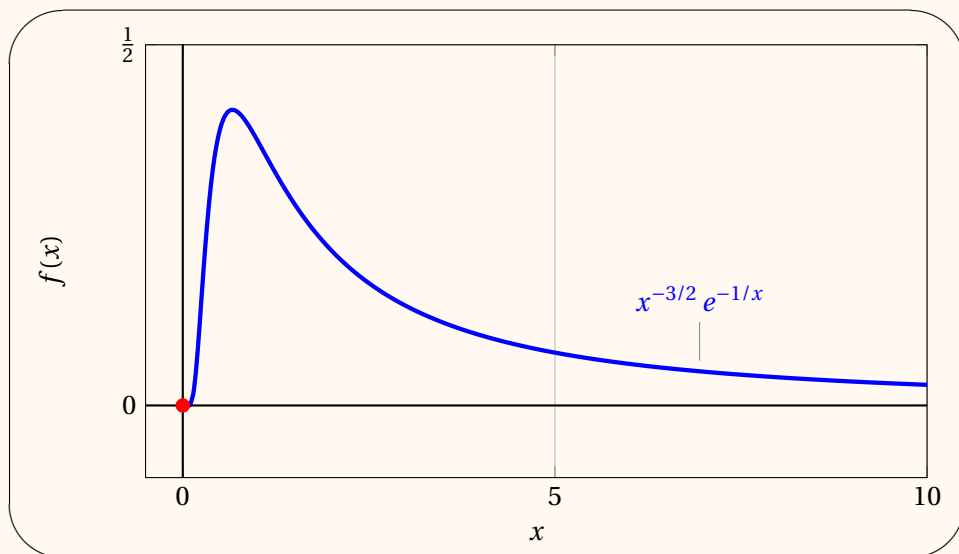
$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = \lim_{x \rightarrow +\infty} \left(\frac{1}{x}\right)^{-\alpha} \frac{1}{b^{\lambda x}} = \lim_{t \rightarrow 0^+} t^{-\alpha} b^{-\lambda/t} = 0$$

□

Example 6: Comparison of infinities

$$\lim_{x \rightarrow 0^+} x^{-3/2} e^{-1/x} = 0$$

since we have $\alpha = \frac{3}{2}, \lambda = 1$ and $b = e > 1$.



Remark 4

$$\lim_{x \rightarrow -\infty} x^\alpha b^{\lambda x} = 0 \quad \left(\alpha = \frac{m}{n} > 0, n, m \in \mathbb{N}, n \text{ odd}, m \neq 0, \lambda > 0, b > 1 \right)$$

Proof. Given $\alpha, \lambda > 0, b > 1$ we have:

$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = 0$$

Setting $t = -x$, $x \rightarrow +\infty$ is equivalent to $t \rightarrow -\infty$, and substituting we have:

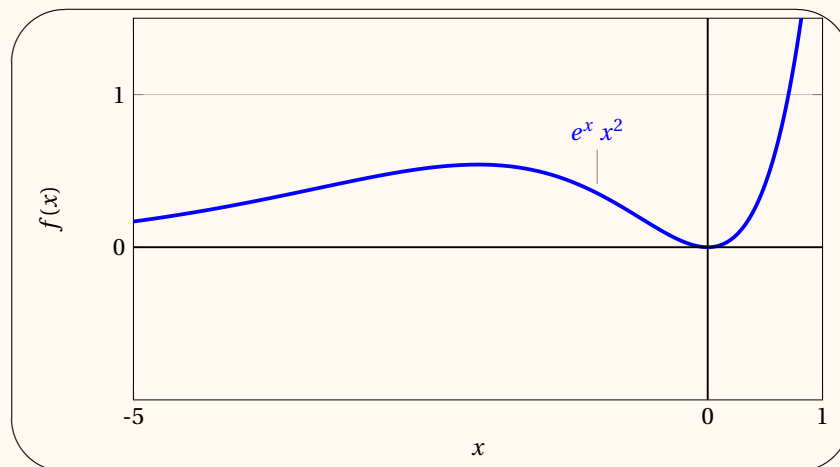
$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = \lim_{t \rightarrow -\infty} \frac{(-t)^\alpha}{b^{-\lambda t}} = (-1)^\alpha \lim_{t \rightarrow -\infty} t^\alpha b^{\lambda t} = 0$$

□

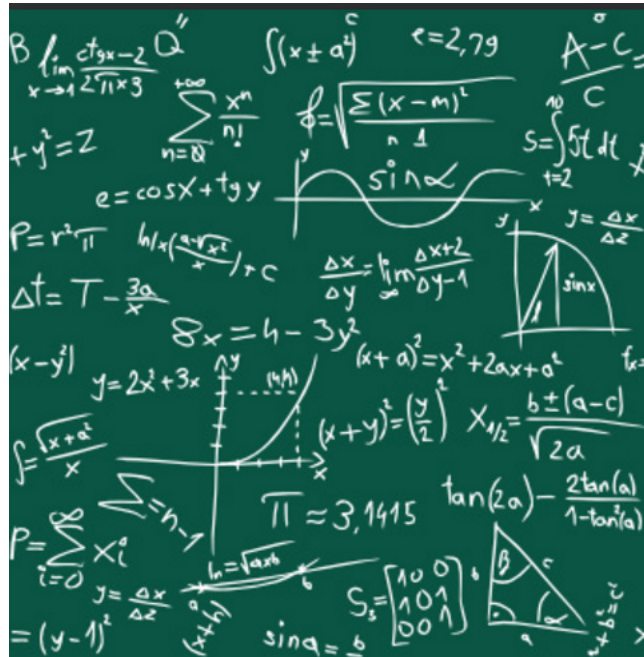
Example 7: Comparison of infinities

$$\lim_{x \rightarrow -\infty} x^2 e^x = 0$$

since we have $\alpha = 2, \lambda = 1$ and $b = e > 1$.



Fundamental limits and asymptotic estimates



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Fundamental limits

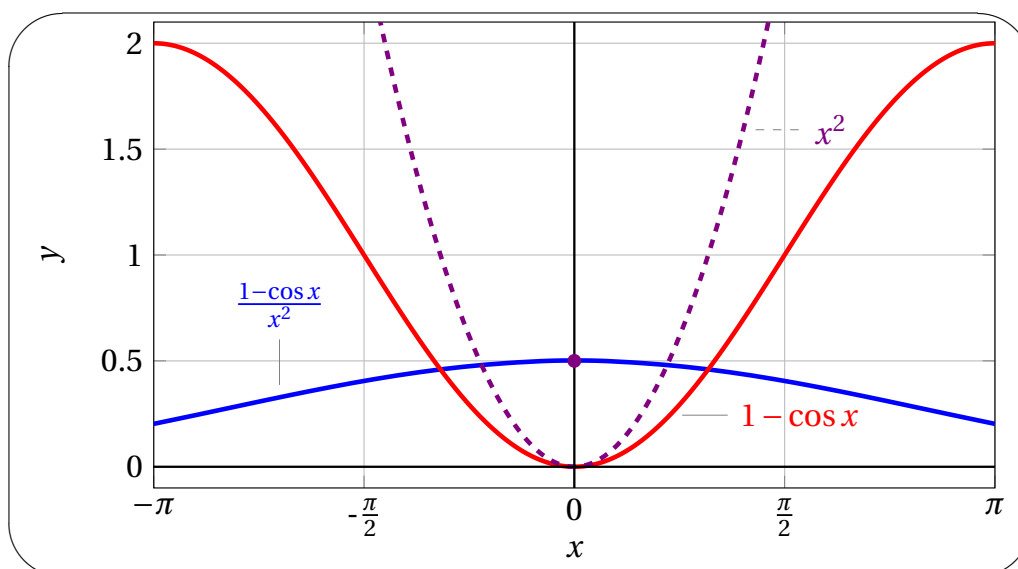
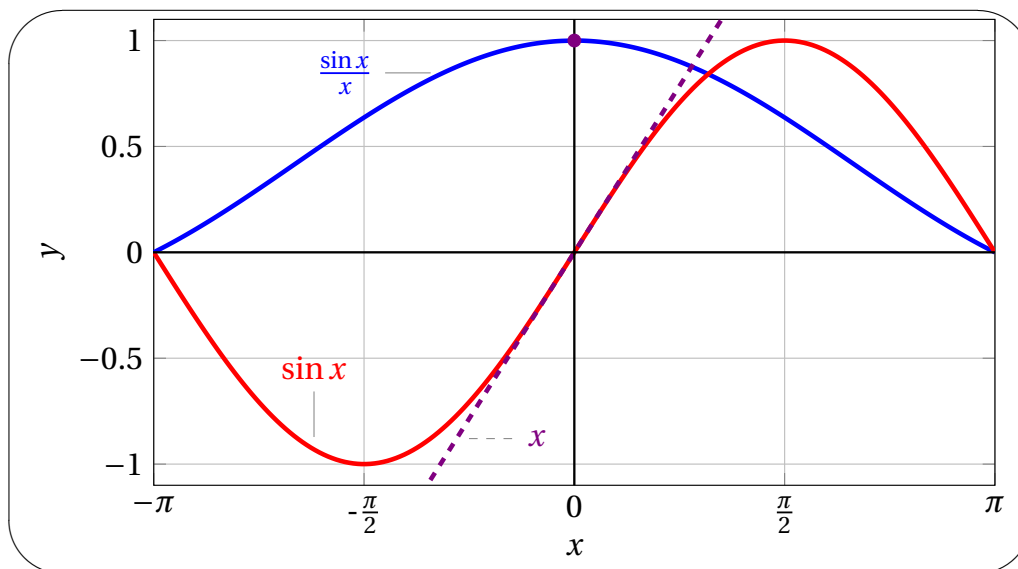
- We now look at some common techniques for computing limits that combine the general theorems on limits with the use of some fundamental limits of certain elementary functions.

1.1 Fundamental limits of sine and cosine

- We want to compute

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \left[\frac{0}{0} \right] \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \left[\frac{0}{0} \right]$$

which are indeterminate forms.



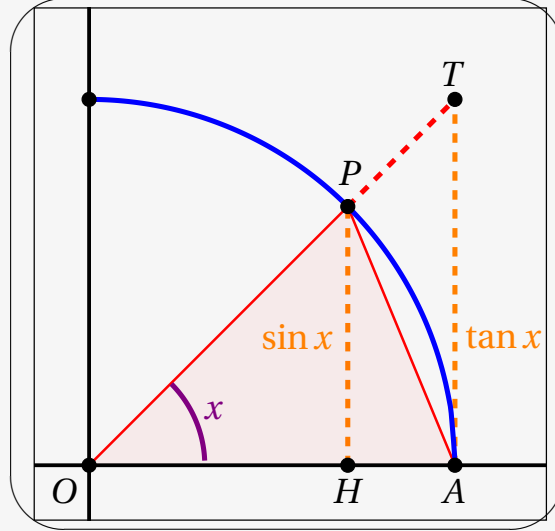
Lemma 1

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad (1)$$

Proof. The functions $\sin x$ and x are odd functions, hence $\frac{\sin x}{x}$ is an even function. Therefore it is sufficient to compute

$$\lim_{x \rightarrow 0^+} \frac{\sin x}{x}$$

We have:



$$HP = \sin x \quad AT = \tan x \quad \widehat{AP} = x$$

The area of the triangle OPA is smaller than that of the circular sector OPA , which in turn is smaller than that of the triangle OTA . It follows that

$$\underbrace{\frac{1}{2} \cdot 1 \cdot \sin x}_{\text{area triangle } OPA} \leq \underbrace{\frac{1}{2} \cdot 1 \cdot x}_{\text{area circ. sector } OPA} \leq \underbrace{\frac{1}{2} \cdot 1 \cdot \tan x}_{\text{area triangle } OTA}$$

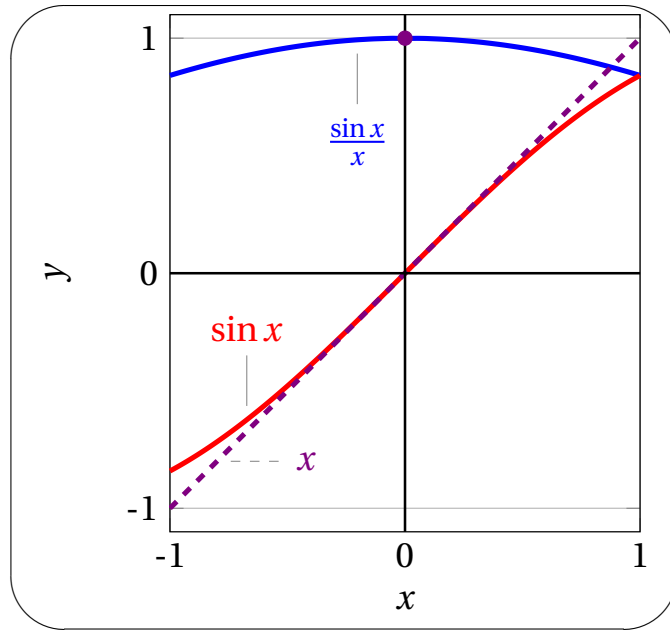
that is, for $x \in (0, \frac{\pi}{2})$:

$$\sin x < x < \tan x$$

Dividing by $\sin x$, which is positive because $x \in (0, \frac{\pi}{2})$, we have

$$1 < \frac{x}{\sin x} < \frac{1}{\cos x}, \quad \forall x \in (0, \frac{\pi}{2}) \quad \text{that is} \quad \cos x < \frac{\sin x}{x} < 1, \quad \forall x \in (0, \frac{\pi}{2})$$

By the comparison theorem, since $\lim_{x \rightarrow 0} \cos x = 1$, the lemma follows. □



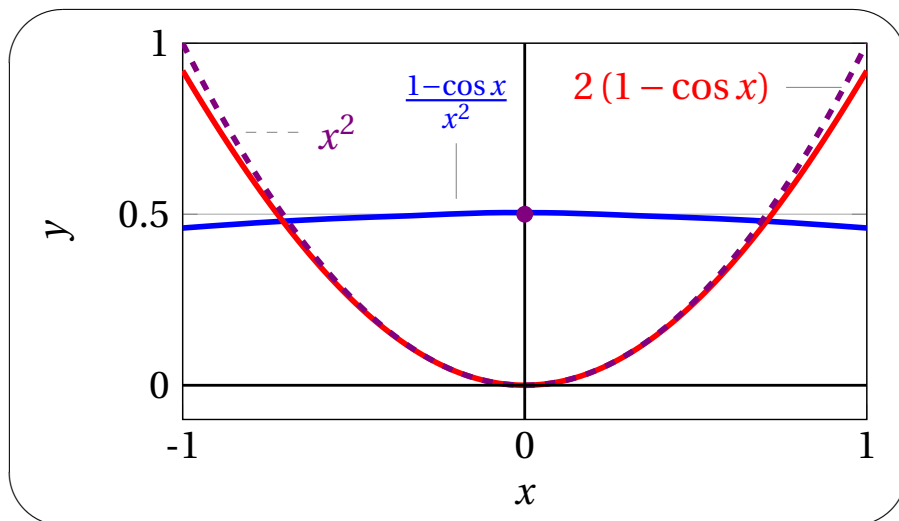
Lemma 2

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \quad (2)$$

Proof. We consider

$$\frac{1 - \cos x}{x^2} = \frac{1 - \cos^2 x}{x^2 (1 + \cos x)} = \left(\frac{\sin x}{x} \right)^2 \frac{1}{1 + \cos x}$$

hence, since $\frac{\sin x}{x} \rightarrow 1$ and $(1 + \cos x) \rightarrow 2$ as $x \rightarrow 0$, the claim follows. $\square$



1.2 Continuous extension of a function

- Based on the limit (1) just proved, the functions $f(x) = \frac{\sin x}{x}$ and $g(x) = \frac{1-\cos x}{x^2}$, initially not defined at $x = 0$, can be extended by continuity also at $x = 0$, by setting

$$f(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases} \quad g(x) = \begin{cases} \frac{1-\cos x}{x^2} & \text{if } x \neq 0 \\ \frac{1}{2} & \text{if } x = 0 \end{cases}$$

The functions f and g defined in this way are continuous also at $x = 0$.

- if a function $f(x)$ is not defined at x_0 but the finite limit exists

$$\lim_{x \rightarrow x_0} f(x) = \ell$$

the function can be extended by continuity also at x_0 , by defining

$$f(x_0) = \ell$$

- If instead the function f has at x_0 a jump discontinuity, a vertical asymptote, or in any case does not have a finite limit, it is not possible to make it continuous at x_0 by changing its definition at a single point.

1.3 Other fundamental limits

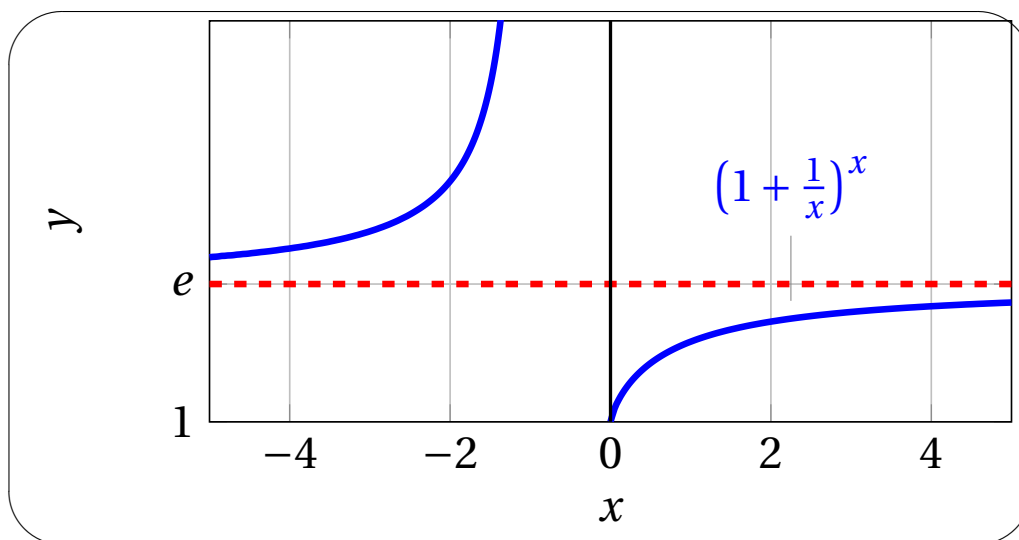
- We know that for every sequence $\{a_n\}$ diverging to $+\infty$ or $-\infty$ we have

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{a_n}\right)^{a_n} = e$$

- By the sequential definition of the limit of a function, this fact immediately implies the next fundamental limit

Lemma 3

$$\lim_{x \rightarrow \pm\infty} \left(1 + \frac{1}{x}\right)^x = e \quad (3)$$



If $\eta(x)$ is a function that tends to infinity^a (i.e., it is an infinity: $\eta(x) \rightarrow \pm\infty$), we have:

$$\left(1 + \frac{1}{\eta(x)}\right)^{\eta(x)} \rightarrow e \quad (4)$$

^ait does not matter what x tends to

Example 1: Limits of functions tending to e

1.

$$\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{2x^2 - 10}\right)^{2x^2 - 10} = e$$

since with $\eta(x) = 2x^2 - 10$ we have

$$\eta(x) \rightarrow +\infty \text{ as } x \rightarrow +\infty.$$

2.

$$\lim_{x \rightarrow 0^+} \left(1 + \frac{1}{1/x}\right)^{1/x} = e$$

since with $\eta(x) = \frac{1}{x}$ we have

$$\eta(x) \rightarrow +\infty \text{ as } x \rightarrow 0^+.$$

From the fundamental limit (3) three more can be deduced

Corollary 1

$$\lim_{y \rightarrow 0} \frac{\log(1+y)}{y} = 1 \quad (5)$$

Proof. Taking logarithms in (3), we obtain

$$\log\left(1 + \frac{1}{x}\right)^x = x \log\left(1 + \frac{1}{x}\right)$$

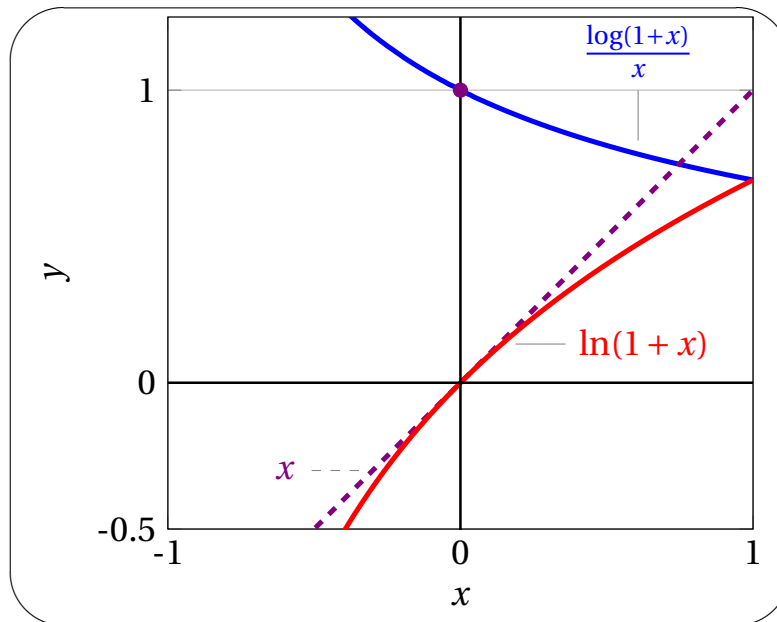
hence

$$\lim_{x \rightarrow \pm\infty} x \log\left(1 + \frac{1}{x}\right) = \log e = 1.$$

Now, if we set $y = \frac{1}{x}$, $x \rightarrow \pm\infty$ is equivalent to $y \rightarrow 0^\pm$, and the last limit can therefore be rewritten in the following form:

$$\frac{\log(1+y)}{y} \rightarrow 1 \text{ as } y \rightarrow 0.$$

□



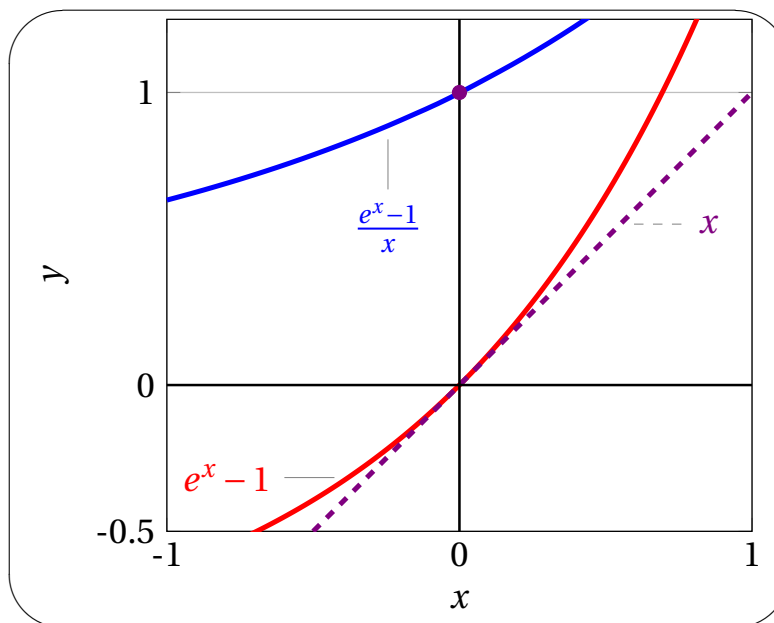
Corollary 2

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \quad (6)$$

Proof. If in (5) we instead set $y = e^x - 1$, $y \rightarrow 0$ is equivalent to $x \rightarrow 0$, and substituting we obtain

$$\frac{\log e^x}{e^x - 1} = \frac{x}{e^x - 1} \rightarrow 1 \text{ as } x \rightarrow 0.$$

Taking reciprocals we obtain the fundamental limit. □



Example 2: Fundamental limit

We compute:

$$\lim_{x \rightarrow 0} \frac{e^{-x} - 1}{x}$$

We set $z = -x$; $x \rightarrow 0$ is equivalent to $z \rightarrow 0$, and substituting we obtain:

$$\lim_{x \rightarrow 0} \frac{e^{-x} - 1}{x} = \lim_{z \rightarrow 0} \frac{e^z - 1}{-z} = - \lim_{z \rightarrow 0} \underbrace{\frac{e^z - 1}{z}}_{\rightarrow 1} = -1$$

Corollary 3

$$\lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} = \alpha \quad \text{with } \alpha \in \mathbb{R} \quad (7)$$

Proof. If in (5) we instead set $y = (1+x)^\alpha - 1$, with α any real exponent, then $x \rightarrow 0$ is equivalent to $y \rightarrow 0$ and we have:

$$\begin{aligned} \frac{\log(1+y)}{y} &= \frac{\log[1 + (1+x)^\alpha - 1]}{(1+x)^\alpha - 1} = \frac{\alpha \log(1+x)}{(1+x)^\alpha - 1} \\ &= \frac{\alpha x}{(1+x)^\alpha - 1} \cdot \frac{\log(1+x)}{x} \rightarrow 1 \text{ as } x \rightarrow 0. \end{aligned}$$

But since also

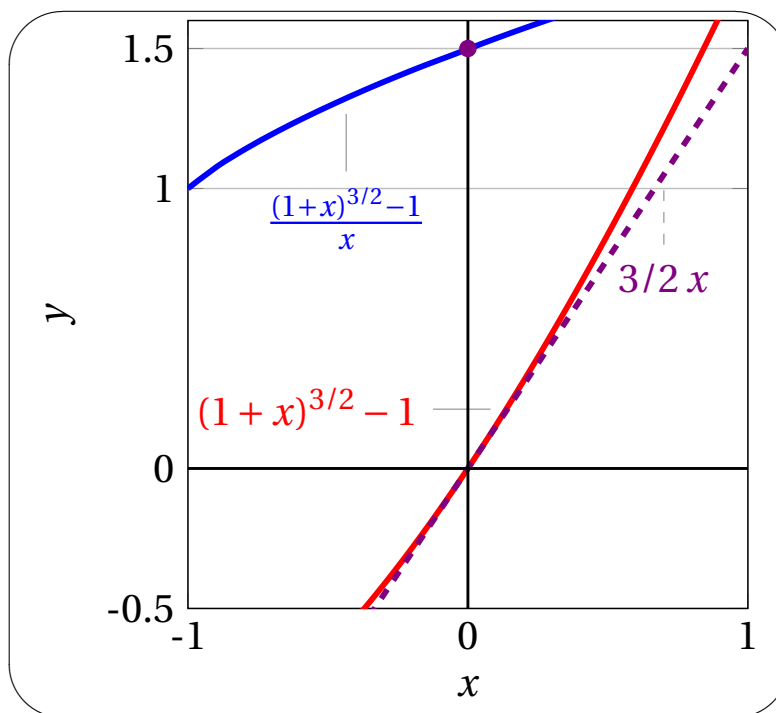
$$\frac{\log(1+x)}{x} \rightarrow 1 \text{ as } x \rightarrow 0,$$

then

$$\frac{\alpha x}{(1+x)^\alpha - 1} \rightarrow 1 \text{ as } x \rightarrow 0.$$

Taking reciprocals and then multiplying by α we obtain the fundamental limit.

□



2 Asymptotic estimates

Definition 1: Asymptotic functions

Two functions f, g are said to be **asymptotic** as $x \rightarrow c$ if

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = 1$$

and we write $f \sim g$ as $x \rightarrow c$.

- The asymptotic symbol for functions enjoys all the properties stated for sequences

As $x \rightarrow 0$ we have (from the fundamental limits):

$$\sin x \sim x \quad (8a)$$

$$\log(1+x) \sim x \quad (8b)$$

$$\cos x \sim 1 - \frac{1}{2}x^2 \quad (8c)$$

$$e^x \sim 1+x \quad (8d)$$

$$(1+x)^\alpha \sim 1+\alpha x \quad \text{with } \alpha \in \mathbb{R} \quad (8e)$$

- The functions $\sim x$ behave, to a first approximation or to first order, like x as $x \rightarrow 0$.

If $\varepsilon(x)$ is a function that tends to zero^a (i.e., it is an infinitesimal: $\varepsilon(x) \rightarrow 0$), we have:

$$\sin(\varepsilon(x)) \sim \varepsilon(x) \quad (9a)$$

$$\log(1+\varepsilon(x)) \sim \varepsilon(x) \quad (9b)$$

$$\cos(\varepsilon(x)) \sim 1 - \frac{1}{2}\varepsilon^2(x) \quad (9c)$$

$$e^{\varepsilon(x)} \sim 1+\varepsilon(x) \quad (9d)$$

$$(1+\varepsilon(x))^\alpha \sim 1+\alpha \varepsilon(x) \quad \text{with } \alpha \in \mathbb{R} \quad (9e)$$

^ait does not matter what x tends to

- Formulas (9) follow from formulas (8) simply by a change of variable

$$x = \varepsilon(x)$$

Example 3: Limits with asymptotic functions

$$\lim_{x \rightarrow 1} \frac{(x-1)^2}{e^{3(x-1)^2} - 1} = \left[\frac{0}{0} \right]$$

We use the estimate

$$e^{\varepsilon(x)} - 1 \sim \varepsilon(x)$$

with

$$\varepsilon(x) = 3(x-1)^2 \text{ and } \varepsilon(x) \rightarrow 0 \text{ as } x \rightarrow 1$$

Then, as $x \rightarrow 1$, we have

$$e^{3(x-1)^2} - 1 \sim 3(x-1)^2.$$

and

$$\lim_{x \rightarrow 1} \frac{(x-1)^2}{e^{3(x-1)^2} - 1} = \lim_{x \rightarrow 1} \frac{(x-1)^2}{3(x-1)^2} = \frac{1}{3}$$

Example 4: Limits with asymptotic functions

$$\lim_{x \rightarrow 0} \frac{\log(1+2x)}{\sin 3x} = \left[\frac{0}{0} \right]$$

We use the estimate

$$\log(1+\varepsilon(x)) \sim \varepsilon(x)$$

with

$$\varepsilon(x) = 2x \text{ and } \varepsilon(x) \rightarrow 0 \text{ as } x \rightarrow 0.$$

Hence, as $x \rightarrow 0$ we have

$$\log(1+2x) \sim 2x$$

Now we also use the estimate

$$\sin \varepsilon(x) \sim \varepsilon(x)$$

with

$$\varepsilon(x) = 3x \text{ and } \varepsilon(x) \rightarrow 0 \text{ as } x \rightarrow 0$$

Hence, as $x \rightarrow 0$ we have

$$\sin 3x \sim 3x$$

Then, as $x \rightarrow 0$, we have

$$\frac{\log(1+2x)}{\sin 3x} \sim \frac{2x}{3x}$$

and

$$\lim_{x \rightarrow 0} \frac{\log(1+2x)}{\sin 3x} = \lim_{x \rightarrow 0} \frac{2x}{3x} = \frac{2}{3}$$

Example 5: Limits with asymptotic functions

$$\lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + 2x^2 + 1} - x = [+ \infty - \infty]$$

We rewrite,

$$\sqrt[3]{x^3 + 2x^2 + 1} - x = x \left(\sqrt[3]{1 + \left(\frac{2}{x} + \frac{1}{x^3} \right)} - 1 \right)$$

We use the estimate

$$\sqrt[3]{1 + \varepsilon(x)} - 1 \sim \frac{1}{3} \varepsilon(x)$$

with

$$\varepsilon(x) = \left(\frac{2}{x} + \frac{1}{x^3} \right) \text{ and } \varepsilon(x) \rightarrow 0 \text{ as } x \rightarrow +\infty.$$

Then, as $x \rightarrow +\infty$, we have

$$\sqrt[3]{x^3 + 2x^2 + 1} - x \sim x \left[\frac{1}{3} \left(\frac{2}{x} + \frac{1}{x^3} \right) \right]$$

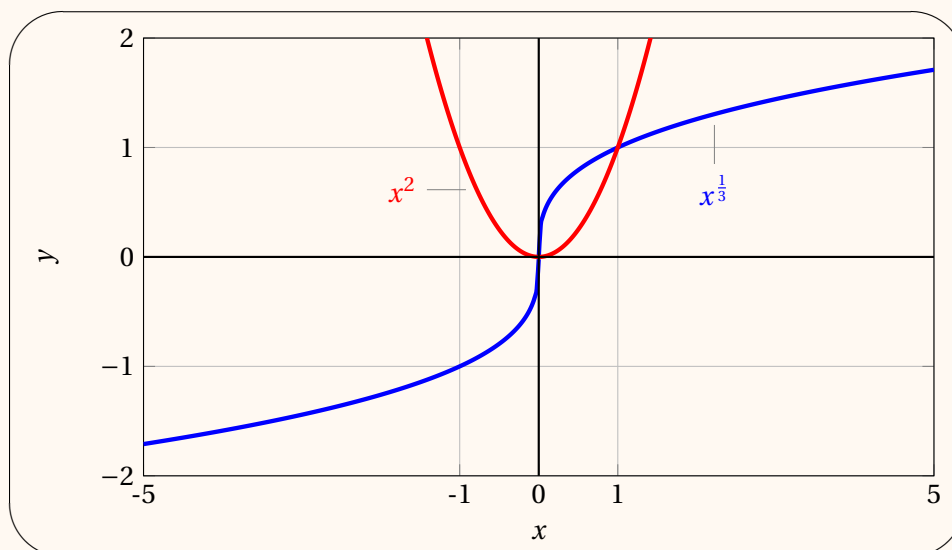
and

$$\lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + 2x^2 + 1} - x = \lim_{x \rightarrow +\infty} x \left[\frac{1}{3} \left(\frac{2}{x} + \frac{1}{x^3} \right) \right] = \frac{2}{3}$$

3 Asymptotic estimates and graphs

- Asymptotic estimates are useful not only for computing limits, but also for sketching the qualitative graph of a function in a neighborhood of a given point, or as $x \rightarrow \pm\infty$.

Example 6: Known graphs



Example 7: Qualitative graph

We now want to study the qualitative graph of the function (the sum of the previous two):

$$f(x) = \sqrt[3]{x} + x^2 = x^{\frac{1}{3}} + x^2$$

- the function is defined and continuous on all of $\mathbb{R}$;
- as $x \rightarrow \pm\infty$, $f(x) \sim x^2$, since

$$\lim_{x \rightarrow \pm\infty} \frac{x^{\frac{1}{3}} + x^2}{x^2} = 1$$

Therefore $f(x) \rightarrow +\infty$ as $x \rightarrow \pm\infty$; moreover, for x large in absolute value, its graph will be similar to that of x^2 .

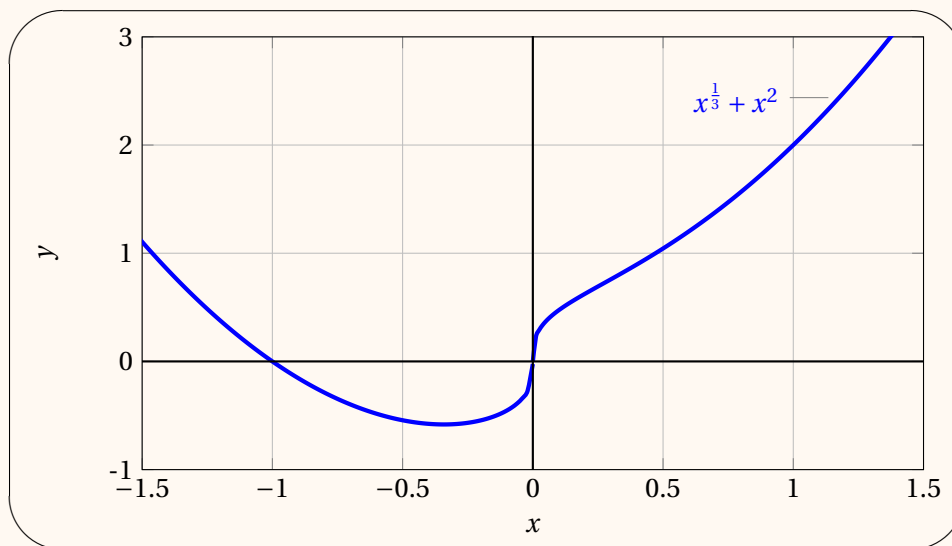
- Moreover, the function vanishes at $x = 0$ and, as $x \rightarrow 0$, $f(x) \sim x^{\frac{1}{3}}$, since

$$\lim_{x \rightarrow 0} \frac{x^{\frac{1}{3}} + x^2}{x^{\frac{1}{3}}} = 1$$

therefore, in a neighborhood of $x = 0$, its graph will be similar to that of $x^{\frac{1}{3}}$; in particular, it will have a vertical tangent at the origin.

Example 8: Actual graph

$$f(x) = \sqrt[3]{x} + x^2 = x^{\frac{1}{3}} + x^2$$



- Often the behavior of a function in a neighborhood of a point (for example, the fact that it has a vertical or horizontal tangent) can be predicted from a suitable asymptotic estimate.
- The asymptotic estimate allows us to sketch the qualitative graph of f (in a neighborhood of the point) by comparison with that of a known function (for example, a power with

rational exponent).

- Analogous estimates are useful as $x \rightarrow \pm\infty$.

Growth of a function at infinity

- Suppose we want to sketch the graph of a function f that, as $x \rightarrow +\infty$ (or $-\infty$), tends to $+\infty$ (or $-\infty$).
- To describe the speed at which the function tends to infinity, the following notions are useful: we say that, as $x \rightarrow +\infty$,

$$f \text{ has } \begin{cases} \text{superlinear growth} \\ \text{linear growth} \\ \text{sublinear growth} \end{cases} \quad \text{if} \quad \lim_{x \rightarrow +\infty} \frac{f(x)}{x} = \begin{cases} \pm\infty \\ m \text{ (finite and different from 0)} \\ 0 \end{cases}$$

- Analogous definitions are given for $x \rightarrow -\infty$.
- Only when a function has linear growth is it possible for it to have an oblique asymptote

Example 9: Growth of a function at infinity

For example, as $x \rightarrow +\infty$

- Superlinear growth: exponentials a^x and powers x^a with $a > 1$.
- Sublinear growth: logarithms $\log_a x$ and powers x^a with $0 < a < 1$.

Example 10: Growth of a function at infinity

The function

$$f(x) = 2x + e^x + e^{\frac{1}{x}}$$

- as $x \rightarrow +\infty$ it is asymptotic to e^x ; therefore it tends to $+\infty$ with superlinear growth;
- as $x \rightarrow -\infty$ it is asymptotic to $2x$; therefore it tends to $-\infty$ linearly
- since

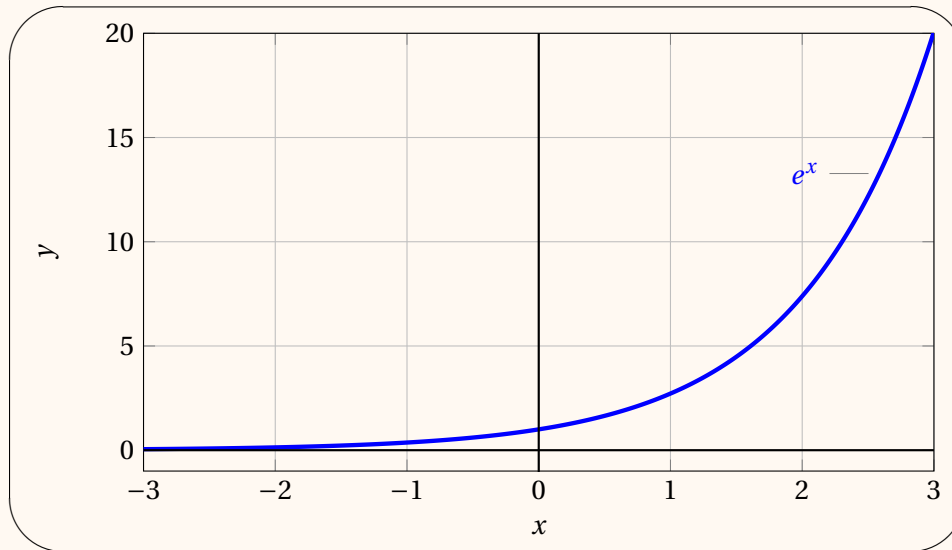
$$\lim_{x \rightarrow -\infty} [f(x) - 2x] = 1$$

the function has the oblique asymptote

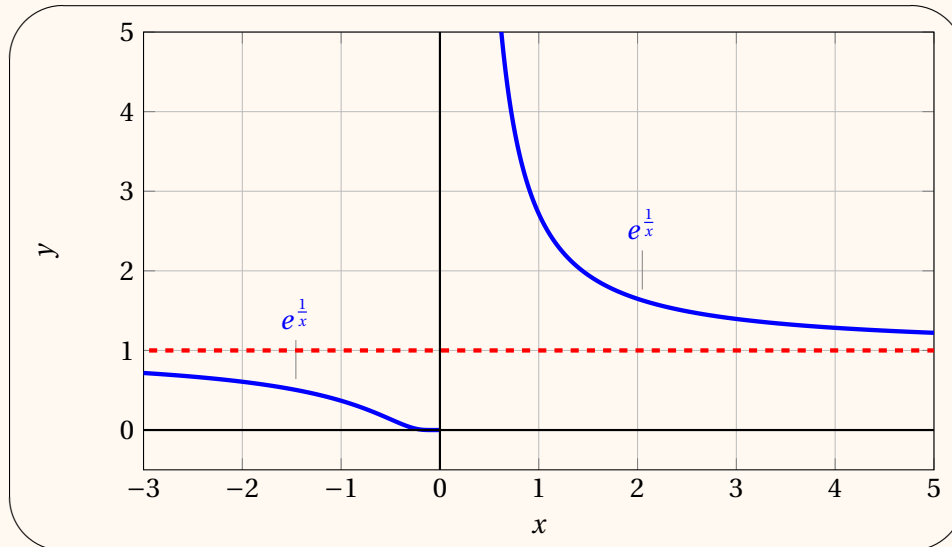
$$y = 2x + 1 \text{ as } x \rightarrow -\infty$$

Example 11: Graphs

$$f(x) = e^x, \quad \lim_{x \rightarrow +\infty} e^x = +\infty, \quad \lim_{x \rightarrow -\infty} e^x = 0, \quad \lim_{x \rightarrow 0} e^x = 1$$

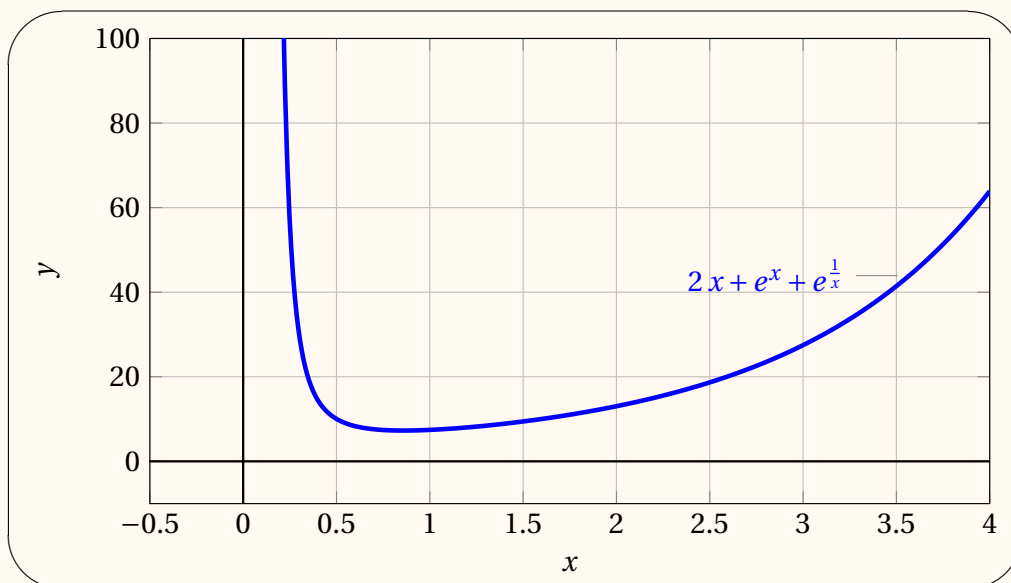


$$f(x) = e^{\frac{1}{x}}, \quad \lim_{x \rightarrow +\infty} e^{\frac{1}{x}} = 1, \quad \lim_{x \rightarrow -\infty} e^{\frac{1}{x}} = 1, \quad \lim_{x \rightarrow 0^-} e^{\frac{1}{x}} = 0, \quad \lim_{x \rightarrow 0^+} e^{\frac{1}{x}} = +\infty$$

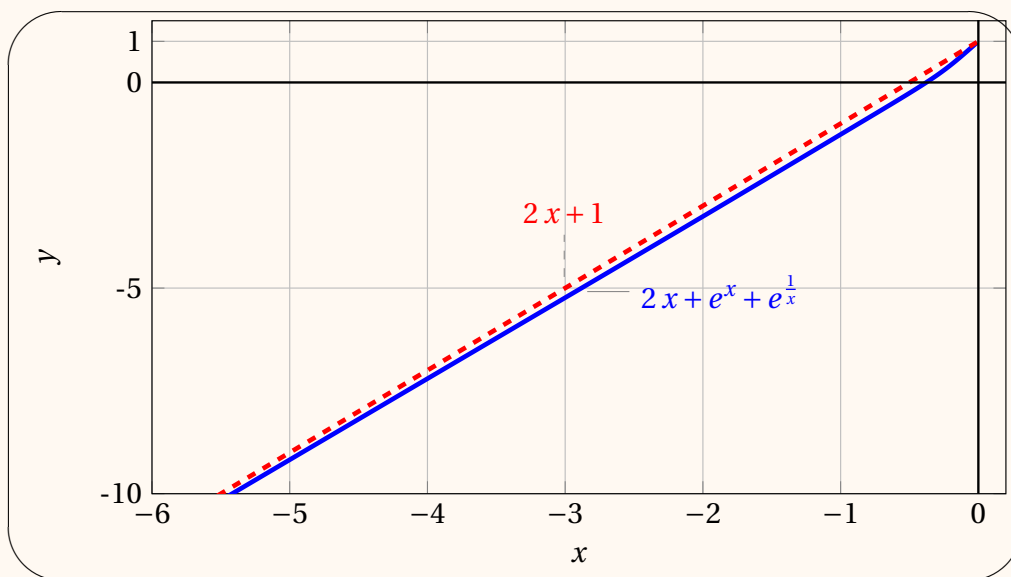


Example 12: Actual graph

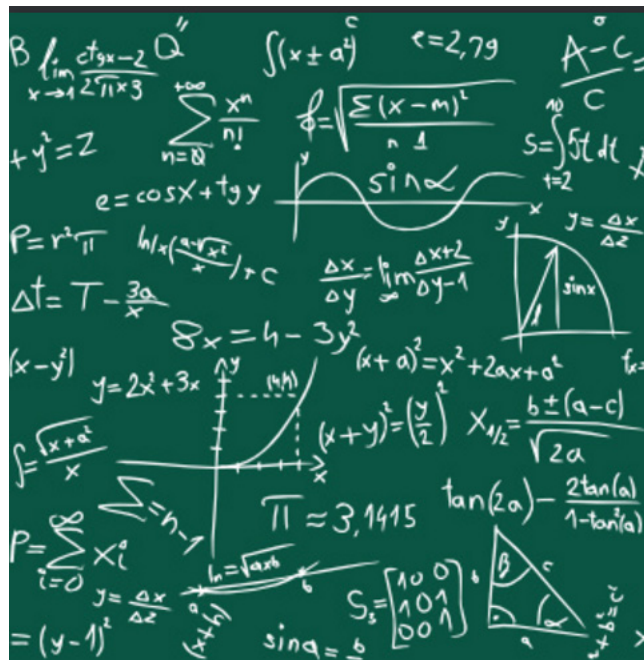
$$f(x) = 2x + e^x + e^{\frac{1}{x}}, \quad f(x) \sim e^x \text{ as } x \rightarrow +\infty, \quad \lim_{x \rightarrow 0^+} f(x) = +\infty$$



$$f(x) = 2x + e^x + e^{\frac{1}{x}}, \quad f(x) \sim 2x \text{ as } x \rightarrow -\infty, \quad \lim_{x \rightarrow 0^-} f(x) = 1$$



Asymptotic expansions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 The “little o ” symbol and asymptotic expansions

Definition 1: of little o

Given two functions $f(x)$ and $g(x)$, defined in a neighborhood of $c \in \mathbb{R}^*$, we say that

$$f(x) = o(g(x)) \text{ as } x \rightarrow c$$

(read “ $f(x)$ is little o of $g(x)$ ” as $x \rightarrow c$) if and only if

$$\frac{f(x)}{g(x)} \rightarrow 0 \text{ as } x \rightarrow c$$

The symbol $o(g(x))$ as x tends to c does not denote a particular function $f(x)$, but any function $f(x)$ such that the ratio between $f(x)$ and $g(x)$ tends to 0 as x tends to c .

Example 1: Little o

For example:

$$x^2 = o(x) \text{ as } x \rightarrow 0 \text{ since } \frac{x^2}{x} \rightarrow 0 \text{ as } x \rightarrow 0$$

$$x^3 = o(x) \text{ as } x \rightarrow 0 \text{ since } \frac{x^3}{x} \rightarrow 0 \text{ as } x \rightarrow 0$$

$$x^3 = o(x^2) \text{ as } x \rightarrow 0 \text{ since } \frac{x^3}{x^2} \rightarrow 0 \text{ as } x \rightarrow 0$$

$$e^{-1/x^2} = o(x^4) \text{ as } x \rightarrow 0 \text{ since } \frac{e^{-1/x^2}}{x^4} \rightarrow 0 \text{ as } x \rightarrow 0$$

- From the definition of asymptotically equivalent functions, we have:

$$f(x) \sim g(x) \text{ as } x \rightarrow c \iff \frac{f(x)}{g(x)} \rightarrow 1 \text{ as } x \rightarrow c$$

in these cases we have:

$$\frac{f(x)}{g(x)} - 1 \rightarrow 0 \text{ as } x \rightarrow c \quad \text{and} \quad \frac{f(x) - g(x)}{g(x)} \rightarrow 0 \text{ as } x \rightarrow c$$

from the definition of little o we can then write:

$$f(x) - g(x) = o(g(x)) \text{ as } x \rightarrow c$$

We have the following *asymptotic expansion*:

$$f(x) \sim g(x) \text{ as } x \rightarrow c \iff f(x) = g(x) + o(g(x)) \text{ as } x \rightarrow c$$

- Moreover, we have:

$$g(x) + f(x) \sim g(x) \text{ as } x \rightarrow c \iff \frac{f(x)}{g(x)} \rightarrow 0 \text{ as } x \rightarrow c,$$

since:

$$\frac{g(x) + f(x)}{g(x)} = 1 + \frac{f(x)}{g(x)}$$

which tends to 1 as $x \rightarrow c$ if and only if $f(x)/g(x)$ tends to 0 as $x \rightarrow c$.

In such situations we say that $g(x)$ is the **principal part** of the sum $g(x) + f(x)$ and that $f(x)$ is **negligible** with respect to $g(x)$ as $x \rightarrow c$; that is:

$$f(x) = o(g(x)) \text{ as } x \rightarrow c$$

We have the following *asymptotic expansion*:

$$g(x) + f(x) \sim g(x) \text{ as } x \rightarrow c \iff f(x) = o(g(x)) \text{ as } x \rightarrow c$$

- It follows from the definition of little o that with three functions we have:

$$f(x) = h(x) + o(g(x)) \text{ as } x \rightarrow c \iff \frac{f(x) - h(x)}{g(x)} \rightarrow 0 \text{ as } x \rightarrow c$$

- The “little o ” symbol behaves as follows with products:

$$f \cdot o(g) = o(f \cdot g)$$

$$o(f) \cdot o(g) = o(f \cdot g)$$

Example 2: Little o and products

For example, as $x \rightarrow c$:

$$x \cdot o(x^2) = o(x^3), \quad \frac{o(x^3)}{x} = o(x^2), \quad o(x) \cdot o(x^2) = o(x^3)$$

Some properties of the little o symbol, shown through examples:

$$o(x) \pm o(x) = o(x), \quad \text{as } x \rightarrow c$$

$$o(ax) = o(x), \quad \forall a \in \mathbb{R}, a \neq 0, \quad \text{as } x \rightarrow c$$

$$ao(x) = o(x), \quad \forall a \in \mathbb{R}, \quad \text{as } x \rightarrow c$$

$$o(x) + o(x^2) = o(x), \quad \text{as } x \rightarrow 0$$

$$o(x) + o(x^2) = o(x^2), \quad \text{as } x \rightarrow \infty$$

Definition 2: of $o(1)$

A function f is said to be infinitesimal as $x \rightarrow c$, and we write

$$f(x) = o(1) \text{ as } x \rightarrow c \text{ if } f(x) \rightarrow 0 \text{ as } x \rightarrow c$$

Example 3: $o(1)$

For example:

$$\sqrt{x} = o(1) \text{ as } x \rightarrow 0^+ \text{ since } \frac{\sqrt{x}}{1} \rightarrow 0 \text{ as } x \rightarrow 0^+$$

$$x^{-2} = o(1) \text{ as } x \rightarrow +\infty \text{ since } \frac{x^{-2}}{1} \rightarrow 0 \text{ as } x \rightarrow +\infty$$

- We have:

$$\lim_{x \rightarrow c} f(x) = \ell \iff \lim_{x \rightarrow c} (f(x) - \ell) = 0 \iff \lim_{x \rightarrow c} |f(x) - \ell| = 0$$

hence

$$\lim_{x \rightarrow c} f(x) = \ell \iff f(x) = \ell + o(1) \text{ as } x \rightarrow c$$

the last expression reads “ $f(x)$ is the sum of ℓ and of a function that is infinitesimal as $x \rightarrow c$ ”

Example 4: $o(1)$

For example:

$$x^2 = x(1 + o(1)) \text{ as } x \rightarrow 1 \text{ since } \frac{x^2}{x} = x \rightarrow 1 \text{ as } x \rightarrow 1$$

$$x^2 + 3x = x(3 + o(1)) \text{ as } x \rightarrow 0 \text{ since } \frac{x^2 + 3x}{x} = x + 3 \rightarrow 3 \text{ as } x \rightarrow 0$$

$$x^2 + 3x = x^2(1 + o(1)) \text{ as } x \rightarrow +\infty \text{ since } \frac{x^2 + 3x}{x^2} = 1 + \frac{3}{x} \rightarrow 1 \text{ as } x \rightarrow +\infty$$

$$\sqrt{x+5} = \sqrt{x}(1 + o(1)) \text{ as } x \rightarrow +\infty \text{ since } \frac{\sqrt{x+5}}{\sqrt{x}} = \sqrt{1 + \frac{5}{x}} \rightarrow 1 \text{ as } x \rightarrow +\infty$$

Some properties of the $o(1)$ symbol:

$$o(1) \pm o(1) = o(1), \text{ as } x \rightarrow c$$

$$o(1) \cdot o(1) = o(1), \text{ as } x \rightarrow c$$

$$c \cdot o(1) = o(1), \text{ as } x \rightarrow c, c \neq 0$$

$$(1 + o(1)) \cdot (1 + o(1)) = 1 + o(1), \text{ as } x \rightarrow c$$

$$\frac{1}{1 + o(1)} = 1 + o(1), \text{ as } x \rightarrow c$$

$$\frac{c}{1 + o(1)} - c = o(1), \text{ as } x \rightarrow c, c \neq 0$$

Since

$$\frac{c}{1 + o(1)} - c = c \left(\frac{1}{1 + o(1)} - 1 \right) = c(1 + o(1) - 1) = o(1)$$

Example 5: $o(1)$

We have

$$(x + o(1))^2 = x^2(1 + o(1)) \text{ as } x \rightarrow +\infty \text{ since } \frac{(x + o(1))^2}{x^2} \rightarrow 1 \text{ as } x \rightarrow +\infty$$

2 Asymptotic expansions derived from the fundamental limits

We have seen that the following fundamental limits hold:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}, \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1, \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} = \alpha$$

and consequently the following **asymptotic equivalences** hold as $x \rightarrow 0$:

$$\sin x \sim x, \quad 1 - \cos x \sim \frac{1}{2}x^2, \quad e^x - 1 \sim x, \quad \log(1+x) \sim x, \quad (1+x)^\alpha - 1 \sim \alpha x$$

Let us now see how to obtain the **asymptotic expansions** starting from the fundamental limits:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Hence, as $x \rightarrow 0$:

$$\frac{\sin x}{x} - 1 \rightarrow 0, \quad \frac{\sin x - x}{x} \rightarrow 0 \text{ hence } \sin x - x = o(x)$$

which gives the asymptotic expansion:

$$\sin x = x + o(x)$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

Hence, as $x \rightarrow 0$:

$$\frac{1 - \cos x}{x^2} - \frac{1}{2} \rightarrow 0, \quad \frac{2(1 - \cos x) - x^2}{2x^2} \rightarrow 0 \text{ hence } 2(1 - \cos x) - x^2 = o(2x^2)$$

which gives the asymptotic expansion:

$$\cos x = 1 - \frac{1}{2}x^2 + o(x^2)$$

$$\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$$

Hence, as $x \rightarrow 0$:

$$\frac{\log(1+x)}{x} - 1 \rightarrow 0, \quad \frac{\log(1+x) - x}{x} \rightarrow 0 \text{ hence } \log(1+x) - x = o(x)$$

which gives the asymptotic expansion:

$$\log(1+x) = x + o(x)$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

Hence, as $x \rightarrow 0$:

$$\frac{e^x - 1}{x} - 1 \rightarrow 0, \quad \frac{e^x - 1 - x}{x} \rightarrow 0 \text{ hence } e^x - 1 - x = o(x)$$

which gives the asymptotic expansion:

$$e^x = 1 + x + o(x)$$

$$\lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} = \alpha \quad \text{with } \alpha \in \mathbb{R}$$

Hence, as $x \rightarrow 0$:

$$\frac{(1+x)^\alpha - 1}{x} - \alpha \rightarrow 0 \quad \frac{(1+x)^\alpha - 1 - \alpha x}{x} \rightarrow 0 \text{ hence } (1+x)^\alpha - 1 - \alpha x = o(x)$$

which gives the asymptotic expansion:

$$(1+x)^\alpha = 1 + \alpha x + o(x)$$

- These asymptotic expansions can be generalized

If $\varepsilon(x)$ is a function that tends to zero, i.e., it is an infinitesimal (it does not matter what x tends to), we have the following **asymptotic expansions** derived from the fundamental limits.

As $\varepsilon(x) \rightarrow 0$, we have:

$$\sin(\varepsilon(x)) = \varepsilon(x) + o(\varepsilon(x)) \tag{1a}$$

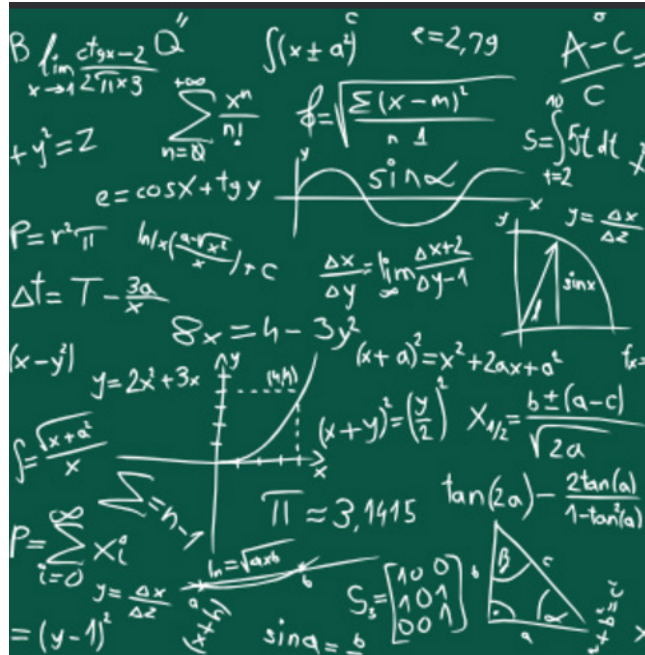
$$\cos(\varepsilon(x)) = 1 - \frac{1}{2} \varepsilon^2(x) + o(\varepsilon^2(x)) \tag{1b}$$

$$\log(1 + \varepsilon(x)) = \varepsilon(x) + o(\varepsilon(x)) \tag{1c}$$

$$e^{\varepsilon(x)} = 1 + \varepsilon(x) + o(\varepsilon(x)) \tag{1d}$$

$$(1 + \varepsilon(x))^\alpha = 1 + \alpha \varepsilon(x) + o(\varepsilon(x)) \quad \text{with } \alpha \in \mathbb{R} \tag{1e}$$

Intermediate zero theorem and bisection method



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Zeros of a function

Given a function f , we are interested in solving the equation:

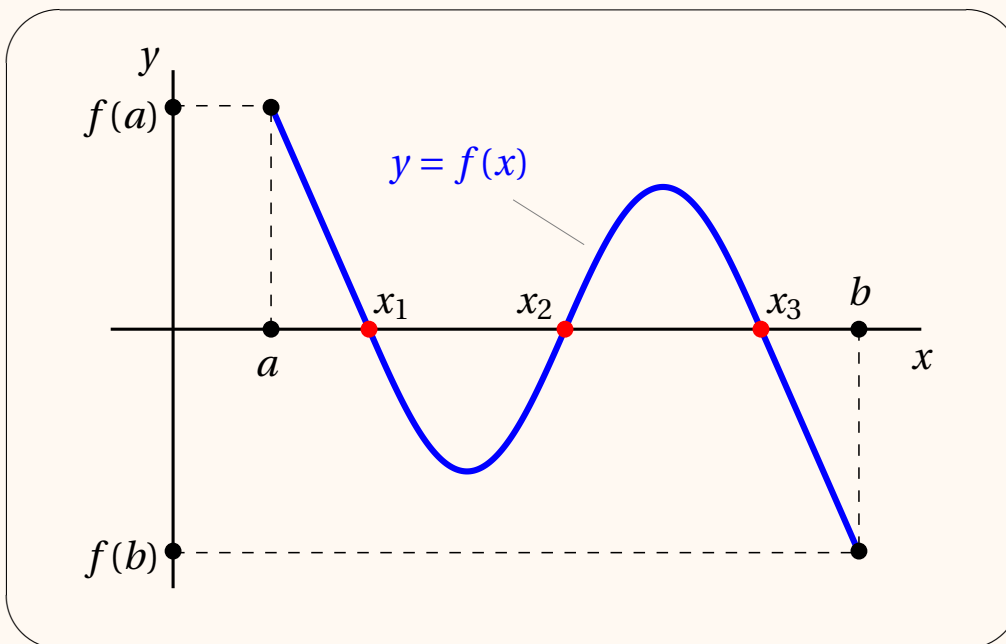
$$f(x) = 0 \quad (1)$$

that is, in finding the **zeros** of f . The zeros are the solutions of equation (1), that is, the points c of the domain of the function where $f(c) = 0$.

- When f is a polynomial of degree ≤ 4 there are formulas that give the solutions of (1). However, if f is a polynomial of degree > 4 or a more complicated function, except in particularly lucky cases, there are no formulas for the solutions of equation (1).
- Geometrically, solving equation (1) means determining the x -coordinates of the intersection points between the graph of $y = f(x)$ and the x -axis. There can be: **infinitely many solutions, a finite number of solutions, no solutions**.

Example 1: Zeros of a function

Consider the function f with the following graph:

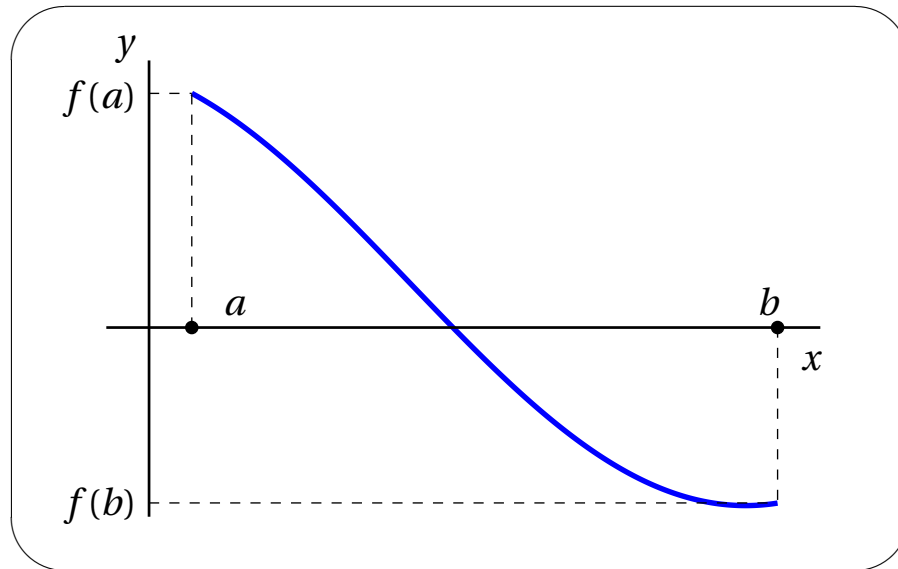


the equation $f(x) = 0$ has 3 solutions in the interval $[a, b]$, that is, the function f has 3 zeros, x_1 , x_2 and x_3 .

Theorem 1: Intermediate zero theorem (Bolzano's theorem)

If a function $f : [a, b] \rightarrow \mathbb{R}$ is continuous on the interval $[a, b]$ and $f(a) \cdot f(b) < 0$, then there exists $c \in (a, b)$ such that $f(c) = 0$. If f is also strictly monotonic then the zero c is unique.

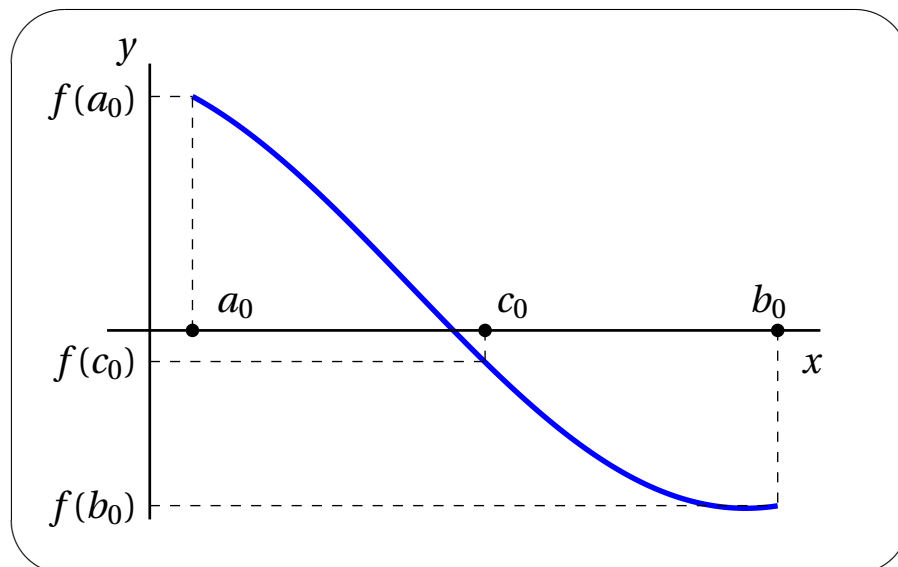
- The idea of the proof is to construct two sequences that converge to the zero of the function. To understand how to construct them, consider for example a function with $f(a) > 0$ and $f(b) < 0$:



- We set $a_0 = a$ and $b_0 = b$ and compute

$$c_0 = \frac{a_0 + b_0}{2}, \text{ the midpoint of the interval } [a_0, b_0].$$

In the previous example we have:



If $f(c_0) = 0$ we have found a zero. If $f(c_0) \neq 0$ we look at the sign of $f(a_0) \cdot f(c_0)$ and proceed as follows:

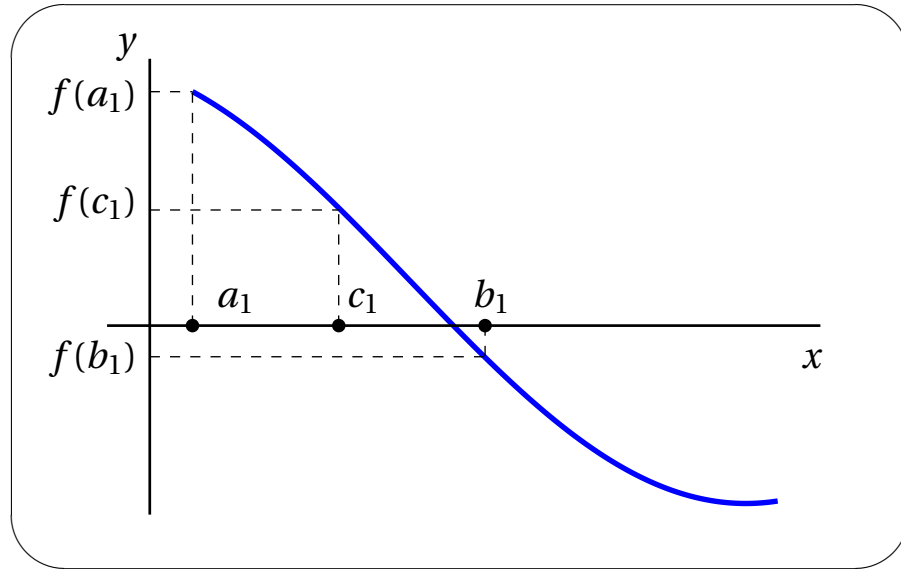
$$\left\{ \begin{array}{l} \text{if } f(a_0) \cdot f(c_0) < 0 \\ \text{if } f(a_0) \cdot f(c_0) > 0 \end{array} \right. \quad \text{we create } [a_1, b_1] \quad \left\{ \begin{array}{l} \text{with } a_1 = a_0, \ b_1 = c_0 \\ \text{with } a_1 = c_0, \ b_1 = b_0 \end{array} \right.$$

In the previous example we have $f(a_0) \cdot f(c_0) < 0$, hence we have $a_1 = a_0$, $b_1 = c_0$.

- With a_1 and b_1 , we compute:

$$c_1 = \frac{a_1 + b_1}{2}, \text{ the midpoint of the interval } [a_1, b_1].$$

In the previous example we have:



If $f(c_1) = 0$ we have found a zero. If $f(c_1) \neq 0$ we look at the sign of $f(a_1) \cdot f(c_1)$ and proceed as follows:

$$\left\{ \begin{array}{l} \text{if } f(a_1) \cdot f(c_1) < 0 \\ \text{if } f(a_1) \cdot f(c_1) > 0 \end{array} \right. \quad \text{we create } [a_2, b_2] \quad \left\{ \begin{array}{l} \text{with } a_2 = a_1, \ b_2 = c_1 \\ \text{with } a_2 = c_1, \ b_2 = b_1 \end{array} \right.$$

In the previous example we have $f(a_1) \cdot f(c_1) > 0$, hence we have $a_2 = c_1$, $b_2 = b_1$.

- We now generalize this idea in the proof of the theorem.

Proof. Consider the following two recursively defined sequences:

$$a_0 = a, \quad a_{n+1} = \begin{cases} a_n & \text{if } f(a_n) \cdot f(c_n) < 0 \\ c_n & \text{if } f(a_n) \cdot f(c_n) > 0 \end{cases} \quad \forall n \in \mathbb{N}$$

$$b_0 = b, \quad b_{n+1} = \begin{cases} c_n & \text{if } f(a_n) \cdot f(c_n) < 0 \\ b_n & \text{if } f(a_n) \cdot f(c_n) > 0 \end{cases} \quad \forall n \in \mathbb{N}$$

where

$$c_n = \frac{a_n + b_n}{2}, \quad \forall n \in \mathbb{N}$$

The two sequences create a sequence of intervals $[a_n, b_n]$ with the following properties:

- 1) We have $a_n \leq a_{n+1}$, hence the sequence $\{a_n\}$ is increasing and, since $a_n \leq b, \forall n$, it is also bounded. Moreover, we have $b_n \geq b_{n+1}$, hence the sequence $\{b_n\}$ is decreasing and, since $b_n \geq a, \forall n$, it is also bounded.
- 2) $b_n - a_n = \frac{b-a}{2^n}$ (each interval is half as long as the previous one)
- 3) $f(a_n) \cdot f(b_n) < 0$ (because of how a_n and b_n were chosen at each step)

By point 1), we can then deduce that the sequences $\{a_n\}$ and $\{b_n\}$ have a finite limit, thanks to the monotone sequence theorem. Hence:

$$a_n \rightarrow \ell_1 \in \mathbb{R} \text{ and } b_n \rightarrow \ell_2 \in \mathbb{R} \text{ as } n \rightarrow +\infty.$$

From point 2) we deduce that:

$$b_n - a_n = \frac{b-a}{2^n} \rightarrow 0 \text{ as } n \rightarrow +\infty, \text{ and therefore } \ell_2 = \ell_1 = \ell$$

By the continuity of f , we then have that:

$$f(a_n) \cdot f(b_n) \rightarrow (f(\ell))^2 \text{ as } n \rightarrow +\infty$$

On the other hand, from point 3) and the sign-preservation theorem we deduce $(f(\ell))^2 \leq 0$. Therefore it must be $f(\ell) = 0$ and thus ℓ is the zero we are looking for, that is, $c = \ell$.

□

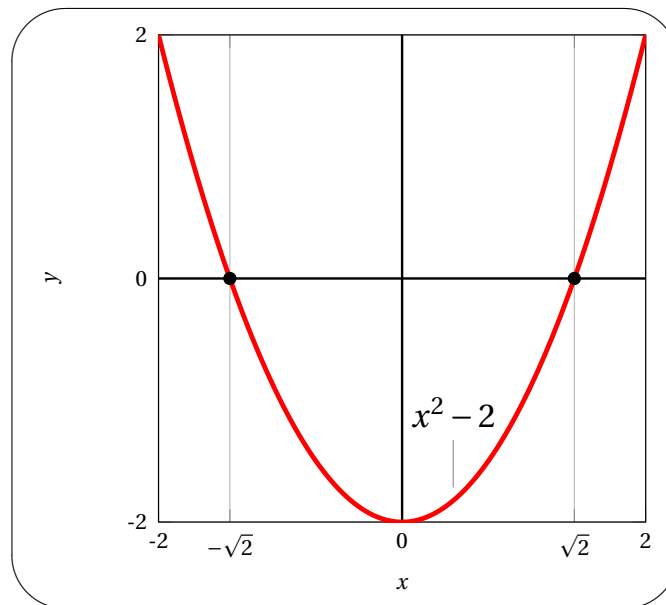
- We have proved:

$f : [a, b] \rightarrow \mathbb{R}$ continuous on $[a, b]$ and $f(a) \cdot f(b) < 0 \Rightarrow$ there exists $c \in [a, b]$ such that $f(c) = 0$

hence “ $f : [a, b] \rightarrow \mathbb{R}$ continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$ ” is a sufficient condition for “there exists $c \in [a, b]$ such that $f(c) = 0$ ” and “there exists $c \in [a, b]$ such that $f(c) = 0$ ” is a necessary condition for “ $f : [a, b] \rightarrow \mathbb{R}$ continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$ ”. The implication does not work in the other direction:

there exists $c \in [a, b]$ such that $f(c) = 0 \not\Rightarrow f : [a, b] \rightarrow \mathbb{R}$ continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$

It suffices, for example, to consider $f(x) = x^2 - 2$ on the interval $[-2, 2]$. We have $f(-2) = 2$ and $f(2) = 2$, hence $f(a) \cdot f(b) \not< 0$, but the function does have zeros:



Hence the theorem provides **sufficient** (but not necessary) conditions for the existence of a zero of a function.

- The proof of the theorem is constructive, that is, we have constructed two sequences that tend to a zero of the function (**constructive proof/algorithm**). This algorithm is called the **bisection method**.

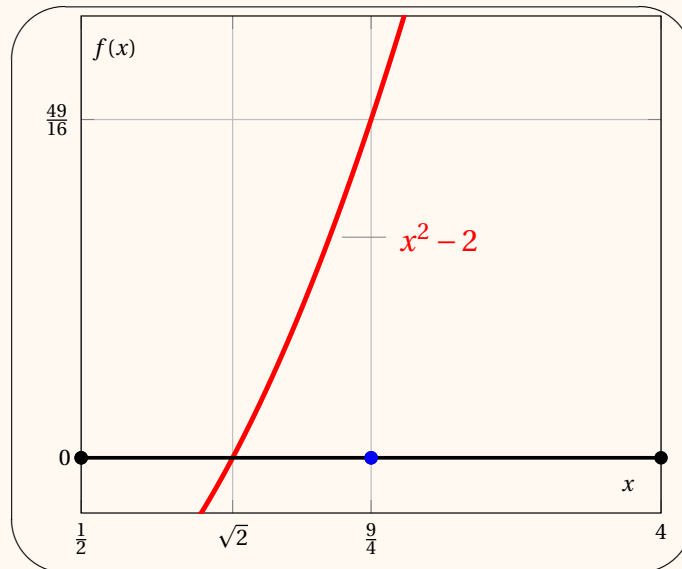
If the function f has several zeros in $[a, b]$, the procedure does not indicate which of them will be determined. The zero found depends on the interval $[a, b]$ given as input, and finding it may require infinitely many iterations. However, by stopping the procedure, as we will see, we obtain an estimate of the zero and of the error made.

2 Bisection method

Example 2: Bisection method

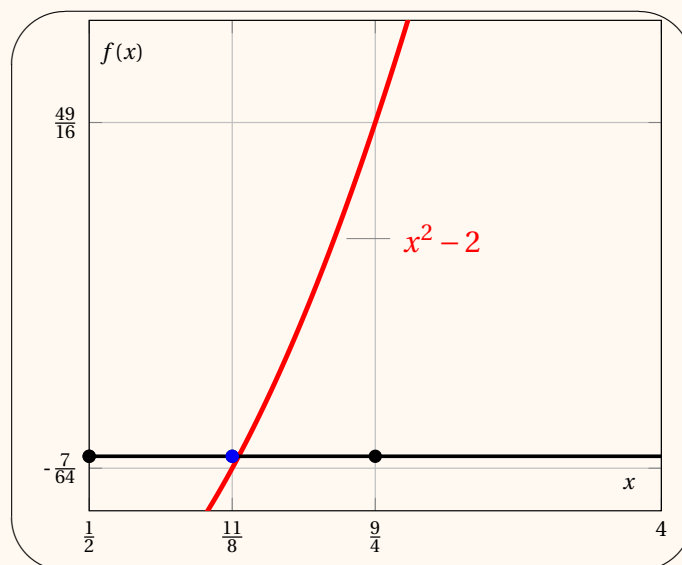
- We look for a zero of the continuous function $f(x) = x^2 - 2$ on the interval $[\frac{1}{2}, 4]$, that is, we look for the value of $\sqrt{2}$. Initialization, with $n = 0$ we have:

$$[a_0, b_0] = \left[\frac{1}{2}, 4\right], \quad f(a) = -\frac{7}{4}, \quad f(b) = 14 \Rightarrow c_0 = \frac{9}{4} \text{ and } f(c_0) = \frac{49}{16}$$



- Iteration $n = 1$, we have:

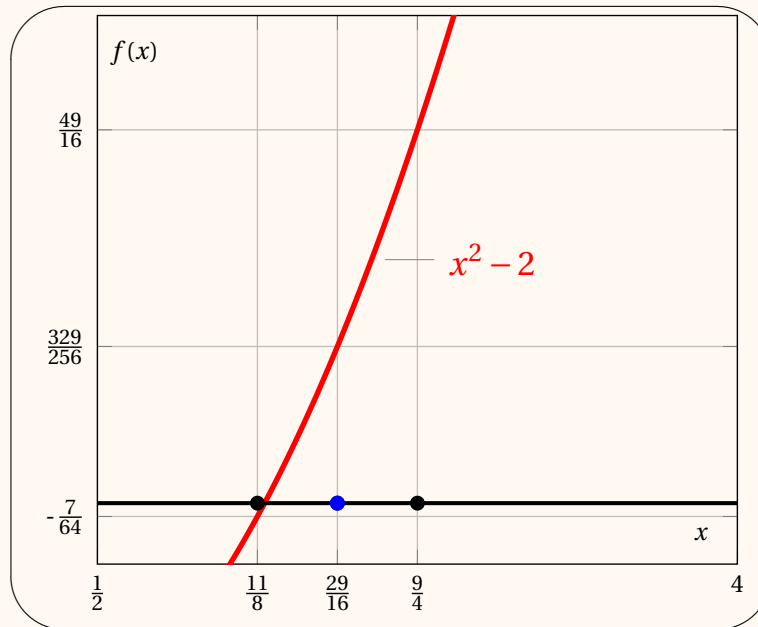
$$[a_1, b_1] = \left[\frac{1}{2}, \frac{9}{4}\right], \quad f(a_1) = -\frac{7}{4}, \quad f(b_1) = \frac{49}{16} \Rightarrow c_1 = \frac{11}{8} \text{ and } f(c_1) = -\frac{7}{64}$$



Example 3: Bisection method

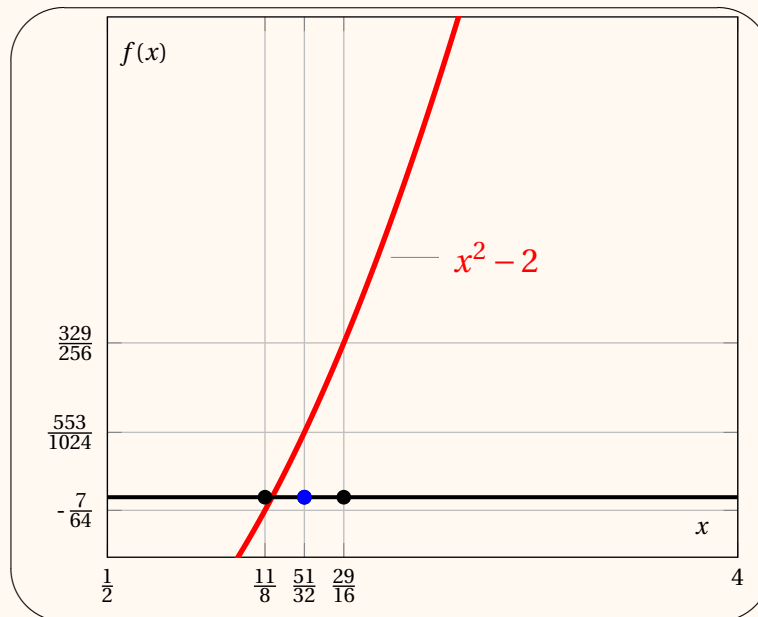
- Iteration $n = 2$, we have:

$$[a_2, b_2] = \left[\frac{11}{8}, \frac{9}{4} \right], \quad f(a_2) = -\frac{7}{64}, \quad f(b_2) = \frac{49}{16} \Rightarrow c_2 = \frac{29}{16} \text{ and } f(c_2) = \frac{329}{256}$$



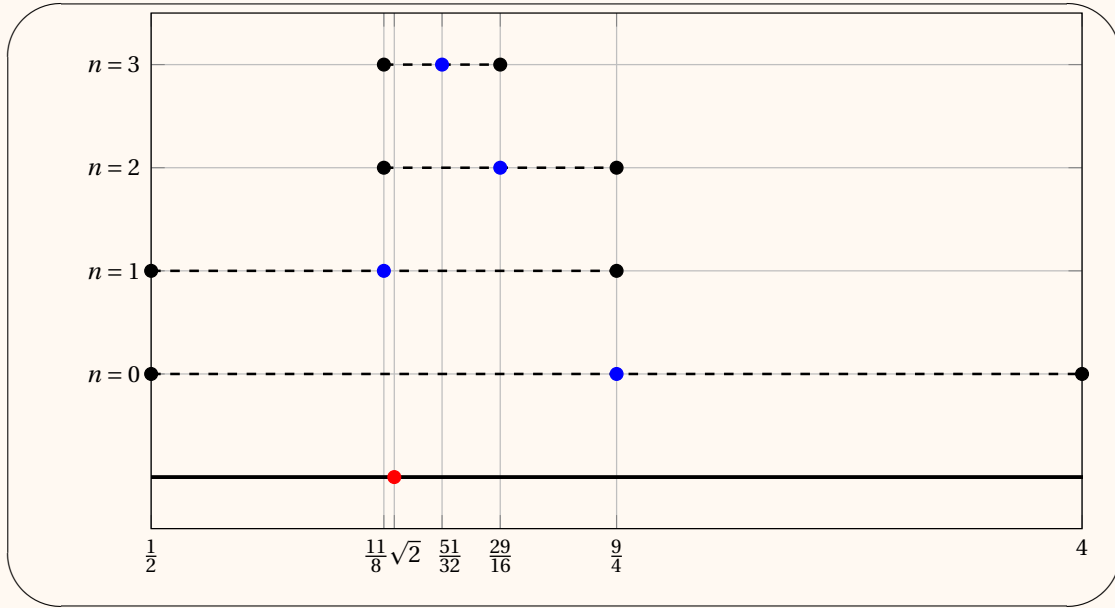
- Iteration $n = 3$, we have:

$$[a_3, b_3] = \left[\frac{11}{8}, \frac{29}{16} \right], \quad f(a_3) = -\frac{7}{64}, \quad f(b_3) = \frac{329}{256} \Rightarrow c_3 = \frac{51}{32} \text{ and } f(c_3) = \frac{553}{1024}$$



Example 4: Intervals of the bisection method

Plotting the values obtained in the previous exercise, we have the following intervals and estimates of $\sqrt{2}$:



2.1 Error estimates for the bisection method

- At iteration n , we have:

$$c \in [a_n, b_n]$$

that is, the zero lies inside the n -th interval. Hence, stopping the procedure after n steps, a_n approximates the zero from below and b_n approximates the zero from above. Therefore:

$$a_n \leq c \leq b_n$$

- The estimate of the value c at iteration n is:

$$c_n = \frac{a_n + b_n}{2}$$

that is, the midpoint of the interval $[a_n, b_n]$.

- The absolute error ε_n made at iteration n is:

$$\varepsilon_n = |c_n - c|$$

The value of c is unknown, hence we need a way to analyze the values of the errors that is independent of c .

- We can estimate the error at iteration n as follows:

$$\varepsilon_n = |c_n - c| \leq \frac{b-a}{2^n}$$

this follows from:

- the length of the interval at iteration n is $\frac{b-a}{2^n}$
- the midpoint of this interval is c_n and c lies inside the interval itself
- the distance of any point c from the center of the interval is less than or equal to the length of the interval itself

If $n \rightarrow +\infty$ then $\varepsilon_n \rightarrow 0$, that is, the error tends to zero as n tends to infinity.

- At iteration n , we can improve the error estimate as follows:

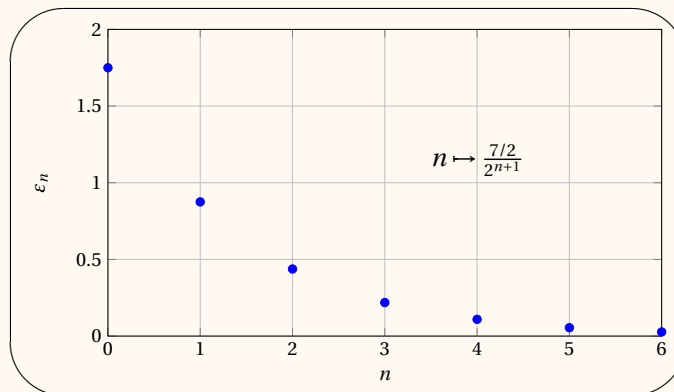
$$\varepsilon_n = |c_n - c| \leq \frac{1}{2} \frac{b-a}{2^n} = \frac{b-a}{2^{n+1}}$$

that is, half the length of the interval $[a_n, b_n]$, since $c \in (a_n, c_n)$ or $c \in (c_n, b_n)$.

Example 5: Error estimates for the bisection method ($\sqrt{2} = 1.414213562\dots$)

Let us go back to the previous exercise with $b-a = \frac{7}{2}$; in the first 6 iterations we have:

iteration	a_n	c_n	b_n	estimate of $\sqrt{2}$	ε_n
$n=0$	$\frac{1}{2}$	$\frac{9}{4}$	4	2.25	$\leq \frac{7/2}{2^1} = 1.75$
$n=1$	$\frac{1}{2}$	$\frac{11}{8}$	$\frac{9}{4}$	1.375	$\leq \frac{7/2}{2^2} = 0.875$
$n=2$	$\frac{11}{8}$	$\frac{29}{16}$	$\frac{9}{4}$	1.8125	$\leq \frac{7/2}{2^3} = 0.4375$
$n=3$	$\frac{11}{8}$	$\frac{51}{32}$	$\frac{29}{16}$	1.59375	$\leq \frac{7/2}{2^4} = 0.21875$
$n=4$	$\frac{11}{8}$	$\frac{95}{64}$	$\frac{51}{32}$	1.484375	$\leq \frac{7/2}{2^5} = 0.109375$
$n=5$	$\frac{11}{8}$	$\frac{183}{128}$	$\frac{95}{64}$	1.4296875	$\leq \frac{7/2}{2^6} = 0.0546875$



- In order to guarantee that the error made does not exceed a given tolerance δ , that is, to impose $\varepsilon_n \leq \delta$, we need to perform $n(\delta)$ iterations, where $n(\delta)$ is the smallest integer that satisfies the inequality:

$$n(\delta) > \log_2 \left(\frac{b-a}{\delta} \right) - 1$$

obtained by solving for $n(\delta)$ in the maximum error estimate:

$$\delta = \frac{b-a}{2^{(n(\delta)+1)}}$$

hence

$$2^{(n(\delta)+1)} = \frac{b-a}{\delta} \quad \text{and} \quad n(\delta) + 1 = \log_2 \left(\frac{b-a}{\delta} \right)$$

Clearly, if $\delta \rightarrow 0$ then $n(\delta) \rightarrow +\infty$, and the smaller the error tolerance, the larger the number of iterations needed to guarantee it.

- If we wanted to reduce the tolerance by one decimal digit, that is, to go from δ to $\frac{\delta}{10}$, we would have:

$$n(\delta) \approx \log_2 \left(\frac{b-a}{\delta} \right) - 1 \quad \text{and} \quad n\left(\frac{\delta}{10}\right) \approx \log_2 \left(\frac{b-a}{\frac{\delta}{10}} \right) - 1$$

since

$$\log_2 \left(\frac{b-a}{\frac{\delta}{10}} \right) - 1 = \log_2 \left(\frac{b-a}{\delta} \cdot 10 \right) - 1 = \underbrace{\log_2 \left(\frac{b-a}{\delta} \right) - 1}_{\approx n(\delta)} + \log_2 10$$

the number of additional iterations is independent of the interval $[a, b]$, and it is equal to

$$\log_2 10 \approx 3,32$$

On average, more than three bisections are needed to improve the accuracy of the estimate by one significant digit; **convergence is therefore slow.**

Example 6: Bisection method (error estimates)

Again for the previous example:

- if we want a tolerance (maximum error) $\delta = 0.00001 = 10^{-5}$:

$$n(10^{-5}) > \log_2 \left(\frac{7/2}{10^{-5}} \right) - 1 = 17.416... \quad \text{hence} \quad n(10^{-5}) = 18$$

- if we want a tolerance (maximum error) $\delta = 0.000001 = 10^{-6}$:

$$n(10^{-6}) > \log_2 \left(\frac{7/2}{10^{-6}} \right) - 1 = 20.738... \quad \text{hence} \quad n(10^{-6}) = 21$$

We now want to understand how “fast” the error decreases by comparing the errors of two consecutive iterations. This estimate gives us information about the quality of the algorithm, that is, we want to determine the “**rate of convergence**” of the algorithm.

- Consider the maximum error, which we denote by ε_n^* ; we have:

$$\varepsilon_n^* = \frac{b-a}{2^{n+1}} \quad \text{and} \quad \varepsilon_n \leq \varepsilon_n^*$$

hence

$$\frac{\varepsilon_{n+1}^*}{\varepsilon_n^*} = \frac{1}{2} \quad \text{since} \quad \frac{\frac{b-a}{2^{n+2}}}{\frac{b-a}{2^{n+1}}} = \frac{1}{2}$$

that is, the maximum error is halved at each iteration. Consequently, we have:

$$\underbrace{\varepsilon_{n+1}^*}_{\geq \varepsilon_{n+1}} = \frac{1}{2} \underbrace{\varepsilon_n^*}_{\geq \varepsilon_n}$$

The maximum absolute error of the bisection method at each step is proportional to the maximum absolute error at the previous step. **The rate of convergence is linear.**

- For every triple c_{n+1}, c_n and c_{n-1} , with $n \geq 1$, we have the following relation:

$$c_{n+1} = \frac{c_{n-1} + c_n}{2} \tag{2}$$

that is, the estimate c_{n+1} is the midpoint of the interval with endpoints c_{n-1} and c_n . We now write:

$$c_{n+1} = c + \varepsilon_{n+1}, \quad c_n = c + \varepsilon_n, \quad c_{n-1} = c + \varepsilon_{n-1}$$

now, substituting into equation (2), we obtain:

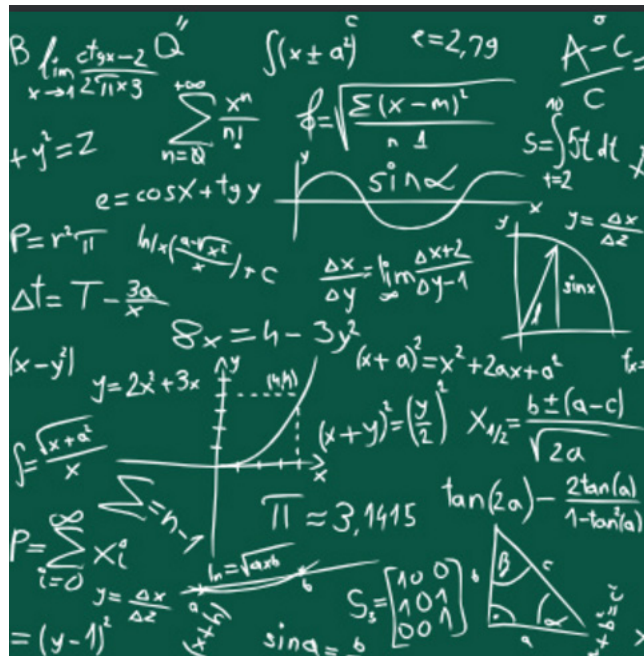
$$\begin{aligned} c + \varepsilon_{n+1} &= \frac{c + \varepsilon_n + c + \varepsilon_{n-1}}{2} = \frac{2c + \varepsilon_n + \varepsilon_{n-1}}{2} \\ &= c + \frac{\varepsilon_n + \varepsilon_{n-1}}{2} \end{aligned}$$

hence

$$\varepsilon_{n+1} = \frac{\varepsilon_n + \varepsilon_{n-1}}{2} = \frac{1}{2} \varepsilon_n \left(\frac{\varepsilon_{n-1}}{\varepsilon_n} + 1 \right)$$

The term $\varepsilon_{n-1}/\varepsilon_n$, that is, the ratio between the errors of two consecutive iterations, can be an arbitrarily small or large number (indeed, the error at each iteration can increase or decrease).

Weierstrass theorem and intermediate value theorem



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Weierstrass theorem

- The following theorem establishes **sufficient but not necessary** conditions for a function to have a maximum and a minimum.

Theorem 1: Weierstrass theorem

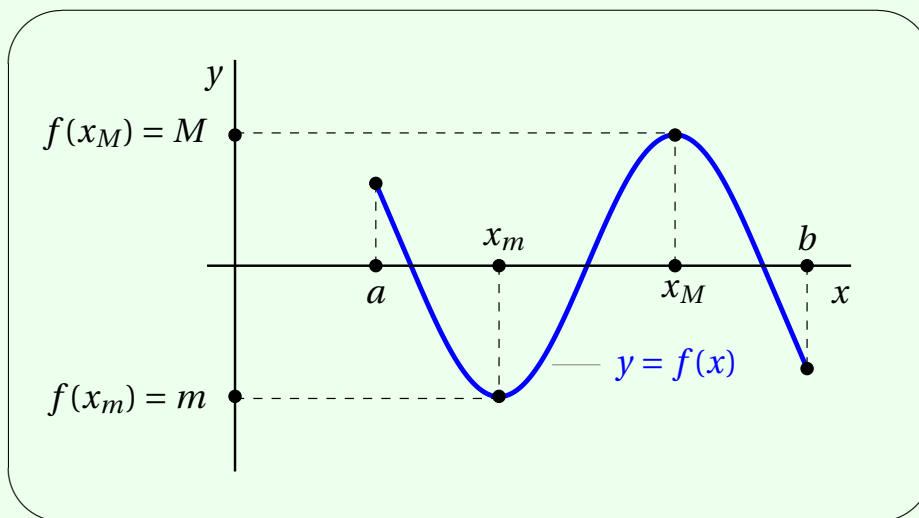
If a function $f : [a, b] \rightarrow \mathbb{R}$ is continuous on the interval $[a, b]$, then it has a maximum M and a minimum m in $[a, b]$.

Under the hypotheses of the theorem there exist:

x_m and $x_M \in [a, b]$ such that $f(x_m) \leq f(x) \leq f(x_M)$ for every $x \in [a, b]$

We say that x_m is a **minimum point** of f and $m = f(x_m)$ is the **minimum** of f .

We say that x_M is a **maximum point** of f and $M = f(x_M)$ is the **maximum** of f .



Properties of the supremum/infimum

- Given two non-empty subsets E_1, E_2 of $\mathbb{R}$ we have:

$$\sup(E_1 \cup E_2) = \max(\sup E_1, \sup E_2)$$

This property holds for both bounded and unbounded sets. If one or both sets are unbounded above we have: $\sup(E_1 \cup E_2) = +\infty$. For the infimum we have:

$$\inf(E_1 \cup E_2) = \min(\inf E_1, \inf E_2)$$

Proof.

We prove that f has a maximum in $[a, b]$. Consider the function:

$$f : [a, b] \rightarrow \mathbb{R}, \text{ and set } \ell = \sup_{[a, b]} f$$

We now need to prove that:

- i) ℓ is finite, that is, $\ell \in \mathbb{R}$;
- ii) ℓ is equal to $f(x_0)$ for some $x_0 \in [a, b]$ (and hence ℓ is the maximum and x_0 is the maximum point).

We split the interval $[a, b]$ into two equal intervals: I_1 and I_2 ; by the properties of the supremum we have:

$$\sup_{[a, b]} f = \max \left(\sup_{I_1} f, \sup_{I_2} f \right) \text{ that is } \sup_{[a, b]} f = \sup_{I_1} f \text{ or } \sup_{[a, b]} f = \sup_{I_2} f$$

Hence for one of the two intervals, which we call $[a_1, b_1]$, it will be true that:

$$\ell = \sup_{[a_1, b_1]} f$$

We now split $[a_1, b_1]$ into two equal intervals; for one of them, which we call $[a_2, b_2]$, it will be true that:

$$\ell = \sup_{[a_2, b_2]} f$$

Proceeding by bisection in this way, we construct a sequence of intervals $[a_n, b_n]$, each contained in the previous ones, with the properties:

- 1) the sequence $\{a_n\}$ is monotone increasing and bounded, and the sequence $\{b_n\}$ is monotone decreasing and bounded;
- 2) $b_n - a_n = \frac{b-a}{2^n} \rightarrow 0$ as $n \rightarrow +\infty$;
- 3) $\ell = \sup_{[a_n, b_n]} f$ (the supremum of f is attained in the interval $[a_n, b_n]$)

By the same reasoning used in the proof of the intermediate zero theorem (which uses the monotonicity theorem for sequences), from points 1) and 2) it follows that the sequences $\{a_n\}$ and $\{b_n\}$ converge to the same limit $x_0 \in [a, b]$:

$$a_n \rightarrow x_0 \text{ and } b_n \rightarrow x_0 \text{ as } n \rightarrow +\infty.$$

We now proceed by cases.

□

Proof. Case 1: $\ell \in \mathbb{R}$

If $\ell \in \mathbb{R}$, for every n there exists a point $t_n \in [a_n, b_n]$ such that:

$$\ell - \frac{1}{n} < f(t_n) \leq \ell \quad (1)$$

Indeed, since $\ell - \frac{1}{n}$ is less than ℓ , which is the least upper bound of the values of $f(x)$ in $[a_n, b_n]$, $\ell - \frac{1}{n}$ is not an upper bound; hence there exists $t_n \in [a_n, b_n]$ with property (1).

Since $t_n \in [a_n, b_n]$, and we have:

$$a_n \rightarrow x_0 \text{ and } b_n \rightarrow x_0 \text{ as } n \rightarrow +\infty \text{ then } t_n \rightarrow x_0 \text{ as } n \rightarrow +\infty$$

by the comparison (squeeze) theorem. Again by the comparison theorem, (1) then gives

$$\lim_{n \rightarrow +\infty} f(t_n) = \ell$$

On the other hand, since f is continuous and $t_n \rightarrow x_0$, we have:

$$\lim_{n \rightarrow +\infty} f(t_n) = f(x_0) \text{ therefore } f(x_0) = \ell.$$

Then, since ℓ is the sup of the values of f in $[a, b]$, ℓ is the **maximum** of f in $[a, b]$, and it is attained at the point x_0 , the **maximum point**. Therefore, in this case the theorem is proved.

Case 2: $\ell = +\infty$

If $\ell = +\infty$, for every n there exists a point $t_n \in [a_n, b_n]$ such that:

$$f(t_n) \geq n \quad (2)$$

Reasoning as above, one proves that $t_n \rightarrow x_0$ for some $x_0 \in [a, b]$. Since f is continuous at x_0 :

$$\lim_{n \rightarrow +\infty} f(t_n) = f(x_0)$$

but by (2) we have

$$\lim_{n \rightarrow +\infty} f(t_n) = +\infty$$

which is a **contradiction**, since at x_0 the function must have a finite value. Hence this case cannot occur.

Similarly, one proves that the function has a minimum in $[a, b]$.

□

The proof is a constructive existence proof (similar to that of the intermediate zero theorem), which, however, cannot be used algorithmically since, after each bisection, it is not known a priori which interval the supremum or infimum is associated with.

The hypotheses of the Weierstrass theorem are all essential

1. The interval must be closed

Consider as a counterexample:

$$f(x) = x \text{ with } x \in (0, 1)$$

The function is continuous on a bounded, but not closed, interval. In this case, the function has neither a maximum nor a minimum (its supremum, 1, and its infimum, 0, are not attained by the function).

2. The interval must be bounded

Consider as a counterexample:

$$f(x) = x \text{ with } x \in \mathbb{R}$$

The function is continuous on an unbounded interval but has neither a maximum nor a minimum (it is not even bounded).

3. The function must be continuous

Consider as a counterexample:

$$f(x) = \begin{cases} x & \text{for } x \in (0, 1) \\ \frac{1}{2} & \text{for } x = 0 \text{ and } x = 1 \end{cases}$$

The function is defined on a closed and bounded interval $[0, 1]$ but it is not continuous. The function has neither a maximum nor a minimum (its supremum, 1, and its infimum, 0, are not attained by the function).

2 Intermediate value theorem

Theorem 2: Intermediate value theorem

If a function $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, then it has a maximum M and a minimum m in $[a, b]$ and it takes all the values between m and M .

We are under the hypotheses of the Weierstrass theorem, hence we have:

$$x_m \text{ and } x_M \in [a, b] \text{ such that } f(x_m) \leq f(x) \leq f(x_M) \text{ for every } x \in [a, b]$$

The intermediate value theorem further tells us that:

$$\forall \lambda \in (m, M), \text{ there exists } x(\lambda) \in [x_m, x_M] \text{ such that } f(x(\lambda)) = \lambda$$

This property is called the **intermediate value property**.

Proof. From the Weierstrass theorem we have a maximum point x_M and a minimum point x_m such that $f(x_M) = M$ (maximum) and $f(x_m) = m$ (minimum) in $[a, b]$. Let

$$m < \lambda < M$$

and consider the function

$$g(x) = f(x) - \lambda \text{ with } x \in [x_m, x_M]$$

which is continuous since $f(x)$ is continuous on $[a, b]$ and $x_m, x_M \in [a, b]$.

Hence:

$$g(x_M) = f(x_M) - \lambda = M - \lambda > 0$$

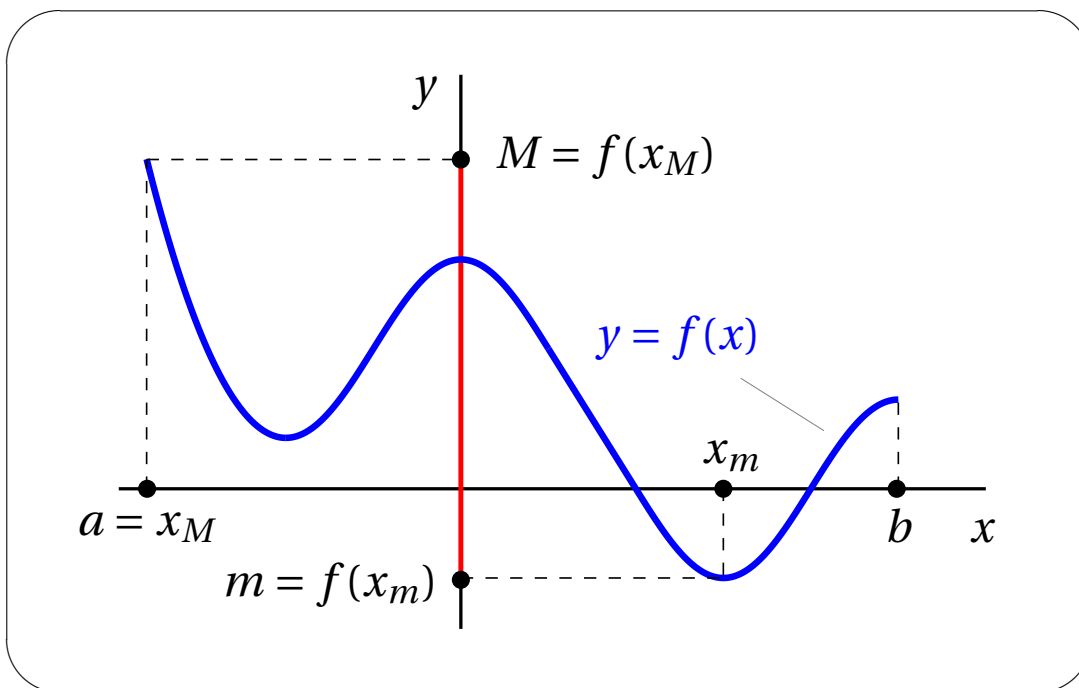
$$g(x_m) = f(x_m) - \lambda = m - \lambda < 0$$

Then, by the intermediate zero theorem, there exists $x(\lambda) \in (x_m, x_M)$ such that:

$$g(x(\lambda)) = 0 \text{ that is } f(x(\lambda)) = \lambda$$

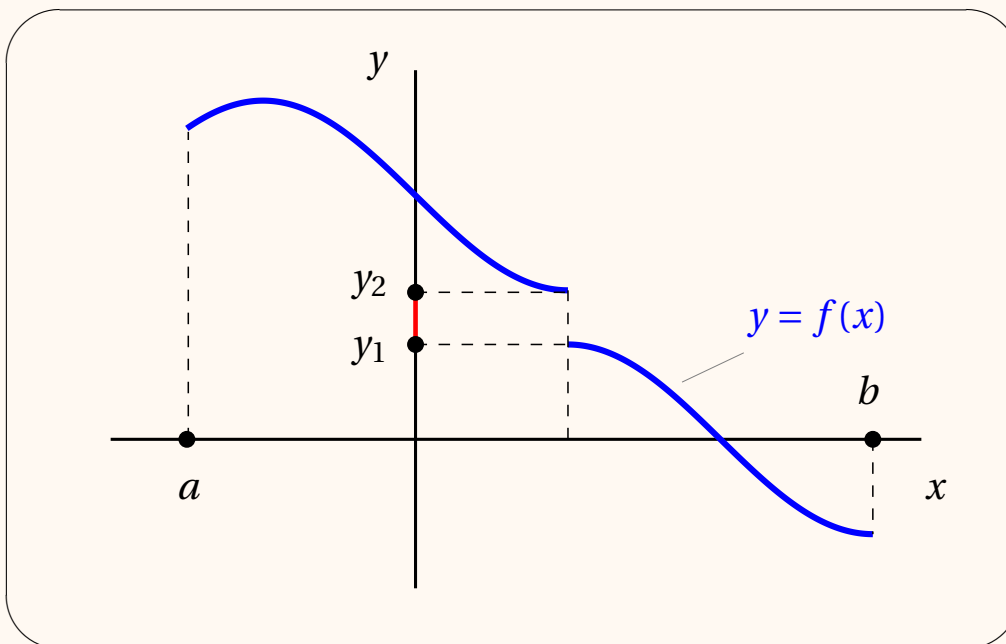
□

- Given a function continuous on an interval $[a, b]$, graphically we have:



Example 1: Discontinuous function without the intermediate value property

Consider for example the graph of the following function, discontinuous on $[a, b]$:



This function does not have the intermediate value property, that is, the values $\lambda \in (y_1, y_2)$ are not outputs of f .

- The properties of the two previous theorems can be summarized in the following single statement:

Corollary 1

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then

$$f([a, b]) = [m, M];$$

i.e., the image of an interval $[a, b]$ is the interval with endpoints:

$$m = \min_{[a, b]} f \quad \text{and} \quad M = \max_{[a, b]} f.$$

Proof. It follows from the Weierstrass theorem and the intermediate value theorem. □

Example 2: Failure of the intermediate value theorem in $\mathbb{Q}$

- Let

$$f(x) = x^2$$

and consider f as a function from the set $\mathbb{Q}$ of rational numbers to $\mathbb{Q}$ itself (this is legitimate because the square of a rational number is rational).

- Then f does not have the intermediate value property.
- Indeed, for example,

$$f(1) = 1, f(2) = 4,$$

but f does not take all the rational values between 1 and 4: for example, it never takes the value 2, or 3.

- In other words, the intermediate value property holds for continuous functions **thanks to the properties of the set of real numbers**.
- This is a further reason why it is useful to work in the set of real numbers rather than in the set of rational numbers.

3 Existence theorem for the n -th root

Theorem 3

For every $y \in \mathbb{R}$, $y > 0$ and $n \in \mathbb{N}$, $n \geq 1$, there exists one and only one $x \in \mathbb{R}$, $x > 0$, such that $x^n = y$.

This number x is called the n -th root of y

Proof. Consider the function $f(x) = x^n$.

This function is continuous on all of $\mathbb{R}$, because it is the product of n continuous functions $g(x) = x$.

We show that there exist

$$x_2 > x_1 > 0 \text{ such that } x_1^n < y < x_2^n.$$

If $y = 1$ the n -th root exists and is unique; moreover:

1. If $y > 1$: it suffices to choose

$$x_1 = 1, x_2 = y \text{ and we have } x_1^n < y < x_2^n \quad (1 < y < y^n)$$

2. If $y < 1$: it suffices to choose

$$x_1 = y, x_2 = 1 \text{ and we have } x_1^n < y < x_2^n \quad (y^n < y < 1)$$

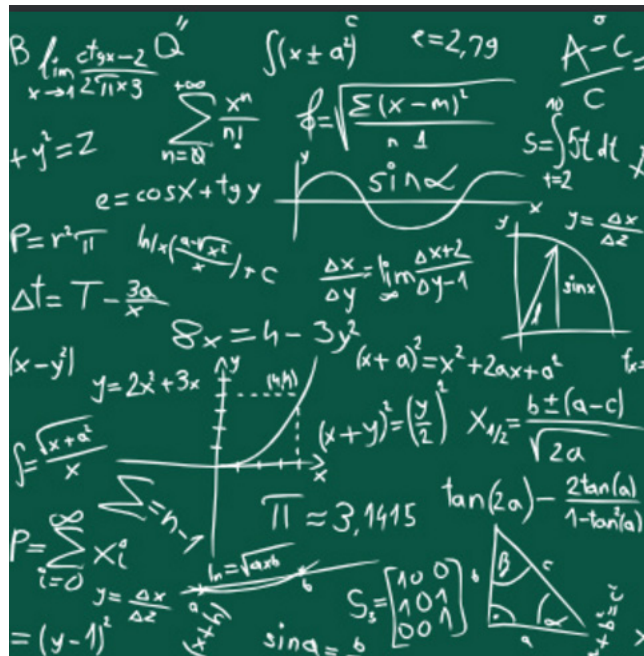
Then we can apply the intermediate value theorem to the continuous function $f(x) = x^n$ on the interval $[x_1, x_2]$. Here its minimum is x_1^n and its maximum is x_2^n .

Since:

$$x_1^n < y < x_2^n, \text{ there exists } x_0 \in [x_1, x_2] \text{ such that } x_0^n = y.$$

Uniqueness follows from the fact that the function is strictly increasing. □

Monotone functions on an interval and invertibility



- **Author:** Fabio Furini.
- **Book:** "Analisi matematica 1", Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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2	Continuity and invertibility	4

1 Monotone functions on an interval

- We now deal with monotone functions on an interval (and **not necessarily continuous**), and the next theorem (based on the completeness axiom of $\mathbb{R}$) can be seen as an extension to functions of the monotonicity theorem for sequences.

Theorem 1: Monotonicity theorem for functions

Let $f : (a, b) \rightarrow \mathbb{R}$ be a monotone function. Then for every $c \in (a, b)$ the right and left limits as $x \rightarrow c$ exist and are finite; at the two endpoints a, b the right limit (at a) and the left limit (at b) exist, possibly infinite.

Proof. Suppose f is increasing in (a, b) and let $c \in (a, b)$, that is, a point inside the interval.

We prove that the following limit exists and is finite:

$$\lim_{x \rightarrow c^-} f(x) = \sup \{f(x) : x \in (a, c)\}.$$

We set

$$\ell = \sup \{f(x) : x \in (a, c)\}$$

Note that ℓ exists and is finite by the supremum property, since $f(c)$ is an upper bound of the set $\{f(x) : x \in (a, c)\}$.

Hence we need to prove that

$$\lim_{x \rightarrow c^-} f(x) = \ell$$

So let $\{x_n\}$ be any sequence in (a, c) such that $x_n \rightarrow c$, and let us prove that $f(x_n) \rightarrow \ell$, i.e., that for every $\varepsilon > 0$ we eventually have

$$\ell - \varepsilon < f(x_n) < \ell + \varepsilon.$$

The second inequality is obvious, since $f(x_n) \leq \ell$ by definition of ℓ , because $x_n \in (a, c)$.

□

Proof. To prove the first one, we observe that, since $\ell - \varepsilon$ is less than ℓ , i.e., less than the least upper bound of $\{f(x) : x \in (a, c)\}$, it is not an upper bound of this set; therefore there exists a point

$$\tilde{x} \in (a, c) \text{ such that } f(\tilde{x}) > \ell - \varepsilon$$

Since f is increasing, it follows that

$$f(x) > \ell - \varepsilon, \forall x \in (\tilde{x}, c)$$

On the other hand, by hypothesis

$$x_n < c, \forall n \text{ and } x_n \rightarrow c \text{ as } n \rightarrow +\infty$$

therefore $x_n \in (\tilde{x}, c)$ eventually. Hence

$$f(x_n) \geq \ell - \varepsilon \quad (\text{eventually})$$

which is what we needed to prove.

Similarly, one can prove that the following limit exists and is finite:

$$\lim_{x \rightarrow c^+} f(x) = \inf \{f(x) : x \in (c, b)\}.$$

As for the limits at the two endpoints of the interval, we prove that the following limit exists:

$$\lim_{x \rightarrow b^-} f(x)$$

We set (as before)

$$\ell = \sup \{f(x) : x \in (a, b)\}$$

however, in this case ℓ could also be $+\infty$ (we cannot claim that $f(b)$ is an upper bound of the set, because the function is not defined at b). We therefore proceed by cases.

In the case $\ell < \infty$ the previous proof can be repeated. □

Proof. If instead $\ell = \infty$, the argument changes. From the hypothesis

$$\sup \{f(x) : x \in (a, b)\} = +\infty$$

it follows that for every $M > 0$ there exists $\tilde{x} \in (a, b)$ such that $f(\tilde{x}) > M$.

By the monotonicity of f , then

$$f(x) > M, \quad \forall x \in (\tilde{x}, b)$$

hence, taking any sequence $x_n \rightarrow b$, we have $x_n \in (\tilde{x}, b)$ eventually, and therefore

$$f(x_n) > M, \quad \text{eventually,}$$

therefore

$$\lim_{n \rightarrow +\infty} f(x_n) = +\infty$$

In the same way one proves that the following limit exists:

$$\lim_{x \rightarrow a^+} f(x)$$

□

A consequence of the monotonicity theorem is that if a function is monotone on an interval (a, b) , its possible points of discontinuity in (a, b) are necessarily jump discontinuities, except for the endpoints a, b , where there may also be a vertical asymptote.

2 Continuity and invertibility

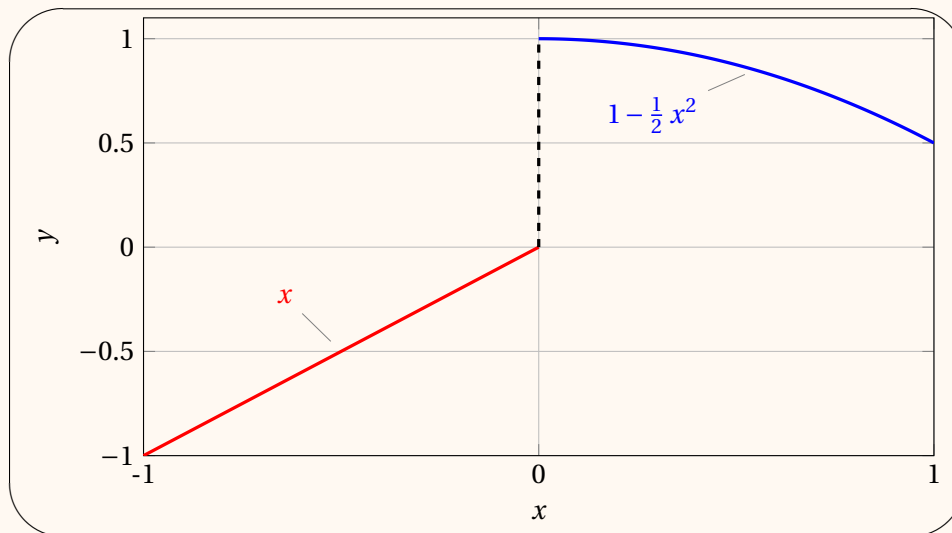
- We have seen the theorem stating that if a generic function with domain D is strictly monotone, then it is invertible.
- We also know that the converse is not true in general: there exist functions that are invertible on an interval but not monotone.

Example 1: Invertible but not monotone function

Consider for example

$$f(x) = \begin{cases} 1 - \frac{1}{2}x^2 & \text{if } 0 < x \leq 1, \\ x & \text{if } x \leq 0 \end{cases}$$

its graph is:



This function satisfies the invertibility condition, which requires the graph of f to be intersected at most at one point by every line parallel to the x -axis, but it is not monotone.

- If we add the hypothesis of continuity and require the domain to be an interval, being strictly monotone becomes a necessary and sufficient condition for invertibility, as stated by the following theorem.

Theorem 2: Invertibility of monotone and continuous functions

Let $f : I \rightarrow \mathbb{R}$ be a function defined on an interval I .

If the function f is continuous, then it is invertible on the interval I if and only if it is strictly monotone.

In this case its inverse function is also strictly monotone and continuous.

Proof. We already know that if f is strictly monotone, it is invertible (regardless of the hypotheses that f is continuous and that it is defined on an interval).

We show that the converse holds, i.e., that if f is continuous and invertible, then it is strictly monotone.

Suppose by contradiction that the function is not strictly monotone; then there exist three points

$$x_1 < x_2 < x_3 \text{ (in the interval } I \text{)}$$

such that

$$f(x_1) < f(x_2) \text{ and } f(x_2) > f(x_3)$$

or such that

$$f(x_1) > f(x_2) \text{ and } f(x_2) < f(x_3).$$

Suppose the first of the two alternatives holds (the other case is handled similarly). Let us compare the values $f(x_1)$ and $f(x_3)$; they cannot be equal because f is invertible by hypothesis, hence

$$f(x_1) < f(x_3) \text{ or } f(x_1) > f(x_3).$$

Again, suppose the first of the two alternatives holds (the other case is handled similarly). So we know that:

$$x_1 < x_2 < x_3 \text{ and } f(x_1) < f(x_3) < f(x_2).$$

Since f is continuous, by the intermediate value theorem there exists

$$x_0 \in (x_1, x_2) \text{ such that } f(x_0) = f(x_3).$$

Since $x_0 \neq x_3$ (because $x_1 < x_2 < x_3$), it follows that f cannot be invertible, a **contradiction**. This proves the first part of the theorem.

Now let f be a continuous, strictly monotone and hence invertible function on I , and let g be its inverse function, which is also strictly monotone and invertible. We prove that g is continuous.

- By what we observed after the monotonicity theorem, the strictly monotone function g is either continuous or has jump discontinuity points.
- In the latter case the image of g is not an interval (but the union of at least two disjoint intervals), which is a **contradiction** because this image is I . Hence g is continuous.

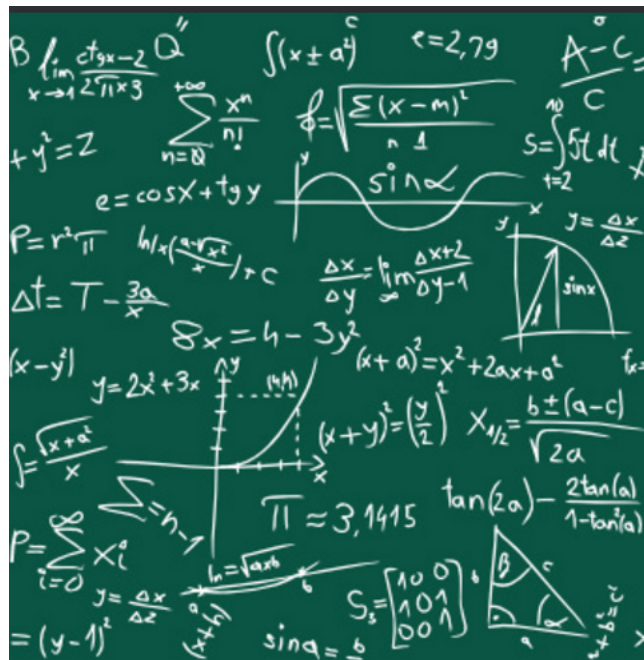
□

- Note that in the proof of the previous theorem we used both the intermediate value theorem and the monotonicity theorem for functions. Moreover, we implicitly used the completeness axiom of $\mathbb{R}$.
- The theorem just proved means, in particular, that:

a continuous and invertible function on an interval has a continuous inverse function.

- This fact completes the proof of the continuity theorem for elementary functions:
 1. the continuity of the function a^x implies the continuity of the function $\log_a x$.
 2. the continuity of the functions $\sin x$, $\cos x$ and $\tan x$ implies the continuity of the functions $\arcsin x$, $\arccos x$ and $\arctan x$.

The derivative function



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani e Sandro Salsa, Zanichelli.

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1 Line through two points

- Starting from two points

$$(x_1, y_1) \text{ and } (x_2, y_2)$$

it is possible to compute the function

$$y = m x + q$$

whose graph is the line through the two points (equation of the line in explicit form). The **slope** of the line is the value m and the **y-intercept** of the line is the value q .

- Requiring the line to pass through the two points we have:

$$\begin{cases} y_1 = m x_1 + q \\ y_2 = m x_2 + q \end{cases}$$

and we can determine m and q as follows (two equations in two unknowns):

$$\begin{cases} q = y_1 - m x_1 \\ q = y_2 - m x_2 \end{cases} \quad \begin{cases} q = y_1 - m x_1 \\ y_1 - m x_1 = y_2 - m x_2 \end{cases} \text{ which becomes } m x_2 - m x_1 = y_2 - y_1$$

$$\begin{cases} q = y_1 - m x_1 \\ m = \frac{y_2 - y_1}{x_2 - x_1} \end{cases} \quad \begin{cases} q = y_1 - \frac{y_2 - y_1}{x_2 - x_1} x_1 \\ m = \frac{y_2 - y_1}{x_2 - x_1} \end{cases}$$

And we obtain the line:

$$y = \underbrace{\frac{y_2 - y_1}{x_2 - x_1}}_m x + y_1 - \underbrace{\frac{y_2 - y_1}{x_2 - x_1} x_1}_q \quad \text{clearly} \quad \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}$$

The line through two points (x_1, y_1) and (x_2, y_2) :

$$y = \underbrace{\frac{y_2 - y_1}{x_2 - x_1}}_m x + y_1 - \underbrace{\frac{y_2 - y_1}{x_2 - x_1} x_1}_m \text{ can be rewritten as } y = m x + y_1 - m x_1.$$

Hence the line can be written only in terms of the slope m and the point (x_1, y_1) as follows:

$$y = y_1 + m (x - x_1) \quad \text{with} \quad m = \frac{y_1 - y_2}{x_1 - x_2}$$

The same holds for any point $(\tilde{x}, \tilde{y})$ of the line; in this case we would have:

$$y = \tilde{y} + m(x - \tilde{x})$$

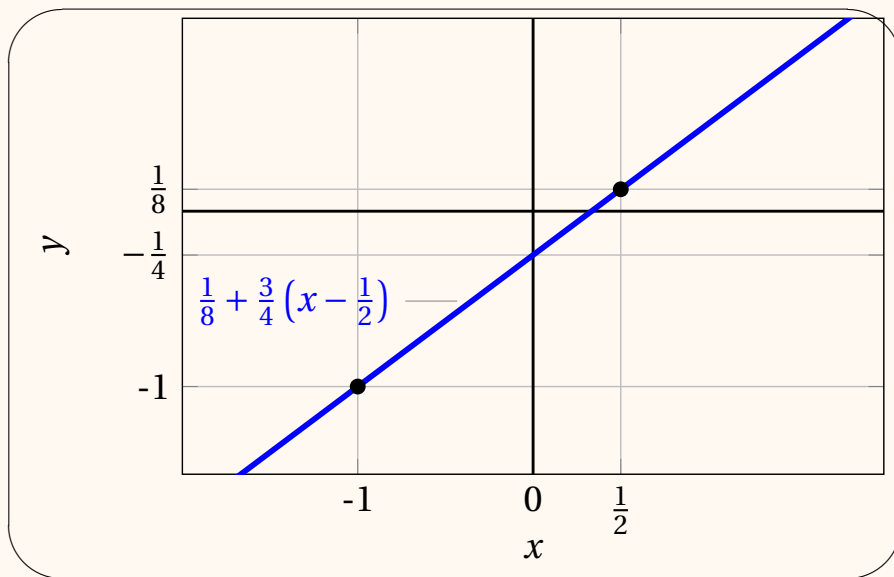
Example 1: Line through two points

We compute the line through the points: $(x_1, y_1) = (-1, -1)$ and $(x_2, y_2) = (\frac{1}{2}, \frac{1}{8})$. We compute the slope:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\frac{1}{8} + 1}{\frac{1}{2} + 1} = \frac{3}{4}$$

Using the point $(x_2, y_2) = (\frac{1}{2}, \frac{1}{8})$ we obtain:

$$y = \frac{1}{8} + \frac{3}{4} \left(x - \frac{1}{2} \right) = \frac{3}{4}x - \frac{1}{4}$$



Now using the point $(x_1, y_1) = (-1, -1)$ we obtain the same line:

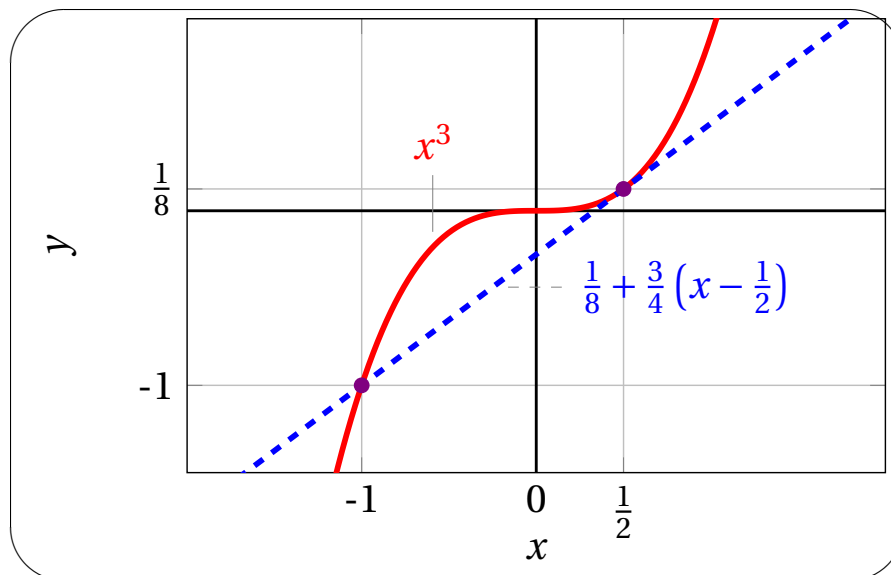
$$y = -1 + \frac{3}{4} (x + 1) = \frac{3}{4}x - \frac{1}{4}$$

2 Tangent line

What is the **tangent** line to the graph of a function at a point?

- The intuitive definition of a line tangent to the graph is that of a line that “touches” the graph without “cutting” or “crossing” it (imagining the curve as if it were an impenetrable physical object)
- This intuitive definition works, for example, for parabolas or hyperbolas. However, take for example the function $f(x) = x^3$

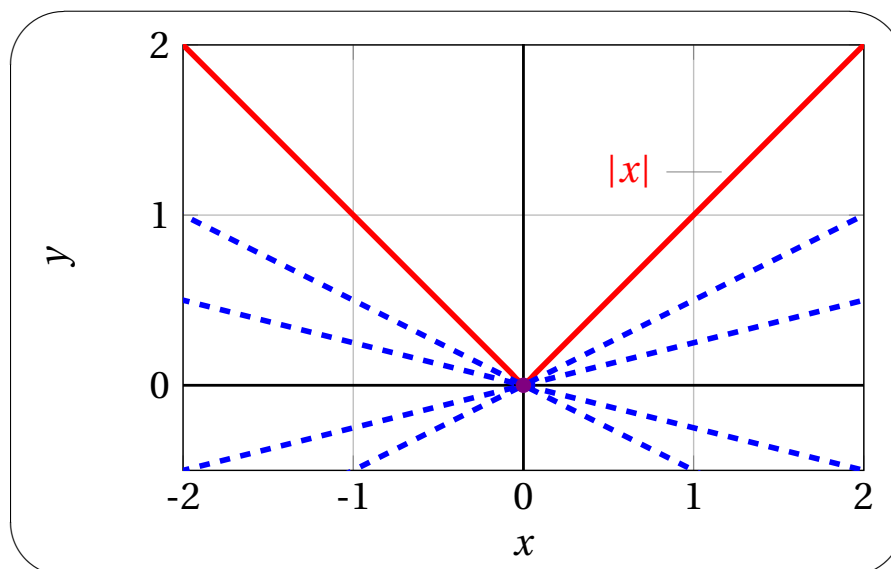
$$\text{the graph } y = x^3 \text{ and the line } y = \frac{1}{8} + \frac{3}{4} \left(x - \frac{1}{2} \right)$$



the intuitive definition works at the point $(\frac{1}{2}, \frac{1}{8})$, but the line also intersects the graph of the function at the point $(-1, -1)$.

- Conversely, there are functions, such as $f(x) = |x|$, whose graph is

$$y = |x|$$



This graph has a point (the origin) at which there are infinitely many lines satisfying the intuitive definition, but clearly none of them can be called tangent.

Geometrically, the tangent line can be approximated by the **line through two points** very close to each other on the curve, the first one **fixed** (the point of tangency) and the other one “**moving**” (and chosen closer and closer to the point of tangency).

- Two points determine a line; the closer the two points get, the closer this line gets to the tangent. We therefore need to understand what happens to the line through the two points when the second, moving point gets closer and closer to the first one, without ever coinciding with it. The “**limit line**” - if it exists - will be called the **tangent** line.

A further way of looking at the concept of tangency is to think of the tangent (to a curve at a point) as the line that **best approximates** the curve in a neighborhood of the point.

2.1 Practical use of the tangent line

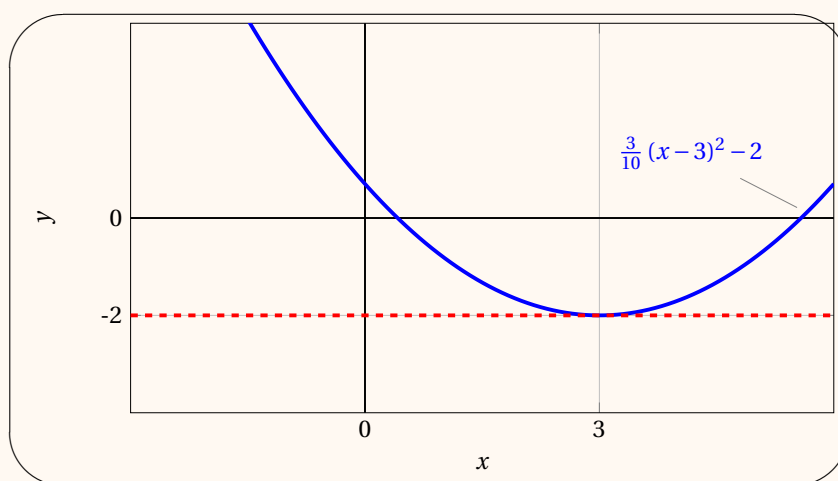
Computing the tangent is important to determine the points where the graph of a function has a **horizontal tangent** (local or global maximum and minimum points, and possibly other points as well).

- It is therefore useful to be able to write analytically the equation of the tangent line to the curve at a generic point, and then see at which points it is horizontal. This idea is due first to Fermat, who developed it around 1630.

Example 2: Graph of a function – point with horizontal tangent

Consider the function:

$$f(x) = \frac{3}{10}(x-3)^2 - 2$$

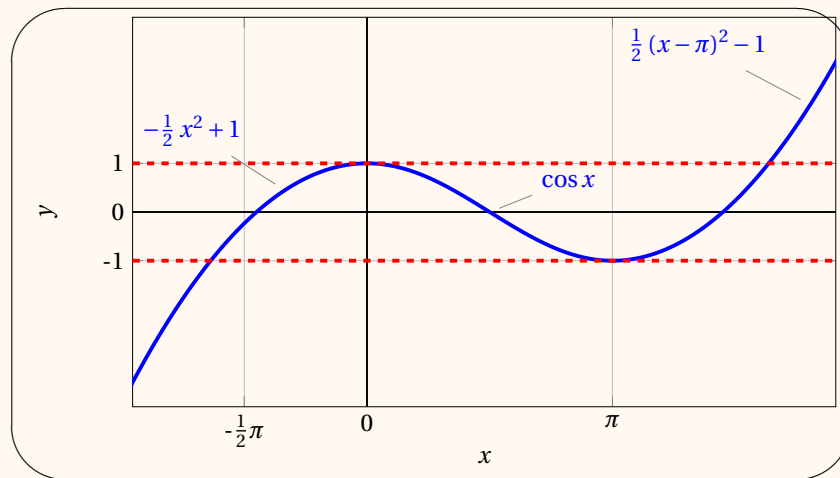


The function has a point with horizontal tangent, which is a global minimum point.

Example 3: Graph of a function – points with horizontal tangent

Consider the (piecewise-defined) function:

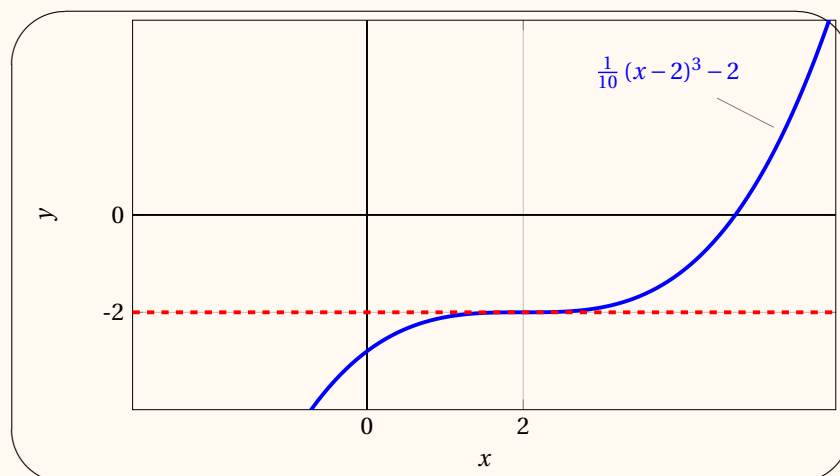
$$f(x) = \begin{cases} -\frac{1}{2}x^2 + 1 & x \leq 0 \\ \cos x & 0 < x \leq \pi \\ \frac{1}{2}(x - \pi)^2 - 1 & x > \pi \end{cases}$$



The function has two points with horizontal tangent, which are local minimum and maximum points, for example in the interval $[-\frac{1}{2}\pi, \pi]$.

Now consider the function:

$$f(x) = \frac{1}{10}(x - 2)^3 - 2$$



The function has a point with horizontal tangent which, however, is neither a maximum nor a minimum point.

3 Derivative of a function at a point and derivative function

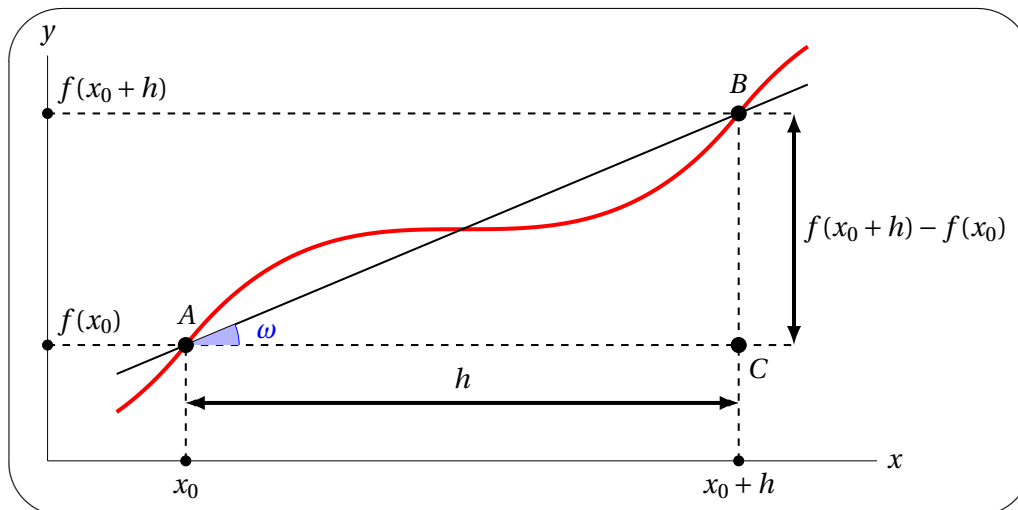
- **Differential calculus** is the study of the notion of derivative, and it is part of infinitesimal calculus.

3.1 Derivative and tangent line

We look for the **equation of the tangent line** to the graph of the function at a **generic point A** of its graph.

- Consider the graph of a generic function $y = f(x)$; the coordinates of the points A and B in the figure are respectively:

$$(x_0, f(x_0)) \quad \text{and} \quad (x_0 + h, f(x_0 + h))$$



- The **slope** of the line through the point A and the point B is:

$$\frac{f(x_0 + h) - f(x_0)}{h}$$

Note that a positive slope indicates (on average) a rise, while a negative one indicates a fall (on average).

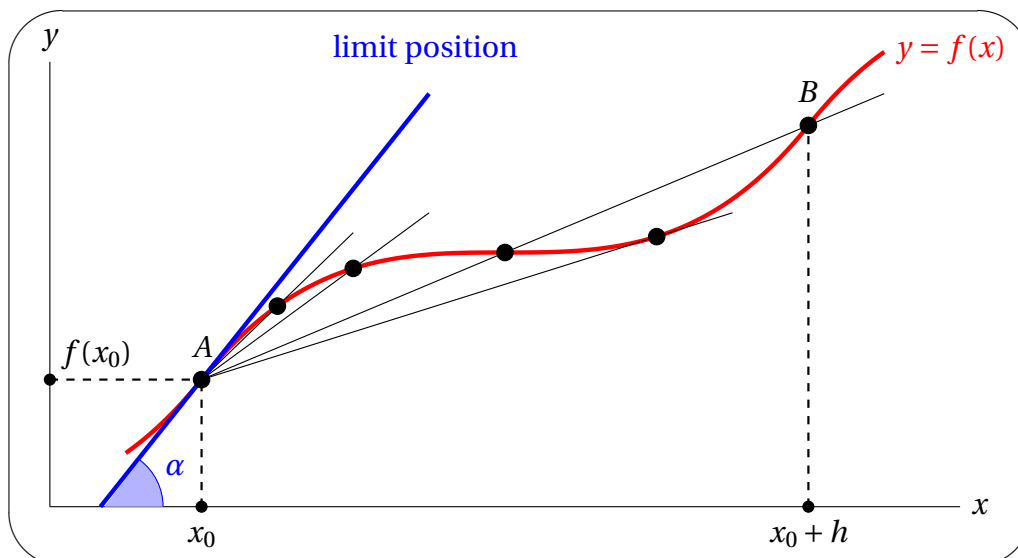
This ratio is called the **difference quotient** of the function f relative to the interval $[x_0, x_0 + h]$.

- Geometrically, since the triangle ABC is a right triangle, we have:

$$\frac{f(x_0 + h) - f(x_0)}{h} = \tan \omega \quad (\text{slope of the line through } A \text{ and } B)$$

- Consider the difference quotient and take the limit (assuming it exists) as $h \rightarrow 0$. Geometrically we have:

- the point A , with coordinates $(x_0, f(x_0))$, stays fixed
- while the point B , with coordinates $(x_0 + h, f(x_0 + h))$, moves towards A (staying on the graph of f).



- Moving B towards A along the graph of the function, the line AB changes its slope and settles into a **limit position**.

- the **limit line** is called the **tangent line to the graph** of the function f at the point with abscissa x_0 ;
- its **slope** is given by $\tan \alpha$ and is called the **first derivative** of the function f at the point x_0 .
- The angle α is the angle between the tangent line and the x -axis.

Definition 1: of derivative

Let $f : (a, b) \rightarrow \mathbb{R}$; f is said to be differentiable at $x_0 \in (a, b)$ if the following limit exists and is finite

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

this limit is called the first derivative (or simply the derivative) of f at x_0 .

With $h = x - x_0$, we have $h \rightarrow 0 \Leftrightarrow x - x_0 \rightarrow 0 \Leftrightarrow x \rightarrow x_0$ and $x = x_0 + h$, hence:

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

- To denote the derivative the following symbols are used:

$\underbrace{f'(x_0)}_{\text{Lagrange's notation}}$

$\underbrace{\dot{f}(x_0)}_{\text{Newton's notation}}$

$\underbrace{\left. \frac{df}{dx} \right|_{x=x_0} \text{ and } \left. \frac{dy}{dx} \right|_{x=x_0}}_{\text{Leibniz's notation}}$

Definition 2: tangent line

Given a function $f : (a, b) \rightarrow \mathbb{R}$ with f differentiable at $x_0 \in (a, b)$, the line with equation:

$$y = f(x_0) + f'(x_0)(x - x_0)$$

is called the **tangent line** to the graph of the function f at the point $(x_0, f(x_0))$.

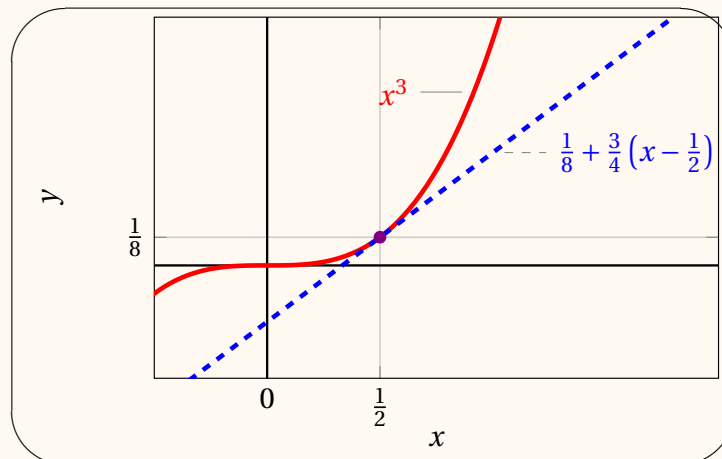
Example 4: Tangent line

We compute the equation of the tangent line to the graph of the function $f(x) = x^3$ at the point with abscissa $x_0 = \frac{1}{2}$. The derivative at the point $x_0 = \frac{1}{2}$ is:

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0} \frac{\left(\frac{1}{2} + h\right)^3 - \left(\frac{1}{2}\right)^3}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{8} + \frac{3}{4}h + \frac{3}{2}h^2 + h^3 - \frac{1}{8}}{h} = \lim_{h \rightarrow 0} \left(\frac{3}{4} + \frac{3}{2}h + h^2\right) = \frac{3}{4}$$

We have $f\left(\frac{1}{2}\right) = \frac{1}{8}$, $f'\left(\frac{1}{2}\right) = \frac{3}{4}$ and the tangent line at the point $\left(\frac{1}{2}, \frac{1}{8}\right)$ is:

$$y = f(x_0) + f'(x_0)(x - x_0) = \frac{1}{8} + \frac{3}{4}\left(x - \frac{1}{2}\right)$$



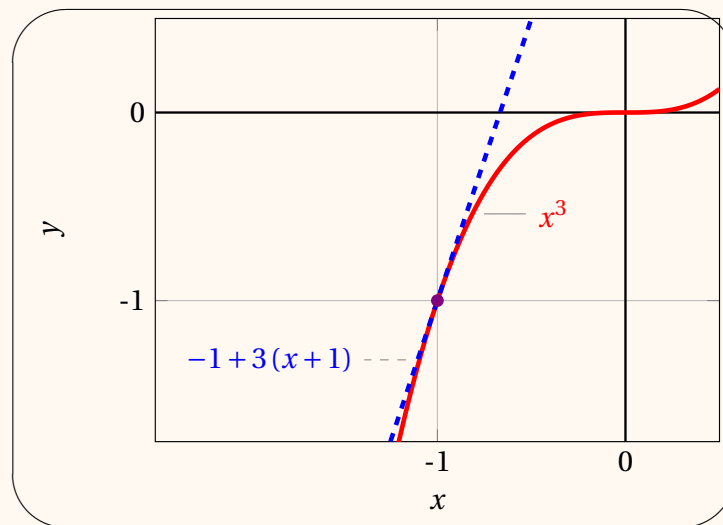
Example 5: Tangent line

We compute the equation of the tangent line to the graph of the function $f(x) = x^3$ at the point with abscissa $x_0 = -1$. The derivative at the point $x_0 = -1$ is:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} &= \lim_{h \rightarrow 0} \frac{(-1 + h)^3 - (-1)^3}{h} = \lim_{h \rightarrow 0} \frac{-1 + 3h - 3h^2 + h^3 + 1}{h} \\ &= \lim_{h \rightarrow 0} (3 - 3h + h^2) = 3 \end{aligned}$$

We have $f(-1) = -1$, $f'(-1) = 3$ and the tangent line at the point $(-1, -1)$ is:

$$y = f(x_0) + f'(x_0)(x - x_0) = -1 + 3(x + 1)$$



- We can now define a function that associates to each x the derivative of a function f at the point x (of course, if f is differentiable).

Definition 3: of derivative function

If a function f is differentiable at every point of an interval (a, b) , the function:

$$f' : (a, b) \rightarrow \mathbb{R}, \quad f' : x \mapsto f'(x)$$

is called the **derivative function** of f .

- To denote the first derivative function the following notations are used:

$\underbrace{f'(x)}_{\text{Lagrange's notation}}$

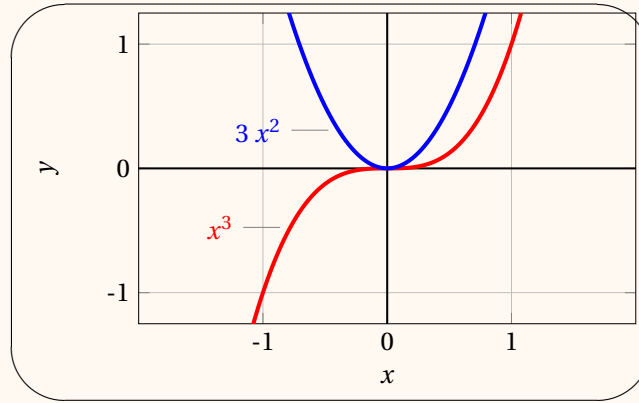
$\underbrace{\dot{f}(x)}_{\text{Newton's notation}}$

$\underbrace{\frac{df}{dx}, \frac{df(x)}{dx} \text{ and } \frac{dy}{dx}}_{\text{Leibniz's notation}}$

Example 6: derivative function

We compute the derivative function of the function $f(x) = x^3$. For every $x \in \mathbb{R}$ we have:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} &= \lim_{h \rightarrow 0} \frac{(x+h)^3 - (x)^3}{h} = \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) = 3x^2 \quad \text{hence} \quad f'(x) = 3x^2 \end{aligned}$$



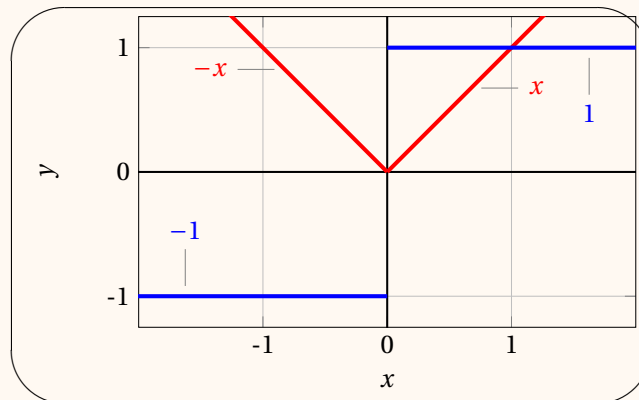
We compute the derivative function of the function $f(x) = |x|$. For $x \neq 0$, we have:

$$f(x) = \begin{cases} x & \text{if } x > 0 \text{ and } \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{x+h-x}{h} = 1 \text{ hence } f'(x) = 1 \\ -x & \text{if } x < 0 \text{ and } \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{-(x+h)+x}{h} = -1 \text{ hence } f'(x) = -1 \end{cases}$$

For $x = 0$ we have:

$$\frac{f(x+h) - f(x)}{h} = \frac{f(h) - f(0)}{h} = \frac{|h|}{h} \quad \text{hence} \quad \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1, \quad \lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$$

since if $h \rightarrow 0^+$ then $|h| = h$ and if $h \rightarrow 0^-$ then $|h| = -h$. We conclude that, since the limit of the difference quotient does not exist, f is not differentiable at $x = 0$. The function is continuous at the origin but the tangent is not well defined.



3.2 Continuity and differentiability

Theorem 1

Given a function $f : [a, b] \rightarrow \mathbb{R}$, if f is differentiable at a point $x_0 \in (a, b)$ then it is continuous at x_0 .

Proof. The function is differentiable at $x_0 \in (a, b)$, so we write:

$$f(x_0 + h) - f(x_0) = \frac{f(x_0 + h) - f(x_0)}{h} \cdot h \sim f'(x_0) \cdot h \text{ for } h \rightarrow 0$$

Moreover we have

$$f'(x_0) \cdot h \rightarrow 0 \text{ for } h \rightarrow 0$$

Therefore

$$\lim_{h \rightarrow 0} (f(x_0 + h) - f(x_0)) = 0 \quad \text{hence} \quad \lim_{h \rightarrow 0} f(x_0 + h) = f(x_0)$$

Setting $x_0 + h = x$ we have $h \rightarrow 0$ if and only if $x \rightarrow x_0$, and consequently we have:

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

□

- A possible alternative but equivalent proof is the following:

Proof. The function is differentiable at $x_0 \in (a, b)$, so we write:

$$\begin{aligned} \lim_{h \rightarrow 0} f(x_0 + h) &= \lim_{h \rightarrow 0} f(x_0 + h) - f(x_0) + f(x_0) \\ &= \lim_{h \rightarrow 0} \underbrace{\frac{f(x_0 + h) - f(x_0)}{h}}_{\substack{\rightarrow f'(x_0) \\ \rightarrow 0}} \cdot h + f(x_0) = f(x_0) \end{aligned}$$

Setting $x_0 + h = x$ we have $h \rightarrow 0$ if and only if $x \rightarrow x_0$, and consequently we have:

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

□

Given a function f , differentiability at a point x_0 in the interior of its domain implies continuity at the point x_0 :

$$f \text{ is differentiable at } x_0 \Rightarrow f \text{ is continuous at } x_0 \quad (1)$$

but the converse is not true:

$$f \text{ is continuous at } x_0 \not\Rightarrow f \text{ is differentiable at } x_0 \quad (2)$$

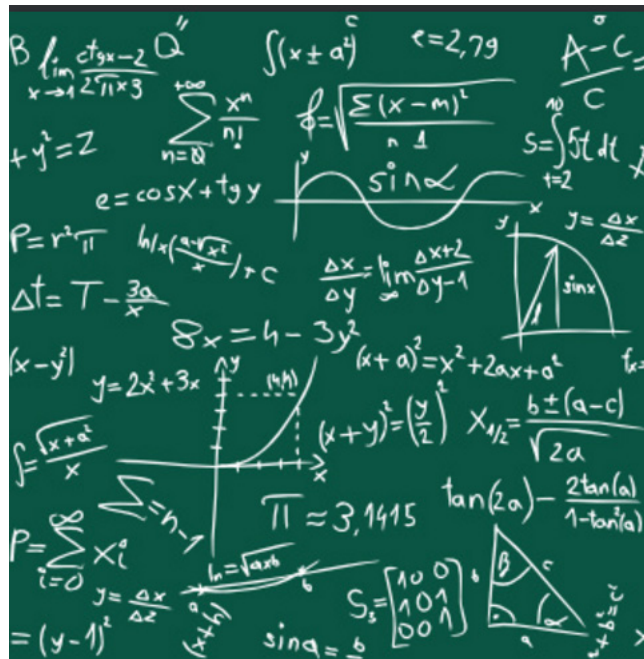
a counterexample is the function $f(x) = |x|$, which is continuous at $x_0 = 0$ but not differentiable at $x_0 = 0$. Consequently, **if a function is continuous at x_0 , it is not necessarily differentiable at x_0 .**

- The fact that f is differentiable at x_0 is a sufficient but not necessary condition for f to be continuous at x_0 . Moreover, the fact that f is continuous at x_0 is a necessary but not sufficient condition for f to be differentiable at x_0 .
- From the contrapositive of (1) we have:

$$f \text{ is not continuous at } x_0 \Rightarrow f \text{ is not differentiable at } x_0$$

that is, if a function is discontinuous at x_0 it cannot be differentiable at x_0 .

Derivatives of elementary functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani e Sandro Salsa, Zanichelli.

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1	Derivatives of elementary functions	2
1.1	Derivatives of power functions	2
1.2	Derivatives of elementary trigonometric functions	4
1.3	Derivatives of the exponential and logarithmic functions with base e	5

1 Derivatives of elementary functions

- In what follows we derive the derivative functions of some of the main elementary functions.

Remark 1

Given the function $f(x) = c$ with $c \in \mathbb{R}$ constant, the derivative function is $f'(x) = 0$.

Proof. We have:

$$\frac{f(x+h) - f(x)}{h} = \frac{c - c}{h} = 0$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} 0 = 0$$

□

1.1 Derivatives of power functions

Remark 2

Given the function $f(x) = x^n$ with $n \in \mathbb{N}, n \geq 1$, the derivative function is $f'(x) = n x^{n-1}$.

Proof. From the binomial theorem (Newton's binomial formula) we have:

$$\begin{aligned} \frac{f(x+h) - f(x)}{h} &= \frac{(x+h)^n - x^n}{h} = \frac{\sum_{k=0}^n \binom{n}{k} x^{n-k} h^k - x^n}{h} \\ &= \frac{x^n + n x^{n-1} h + \sum_{k=2}^n \binom{n}{k} x^{n-k} h^k - x^n}{h} \\ &= n x^{n-1} + \frac{\sum_{k=2}^n \binom{n}{k} x^{n-k} h^k}{h} \end{aligned}$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} n x^{n-1} + \underbrace{\frac{\sum_{k=2}^n \binom{n}{k} x^{n-k} h^k}{h}}_{\rightarrow 0} = n x^{n-1}$$

□

Proof. With $n = 2$, we have:

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = 2x + h$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} 2x + h = 2x$$

□

Remark 3

Given the function $f(x) = x^\alpha$ with $\alpha \in \mathbb{R}$, the derivative function is $f'(x) = \alpha x^{\alpha-1}$ for $x > 0$.

Proof. Let $x > 0$. We have:

$$\begin{aligned} \frac{f(x+h) - f(x)}{h} &= \frac{(x+h)^\alpha - x^\alpha}{h} = \frac{\left(x\left(1 + \frac{h}{x}\right)\right)^\alpha - x^\alpha}{h} \\ &= x^\alpha \cdot \frac{\left(1 + \frac{h}{x}\right)^\alpha - 1}{h} \sim x^\alpha \cdot \frac{\alpha \frac{h}{x}}{h} = \alpha x^{\alpha-1} \text{ for } h \rightarrow 0 \end{aligned}$$

where we used the fundamental limit

$$(1 + \varepsilon(h))^\alpha - 1 \sim \alpha \varepsilon(h) \text{ for } \varepsilon(h) \rightarrow 0$$

where

$$\varepsilon(h) = \frac{h}{x} \rightarrow 0 \text{ for } h \rightarrow 0.$$

Hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \alpha x^{\alpha-1} = \alpha x^{\alpha-1}$$

□

Example 1: Derivative function

$$f(x) = x^{10} \qquad f'(x) = 10x^9 \quad (\text{for } x \in \mathbb{R})$$

$$f(x) = \frac{1}{x} = x^{-1} \qquad f'(x) = -\frac{1}{x^2} \quad (\text{for } x > 0)$$

$$f(x) = \sqrt{x} = x^{\frac{1}{2}} \qquad f'(x) = \frac{1}{2\sqrt{x}} \quad (\text{for } x > 0)$$

1.2 Derivatives of elementary trigonometric functions

Remark 4

Given the function $f(x) = \sin x$, the derivative function is $f'(x) = \cos x$.

Proof. Using the addition formulas we have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{\sin(x+h) - \sin x}{h} = \frac{\sin x \cos h + \sin h \cos x - \sin x}{h} = \\ &= \sin x \frac{\cos h - 1}{h} + \frac{\sin h}{h} \cos x\end{aligned}$$

Using the fundamental limit

$$\frac{1 - \cos h}{h^2} \rightarrow \frac{1}{2} \text{ for } h \rightarrow 0$$

we have

$$\frac{\cos h - 1}{h} = h \cdot \left(\underbrace{-\frac{1 - \cos h}{h^2}}_{\rightarrow -\frac{1}{2} \text{ for } h \rightarrow 0} \right) \sim -\frac{1}{2} h \text{ for } h \rightarrow 0.$$

Using the fundamental limit

$$\frac{\sin h}{h} \rightarrow 1 \text{ for } h \rightarrow 0$$

we have

$$\frac{\sin h}{h} \cos x \sim \cos x \text{ for } h \rightarrow 0.$$

Then

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \sin x \frac{\cos h - 1}{h} + \frac{\sin h}{h} \cos x \\ &= \lim_{h \rightarrow 0} \sin x \left(-\frac{1}{2} h \right) + \cos x = \cos x\end{aligned}$$

□

Remark 5

Given the function $f(x) = \cos x$, the derivative function is $f'(x) = -\sin x$.

Proof. Using the addition formulas we have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{\cos(x+h) - \cos x}{h} = \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} = \\ &= \cos x \frac{\cos h - 1}{h} - \frac{\sin h}{h} \sin x\end{aligned}$$

and, using the arguments of the previous proof, we have:

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \cos x \frac{\cos h - 1}{h} - \frac{\sin h}{h} \sin x \\ &= \lim_{h \rightarrow 0} \cos x \left(-\frac{1}{2} h \right) - \sin x = -\sin x\end{aligned}$$

□

1.3 Derivatives of the exponential and logarithmic functions with base e

Remark 6

Given the function $f(x) = e^x$, the derivative function is $f'(x) = e^x$.

Proof. We have

$$\frac{f(x+h) - f(x)}{h} = \frac{e^{x+h} - e^x}{h} = e^x \cdot \frac{e^h - 1}{h} \sim e^x \text{ for } h \rightarrow 0$$

using the fundamental limit

$$\frac{e^h - 1}{h} \rightarrow 1 \text{ for } h \rightarrow 0.$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} e^x = e^x$$

□

Remark 7

Given the function $f(x) = \log x$, the derivative function is $f'(x) = \frac{1}{x}$.

Proof. We have:

$$\frac{f(x+h) - f(x)}{h} = \frac{\log(x+h) - \log x}{h} = \frac{\log\left(1 + \frac{h}{x}\right)}{h} \sim \frac{h}{x} \cdot \frac{1}{h} = \frac{1}{x} \text{ for } h \rightarrow 0$$

where we used the fundamental limit

$$\log(1 + \varepsilon(h)) \sim \varepsilon(h) \text{ for } \varepsilon(h) \rightarrow 0,$$

and

$$\varepsilon(h) = \frac{h}{x} \rightarrow 0 \text{ for } h \rightarrow 0.$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{x} = \frac{1}{x}$$

□

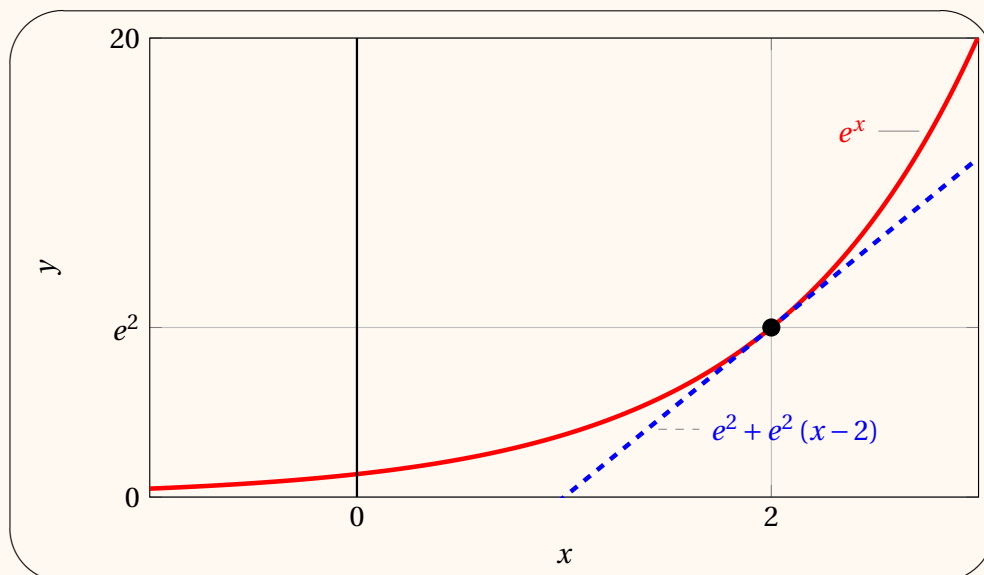
Example 2: Tangent line

We compute the equation of the tangent line to the graph of the function

$$f(x) = e^x \text{ at the point with abscissa } x = 2$$

We have $f(2) = e^2$, $f'(x) = e^x$, $f'(2) = e^2$, hence the tangent line at the point $(2, e^2)$ is:

$$y = f(2) + f'(2)(x-2) = e^2 + e^2(x-2)$$



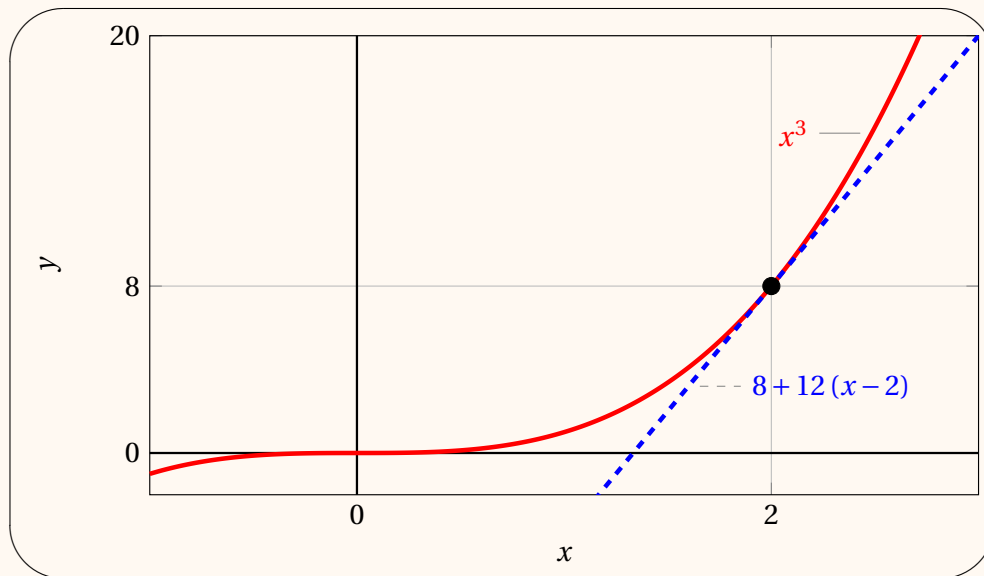
Example 3: Tangent line

We compute the equation of the tangent line to the graph of the function

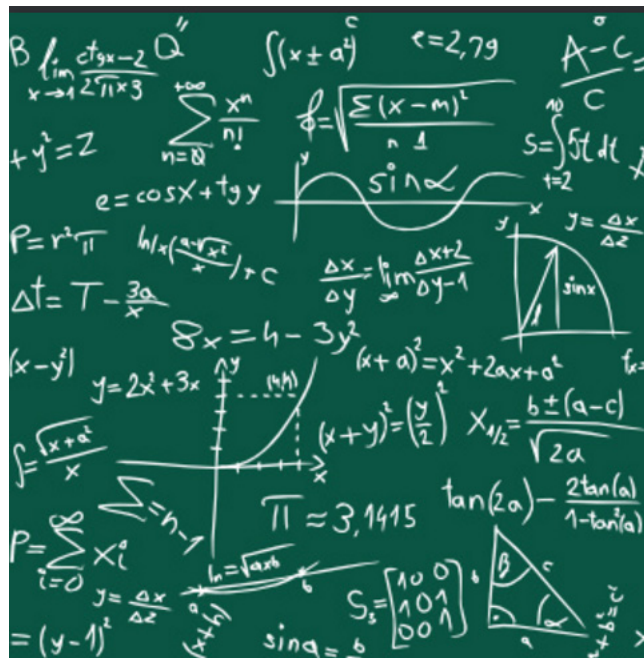
$$f(x) = x^3 \text{ at the point with abscissa } x = 2$$

We have $f(2) = 8$, $f'(x) = 3x^2$, $f'(2) = 12$, hence the tangent line at the point $(2, 8)$ is:

$$y = f(2) + f'(2)(x - 2) = 8 + 12(x - 2)$$



Corner points, cusps, points with vertical/horizontal tangent



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani e Sandro Salsa, Zanichelli.

Contents

1	Right derivative and left derivative	2
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1 Right derivative and left derivative

Definition 1: of right (left) derivative

Let $f : (a, b) \rightarrow \mathbb{R}$; the function f is said to be differentiable at $x_0 \in (a, b)$ from the right (from the left) if the following limit exists and is finite

$$\lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h} \quad \left(\lim_{h \rightarrow 0^-} \frac{f(x_0 + h) - f(x_0)}{h} \right)$$

then f is differentiable from the right (from the left) and the limit is called the **right derivative (left derivative)**.

- The right derivative is denoted by the symbol $f'_+(x_0)$, while the left derivative is denoted by the symbol $f'_-(x_0)$

2 Corner points

Definition 2: of corner point

If f is continuous and differentiable from the right and from the left (but not differentiable) at x_0 , we say that f has a **corner point** at $x = x_0$.

Example 1: Corner points

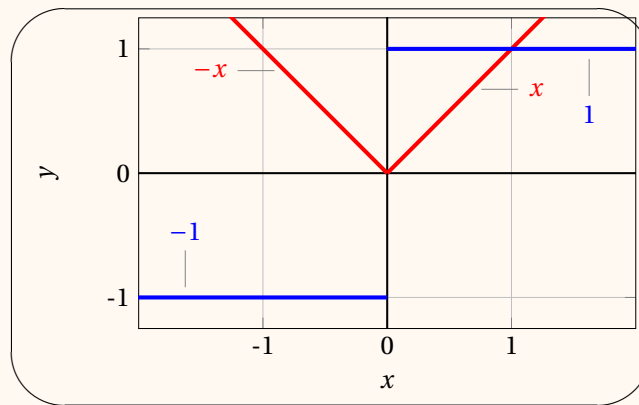
Consider the function $f(x) = |x|$ and the point $x_0 = 0$. We have:

$$f'_+(0) = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1 \quad \text{and} \quad f'_-(0) = \lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$$

Since the limit of the difference quotient does not exist, f is not differentiable at $x = 0$. The function is continuous at $x = 0$ (the origin) since

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} -x = 0 \quad \text{hence} \quad \lim_{x \rightarrow 0} f(x) = f(0) = 0$$

Since the right and left limits of the difference quotient at $x = 0$ exist and are finite, the graph therefore has a corner point at $x = 0$.



The formula that concisely expresses the derivative of the absolute value function (away from the origin) is the following:

$$f(x) = |x|, \quad f'(x) = \operatorname{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

3 Points with vertical/horizontal tangent

- If f is continuous at a point x_0 and

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = \pm\infty$$

then f is not differentiable at x_0 but, geometrically, the graph of f has a well-defined tangent line parallel to the y -axis.

In this case we will allow the notation

$$f'(x_0) = +\infty, \quad f'(x_0) = -\infty$$

and we will speak of a **point with vertical tangent**.

Definition 3: of point with vertical tangent

If

$$f'(x_0) = +\infty \quad \text{or} \quad f'(x_0) = -\infty$$

*we say that f has a **point with vertical tangent** at $x = x_0$.*

- Similarly, if f is continuous at a point x_0 and

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = 0$$

the graph of f has a well-defined tangent line parallel to the x -axis. In this case we will speak of a **point with horizontal tangent**.

Definition 4: of point with horizontal tangent

If

$$f'(x_0) = 0$$

*we say that f has a **point with horizontal tangent** at $x = x_0$.*

Example 2: Point with vertical tangent

Let

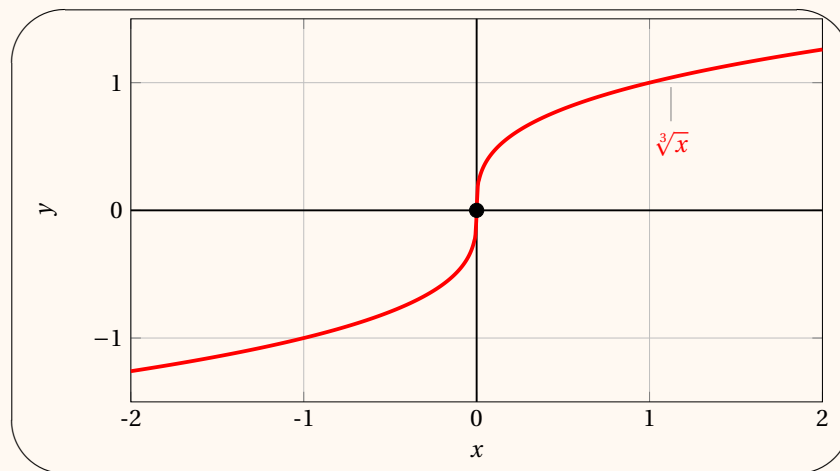
$$f(x) = \sqrt[3]{x}$$

then for $x_0 = 0$

$$\frac{f(h) - f(0)}{h} = \frac{\sqrt[3]{h}}{h} = \frac{1}{h^{2/3}}$$

and the limits are:

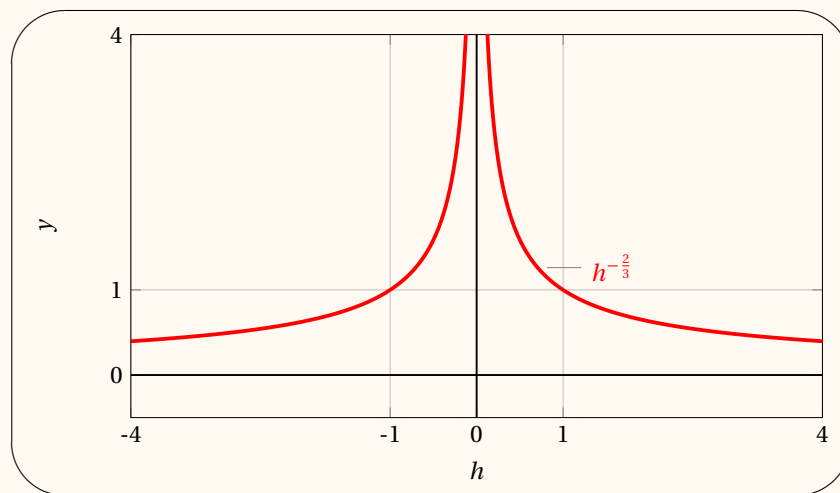
$$\lim_{h \rightarrow 0^-} \frac{1}{h^{2/3}} = +\infty \text{ and } \lim_{h \rightarrow 0^+} \frac{1}{h^{2/3}} = +\infty \text{ hence } \lim_{h \rightarrow 0} \frac{1}{h^{2/3}} = +\infty \text{ and } f'(0) = +\infty.$$



The function has a point with vertical tangent at $x_0 = 0$. The function:

$$f(h) = \frac{1}{h^{2/3}} = h^{-\frac{2}{3}}$$

is a power with negative rational exponent $\frac{m}{n}$ with n odd (hence defined on $\mathbb{R} \setminus \{0\}$) and m even (hence an even function). Its graph is:



Example 3: Point with vertical tangent

Let

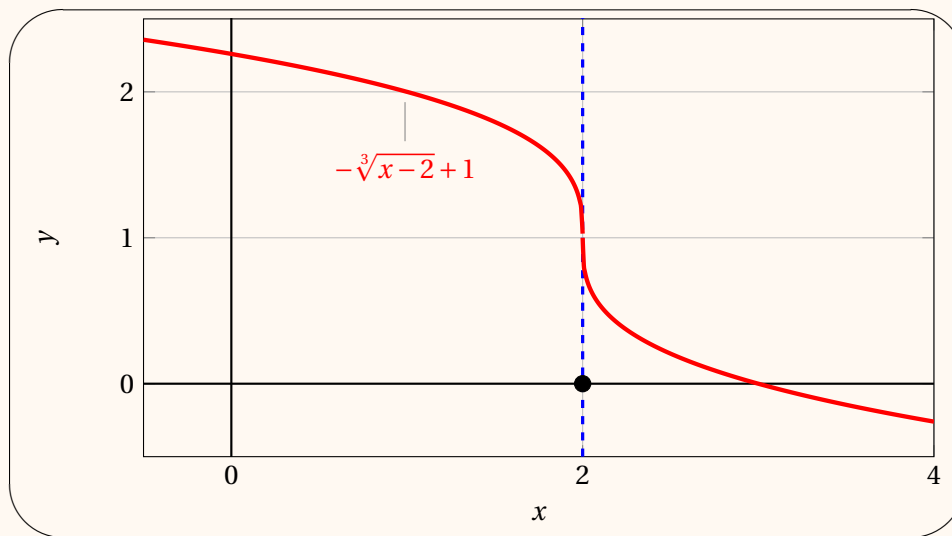
$$f(x) = -\sqrt[3]{x-2} + 1$$

then for $x_0 = 2$

$$\frac{f(2+h) - f(2)}{h} = \frac{-\sqrt[3]{(2+h)-2} + 1 - 1}{h} = -\frac{\sqrt[3]{h}}{h} = -\frac{1}{h^{2/3}}$$

and the limits are

$$\lim_{h \rightarrow 0^-} -\frac{1}{h^{2/3}} = -\infty \quad \text{and} \quad \lim_{h \rightarrow 0^+} -\frac{1}{h^{2/3}} = -\infty \quad \text{hence} \quad \lim_{h \rightarrow 0} -\frac{1}{h^{2/3}} = -\infty \quad \text{and} \quad f'(2) = -\infty.$$



The function has a point with vertical tangent at $x_0 = 2$.

4 Cusps

Definition 5: of cusp

Let f be a function continuous at x_0 ; if

$$f'_+(x_0) = \pm\infty \quad \text{and} \quad f'_-(x_0) = \mp\infty$$

we say that f has a cusp at x_0 .

Example 4: Cusp

Let

$$f(x) = \sqrt[3]{|x|}$$

then for $x_0 = 0$

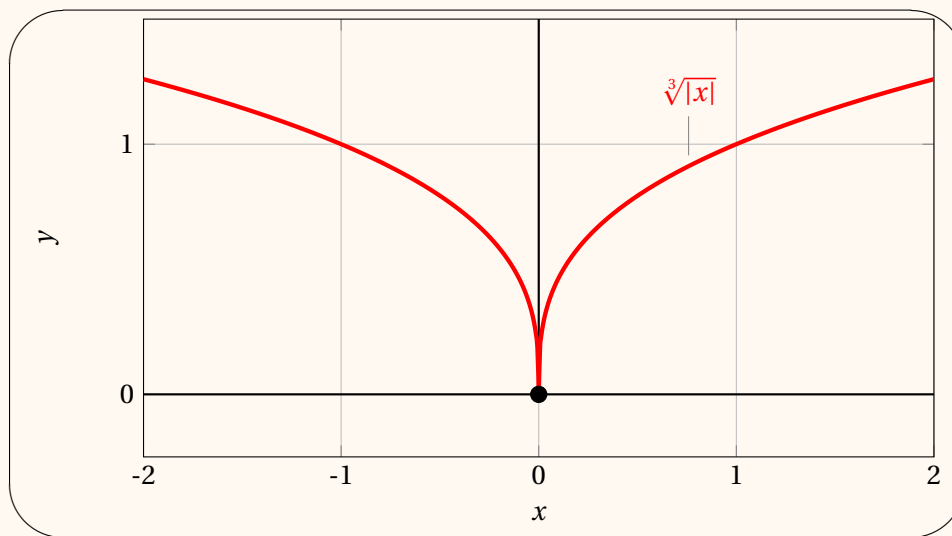
$$\frac{f(h) - f(0)}{h} = \frac{\sqrt[3]{|h|}}{h}$$

and the limits are:

$$\lim_{h \rightarrow 0^-} \frac{\sqrt[3]{-h}}{h} = \lim_{h \rightarrow 0^-} -\frac{1}{h^{2/3}} = -\infty \quad \text{and} \quad \lim_{h \rightarrow 0^+} \frac{\sqrt[3]{h}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{h^{2/3}} = +\infty,$$

hence:

$$f'_-(0) = -\infty \quad \text{and} \quad f'_+(0) = +\infty.$$



The function has a cusp at $x_0 = 0$.

- In the mixed case where one of the two derivatives is finite and the other is infinite (with f continuous) we still speak of a corner point.
- Finally, if the function is defined only for $x \geq x_0$ and at that point has an infinite (right) derivative, we will simply say that at that point it has a vertical tangent, without speaking of either a cusp or a point with vertical tangent.

Example 5: Point with vertical tangent

Let

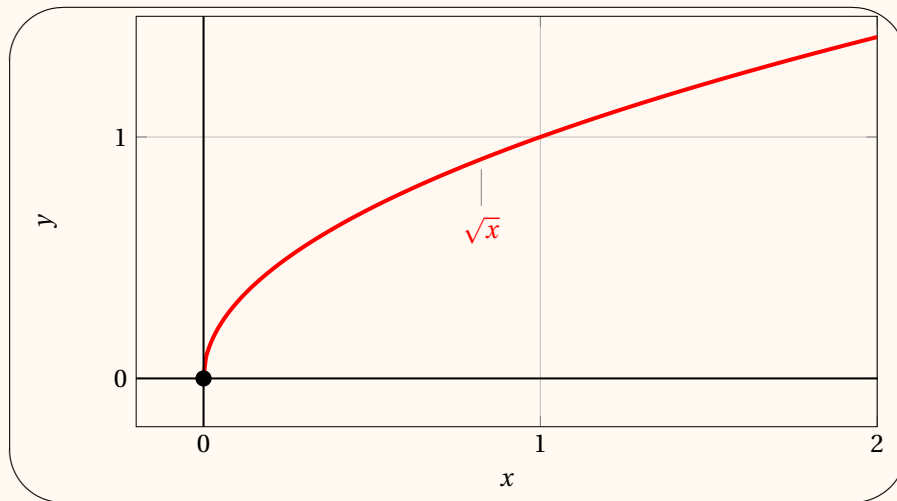
$$f(x) = \sqrt{x}$$

then for $x_0 = 0$

$$\frac{f(h) - f(0)}{h} = \frac{\sqrt{h}}{h} = \frac{1}{h^{1/2}}$$

we only have the right limit

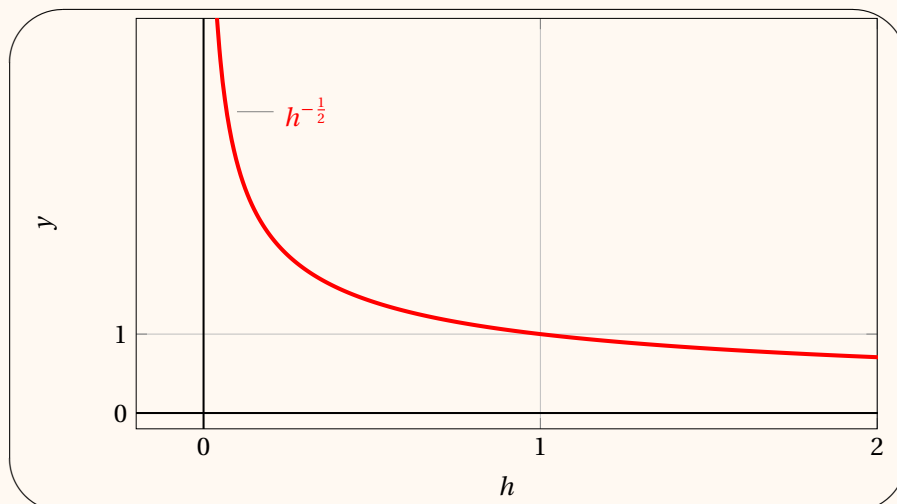
$$\lim_{h \rightarrow 0^+} \frac{1}{h^{1/2}} = +\infty \quad \text{and} \quad f'_+(0) = +\infty.$$



The function has a point with vertical tangent at $x_0 = 0$. The function:

$$f(h) = \frac{1}{h^{1/2}} = h^{-\frac{1}{2}}$$

is a power with negative rational exponent $\frac{m}{n}$ with n even (hence defined only on $\mathbb{R}_+$). Its graph is:



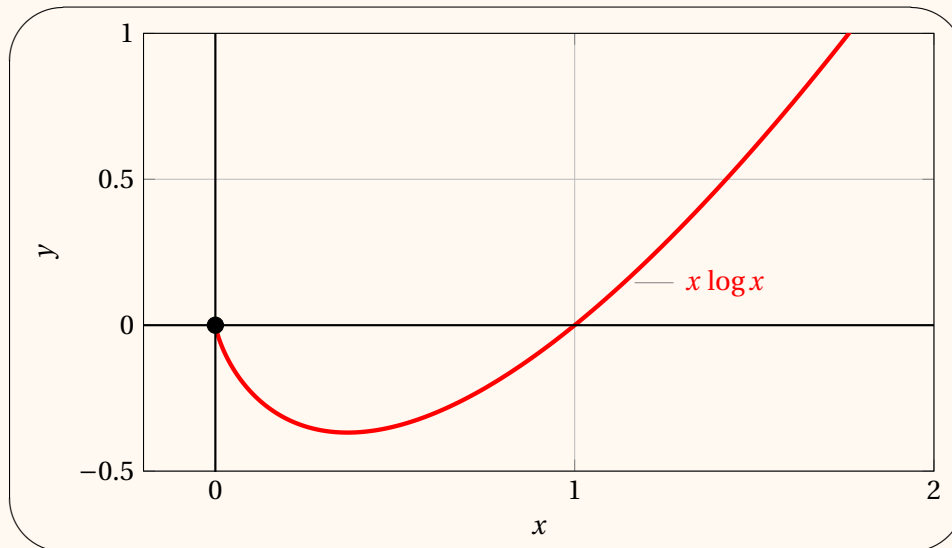
Example 6: Continuous extension from the right and behavior at the origin

Let

$$f(x) = x \log x, \text{ for } x > 0 \text{ we have } \lim_{x \rightarrow 0^+} x \log x = 0$$

hence the function can be extended by continuity from the right at $x = 0$, by setting:

$$f(x) = \begin{cases} x \log x & \text{if } x > 0 \\ 0 & \text{if } x = 0 \end{cases}$$



We compute the right derivative at $x_0 = 0$ of $f(x) = x \log x$ extended by continuity from the right at $x = 0$. We have, for $x_0 = 0$:

$$\frac{f(x_0 + h) - \overbrace{f(x_0)}^{=0}}{h} = \frac{f(x_0 + h)}{h} = \frac{f(h)}{h}$$

and consequently

$$\lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0^+} \frac{h \log h}{h} = \lim_{h \rightarrow 0^+} \log h = -\infty \quad \text{hence} \quad f'_+(x_0) = -\infty$$

The function therefore has a point with vertical tangent at $x_0 = 0$.

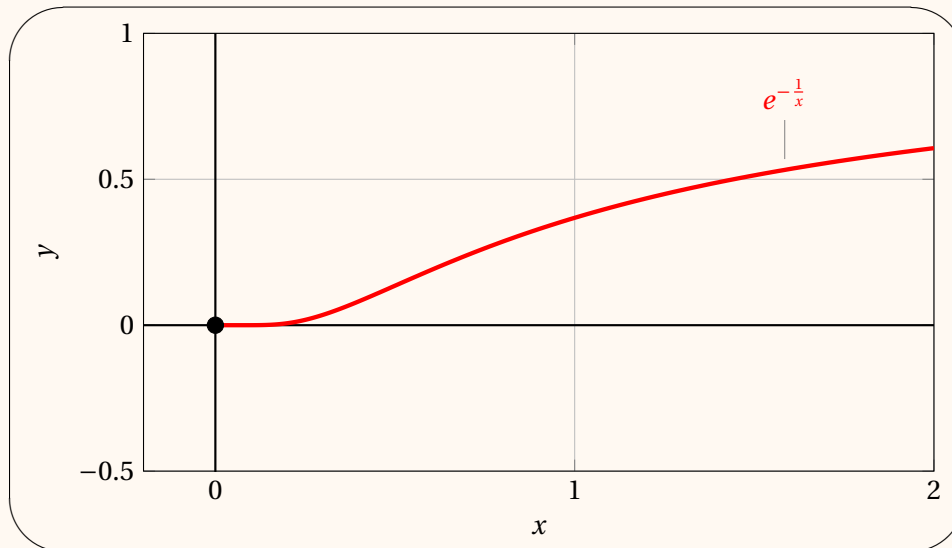
Example 7: Continuous extension from the right and behavior at the origin

Let

$$f(x) = e^{-\frac{1}{x}}, \text{ for } x > 0 \text{ we have } \lim_{x \rightarrow 0^+} e^{-\frac{1}{x}} = 0$$

hence the function can be extended by continuity from the right at $x = 0$, by setting:

$$f(x) = \begin{cases} e^{-\frac{1}{x}} & \text{if } x > 0 \\ 0 & \text{if } x = 0 \end{cases}$$



We compute the right derivative at $x_0 = 0$ of $f(x) = e^{-\frac{1}{x}}$ extended by continuity from the right at $x = 0$.

We have:

$$\lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0^+} \frac{e^{-\frac{1}{h}}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{e^{\frac{1}{h}} h}$$

Changing the variable:

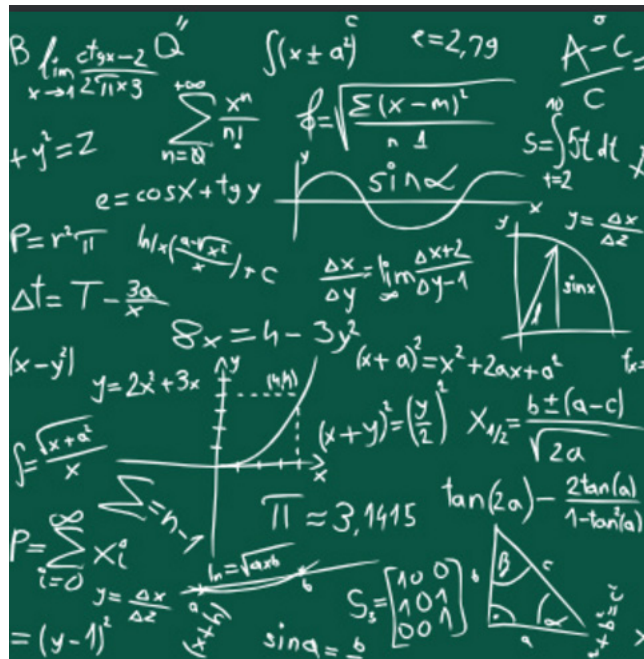
$$y = \frac{1}{h}, \text{ if } h \rightarrow 0^+ \text{ then } y \rightarrow +\infty$$

We obtain:

$$\lim_{h \rightarrow 0^+} \frac{1}{e^{\frac{1}{h}} h} = \lim_{y \rightarrow +\infty} \frac{y}{e^y} = 0 \text{ hence } f'_+(x_0) = 0$$

and the function has a point with horizontal tangent (from the right) at $x_0 = 0$.

Rules for computing derivatives



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani e Sandro Salsa, Zanichelli.

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1.1	Algebra of derivatives	2
1.2	Derivative of a composite function	5
1.3	Derivative of the inverse function	14
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1 Rules for computing derivatives

- We now look at the relation between the derivative operation and the main operations on functions that we already know; in particular we will show the relation between:
 1. differentiation and algebraic operations
 2. differentiation and composition
 3. differentiation and inversion

1.1 Algebra of derivatives

Theorem 1: algebra of derivatives

Let $f, g: (a, b) \rightarrow \mathbb{R}$ be two functions differentiable in (a, b) .

Then $f \pm g, f \cdot g, f/g$ ($g \neq 0$) are differentiable in (a, b) and the following formulas hold:

$$(f \pm g)' = f' \pm g' \quad (1)$$

$$(f \cdot g)' = f' \cdot g + f \cdot g' \quad (2)$$

$$\left(\frac{f}{g}\right)' = \frac{f' \cdot g - f \cdot g'}{g^2} \quad (g \neq 0) \quad (3)$$

- From rule (2) we deduce

$$(k \cdot g)' = k \cdot g' \quad \text{with } k \in \mathbb{R} \text{ constant} \quad (4)$$

since the derivative of a constant is equal to zero.

- From rule (3) we deduce

$$\left(\frac{1}{g}\right)' = -\frac{g'}{g^2} \quad (g \neq 0) \quad (5)$$

- Rule (2) is called the **Leibniz rule** (product rule) and it extends to the product of n functions:

$$(f_1 f_2 \cdots f_n)' = f_1' f_2 \cdots f_n + f_1 f_2' \cdots f_n + \cdots + f_1 f_2 \cdots f_n' \quad (6)$$

- The theorem actually has a pointwise character: that is, if f and g are differentiable at a point $x_0 \in (a, b)$, then at that point $f \pm g$, $f \cdot g$, f/g are also differentiable, and the formulas above hold (and they extend to intervals).

Proof. We prove rule (2). Fixing $x \in (a, b)$, we have

$$\begin{aligned} & f(x+h)g(x+h) - f(x)g(x) = \\ &= f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x), \end{aligned}$$

and hence the difference quotient can be written as:

$$\begin{aligned} & \frac{f(x+h)g(x+h) - f(x)g(x)}{h} = \\ &= f(x+h) \underbrace{\frac{g(x+h) - g(x)}{h}}_{=g'(x) \text{ for } h \rightarrow 0} + g(x) \underbrace{\frac{f(x+h) - f(x)}{h}}_{=f'(x) \text{ for } h \rightarrow 0} \end{aligned}$$

We have

$$f(x+h) \rightarrow f(x) \text{ for } h \rightarrow 0,$$

since f is continuous, being differentiable.

Hence

$$\frac{f(x+h)g(x+h) - f(x)g(x)}{h} \rightarrow f(x)g'(x) + f'(x)g(x) \text{ for } h \rightarrow 0.$$

□

Proof. We prove rule (5). Fixing $x \in (a, b)$, the difference quotient can be written as:

$$\begin{aligned} & \frac{1}{h} \left[\frac{1}{g(x+h)} - \frac{1}{g(x)} \right] = \frac{g(x) - g(x+h)}{h g(x) g(x+h)} = \\ &= - \frac{g(x+h) - g(x)}{h} \cdot \frac{1}{g(x) g(x+h)} \end{aligned}$$

We have

$$g(x+h) \rightarrow g(x) \text{ for } h \rightarrow 0,$$

since g is continuous, being differentiable.

Hence

$$\frac{1}{h} \left[\frac{1}{g(x+h)} - \frac{1}{g(x)} \right] \rightarrow - \frac{g'(x)}{g^2(x)} \text{ for } h \rightarrow 0.$$

□

Proof. Note that rule (3) follows from rules (2) and (5); indeed:

$$\begin{aligned}\left(\frac{f(x)}{g(x)}\right)' &= \left(f(x) \cdot \frac{1}{g(x)}\right)' = \underbrace{f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \left(\frac{1}{g(x)}\right)'}_{\text{rule (2)}} = \\ &= f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \underbrace{\left(-\frac{g'(x)}{g^2(x)}\right)}_{\text{rule (5)}} = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{g^2(x)}.\end{aligned}$$

□

Remark 1

Given the function $f(x) = \tan x$, the derivative function is $f'(x) = \frac{1}{\cos^2 x} = 1 + \tan^2 x$.

Proof. Using formula (3) we have

$$f'(x) = \left(\frac{\sin x}{\cos x}\right)' = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = 1 + \tan^2 x$$

□

Remark 2

Given the function $f(x) = \cot x$, the derivative function is $f'(x) = -\frac{1}{\sin^2 x} = -(1 + \cot^2 x)$.

Proof. Using formula (3) we have

$$\begin{aligned}f'(x) &= \left(\frac{\cos x}{\sin x}\right)' = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = -\frac{\sin^2 x + \cos^2 x}{\sin^2 x} \\ &= -\frac{1}{(\sin^2 x)} = -(1 + \cot^2 x)\end{aligned}$$

□

1.2 Derivative of a composite function

Theorem 2: chain rule

Let $g \circ f$ be the composite function of two functions f and g . If f is differentiable at a point x and g is differentiable at $y = f(x)$, then $g \circ f$ is differentiable at x and the following formula holds:

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) \quad (7)$$

Proof. We have

$$(g \circ f)(x+h) - (g \circ f)(x) = g(f(x+h)) - g(f(x))$$

If we set

$$k = f(x+h) - f(x) \quad \text{and} \quad y = f(x) \quad \text{then} \quad f(x+h) = y + k$$

By the continuity of f , $h \rightarrow 0$ implies $k \rightarrow 0$. With the new notation we have:

$$g(f(x+h)) - g(f(x)) = g(y+k) - g(y)$$

We now observe that the definition of derivative:

$$g'(y) = \lim_{k \rightarrow 0} \frac{g(y+k) - g(y)}{k}$$

can be rewritten, for $k \neq 0$, as

$$\frac{g(y+k) - g(y)}{k} = g'(y) + \varepsilon(k)$$

where $\varepsilon(k)$ denotes a quantity that tends to zero as $k \rightarrow 0$.

Multiplying both sides of the previous equation by k we find

$$g(y+k) - g(y) = k \cdot g'(y) + k \cdot \varepsilon(k)$$

a relation that also holds for $k = 0$. Therefore:

$$g(f(x+h)) - g(f(x)) = k \cdot g'(y) + k \cdot \varepsilon(k)$$

Dividing by h , and observing that

$$\frac{k}{h} \rightarrow f'(x) \quad \text{for} \quad h \rightarrow 0$$

which implies, as seen above, also $k \rightarrow 0$, we obtain the claim. □

- As the name “chain rule” suggests, (7) can be generalized to the composition of any number of functions, composed one with another. For example, for three functions we have:

$$\left(f(g(h(x))) \right)' = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$$

Example 1: Derivative of a composite function

We compute the derivative function of the composite function:

$$h(x) = \underbrace{\sin^3 x}_{=(g \circ f)(x)}$$

The two functions are:

$$f(x) = \sin x \quad \text{and} \quad g(y) = y^3 \quad \text{with} \quad y = \sin x.$$

Both functions are differentiable on all of $\mathbb{R}$. We have

$$g'(y) = 3y^2 \quad \text{and} \quad f'(x) = \cos x$$

hence, using the chain rule (7) and substituting back, we obtain:

$$h'(x) = 3 \sin^2 x \cdot \cos x.$$

Consider the composite function $g(f(x))$, equivalently written $(g \circ f)(x)$. Setting $y = f(x)$ and $w = g(y)$, and using the (Leibniz) notation

$$\frac{df}{dx} \quad \text{and} \quad \frac{dg}{dy} \quad \text{for the derivatives of } f \text{ and } g \text{ with respect to } x$$

the (chain) rule (7) can be rewritten as:

$$\frac{dw}{dx} = \frac{dw}{dy} \cdot \frac{dy}{dx} \tag{8}$$

Rule (8) expresses the fact that the rate of change of w with respect to x is the product of the “intermediate” rates of change, of w with respect to y and of y with respect to x .

Example 2: Derivative of a composite function

Consider the composite function of the previous exercise. Setting

$$y = \sin x \quad (y = f(x)) \quad \text{and} \quad w = y^3 \quad (w = g(y))$$

and using Leibniz's notation, rule (8) would be written:

$$\frac{dw}{dx} = \frac{dw}{dy} \cdot \frac{dy}{dx} = 3y^2 \cdot \cos x.$$

And hence, substituting $y = \sin x$, we obtain:

$$\frac{dw}{dx} = 3 \sin^2 x \cdot \cos x.$$

Example 3: Derivative of a composite function

We compute the derivative function of the composite function (multiplied by a constant):

$$h(x) = A \cdot \underbrace{\sin(\omega x + \varphi)}_{=(g \circ f)(x)} \quad \text{with} \quad A, \omega, \varphi \in \mathbb{R}.$$

The two functions are:

$$f(x) = \omega x + \varphi \quad \text{and} \quad g(y) = \sin y \quad \text{with} \quad y = \omega x + \varphi.$$

Both functions are differentiable on all of $\mathbb{R}$. We have:

$$f'(x) = \omega \quad \text{and} \quad g'(y) = \cos y$$

hence, using rule (7), rule (4) and substituting back, we obtain:

$$h'(x) = A \cdot (\cos(\omega x + \varphi) \cdot \omega)$$

Example 4: Derivative of a product of composite functions

We compute the derivative function of the product of two composite functions times a constant:

$$h(x) = A \cdot \underbrace{e^{-\alpha x}}_{=(r \circ s)(x)} \cdot \underbrace{\cos(\omega x + \varphi)}_{=(g \circ f)(x)} \quad \text{with } A, \omega, \varphi \in \mathbb{R}, \alpha \in \mathbb{R}_+.$$

The functions are:

$$s(x) = -\alpha x \quad \text{and} \quad r(z) = e^z \quad \text{with } z = -\alpha x; \quad f(x) = \omega x + \varphi \quad \text{and} \quad g(y) = \cos y \quad \text{with } y = \omega x + \varphi.$$

The functions are differentiable on all of $\mathbb{R}$. We have:

$$s'(x) = -\alpha \quad \text{and} \quad r'(z) = e^z; \quad f'(x) = \omega \quad \text{and} \quad g'(y) = -\sin y$$

consequently, using the chain rule (7) and substituting back, we obtain:

$$(r \circ s)'(x) = -\alpha e^{-\alpha x}; \quad (g \circ f)'(x) = -\sin(\omega x + \varphi) \cdot \omega$$

Hence, using rule (2) and rule (4):

$$h'(x) = A \cdot \left(\underbrace{-\alpha e^{-\alpha x}}_{(r \circ s)'(x)} \cdot \underbrace{\cos(\omega x + \varphi)}_{(g \circ f)(x)} + \underbrace{e^{-\alpha x}}_{(r \circ s)(x)} \cdot \underbrace{(-\sin(\omega x + \varphi) \cdot \omega)}_{(g \circ f)'(x)} \right)$$

Example 5: Derivative of a composite function

Let $f(x) > 0$ be differentiable; we compute the derivative of the function

$$h(x) = \log(f(x))$$

using the chain rule (7) we have:

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$

Remark 3

Given the function $f(x) = a^x$, with $a > 0$, the derivative function is $f'(x) = a^x \cdot \log a$.

Proof. We have

$$f'(x) = \left(\exp(x \cdot \log a) \right)'$$

hence, using the formulas for the derivative of e^y (with $y = x \log a$) and for the derivative of a composite function, we have

$$f'(x) = \exp(x \cdot \log a) \cdot \log a = a^x \cdot \log a$$

□

Remark 4

Given the function $f(x) = \log_a x$, with $a > 0$, $a \neq 1$, the derivative function is $f'(x) = \frac{1}{x \log a}$.

Proof. We have

$$f'(x) = \left(\frac{\log x}{\log a} \right)'$$

hence, using the formulas for the derivative of $\log x$ and for the derivative of $k f(x)$, we have:

$$f'(x) = \frac{1}{\log a} \cdot \frac{1}{x} = \frac{1}{x \log a}$$

□

Remark 5

Given the function $f(x) = \sinh x$, the derivative function is $f'(x) = \cosh x$.

Proof. We have:

$$f'(x) = \left(\frac{e^x - e^{-x}}{2} \right)' = \frac{1}{2} \left(e^x - (-e^{-x}) \right) = \frac{e^x + e^{-x}}{2} = \cosh x$$

□

Remark 6

Given the function $f(x) = \cosh x$, the derivative function is $f'(x) = \sinh x$.

Proof. We have:

$$f'(x) = \left(\frac{e^x + e^{-x}}{2} \right)' = \frac{1}{2} (e^x + (-e^{-x})) = \frac{e^x - e^{-x}}{2} = \sinh x$$

□

The derivatives of functions of the type:

$$h(x) = f(x)^{g(x)}$$

are based on rewriting the function (with $f(x) > 0$) in the following form:

$$f(x)^{g(x)} = \exp \left(g(x) \cdot \log(f(x)) \right)$$

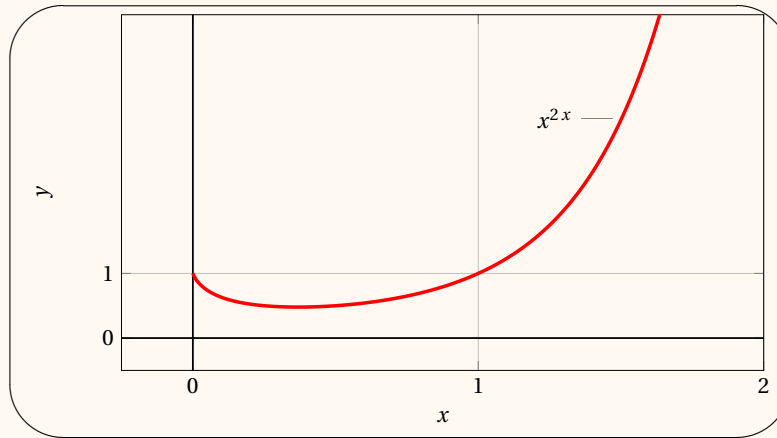
We have:

$$\begin{aligned} \left(f(x)^{g(x)} \right)' &= \left(\exp \left(g(x) \cdot \log(f(x)) \right) \right)' \\ &= \exp \left(g(x) \cdot \log(f(x)) \right) \cdot \left(g(x) \cdot \log(f(x)) \right)' \\ &= f(x)^{g(x)} \cdot \left(g'(x) \cdot \log(f(x)) + g(x) \cdot \frac{f'(x)}{f(x)} \right) \end{aligned}$$

Example 6: Derivative of a function raised to another function

We compute the derivative of the function

$$h(x) = x^{2x} = \exp(2x \cdot \log x) \text{ with } x > 0$$



The two functions are:

$$f(x) = x \quad \text{and} \quad g(x) = 2x$$

Hence:

$$h'(x) = \left(\exp(2x \cdot \log x) \right)' = \exp(2x \cdot \log x) \cdot (2x \cdot \log x)' = x^{2x} \cdot \left(2 \log x + 2x \frac{1}{x} \right) = x^{2x} (2 \log x + 2)$$

Consider the absolute value of a function: $|f(x)|$. At the points where $f(x) \neq 0$, the differentiation of a composite function gives:

$$(|f(x)|)' = \operatorname{sgn}(f(x)) \cdot f'(x) = \begin{cases} f'(x) & \text{for } f(x) > 0 \\ -f'(x) & \text{for } f(x) < 0 \end{cases}$$

In general, we expect the function $|f(x)|$ to have corner points at the points where $f(x)$ vanishes.

Example 7: Derivative of a function with absolute value

We compute the derivative function of the function:

$$f(x) = |x^2 - 4x + 3|, \text{ with } x \neq 1 \text{ and } x \neq 3, \text{ we have } f'(x) = \operatorname{sgn}(x^2 - 4x + 3) \cdot (2x - 4)$$

equivalently:

$$f'(x) = \begin{cases} 2x - 4, & x < 1 \text{ or } x > 3 \quad \text{i.e. for } f(x) > 0 \\ -(2x - 4) & 1 < x < 3 \quad \text{i.e. for } f(x) < 0 \end{cases}$$

At $x_0 = 1$, we need to use the limit of the difference quotient:

$$\frac{f(x_0 + h) - f(x_0)}{h} = \frac{f(1 + h) - f(1)}{h} = \frac{|(1 + h)^2 - 4(1 + h) + 3|}{h} = \frac{|h^2 - 2h|}{h}$$

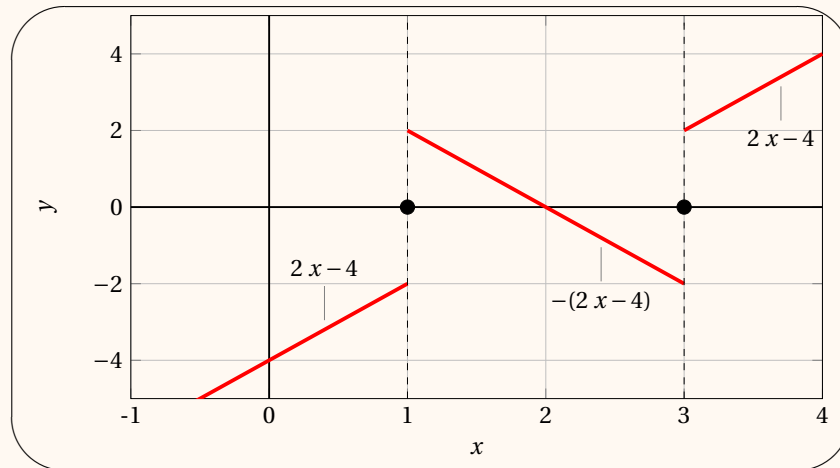
If $h \rightarrow 0^-$ we have $|h^2 - 2h| = h^2 - 2h$, hence:

$$\lim_{h \rightarrow 0^-} \frac{h^2 - 2h}{h} = -2$$

If $h \rightarrow 0^+$ we have $|h^2 - 2h| = -h^2 + 2h$, hence:

$$\lim_{h \rightarrow 0^+} \frac{-h^2 + 2h}{h} = 2$$

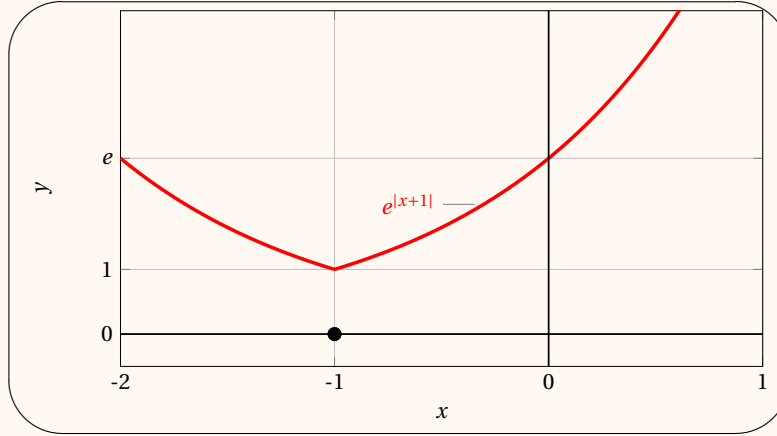
The function is not differentiable at $x_0 = 1$ and $x_0 = 3$ (by arguments analogous to those for $x_0 = 1$), where it has corner points.



Example 8: Derivative of a function with absolute value

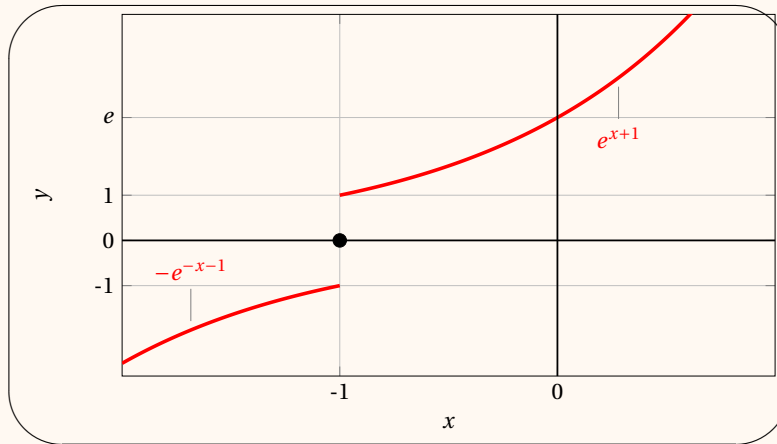
We compute the derivative of the function:

$$f(x) = e^{|x+1|}$$



For $x \neq -1$, we have:

$$f(x) = \begin{cases} e^{x+1}, & x > -1 \\ e^{-x-1}, & x < -1 \end{cases} \quad \text{hence} \quad f'(x) = \begin{cases} e^{x+1}, & x > -1 \\ -e^{-x-1}, & x < -1 \end{cases}$$



At $x_0 = -1$, we need to use the limit of the difference quotient:

$$\frac{f(x_0 + h) - f(x_0)}{h} = \frac{f(-1 + h) - f(-1)}{h} = \frac{e^{|-1+h+1|} - 1}{h} = \frac{e^{|h|} - 1}{h}$$

We have:

$$\text{if } h \rightarrow 0^+, e^{|h|} = e^h \text{ and } \lim_{h \rightarrow 0^+} \frac{e^h - 1}{h} = 1; \quad \text{if } h \rightarrow 0^-, e^{|h|} = e^{-h} \text{ and } \lim_{h \rightarrow 0^-} \frac{e^{-h} - 1}{h} = -1$$

We conclude that, since the limit of the difference quotient does not exist, the function is not differentiable at $x_0 = -1$. The function has a corner point at $x_0 = -1$.

We now consider the derivatives of some logarithmic functions. We compute:

$$\left(\log(|x|) \right)' = \frac{1}{|x|} \cdot \operatorname{sgn}(x) = \frac{1}{x}$$

Moreover, by the differentiation of composite functions we have:

$$\left(\log(|f(x)|) \right)' = \frac{1}{|f(x)|} \cdot \operatorname{sgn}(f(x)) \cdot f'(x) = \frac{f'(x)}{f(x)}$$

Sometimes it is convenient to use the properties of logarithms to transform a logarithmic function before computing its derivative function:

$$(\log(cx))' = (\log(c) + \log(x))' = \frac{1}{x} \quad \text{with } c \in \mathbb{R}$$

$$\left(\log\left(\frac{ax+b}{cx+d}\right) \right)' = (\log(ax+b) - \log(cx+d))' = \frac{a}{ax+b} - \frac{c}{cx+d} \quad \text{with } a, b, c, d \in \mathbb{R}$$

1.3 Derivative of the inverse function

Theorem 3: derivative of the inverse function

Let $f : (a, b) \rightarrow \mathbb{R}$ be a continuous and invertible function in (a, b) and $g = f^{-1}$ its inverse function, defined in $f((a, b))$.

Assume moreover that $f'(x_0) \neq 0$ exists for some $x_0 \in (a, b)$.

Then g is differentiable at $y_0 = f(x_0)$ and

$$g'(y_0) = \frac{1}{f'(x_0)} \tag{9}$$

Proof. Let:

$$f(x_0) = y_0; \quad g(y_0) = x_0$$

$$f(x_0 + h) = y_0 + k; \quad g(y_0 + k) = x_0 + h$$

Consider the difference quotient of g at y_0 :

$$\frac{g(y_0 + k) - g(y_0)}{k} = \frac{h}{f(x_0 + h) - f(x_0)}$$

If $k \neq 0$, then $f(x_0 + h) - f(x_0) \neq 0$ and hence also $h \neq 0$; therefore the last quotient can also be rewritten in the form:

$$\frac{1}{\frac{f(x_0 + h) - f(x_0)}{h}}$$

Moreover, for $k \rightarrow 0$ we have $g(y_0 + k) \rightarrow g(y_0)$ because g is continuous, being the inverse of a continuous function on an interval (theorem on the continuity of inverse functions); on the other hand $h = g(y_0 + k) - g(y_0)$, hence for $k \rightarrow 0$ also $h \rightarrow 0$, and by hypothesis

$$\frac{1}{\frac{f(x_0 + h) - f(x_0)}{h}} \rightarrow \frac{1}{f'(x_0)} \quad \text{for } h \rightarrow 0$$

hence the following limit exists

$$\lim_{k \rightarrow 0} \frac{g(y_0 + k) - g(y_0)}{k} = \frac{1}{f'(x_0)}$$

□

- We observe that, assuming the differentiability of f^{-1} , (9) follows immediately from the identity

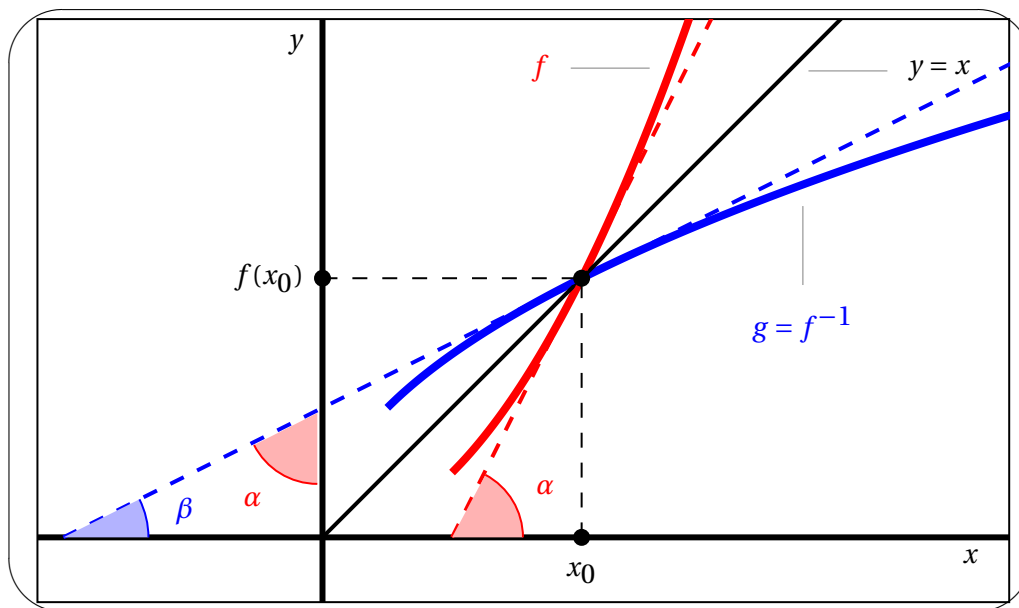
$$g(f(x)) = x$$

and from the chain rule:

$$g'(f(x)) \cdot f'(x) = 1$$

from which, if $f'(x) \neq 0$, (9) follows. The proof we have given, however, is necessary in order to deduce the differentiability of g from our hypotheses on f .

- (9) has a simple geometric meaning, recalling that the graphs of f and $g = f^{-1}$ are symmetric with respect to the bisector $y = x$. Geometrically we have:



hence the angles α and β are complementary, that is:

$$\left(\alpha + \beta = \frac{\pi}{2}\right)$$

and hence

$$f'(x) = \tan \alpha = \tan \left(\frac{\pi}{2} - \beta\right) = \frac{1}{\tan \beta} = \frac{1}{g'(y)}$$

With Leibniz's notation, setting $y = f(x)$ and $x = g(y)$, (9) is written in the form

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$$

- Pay attention to the fact that in the formula for the derivative of the inverse function, the derivatives f' and g' are computed at two different points: this is the main thing to be careful about when applying this theorem.

Remark 7

Given the function $f(y) = \arctan y$, the derivative function is $f'(y) = \frac{1}{1+y^2}$.

Proof. We set $y = \tan x$, $x = \arctan y$ with $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $y \in \mathbb{R}$. We have:

$$(\arctan y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{1 + \tan^2 x} = \frac{1}{1 + y^2}$$

□

Remark 8

Given the function $f(y) = \arcsin y$, the derivative function is $f'(y) = \frac{1}{\sqrt{1-y^2}}$

Proof. We set $y = \sin x$, $x = \arcsin y$, with $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, $y \in [-1, 1]$. Since for those values of x we have

$$\cos x = \sqrt{1 - \sin^2 x} = \sqrt{1 - y^2}$$

we have:

$$(\arcsin y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{\cos x} = \frac{1}{\sqrt{1-y^2}}$$

□

Remark 9

Given the function $f(y) = \arccos y$, the derivative function is $f'(y) = -\frac{1}{\sqrt{1-y^2}}$

Proof. Now we set $y = \cos x$, $x = \arccos y$, with $x \in [0, \pi]$, $y \in [-1, 1]$. Since for those values of x we have

$$\sin x = \sqrt{1 - \cos^2 x} = \sqrt{1 - y^2}$$

we have

$$(\arccos y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{-\sin x} = -\frac{1}{\sqrt{1-y^2}}$$

□

- Note that the functions $\arcsin x$, $\arccos x$, although defined and continuous on $[-1, 1]$, are not differentiable at the endpoints of the interval: precisely, at these points they have a vertical tangent.

The usefulness of the theorem on the derivative of the inverse function lies in the fact that it allows us to compute the derivative of the inverse function g even in situations where we do not know how to write g explicitly.

Example 9: Derivative of the inverse function

Let

$$f(x) = x + e^x$$

The function is strictly increasing on all of $\mathbb{R}$, hence invertible; let g be its inverse. We compute, for example, $g'(y_0)$ for $y_0 = f(0) = 1$. We have:

$$f'(x) = 1 + e^x, \quad f'(0) = 2 \neq 0$$

Hence:

$$g'(1) = \frac{1}{f'(0)} = \frac{1}{2}.$$

Remark 10

Given the function $f(y) = \text{SettSh } y$ (the inverse hyperbolic sine), the derivative function is $f'(y) = \frac{1}{\sqrt{y^2+1}}$

Proof. Now we set $y = \text{Sh } x$, $x = \text{SettSh } y$, with $x \in \mathbb{R}$, $y \in \mathbb{R}$. Since for those values of x we have

$$\text{Ch } x = \sqrt{\text{Sh}^2 x + 1} = \sqrt{y^2 + 1}$$

we have:

$$(\text{SettSh } y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{\text{Ch } x} = \frac{1}{\sqrt{y^2 + 1}}$$

□

Remark 11

Given the function $f(y) = \text{SettCh } y$ (the inverse hyperbolic cosine), the derivative function is $f'(y) = \frac{1}{\sqrt{y^2-1}}$

Proof. Now we set $y = \text{Ch } x$, $x = \text{SettCh } y$, with $x \in [0, +\infty)$, $y \in [1, +\infty)$. Since for those values of x we have

$$\text{Sh } x = \sqrt{\text{Ch}^2 x - 1} = \sqrt{y^2 - 1}$$

we have:

$$(\text{SettCh } y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{\text{Sh } x} = \frac{1}{\sqrt{y^2 - 1}}$$

□

1.4 Logarithmic derivative and elasticity

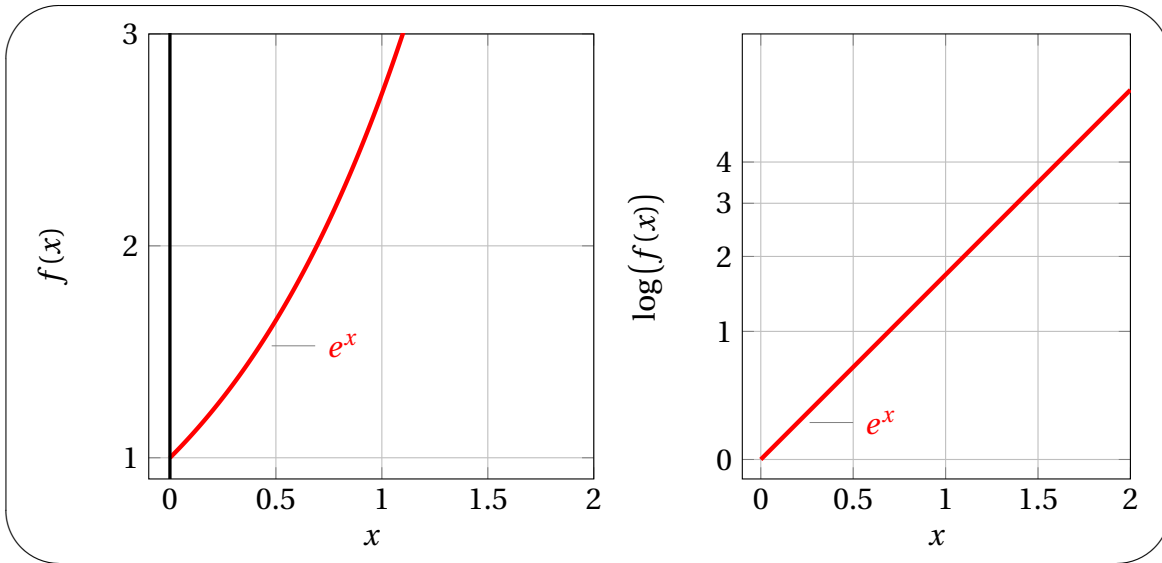
Definition 1: of logarithmic derivative

Given $f > 0$, the logarithmic derivative function of f is the derivative function of $\log f$; in formulas:

$$\frac{d}{dx} \log(f(x)) = \frac{f'(x)}{f(x)}$$

- The logarithmic derivative has the meaning of **relative growth rate** of f with respect to x .
- In some cases the relative growth rate $\frac{f'(x)}{f(x)}$ is often more meaningful than the absolute rate $f'(x)$.
- For example, a yearly capital increase of 1 billion out of a total of 10 has a very different effect from an increase of 1 billion out of 1000 billion. In the first case the relative rate of change is $1/10$, in the second $1/1000$, while the absolute increases are equal.

When we are interested in visualizing relative increases, we use graphs on a **semi-logarithmic scale**: the values of x are placed on the x -axis, while the values of $\log(f(x))$ are placed on the y -axis



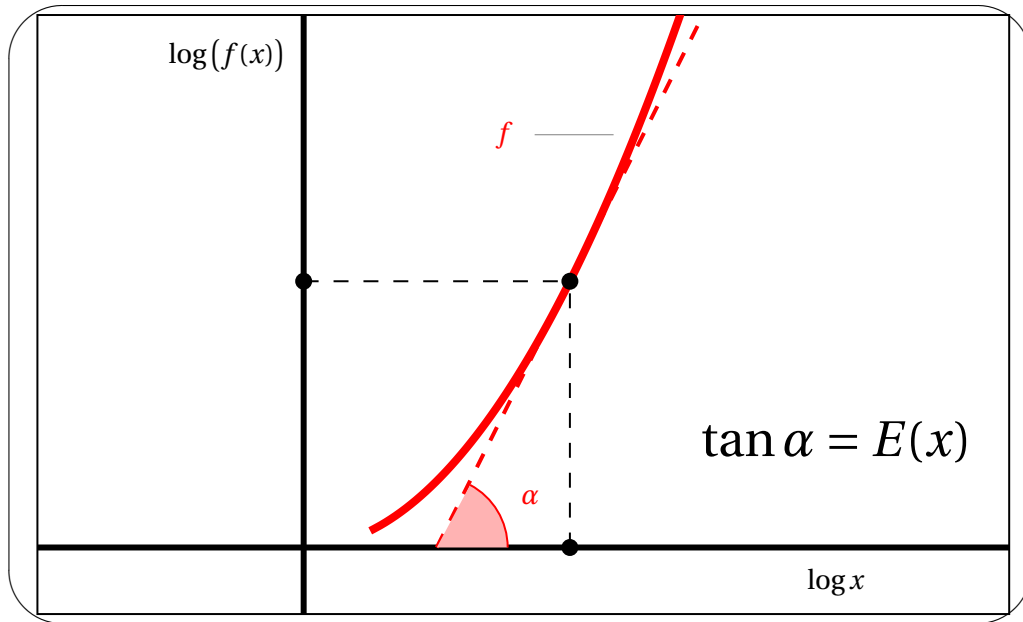
- This kind of representation is also used when $f(x)$ grows so rapidly that it would require an excessive compression of the scale (unit of measure) on the y -axis (or on the x -axis).

A **log-log graph** (graph on a logarithmic scale) is a graph in which, instead of x , the values of $\log x$ are placed on the x -axis and, instead of $f(x)$, the values of $\log f(x)$ are placed on the y -axis

Definition 2: of elasticity

The slope of the tangent line to a graph on a logarithmic scale is called the elasticity of f and is denoted by $E(x)$.

- Elasticity represents the relative rate of change of f with respect to relative changes of x , and it equals the derivative of $\log f$ with respect to $\log x$



- To find the analytic expression of $E(x)$, we observe that, by the theorem on the differentiation of composite functions, setting $u = \log x$ we have:

$$\frac{d \log f}{dx} = \frac{d \log f}{du} \frac{du}{dx} \quad \text{i.e.} \quad \frac{f'(x)}{f(x)} = E(x) \cdot \frac{1}{x}$$

from which we obtain:

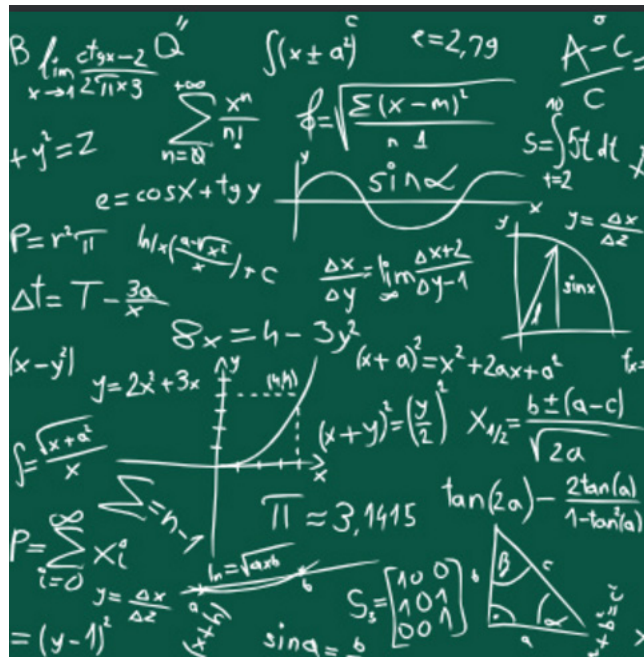
$$E(x) = x \cdot \frac{f'(x)}{f(x)}$$

Example 10: elasticity

We compute the elasticity of the power function $f(x) = x^\alpha$, $x > 0$, $\alpha \in \mathbb{R}$; we have:

$$E(x) = x \cdot \frac{\alpha x^{\alpha-1}}{x^\alpha} = \alpha \quad (\text{constant elasticity})$$

Mean value theorem, maxima and minima



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani e Sandro Salsa, Zanichelli.

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1 Local and global maxima/minima

- One of the uses of differential calculus is the **search for maxima and minima**, that is, the **optimization** of a function defined on an interval I (closed/open, bounded/unbounded).

Definition 1: of maximum point and maximum

Given a function $f : I \rightarrow \mathbb{R}$, if there exists a point $\tilde{x}_M \in I$ such that:

$$f(\tilde{x}_M) \geq f(x), \quad \forall x \in I$$

then $\tilde{x}_M$ is a (global) **maximum point** and $f(\tilde{x}_M)$ is the (global) **maximum** of f in I .

Definition 2: of minimum point and minimum

Given a function $f : I \rightarrow \mathbb{R}$, if there exists a point $\tilde{x}_m \in I$ such that:

$$f(\tilde{x}_m) \leq f(x), \quad \forall x \in I$$

then $\tilde{x}_m$ is a (global) **minimum point** and $f(\tilde{x}_m)$ is the (global) **minimum** of f in I .

We call an **extremum** a maximum or a minimum, and an **extremum point** a maximum or minimum point. An extremum, if it exists, is unique, while there may be several extremum points (even infinitely many).

Definition 3: of local maximum point and local maximum

Given a function $f : I \rightarrow \mathbb{R}$, if there exist a point $\bar{x}_M \in I$ and a neighborhood $(\bar{x}_M - \delta, \bar{x}_M + \delta)$ with $\delta > 0$, such that:

$$f(\bar{x}_M) \geq f(x), \quad \forall x \in (\bar{x}_M - \delta, \bar{x}_M + \delta) \cap I$$

then $\bar{x}_M$ is a **local maximum point** and $f(\bar{x}_M)$ is the **local maximum** of f in $(\bar{x}_M - \delta, \bar{x}_M + \delta) \cap I$.

Definition 4: of local minimum point and local minimum

Given a function $f : I \rightarrow \mathbb{R}$, if there exist a point $\bar{x}_m \in I$ and a neighborhood $(\bar{x}_m - \delta, \bar{x}_m + \delta)$ with $\delta > 0$, such that:

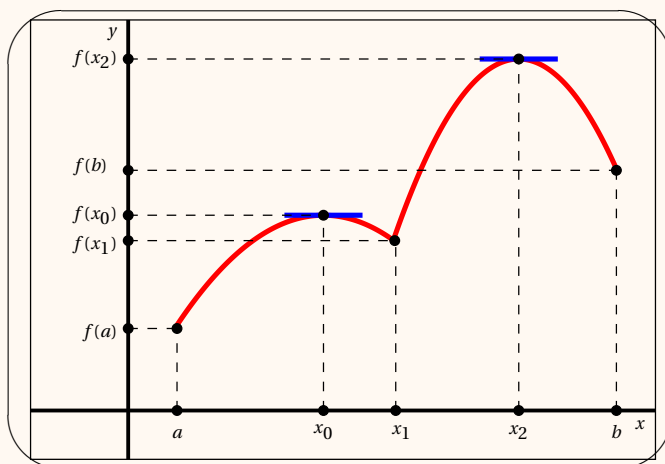
$$f(\bar{x}_m) \leq f(x), \quad \forall x \in (\bar{x}_m - \delta, \bar{x}_m + \delta) \cap I$$

then $\bar{x}_m$ is a **local minimum point** and $f(\bar{x}_m)$ is the **local minimum** of f in $(\bar{x}_m - \delta, \bar{x}_m + \delta) \cap I$.

- Global extremum points are also local extremum points, and global extrema are also local extrema.

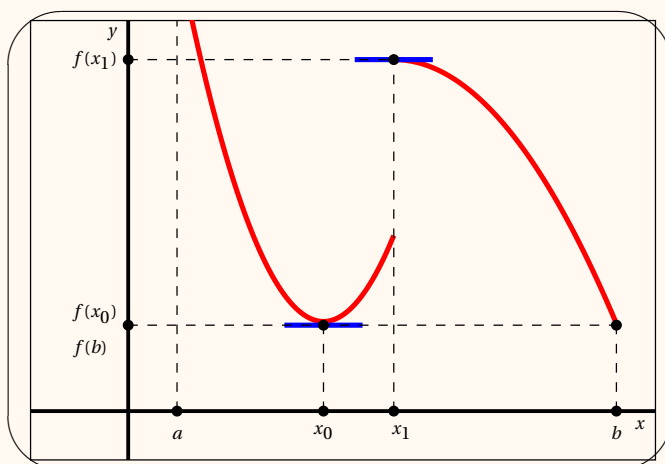
Example 1: extrema and extremum points (global and local)

Consider the function f with the following graph:



- the global maximum is $f(x_2)$ and x_2 is a global maximum point (it is also unique); $f(x_0)$ is a local (not global) maximum and x_0 is a local (not global) maximum point
- the global minimum is $f(a)$ and a is a global minimum point (it is also unique); $f(b)$ and $f(x_1)$ are two local (not global) minima and b and x_1 are two local (not global) minimum points

Now consider the function f with the following graph:



- the global maximum does not exist ($\lim_{x \rightarrow a^+} f(x) = +\infty$); $f(x_1)$ is a local (not global) maximum and x_1 is a local (not global) maximum point
- the global minimum is $f(x_0)$, which is also equal to $f(b)$; x_0 and b are two global minimum points

2 Fermat's theorem and stationary points

- At a local or global extremum point the function may fail to be differentiable and may even be discontinuous. However, the following theorem tells us that if a function is differentiable at a local extremum point, then at that point the derivative vanishes and hence the tangent to the graph is horizontal.

Theorem 1: Fermat's theorem

Given a function $f : (a, b) \rightarrow \mathbb{R}$ differentiable at $x_0 \in (a, b)$, if f has a local extremum at x_0 then $f'(x_0) = 0$.

Proof. Consider the case where x_0 is a local maximum point. Then:

$$\exists(x_0 - \delta, x_0 + \delta) \text{ with } \delta > 0: f(x_0) \geq f(x), \forall x \in (x_0 - \delta, x_0 + \delta) \cap (a, b)$$

Therefore for every $x \in (x_0 - \delta, x_0 + \delta) \cap (a, b)$ we have:

$$x < x_0 \Rightarrow \frac{f(x) - f(x_0)}{x - x_0} \geq 0 \text{ and hence } f'_-(x_0) = \lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0} \geq 0$$

where for the non-negativity of the limit we used the sign-preservation theorem. On the other hand:

$$x > x_0 \Rightarrow \frac{f(x) - f(x_0)}{x - x_0} \leq 0 \text{ and hence } f'_+(x_0) = \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0} \leq 0$$

Since f is differentiable at x_0 , we have:

$$f'(x_0) = f'_-(x_0) = f'_+(x_0) = 0$$

The case where x_0 is a local minimum point is handled in the same way. $\square$

Proof. Suppose, by contradiction, that $f'(x_0) > 0$; then by the sign-preservation theorem we would have $\frac{f(x) - f(x_0)}{x - x_0} > 0$ eventually as $x \rightarrow x_0$. Taking into account the sign of $x - x_0$, this implies that $f(x) > f(x_0)$ eventually as $x \rightarrow x_0^+$ and $f(x) < f(x_0)$ eventually as $x \rightarrow x_0^-$. But this contradicts the hypothesis that x_0 is a local extremum point of f . Similarly, we rule out the case $f'(x_0) < 0$. Therefore we must have $f'(x_0) = 0$. $\square$

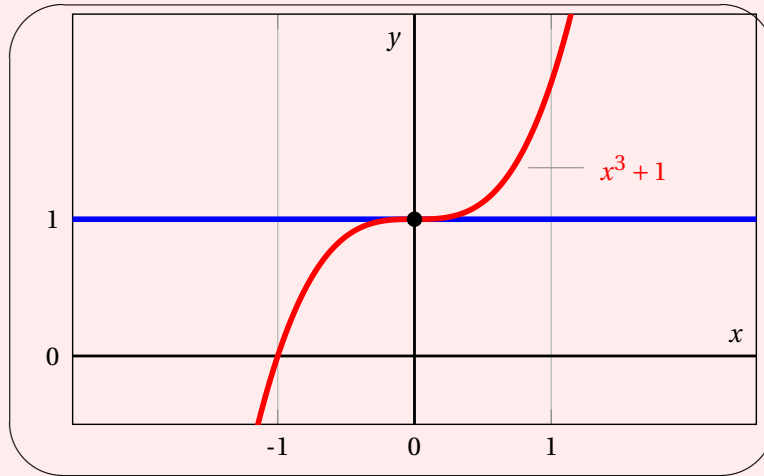
Definition 5: stationary point

A point x_0 is called a stationary point of f if f is differentiable at x_0 and $f'(x_0) = 0$.

Fermat's theorem says that for a function $f : (a, b) \rightarrow \mathbb{R}$ differentiable at $x_0 \in (a, b)$ we have:

$$x_0 \text{ is a local extremum point} \Rightarrow \underbrace{x_0 \text{ is a stationary point}}_{f'(x_0)=0} \quad (1)$$

Hence “ x_0 is a local extremum point” is a sufficient condition for “ x_0 is a stationary point”. But the converse is not true. A counterexample is given by the function $f(x) = x^3 + 1, \forall x \in \mathbb{R}$, whose derivative function is $f'(x) = 3x^2, \forall x \in \mathbb{R}$. With $x_0 = 0$ we have $f'(0) = 0$ but $x_0 = 0$ is not a local extremum point.



Hence we have:

$$\underbrace{x_0 \text{ is a stationary point}}_{f'(x_0)=0} \not\Rightarrow x_0 \text{ is a local extremum point}$$

Finally, from the contrapositive of (1), we have:

$$x_0 \text{ is not a stationary point} \Rightarrow x_0 \text{ is not a local extremum point}$$

3 Mean value theorem (Lagrange's theorem)

Theorem 2: mean value theorem (Lagrange)

Given a function f differentiable in (a, b) and continuous in $[a, b]$, then:

$$\exists c \in (a, b) : \frac{f(b) - f(a)}{b - a} = f'(c) \quad (2)$$

In the case $f(b) = f(a)$, the theorem tells us that there exists $c \in (a, b)$ such that $f'(c) = 0$, that is, there exists at least one point where the derivative vanishes. This corollary of Lagrange's theorem is called **Rolle's theorem**.

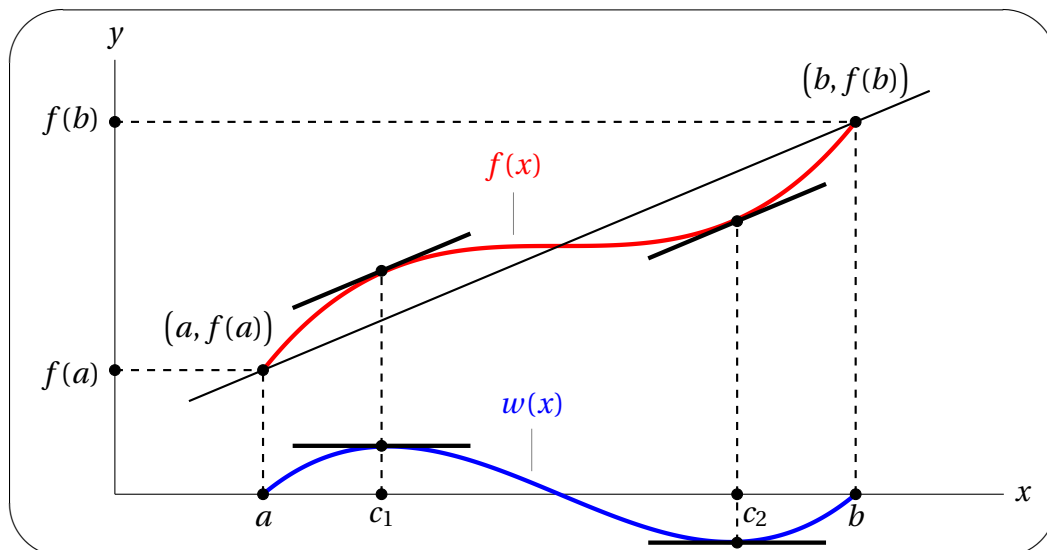
- Differentiability is required only on the open interval (a, b) in order to include also functions that are continuous at one of the endpoints but not differentiable there. For example, with $a = 0$:

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) x & \text{if } x \in (0, b) \\ 0 & \text{if } x = 0 \end{cases} \quad \text{or} \quad g(x) = \sqrt{x}$$

- Geometrically, considering the graph of f we have:

1. $\frac{f(b) - f(a)}{b - a}$ is the slope of the line through $(a, f(a))$ and $(b, f(b))$
2. $f'(c)$ is the slope of the tangent line to the graph of f at the point $(c, f(c))$

The mean value theorem therefore expresses the fact that at the point $(c, f(c))$ the tangent to the graph of f is parallel to the line through $(a, f(a))$ and $(b, f(b))$. There may be more than one point with this property, as the following figure shows:



- The figure also shows the function

$$w(x) = f(x) - \left(f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right)$$

given by the difference between the values of the function and the line through the points $(a, f(a))$ and $(b, f(b))$, which has equation:

$$y = f(a) + \frac{f(b) - f(a)}{b - a} (x - a)$$

The function $w(x)$ is the key to the proof of the theorem.

Proof. Consider the function:

$$w(x) = f(x) - \left(f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right)$$

Clearly $w(a) = w(b) = 0$; moreover w is continuous in $[a, b]$ and differentiable in (a, b) . Since

$$w'(x) = f'(x) - \frac{f(b) - f(a)}{b - a} \quad \text{then} \quad \frac{f(b) - f(a)}{b - a} = f'(c) \iff \exists c \in (a, b) : w'(c) = 0$$

Since w is continuous in $[a, b]$, by the Weierstrass theorem there exist two points x_1 and x_2 in $[a, b]$ such that:

$w(x_1) = M$, the maximum of w in $[a, b]$; $w(x_2) = m$, the minimum of w in $[a, b]$.

If $M = m$, then $w(x)$ is constant in $[a, b]$, and hence $w'(x) = 0, \forall x \in (a, b)$.

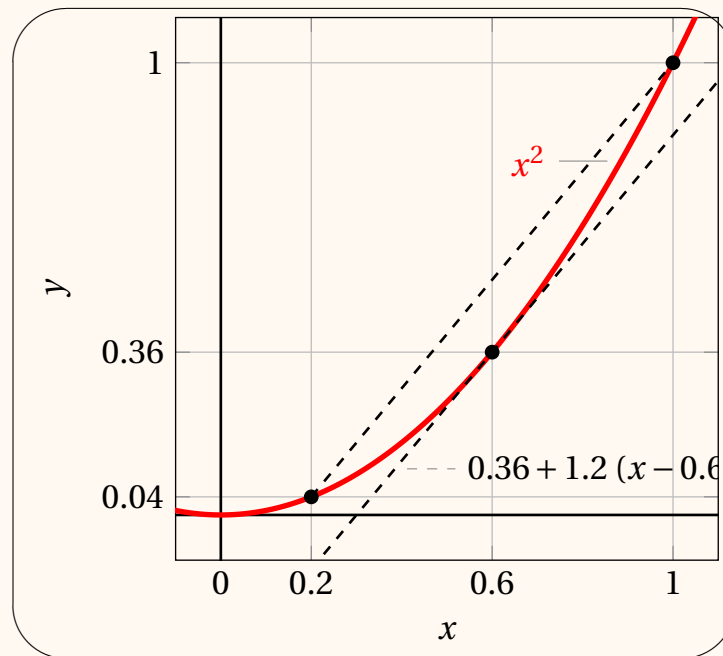
If $M > m$, at least one of the two points x_1 or x_2 is not at the endpoints of the interval, since $w(a) = w(b) = 0$. Fermat's theorem then implies that at the maximum or minimum point that lies in the interior (possibly both) the derivative of w vanishes, and the theorem is thus proved. $\square$

Example 2: using the mean value theorem

Let $f(x) = x^2$. Then $f'(x) = 2x$ and the mean value theorem states that in every interval $[a, b]$ there exists a number c such that:

$$\frac{b^2 - a^2}{b - a} = 2c \quad \text{from which} \quad c = \frac{a + b}{2} \quad (\text{arithmetic mean})$$

That is, every chord AB of the parabola $y = x^2$ is parallel to the tangent at the point whose abscissa equals the arithmetic mean of the abscissas of A and B . Taking for example the interval $[0.2, 1]$ ($a = 0.2, b = 1$ and $c = 0.6$), we have:

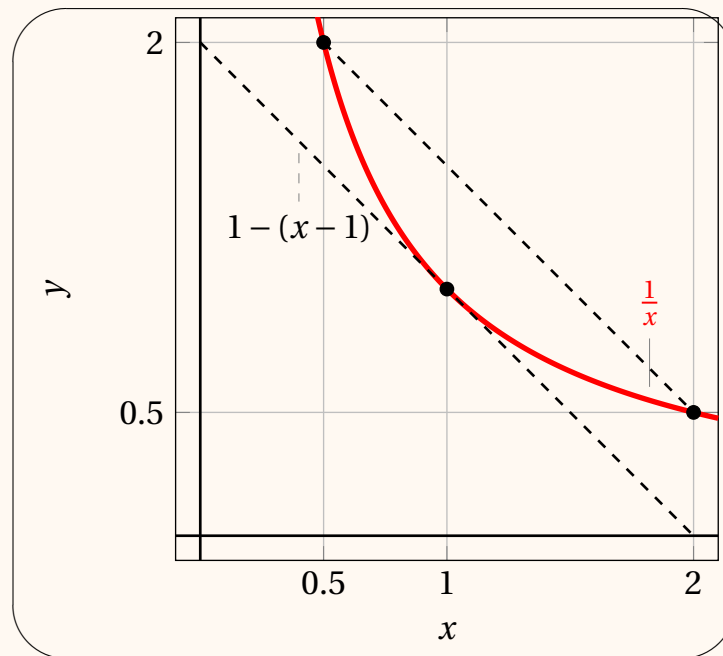


Example 3: using the mean value theorem

Let $f(x) = \frac{1}{x}$. Then $f'(x) = -\frac{1}{x^2}$ and the mean value theorem states that in every interval $[a, b]$ there exists a number c such that:

$$\frac{\frac{1}{b} - \frac{1}{a}}{b - a} = -\frac{1}{c^2} \quad \text{from which} \quad c = \sqrt{a \cdot b} \quad (\text{geometric mean})$$

That is, every chord AB of the hyperbola $y = \frac{1}{x}$ is parallel to the tangent at the point whose abscissa equals the geometric mean of the abscissas of A and B . Taking for example the interval $[0.5, 2]$ ($a = 0.5, b = 2$ and $c = 1$), we have:



4 Differential monotonicity test theorem

Theorem 3: Differential Monotonicity Test

Given a function $f : I \rightarrow \mathbb{R}$, continuous in I and differentiable at the interior points of I , then:

$$f'(x) \geq 0, \forall x \text{ in the interior of } I \iff f \text{ is non-decreasing in } I \quad (3)$$

$$f'(x) \leq 0, \forall x \text{ in the interior of } I \iff f \text{ is non-increasing in } I \quad (4)$$

$$f'(x) > 0, \forall x \text{ in the interior of } I \implies f \text{ is increasing in } I \quad (5)$$

$$f'(x) < 0, \forall x \text{ in the interior of } I \implies f \text{ is decreasing in } I \quad (6)$$

Proof.

($\Rightarrow$) Taking two points $x_1 < x_2$ in the interval I , we can apply Lagrange's theorem on the interval $[x_1, x_2]$, hence there exists $c \in (x_1, x_2)$ such that

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

If $f'(c) \geq 0$ (resp. > 0), then $f(x_2) \geq f(x_1)$ (resp. $f(x_2) > f(x_1)$). Since x_1 and x_2 are arbitrary, we have proved that f is non-decreasing (resp. increasing) in I . If $f'(c) \leq 0$ (resp. < 0) the argument is analogous.

($\Leftarrow$) If f is non-decreasing, then we have

$$\frac{f(z) - f(x)}{z - x} \geq 0 \quad \text{for all } z, x \in I, z \neq x$$

and, by the sign-preservation theorem, the limit of the ratio as $z \rightarrow x$ is also non-negative; by hypothesis this limit exists and equals $f'(x)$ for every x in the interior of I . If f is non-increasing the argument is analogous.

□

The Differential Monotonicity Test theorem says that for a function $f : I \rightarrow \mathbb{R}$ differentiable at the interior points of I we have:

$$f'(x) \geq 0, \forall x \text{ in the interior of } I \iff f \text{ is non-decreasing in } I$$

Hence “ $f'(x) \geq 0, \forall x$ in the interior of I ” is a necessary and sufficient condition for “ f is non-decreasing in I ”. The same holds for the case of non-increasing functions.

The Differential Monotonicity Test theorem says that for a function $f : I \rightarrow \mathbb{R}$ differentiable at the interior points of I we have:

$$f'(x) > 0, \forall x \text{ in the interior of } I \implies f \text{ is increasing in } I$$

Hence “ $f'(x) > 0, \forall x$ in the interior of I ” is a sufficient condition for “ f is increasing in I ”. But the converse is not true. A counterexample is given by the function $f(x) = x^3 + 1, \forall x \in \mathbb{R}$, which is increasing and differentiable in $\mathbb{R}$, but $f'(x) = 3x^2$ vanishes at $x_0 = 0$.

Hence we have:

$$f \text{ is increasing in } I \not\Rightarrow f'(x) > 0, \forall x \text{ in the interior of } I$$

Finally, from the contrapositive of (5), we have:

$$f \text{ is not increasing in } I \Rightarrow \exists x \text{ in the interior of } I : f'(x) \leq 0,$$

To be checked. The same holds for the case of decreasing functions.

- It follows immediately from the theorem that with $I = (a, b)$ we have:

$$f'(x) = 0, \forall x \in (a, b) \implies f \text{ constant in } (a, b)$$

The converse implication is obvious. We then have

$$f'(x) = 0, \forall x \in (a, b) \iff f \text{ constant in } (a, b) \quad (7)$$

Hence “ $f'(x) = 0, \forall x \in (a, b)$ ” is a necessary and sufficient condition for “ f constant in (a, b) ”.

Example 4: functions with zero derivative

Consider the function:

$$f(x) = \arctan x + \arctan \frac{1}{x}, \quad \forall x \neq 0$$

we have

$$f'(x) = \frac{1}{1+x^2} + \frac{1}{1+\frac{1}{x^2}} \left(-\frac{1}{x^2}\right) = 0, \quad \forall x \neq 0$$

Implication (7) allows us to say that f is constant on the interval $(-\infty, 0)$ and on the interval $(0, \infty)$. To find its value, it is enough to compute f at one point of each interval, for example:

$$f(1) = \arctan(1) + \arctan(1) = 2 \frac{\pi}{4} = \frac{\pi}{2}$$

$$f(-1) = \arctan(-1) + \arctan(-1) = -\frac{\pi}{2}$$

We have therefore proved that:

$$\arctan x + \arctan \frac{1}{x} = \begin{cases} \frac{\pi}{2} & \text{if } x > 0 \\ -\frac{\pi}{2} & \text{if } x < 0 \end{cases}$$

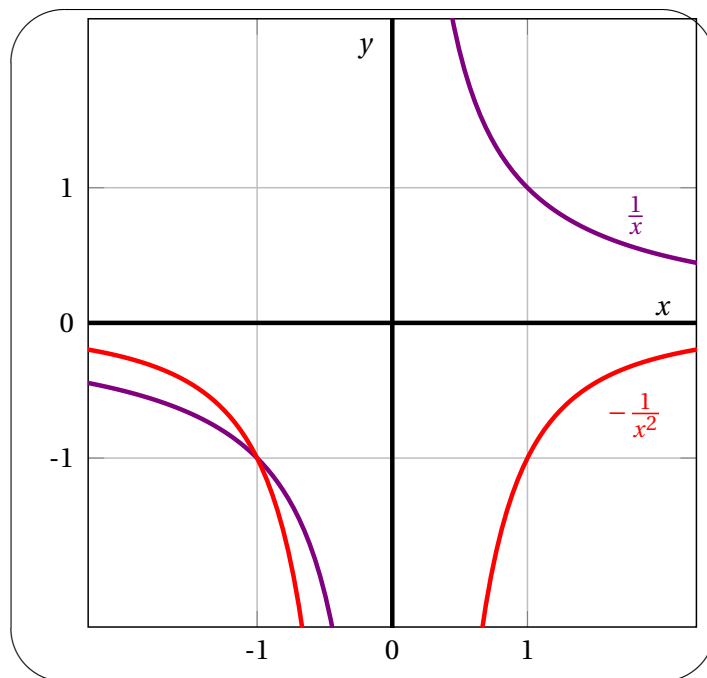
- Both in the Differential Monotonicity Test theorem and in implication (7) it is essential that the set I is an interval. For example, the function

$$f(x) = \frac{1}{x}, \quad \forall x \in \mathbb{R} \setminus \{0\}$$

has derivative

$$f'(x) = -\frac{1}{x^2} < 0, \quad \forall x \in \mathbb{R} \setminus \{0\}$$

but the function is not decreasing on its domain. The function is decreasing on the interval $(-\infty, 0)$ and on the interval $(0, \infty)$, as the figure shows:



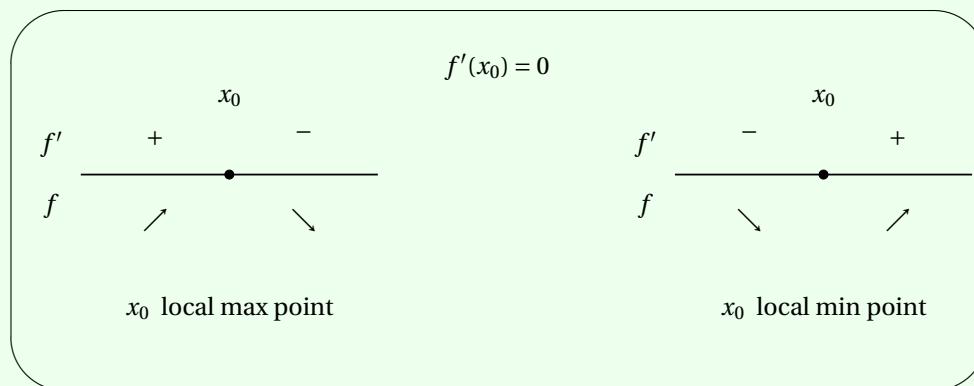
5 Finding maxima and minima

Suppose we have a function $f : [a, b] \rightarrow \mathbb{R}$ and we want to find its local and global maxima and minima in $[a, b]$. If f is differentiable in (a, b) we can proceed as follows:

Step 1 Compute $f(a)$ and $f(b)$.

Step 2 Compute $f'(x)$ and solve the equation $f'(x) = 0$. In this way we find the stationary points, among which are the possible local extremum points in the interval (a, b) .

Step 3 If there are no stationary points, $f(a)$ and $f(b)$ are the global extrema and a and b are global extremum points. Otherwise we need to determine the nature of the stationary points found. The function f has a local extremum at a stationary point if and only if the sign of f' changes there, which we check by studying the sign of f' in a neighborhood of the point. We have the following cases:



Step 4 Having found the possible local extremum points, compute the value of f at these points and compare it with $f(a)$ and $f(b)$ to understand whether or not they are global extremum points; in this way the global extrema are determined.

Example 5: finding maxima and minima

Consider the function:

$$f(x) = x e^{-x^2}, \text{ with } x \in [0, 2]$$

Step 1:

$$f(0) = 0, \quad f(2) = 2 e^{-4}$$

Example 6: finding maxima and minima

Step 2:

$$f'(x) = e^{-x^2} + x e^{-x^2} (-2x) = e^{-x^2} (1 - 2x^2)$$

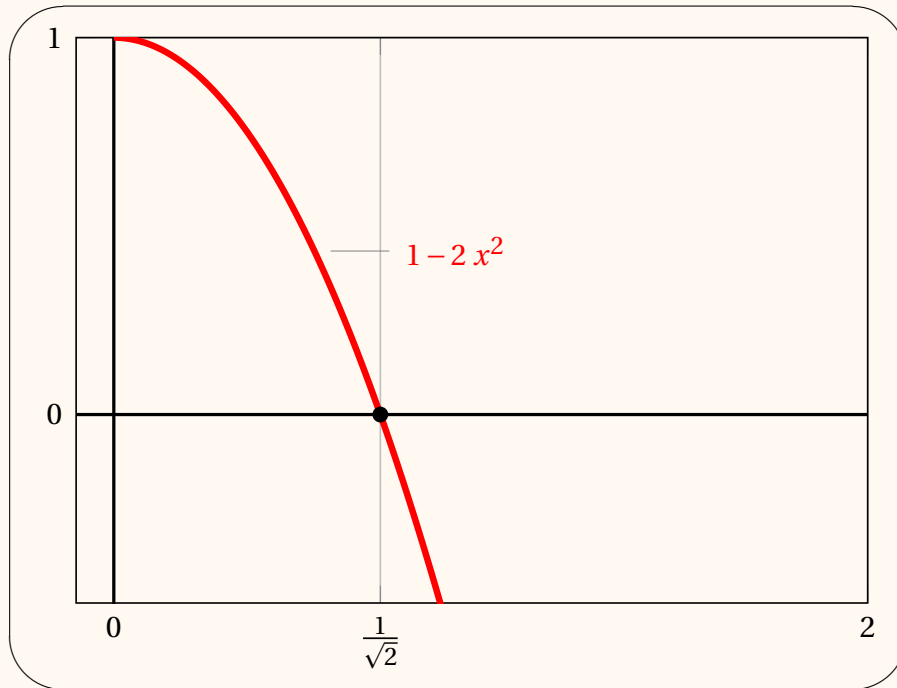
We have $e^{-x^2} > 0, \forall x \in \mathbb{R}$, hence:

$$f'(x) = 0 \iff (1 - 2x^2) = 0 \iff x = \pm \frac{1}{\sqrt{2}}$$

Only $\frac{1}{\sqrt{2}} \in [0, 2]$, hence we have the stationary point $x_0 = \frac{1}{\sqrt{2}}$.

Step 3:

We study the sign of f' near $x_0 = \frac{1}{\sqrt{2}}$. We have $f'(x) \geq 0$ for $2x^2 \leq 1$, that is, for $-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$



We therefore conclude that:

$$\frac{1}{\sqrt{2}} \text{ is a local maximum point and } f\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}} e^{-1/2} = \frac{1}{\sqrt{2}e} \text{ is a local maximum}$$

We also note that:

$$\lim_{x \rightarrow 0^+} \frac{x e^{-x^2}}{x} = \lim_{x \rightarrow 0^+} \frac{1}{e^{x^2}} = 1 \quad \text{hence} \quad f(x) \sim x \text{ for } x \rightarrow 0^+$$

Example 7: finding maxima and minima

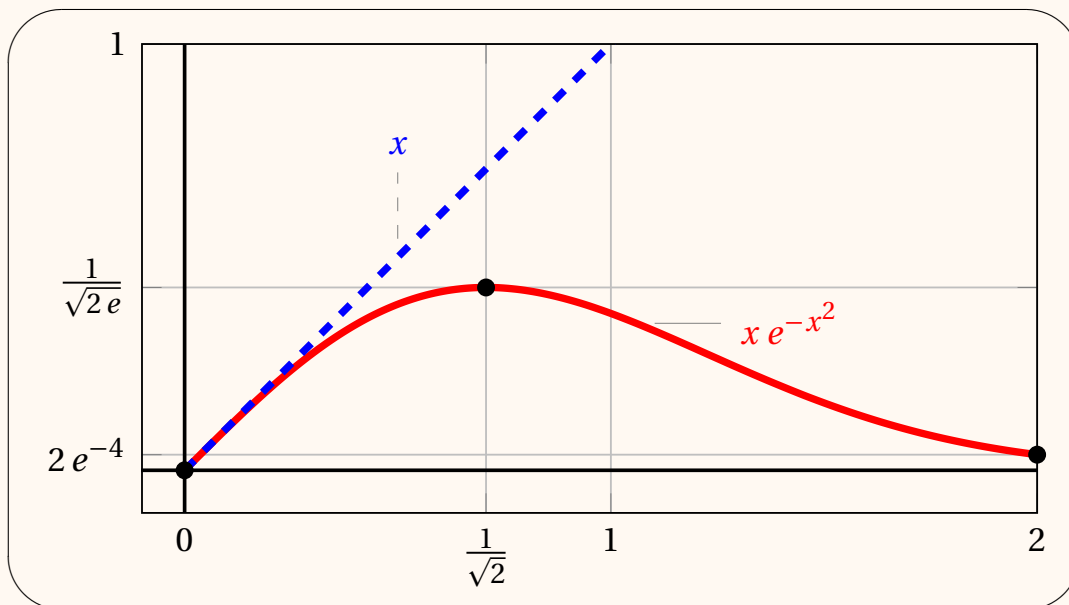
Step 4:

$$\frac{1}{\sqrt{2}e} > f(0) = 0, \quad \frac{1}{\sqrt{2}e} > f(2) = 2e^{-4}$$

We therefore conclude that:

$\frac{1}{\sqrt{2}}$ is the (unique) global maximum point and 0 is the (unique) global minimum point

$\frac{1}{\sqrt{2}e}$ is the global maximum and 0 is the global minimum



5.1 From the discrete to the continuous

Passing “from the discrete to the continuous” for sequences (i.e., from the natural numbers to the reals) is a way to make the tools of differential calculus available

Example 8: proving monotonicity of sequences by passing to the continuous

Consider the sequence

$$a_n = \frac{\log n}{n}, \quad \text{we have } a_n \geq 0, \forall n \in \mathbb{N}_{>0} \text{ and } a_n \rightarrow 0 \text{ for } n \rightarrow +\infty$$

To prove that the sequence is eventually monotone non-increasing, one way is to use the definition and prove that:

$$a_n \geq a_{n+1}, \quad \forall n \in \mathbb{N}_{>0} \quad \text{i.e.} \quad \frac{\log n}{n} \geq \frac{\log(n+1)}{n+1}, \quad \forall n \in \mathbb{N}_{>0}$$

Since both the numerator and the denominator grow as n grows, it is not easy to prove this inequality algebraically.

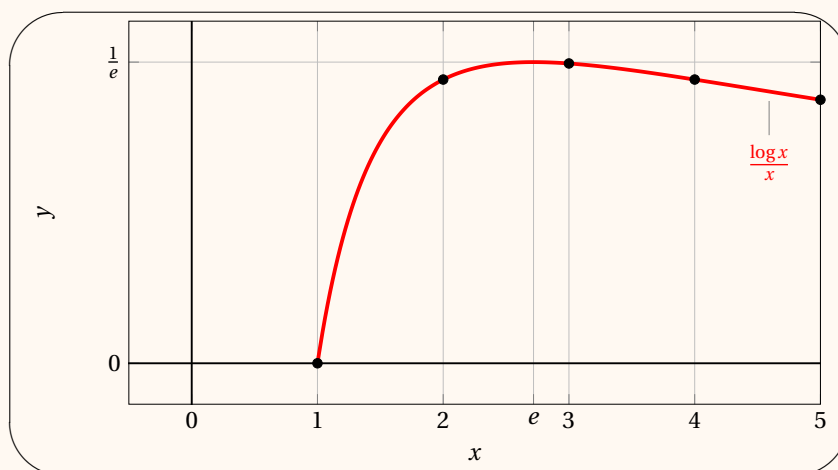
We pass from the discrete to the continuous and define the function

$$f(x) = \frac{\log x}{x}, \quad \forall x > 1$$

We have

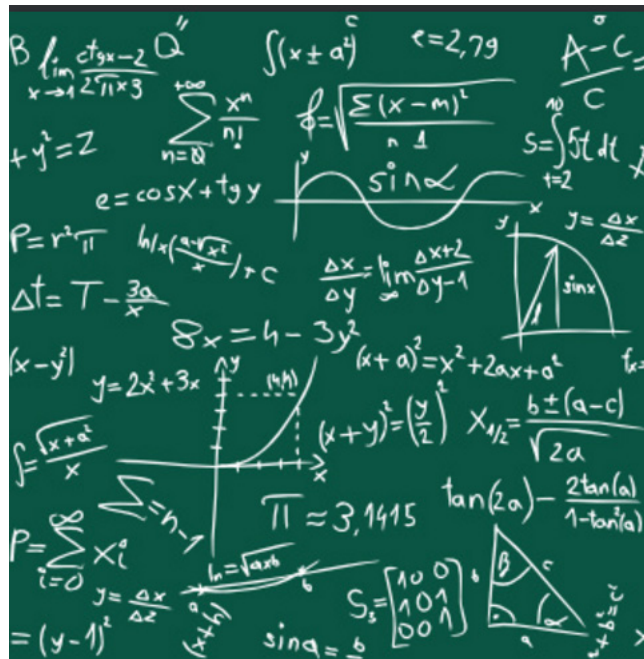
$$f'(x) = \frac{1 - \log x}{x^2} \leq 0, \quad \forall x \geq e$$

and it follows that f is non-increasing $\forall x \geq e$. Consequently the sequence a_n is non-increasing for $n \geq 3$ (the first natural number $> e$).



- Be careful not to use the passage from the discrete to the continuous indiscriminately. For example, it cannot be done for the sequences $\frac{n!}{2^n}$ or $\frac{n^2 + (-1)^n n}{n^3 + 1}$, since they are defined only for natural numbers.

L'Hôpital's rule and differentiability



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 L'Hôpital's rule

- A remarkable application of differential calculus is the computation of limits that appear in the indeterminate forms:

$$\left[\frac{0}{0} \right] \text{ and } \left[\frac{\infty}{\infty} \right]$$

Theorem 1: L'Hôpital's rule

Let f, g be differentiable functions on an interval (a, b) with $g, g' \neq 0$ in (a, b) . If

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0 \text{ (or } \pm \infty) \quad \text{and} \quad \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = \ell \in \mathbb{R}^*$$

then:

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = \ell$$

The theorem still holds if $a = -\infty$ or if we consider the limit as $x \rightarrow b^-$, with $b \leq +\infty$ (a case we do not prove). Hence, if the hypotheses hold for a^- and a^+ , it can clearly be applied also at an interior point of an interval.

Example 1: using L'Hôpital's rule

We want to compute:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \left[\frac{0}{0} \right] \quad \text{and} \quad f(x) = \sin x, f'(x) = \cos x, \quad g(x) = x, g'(x) = 1$$

applying L'Hôpital's rule, for example with $x \in (-1, 0) \cup (0, 1)$, we obtain:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} =^H \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$$

We want to compute:

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{\cos x} = \left[\frac{0}{0} \right] \quad \text{and} \quad f(x) = 1 - \sin x, f'(x) = -\cos x, \quad g(x) = \cos x, g'(x) = -\sin x$$

applying L'Hôpital's rule, for example with $x \in (\frac{\pi}{4}, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \frac{3\pi}{4})$, we obtain:

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{\cos x} =^H \lim_{x \rightarrow \frac{\pi}{2}} \frac{-\cos x}{-\sin x} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\sin x} = \frac{0}{1} = 0$$

Proof. We consider the case:

$$f(x), g(x) \rightarrow 0, \text{ as } x \rightarrow a^+$$

Let $\{x_n\}$ be a sequence such that $x_n \rightarrow a^+$ as $n \rightarrow +\infty$ and $x_n \neq a, \forall n$, and let us extend f and g by continuity at a by setting $f(a) = g(a) = 0$.

We now define the following function:

$$h(x) = f(x_n) g(x) - g(x_n) f(x)$$

The function h satisfies the hypotheses of Lagrange's theorem on the interval $[a, x_n]$, hence there exists $t_n \in (a, x_n)$ such that

$$h'(t_n) = \frac{h(x_n) - h(a)}{x_n - a} = 0 \quad \text{since} \quad h(a) = h(x_n) = 0$$

Computing

$$h'(x) = f(x_n) g'(x) - g(x_n) f'(x), \quad \text{we have} \quad \underbrace{f(x_n) g'(t_n) - g(x_n) f'(t_n)}_{=h'(t_n)} = 0$$

Hence for every x_n there exists a point $t_n \in (a, x_n)$ such that:

$$\frac{f(x_n)}{g(x_n)} = \frac{f'(t_n)}{g'(t_n)} \tag{1}$$

As $n \rightarrow +\infty$ we have $t_n \rightarrow a^+$, since $a^+ < t_n < x_n$ and $x_n \rightarrow a^+$ as $n \rightarrow +\infty$, therefore by the hypothesis of the theorem we have:

$$\frac{f'(t_n)}{g'(t_n)} \rightarrow \ell \quad \text{and consequently} \quad \frac{f(x_n)}{g(x_n)} \rightarrow \ell$$

which is what we wanted to prove. □

With L'Hôpital's rule we can easily prove the theorem on the hierarchy of infinities for functions.

Proof. We prove:

$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = 0, \text{ with } \alpha > 0, \lambda > 0, b > 1$$

Setting $f(x) = x$, $g(x) = b^{\mu x}$ ($\mu > 0$), we have $f'(x) = 1$, $g'(x) = \mu \log b \cdot b^{\mu x}$ and since:

$$\lim_{x \rightarrow +\infty} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow +\infty} \frac{1}{\mu \log b \cdot b^{\mu x}} = 0 \quad \text{then} \quad \lim_{x \rightarrow +\infty} \frac{x}{b^{\mu x}} = 0$$

Observing now that

$$\frac{x^\alpha}{b^{\lambda x}} = \left(\frac{x}{b^{(\lambda/\alpha)x}} \right)^\alpha = \left(\frac{x}{b^{\mu x}} \right)^\alpha \text{ with } \mu = \frac{\lambda}{\alpha} > 0 \quad \text{then} \quad \lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = \left(\lim_{x \rightarrow +\infty} \frac{x}{b^{\mu x}} \right)^\alpha = 0$$

We prove:

$$\lim_{x \rightarrow +\infty} \frac{(\log_a x)^\beta}{x^\alpha} = 0, \text{ with } \beta > 0, \alpha > 0, a > 1$$

it suffices to perform the change of variable

$$t = \log_a x, \quad x = a^t, \text{ we have that as } x \rightarrow +\infty \text{ also } t \rightarrow +\infty$$

hence

$$\lim_{x \rightarrow +\infty} \frac{(\log_a x)^\beta}{x^\alpha} = \lim_{t \rightarrow +\infty} \frac{t^\beta}{a^{\alpha t}} = 0$$

since the second limit has the same form as the one just proved. □

- L'Hôpital's rule can be useful for limits that cannot be solved using only fundamental limits or the asymptotic expansions derived from them.

Example 2: L'Hôpital's rule and fundamental limits

We compute

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = \left[\frac{0}{0} \right]$$

Fundamental limits are not sufficient to resolve the indeterminate form, since:

$$\frac{x - \sin x}{x^3} = \frac{1}{x^2} \left(1 - \frac{\sin x}{x} \right) \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{1}{x^2} \underbrace{\left(1 - \frac{\sin x}{x} \right)}_{\rightarrow 1} = [\infty \cdot 0]$$

L'Hôpital's rule gives:

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{(x - \sin x)'}{(x^3)'} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2} = \frac{1}{3} \cdot \lim_{x \rightarrow 0} \underbrace{\frac{1 - \cos x}{x^2}}_{\rightarrow \frac{1}{2}} = \frac{1}{6}$$

Example 3: L'Hôpital's rule and asymptotic expansions

We now treat the same limit with the first-order asymptotic expansion:

$$\sin x = x + o(x) \text{ as } x \rightarrow 0$$

we obtain:

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{x - (x + o(x))}{x^3} = \lim_{x \rightarrow 0} \frac{o(x)}{x^3}$$

That is, we still have an indeterminate form, since:

$$\lim_{x \rightarrow 0} \frac{o(x)}{x^3} = \lim_{x \rightarrow 0} \frac{1}{x^2} \frac{o(x)}{x} = \lim_{x \rightarrow 0} \frac{1}{x^2} o(1) = [+\infty \cdot 0]$$

where $o(1)$ denotes a generic function that tends to 0 as $x \rightarrow 0$.

We reach the same conclusion also with the following reasoning. The symbol $o(x)$ as $x \rightarrow 0$ denotes the set of functions that, divided by x , tend to 0 as $x \rightarrow 0$. Hence, for example, $5x^2 = o(x)$ but also $x^3 = o(x)$, and the value of the limit cannot be determined with the first-order asymptotic expansion.

As we will see later, to determine this limit with asymptotic expansions we need an asymptotic expansion of order higher than the first.

Example 4: L'Hôpital's rule and asymptotic estimates

We compute

$$\lim_{x \rightarrow +\infty} \frac{x - \sin x}{x + \sin x} = \left[\frac{+\infty}{+\infty} \right]$$

L'Hôpital's rule gives:

$$\lim_{x \rightarrow +\infty} \frac{x - \sin x}{x + \sin x} \stackrel{H}{=} \lim_{x \rightarrow +\infty} \frac{(x - \sin x)'}{(x + \sin x)'} = \lim_{x \rightarrow +\infty} \frac{1 - \cos x}{1 + \cos x} \text{ does not exist}$$

the fact that this limit does not exist does not allow us to conclude that the limit we are looking for does not exist.

Let us proceed instead with asymptotic estimates:

$$\frac{x - \sin x}{x + \sin x} \sim \frac{x}{x} \text{ as } x \rightarrow +\infty$$

Since

$$\lim_{x \rightarrow +\infty} \frac{x - \sin x}{x} = \lim_{x \rightarrow +\infty} 1 - \underbrace{\frac{\sin x}{x}}_{\rightarrow 0} \quad \text{and} \quad \lim_{x \rightarrow +\infty} \frac{x + \sin x}{x} = \lim_{x \rightarrow +\infty} 1 + \underbrace{\frac{\sin x}{x}}_{\rightarrow 0}$$

hence:

$$\lim_{x \rightarrow +\infty} \frac{x - \sin x}{x + \sin x} = \lim_{x \rightarrow +\infty} \frac{x}{x} = 1$$

- The theorem is useful when its application simplifies the limit instead of complicating it, that is, when the order of the infinitesimal (or of the infinity) in the numerator or in the denominator decreases.

Example 5: L'Hôpital's rule

We compute

$$\lim_{x \rightarrow 0^+} \frac{e^{-1/x^2}}{x} = \left[\frac{0}{0} \right]$$

L'Hôpital's rule gives:

$$\lim_{x \rightarrow 0^+} \frac{e^{-1/x^2}}{x} \stackrel{H}{=} \lim_{x \rightarrow 0^+} \frac{(e^{-1/x^2})'}{(x)'} = \lim_{x \rightarrow 0^+} \frac{2e^{-1/x^2}}{x^3} = \left[\frac{0}{0} \right]$$

Applying instead the change of variable:

$$x^2 = \frac{1}{t}, \quad x \rightarrow 0^+ \text{ is equivalent to } t \rightarrow +\infty$$

substituting we have:

$$\lim_{x \rightarrow 0^+} \frac{e^{-1/x^2}}{(x^2)^{1/2}} = \lim_{t \rightarrow +\infty} \frac{e^{-t}}{(\frac{1}{t})^{1/2}} = \lim_{t \rightarrow +\infty} \frac{e^{-t}}{t^{-1/2}} = \lim_{t \rightarrow +\infty} \frac{\sqrt{t}}{e^t} = 0$$

where the last limit is obtained thanks to the theorem on the hierarchy of infinities.

- Sometimes the theorem must be applied several times consecutively to resolve the indeterminate form. Also in this case, already after the first application, one should notice that the order of the infinitesimal (or of the infinity) in the numerator or in the denominator has decreased.

Example 6: L'Hôpital's rule

We compute

$$\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x^3 - x^2} = \left[\frac{0}{0} \right]$$

L'Hôpital's rule gives:

$$\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x^3 - x^2} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{3 \sin x \cos^2 x}{3x^2 - 2x} = \left[\frac{0}{0} \right]$$

L'Hôpital's rule applied twice gives:

$$\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x^3 - x^2} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{3 \sin x \cos^2 x}{3x^2 - 2x} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\overbrace{3 \cos x}^{-1} (\overbrace{\cos^2 x}^{-1} - \overbrace{2 \sin^2 x}^{-0})}{\underbrace{6x - 2}_{\rightarrow 0}} = -\frac{3}{2}$$

- The theorem is used for quotients, not for products
- The theorem is used for quotients that are actual indeterminate forms:

$$\left[\frac{0}{0} \right] \quad \text{and} \quad \left[\frac{\infty}{\infty} \right]$$

- The theorem prescribes computing the quotient of the derivatives, not the derivative of the quotient
- If the limit of f'/g' does not exist, nothing can be said about the limit of f/g .

2 Limit of the derivative and differentiability

Theorem 2: limit of the derivative

If $f : [a, b) \rightarrow \mathbb{R}$ is continuous at a , differentiable in (a, b) , and $\lim_{x \rightarrow a^+} f'(x) = m \in \mathbb{R}^*$, then $f'_+(a) = m$.

If the function is continuous at a and the right-hand limit of the derivative exists, then the right derivative exists and coincides with that limit. An analogous statement of the theorem holds for the left derivative and hence for the derivative.

Proof. Let $h > 0$. Applying Lagrange's theorem to f on the interval $[a, a + h]$, we obtain that there exists $t_h \in [a, a + h]$ such that

$$\frac{f(a + h) - f(a)}{h} = f'(t_h)$$

As $h \rightarrow 0^+$ we have $t_h \rightarrow a^+$, hence by the hypothesis of the theorem $f'(t_h) \rightarrow m$ as $t_h \rightarrow a^+$. Consequently, there exists

$$f'_+(a) = \lim_{h \rightarrow 0^+} \frac{f(a + h) - f(a)}{h} = m$$

□

Under the hypotheses of the theorem it is therefore possible to compute the right derivative, the left derivative or the derivative without using the limit of the difference quotient

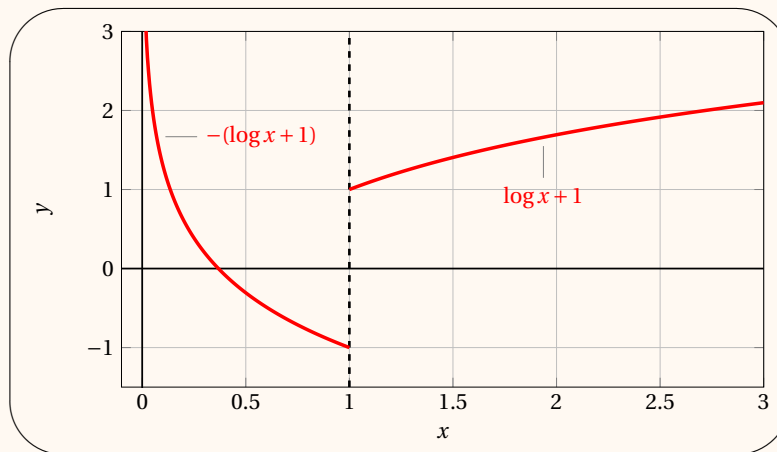
Example 7: differentiability

Suppose we want to study whether or not the following function is differentiable:

$$f(x) = x \cdot |\log x|$$

defined for $x > 0$. For $x \neq 1$, where $\log x = 0$, we have:

$$f(x) = \begin{cases} x \cdot \log x, & x > 1 \\ -x \cdot \log x, & 0 < x < 1 \end{cases} \quad \text{hence} \quad f'(x) = \begin{cases} \log x + 1, & x > 1 \\ -(\log x + 1), & 0 < x < 1 \end{cases}$$

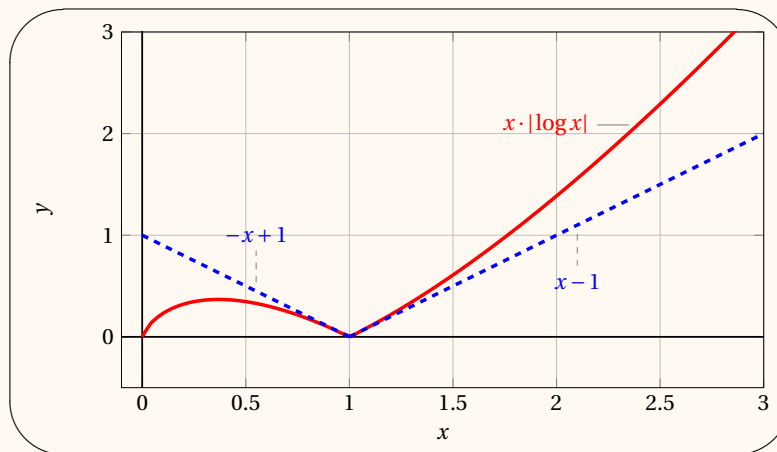


At $x = 1$ the function $f(x)$ is continuous, and we have:

$$\lim_{x \rightarrow 1^+} f'(x) = \lim_{x \rightarrow 1^+} (\log x + 1) = 1 \quad \text{and} \quad \lim_{x \rightarrow 1^-} f'(x) = \lim_{x \rightarrow 1^-} (-\log x - 1) = -1$$

hence the theorem is applicable (from the right and from the left) and we have:

$f'_+(1) = 1, f'_-(1) = -1$, so that $x = 1$ is a corner point and $f'(1)$ does not exist



Example 8: differentiability

Suppose we want to study whether or not the following function is differentiable:

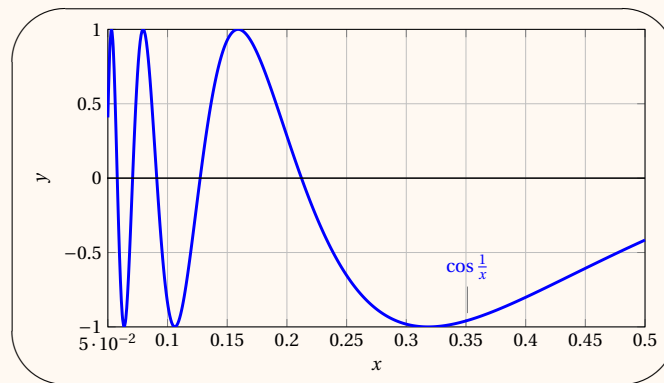
$$f(x) = x^2 \cdot \sin\left(\frac{1}{x}\right)$$

defined for $x \neq 0$. We know that

$$\lim_{x \rightarrow 0} x^2 \cdot \sin\left(\frac{1}{x}\right) = 0$$

and we can extend f by continuity by defining $f(0) = 0$, and f turns out to be continuous also at 0. For $x \neq 0$, we have:

$$f'(x) = 2x \cdot \sin\left(\frac{1}{x}\right) + x^2 \cdot \cos\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right) = \underbrace{2x \cdot \sin\left(\frac{1}{x}\right)}_{\rightarrow 0 \text{ as } x \rightarrow 0} - \cos\left(\frac{1}{x}\right)$$

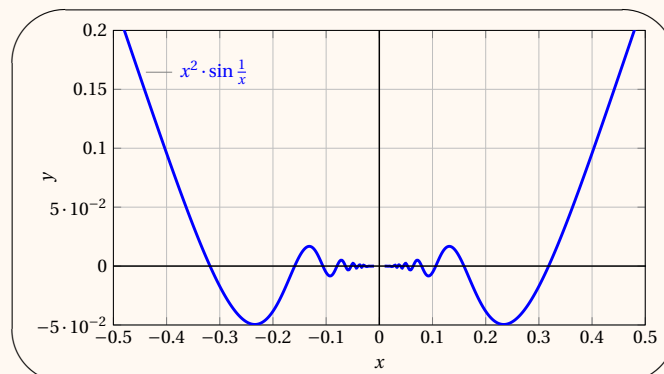


Near zero, the function $\cos \frac{1}{x}$, like the function $\sin \frac{1}{x}$, cannot be drawn since it has infinitely many oscillations. Hence the hypotheses of the theorem are not satisfied, since

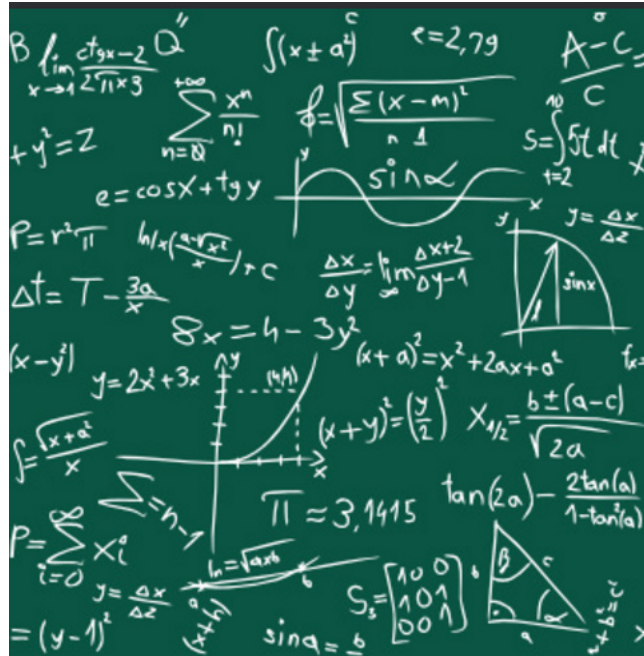
$$\lim_{x \rightarrow 0} f'(x) \text{ does not exist, } \lim_{x \rightarrow 0^-} f'(x) \text{ does not exist, } \lim_{x \rightarrow 0^+} f'(x) \text{ does not exist}$$

However, this does not allow us to conclude that the function is not differentiable at $x_0 = 0$. To compute the derivative we need to use the limit of the difference quotient. We have

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - 0}{h} = \lim_{h \rightarrow 0} h \cdot \sin\left(\frac{1}{h}\right) = 0 \text{ hence the function is differentiable at } x = 0$$



Second derivative



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Second derivative and second derivative function

- We can now ask whether the function $f'(x)$ is in turn differentiable (at a point or on an interval).

Definition 1: second derivative

Let $f : (a, b) \rightarrow \mathbb{R}$, $x_0 \in (a, b)$ and $f' : (x_0 - \delta, x_0 + \delta) \rightarrow \mathbb{R}$ with $\delta > 0$; f is said to be twice differentiable at x_0 if the following limit exists and is finite

$$\lim_{h \rightarrow 0} \frac{f'(x_0 + h) - f'(x_0)}{h}$$

this limit is called the second derivative of f at x_0 .

- To denote the second derivative function the following notations are used:

$$\underbrace{f''(x_0)}_{\text{Lagrange's notation}} \quad \underbrace{\ddot{f}(x_0)}_{\text{Newton's notation}} \quad \underbrace{\left. \frac{d^2 f}{dx^2} \right|_{x=x_0} \quad \text{and} \quad \left. \frac{d^2 y}{dx^2} \right|_{x=x_0}}_{\text{Leibniz's notation}}$$

Definition 2: second derivative function

If a derivative function f' is differentiable at every point of the interval (a, b) , the function

$$f'' : (a, b) \rightarrow \mathbb{R}, \quad f'' : x \mapsto f''(x)$$

is called the **second derivative function** of f .

- To denote the second derivative function the following notations are used:

$$\underbrace{f''(x)}_{\text{Lagrange's notation}} \quad \underbrace{\ddot{f}(x)}_{\text{Newton's notation}} \quad \underbrace{\frac{d^2 f}{dx^2}, \quad \frac{d^2 f(x)}{dx^2} \quad \text{and} \quad \frac{d^2 y}{dx^2}}_{\text{Leibniz's notation}}$$

1.1 Derivative function of order n

- In a completely analogous way one defines the derivative of order n , or n -th derivative, and the n -th derivative function, which are denoted by the symbols:

$$\underbrace{f^{(n)}(x_0), \quad f^{(n)}(x)}_{\text{Lagrange's notation}} \quad \underbrace{\left. \frac{d^n f}{dx^n} \right|_{x=x_0} \quad \text{and} \quad \left. \frac{d^n y}{dx^n} \right|_{x=x_0}, \quad \frac{d^n f}{dx^n}, \quad \frac{d^n f(x)}{dx^n} \quad \text{and} \quad \frac{d^n y}{dx^n}}_{\text{Leibniz's notation}}$$

1.2 Geometric meaning of the second derivative

While the first derivative has, as its geometric meaning, the slope of the graph, the second derivative represents the **rate of change of this slope** and therefore is a measure of the **degree to which the graph deviates from a straight line**.

- We start by considering the family of functions that satisfy the conditions:

$$f(0) = f'(0) = 0, \quad f''(0) \geq 0$$

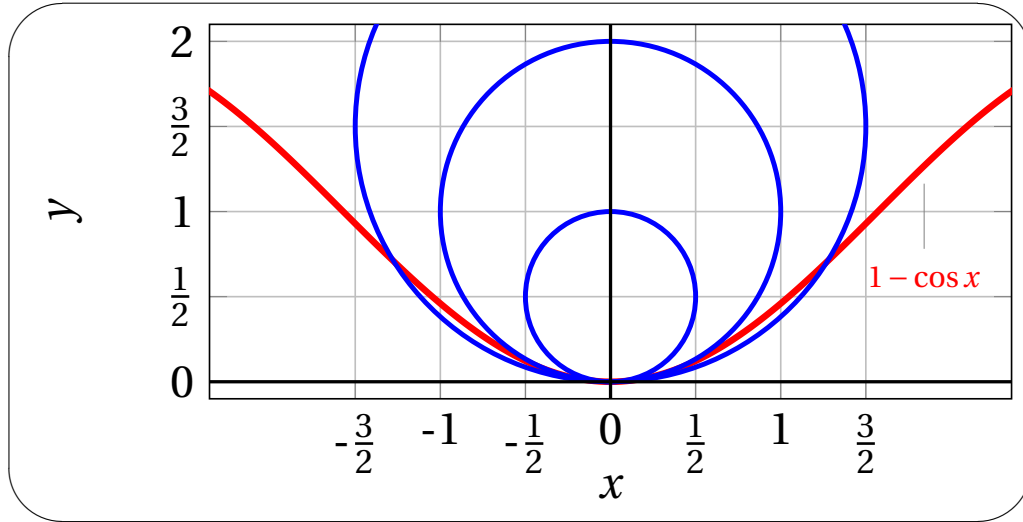
and the family of semicircles with center on the y -axis, tangent to the graph of f at the origin

$$c_r(x) = r - \sqrt{r^2 - x^2}, \quad \text{where } r > 0 \text{ is the radius}$$

For every $r \in \mathbb{R}_+$, we have:

$$c'_r(0) = c_r(0) = 0 \quad \text{since} \quad c'_r(x) = \frac{x}{\sqrt{r^2 - x^2}}$$

For example, the function $f(x) = 1 - \cos x$ with $f(0) = f'(0) = 0$ and the circles with radii $r \in \{0.5, 1, 1.5\}$ are:



- Among these semicircles we want to select the one that not only has the same tangent at $x = 0$ but also the same rate of change of the slope at $x = 0$. That is, we want to choose r so that:

$$c''_r(0) = f''(0) \tag{1}$$

We have:

$$c'_r(x) = \frac{x}{\sqrt{r^2 - x^2}} \quad \text{and} \quad c''_r(x) = \frac{r^2}{(r^2 - x^2)^{3/2}}$$

hence, if we want (1) to be satisfied, we need to choose r so that:

$$\underbrace{\frac{1}{r}}_{=c_r''(0)} = f''(0) \quad (2)$$

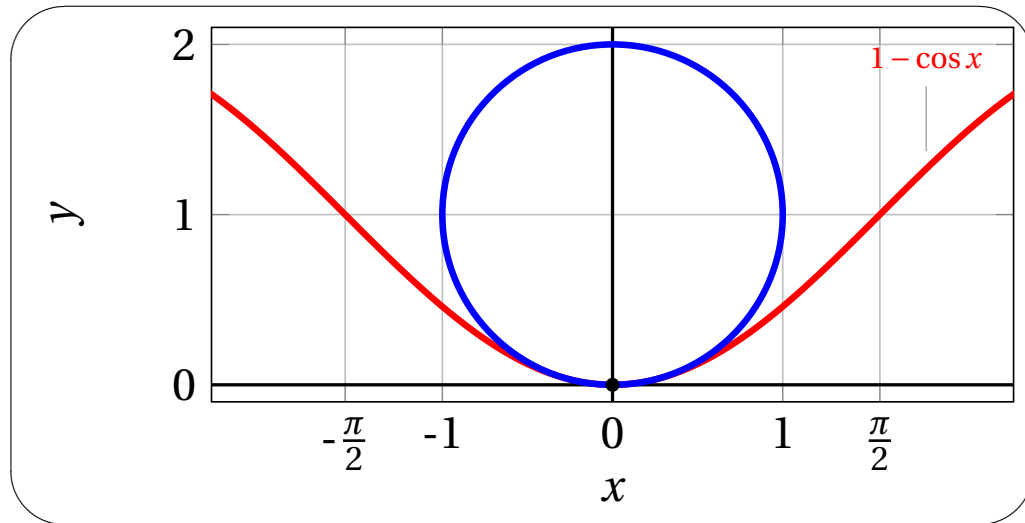
Equation (2) expresses the geometric meaning of the second derivative at $x = 0$ for the family of functions that satisfy the conditions $f(0) = f'(0) = 0$, $f''(0) \geq 0$. That is, $f''(0)$ represents the reciprocal of the radius of the semicircle that best approximates f at $x = 0$.

- Let us go back to the function $f(x) = 1 - \cos x$; we have

$$f'(x) = \sin x \quad \text{and} \quad f''(x) = \cos x$$

hence

$$f''(0) = 1 \quad \text{and} \quad \frac{1}{r} = 1 \quad \text{hence} \quad r = 1$$



Repeating an analogous construction at a generic point $(x, f(x))$ of the graph of f (without assuming a priori $f(x) = f'(x) = 0$) one finds that the **circle that best approximates the graph of the function at the point has radius $r(x)$** given by:

$$\frac{1}{r(x)} = \frac{|f''(x)|}{\left(1 + (f'(x))^2\right)^{3/2}} \quad (3)$$

The value $\frac{1}{r(x)}$ is called the **curvature** (of the graph) of f at x and the value $r(x)$ is the **radius of curvature**.

Example 1: radius of curvature and curvature

Consider the (parametric) parabola:

$$f(x) = a x^2 \text{ with } a \in \mathbb{R} \text{ we have } f'(x) = 2 a x \quad \text{and} \quad f''(x) = 2 a$$

and the following curvature:

$$\frac{1}{r(x)} = \frac{2|a|}{(1 + 4 a^2 x^2)^{3/2}}$$

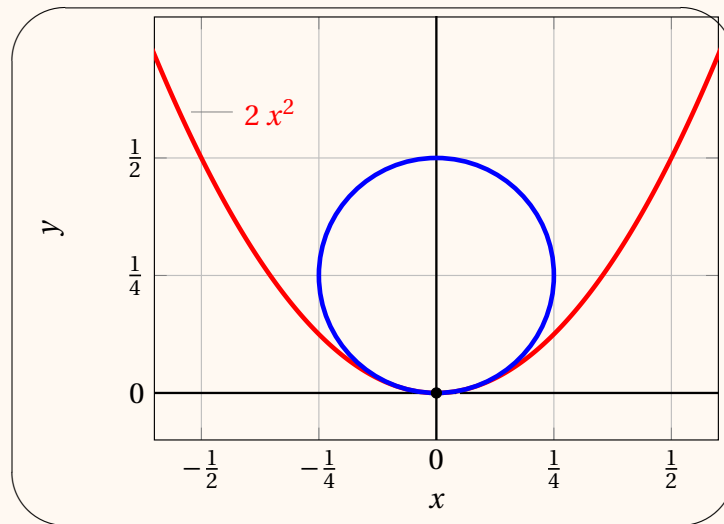
Note that the curvature is maximum at $x = 0$, i.e., at the vertex.

Let us now consider the parabola

$$y = 2 x^2$$

and compute the radius $r(x)$ of the circle that best approximates the graph at the points with abscissa $x = 0$ and $x = \frac{1}{2}$. We have:

$$\frac{1}{r(0)} = \frac{2|2|}{(1 + 4 \cdot 2^2 \cdot 0^2)^{3/2}} = 4 \quad \text{hence} \quad r(0) = \frac{1}{4}$$



$$\frac{1}{r\left(\frac{1}{2}\right)} = \frac{2|2|}{\left(1 + 4 \cdot 2^2 \left(\frac{1}{2}\right)^2\right)^{3/2}} = \frac{4}{5^{3/2}} \approx 0.3577 \quad \text{hence} \quad r\left(\frac{1}{2}\right) \approx 2.7950$$

2 Second derivative, concavity and convexity

- We will now see how to make this idea of curvature of the graph more precise, qualitatively and quantitatively, through the concept of **convexity**.

2.1 Convex sets

Definition 3: convex set

A set $F \subseteq \mathbb{R}^2$ (a portion of the Euclidean space) is **convex** if for every pair of points $P_a, P_b \in F$ the segment joining P_a to P_b (called a **chord**) is entirely contained in F .

Given two points P_a and P_b and a value λ between 0 and 1, i.e., $0 \leq \lambda \leq 1$, the point:

$$P_c = \lambda P_a + (1 - \lambda) P_b$$

is called a **convex (linear) combination** of the points P_a and P_b . As λ varies, the points P_c move along the segment joining P_a to P_b .

Definition 4: convex set – equivalent definition

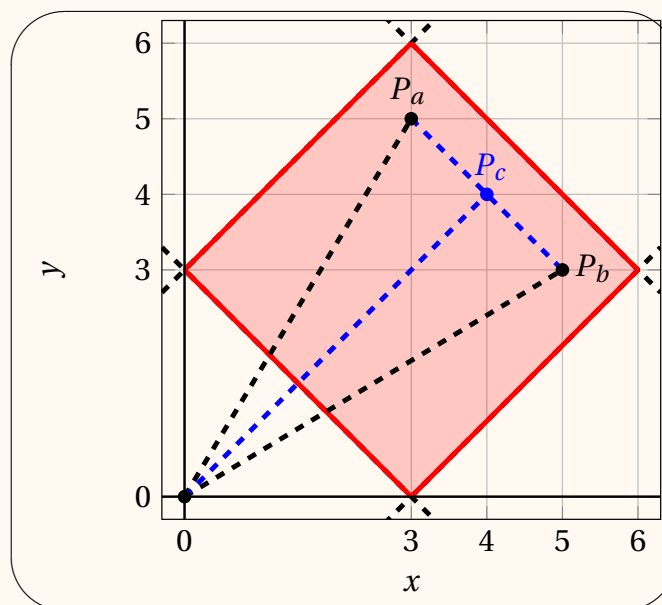
A set $F \subseteq \mathbb{R}^2$ is **convex** if:

$$\forall P_a, P_b \in F, \quad 0 \leq \lambda \leq 1, \quad P_c = \lambda P_a + (1 - \lambda) P_b \in F$$

Example 2: convex set

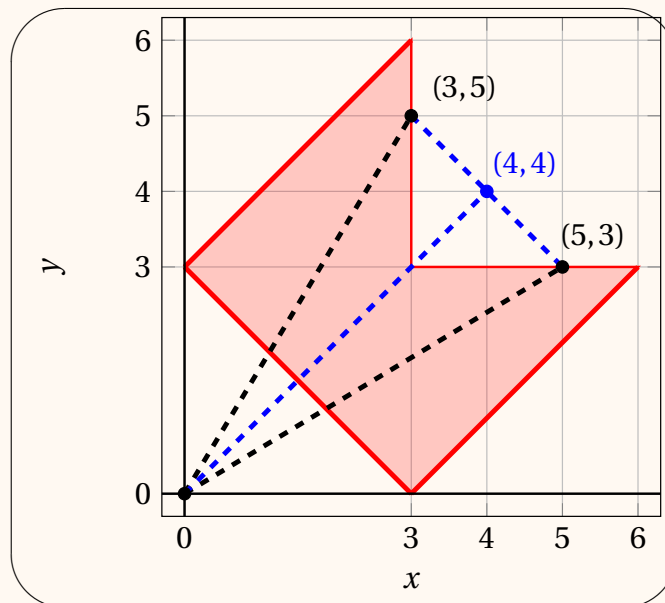
Consider, for example, the following convex set given by the intersection of 4 half-planes:

$$F = \{ (x, y) \in \mathbb{R}^2 : \begin{array}{l} x + y \geq 3, \quad x + y \leq 9, \\ -x + y \leq 3, \quad -x + y \geq -3 \end{array} \}$$



Example 3: non-convex set

Let us now consider the following set:



The set is not convex since, for example, the convex combination of the points $(5,3)$ and $(3,5)$ with $\lambda = \frac{1}{2}$, that is, the point:

$$\left(\frac{1}{2} \cdot 5 + \frac{1}{2} \cdot 3, \frac{1}{2} \cdot 3 + \frac{1}{2} \cdot 5 \right) = (4, 4)$$

does not belong to the set.

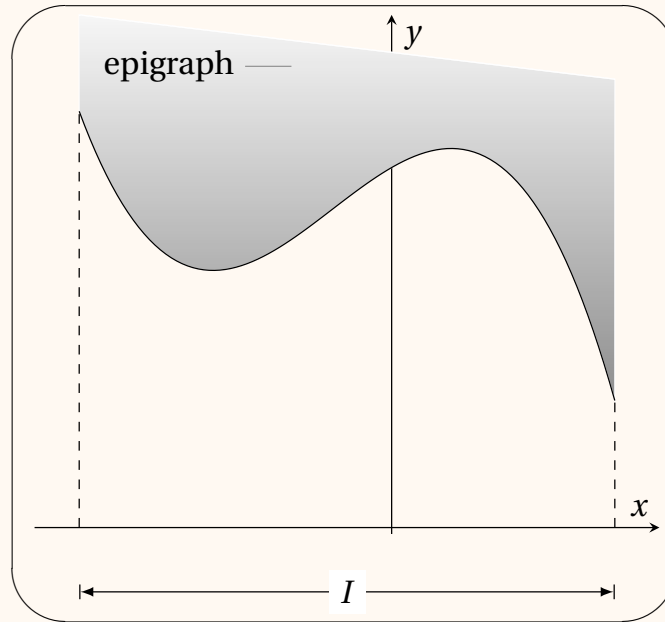
2.2 Convex/concave functions

Definition 5: epigraph

Consider a function $f : I \rightarrow \mathbb{R}$. The **epigraph** of f is the set:

$$\text{epi } f = \{(x, y) \in \mathbb{R}^2 : x \in I \text{ and } y \geq f(x)\}$$

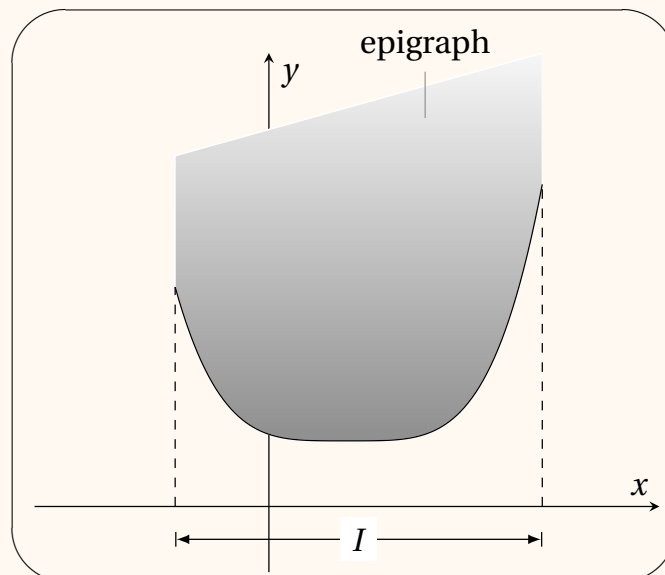
Example 4: epigraph



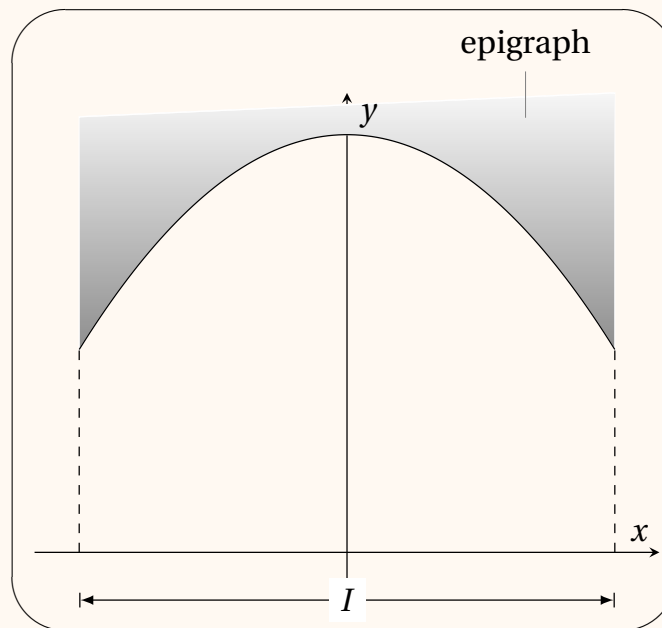
Definition 6: convex (concave) function

A function $f : I \rightarrow \mathbb{R}$ is **convex** on I if its epigraph is a convex set. A function is **concave** on I if $-f$ is convex on I .

Example 5: convex function



Example 6: concave function

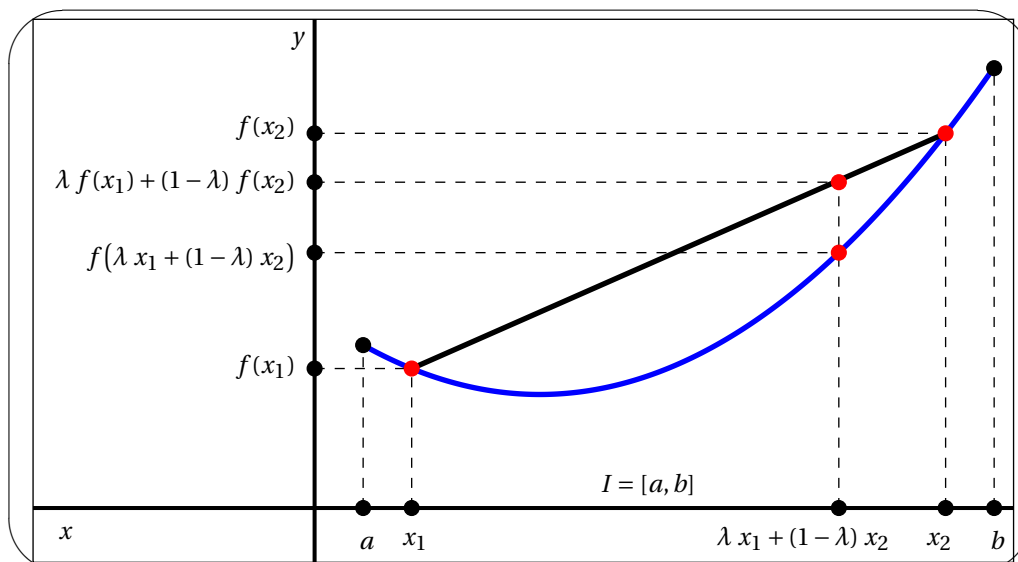


Definition 7: convex (concave) function – equivalent definition

A function $f : I \rightarrow \mathbb{R}$ is **convex** (**concave**) on I if for every pair of points $x_1, x_2 \in I$ the segment (“chord”) with endpoints $(x_1, f(x_1))$, $(x_2, f(x_2))$ has no points below (above) the graph of f .

The definition of convex function translates into the analytic inequality:

$$f(\lambda x_1 + (1 - \lambda) x_2) \leq \lambda f(x_1) + (1 - \lambda) f(x_2), \quad \forall x_1, x_2 \in I, 0 \leq \lambda \leq 1$$



- If in the previous inequality the strict $<$ always holds (with $\lambda \neq 0, 1$), the function is called **strictly convex**. For concave (or strictly concave) functions an analogous inequality holds with the sign $\geq$ (or $>$).
- Observe that as $0 \leq \lambda \leq 1$ varies: the point $\lambda x_1 + (1 - \lambda) x_2$ runs along the segment $[x_1, x_2]$ on the x -axis; the point $\lambda f(x_1) + (1 - \lambda) f(x_2)$ runs along the segment $[f(x_1), f(x_2)]$ on the y -axis; the point:

$$\left(\lambda x_1 + (1 - \lambda) x_2, f(\lambda x_1 + (1 - \lambda) x_2) \right)$$

runs along the graph of the function; the point:

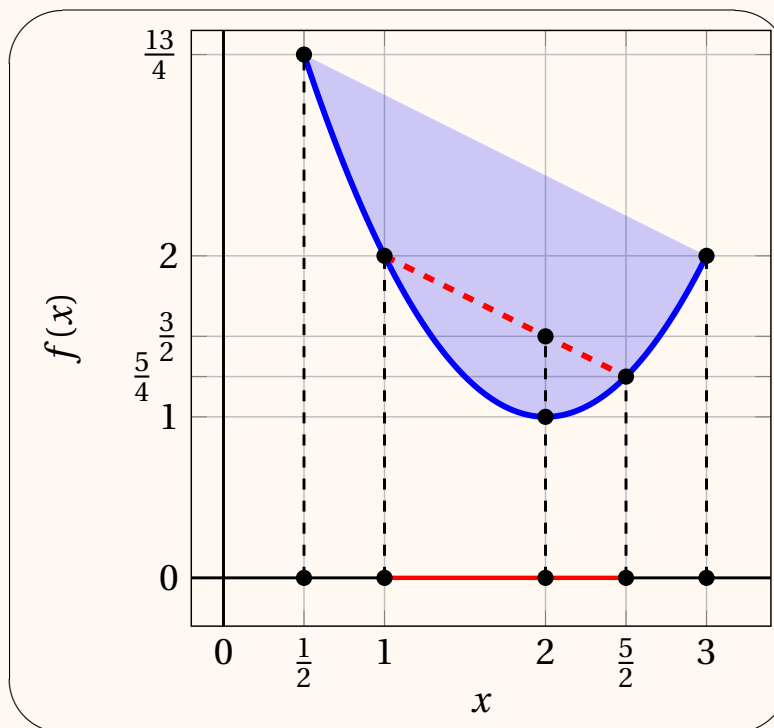
$$\left(\lambda x_1 + (1 - \lambda) x_2, \lambda f(x_1) + (1 - \lambda) f(x_2) \right)$$

runs along the segment with endpoints $(x_1, f(x_1))$ and $(x_2, f(x_2))$.

Example 7: graph of a convex function

Consider the convex function $f(x) = (x - 2)^2 + 1$ with domain $[\frac{1}{2}, 3]$ and take, for example, $x_1 = 1$ and $x_2 = \frac{5}{2}$ and $\lambda = \frac{1}{3}$. We have the following convex (linear) combination:

$$\left(\frac{1}{3} \cdot 1 + \frac{2}{3} \cdot \frac{5}{2}, \frac{1}{3} \cdot 2 + \frac{2}{3} \cdot \frac{5}{4} \right) = \left(2, \frac{3}{2} \right)$$



- Note that the definition of convex function does not require a priori that the function be continuous or differentiable on an interval.

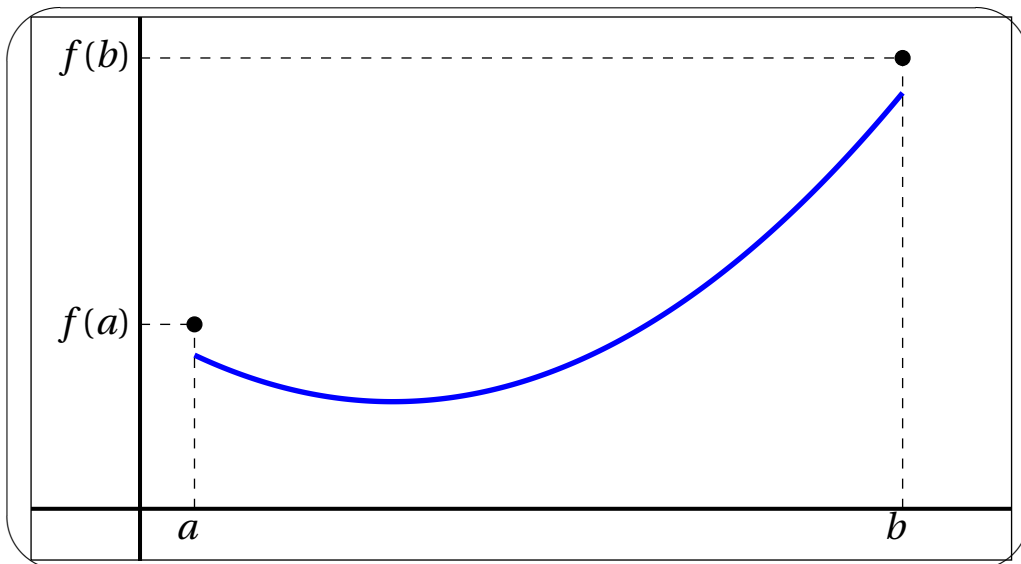
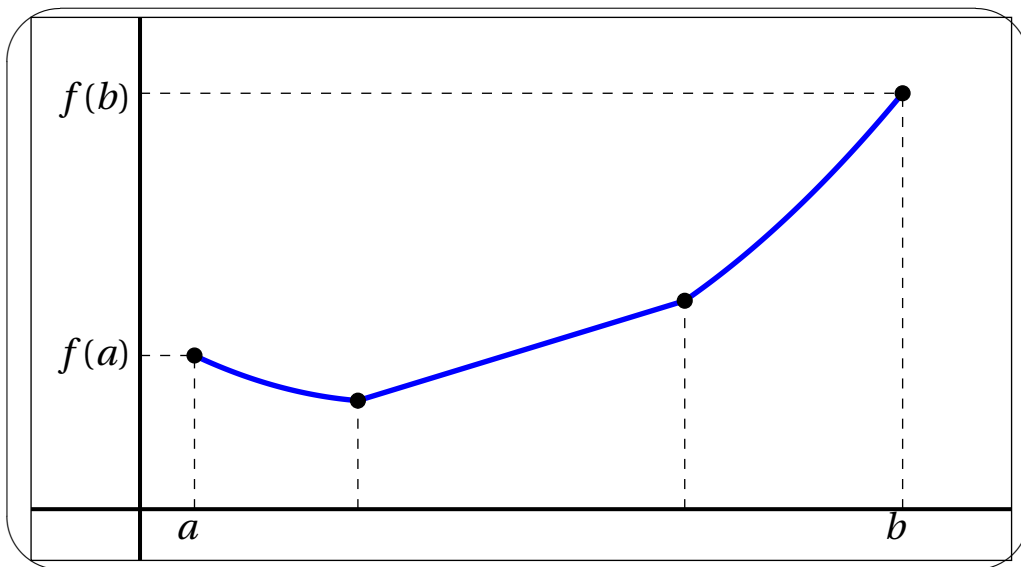
Theorem 1

A convex (or concave) function on an interval I is continuous on I , except possibly at the endpoints of the interval. Moreover, it has a right and a left derivative at every interior point of the interval.

Proof. Omitted



- Corner points inside the interval and points of discontinuity at the endpoints of the interval are the only irregular behaviors allowed for a convex or concave function, as shown by the following examples:



3 Convexity and derivatives

- If we know a priori that the function is differentiable once or twice on the interval considered, then convexity is related to the first and second derivative of the function.

Theorem 2

Let $f : (a, b) \rightarrow \mathbb{R}$.

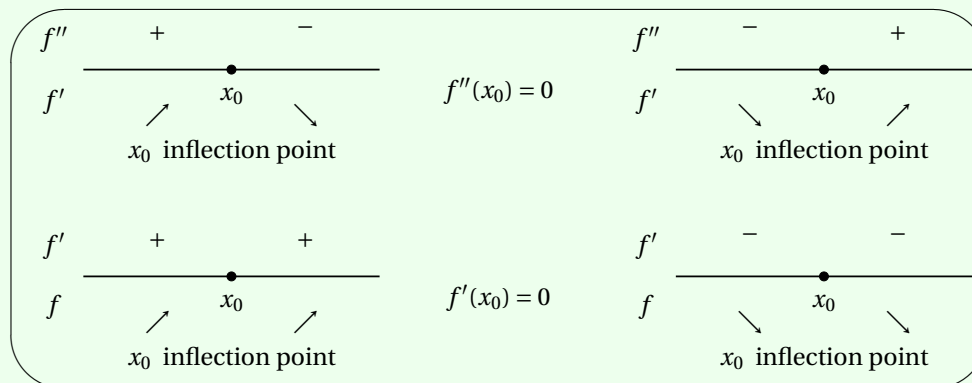
- If f is differentiable in (a, b) , then f is convex (concave) on (a, b) if and only if f' is increasing (decreasing) on (a, b) .
- If f is twice differentiable in (a, b) , then f is convex (concave) on (a, b) if and only if:

$$f''(x) \geq 0 \ (\leq 0), \quad \forall x \in (a, b)$$

- The theorem is modified in the obvious way for strictly convex or concave functions.

Proof. We do not prove part (a). Part (b) follows from (a) by the monotonicity test applied to f' . □

As a consequence of this theorem, the study of the sign of the second derivative allows us to decide on the convexity or concavity of a function (**check**).



Example 8: convexity of exponential functions

The exponential functions

$$f(x) = a^x$$

are convex on $\mathbb{R}$, for any base $a > 0, a \neq 1$, since:

$$f'(x) = a^x \log a; \quad f''(x) = a^x \log^2 a > 0, \quad \forall x \in \mathbb{R}, \forall a > 0, a \neq 1$$

Example 9: convexity/concavity of logarithmic functions

The logarithmic functions

$$f(x) = \log_a x$$

are concave on $(0, +\infty)$ if $a > 1$ and convex on $(0, +\infty)$ if $0 < a < 1$, since:

$$f'(x) = \frac{1}{x \log a}; \quad f''(x) = -\frac{1}{x^2 \log a} \quad \begin{cases} < 0, \forall x > 0, & \text{if } a > 1 \\ > 0, \forall x > 0, & \text{if } 0 < a < 1 \end{cases}$$

Example 10: convexity/concavity of power functions

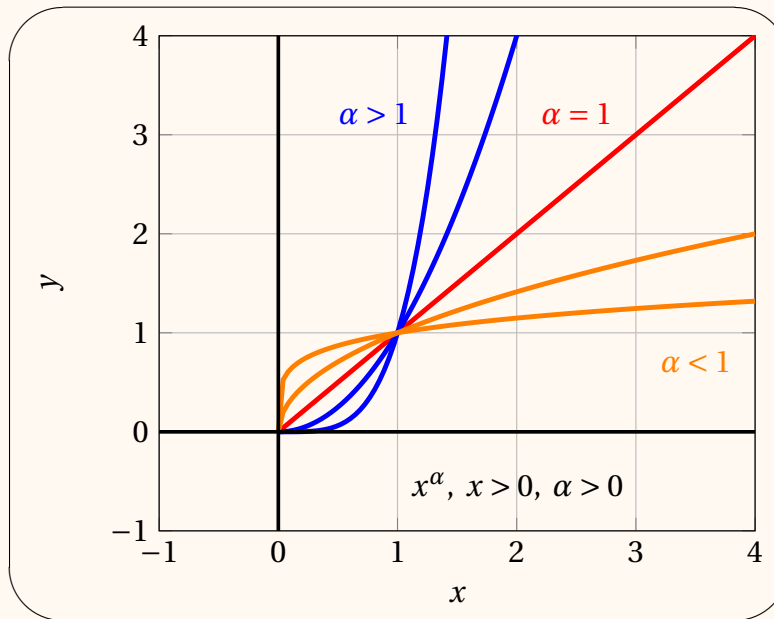
The power functions with real exponent α :

$$f(x) = x^\alpha$$

are convex on $(0, +\infty)$ if $\alpha > 1$ or $\alpha < 0$, and concave on $(0, +\infty)$ if $0 < \alpha < 1$, since:

$$f'(x) = \alpha x^{\alpha-1}; \quad f''(x) = \alpha(\alpha-1)x^{\alpha-2}$$

which has, for every $x > 0$, the sign of $\alpha(\alpha-1)$.



4 Convexity and tangent lines

- A useful geometric characterization of convexity involves the tangent lines to the graph of the function

Theorem 3

A function $f : (a, b) \rightarrow \mathbb{R}$, differentiable in (a, b) , is convex (concave) on (a, b) if and only if, for any choice of a point $x_0 \in (a, b)$, the graph of f lies on the whole of (a, b) above (below) the graph of its tangent line at $(x_0, f(x_0))$.

Proof. Omitted

□

Example 11: tangent lines and graphs of convex functions

Consider the function:

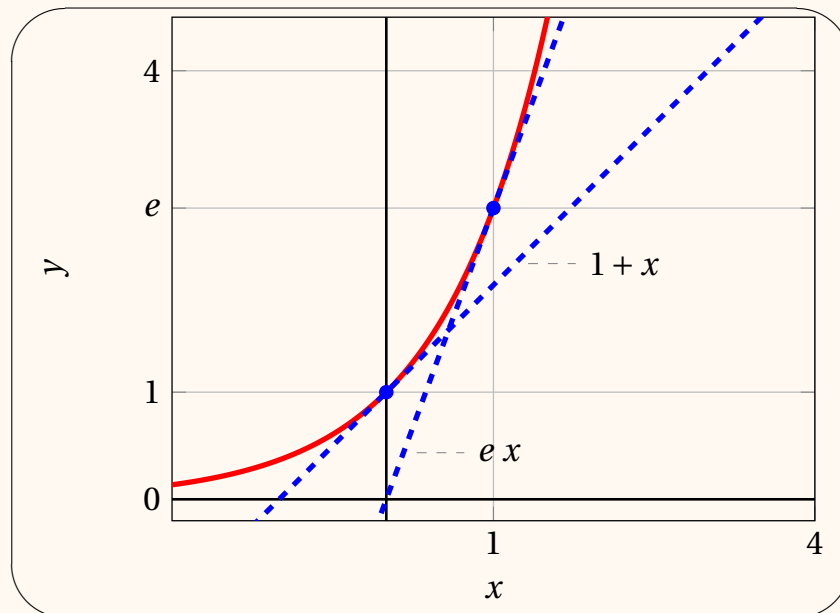
$$f(x) = e^x, \quad f'(x) = e^x, \quad f''(x) = e^x \text{ is convex on all of } \mathbb{R}$$

The tangent line to the graph of f , for example at $x = 0$, that is, at the point $(0, 1)$, is:

$$y = 1 + x \quad \text{hence} \quad e^x \geq 1 + x, \quad x \in \mathbb{R}$$

The tangent line to the graph of f , for example at $x = 1$, that is, at the point $(1, e)$, is:

$$y = e + e(x - 1) \quad \text{hence} \quad e^x \geq ex, \quad x \in \mathbb{R}$$



A (differentiable) convex function lies above its tangent lines and, at the same time, below its chords (definition of convexity).

- This allows us to conclude that, given any two points on the graph of a convex function, the graph between those two points lies entirely in the **triangle** whose sides are the chord joining them and the tangent lines to the graph at the two points.

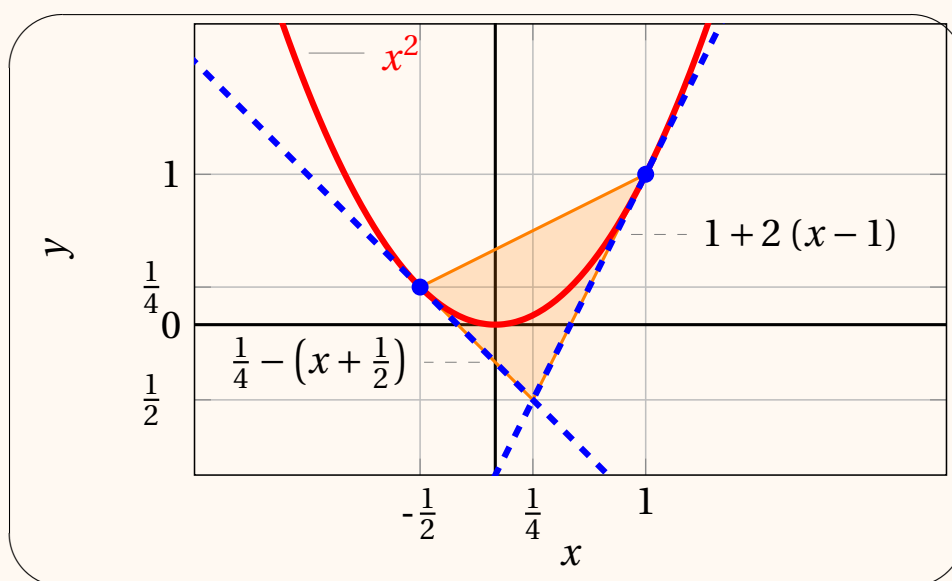
Example 12: tangent lines/chords and graphs of convex functions

Consider the function:

$$f(x) = x^2, \quad f'(x) = 2x, \quad f''(x) = 2 \quad \text{is convex on all of } \mathbb{R}$$

The tangent lines to the graph of f , for example at the points $x_1 = -0.5$ and $x_2 = 1$, are:

$$y = \frac{1}{4} - \left(x + \frac{1}{2}\right) \quad \text{and} \quad y = 1 + 2(x - 1)$$



5 Inflection points

- The direction of concavity of a function (that is, whether it is convex or concave) can change within its domain; this leads us to the concept of inflection point.

Definition 8: inflection point

Let $f : (a, b) \rightarrow \mathbb{R}$ be a function and let $x_0 \in (a, b)$ be a point of differentiability for f , or let $f'(x_0) = \pm\infty$. The point x_0 is called an **inflection point** of f if there exist a right neighborhood $(x_0, x_0 + h)$, $h > 0$, in which f is convex (concave) and a left neighborhood $(x_0 - h, x_0)$, $h > 0$, in which f is concave (convex).

- When crossing an inflection point, the second derivative of f (if it exists) changes sign. We then expect f'' to vanish at this point.

Theorem 4

Let x_0 be an inflection point of f ; if $f''(x_0)$ exists, then $f''(x_0) = 0$.

Proof. Note that, if we knew that f'' exists in a neighborhood of x_0 **and is continuous** at x_0 , then the thesis of the theorem would follow from the intermediate value theorem for continuous functions (applied to f''). The theorem can also be proved without these additional hypotheses (omitted). $\square$

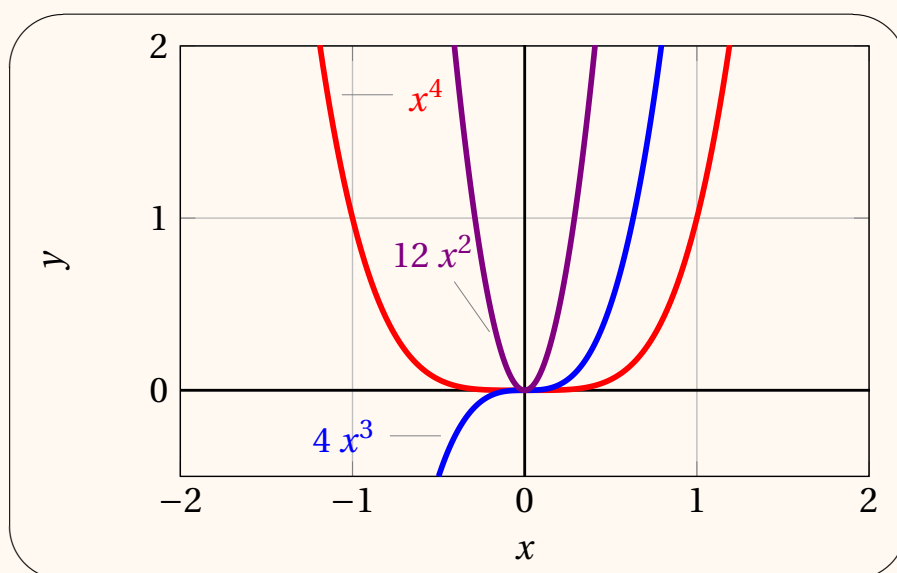
The converse of the implication stated by the theorem is not true: a point where the second derivative vanishes may not be an inflection point.

Example 13: points with zero second derivative that are not inflection points

Consider the function:

$$f(x) = x^4, \quad f'(x) = 4x^3, \quad f''(x) = 12x^2 \quad \text{is convex on all of } \mathbb{R}$$

Since $f'(x) > 0$ for $x > 0$, the function is increasing for $x > 0$, decreasing for $x < 0$ and has a minimum point at $x = 0$. We have $f''(0) = 0$, but the function is convex on all of $\mathbb{R}$, hence $x = 0$ **is not an inflection point**.



- The geometric meaning of inflection points is clarified by the following theorem.

Theorem 5

If $f : (a, b) \rightarrow \mathbb{R}$ is differentiable in (a, b) and $x_0 \in (a, b)$ is an inflection point, then the graph of $f(x)$ crosses its tangent line at $(x_0, f(x_0))$.

Proof. Let us draw the tangent line to the graph of $f(x)$ at the point with abscissa x_0 . If f is (for example) concave on (a, x_0) , since f is differentiable, the graph of f lies below the line on (a, x_0) ; on the other hand f is convex on (x_0, b) , so its graph lies above the line on (x_0, b) . Consequently, at x_0 the graph crosses the tangent line. $\square$

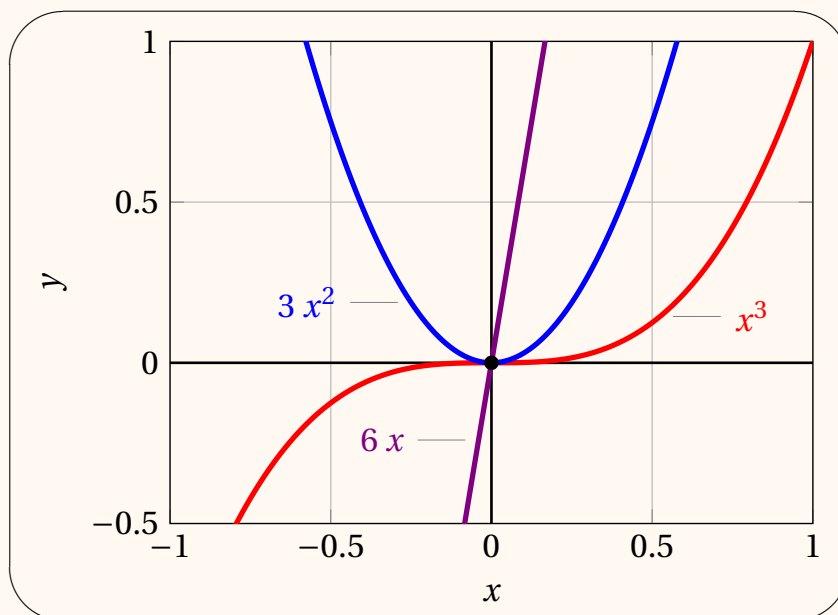
Example 14: inflection points (with horizontal tangent)

Consider the function:

$$f(x) = x^3, \quad f'(x) = 3x^2, \quad f''(x) = 6x$$

Since $f'(x) > 0$ for $x \neq 0$, the function is increasing on all of $\mathbb{R}$. It has a stationary point at $x = 0$, which, however, is not a maximum or minimum point, because the function is always increasing.

We have $f''(0) = 0$. Moreover, $f''(x) < 0$ for $x < 0$, hence it is concave for $x < 0$, and $f''(x) > 0$ for $x > 0$, hence it is convex for $x > 0$. Consequently, the point x_0 is **an inflection point with horizontal tangent**, and the graph of the function crosses its tangent line at $(0, 0)$, that is, the line $y = 0$.



Example 15: inflection points

Consider the function:

$$f(x) = e^{-x^2}, \quad f'(x) = -2x e^{-x^2}, \quad f''(x) = e^{-x^2} (4x^2 - 2)$$

We have $f'(x) > 0$ for $x < 0$, hence the function increases for $x \leq 0$; and $f'(x) < 0$ for $x > 0$, hence it decreases for $x \geq 0$, and moreover $f'(0) = 0$. Therefore it has a relative maximum point at $x = 0$.

We have:

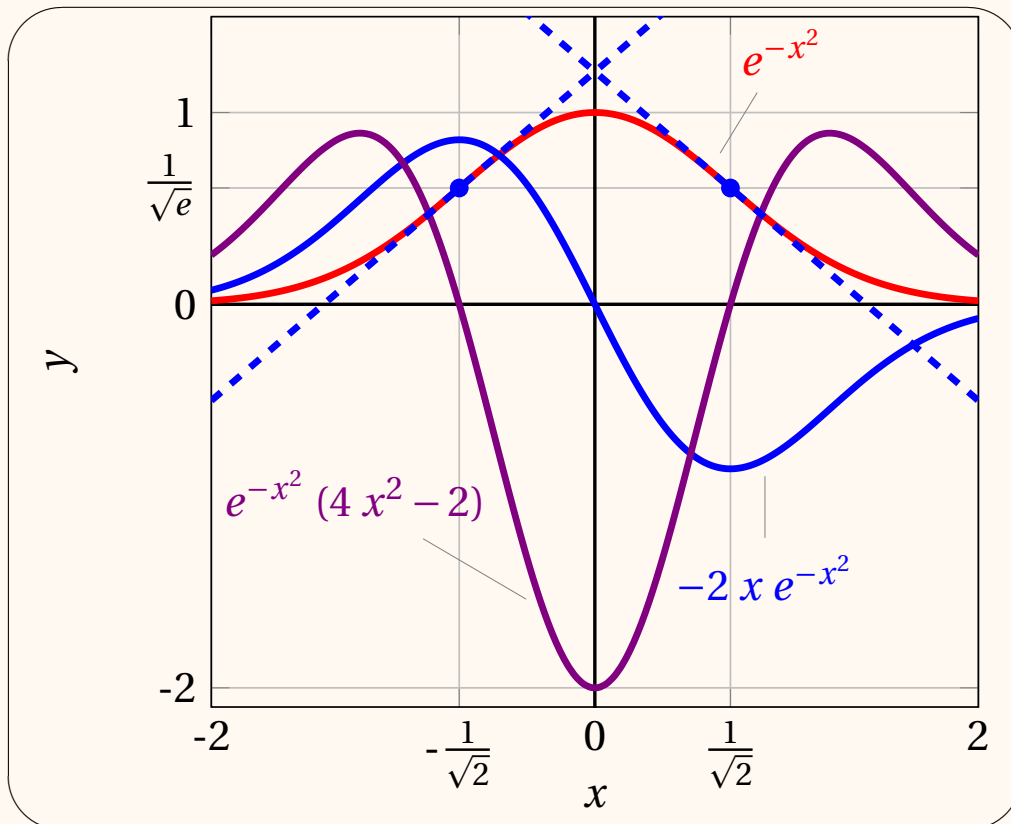
$$f''(x) = e^{-x^2} (4x^2 - 2) \geq 0 \text{ for } x^2 \geq \frac{1}{2} \text{ that is } x \in \left(-\infty, -\frac{1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, +\infty\right)$$

The function is convex for these values, and concave for $-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$. Hence it has inflection points at $x = \pm \frac{1}{\sqrt{2}}$, with tangent line of slope:

$$f' \left(\pm \frac{1}{\sqrt{2}} \right) = \mp \sqrt{\frac{2}{e}} \text{ since } -2 \left(\pm \frac{1}{\sqrt{2}} \right) e^{-\left(\pm \frac{1}{\sqrt{2}}\right)^2} = \mp \sqrt{2} \cdot \frac{1}{\sqrt{e}}$$

The tangent lines are:

$$y = \frac{1}{\sqrt{e}} + \sqrt{\frac{2}{e}} \left(x + \frac{1}{\sqrt{2}} \right) \quad \text{and} \quad y = \frac{1}{\sqrt{e}} - \sqrt{\frac{2}{e}} \left(x - \frac{1}{\sqrt{2}} \right)$$

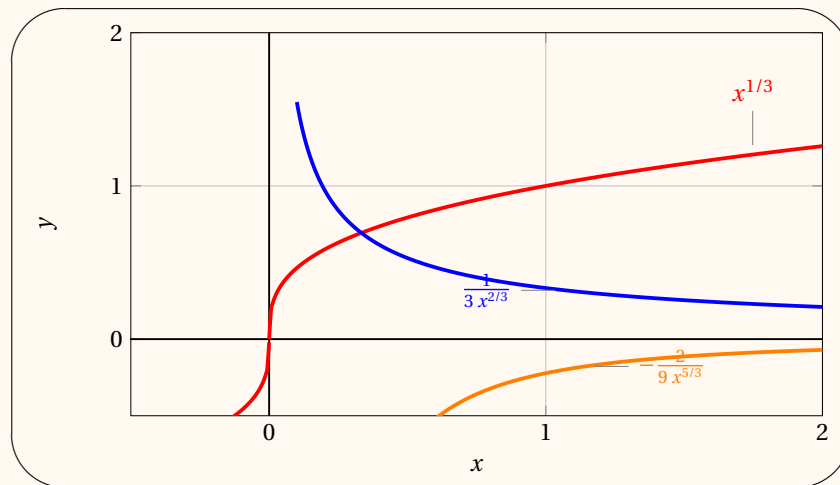


Example 16: inflection points (with vertical tangent)

Consider the function:

$$f(x) = x^{1/3}, \quad f'(x) = \frac{1}{3x^{2/3}}, \quad \text{for } x > 0, \quad f''(x) = -\frac{2}{9x^{5/3}}, \quad \text{for } x > 0$$

For $x_0 = 0$, we have seen that $f'(0) = +\infty$. In this case, however, $f''(0)$ does not exist and the function has an inflection point with vertical tangent.

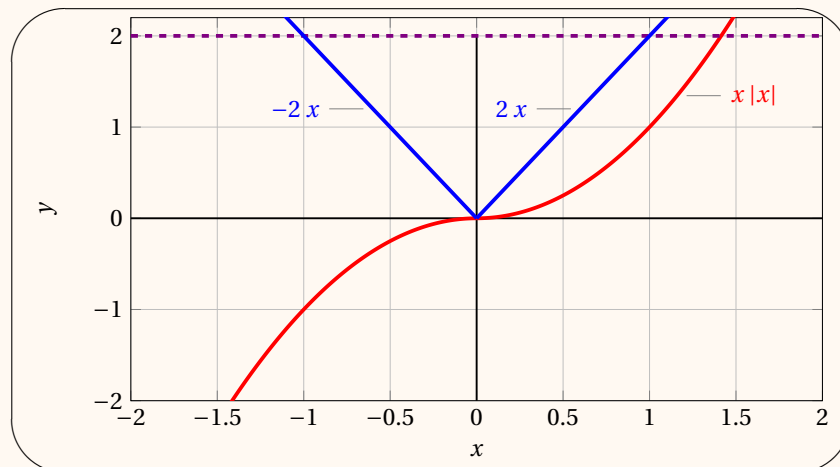


Example 17: inflection points (with horizontal tangent)

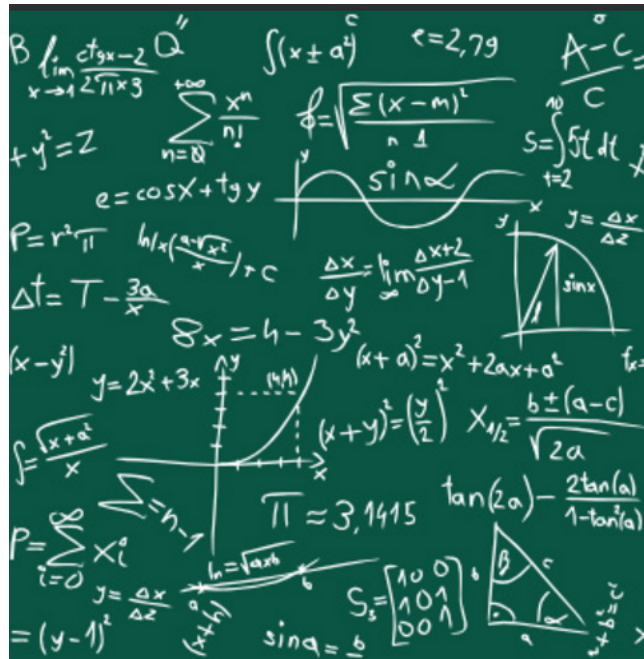
Consider the function:

$$f(x) = x|x|, \quad \text{with } x \neq 0, \quad f(x) = \begin{cases} x^2, & x > 0 \\ -x^2, & x < 0 \end{cases} \quad f'(x) = \begin{cases} 2x, & x > 0 \\ -2x, & x < 0 \end{cases} \quad f''(x) = \begin{cases} 2, & x > 0 \\ -2, & x < 0 \end{cases}$$

Hence, for $x > 0$ the function is convex and for $x < 0$ it is concave, and $f'(0) = 0$; therefore x_0 is an inflection point (but $f''(0)$ does not exist).



Differential calculus and approximations



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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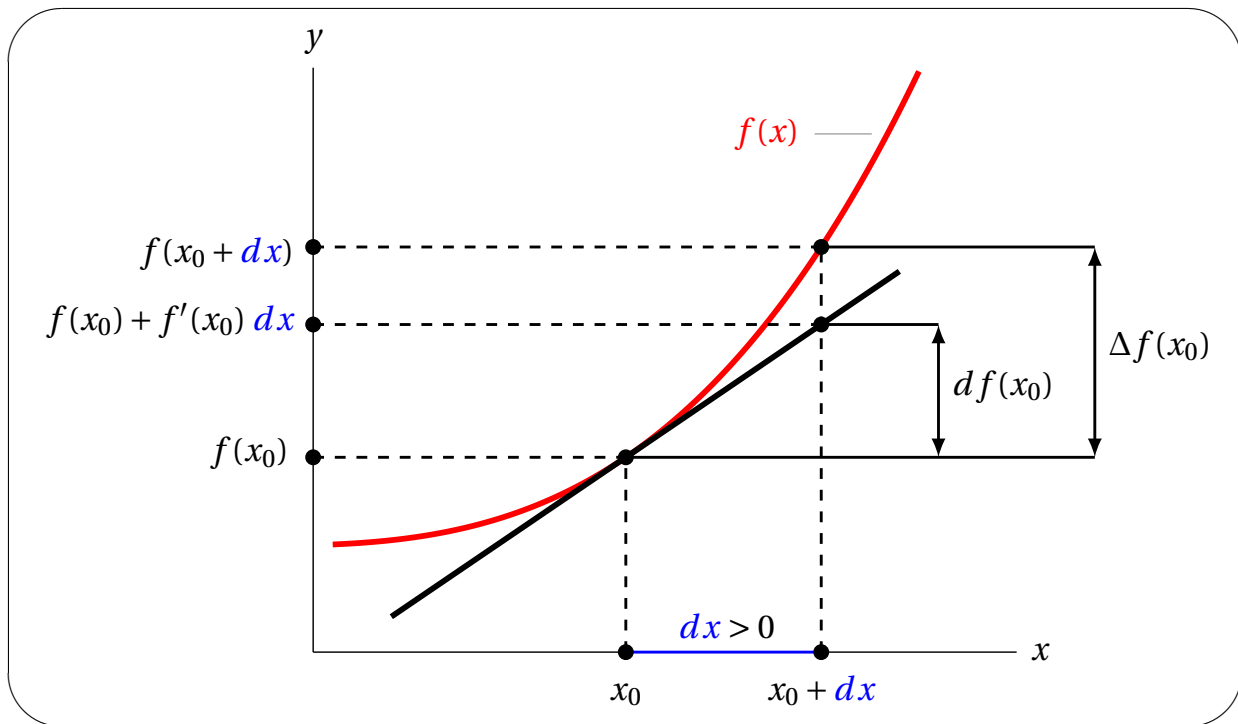
1 First-order asymptotic expansions

- A very frequent operation, both in mathematics and in its applications, is the linear approximation of a given differentiable function (an operation also called first-order asymptotic expansion).
- Let $f : (a, b) \rightarrow \mathbb{R}$ be a function differentiable at a point $x_0 \in (a, b)$. Consider the argument $x_0 + dx$ due to the increment/decrement dx with respect to x_0 . As a consequence, the values of f undergo the increment/decrement:

$$\Delta f(x_0) = f(x_0 + dx) - f(x_0)$$

which, in general, for fixed x_0 , is not proportional to dx , i.e., it is not linear with respect to dx . The increment/decrement evaluated along the tangent line is:

$$df(x_0) = f(x_0) + f'(x_0) dx - f(x_0) = f'(x_0) dx$$



- Given a function f differentiable at a point x_0 in the interior of its domain, the **linear approximation** consists in approximating the increment/decrement $\Delta f(x_0)$ of the values of the function due to an increment/decrement of the argument from x_0 to $x_0 + dx$, by replacing the function with its tangent line at the point $(x_0, f(x_0))$.
- The idea behind linear approximation therefore consists in estimating $\Delta f(x_0)$ with $df(x_0)$. To obtain good approximations, we consider increments that are very “small” in absolute value, that is, $|dx| \ll 1$.

Definition 1: differential of a function

Given a function $f : (a, b) \rightarrow \mathbb{R}$, differentiable at a point $x_0 \in (a, b)$, the **differential** of f at the point x_0 is the increment/decrement $df(x_0)$ of the values of the function from x_0 to $x_0 + dx$ evaluated along the tangent line:

$$df(x_0) = f'(x_0) dx$$

- Given two functions f and g , the rules for computing differentials are analogous to the differentiation rules:

$$d(f \pm g)(x_0) = df(x_0) \pm dg(x_0), \quad d(f \cdot g)(x_0) = g(x_0) df(x_0) + f(x_0) dg(x_0)$$

$$d\left(\frac{f}{g}\right)(x_0) = \frac{g(x_0) df(x_0) - f(x_0) dg(x_0)}{g^2(x_0)}$$

- In all types of approximation it is necessary to provide information (qualitative or quantitative) on the **error made**. What is the error made in the approximation? What is the value of $\Delta f(x_0) - df(x_0)$?

To answer, we observe that dx is equivalent to $x - x_0$, hence we have:

$$\Delta f(x_0) = f(x_0 + x - x_0) - f(x_0) = f(x) - f(x_0) \quad \text{and} \quad df(x_0) = f'(x_0) (x - x_0)$$

Moreover, from the definition of derivative we have:

$$\begin{aligned} \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0) &\Rightarrow \lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} - f'(x_0) \right) = 0 \\ &\Rightarrow \lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0) - f'(x_0)(x - x_0)}{x - x_0} \right) = 0 \end{aligned}$$

and from the definition of little- o we have:

$$\underbrace{f(x) - f(x_0)}_{=\Delta f(x_0)} - \underbrace{f'(x_0)(x - x_0)}_{=df(x_0)} = o(x - x_0) \quad \text{as } x \rightarrow x_0 \quad (1)$$

Hence the error $\Delta f(x_0) - df(x_0)$ tends to zero faster than $(x - x_0)$ as $x \rightarrow x_0$.

Definition 2: first-order asymptotic expansion

Given a function $f : (a, b) \rightarrow \mathbb{R}$, differentiable at a point $x_0 \in (a, b)$, the **first-order asymptotic expansion** (or linear expansion) of f at the point x_0 is the following expression:

$$f(x) = f(x_0) + f'(x_0) (x - x_0) + o(x - x_0) \quad \text{as } x \rightarrow x_0 \quad (2)$$

Expression (2) follows directly from (1), and in the case $x_0 = 0$ we have:

$$f(x) = f(0) + f'(0) x + o(x) \quad \text{as } x \rightarrow 0 \quad (3)$$

Example 1: first-order (linear) asymptotic expansions and approximation errors

We compute the first-order (linear) asymptotic expansion as $x \rightarrow 0$ of:

$$f(x) = \sqrt{1+x}$$

We have $x_0 = 0$ and moreover:

$$f'(x) = \frac{1}{2\sqrt{1+x}}, \quad f'(0) = \frac{1}{2} \quad \text{and} \quad f(0) = 1$$

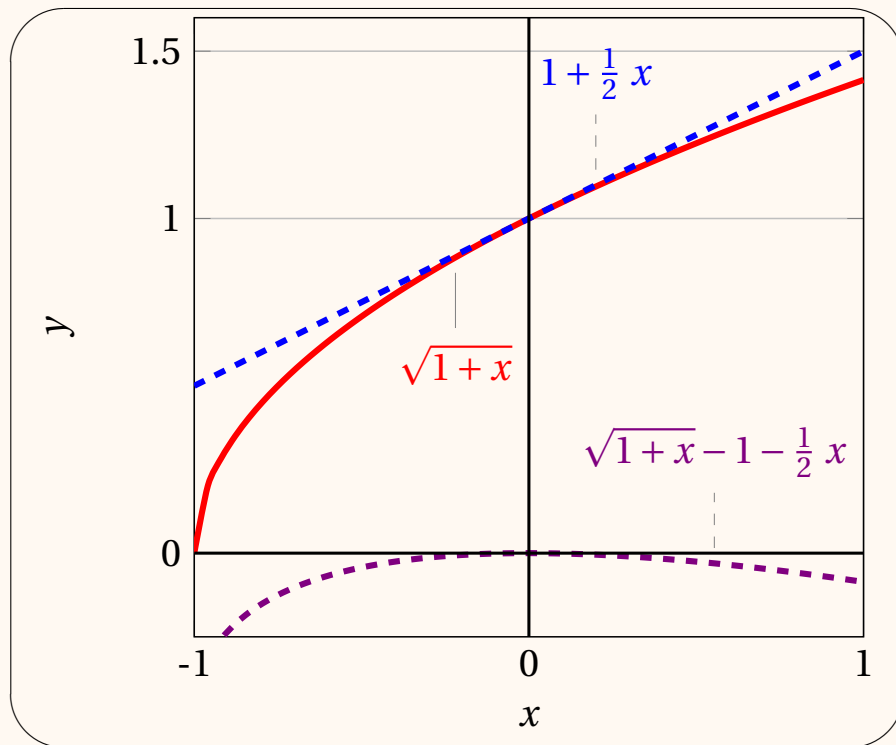
Hence we have the following first-order (linear) asymptotic expansion:

$$\sqrt{1+x} = 1 + \frac{1}{2}x + o(x) \quad \text{as } x \rightarrow 0$$

Moreover, the error made by the approximation is:

$$\underbrace{\sqrt{1+x} - 1}_{=\Delta f(0)=f(x)-f(0)} - \underbrace{\frac{1}{2}x}_{=df(0)=f'(0)x} = o(x) \quad \text{as } x \rightarrow 0$$

Graphically we have:



Example 2: first-order (linear) asymptotic expansions

We compute the first-order (linear) asymptotic expansion as $x \rightarrow 1$ of:

$$f(x) = \sqrt{1+x}$$

We have $x_0 = 1$ and moreover:

$$f'(x) = \frac{1}{2\sqrt{1+x}}, \quad f'(1) = \frac{1}{2\sqrt{2}} \quad \text{and} \quad f(1) = \sqrt{2}$$

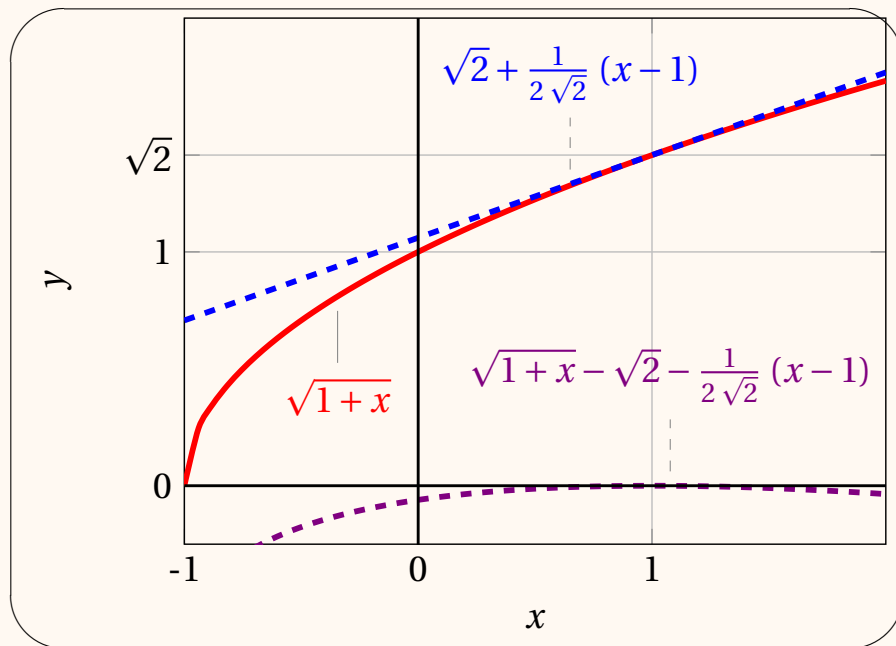
Hence we have the following first-order (linear) asymptotic expansion:

$$\sqrt{1+x} = \sqrt{2} + \frac{1}{2\sqrt{2}}(x-1) + o(x-1) \quad \text{as } x \rightarrow 1$$

Moreover, the error made by the approximation is:

$$\underbrace{\sqrt{1+x} - \sqrt{2}}_{=\Delta f(1)=f(x)-f(1)} - \underbrace{\frac{1}{2\sqrt{2}}(x-1)}_{=df(1)=f'(1)(x-1)} = o(x-1) \quad \text{as } x \rightarrow 1$$

Graphically we have:



2 Derivatives of polynomials

- A polynomial of degree n can be written as:

$$P_n(x) = \sum_{i=0}^n a_i \cdot x^i \quad \text{with } a_i \in \mathbb{R}, \text{ for } i \in \{0, 1, \dots, n\}, \text{ and } a_n \neq 0.$$

The value a_i is the **coefficient** of the monomial of degree i , with $i \in \{0, 1, \dots, n\}$, while x^i is the **literal part** of the monomial, a positive integer power of x . The value a_0 is the **constant term** of the polynomial.

The first and second derivative functions of a polynomial of degree $n \geq 2$ are:

$$P'_n(x) = \sum_{i=1}^n i a_i x^{i-1} \qquad P''_n(x) = \sum_{i=2}^n (i-1) \cdot i \cdot a_i x^{i-2}$$

$$P'_n(0) = a_1 \qquad P''_n(0) = 2 a_2$$

Example 3: first and second derivative of a polynomial

Consider the following polynomial of degree 5:

$$P_5(x) = 10 + 7x + 3x^2 - 6x^3 + 4x^4 - 2x^5$$

the first derivative function is:

$$P'_5(x) = 1 \cdot 7x^{1-1} + 2 \cdot 3x^{2-1} + 3 \cdot (-6)x^{3-1} + 4 \cdot 4x^{4-1} + 5 \cdot (-2)x^{5-1} = 7 + 6x - 18x^2 + 16x^3 - 10x^4$$

$$P'_5(0) = 7$$

the second derivative function is:

$$P''_5(x) = 1 \cdot 2 \cdot 3x^{2-2} + 2 \cdot 3 \cdot (-6)x^{3-2} + 3 \cdot 4 \cdot 4x^{4-2} + 4 \cdot 5 \cdot (-2)x^{5-2} = 6 - 36x + 48x^2 - 40x^3$$

$$P''_5(0) = 2 \cdot 3 = 6$$

The k -th derivative function of a polynomial of degree n with $k \in \{0, 1, \dots, n\}$ is:

$$P_n^{(k)}(x) = \sum_{i=k}^n \underbrace{(i-k+1) \cdots (i-2) \cdot (i-1) \cdot i}_{=\prod_{j=1}^k (i-j+1)} a_i x^{i-k}$$

$$P_n^{(k)}(0) = k! a_k$$

Example 4: third, fourth and fifth derivative of a polynomial

Consider the following polynomial of degree 5:

$$P_5(x) = 10 + 7x + 3x^2 - 6x^3 + 4x^4 - 2x^5$$

the third derivative function is:

$$P_5^{(3)}(x) = 1 \cdot 2 \cdot 3 \cdot (-6) x^{3-3} + 2 \cdot 3 \cdot 4 \cdot 4 x^{4-3} + 3 \cdot 4 \cdot 5 \cdot (-2) x^{5-3} = -36 + 96x - 120x^2$$

$$P_5^{(3)}(0) = 3! \cdot (-6) = 6 \cdot (-6) = -36$$

the fourth derivative function is:

$$P_5^{(4)}(x) = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 4 x^{4-4} + 2 \cdot 3 \cdot 4 \cdot 5 \cdot (-2) x^{5-4} = 96 - 240x$$

$$P_5^{(4)}(0) = 4! \cdot 4 = 24 \cdot 4 = 96$$

the fifth derivative function is:

$$P_5^{(5)}(x) = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot (-2) x^{5-5} = -240$$

$$P_5^{(5)}(0) = 5! \cdot (-2) = 120 \cdot (-2) = -240$$

- Given $x_0 \in \mathbb{R}$, equivalently, a polynomial of degree n can be written as:

$$P_n(x) = \sum_{i=0}^n a_i \cdot (x - x_0)^i \quad \text{with } a_i \in \mathbb{R}, \text{ for } i \in \{0, 1, \dots, n\}, \text{ and } a_n \neq 0.$$

The k -th derivative function, $k \in \{0, 1, \dots, n\}$, is in this case:

$$P_n^{(k)}(x) = \sum_{i=k}^n (i - k + 1) \cdots (i - 2) \cdot (i - 1) \cdot i \cdot a_i \cdot (x - x_0)^{i-k}$$

$$P_n^{(k)}(x_0) = k! \cdot a_k$$

3 Taylor formula/expansion with Peano remainder

- Linear approximation (or linearization) approximates a function with its tangent line at a point $(x_0, f(x_0))$, that is, a first-degree polynomial whose derivative equals that of the function at the point x_0 and whose constant term is the value of the function at x_0 .

For simplicity, we start by reasoning with $x_0 = 0$.

- With first-degree polynomials we have:

$$P_1(x) = a_0 + a_1 x, \quad P_1'(x) = a_1, \quad \text{setting } a_0 = f(0) \text{ and } a_1 = f'(0)$$

from (2) we obtain the first-order asymptotic expansion:

$$f(x) = \underbrace{a_0}_{=f(0)} + \underbrace{a_1}_{=f'(0)} x + o(x)$$

and the error made by using the first-degree polynomial instead of the function:

$$\underbrace{f(x) - a_0}_{\Delta f(0)} - \underbrace{a_1 x}_{df(0)} = o(x)$$

tends to zero faster than any linear function.

- We now want to generalize the procedure of “**linear approximation**” to that of “**polynomial approximation**”.

We ask: given a function, differentiable as many times as necessary, does there exist a polynomial of degree $n \geq 2$ that, in a neighborhood of a fixed point, approximates the function “better” than its tangent line? **We proceed in two steps.**

First step

- We identify a candidate polynomial to approximate the function “well”, by looking for a polynomial whose derivatives up to order n are all equal to those of the function f at the point $x_0 = 0$. The polynomial must be of degree n in order to have its n -th derivative equal to $f^{(n)}(0)$.

Theorem 1: Maclaurin polynomial

Given a function $f : (a, b) \rightarrow \mathbb{R}$ differentiable $n - 1$ times in (a, b) and n times at $0 \in (a, b)$, there exists one and only one polynomial $T_{n,f}$ of degree $\leq n$ with the property that:

$$T_{n,f}(0) = f(0), \quad T'_{n,f}(0) = f'(0), \quad \dots, \quad T_{n,f}^{(n)}(0) = f^{(n)}(0)$$

and this polynomial, called the **Maclaurin polynomial** of f of degree n , is:

$$T_{n,f}(x) = \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k \quad (\text{setting } f^{(0)} = f)$$

Proof. We prove that the Maclaurin polynomial of a function f satisfying the hypotheses of the theorem has the same derivatives at 0 as the function, up to order n . The k -th derivative at $x_0 = 0$ of the Maclaurin polynomial of degree n , with $k \in \{0, 1, \dots, n\}$, is:

$$T_{n,f}^{(k)}(0) = k! \cdot \frac{f^{(k)}(0)}{k!} = f^{(k)}(0) \quad \text{and with } k = 0 \text{ we have } T_{n,f}(0) = f(0)$$

Hence all the derivatives at 0 up to order n are equal to those of the function f (and it has the same value as the function at 0).

We now prove that this polynomial is unique. Let us now consider a generic polynomial of degree n :

$$P_n(x) = \sum_{i=0}^n a_i \cdot x^i \quad \text{with } a_i \in \mathbb{R}, \text{ for } i = 0, 1, \dots, n, \text{ and } a_n \neq 0.$$

Its k -th derivative at $x_0 = 0$, with $k \in \{0, 1, \dots, n\}$, is

$$P_n^{(k)}(0) = k! \cdot a_k$$

Hence, if the polynomial has the same k -th derivative as the function, we have:

$$k! \cdot a_k = f^{(k)}(0) \quad \text{which implies } a_k = \frac{f^{(k)}(0)}{k!}$$

that is, the coefficients of the Maclaurin polynomial. □

The expanded form of the Maclaurin polynomial of degree n of a function f is:

$$T_{n,f}(x) = f(0) + f'(0)x + \frac{1}{2} f''(0)x^2 + \frac{1}{3!} f'''(0)x^3 + \dots + \frac{1}{n!} f^{(n)}(0)x^n$$

- When the function we refer to is clear, for convenience we omit the subscript f of the Maclaurin polynomial and simply write $T_n(x)$.

Example 5: Computing the Maclaurin polynomial

We compute the Maclaurin polynomial of degree 3 of the function

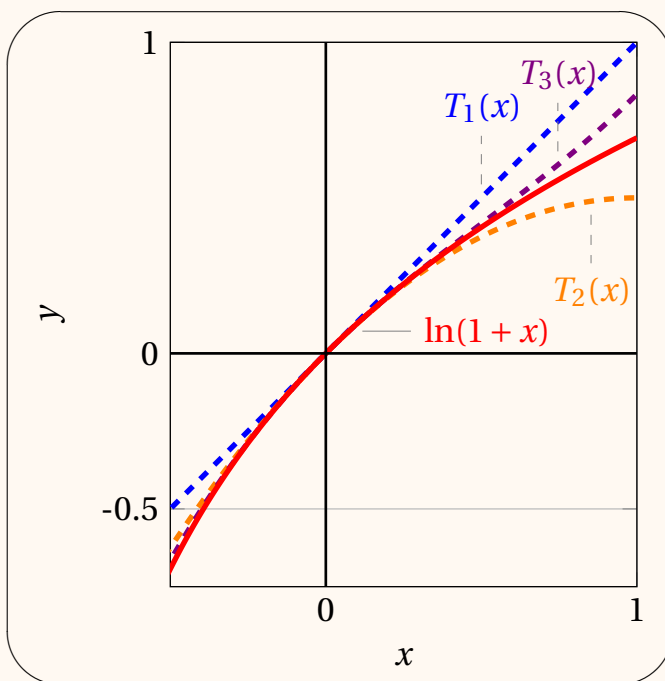
$$f(x) = \log(1+x)$$

We have $f(0) = 0$ and for the derivatives we have:

$$f^{(1)}(x) = \frac{1}{1+x}, \quad f^{(1)}(0) = 1, \quad f^{(2)}(x) = -\frac{1}{(1+x)^2}, \quad f^{(2)}(0) = -1, \quad f^{(3)}(x) = \frac{2}{(1+x)^3}, \quad f^{(3)}(0) = 2$$

Hence the Maclaurin polynomial of degree 3 of the function is:

$$T_3(x) = 0 + 1x + \frac{1}{2}(-1)x^2 + \frac{1}{6}2x^3 = x - \frac{1}{2}x^2 + \frac{1}{3}x^3$$



The fourth derivative is:

$$f^{(4)}(x) = -\frac{6}{(1+x)^4}, \quad f^{(4)}(0) = -6$$

Generalizing to the n -th derivative, it follows that the Maclaurin polynomial of degree n of the function is:

$$T_n(x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^{n-1} \frac{x^n}{n}$$

Example 6: Computing the Maclaurin polynomial

We compute the Maclaurin polynomial of degree 3 of the function

$$f(x) = (1+x)^\alpha$$

We have $f(0) = 1$ and for the derivatives we have:

$$f^{(1)}(x) = \alpha (1+x)^{\alpha-1}, \quad f^{(1)}(0) = \alpha, \quad f^{(2)}(x) = (\alpha-1)\alpha (1+x)^{\alpha-2}, \quad f^{(2)}(0) = (\alpha-1)\alpha$$

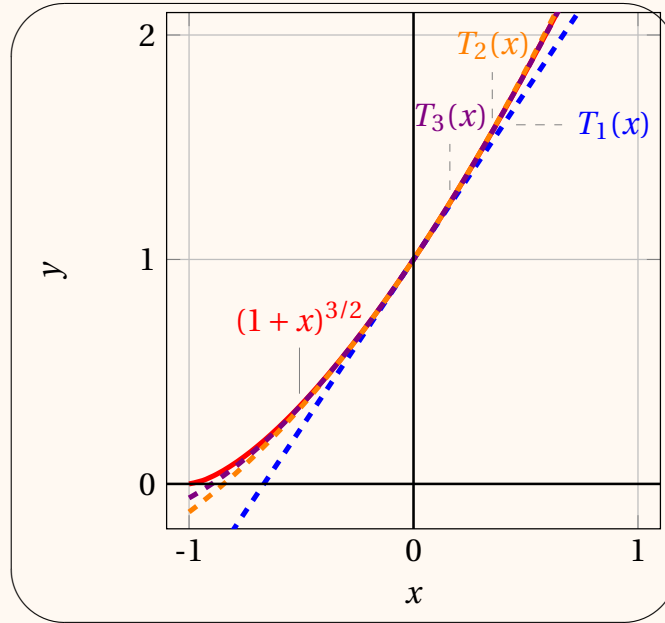
$$f^{(3)}(x) = (\alpha-2)(\alpha-1)\alpha (1+x)^{\alpha-3}, \quad f^{(3)}(0) = (\alpha-2)(\alpha-1)\alpha$$

Hence the Maclaurin polynomial of degree 3 of the function is:

$$T_3(x) = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2} x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!} x^3$$

For example, with $\alpha = \frac{3}{2}$ we have:

$$T_3(x) = 1 + \frac{3}{2} x + \frac{3}{8} x^2 - \frac{1}{16} x^3$$



The fourth derivative is:

$$f^{(4)}(x) = (\alpha-3)(\alpha-2)(\alpha-1)\alpha (1+x)^{\alpha-4}, \quad f^{(4)}(0) = (\alpha-3)(\alpha-2)(\alpha-1)\alpha$$

Generalizing to the n -th derivative, it follows that the Maclaurin polynomial of degree n of the function is:

$$T_n(x) = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2} x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!} x^3 + \dots + \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!} x^n$$

Second step

- We now prove that the Maclaurin polynomial approximates $f(x)$ “well” in a neighborhood of $x_0 = 0$. Precisely, the following theorem holds:

Theorem 2: Maclaurin formula of order n with Peano remainder

Let $f : (a, b) \rightarrow \mathbb{R}$ be differentiable $n - 1$ times in (a, b) and n times at $0 \in (a, b)$. Then

$$f(x) = T_{n,f}(x) + o(x^n) \text{ as } x \rightarrow 0$$

- The formula has the structure: **function to be approximated equals approximating polynomial plus approximation error**. The error is the term $o(x^n)$ and is called the Peano remainder. As $x \rightarrow 0$, the larger n is, the smaller the error.

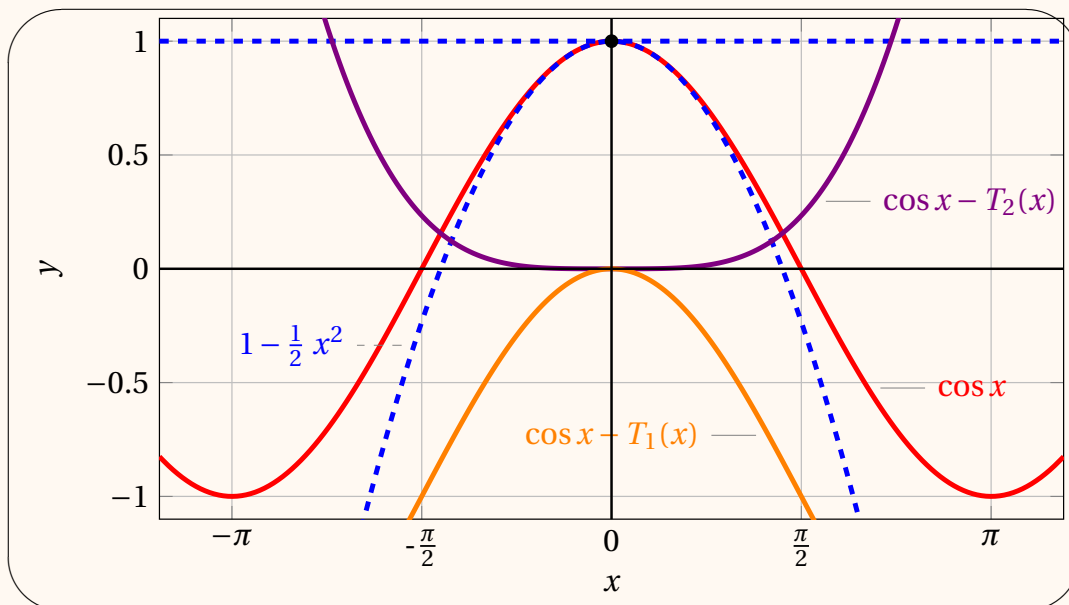
Example 7: errors of polynomial approximations

Consider $f(x) = \cos x$; we have:

$$f(0) = 1, \quad f^{(1)}(x) = -\sin x, \quad f^{(1)}(0) = 0, \quad f^{(2)} = -\cos x, \quad f^{(2)}(0) = -1$$

The Maclaurin polynomials of first and second degree and the approximation errors are:

$$T_1(x) = 1, \quad T_2(x) = 1 - \frac{1}{2}x^2, \quad \cos x - 1 = o(x) \text{ and } \cos x - 1 + \frac{1}{2}x^2 = o(x^2)$$



The second-degree polynomial $T_2(x)$ approximates the function $\cos x$ near $x_0 = 0$ “better” than the first-degree polynomial $T_1(x)$. The difference between the function and $T_2(x)$ tends to zero faster than x^2 , while the difference with $T_1(x)$ tends to zero only faster than x .

Proof. We prove the theorem in the case $n = 2$, i.e.:

$$f(x) = f(0) + f'(0) x + \frac{1}{2} f''(0) x^2 + o(x^2) \text{ as } x \rightarrow 0$$

We therefore need to prove that:

$$f(x) - \left(f(0) + f'(0) x + \frac{1}{2} f''(0) x^2 \right) = o(x^2) \text{ as } x \rightarrow 0$$

i.e. (by definition of little- o) that:

$$\lim_{x \rightarrow 0} \frac{f(x) - \left(f(0) + f'(0) x + \frac{1}{2} f''(0) x^2 \right)}{x^2} = 0$$

This limit gives an indeterminate form $[0/0]$; we now apply L'Hôpital's rule and obtain:

$$\lim_{x \rightarrow 0} \frac{f(x) - \left(f(0) + f'(0) x + \frac{1}{2} f''(0) x^2 \right)}{x^2} = \lim_{x \rightarrow 0} \frac{f'(x) - \left(f'(0) + f''(0) x \right)}{2x}$$

which is still in the indeterminate form $[0/0]$. We apply L'Hôpital's rule again and obtain:

$$\lim_{x \rightarrow 0} \frac{f'(x) - \left(f'(0) + f''(0) x \right)}{2x} = \lim_{x \rightarrow 0} \frac{f''(x) - f''(0)}{2} = 0$$

the limit equals 0 if we add the hypothesis that $f''(x)$ is continuous at $x = 0$.

Let us see how to proceed also without requiring the continuity of f'' at 0, and prove directly that:

$$\frac{1}{2} \lim_{x \rightarrow 0} \frac{f'(x) - \left(f'(0) + f''(0) x \right)}{x} = 0$$

By hypothesis, f is twice differentiable at 0, hence f' is differentiable; if we apply the linear approximation (equation (3)) to the function $f'(x)$ we obtain:

$$f'(x) = f'(0) + f''(0) x + o(x) \text{ as } x \rightarrow 0$$

hence it follows that (definition of little- o):

$$\frac{f'(x) - \left(f'(0) + f''(0) x \right)}{x} \rightarrow 0 \text{ as } x \rightarrow 0$$

□

Proof. The general case (any n) can be proved by induction on n . We have already proved that the thesis holds for $n = 2$.

Suppose now that we know that for any function differentiable $n - 1$ times at $x = 0$ the Maclaurin formula holds, and let us prove that

$$f(x) = T_{n,f}(x) + o(x^n) \text{ as } x \rightarrow 0 \text{ i.e. } \lim_{x \rightarrow 0} \frac{f(x) - T_{n,f}(x)}{x^n} = 0 \quad (4)$$

Applying L'Hôpital's rule, we have:

$$\lim_{x \rightarrow 0} \frac{f'(x) - T'_{n,f}(x)}{n x^{n-1}} \quad (5)$$

We now observe that the derivative of the polynomial $T_{n,f}(x)$ of f is nothing but the polynomial $T_{n-1,f'}(x)$ of the function f' :

$$T'_{n,f}(x) = T_{n-1,f'}(x)$$

as can be verified directly from the definition of $T_{n,f}$. On the other hand, by the inductive hypothesis applied to the function f' , we know that

$$f'(x) = T_{n-1,f'}(x) + o(x^{n-1}) \text{ as } x \rightarrow 0 \text{ i.e. } \lim_{x \rightarrow 0} \frac{f'(x) - T_{n-1,f'}(x)}{x^{n-1}} = 0$$

hence the limit (5) is zero and, by L'Hôpital's rule, the limit (4) is also zero, as we wanted to prove. $\square$

Example 8: computing Maclaurin polynomials

To better understand the equality:

$$T'_{n,f}(x) = T_{n-1,f'}(x)$$

consider a function f ; its Maclaurin polynomial, for example of degree 3, is:

$$T_{3,f}(x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 + \frac{1}{3!}f'''(0)x^3$$

and the derivative of the polynomial $T_{3,f}(x)$ is:

$$T'_{3,f}(x) = f'(0) + f''(0)x + \frac{1}{2}f'''(0)x^2$$

Let us now consider its derivative function $f'(x)$ and compute its Maclaurin polynomial of degree 2:

$$T_{2,f'}(x) = f'(0) + f''(0)x + \frac{1}{2}f'''(0)x^2$$

that is, we obtain the same polynomial; the reasoning easily extends to degree n .

- The derivative $T'_{n,f}(x)$ of the Maclaurin polynomial $T_{n,f}(x)$ of f is the Maclaurin polynomial $T_{n-1,f'}(x)$ of degree $n-1$ of the derivative function f' , that is:

$$T'_{n,f}(x) = \sum_{k=1}^n k \frac{f^{(k)}(0)}{k!} x^{k-1} = \sum_{k=1}^n \frac{f^{(k)}(0)}{(k-1)!} x^{k-1} = \sum_{k=0}^{n-1} \frac{f^{(k+1)}(0)}{(k)!} x^k = T_{n-1,f'}(x)$$

Example 9: Maclaurin formula or expansion of the sine

Consider

$$f(x) = \sin x$$

$f(0) = 0$ and moreover we have:

$$f^{(1)}(x) = \cos x, \quad f^{(2)}(x) = -\sin x, \quad f^{(3)}(x) = -\cos x, \quad f^{(4)}(x) = \sin x = f(x), \quad f^{(5)}(x) = \cos x = f^{(1)}(x) \dots$$

It follows that the derivatives of order $2k$ (even) vanish at $x = 0$, while those of order $2k+1$ (odd) are alternately $+1$ and -1 at $x = 0$. The Maclaurin polynomials of the sine function therefore contain only terms of odd degree:

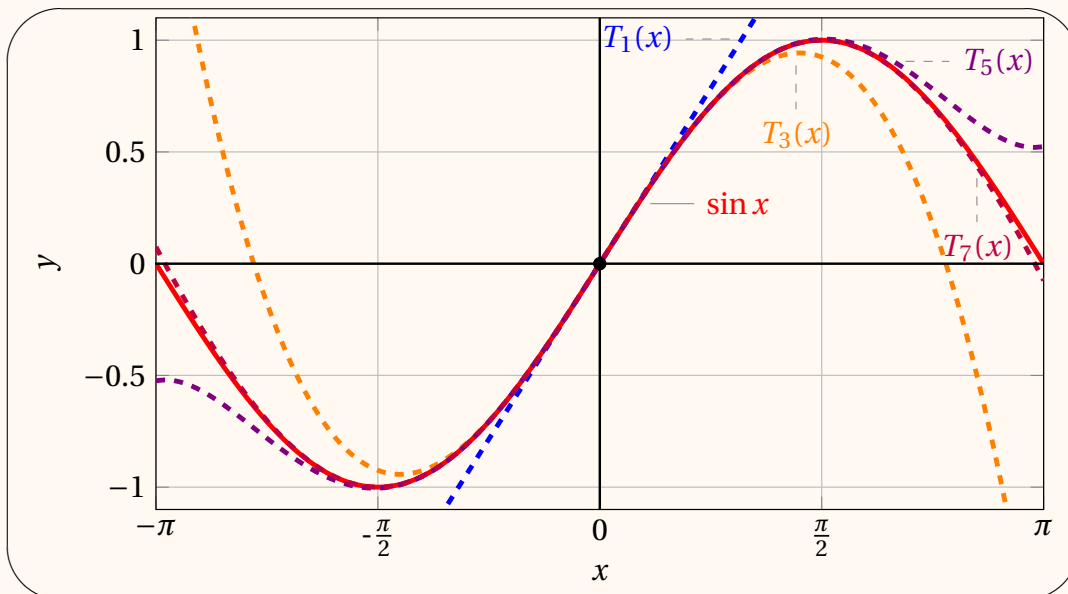
$$\sin x = T_1(x) + o(x) = x + o(x) \quad \text{as } x \rightarrow 0$$

$$\sin x = T_3(x) + o(x^3) = x - \frac{x^3}{6} + o(x^3) \quad \text{as } x \rightarrow 0$$

$$\sin x = T_5(x) + o(x^5) = x - \frac{x^3}{6} + \frac{x^5}{120} + o(x^5) \quad \text{as } x \rightarrow 0$$

$$\sin x = T_7(x) + o(x^7) = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + o(x^7) \quad \text{as } x \rightarrow 0$$

Graphically, the Maclaurin polynomials of the function $f(x) = \sin x$ are:



Example 10: Maclaurin formula or expansion of the cosine

Consider

$$f(x) = \cos x$$

$f(0) = 1$ and moreover we have:

$$f^{(1)}(x) = -\sin x, \quad f^{(2)}(x) = -\cos x, \quad f^{(3)}(x) = \sin x, \quad f^{(4)}(x) = \cos x = f(x), \quad f^{(5)}(x) = -\sin x = f^{(1)}(x) \dots$$

It follows that the derivatives of order $2k+1$ (odd) vanish at $x=0$, while those of order $2k$ (even) are alternately -1 and $+1$ at $x=0$. The Maclaurin polynomials of the cosine function therefore contain only terms of even degree:

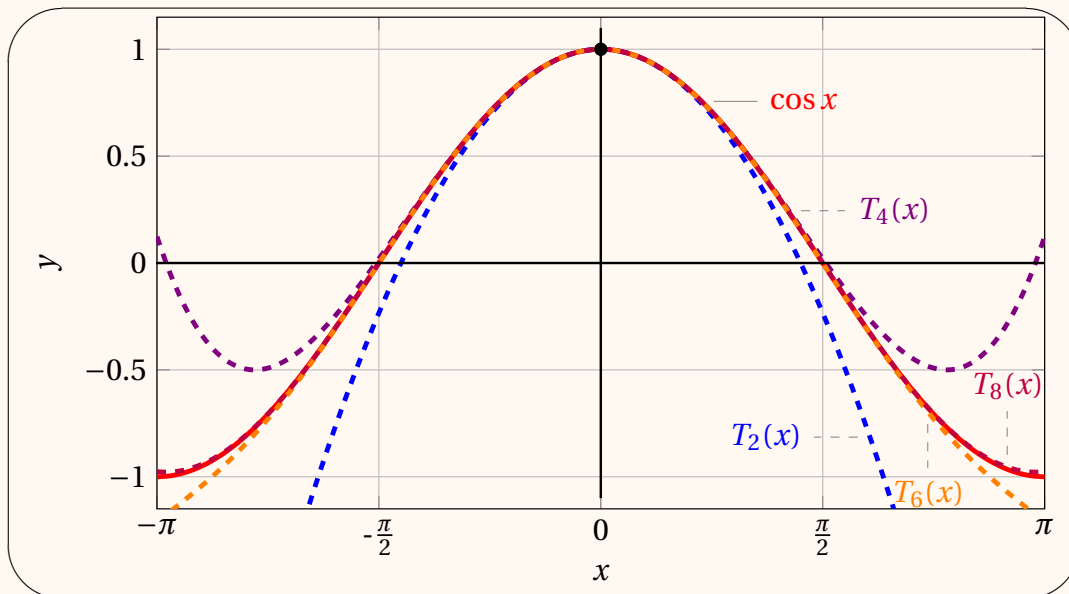
$$\cos x = T_2(x) + o(x^2) = 1 - \frac{1}{2}x^2 + o(x^2) \quad \text{as } x \rightarrow 0$$

$$\cos x = T_4(x) + o(x^4) = 1 - \frac{1}{2}x^2 + \frac{x^4}{24} + o(x^4) \quad \text{as } x \rightarrow 0$$

$$\cos x = T_6(x) + o(x^6) = 1 - \frac{1}{2}x^2 + \frac{x^4}{24} - \frac{x^6}{720} + o(x^6) \quad \text{as } x \rightarrow 0$$

$$\cos x = T_8(x) + o(x^8) = 1 - \frac{1}{2}x^2 + \frac{x^4}{24} - \frac{x^6}{720} + \frac{x^8}{40320} + o(x^8) \quad \text{as } x \rightarrow 0$$

Graphically, the Maclaurin polynomials of the function $f(x) = \cos x$ are:



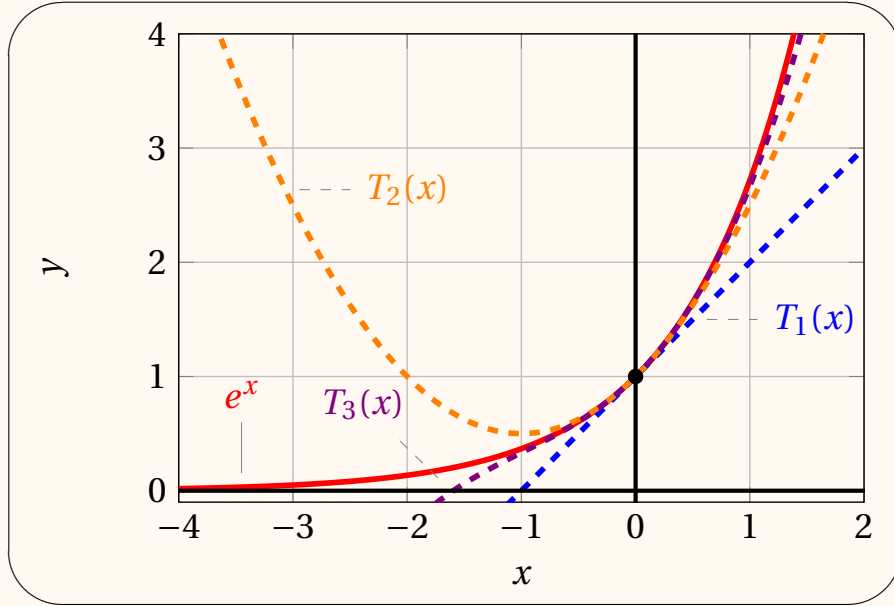
Example 11: Maclaurin formula or expansion

Consider:

$$f(x) = e^x \text{ we have } f^{(n)}(x) = e^x \text{ and } f^{(n)}(0) = 1, \forall n \in \mathbb{N}, n \geq 1$$

Hence:

$$e^x = T_n(x) + o(x^n) = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + o(x^n) \text{ as } x \rightarrow 0$$



Example 12: Computing limits with the Maclaurin expansion

We want to compute:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \left[\frac{0}{0} \right]$$

Expanding the function to first order, $\cos x = 1 + o(x)$, we have:

$$\lim_{x \rightarrow 0} \frac{1 - (1 + o(x))}{x^2} = \lim_{x \rightarrow 0} \frac{o(x)}{x^2}$$

The symbol $o(x)$ as $x \rightarrow 0$ denotes the set of functions that, divided by x , tend to 0 as $x \rightarrow 0$. Hence, for example, $5x^2 = o(x)$ but also $x^3 = o(x)$, and the value of the limit cannot yet be determined. Expanding the function to second order, $\cos x = 1 - \frac{1}{2}x^2 + o(x^2)$, we have:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{1 - (1 - \frac{1}{2}x^2 + o(x^2))}{x^2} = \lim_{x \rightarrow 0} \frac{\frac{1}{2}x^2 + o(x^2)}{x^2} = \lim_{x \rightarrow 0} \frac{\frac{1}{2}x^2 + o(x^2)}{x^2} = \lim_{x \rightarrow 0} \underbrace{\frac{1}{2}}_{\rightarrow 0} + \frac{o(x^2)}{x^2} = \frac{1}{2}$$

The symbol $o(x^2)$ as $x \rightarrow 0$ denotes the set of functions that, divided by x^2 , tend to 0 as $x \rightarrow 0$; hence the second term tends to 0 as $x \rightarrow 0$. Equivalently, $\frac{o(x^2)}{x^2} = o(1)$, that is, a generic function that tends to 0 as $x \rightarrow 0$.

Example 13: Computing limits with the Maclaurin expansion

We want to compute:

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x} = \left[\frac{0}{0} \right]$$

Expanding the functions to first order we have:

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x} = \lim_{x \rightarrow 0} \frac{1 + x + o(x) - (1 - x + o(x)) - 2x}{x - (x + o(x))} = \lim_{x \rightarrow 0} \frac{o(x)}{o(x)}$$

The symbol $o(x)$ as $x \rightarrow 0$ denotes the set of functions that, divided by x , tend to 0 as $x \rightarrow 0$. Hence, for example, $5x^2 = o(x)$ but also $x^3 = o(x)$, and the value of the limit cannot yet be determined.

Expanding the functions to second order we have:

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x} = \lim_{x \rightarrow 0} \frac{1 + x + \frac{x^2}{2} + o(x^2) - (1 - x + \frac{x^2}{2} + o(x^2)) - 2x}{x - (x + o(x^2))} = \lim_{x \rightarrow 0} \frac{o(x^2)}{o(x^2)}$$

The limit still cannot be solved; expanding the functions to third order we have:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x} &= \lim_{x \rightarrow 0} \frac{1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3) - (1 - x + \frac{x^2}{2} - \frac{x^3}{6} + o(x^3)) - 2x}{x - (x - \frac{x^3}{6} + o(x^3))} \\ &= \lim_{x \rightarrow 0} \frac{\frac{x^3}{3} + o(x^3)}{\frac{x^3}{6} + o(x^3)} = \lim_{x \rightarrow 0} \frac{x^3 \left(\frac{1}{3} + \frac{o(x^3)}{x^3} \right)}{x^3 \left(\frac{1}{6} + \frac{o(x^3)}{x^3} \right)} = \lim_{x \rightarrow 0} \frac{\frac{1}{3} + o(1)}{\frac{1}{6} + o(1)} = 2 \end{aligned}$$

Example 14: Computing the Maclaurin polynomial

Compute the Maclaurin polynomial of degree 4 of the function

$$f(x) = \log(1 + \sin x)$$

The Maclaurin polynomials of degree 4 of the functions $\sin x$ and $\log(1 + x)$ are:

$$\sin x = x - \frac{x^3}{6}, \quad \log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4}$$

Hence, substituting, the Maclaurin polynomial of the function $f(x)$ is:

$$T_4(x) = \left(x - \frac{x^3}{6} \right) - \frac{\left(x - \frac{x^3}{6} \right)^2}{2} + \frac{\left(x - \frac{x^3}{6} \right)^3}{3} - \frac{\left(x - \frac{x^3}{6} \right)^4}{4}$$

We have $\left(x - \frac{x^3}{6} \right)^2 = x^2 - \frac{2x^4}{6} + \frac{x^6}{36}$ and the last term has degree greater than 4. Since we are looking only for terms up to degree 4, we discard the last summand. Applying the same reasoning to the other numerators we have:

$$T_4(x) = x - \frac{x^3}{6} - \frac{x^2 - \frac{2x^4}{6}}{2} + \frac{x^3}{3} - \frac{x^4}{4} = x - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12}$$

Maclaurin formulas or expansions of some elementary functions, with the Peano remainder, as $x \rightarrow 0$:

$$e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + o(x^n)$$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \dots + (-1)^{n-1} \frac{x^n}{n} + o(x^n)$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2} x^2 + \dots + \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!} x^n + o(x^n), \forall \alpha \in \mathbb{R}$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+1})$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots + (-1)^n \frac{x^{2n}}{(2n)!} + o(x^{2n})$$

$$\operatorname{Sh} x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+1})$$

$$\operatorname{Ch} x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots + \frac{x^{2n}}{(2n)!} + o(x^{2n})$$

The expansions clearly also hold if we replace x with any function $\varepsilon(x)$ that tends to 0, that is, an infinitesimal (it does not matter what x tends to in this case).

- All of this can be generalized to a point $x_0 \neq 0$. The proofs are analogous to the previous ones and are left as an exercise.

Theorem 3: Taylor polynomial

Given a function $f : (a, b) \rightarrow \mathbb{R}$ differentiable $n-1$ times in (a, b) and n times at $x_0 \in (a, b)$, there exists one and only one polynomial T_{n,f,x_0} of degree $\leq n$ with the property that:

$$T_{n,f,x_0}(x_0) = f(x_0), \quad T'_{n,f,x_0}(x_0) = f'(x_0), \quad \dots, \quad T^{(n)}_{n,f,x_0}(x_0) = f^{(n)}(x_0)$$

and this polynomial, called the **Taylor polynomial** of f of degree n , is:

$$T_{n,f,x_0}(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \quad (\text{setting } f^{(0)} = f)$$

The expanded form of the Taylor polynomial of degree n of a function f at x_0 is:

$$T_{n,f,x_0}(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2} f''(x_0)(x - x_0)^2 + \frac{1}{3!} f'''(x_0)(x - x_0)^3 + \dots$$

$$+ \frac{1}{n!} f^{(n)}(x_0)(x - x_0)^n$$

- When the function we refer to is clear, for convenience we omit the subscript f of the Taylor polynomial and simply write $T_{n,x_0}(x)$.

Theorem 4: Taylor formula of order n with Peano remainder

Let $f : (a, b) \rightarrow \mathbb{R}$ be differentiable $n - 1$ times in (a, b) and n times at $x_0 \in (a, b)$. Then

$$f(x) = T_{n,f,x_0}(x) + o((x - x_0)^n) \text{ as } x \rightarrow x_0$$

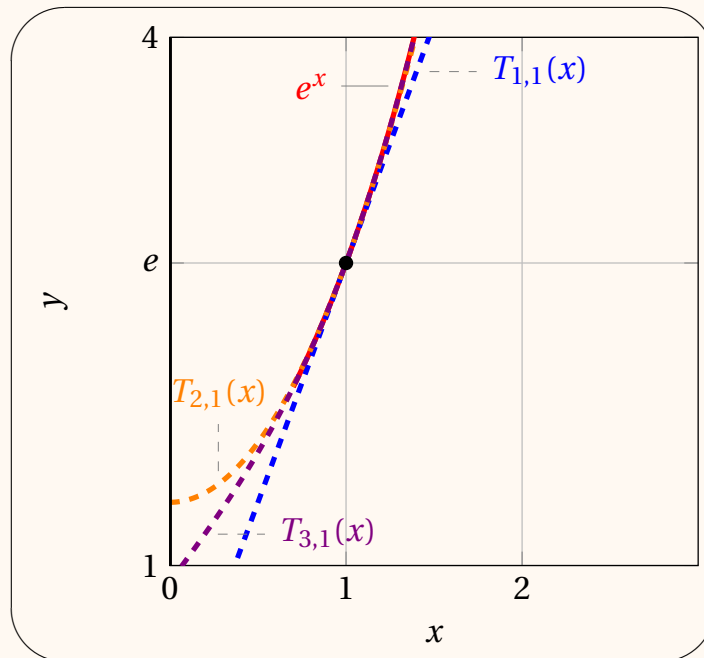
Example 15: Taylor formula or expansion

Consider $x_0 = 1$ and:

$$f(x) = e^x \text{ we have } f^{(n)}(x) = e^x \text{ and } f^{(n)}(1) = e, \quad \forall n \in \mathbb{N}, n \geq 1$$

Hence:

$$e^x = T_{n,1}(x) + o((x-1)^n) = e + e(x-1) + e \frac{(x-1)^2}{2} + e \frac{(x-1)^3}{3!} + \dots + e \frac{(x-1)^n}{n!} + o((x-1)^n) \text{ as } x \rightarrow 1$$



4 Taylor formula/expansion with Lagrange remainder

- In the Taylor-Maclaurin formula with Peano remainder, the information we have on the error made in approximating f with its Taylor-Maclaurin polynomial is of a **dynamic type**: as the increment $(x - x_0)$ tends to zero, we know that the remainder tends to zero faster than $(x - x_0)^n$.

For a fixed value of the increment $(x - x_0)$ the Taylor-Maclaurin formula with Peano remainder says nothing about the size of the error made.

- In various problems of approximate computation, instead, it is essential to estimate the error made in approximating a function with its Taylor polynomial when the increment $(x - x_0)$ has a fixed value, or a value that does not exceed a fixed threshold.
- This type of problem is addressed by the next result, which gives an alternative way of quantifying the approximation error made.

Theorem 5: Taylor formula of order n with Lagrange remainder

Let $f : [a, b] \rightarrow \mathbb{R}$ be differentiable $n + 1$ times in $[a, b]$ and let $x_0 \in [a, b]$. Then there exists a point c between x_0 and x such that:

$$f(x) = T_{n,f,x_0}(x) + \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1} \quad (6)$$

- The formula has the structure: **function to be approximated equals approximating polynomial plus approximation error**. The error is the term $\frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}$ and is called the Lagrange remainder.
- For $n = 0$, the theorem coincides with Lagrange's theorem: setting $x_0 = a$ and $b = x$ we have:

$$\exists c \in [a, b] : f(b) = f(a) + f'(c)(b - a) \text{ that is } \frac{f(b) - f(a)}{b - a} = f'(c)$$

which is Lagrange's theorem. Warning: the interval (a, b) becomes $[a, b]$.

The point c depends on x_0 , x and n , and lies between x_0 and x . If we manage to prove that

$$|f^{(n+1)}(t)| \leq M, \quad \forall t \in [x_0, x] \quad (7)$$

then the Taylor formula with Lagrange remainder says that

$$|f(x) - T_{n,x_0}(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1} \quad (8)$$

Example 16: error estimates with the Lagrange remainder

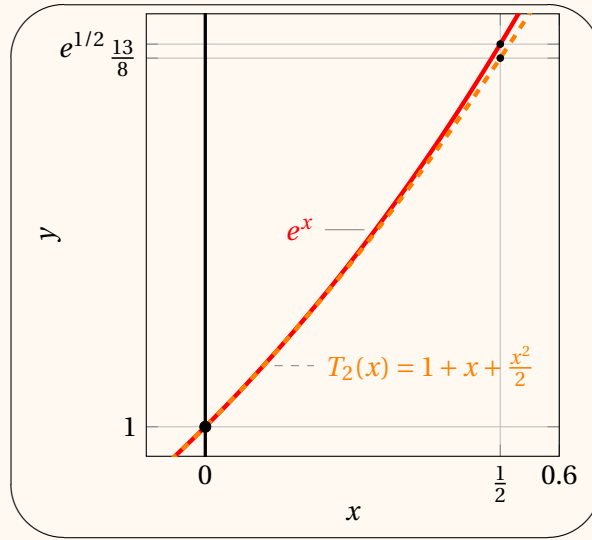
Consider $f(x) = e^x$; we know that the second- and third-order Maclaurin polynomials of e^x are:

$$T_2(x) = 1 + x + \frac{1}{2}x^2, \quad T_3(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$$

Now, if we wanted to use the polynomial to compute an approximate value of $e^{1/2}$, i.e., if we planned to approximate

$$e^{1/2} \quad \text{with} \quad T_2\left(\frac{1}{2}\right) = \frac{13}{8} \quad \text{or with} \quad T_3\left(\frac{1}{2}\right) = \frac{79}{48}$$

how can we estimate a priori (i.e., without already knowing the true value of $e^{1/2}$) how large our error is?



We have $x_0 = 0$, $x = \frac{1}{2}$. In formula (6), expanding up to the third degree ($n = 3$), we obtain:

$$e^{1/2} = \underbrace{T_3\left(\frac{1}{2}\right)}_{= \frac{79}{48}} + \frac{e^c}{4!} \left(\frac{1}{2}\right)^4 \quad \text{with} \quad c \in \left[0, \frac{1}{2}\right]$$

The value c is unknown, hence the value e^c is also unknown; we therefore need to use (7) to bound e^c from above. We have:

$$f^{(4)}(t) = e^t \quad \text{consequently} \quad |e^t| \leq \underbrace{3^{1/2}}_{=M}, \quad \forall t \in \left[0, \frac{1}{2}\right], \quad \text{since} \quad e < 3$$

Hence from (8) we obtain:

$$\left| e^{1/2} - T_3\left(\frac{1}{2}\right) \right| \leq \frac{3^{1/2}}{2^4 \cdot 4!}$$

This means that by approximating $e^{1/2}$ with the value $\frac{79}{48}$ we make an error not greater than $\sqrt{3}/384$, which is approximately 0.0045.

Proof. We prove the theorem in the case $n = 1$. Setting for convenience $x_0 = a$, $x = b$, the statement becomes: if $f : [a, b] \rightarrow \mathbb{R}$ is twice differentiable in $[a, b]$, then there exists a point $c \in [a, b]$ such that:

$$f(b) = f(a) + f'(a)(b-a) + \frac{1}{2} f''(c)(b-a)^2$$

We set

$$f(b) - (f(a) + f'(a)(b-a)) = k(b-a)^2$$

and we try to determine the form of k .

Let us therefore consider the function:

$$g(x) = f(b) - f(x) - f'(x)(b-x) - k(b-x)^2$$

and let us apply to it the mean value (Lagrange's) theorem. Since $g(b) = g(a) = 0$ (the second equality follows from the definition of k), we find that there exists $c \in (a, b)$ such that

$$0 = \frac{g(b) - g(a)}{b-a} = g'(c)$$

But we have:

$$g'(x) = -f'(x) - f''(x)(b-x) + f'(x) + 2k(b-x) = (b-x) \cdot (2k - f''(x))$$

and hence $g'(c) = 0$ implies, since $c \neq b$:

$$k = \frac{1}{2} f''(c)$$

The procedure just seen can be extended to any n .

We now define:

$$g(x) = f(b) - \sum_{j=0}^n \frac{f^{(j)}(x)(b-x)^j}{j!} - k(b-x)^{n+1}$$

with k defined implicitly by the identity

$$f(b) - T_{n,a}(b) = k(b-a)^{n+1} \tag{9}$$

□

Proof. We now proceed as follows:

1. We verify that $g(a) = g(b) = 0$ since

$$g(b) = f(b) - f(b) = 0, \quad g(a) = f(b) - T_{n,a}(b) - k(b-a)^{n+1} = 0 \text{ by def. of } k$$

2. We apply Lagrange's theorem to g on $[a, b]$, and we show that the statement "there exists $c \in (a, b)$ such that $g'(c) = 0$ " is exactly the theorem.

We start by computing g' :

$$\begin{aligned} g'(x) &= - \sum_{j=0}^n \frac{f^{(j+1)}(x) (b-x)^j}{j!} + \sum_{j=1}^n \frac{f^{(j)}(x) (b-x)^{j-1}}{(j-1)!} + k(n+1)(b-x)^n \\ &= - \sum_{j=0}^n \frac{f^{(j+1)}(x) (b-x)^j}{j!} + \sum_{j=0}^{n-1} \frac{f^{(j+1)}(x) (b-x)^j}{j!} + k(n+1)(b-x)^n \\ &= - \frac{f^{(n+1)}(x) (b-x)^n}{n!} + k(n+1)(b-x)^n \end{aligned}$$

Then $g'(c) = 0$ means:

$$- \frac{f^{(n+1)}(c) (b-c)^n}{n!} + k(n+1)(b-c)^n = 0$$

that is

$$k = \frac{f^{(n+1)}(c)}{(n+1)!}$$

which, substituted into (9), gives the thesis. □

4.1 Relations with convexity

- Consider formula (6) with $n = 1$:

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2} f''(c)(x-x_0)^2$$

and suppose that at every point of (a, b) we have $f''(x) \geq 0$, that is, f is convex on (a, b) . Then we have

$$\frac{1}{2} f''(c)(x-x_0)^2 \geq 0$$

and therefore we can write, for every pair of points x_0 and $x \in (a, b)$,

$$f(x) \geq f(x_0) + f'(x_0)(x - x_0) \quad (10)$$

Geometrically, this means that the graph of $f(x)$ lies on the whole of (a, b) above the graph of its tangent line at x_0 (and this holds for every choice of the point $x_0 \in (a, b)$).

We have therefore proved that if a (twice differentiable) function is convex, it is also “convex with respect to tangents”

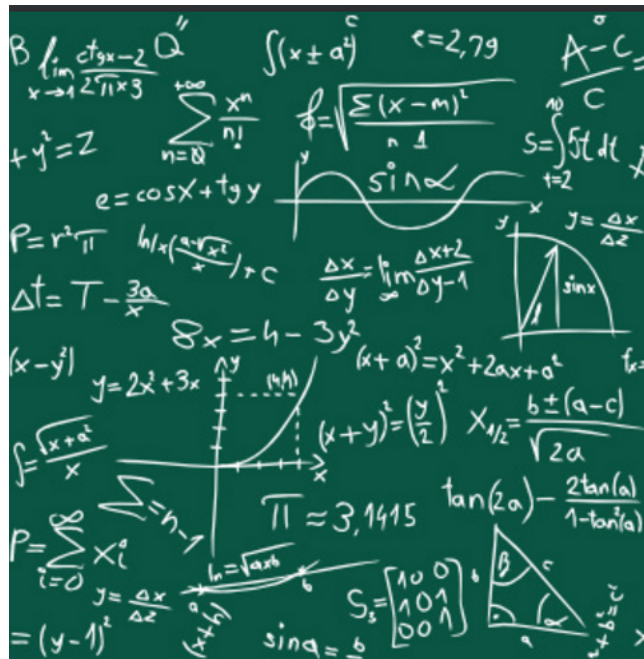
- If instead f is concave, i.e., $f''(x) \leq 0$ on the whole of (a, b) , we have

$$f(x) \leq f(x_0) + f'(x_0)(x - x_0) \quad (11)$$

and the graph of $f(x)$ lies on the whole of (a, b) below the graph of its tangent line at x_0 .

We have therefore proved that if a (twice differentiable) function is concave, it is also “concave with respect to tangents”

Curve sketching



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Graphs of real functions of a real variable

- The infinitesimal and differential calculus developed so far allows us to tackle completely the problem of drawing the graph of a function f , that is, to carry out **curve sketching** (function analysis).

The steps of curve sketching:

1. Determine the (maximal) **domain** of f .
2. Compute the **limits at the boundary** of the domain. Determine any **horizontal/vertical asymptotes** and **points of discontinuity**. Study the **sign of the function** and the **points where it vanishes** (possible only for relatively simple functions).
3. If the function tends to ∞ as $x \rightarrow \infty$, determine any **oblique asymptotes**.
4. Compute the **derivative function** f' at the points where it exists. Study the points where f is continuous but not differentiable and determine their nature (**corner points, inflection points with vertical tangent or cusps**). At corner points or at the endpoints of the domain it is useful to compute the **right or left derivatives**, which determine the **slope of the graph at those points**.
5. Study the sign of f' (on its domain) to obtain information on the **monotonicity** of f and on its **relative maximum and minimum points**. Then determine the **absolute maximum and minimum points**.

Additional information:

1. It can be useful to determine an **asymptotic estimate at infinity** that tells us whether the function tends to ∞ in a *superlinear* or *sublinear* way or in a *linear* way (in which case it may have an oblique asymptote). The asymptotic estimate at infinity normally also gives information on the **concavity** of f at infinity.
2. Compute the **second derivative function** f'' and study its sign, to deduce information on the **concavity** and the **inflection points** of f .
3. Determine any symmetry of f (**even or odd functions**) and restrict the study to $x \geq 0$.
4. Determine any periodicity of f (**periodic functions**) and restrict the study to one period.

2 Examples

Example 1: Function analysis and graph

Let us study the function and draw its graph:

$$f(x) = e^{-|x|} \sqrt{x^2 - 5x + 6}$$

We expect a corner point at $x = 0$ due to the presence of $|x|$, and points with vertical tangent where the radicand vanishes.

Step 1. The (maximal) domain of f is:

$$x^2 - 5x + 6 \geq 0, (x-3)(x-2) \geq 0 \quad \text{that is} \quad x \in (-\infty, 2] \cup [3, +\infty)$$

Step 2. The limits at the boundary are:

$$\lim_{x \rightarrow +\infty} e^{-|x|} \sqrt{x^2 - 5x + 6} = \lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 - 5x + 6}}{e^x} = 0^+$$

$$\lim_{x \rightarrow -\infty} e^{-|x|} \sqrt{x^2 - 5x + 6} = \lim_{x \rightarrow -\infty} \sqrt{e^{2x}(x^2 - 5x + 6)} = \lim_{y \rightarrow +\infty} \sqrt{\frac{y^2 + 5y + 6}{e^{2y}}} = 0^+$$

$$\lim_{x \rightarrow 3^+} e^{-|x|} \sqrt{x^2 - 5x + 6} = \lim_{x \rightarrow 3^+} e^{-x} \sqrt{(x-3)(x-2)} = 0^+$$

$$\lim_{x \rightarrow 2^-} e^{-|x|} \sqrt{x^2 - 5x + 6} = \lim_{x \rightarrow 2^-} e^{-x} \sqrt{(x-3)(x-2)} = 0^+$$

Hence the line $y = 0$ is a horizontal asymptote as $x \rightarrow \pm\infty$ and there are no vertical asymptotes. In the domain there are no points of discontinuity. We have $f(x) \geq 0$ for every x in the domain and $f(x) = 0$ for $x = 2$ and $x = 3$.

Step 3. The function has no oblique asymptotes.

Step 4. We compute the derivative function for $x \neq 0$:

$$\begin{aligned} f'(x) &= e^{-|x|} \left(-\operatorname{sgn}(x) \sqrt{x^2 - 5x + 6} + \frac{2x - 5}{2\sqrt{x^2 - 5x + 6}} \right) \\ &= \frac{e^{-|x|}}{2\sqrt{(x-3)(x-2)}} \left(-2\operatorname{sgn}(x)(x^2 - 5x + 6) + 2x - 5 \right) \end{aligned}$$

Or, equivalently:

$$f'(x) = \begin{cases} e^{-x} \cdot \frac{-2x^2 + 12x - 17}{2\sqrt{(x-3)(x-2)}} & \text{if } x > 0 \\ e^x \cdot \frac{2x^2 - 8x + 7}{2\sqrt{(x-3)(x-2)}} & \text{if } x < 0 \end{cases}$$

Example 2: Function analysis and graph

We compute the right and left derivatives at $x = 0$:

$$f'_+(0) = \lim_{x \rightarrow 0^+} \frac{e^{-x} \overbrace{(-2x^2 + 12x - 17)}^{-1}}{2\sqrt{(x-3)(x-2)}} = -\frac{17}{2\sqrt{6}}$$

$$f'_-(0) = \lim_{x \rightarrow 0^-} \frac{e^x (2x^2 - 8x + 7)}{2\sqrt{(x-3)(x-2)}} = \frac{7}{2\sqrt{6}}$$

hence $x = 0$ is a corner point, that is, $f'(0)$ does not exist. We compute the right-hand limit at $x = 3$ and the left-hand limit at $x = 2$ of the derivative function:

$$f'_+(3) = \lim_{x \rightarrow 3^+} \frac{e^{-x} \overbrace{(-2x^2 + 12x - 17)}^{-1}}{2\sqrt{(x-3)(x-2)}} = +\infty$$

$$f'_-(2) = \lim_{x \rightarrow 2^-} \frac{e^{-x} \overbrace{(-2x^2 + 12x - 17)}^{-1}}{2\sqrt{(x-3)(x-2)}} = -\infty$$

Hence the function has points with vertical tangent at $x = 2$ and $x = 3$.

Step 5. We study the sign of the derivative function:

- For $x > 0$ we have:

$$f'(x) \geq 0 \quad \text{if} \quad 2x^2 - 12x + 17 \leq 0, \quad \text{that is} \quad x \in \left[\underbrace{\frac{6-\sqrt{2}}{2}}_{\approx 2.3}, \underbrace{\frac{6+\sqrt{2}}{2}}_{\approx 3.7} \right]$$

hence:

$$f \text{ is decreasing for } x \in [0, 2] \cup \left[\frac{6+\sqrt{2}}{2}, +\infty \right), \quad f \text{ is increasing for } x \in \left[3, \frac{6+\sqrt{2}}{2} \right]$$

We have $f'\left(\frac{6+\sqrt{2}}{2}\right) = 0$, hence $x = \frac{6+\sqrt{2}}{2}$ is a relative maximum point. The relative maximum is $f\left(\frac{6+\sqrt{2}}{2}\right) \approx 0.027$

- For $x < 0$ we have:

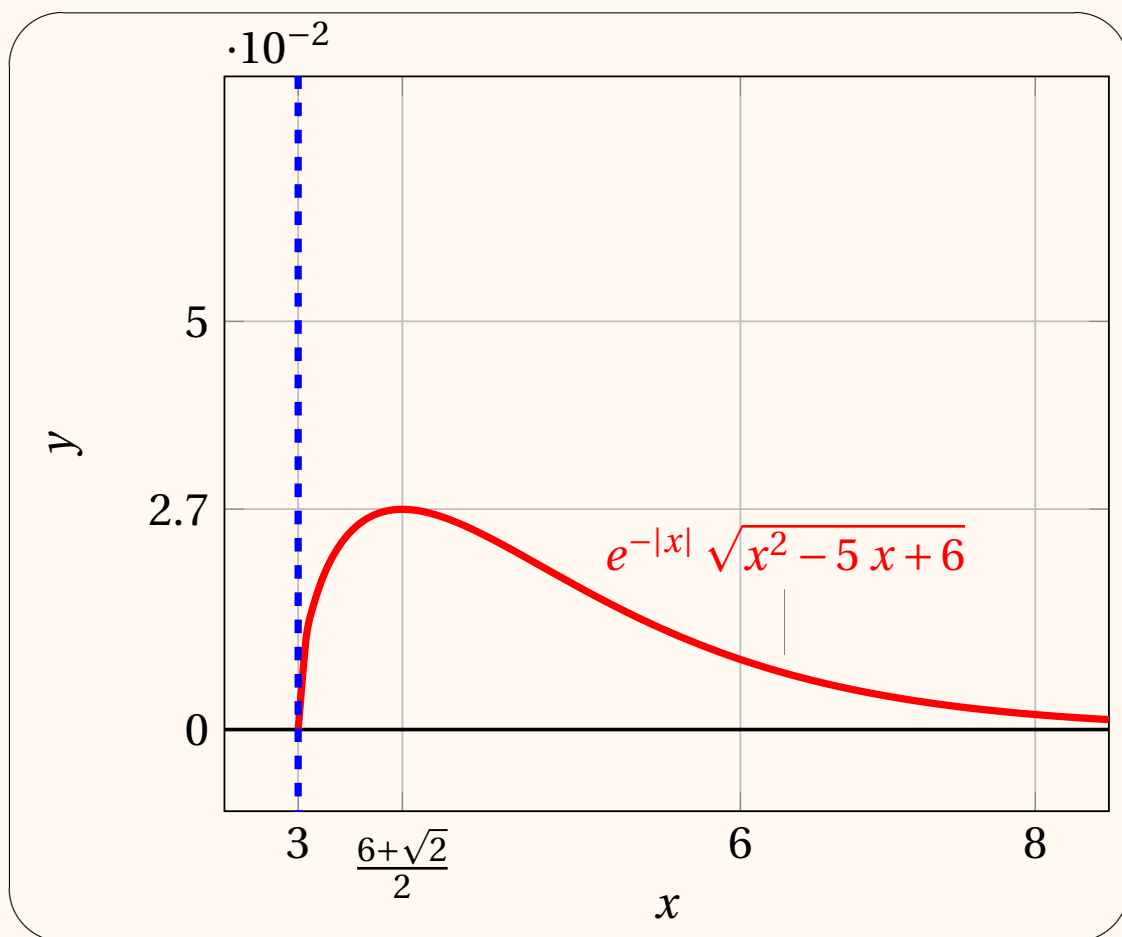
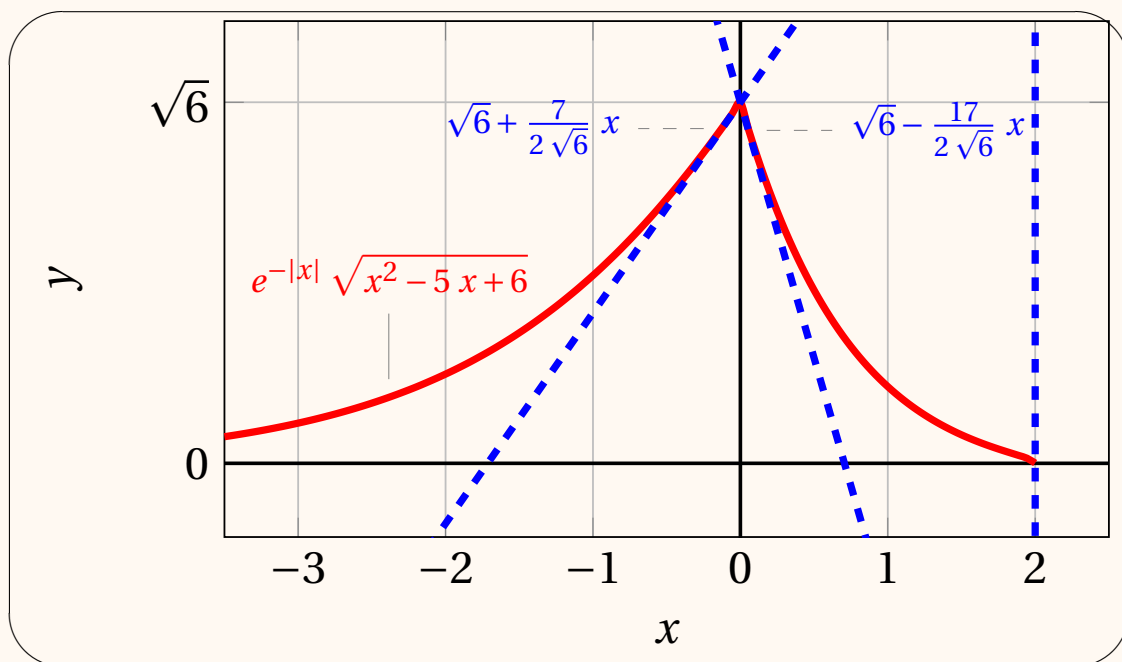
$$f'(x) \geq 0 \quad \text{if} \quad 2x^2 - 8x + 7 \geq 0, \quad \text{that is, if } x \in \left[-\infty, \frac{4-\sqrt{2}}{2} \right] \cup \left[\frac{4+\sqrt{2}}{2}, +\infty \right]$$

hence $f(x)$ is increasing for $x < 0$.

We can deduce that $x = 0$ (where f is not differentiable) is an absolute maximum point; the absolute maximum is $\sqrt{6} \approx 2.44$.

There are also two inflection points, one in $(0, 2)$ and one in $(3, +\infty)$, which can be obtained by studying the sign of the second derivative.

Example 3: Function analysis and graph



Example 4: Function analysis and graph

Let us study the function and draw its graph:

$$f(x) = x \cdot \exp\left(\frac{x+2}{x-1}\right)$$

Step 1. The (maximal) domain of f is:

$$x \in \mathbb{R} \setminus \{1\}$$

Step 2. The limits at the boundary are:

$$\lim_{x \rightarrow +\infty} x \cdot \exp\left(\frac{x+2}{x-1}\right) = +\infty$$

$$\lim_{x \rightarrow -\infty} x \cdot \exp\left(\frac{x+2}{x-1}\right) = -\infty$$

$$\frac{x+2}{x-1} \rightarrow +\infty \text{ as } x \rightarrow 1^+, \text{ hence } \lim_{x \rightarrow 1^+} x \exp\left(\frac{x+2}{x-1}\right) = +\infty$$

$$\frac{x+2}{x-1} \rightarrow -\infty \text{ as } x \rightarrow 1^-, \text{ hence } \lim_{x \rightarrow 1^-} x \exp\left(\frac{x+2}{x-1}\right) = 0$$

Hence there are no horizontal asymptotes and $x = 1$ is a point of discontinuity. Moreover, $x = 1$ is a vertical asymptote as $x \rightarrow 1^+$. We have $f(x) \geq 0$ for $x \geq 0$ and $f(x) = 0$ for $x = 0$.

Step 3. We compute an asymptotic estimate; we have:

$$\frac{x+2}{x-1} \rightarrow 1 \text{ as } x \rightarrow +\infty, \quad \frac{x+2}{x-1} \rightarrow 1 \text{ as } x \rightarrow -\infty$$

hence

$$x \cdot \exp\left(\frac{x+2}{x-1}\right) \sim x \cdot e \text{ as } x \rightarrow \pm\infty \text{ therefore } f \text{ has linear growth}$$

We therefore check for the presence of oblique asymptotes. We try to compute the following limit:

$$\lim_{x \rightarrow \pm\infty} \left(x \cdot \exp\left(\frac{x+2}{x-1}\right) - x e \right) = \lim_{x \rightarrow \pm\infty} x \cdot e \cdot \left(\exp\left(\frac{x+2}{x-1} - 1\right) - 1 \right)$$

We have

$$\frac{x+2}{x-1} - 1 = \frac{3}{x-1} \rightarrow 0 \text{ as } x \rightarrow \pm\infty$$

therefore

$$\exp\left(\frac{x+2}{x-1} - 1\right) - 1 \sim \frac{3}{x-1} \text{ as } x \rightarrow \pm\infty; \quad x \cdot e \cdot \left(\exp\left(\frac{x+2}{x-1} - 1\right) - 1 \right) \sim \frac{3 \cdot x \cdot e}{x-1} \text{ as } x \rightarrow \pm\infty$$

consequently:

$$\lim_{x \rightarrow \pm\infty} \left(x \cdot \exp\left(\frac{x+2}{x-1}\right) - x e \right) = \lim_{x \rightarrow \pm\infty} \frac{3 \cdot x \cdot e}{x-1} = 3 \cdot e$$

Hence the function has the oblique asymptote $y = e x + 3 e$ as $x \rightarrow \pm\infty$.

Example 5: Function analysis and graph

Step 4. We compute the derivative function for $x \neq 1$:

$$\begin{aligned} f'(x) &= \exp\left(\frac{x+2}{x-1}\right) \cdot \left(1 + x \cdot \frac{(x-1) - (x+2)}{(x-1)^2}\right) = \exp\left(\frac{x+2}{x-1}\right) \cdot \frac{(x-1)^2 - 3x}{(x-1)^2} \\ &= \exp\left(\frac{x+2}{x-1}\right) \cdot \frac{x^2 - 5x + 1}{(x-1)^2} \end{aligned}$$

For every $x \neq 1$, f' is defined. We compute the left-hand limit at $x = 1$:

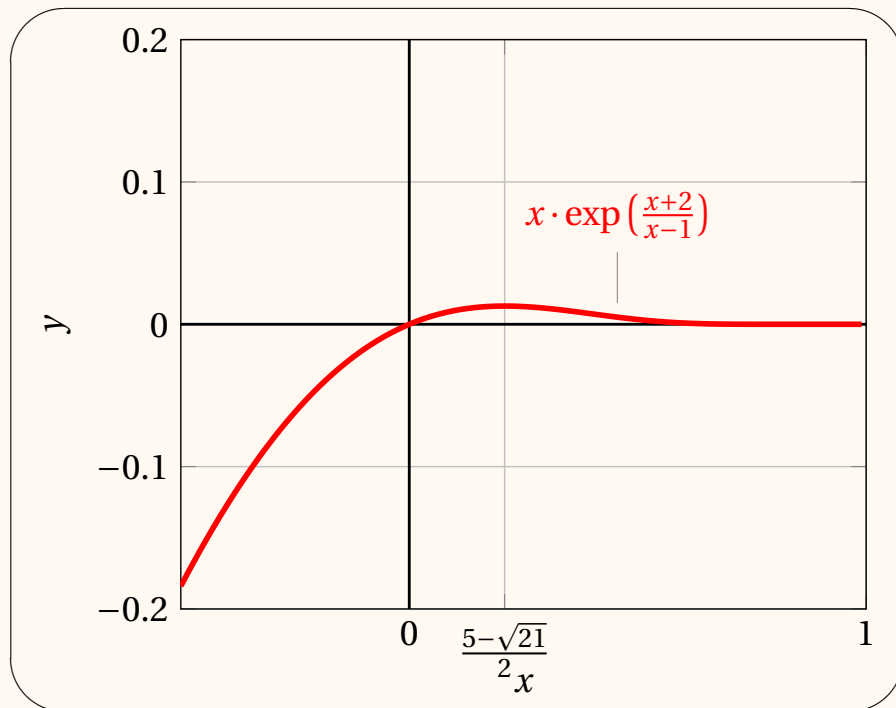
$$f'_-(1) = \lim_{x \rightarrow 1^-} \exp\left(\frac{x+2}{x-1}\right) \cdot \frac{x^2 - 5x + 1}{(x-1)^2} = 0$$

the exponential goes to zero faster than $(x-1)^2$; hence the graph reaches $x = 1$ with a horizontal tangent, from the left.

Step 5. We study the sign of the derivative function:

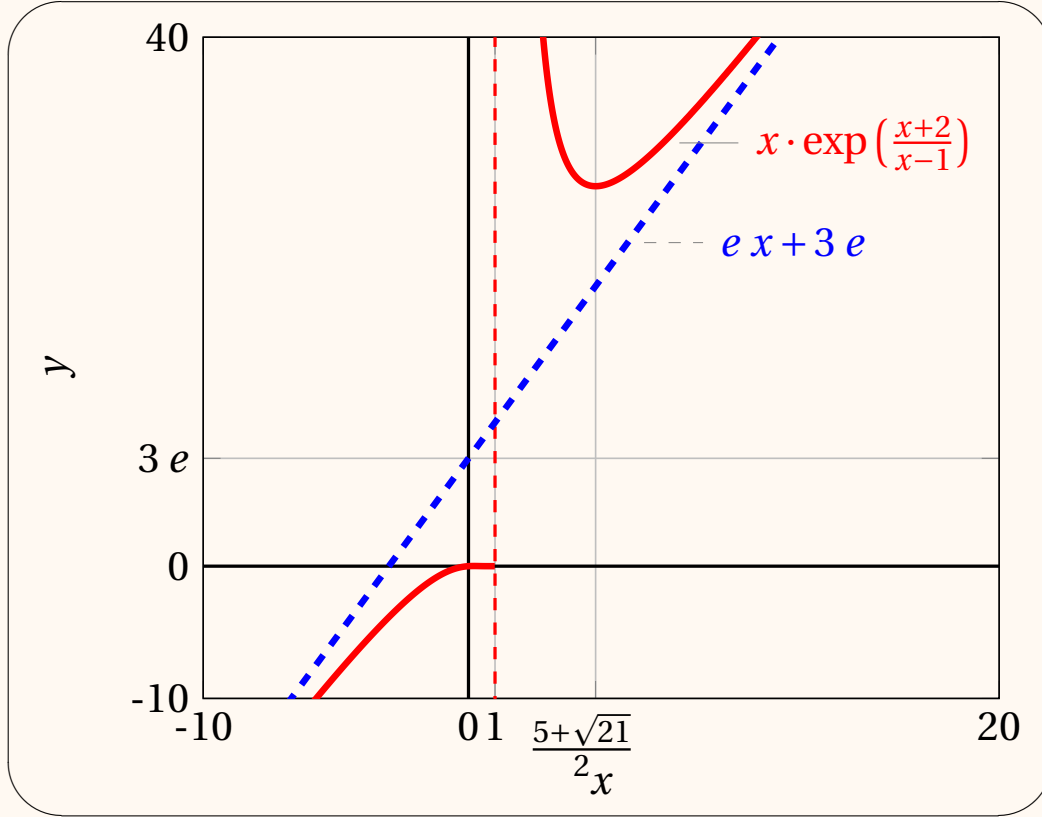
$$f'(x) \geq 0 \quad \text{for} \quad x^2 - 5x + 1 \geq 0 \quad \text{that is} \quad x \in \left[-\infty, \frac{5-\sqrt{21}}{2}\right] \cup \left[\frac{5+\sqrt{21}}{2}, +\infty\right)$$

hence $x = \frac{5+\sqrt{21}}{2}$ is a relative minimum point and $x = \frac{5-\sqrt{21}}{2}$ is a relative maximum point.



There is an inflection point in $\left(\frac{5-\sqrt{21}}{2}, 1\right)$, which can be obtained by studying the sign of the second derivative.

Example 6: Function analysis and graph



We have $f'_-(1) = 0$ since:

$$f'_-(1) = \lim_{x \rightarrow 1^-} \exp\left(\frac{x+2}{x-1}\right) \cdot \frac{x^2 - 5x + 1}{(x-1)^2} = \lim_{x \rightarrow 1^-} \frac{\exp\left(\frac{x+2}{x-1}\right)}{(x-1)^2} \cdot \underbrace{(x^2 - 5x + 1)}_{\rightarrow -3 \text{ as } x \rightarrow 1^-}$$

with the change of variable $y = \frac{x+2}{x-1}$: if $x \rightarrow 1^-$ then $y \rightarrow -\infty$; moreover:

$$y = \frac{x+2}{x-1}, \quad y = \frac{x-1+3}{x-1}, \quad y = 1 + \frac{3}{x-1} \quad \text{and} \quad x = 1 + \frac{3}{y-1}$$

hence we have:

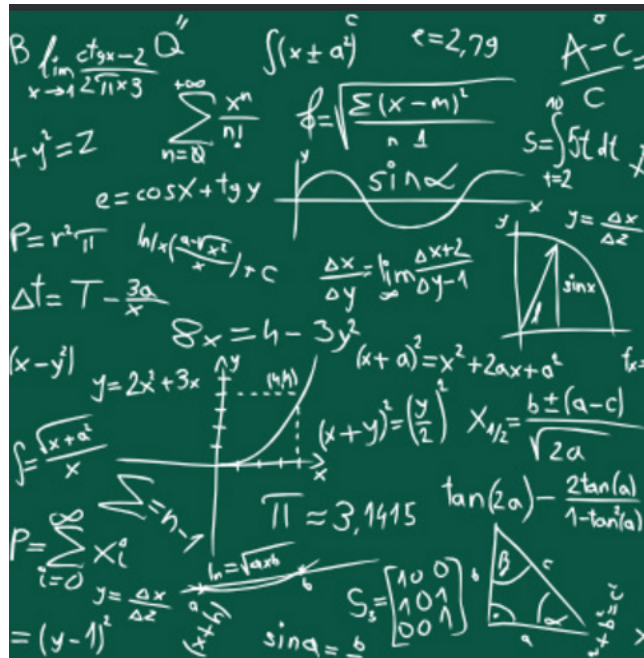
$$\lim_{x \rightarrow 1^-} \frac{\exp\left(\frac{x+2}{x-1}\right)}{(x-1)^2} = \lim_{y \rightarrow -\infty} \frac{e^y}{\left(\frac{3}{y-1}\right)^2} = \frac{1}{9} \lim_{y \rightarrow -\infty} e^y \cdot \underbrace{(y-1)^2}_{\sim y^2 \text{ as } y \rightarrow -\infty}$$

with a second change of variable $z = -y$: if $y \rightarrow -\infty$ then $z \rightarrow +\infty$ and we have

$$\frac{1}{9} \lim_{y \rightarrow -\infty} e^y \cdot y^2 = \frac{1}{9} \lim_{z \rightarrow +\infty} e^{-z} \cdot (-z)^2 = \frac{1}{9} \lim_{z \rightarrow +\infty} \frac{z^2}{e^z} = 0$$

by the theorem on the hierarchy of infinities.

Newton's method



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Newton's method

- Suppose we want to solve the equation

$$f(x) = 0$$

which is equivalent to looking for the intersections of the graph of f with the x -axis, or the zeros of the function. Suppose moreover: i) that the solution $x = c$ exists, is unique and lies inside the interval $[a, b]$; ii) that the function is differentiable in $[a, b]$; iii) that $f'(x) \leq 0$ (decreasing function) and $f''(x) \geq 0$ (convex function) for every $x \in [a, b]$.

- Let us then start from $x_0 = a$ and linearize the equation $f(x) = 0$ by replacing f with the tangent line to its graph at the point $(x_0, f(x_0))$. This line has equation:

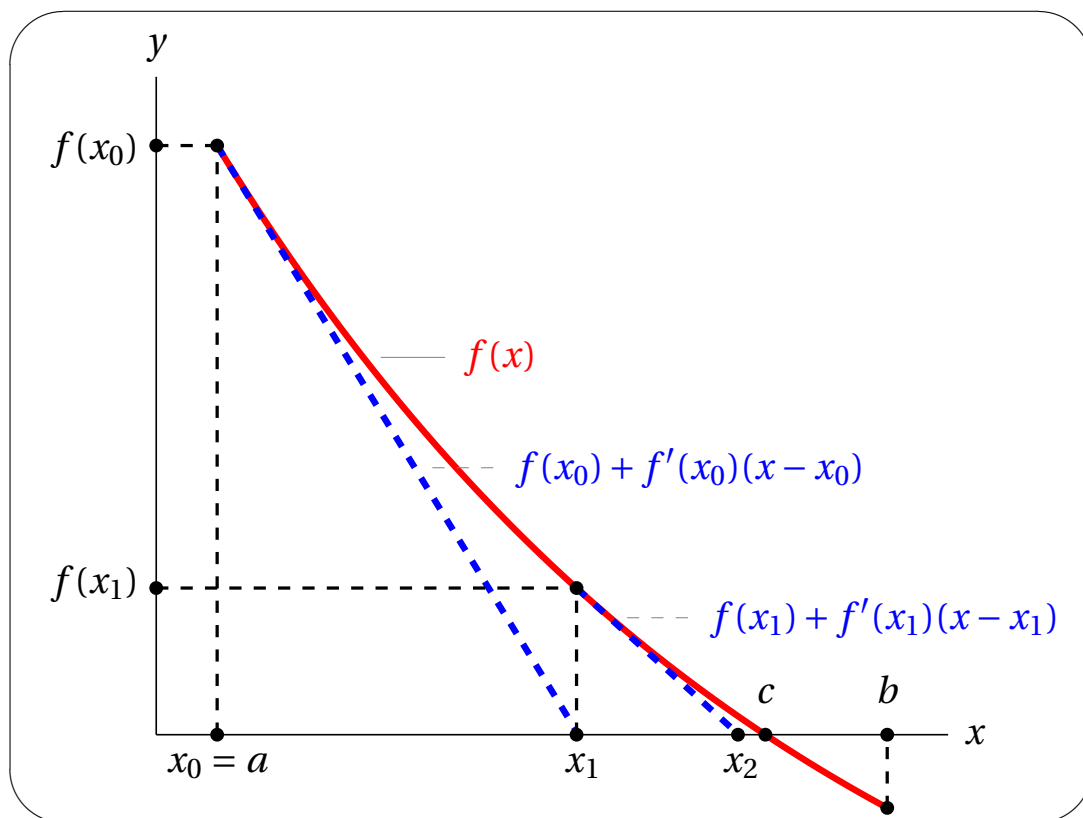
$$y = f(x_0) + f'(x_0)(x - x_0)$$

Instead of solving $f(x) = 0$, we solve:

$$f(x_0) + f'(x_0)(x - x_0) = 0$$

calling the solution x_1 we obtain:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}, \quad \text{the first approximation of } c$$



- Let us now start from x_1 and linearize as before with the tangent line at the new point $(x_1, f(x_1))$. This line has equation:

$$y = f(x_1) + f'(x_1)(x - x_1)$$

As before, we solve:

$$f(x_1) + f'(x_1)(x - x_1) = 0$$

obtaining

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}, \quad \text{the second approximation of } c$$

Proceeding with x_2 as with x_1 and continuing, we obtain the following recursively defined sequence:

$$x_0 = a, \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad \text{with } n \in \mathbb{N}$$

In this way, the term x_n can be constructed starting from x_0 with n iterations of the same algorithm. The method is therefore very well suited to automatic computation.

- The main idea of this method, called **Newton's method**, is therefore to construct a sequence that, **under certain hypotheses**, converges to c .

Theorem 1: Newton's method

Let $f : [a, b] \rightarrow \mathbb{R}$ be twice differentiable in $[a, b]$. If the following three hypotheses hold:

1. $f(a) \cdot f(b) < 0$
2. $f'(x)$ and $f''(x)$ have constant sign in $[a, b]$
3. $f(a) \cdot f''(a) > 0$

then there exists one and only one point $c \in (a, b)$ such that $f(c) = 0$, and the sequence

$$x_0 = a, \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

converges to c from below. If, instead of hypothesis 3, the following hypothesis holds

- 3'. $f(b) \cdot f''(b) > 0$

then the sequence

$$x_0 = b, \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

converges to c from above.

- Note that, under hypotheses 1. and 2., either 3. or 3'. is certainly satisfied. The function f'' has the same sign at a and b by 2., and the function f has opposite signs at a and b by 1. Therefore either 3. or 3'. holds.
- This means that the method is applicable whenever hypotheses 1. and 2. are satisfied: we only need to choose appropriately whether to set $x_0 = a$ or $x_0 = b$.
- Condition 2, apparently restrictive, will in general be satisfied provided we choose an interval $[a, b]$ small enough.

Proof. Since f is continuous, being differentiable in $[a, b]$, and 1 holds, by the intermediate zero theorem there exists at least one $c \in (a, b)$ such that $f(c) = 0$.

Moreover, since $f'(x)$ has constant sign in (a, b) , f is strictly monotone, hence this point c is unique (a strictly monotone function cannot vanish at two distinct points).

This proves the existence and uniqueness of the point c at which f vanishes. We will now prove that the sequence is monotone. From this it will follow that the sequence is convergent, by the theorem on monotone sequences.

We start by showing that if the sequence x_n converges, then $x_n \rightarrow c$. If $x_n \rightarrow \ell \in [a, b]$, passing to the limit in the equality

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad \text{we get} \quad \ell = \ell - \frac{f(\ell)}{f'(\ell)}$$

(we used the fact that f is continuous, and f' is also continuous, since f' is differentiable, because f'' exists by hypothesis). From the last equality it follows that $f(\ell) = 0$, so that $\ell = c$ (the only point where the function vanishes).

Let us therefore prove that x_n is monotone, under hypotheses 1, 2, 3. Without loss of generality, we assume $f(a) < 0$ and, by 1, consequently $f(b) > 0$. Since, by 2, $f'(x)$ has constant sign in $[a, b]$, we must have $f'(x) > 0$ on all of $[a, b]$ (if the other inequality held, f would be decreasing, and we could not have $f(a) < 0 < f(b)$). Since $f(a) < 0$ and 3 holds, $f''(a) < 0$; hence, since by 2 $f''(x)$ has constant sign, $f''(x) < 0$ on all of $[a, b]$.

We are therefore proving the case of an increasing and concave function. The other cases are proved analogously. $\square$

Proof. Consider the function:

$$g(x) = x - \frac{f(x)}{f'(x)}, \text{ we have } x_{n+1} = g(x_n) \text{ and } g(c) = c \text{ since } f(c) = 0$$

We have:

$$g'(x) = 1 - \frac{f'(x)^2 - f(x)f''(x)}{f'(x)^2} = \frac{f(x)f''(x)}{f'(x)^2} > 0 \iff f(x) < 0$$

because $f''(x) < 0$ on all of $[a, b]$. Hence $g'(x) > 0$ for $x \in [a, c]$, i.e., on $[a, c]$ the function g is strictly increasing.

Keeping these facts in mind, we now prove that:

$$x_n < x_{n+1} < c, \quad \forall n \in \mathbb{N} \tag{1}$$

For $n = 0$ we have

$$x_1 = a - \frac{f(a)}{f'(a)} > a$$

because $f(a) < 0$ and $f'(a) > 0$, hence $x_1 > x_0$. Moreover $a < c$ (and $x_0 = a$), and since g is increasing, this implies:

$$g(a) < g(c) \text{ that is } x_1 < c$$

Hence we have proved that

$$x_0 < x_1 < c$$

Applying now g to the previous inequalities (g is strictly increasing on $[a, c]$, and the three points lie in this interval, hence g preserves the inequalities), we have

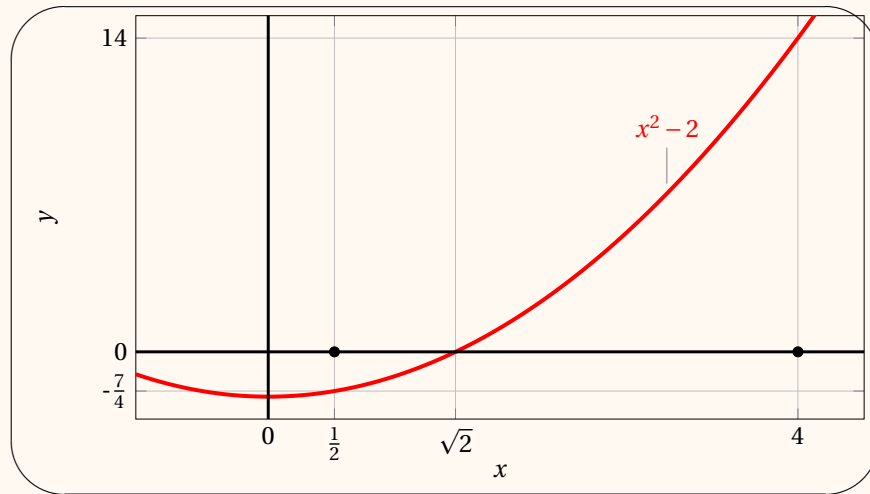
$$g(x_0) < g(x_1) < g(c) \text{ that is } x_1 < x_2 < c$$

Applying g again, we find $x_2 < x_3 < c$, and so on. Therefore (1) is true, and in particular x_n is monotone. This concludes the proof. $\square$

Example 1: Newton's method

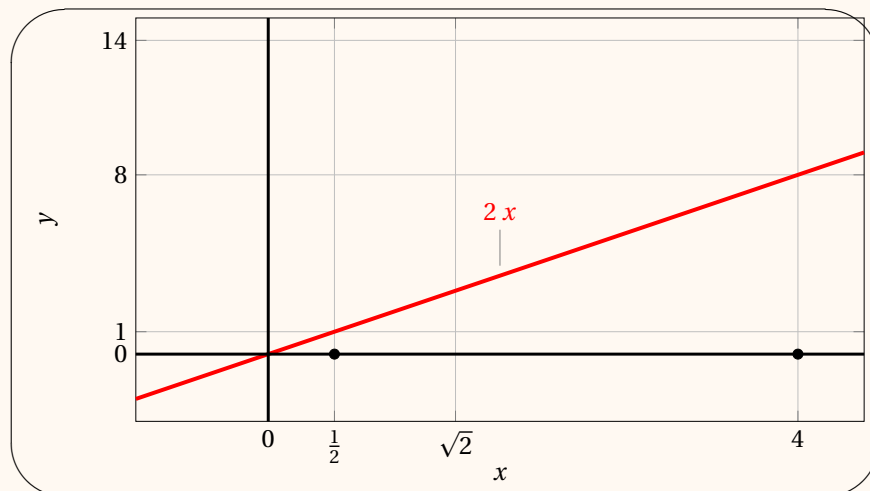
Consider the function:

$$f(x) = x^2 - 2, \quad f'(x) = 2x, \quad f''(x) = 2$$



We have:

$$f\left(\frac{1}{2}\right) = -\frac{7}{4} < 0 \quad \text{and} \quad f(4) = 14 > 0$$



Consider the interval $\left[\frac{1}{2}, 4\right]$; we have:

$$f'(x) > 0 \quad \text{and} \quad f''(x) > 0, \quad \forall x \in \left[\frac{1}{2}, 4\right]$$

Example 2: Newton's method

Hence on the interval $[\frac{1}{2}, 4]$ the function $f(x) = x^2 - 2$ satisfies the hypotheses of the theorem.

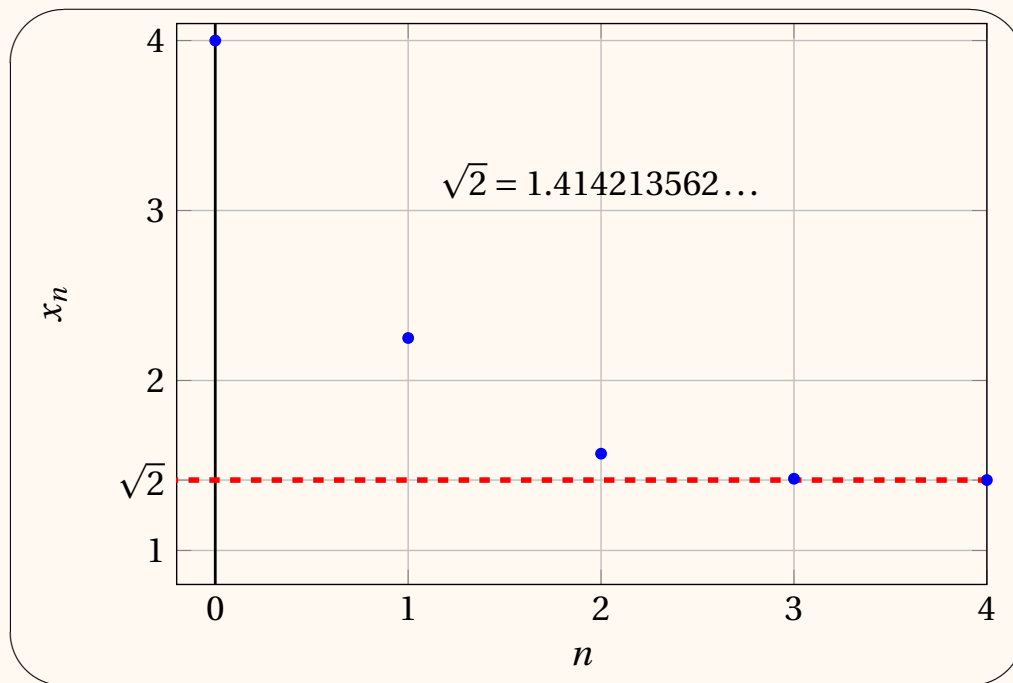
We are in the case of a function that is strictly increasing on $[\frac{1}{2}, 4]$, since $f'(x) > 0, \forall x \in [\frac{1}{2}, 4]$, and convex on $[\frac{1}{2}, 4]$, since $f''(x) > 0, \forall x \in [\frac{1}{2}, 4]$. The endpoints of the chosen interval are $a = \frac{1}{2}$ and $b = 4$.

Hypothesis 3' holds, that is, $f(4) \cdot f''(4) > 0$, and the sequence becomes:

$$x_0 = 4, \quad x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right) \text{ with } n \in \mathbb{N}$$

Computing the first values of the sequence we have:

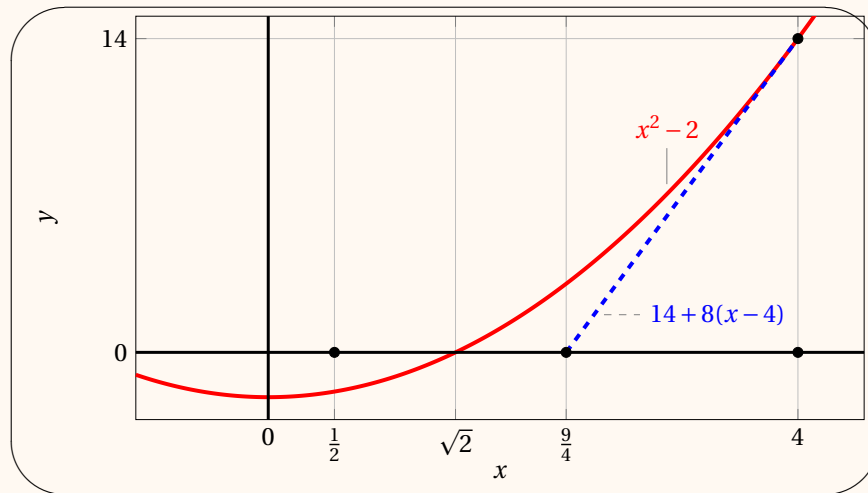
$$\begin{aligned} x_0 = 4, \quad x_1 = \frac{9}{4} = 2.25, \quad x_2 = \frac{113}{72} = 1.569444444\dots \\ x_3 = \frac{23,137}{16,272} = 1.421890363\dots, \quad x_4 = \frac{1,064,876,737}{752,970,528} = 1.414234285\dots \end{aligned}$$



Example 3: Newton's method

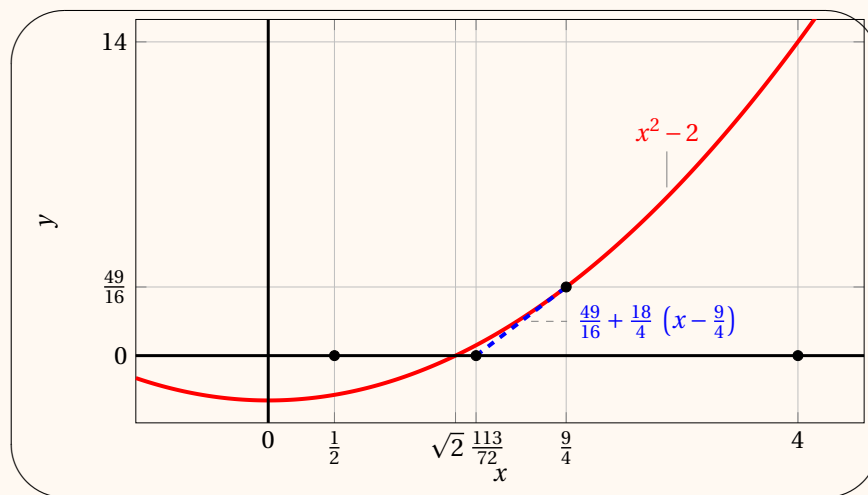
At the first iteration the tangent line is:

$$y = 14 + 8(x - 4) \text{ and } x_1 = \frac{9}{4}$$



At the second iteration the tangent line is:

$$y = \frac{49}{16} + \frac{18}{4} \left(x - \frac{9}{4} \right) \text{ and } x_2 = \frac{113}{72}$$



- Newton's method can be used, for example, to compute the k -th root of a number $c \in \mathbb{R}_+$, by looking for the zeros of the following function:

$$f(x) = x^k - c \text{ that is } x = \sqrt[k]{c}$$

Given a value $b \geq \sqrt[k]{c}$, we have the recursively defined sequence:

$$x_0 = b, \quad x_{n+1} = x_n - \frac{x_n^k - c}{k x_n^{k-1}} \text{ with } n \in \mathbb{N}$$

expanding we obtain

$$x_n - \frac{x_n^k - c}{k x_n^{k-1}} = \frac{k x_n^k - x_n^k + c}{k x_n^{k-1}} = \frac{1}{k} \left((k-1)x_n + \frac{c}{x_n^{k-1}} \right)$$

and hence the sequence becomes:

$$x_0 = b, \quad x_{n+1} = \frac{1}{k} \left((k-1)x_n + \frac{c}{x_n^{k-1}} \right) \text{ with } n \in \mathbb{N}$$

With $k = 2$ we obtain the sequence of Heron's algorithm:

$$x_0 = b, \quad x_{n+1} = \frac{1}{2} \left(x_n + \frac{c}{x_n} \right) \text{ with } n \in \mathbb{N}$$

2 Error estimates for Newton's method

- The **absolute error** of Newton's method is:

$$\varepsilon_n = |x_n - c|$$

where $c \in (a, b)$ is the zero of the function, that is, $f(c) = 0$.

The value of c is unknown, hence we need a way to analyze the values of the absolute errors that is independent of c . We also want to understand how “fast” the error decreases by comparing the errors of two consecutive iterations, that is, we want to determine the “**rate of convergence**” of the algorithm.

- Using the theorem on the Taylor formula with Lagrange remainder we can write:

$$f(x) = f(x_n) + f'(x_n)(x - x_n) + \frac{1}{2} f''(\mu)(x - x_n)^2$$

for some μ between x_n and x . In particular, choosing $x = c$ we have:

$$0 = f(c) = f(x_n) + f'(x_n)(c - x_n) + \frac{1}{2} f''(\mu)(c - x_n)^2 \quad (2)$$

Recalling that x_{n+1} was obtained as the solution of the equation:

$$f(x_n) + f'(x_n)(x - x_n) = 0$$

we have

$$0 = f(x_n) + f'(x_n)(x_{n+1} - x_n) \quad (3)$$

Subtracting (3) from (2), we obtain:

$$0 = f'(x_n)(c - x_{n+1}) + \frac{1}{2} f''(\mu)(c - x_n)^2$$

which in turn implies:

$$x_{n+1} - c = \frac{1}{2} \frac{f''(\mu)}{f'(x_n)} (x_n - c)^2 \quad \text{hence} \quad |x_{n+1} - c| = \frac{1}{2} \frac{|f''(\mu)|}{|f'(x_n)|} |x_n - c|^2 \quad (4)$$

Now, if x_n is close to c (that is, $x_n \approx c$), then μ , which lies between x_n and c , is also close to c ; hence in these cases:

$$f''(\mu) \approx f''(c) \quad \text{and} \quad f'(x_n) \approx f'(c)$$

then, substituting, we obtain:

$$|x_{n+1} - c| \approx \frac{1}{2} \frac{|f''(c)|}{|f'(c)|} |x_n - c|^2 \quad \text{hence} \quad \varepsilon_{n+1} \approx \frac{1}{2} \frac{|f''(c)|}{|f'(c)|} \varepsilon_n^2 \quad (5)$$

The absolute error of Newton's method at each step is proportional to the square of the absolute error at the previous step. **The rate of convergence is quadratic.**

- However, the error is also proportional to the constant:

$$\frac{|f''(c)|}{2|f'(c)|}$$

hence, if $|f'(c)|$ is very small, or zero, or $|f''(c)|$ is very large, the convergence may be very slow or may even fail to occur.

- Let us go back to equation (4), and determine $L, M > 0$ such that:

$$|f'(x_n)| \geq L \quad \text{and} \quad |f''(\mu)| \leq M$$

then we can write:

$$|x_{n+1} - c| \leq \frac{1}{2} \frac{M}{L} |x_n - c|^2 \quad \text{hence} \quad \varepsilon_{n+1} \leq \frac{1}{2} \frac{M}{L} \varepsilon_n^2$$

We have obtained the relation between ε_{n+1} and ε_n , which is independent of both x_n and c (but depends on M and L).

Example 4: Error estimates for Newton's method ($\sqrt{2} = 1.414213562\dots$)

Let us go back to the previous example:

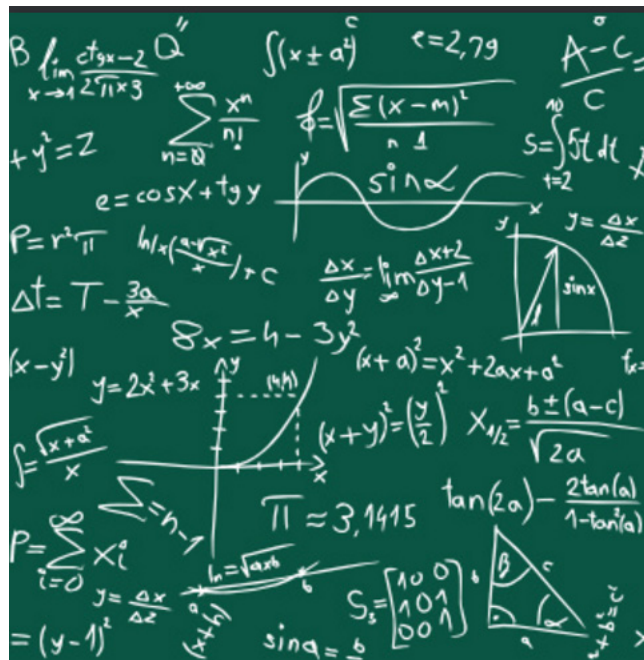
$$f(x) = x^2 - 2, \quad f'(x) = 2x \quad \text{and} \quad f''(x) = 2$$

Hence we can choose $M = 2$ and, since $x_n \geq 1, \forall n \in \mathbb{N}$, we can choose $L = 2$ (in this case $\frac{M}{L} = \frac{2}{2} = 1$).

We know that $\sqrt{2} > 1$, hence with $n = 2$ we have $\varepsilon_2 \leq \left| \frac{113}{72} - 1 \right| \approx 0.56944$, and we have:

iter.	x_n	estimate of $\sqrt{2}$	ε_n
$n = 3$	$\frac{23,137}{16,272}$	1.421890363...	$\leq \frac{1}{2} \varepsilon_2^2 \approx 0.16213\dots$
$n = 4$	$\frac{1,064,876,737}{752,970,528}$	1.414234285...	$\leq \frac{1}{2} \varepsilon_3^2 \approx 0.01314\dots$

Numerical series



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Numerical series

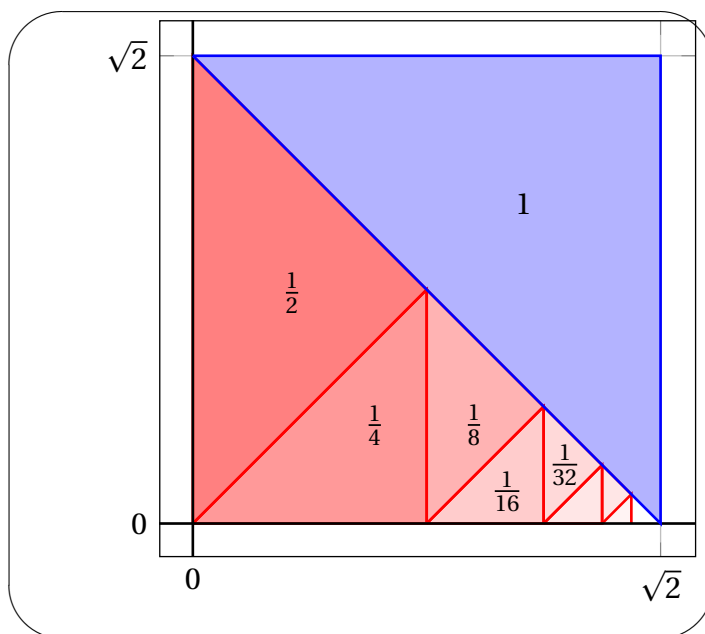
- We now introduce numerical series, which extend the operation of addition to an infinite number of terms.

The sum of infinitely many terms, even if they are all positive, can give a finite result.

- Imagine measuring a square of area 2 using the following procedure: we divide the square in half along the diagonal and measure the first right triangle: we obtain 1; then we divide the second right triangle in half and measure the first of the two resulting right triangles: we obtain $\frac{1}{2}$; the remaining right triangle is again divided in half... and so on indefinitely. We obtain the infinite sum:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots + \frac{1}{2^k} + \dots = \sum_{k=0}^{\infty} \frac{1}{2^k}$$

which, by the way it was constructed, must give 2 as its result.



Definition 1: of numerical series

Given a sequence $\{a_k\}_{k \in \mathbb{N}}$, we call **numerical series** of the terms a_k the expression:

$$\sum_{k=0}^{\infty} a_k$$

- It is read “series (or also sum) for k from 0 to $+\infty$ of a_k ”. The values a_k are called the **general terms of the series**.

Definition 2: of sequence of partial sums

The numbers

$$s_n = \sum_{k=0}^n a_k = a_0 + a_1 + \dots + a_n, \quad \forall n \in \mathbb{N}$$

are called the **n -th partial sums** of the series and they define **the sequence $\{s_n\}$ of partial sums**.

The **behavior (character) of the series** is determined by the limit of the sequence $\{s_n\}$ as n tends to infinity. We say that a series is **convergent, divergent, irregular**, if the **sequence $\{s_n\}$ of partial sums** is **convergent, divergent or irregular**, respectively.

Definition 3: of sum of the series

If the sequence $\{s_n\}$ of partial sums is **convergent**, that is, if

$$s_n \rightarrow s \in \mathbb{R} \quad \text{as } n \rightarrow +\infty$$

we say that s is **the sum of the series**, and we write $\sum_{k=0}^{\infty} a_k = s$.

Hence, if the sequence $\{s_n\}$ is **convergent**, we have:

$$\sum_{k=0}^{\infty} a_k = \lim_{n \rightarrow +\infty} \sum_{k=0}^n a_k = \lim_{n \rightarrow +\infty} s_n = s$$

The series makes precise the idea of a sum of infinitely many terms, that is, we compute the limit as $n \rightarrow +\infty$ of the finite sum of the first n terms.

- If, instead of summing starting from 0, we start from an index $n_0 > 0$, we write $\sum_{k=n_0}^{\infty} a_k$
- To denote a generic numerical series without specifying the starting index n_0 we will use the symbol $\sum a_k$

A generic numerical series $\sum a_k$ always involves **two different sequences**:

1. the sequence $\{a_k\}$ of the **terms of the series**
2. the sequence $\{s_n\}$ of **partial sums**

Example 1: behavior of a series

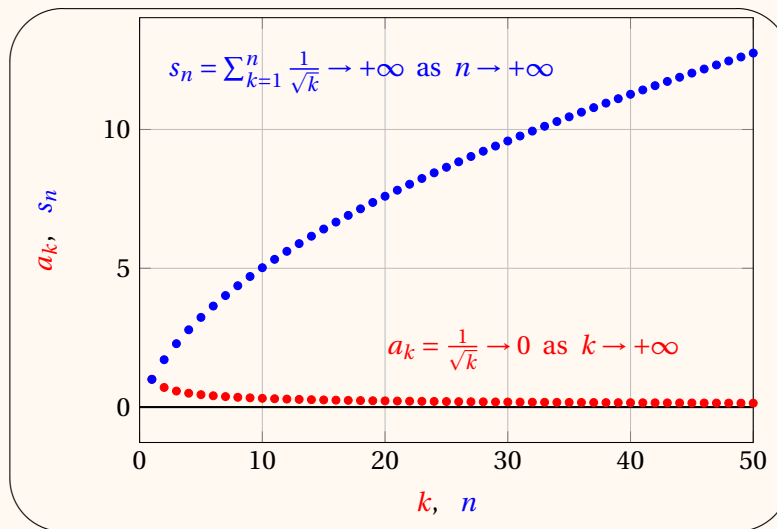
Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$$

We have:

$$s_n = \sum_{k=1}^n \frac{1}{\sqrt{k}} = 1 + \underbrace{\frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}}}_{n \text{ terms, each } \geq \frac{1}{\sqrt{n}}} \geq n \cdot \frac{1}{\sqrt{n}} = \sqrt{n} \rightarrow +\infty \text{ as } n \rightarrow +\infty$$

Hence the sequence of partial sums $\{s_n\}$ is divergent by the comparison theorem for sequences and, consequently, the series is divergent.



1.1 Main properties of numerical series

Remark 1

If a sequence $\{a_k\}$ has non-negative terms, that is, $a_k \geq 0, \forall k$, then the sequence of partial sums $\{s_n\}$ is increasing and regular.

Proof. Given a sequence $\{a_k\}$ with non-negative terms, the sequence of partial sums $\{s_n\}$ is increasing since:

$$s_{n+1} = s_n + \underbrace{a_{n+1}}_{\geq 0} \geq s_n, \quad \forall n, \quad \text{hence} \quad \lim_{n \rightarrow +\infty} s_n = \sup_{n \in \mathbb{N}} \{s_n\}$$

by the monotonicity theorem for sequences. Consequently the sequence $\{s_n\}$ cannot be irregular. It is therefore regular, that is, it either converges or diverges. $\square$

Given a sequence $\{a_k\}$ with non-negative terms:

- If $\{s_n\}$ is **increasing and bounded**, then it has a finite limit and $\sum a_k$ converges.
- If $\{s_n\}$ is **increasing and unbounded**, then $\sum a_k$ diverges to $+\infty$.

- This remark also holds for sequences with eventually non-negative terms, positive terms, or eventually positive terms.

Theorem 1

If a series $\sum a_k$ is convergent, then $\lim_{k \rightarrow +\infty} a_k = 0$

Proof. By definition of convergent series, the sequence of partial sums $\{s_n\}$ converges to a real number s , i.e., $s_n \rightarrow s \in \mathbb{R}$ as $n \rightarrow +\infty$. Without loss of generality we consider $n_0 = 0$.

We observe that the sequence $\{s_n\}$ can be defined recursively:

$$s_0 = a_0, \quad s_n = s_{n-1} + a_n, \quad \forall n \geq 1$$

Consequently we have:

$$a_n = s_n - s_{n-1} \quad \text{and hence} \quad \lim_{n \rightarrow +\infty} a_n = \lim_{n \rightarrow +\infty} (s_n - s_{n-1}) = s - s = 0$$

$\square$

The convergence of a series therefore implies that $a_k \rightarrow 0$ as $k \rightarrow +\infty$, that is:

$$\sum a_k \text{ convergent} \Rightarrow \lim_{k \rightarrow +\infty} a_k = 0 \quad (1)$$

but the converse is not true (Example 1 is a counterexample):

$$\lim_{k \rightarrow +\infty} a_k = 0 \not\Rightarrow \sum a_k \text{ convergent} \quad (2)$$

Hence the fact that $a_k \rightarrow 0$ as $k \rightarrow +\infty$ is a necessary but not sufficient condition for the convergence of a series.

- From the contrapositive of (1) we have:

$$\lim_{k \rightarrow +\infty} a_k \neq 0 \Rightarrow \sum a_k \text{ is divergent or irregular}$$

that is, if the limit is not equal to zero, then the series is not convergent.

1.2 Tails of series

If we modify a finite number of terms of a series, the value of the sum may change, but the behavior of the series remains unchanged.

- If we modify the value of one term, for example a_{n_0} , for $n \geq n_0$ the sequences of partial sums, the original and the modified one, differ only by that term: hence both converge, both diverge, or both are irregular. The same holds if we modify a finite number of terms.

The results on the behavior of series therefore also hold if the hypotheses are satisfied “eventually”, i.e., from a certain index n_0 onward.

Definition 4: of tail of a series

Given a series $\sum_{k=0}^{\infty} a_k$ and a value $m \in \mathbb{N}$, the series $\sum_{k=m+1}^{\infty} a_k$ is called a **tail of the series**.

- Given a series $\sum_{k=0}^{\infty} a_k$ and $m \in \mathbb{N}$ we have:

$$\sum_{k=m+1}^n a_k = \left(\sum_{k=0}^n a_k \right) - \left(\sum_{k=0}^m a_k \right) = s_n - s_m, \quad \forall n > m \quad (3)$$

hence for every tail (that is, for every m) the associated sequence of partial sums is equal to that of the original series minus a constant. Consequently we have the following remark:

Every tail of a series has the same behavior as the original series.

- Given a convergent series, taking the limit as $n \rightarrow +\infty$ in (3) we obtain the relation:

$$\sum_{k=m+1}^{\infty} a_k = \left(\sum_{k=0}^{\infty} a_k \right) - \left(\sum_{k=0}^m a_k \right) = s - s_m \quad \text{as } n \rightarrow +\infty \quad (4)$$

Consequently we have the following remark:

Given $m \in \mathbb{N}$, every tail $\sum_{k=m+1}^{\infty} a_k$ of a convergent series can be interpreted as the **error** that we make when approximating the sum s with the partial sum s_m .

Theorem 2

If a series $\sum a_k$ is convergent, then

$$\underbrace{s - s_m}_{= \sum_{k=m+1}^{\infty} a_k} \rightarrow 0 \quad \text{as } m \rightarrow +\infty$$

Proof. For every $n > m$, the n -th partial sum of the series is:

$$s_n = s_m + \sum_{k=m+1}^n a_k$$

Taking the limit as $n \rightarrow +\infty$, we have:

$$\begin{aligned} s &= s_m + \lim_{n \rightarrow +\infty} \sum_{k=m+1}^n a_k \\ &= \underbrace{\lim_{n \rightarrow +\infty} \sum_{k=m+1}^n a_k}_{= \sum_{k=m+1}^{\infty} a_k} \end{aligned}$$

Now taking the limit as $m \rightarrow +\infty$, we have:

$$s = s + \lim_{m \rightarrow +\infty} \sum_{k=m+1}^{\infty} a_k \Rightarrow \lim_{m \rightarrow +\infty} \sum_{k=m+1}^{\infty} a_k = 0$$

that is, the claim of the theorem. □

This theorem tells us that, if a series is convergent, then the value $s - s_m$, that is, the error that we make when approximating the sum s with the partial sum s_m , tends to zero as $m \rightarrow +\infty$. In other words, the tail of a convergent series tends to zero as m tends to infinity.

1.3 Harmonic series

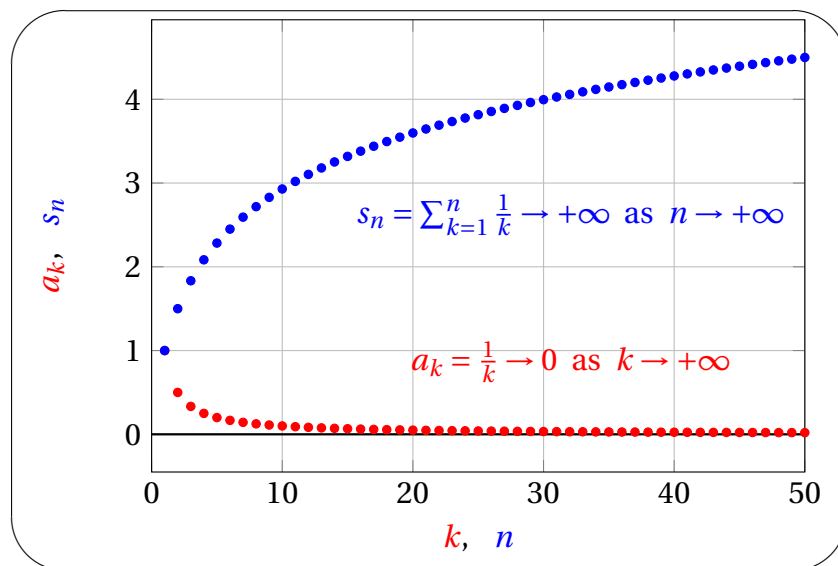
Definition 5: of harmonic series

The **harmonic series** is the series $\sum_{k=1}^{\infty} \frac{1}{k}$

Theorem 3: behavior of the harmonic series

The harmonic series diverges to $+\infty$

- Graphically we have:



The harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ is a counterexample that proves (2), since:

$$\lim_{k \rightarrow +\infty} \frac{1}{k} = 0 \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{1}{k} \text{ diverges to } +\infty$$

- Before proving the theorem, we observe that for certain values of n the partial sums are:

$$\begin{aligned}
\underbrace{s_1}_{=s_{2^0}} &= 1 + \frac{0}{2}, & \underbrace{s_2}_{=s_{2^1}} &= s_1 + \frac{1}{2} = 1 + \frac{0}{2} + \frac{1}{2} = 1 + \frac{1}{2}, & \underbrace{s_4}_{=s_{2^2}} &= s_2 + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{\geq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}} \geq 1 + \frac{1}{2} + \frac{1}{2} = 1 + \frac{2}{2} \\
\underbrace{s_8}_{=s_{2^3}} &= s_4 + \underbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)}_{\geq \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}} \geq 1 + \frac{2}{2} + \frac{1}{2} = 1 + \frac{3}{2} \\
\underbrace{s_{16}}_{=s_{2^4}} &= s_8 + \underbrace{\left(\frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \frac{1}{15} + \frac{1}{16}\right)}_{\geq \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{1}{2}} \geq 1 + \frac{3}{2} + \frac{1}{2} = 1 + \frac{4}{2}
\end{aligned}$$

Proof. We prove by induction on 2^n that

$$s_{2^n} \geq 1 + \frac{n}{2}, \quad \forall n \in \mathbb{N}$$

Base case. Let $n = 0$. Then the statement becomes $s_{2^0} \geq 1 + \frac{0}{2}$, i.e., $1 \geq 1$, which is clearly true.

Inductive step. Assume that it is true for 2^{n-1} and let us prove it for 2^n . By the inductive hypothesis, we have:

$$s_{2^{n-1}} \geq 1 + \frac{n-1}{2}$$

Moreover, we have:

$$\begin{aligned}
s_{2^n} &= s_{2^{n-1}} + \underbrace{\sum_{k=2^{n-1}+1}^{2^n} \left(\frac{1}{k}\right)}_{\geq 2^{n-1} \cdot \frac{1}{2^n} = \frac{1}{2}} \quad \text{hence} \quad s_{2^n} \geq 1 + \frac{n-1}{2} + \frac{1}{2} = 1 + \frac{n}{2}
\end{aligned}$$

which is exactly the desired statement for 2^n .

We therefore have:

$$s_{2^n} \geq 1 + \frac{n}{2} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty$$

and $\{s_n\}$ is unbounded above and also monotonically increasing, since $a_k = 1/k > 0, \forall k > 1$. Consequently $\{s_n\}$ diverges to $+\infty$ by the monotonicity theorem for sequences, and the harmonic series diverges to $+\infty$.

□

The sequence $a_k = 1/k, \forall k \geq 1$, of the harmonic series has positive terms, hence the sequence of partial sums $\{s_n\}$ is increasing, and it is divergent since it is unbounded.

1.4 Geometric series

Definition 6: of geometric series

Given $q \in \mathbb{R}$, the **geometric series** with **ratio** q is the series $\sum_{k=0}^{\infty} q^k$

Theorem 4: behavior and sum of the geometric series

Given the ratio $q \in \mathbb{R}$,

$$\text{the geometric series } \sum_{k=0}^{\infty} q^k \text{ is } \begin{cases} \text{convergent} & \text{if } |q| < 1 \\ \text{divergent to } +\infty & \text{if } q \geq 1 \\ \text{irregular} & \text{if } q \leq -1 \end{cases}$$

If the geometric series is convergent, its sum s is $\frac{1}{1-q}$

Proof. Given $q \in \mathbb{R}$ and $n \in \mathbb{N}$, the n -th partial sum of the geometric series is:

$$\sum_{k=0}^n q^k = \begin{cases} \frac{1 - q^{n+1}}{1 - q} & \text{if } q \neq 1 \\ n + 1 & \text{if } q = 1 \end{cases}$$

since it equals the sum of the first $n + 1$ terms of the geometric progression. Moreover, we have:

$$\lim_{n \rightarrow +\infty} q^n = \begin{cases} +\infty & \text{if } q > 1 \\ 1 & \text{if } q = 1 \\ 0 & \text{if } |q| < 1 \\ \text{does not exist} & \text{if } q \leq -1 \end{cases}$$

Hence, if $q \neq 1$:

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} \frac{1 - q^{n+1}}{1 - q} = \frac{1}{1 - q} \lim_{n \rightarrow +\infty} (1 - q^{n+1}) = \begin{cases} \frac{1}{1 - q} & \text{if } |q| < 1 \\ +\infty & \text{if } q > 1 \\ \text{does not exist} & \text{if } q \leq -1 \end{cases}$$

and if $q = 1$ we have:

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} (n + 1) = +\infty$$

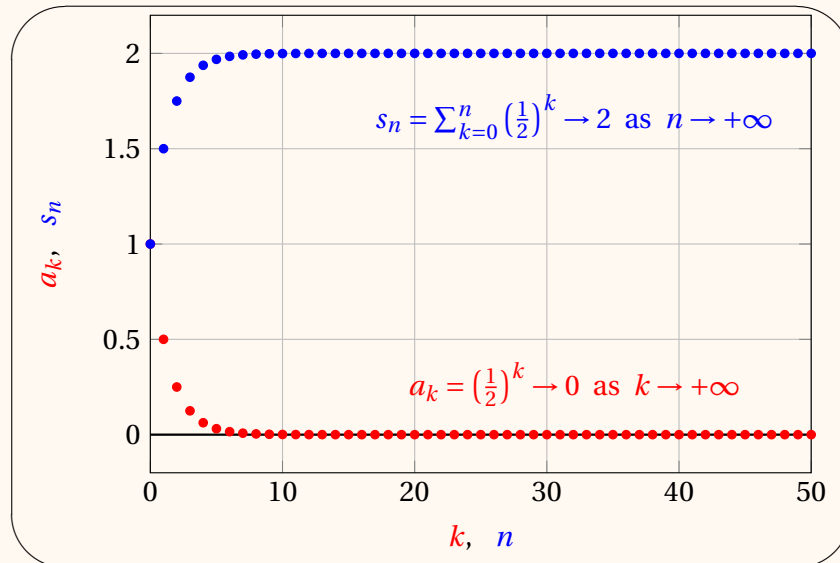
□

Example 2: geometric series

Let us determine the behavior of the series:

$$\sum_{k=0}^{\infty} \frac{1}{2^k} = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k$$

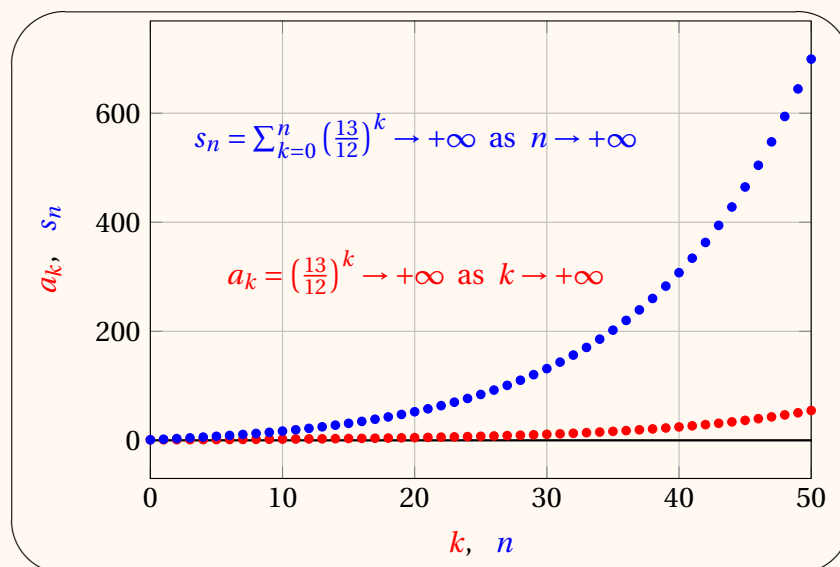
It is a geometric series with ratio $q = 1/2$, hence it is convergent and its sum s is $1/(1 - 1/2) = 2$.



Let us determine the behavior of the series:

$$\sum_{k=0}^{\infty} \left(\frac{13}{12}\right)^k$$

It is a geometric series with ratio $q = 13/12$, hence it is divergent.



1.5 Telescoping series

Definition 7: of telescoping series

A **telescoping series** is a series of the form:

$$\sum_{k=n_0}^{\infty} (b_k - b_{k+1}) \quad \text{where } \{b_k\} \text{ is a sequence}$$

Theorem 5: behavior and sum of the telescoping series

A telescoping series converges, diverges or is irregular according to whether the sequence $\{b_k\}$ converges, diverges or is irregular, respectively.

$$\text{If } \lim_{k \rightarrow +\infty} b_k = \ell \in \mathbb{R}^* \quad \text{then} \quad \sum_{k=n_0}^{\infty} (b_k - b_{k+1}) = b_{n_0} - \underbrace{\lim_{k \rightarrow +\infty} b_k}_{=\ell \in \mathbb{R}^*}$$

Moreover, if $\ell \in \mathbb{R}$, then the telescoping series is convergent and its sum s is $b_{n_0} - \ell$

Proof. We have

$$\begin{aligned} s_n &= \sum_{k=n_0}^{n_0+n} (b_k - b_{k+1}) \\ &= (b_{n_0} - b_{n_0+1}) + (b_{n_0+1} - b_{n_0+2}) + \dots + (b_{n_0+n} - b_{n_0+n+1}) \\ &= b_{n_0} - b_{n_0+n+1} \end{aligned}$$

hence:

$$\sum_{k=n_0}^{\infty} (b_k - b_{k+1}) = \lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} (b_{n_0} - b_{n_0+n+1}) = b_{n_0} - \lim_{n \rightarrow +\infty} b_n$$

□

Remark 2

The series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$, called **Mengoli's series**, is convergent and its sum s is equal to 1.

Proof. We have:

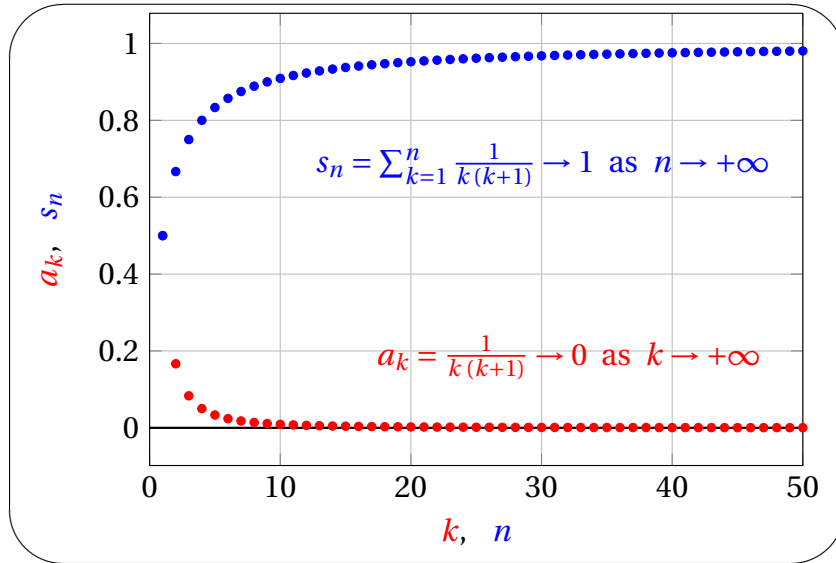
$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \sum_{k=1}^{\infty} \frac{(k+1) - k}{k(k+1)} = \sum_{k=1}^{\infty} \underbrace{\frac{1}{k}}_{=b_k} - \underbrace{\frac{1}{k+1}}_{=b_{k+1}}$$

Mengoli's series therefore has the form:

$$b_k - b_{k+1} \quad \text{with} \quad b_k = \frac{1}{k} \quad \text{and moreover} \quad \lim_{k \rightarrow +\infty} b_k = \lim_{k \rightarrow +\infty} \frac{1}{k} = 0$$

hence it is a convergent telescoping series ($n_0 = 1$) and moreover $s = b_1 = 1$. $\square$

- Graphically we have:



Proof. Alternative proof

For every $n \geq 1$, the value of the partial sum of Mengoli's series is:

$$s_n = \sum_{k=1}^n \frac{1}{k(k+1)} = \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) = \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{n+1}$$

Hence we have

$$\sum_{k=1}^{\infty} a_k = \lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} 1 - \frac{1}{n+1} = 1$$

Consequently Mengoli's series is convergent and its sum s is 1. $\square$

Example 3: telescoping series

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{1}{(3k+2)(3k+5)}$$

We have:

$$\begin{aligned} \frac{1}{(3k+2)(3k+5)} &= \frac{\alpha}{3k+2} - \frac{\beta}{3k+5} = \frac{\alpha(3k+5) - \beta(3k+2)}{(3k+2)(3k+5)} = \frac{3(\alpha-\beta)k + 5\alpha - 2\beta}{(3k+2)(3k+5)} \\ &\Rightarrow \begin{cases} 3(\alpha-\beta) = 0 \\ 5\alpha - 2\beta = 1 \end{cases} \Rightarrow \alpha = \beta, \quad 3\alpha = 1 \text{ hence } \alpha = \beta = \frac{1}{3} \end{aligned}$$

consequently:

$$\frac{1}{(3k+2)(3k+5)} = \frac{1}{3} \left(\frac{1}{3k+2} - \frac{1}{3k+5} \right) = \frac{1}{3} \left(\underbrace{\frac{1}{3k+2}}_{=b_k} - \underbrace{\frac{1}{3(k+1)+2}}_{=b_{k+1}} \right)$$

The series therefore has the form:

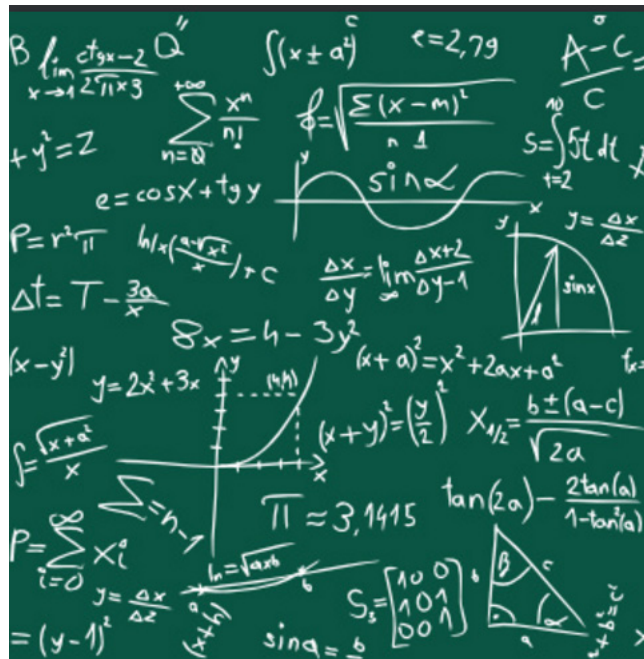
$$b_k - b_{k+1} \quad \text{with} \quad b_k = \frac{1}{3k+2} \quad \text{and moreover} \quad \lim_{k \rightarrow +\infty} b_k = \lim_{k \rightarrow +\infty} \frac{1}{3k+2} = 0$$

that is, a telescoping series ($n_0 = 1$), and we have:

$$\sum_{k=1}^{\infty} \frac{1}{(3k+2)(3k+5)} = \frac{1}{3} \left(b_1 - \underbrace{\lim_{k \rightarrow +\infty} b_k}_{=0} \right) = \frac{1}{3} b_1 = \frac{1}{15}$$

Hence it is a convergent telescoping series and its sum s is $\frac{1}{15}$.

Series with non-negative terms



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Series with non-negative terms

A **series** $\sum a_k$ **with non-negative terms** (or positive terms) is **regular**, that is, it is either convergent or divergent to $+\infty$ (it cannot be irregular, as we have proved). Such a series converges if and only if the sequence of its partial sums is bounded.

1.1 Comparison test

Theorem 1: Comparison test

Let $\{a_k\}$ and $\{b_k\}$ be two sequences with non-negative terms such that $a_k \leq b_k$ eventually; then:

$$i) \quad \sum b_k \text{ convergent} \Rightarrow \sum a_k \text{ convergent}$$

$$ii) \quad \sum a_k \text{ divergent} \Rightarrow \sum b_k \text{ divergent}$$

- The series $\sum b_k$ is called the **dominating series** (majorant), while the series $\sum a_k$ is called the **dominated series** (minorant).

Proof. Since $a_k \leq b_k$ eventually, there exists $m \in \mathbb{N}$ such that:

$$a_k \leq b_k, \quad \forall k \geq m$$

Let us now consider the n -th partial sums of the tails of the two sequences $\{a_k\}$ and $\{b_k\}$, with $n > m$:

$$s_n^a = \sum_{k=m+1}^n a_k \quad \text{and} \quad s_n^b = \sum_{k=m+1}^n b_k$$

Since $0 \leq a_k \leq b_k, \forall k \geq m$, we have

$$0 \leq s_n^a \leq s_n^b, \quad \forall n \geq m \quad (1)$$

The sequences $\{s_n^a\}$ and $\{s_n^b\}$ are regular, since $\{a_k\}$ and $\{b_k\}$ have non-negative terms. Hence the claims *i*) and *ii*) are logically equivalent, so it suffices to prove the second one.

Saying that $\sum a_n$ is divergent means, by the definition of divergent series, that $s_n \rightarrow +\infty$ as $n \rightarrow +\infty$. Moreover, since every tail has the same behavior as the original series, we have $s_n^a \rightarrow +\infty$ as $n \rightarrow +\infty$.

From (1), by the comparison theorem for sequences, also $s_n^b \rightarrow +\infty$ as $n \rightarrow +\infty$. Hence the tail of $\{b_k\}$ is divergent and consequently $\sum b_n$ is divergent. $\square$

Remark 1

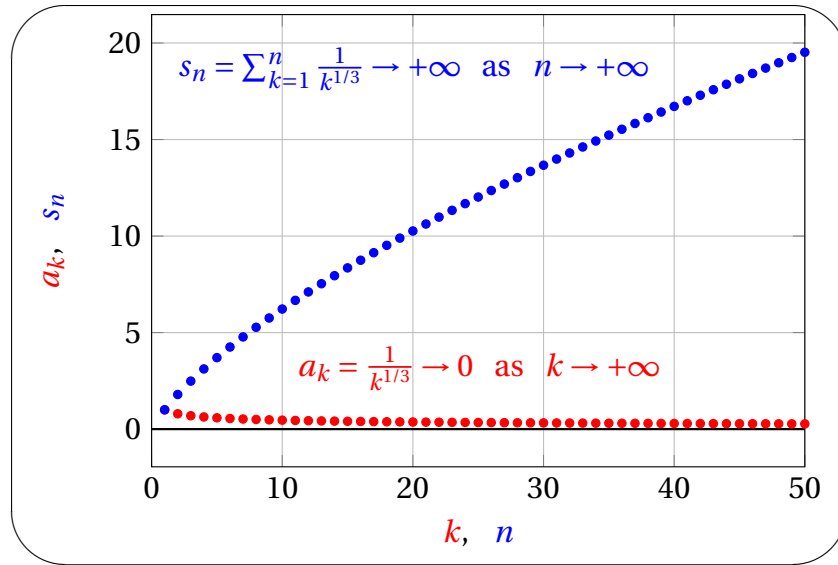
Given $\alpha \leq 1$, the series $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is divergent to $+\infty$.

Proof. With $\alpha = 1$ we have the harmonic series, which diverges to $+\infty$. With $\alpha < 1$, the series dominates the harmonic series, since:

$$\frac{1}{k} \leq \frac{1}{k^\alpha}, \quad \forall k \in \mathbb{N}, k \geq 1 \quad \text{and} \quad \alpha < 1$$

hence $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is divergent by the comparison test. $\square$

- Graphically, for example with $\alpha = \frac{1}{3}$, we have:



1.2 Limit comparison test

Theorem 2: Limit comparison test

Let $\{a_k\}$ and $\{b_k\}$ be two sequences with positive terms. If the sequences are asymptotically equivalent, that is, if:

$$a_k \sim b_k \text{ as } k \rightarrow +\infty$$

then the corresponding series $\sum a_k$ and $\sum b_k$ are regular and have the same behavior, i.e., either they are both convergent or they are both divergent.

Proof. The series $\sum a_k$ and $\sum b_k$ are regular because the sequences $\{a_k\}$ and $\{b_k\}$ have positive terms. Since $a_k \sim b_k$ as $k \rightarrow +\infty$, we have:

$$\frac{a_k}{b_k} \rightarrow 1 \text{ as } k \rightarrow +\infty$$

Hence, for every $\varepsilon > 0$, there exists $m \in \mathbb{N}$ such that $\forall k \geq m$ we have:

$$1 - \varepsilon < \frac{a_k}{b_k} < 1 + \varepsilon \quad \text{and hence} \quad (1 - \varepsilon) b_k < a_k < (1 + \varepsilon) b_k \quad \text{since } b_k > 0, \forall k$$

We have thus proved that $(1 - \varepsilon) b_k < a_k < (1 + \varepsilon) b_k$ eventually. Hence, by the comparison test, the series $\sum a_k$ and $\sum b_k$ have the same behavior.

The first of the two inequalities implies that if $\sum a_k$ is convergent then $\sum b_k$ is also convergent, while the second one implies that if $\sum a_k$ is divergent then $\sum b_k$ is also divergent.

□

Remark 2

Given $\alpha \geq 2$, the series $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is convergent.

Proof. With $\alpha = 2$ we have:

$$\frac{1}{k^2} \sim \frac{1}{k(k+1)} \quad \text{as } k \rightarrow +\infty$$

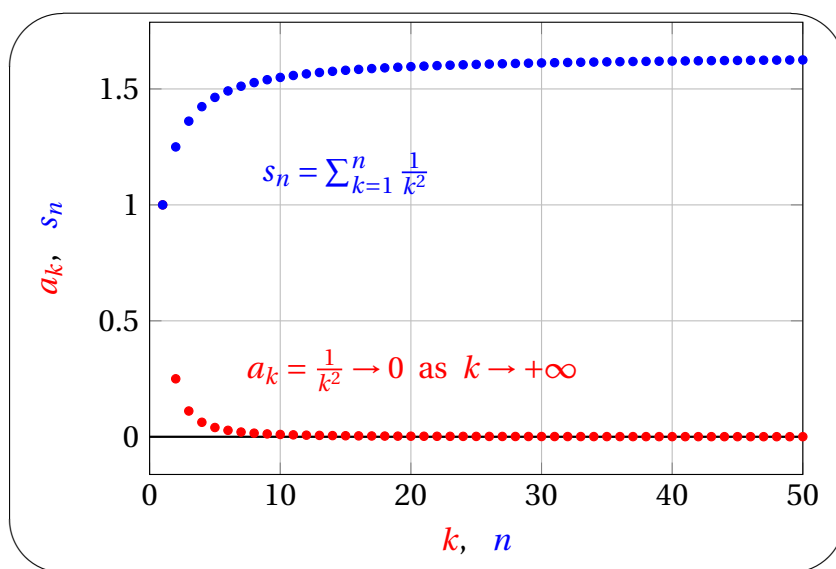
and the series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$ converges (Mengoli's series). Hence, by the limit comparison test, $\sum_{k=1}^{\infty} \frac{1}{k^2}$ also converges.

With $\alpha > 2$, the series is dominated by the series $\sum_{k=1}^{\infty} \frac{1}{k^2}$, since:

$$\frac{1}{k^\alpha} \leq \frac{1}{k^2}, \quad \forall k \in \mathbb{N}, k \geq 1 \quad \text{and} \quad \alpha > 2$$

hence $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is convergent by the comparison test. □

- Graphically, for example with $\alpha = 2$, we have:



Example 1: limit comparison test

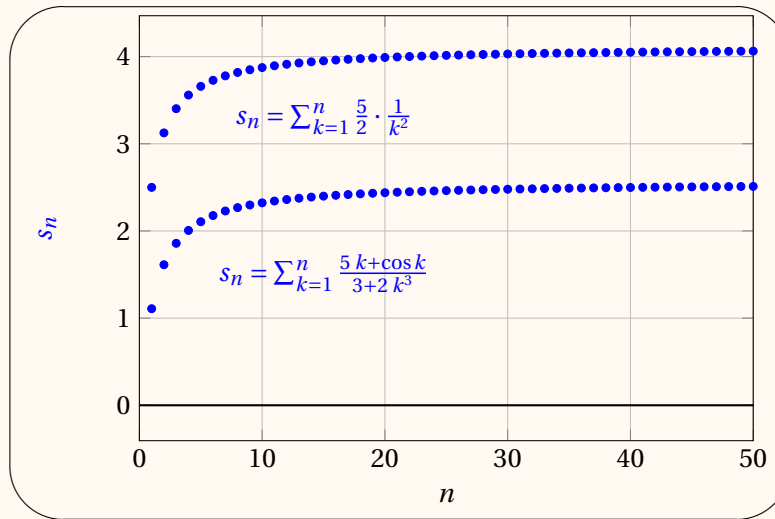
Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{5k + \cos k}{3 + 2k^3}$$

We have:

$$\frac{5k + \cos k}{3 + 2k^3} \sim \frac{5}{2} \cdot \frac{1}{k^2} \text{ as } k \rightarrow +\infty \text{ and } \sum_{k=1}^{\infty} \frac{5}{2} \cdot \frac{1}{k^2} \text{ converges}$$

hence the series is convergent by the limit comparison test.



However, even if the sequences are asymptotically equivalent, the corresponding series may not have the same sum.

1.3 Condensation test

Theorem 3: Condensation test

If $\{a_k\}$ is an eventually decreasing sequence with non-negative terms, then the series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=0}^{\infty} 2^k a_{2^k}$ are regular and have the same behavior, i.e., either they are both convergent or they are both divergent.

- Before proving the theorem, let us consider the sequence $\{\tilde{s}_n\}$ of the partial sums of the sequence $\{2^k a_{2^k}\}$, that is, the sequence:

$$\tilde{s}_n = \sum_{k=0}^n 2^k a_{2^k} \quad \forall n \geq 0$$

and let us derive two important relations with the sequence $\{s_n\}$ of the partial sums of the sequence $\{a_k\}$, that is, the sequence:

$$s_n = \sum_{k=1}^n a_k \quad \forall n \geq 1$$

- We observe that, for certain values of n , the partial sums s_n are:

$$\begin{aligned} \underbrace{s_1}_{= s_{2^1-1}} &= a_1, & \underbrace{s_3}_{= s_{2^2-1}} &= s_1 + a_2 + \underbrace{a_3}_{\leq a_2} \leq a_1 + 2 a_2 \\ \underbrace{s_7}_{= s_{2^3-1}} &= s_3 + a_4 + \underbrace{a_5}_{\leq a_4} + \underbrace{a_6}_{\leq a_4} + \underbrace{a_7}_{\leq a_4} \leq a_1 + 2 a_2 + 4 a_4 \\ \underbrace{s_{15}}_{= s_{2^4-1}} &= s_7 + a_8 + \underbrace{a_9}_{\leq a_8} + \underbrace{a_{10}}_{\leq a_8} + \underbrace{a_{11}}_{\leq a_8} + \underbrace{a_{12}}_{\leq a_8} + \underbrace{a_{13}}_{\leq a_8} + \underbrace{a_{14}}_{\leq a_8} + \underbrace{a_{15}}_{\leq a_8} \leq a_1 + 2 a_2 + 4 a_4 + 8 a_8 \end{aligned}$$

Remark 3

Given an eventually decreasing sequence $\{a_k\}$ with non-negative terms, we have:

$$s_{2^n-1} \leq \underbrace{\sum_{k=0}^{n-1} 2^k a_{2^k}}_{= \tilde{s}_{n-1}} \quad \forall n \geq 1 \quad (2)$$

Proof. We prove by induction on 2^n that

$$s_{2^n-1} \leq \sum_{k=0}^{n-1} 2^k a_{2^k} \quad \forall n \geq 1$$

Base case. Let $n = 1$. Then the claim becomes $s_{2^1-1} \leq 2^0 a_1$, i.e., $a_1 \leq a_1$, which is clearly true.

Inductive step. Assume that it is true for 2^{n-1} and let us prove it for 2^n . By the inductive hypothesis, we have $s_{2^{n-1}-1} \leq \sum_{k=0}^{n-2} 2^k a_{2^k}$. Moreover, we have:

$$s_{2^n-1} = s_{2^{n-1}-1} + \underbrace{\sum_{i=2^{n-1}}^{2^n-1} a_i}_{\leq 2^{n-1} a_{2^{n-1}}} \quad \text{hence} \quad s_{2^n-1} \leq \sum_{k=0}^{n-2} 2^k a_{2^k} + 2^{n-1} a_{2^{n-1}} = \sum_{k=0}^{n-1} 2^k a_{2^k}$$

which is exactly the desired claim for 2^n .

□

- We now observe that the values of $\tilde{s}_n$ are:

$$\tilde{s}_0 = a_1 \leq 2 a_1 = 2 \underbrace{s_1}_{= s_{2^0}}, \quad \tilde{s}_1 = \tilde{s}_0 + 2 a_2 \leq 2 s_1 + 2 a_2 = 2 \underbrace{s_2}_{= s_{2^1}}$$

$$\tilde{s}_2 = \tilde{s}_1 + 4 a_4 \leq 2 s_2 + 2 a_3 + 2 a_4 = 2 \underbrace{s_4}_{= s_{2^2}} \quad (\text{since } a_4 \leq a_3)$$

$$\tilde{s}_3 = \tilde{s}_2 + 8 a_8 \leq 2 s_4 + 2 a_5 + 2 a_6 + 2 a_7 + 2 a_8 = 2 \underbrace{s_8}_{= s_{2^3}} \quad (\text{since } a_8 \leq a_7 \leq a_6 \leq a_5)$$

Remark 4

Given an eventually decreasing sequence $\{a_k\}$ with non-negative terms, we have:

$$\underbrace{\sum_{k=0}^n 2^k a_{2^k}}_{= \tilde{s}_n} \leq 2 s_{2^n} \quad \forall n \geq 0 \quad (3)$$

Proof. We prove by induction on 2^n that

$$\tilde{s}_n \leq 2 s_{2^n} \quad \forall n \geq 0$$

Base case. Let $n = 0$. Then the claim becomes $\tilde{s}_0 \leq 2 s_{2^0} = 2s_1$, i.e., $a_1 \leq 2 a_1$, which is clearly true.

Inductive step. Assume that it is true for $n - 1$ and let us prove it for n . By the inductive hypothesis, we have $\tilde{s}_{n-1} \leq 2 s_{2^{n-1}}$. Moreover, we have:

$$\begin{aligned} \tilde{s}_n &= \tilde{s}_{n-1} + \underbrace{2^n a_{2^n}}_{\substack{2^n \\ i=2^{n-1}+1}} \quad \text{hence} \quad \tilde{s}_n \leq 2 s_{2^{n-1}} + \sum_{i=2^{n-1}+1}^{2^n} 2 a_i = 2 s_{2^n} \\ &\leq \sum_{i=2^{n-1}+1}^{2^n} 2 a_i \end{aligned}$$

which is exactly the desired claim for n . □

- We are now able to prove the theorem.

Proof. From relation (2) we have:

$$s_{2^n-1} \leq \tilde{s}_{n-1}, \quad \forall n \geq 1$$

Hence, if $\{\tilde{s}_n\}$ is convergent, that is, if $\sum_{k=0}^{\infty} 2^k a_{2^k}$ is convergent, the subsequence $\{s_{2^n-1}\}$ is bounded. Since $\{s_n\}$ is monotone, the whole sequence $\{s_n\}$ is bounded and convergent by the monotone sequence theorem. Consequently, if $\sum_{k=0}^{\infty} 2^k a_{2^k}$ is convergent, then $\sum_{k=1}^{\infty} a_k$ is also convergent.

From relation (3) we have:

$$\tilde{s}_n \leq 2 s_{2^n}, \quad \forall n \geq 0$$

Hence, if $\{s_n\}$ is convergent, that is, if $\sum_{k=1}^{\infty} a_k$ is convergent, then the subsequence $\{s_{2^n}\}$ is convergent (since $\{s_n\}$ is monotone). Consequently, by comparison, $\{\tilde{s}_n\}$ is also convergent, that is, $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

Moreover, since $\{a_k\}$ has non-negative terms, the series cannot be irregular; consequently, we also have:

$$\sum_{k=1}^{\infty} a_k \text{ divergent} \iff \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ divergent}$$

which completes the proof of the theorem. □

Given an eventually decreasing sequence $\{a_k\}$ with non-negative terms, we have proved that:

$$\sum_{k=1}^{\infty} a_k \text{ convergent/divergent} \iff \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ convergent/divergent}$$

Hence the convergence/divergence of $\sum_{k=0}^{\infty} 2^k a_{2^k}$ is a necessary and sufficient condition for the convergence/divergence of $\sum_{k=1}^{\infty} a_k$.

1.4 Generalized harmonic series

Definition 1: Generalized harmonic series

Given $\alpha \in \mathbb{R}$, the series $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is called the **generalized harmonic series**

Theorem 4: Behavior of the generalized harmonic series

Given $\alpha \in \mathbb{R}$,

the generalized harmonic series $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is $\begin{cases} \text{divergent to } +\infty & \text{if } \alpha \leq 1 \\ \text{convergent} & \text{if } \alpha > 1 \end{cases}$

Proof. We have already established that it diverges for $\alpha \leq 1$. For $\alpha > 1$, the sequence $\{\frac{1}{k^\alpha}\}$ is a decreasing sequence with positive terms. We have:

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = \sum_{k=0}^{\infty} 2^k \frac{1}{(2^k)^\alpha} = \sum_{k=0}^{\infty} (2^{1-\alpha})^k$$

that is, a geometric series with ratio $2^{1-\alpha}$, which converges if and only if $2^{1-\alpha} < 1$, that is, if $1 - \alpha < 0$. Hence, for $\alpha > 1$, by the condensation test, the series converges. $\square$

1.5 Passing from the discrete to the continuous

- The tools used to establish asymptotic estimates of functions can also be used to obtain asymptotic estimates of sequences (**passing from the discrete to the continuous**), and they provide useful tools for studying the behavior of a series with positive terms.

Example 2: passing from the discrete to the continuous

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{e^{1/k} - 1}{k}$$

From the asymptotic equivalence

$$e^{\varepsilon(x)} - 1 \sim \varepsilon(x) \quad \text{as } \varepsilon(x) \rightarrow 0$$

we have

$$\frac{e^{1/k} - 1}{k} \sim \frac{1}{k^2} \quad \text{as } k \rightarrow +\infty$$

Therefore the series, which has positive terms, converges by limit comparison with the series of $1/k^2$.

Example 3: passing from the discrete to the continuous

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \log\left(\frac{k+3}{k+2}\right)$$

We have

$$\frac{k+3}{k+2} = 1 + \frac{1}{k+2} = 1 + \frac{1}{k+2}$$

From the asymptotic equivalence

$$\log(1 + \varepsilon(x)) \sim \varepsilon(x) \quad \text{as } \varepsilon(x) \rightarrow 0$$

we have

$$\log\left(1 + \frac{1}{k+2}\right) \sim \frac{1}{k+2} \sim \frac{1}{k} \quad \text{as } k \rightarrow +\infty$$

Thus it is a series with positive terms whose general term is asymptotically equivalent to $1/k$. By comparison with the harmonic series, this series diverges to $+\infty$.

Example 4: passing from the discrete to the continuous

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \left(\frac{1}{k} - \sin \frac{1}{k} \right)$$

From the third-order Maclaurin expansion

$$\sin(\varepsilon(x)) = \varepsilon(x) - \frac{(\varepsilon(x))^3}{3!} + o((\varepsilon(x))^3) \quad \text{as } \varepsilon(x) \rightarrow 0$$

we have

$$\frac{1}{k} - \sin \frac{1}{k} = \frac{1}{k} - \left(\frac{1}{k} - \frac{1}{6k^3} + o\left(\frac{1}{k^3}\right) \right) \sim \frac{1}{6k^3} \quad \text{as } k \rightarrow +\infty$$

Therefore the series, which has positive terms, converges by limit comparison with the series of $1/(6k^3)$.

1.6 Root test

Theorem 5: Root test

Let $\{a_k\}$ be a sequence with non-negative terms. If the following limit exists:

$$\lim_{k \rightarrow +\infty} \sqrt[k]{a_k} = \ell \in \mathbb{R}^* \quad \text{then the series } \sum a_k \quad \text{is} \quad \begin{cases} \text{convergent} & \text{if } \ell < 1 \text{ or } \ell = -\infty \\ \text{divergent to } +\infty & \text{if } \ell > 1 \text{ or } \ell = +\infty \end{cases}$$

Proof. Suppose that $\lim_{k \rightarrow +\infty} \sqrt[k]{a_k} = \ell < 1$. Since $\sqrt[k]{a_k} \rightarrow \ell \in \mathbb{R}$, for every $\varepsilon > 0$ there exists $m \in \mathbb{N}$ such that $\forall k \geq m$:

$$\sqrt[k]{a_k} \leq \ell + \frac{\varepsilon}{2}$$

Moreover, since $\ell < 1$, we have $\ell < 1 - \varepsilon$ for a suitable $\varepsilon > 0$. For this ε we therefore have, eventually:

$$\sqrt[k]{a_k} \leq \ell + \frac{\varepsilon}{2} < (1 - \varepsilon) + \frac{\varepsilon}{2} = 1 - \frac{\varepsilon}{2} \quad \text{hence} \quad a_k < \left(1 - \frac{\varepsilon}{2}\right)^k, \quad \forall k \geq m$$

We have thus proved that $a_k < \left(1 - \frac{\varepsilon}{2}\right)^k$ eventually. The geometric series:

$$\sum_{k=0}^{\infty} \left(1 - \frac{\varepsilon}{2}\right)^k \quad \text{is convergent since } 1 - \frac{\varepsilon}{2} < 1$$

Hence, by the comparison test, the original series converges.

Suppose that $\lim_{k \rightarrow +\infty} \sqrt[k]{a_k} = \ell > 1$. Since $\sqrt[k]{a_k} \rightarrow \ell \in \mathbb{R}$, for every $\varepsilon > 0$ there exists $m \in \mathbb{N}$ such that $\forall k \geq m$:

$$\sqrt[k]{a_k} \geq \ell - \frac{\varepsilon}{2}$$

Moreover, since $\ell > 1$, we have $\ell > 1 + \varepsilon$ for a suitable $\varepsilon > 0$. For this ε we therefore have, eventually:

$$\sqrt[k]{a_k} \geq \ell - \frac{\varepsilon}{2} > (1 + \varepsilon) - \frac{\varepsilon}{2} = 1 + \frac{\varepsilon}{2} \quad \text{hence} \quad a_k > \left(1 + \frac{\varepsilon}{2}\right)^k, \quad \forall k \geq m$$

We have thus proved that $a_k > \left(1 + \frac{\varepsilon}{2}\right)^k$ eventually. The geometric series:

$$\sum_{k=0}^{\infty} \left(1 + \frac{\varepsilon}{2}\right)^k \quad \text{is divergent since } 1 + \frac{\varepsilon}{2} > 1$$

Hence, by the comparison test, the original series diverges.

□

Clearly, these arguments remain valid also if $\ell = -\infty$ or $+\infty$, respectively.

Remark 5

The series $\sum_{k=1}^{\infty} k^{\beta} \cdot b^k$ with $\beta \in \mathbb{R}, b \geq 0$ is

{	convergent	if $b < 1$
	divergent	if $b > 1$
	convergent	if $b = 1$ and $\beta < -1$
	divergent	if $b = 1$ and $\beta \geq -1$

Proof. It is a series with non-negative terms, and we have:

$$\sqrt[k]{k^{\beta} \cdot b^k} = b \cdot k^{\beta/k} \quad \text{and} \quad \lim_{k \rightarrow +\infty} k^{\beta/k} = \lim_{k \rightarrow +\infty} \exp \left(\underbrace{\frac{\beta}{k} \cdot \log k}_{\rightarrow 0} \right) = 1$$

hence, by the root test, it converges for $b < 1$ and diverges for $b > 1$. If $b = 1$, the series becomes:

$$\sum_{k=1}^{\infty} k^{\beta} = \sum_{k=1}^{\infty} \frac{1}{k^{-\beta}}$$

that is, a generalized harmonic series; it is convergent for $\beta < -1$, while it is divergent for $\beta \geq -1$. □

Remark 6

The series $\sum_{k=1}^{\infty} b^k / k^k$ with $b \geq 0$ is convergent

Proof. It is a series with non-negative terms, and we have:

$$\sqrt[k]{\frac{b^k}{k^k}} = \frac{b}{k} \rightarrow 0 \quad \text{as } k \rightarrow +\infty$$

hence the series converges by the root test. □

1.7 Ratio test

Theorem 6: Ratio test

Let $\{a_k\}$ be a sequence with positive terms. If the following limit exists:

$$\lim_{k \rightarrow +\infty} \frac{a_{k+1}}{a_k} = \ell \in \mathbb{R}^* \text{ then the series } \sum a_k \text{ is } \begin{cases} \text{convergent} & \text{if } \ell < 1 \text{ or } \ell = -\infty \\ \text{divergent to } +\infty & \text{if } \ell > 1 \text{ or } \ell = +\infty \end{cases}$$

Proof. Suppose that $\lim_{k \rightarrow +\infty} a_{k+1}/a_k = \ell < 1$. Reasoning as in the proof of the root test, there exists $m \in \mathbb{N}$ such that $\forall k \geq m$:

$$\frac{a_{k+1}}{a_k} < \left(1 - \frac{\varepsilon}{2}\right)$$

for a suitable $\varepsilon > 0$. Reasoning iteratively, this implies that:

$$a_{k+1} < \left(1 - \frac{\varepsilon}{2}\right) a_k < \left(1 - \frac{\varepsilon}{2}\right) \left(1 - \frac{\varepsilon}{2}\right) a_{k-1} < \dots < \left(1 - \frac{\varepsilon}{2}\right)^{k-m+1} a_m$$

We have thus proved that $a_k < \left(1 - \frac{\varepsilon}{2}\right)^{k-m} a_m$ eventually. The geometric series:

$$\sum_{k=m}^{\infty} \left(1 - \frac{\varepsilon}{2}\right)^{k-m} a_m \text{ is convergent since } 1 - \frac{\varepsilon}{2} < 1$$

Hence, by the comparison test, the original series converges.

Suppose that $\lim_{k \rightarrow +\infty} a_{k+1}/a_k = \ell > 1$. Reasoning as in the proof of the root test, there exists $m \in \mathbb{N}$ such that $\forall k \geq m$:

$$\frac{a_{k+1}}{a_k} > \left(1 + \frac{\varepsilon}{2}\right)$$

for a suitable $\varepsilon > 0$. Reasoning iteratively, this implies that:

$$a_{k+1} > \left(1 + \frac{\varepsilon}{2}\right) a_k > \left(1 + \frac{\varepsilon}{2}\right) \left(1 + \frac{\varepsilon}{2}\right) a_{k-1} > \dots > \left(1 + \frac{\varepsilon}{2}\right)^{k-m+1} a_m$$

We have thus proved that $a_k > \left(1 + \frac{\varepsilon}{2}\right)^{k-m} a_m$ eventually. The geometric series:

$$\sum_{k=m}^{\infty} \left(1 + \frac{\varepsilon}{2}\right)^{k-m} a_m \text{ is divergent since } 1 + \frac{\varepsilon}{2} > 1$$

Hence, by the comparison test, the original series diverges. □

Clearly, these arguments remain valid also if $\ell = -\infty$ or $+\infty$, respectively.

Remark 7

$$\sum_{k=0}^{\infty} \frac{1}{k!} = e$$

Proof. We have

$$\frac{1/(k+1)!}{1/k!} = \frac{1}{k+1} \rightarrow 0 \text{ as } k \rightarrow +\infty$$

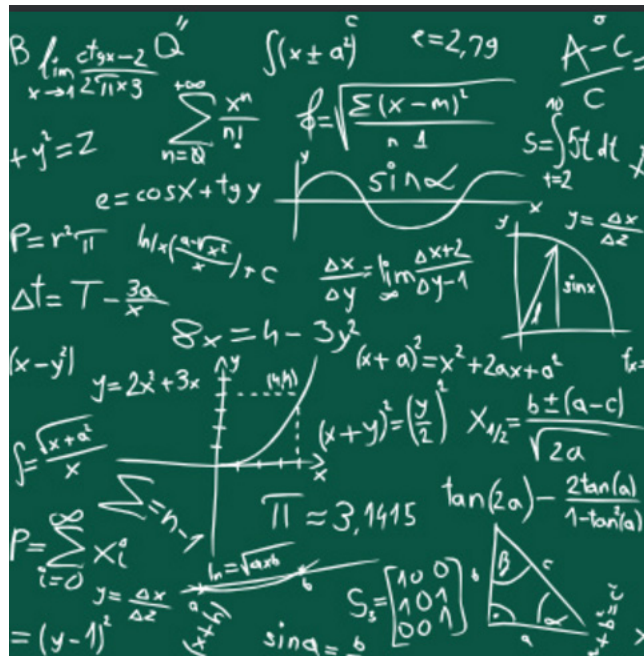
Therefore the series, which has non-negative terms, converges by the ratio test. The proof that the sum of the series is equal to e will be given later using series of functions. $\square$

1.8 Series with non-positive terms

- We know that the behavior of a series does not change if we alter a finite number of its terms. Consequently, the tests for series with non-negative terms also apply to series whose terms are eventually non-negative.
- Moreover, by factoring a minus sign out of the whole series, we see that these tests can also be applied to series with non-positive terms, and hence to series whose terms are eventually non-positive.

In summary, therefore, the tests we have seen apply to series all of whose terms (except for a finite number) have the same sign.

Series with terms of variable sign



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Series with terms of variable sign

Definition 1: absolutely convergent series

A series $\sum a_k$ is absolutely convergent if the series $\sum |a_k|$ converges.

Theorem 1

If a series $\sum a_k$ converges absolutely, then it converges.

Proof. Without loss of generality, let $n_0 = 0$ and consider the following series:

$$\sum_{k=0}^{\infty} (|a_k| - a_k)$$

It is a series with nonnegative terms since, for every $k \in \mathbb{N}$, we have:

$$\begin{cases} |a_k| - a_k = -2a_k \geq 0 & \text{if } a_k < 0 \\ |a_k| - a_k = 0 & \text{if } a_k \geq 0 \end{cases}$$

Moreover, by the triangle inequality, for every $k \in \mathbb{N}$, we have:

$$\underbrace{|a_k| - a_k}_{\geq 0} = ||a_k| - a_k| \leq |a_k| + |a_k| = 2|a_k|$$

Hence, by the comparison test for series with nonnegative terms, we have

$$\sum_{k=0}^{\infty} |a_k| \text{ convergent} \implies \sum_{k=0}^{\infty} (|a_k| - a_k) \text{ convergent}$$

Since for every $k \in \mathbb{N}$ we have $a_k = |a_k| - (|a_k| - a_k)$ and the series $\sum_{k=0}^{\infty} |a_k|$ and $\sum_{k=0}^{\infty} (|a_k| - a_k)$ converge, we have:

$$\sum_{k=0}^{\infty} a_k = \sum_{k=0}^{\infty} |a_k| - \sum_{k=0}^{\infty} (|a_k| - a_k)$$

that is, the series is the difference of two convergent series. Consequently, the series $\sum_{k=0}^{\infty} a_k$ is convergent as well. $\square$

- The following is an alternative proof.

Proof. Without loss of generality, let $n_0 = 0$ and split the sequence $\{s_n\}$ of partial sums into two sequences, the first containing only the positive terms and the second only the negative terms:

$$s_n^+ = \sum_{k \in \{0,1,\dots,n\}: a_k > 0} a_k \quad \text{and} \quad s_n^- = \sum_{k \in \{0,1,\dots,n\}: a_k < 0} -a_k \quad \text{hence} \quad s_n = s_n^+ - s_n^-$$

Consequently, it is enough to show that the sequences $\{s_n^+\}$ and $\{s_n^-\}$ are convergent to conclude that $\{s_n\}$ converges, and hence that the series $\sum a_k$ converges.

We observe that $\{s_n^+\}$ and $\{s_n^-\}$ are monotone non-decreasing sequences and moreover, by the triangle inequality, we have:

$$s_n^+ \leq \sum_{k=0}^n |a_k| \quad \text{and} \quad s_n^- \leq \sum_{k=0}^n |a_k|$$

On the other hand, by hypothesis the series $\sum a_k$ converges absolutely, i.e., $\sum |a_k|$ converges, and hence the quantity $\sum_{k=0}^n |a_k|$ is bounded; therefore the sequences $\{s_n^+\}$ and $\{s_n^-\}$ are bounded above and non-decreasing, and hence they converge, by the monotone sequence theorem. $\square$

Example 1: absolute convergence

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k^\alpha} \quad \text{with } \alpha > 1$$

We have:

$$\left| \frac{(-1)^k}{k^\alpha} \right| = \frac{1}{k^\alpha}, \quad \forall k \in \mathbb{N}, k > 1 \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{1}{k^\alpha} \text{ is convergent for } \alpha > 1$$

Therefore the series converges absolutely and hence it converges.

Absolute convergence implies ordinary convergence, also called simple convergence:

$$\sum |a_k| \text{ convergent} \Rightarrow \sum a_k \text{ convergent} \quad (1)$$

but the converse is not true (we will give a counterexample later):

$$\sum a_k \text{ convergent} \not\Rightarrow \sum |a_k| \text{ convergent} \quad (2)$$

Hence absolute convergence is a sufficient but not necessary condition for ordinary convergence.

1.1 Alternating series and the Leibniz test

- Among series with terms of variable sign, a particularly simple case is given by alternating series, for which the following convergence test holds.

Theorem 2: Leibniz (alternating series) test

Consider the series

$$\sum_{k=n_0}^{\infty} (-1)^k a_k \quad \text{with } a_k \geq 0, \forall k$$

If the sequence $\{a_k\}$ is decreasing and $a_k \rightarrow 0$ as $k \rightarrow +\infty$, then the series is convergent.

Moreover:

$$s_{2n} = \sum_{k=n_0}^{2n} (-1)^k a_k \downarrow s \quad \text{and} \quad s_{2n+1} = \sum_{k=n_0}^{2n+1} (-1)^k a_k \uparrow s \quad \text{as } n \rightarrow +\infty$$

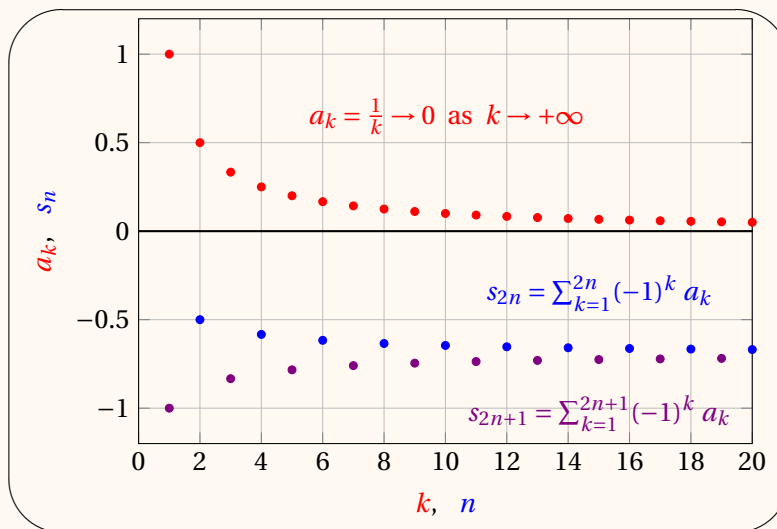
- The partial sums with even index approximate the sum s from above and those with odd index from below.
- The Leibniz test can clearly be applied also if the terms eventually have alternating signs and the sequence $\{a_k\}$ is eventually decreasing.

Example 2: Leibniz test

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} (-1)^k \frac{1}{k}$$

The sequence $a_k = \frac{1}{k}$ is decreasing and non-negative. Moreover $\frac{1}{k} \rightarrow 0$ as $k \rightarrow +\infty$, hence it satisfies the two conditions of the Leibniz test theorem. Consequently, the series is convergent.



The series $\sum_{k=1}^{\infty} \frac{(-1)^k}{k}$ converges and is a counterexample to (2), that is, a series that converges but does not converge absolutely, since:

$$\sum_{k=1}^{\infty} \left| \frac{(-1)^k}{k} \right| = \sum_{k=1}^{\infty} \frac{1}{k} \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{1}{k} \text{ diverges (harmonic series)}$$

Proof. Consider the sequence of partial sums $\{s_n\}$ with $n_0 = 0$ and the two subsequences extracted from it, $\{s_{2n}\}$ and $\{s_{2n+1}\}$.

For $\{s_{2n}\}$ we have:

$$s_0 = a_0, \quad s_2 = s_0 - a_1 + \underbrace{a_2}_{\leq a_1} \leq s_0, \quad s_4 = s_2 - a_3 + \underbrace{a_4}_{\leq a_3} \leq s_2, \quad \dots$$

hence the sequence $\{s_{2n}\}$ is monotone decreasing.

For $\{s_{2n+1}\}$ we have:

$$s_1 = a_0 - a_1, \quad s_3 = s_1 + a_2 - \underbrace{a_3}_{\leq a_2} \geq s_1, \quad s_5 = s_3 + a_4 - \underbrace{a_5}_{\leq a_4} \geq s_3, \quad \dots$$

hence the sequence $\{s_{2n+1}\}$ is monotone increasing.

Moreover, we have:

$$s_1 \leq s_{2n+1} = s_{2n} - a_{2n+1} \leq s_{2n} \leq s_0$$

therefore $\{s_{2n+1}\}$ is bounded above and $\{s_{2n}\}$ is bounded below. The two sequences are hence convergent, by the monotone sequence theorem.

The two sequences converge to the same limit, because

$$0 \leq s_{2n} - s_{2n+1} \leq a_{2n+1} \rightarrow 0 \quad \text{as } n \rightarrow +\infty$$

Calling s this limit, since $\{s_{2n}\}$ is monotone decreasing and $\{s_{2n+1}\}$ is monotone increasing, we have:

$$s_{2n} = \sum_{k=n_0}^{2n} (-1)^k a_k \downarrow s \quad \text{and} \quad s_{2n+1} = \sum_{k=n_0}^{2n+1} (-1)^k a_k \uparrow s \quad \text{as } n \rightarrow +\infty$$

Hence the series is convergent since $s_{2n+1} \rightarrow s$ and $s_{2n} \rightarrow s$ as $n \rightarrow +\infty$, and consequently we have:

$$s_n \rightarrow s \quad \text{as } n \rightarrow +\infty$$

□

Corollary 1: of the Leibniz test theorem

Given a series satisfying the hypotheses of the Leibniz test theorem, we have:

$$\underbrace{|s - s_m|}_{=|\sum_{k=m+1}^{\infty} (-1)^k a_k|} \leq a_{m+1}, \quad \forall m \in \mathbb{N}$$

- For every m , the error made by approximating s with s_m is, in absolute value, bounded above by the value of the first omitted term. In other words, the tail of the series tends to a value less than or equal to a_{m+1} .

Proof. Since

$$s_{2n+1} \uparrow s \quad \text{and} \quad s_{2n} \downarrow s \quad \text{as } n \rightarrow +\infty$$

we therefore have, for every $n \in \mathbb{N}$:

$$s_{2n-1} \leq s \leq s_{2n} \quad \text{and} \quad s_{2n+1} \leq s \leq s_{2n}$$

from which we deduce

$$0 \leq s - s_{2n-1} \leq s_{2n} - s_{2n-1} = a_{2n} \quad \text{and} \quad 0 \leq s_{2n} - s \leq s_{2n} - s_{2n+1} = a_{2n+1}$$

Therefore, for every m , whether even or odd, we have:

$$|s - s_m| = \left| \sum_{k=m+1}^{\infty} (-1)^k a_k \right| \leq a_{m+1}$$

□

Example 3: Leibniz test

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} (-1)^k \frac{k-1}{k^2+k}$$

We have:

$$a_k = \frac{k-1}{k^2+k} = \frac{k-1}{k(k+1)} \geq 0, \quad \forall k \in \mathbb{N}, k \geq 1 \quad \text{and} \quad \frac{k-1}{k(k+1)} \sim \frac{1}{k} \rightarrow 0 \quad \text{as } k \rightarrow +\infty$$

hence the series does not converge absolutely. However, the series is decreasing for $k \geq 2$ since:

$$\underbrace{\frac{k}{(k+1)(k+2)}}_{=a_{k+1}} \leq \underbrace{\frac{k-1}{k(k+1)}}_{=a_k} \iff k^2 \leq k^2 + k - 2 \iff k \geq 2$$

hence the series converges by the Leibniz test, since it is eventually decreasing.

Example 4: Leibniz test

Let us determine the behavior of the series:

$$\sum_{k=2}^{\infty} (-1)^k \frac{\log k}{k}$$

We have:

$$a_k = \frac{\log k}{k} \geq 0, \forall k \geq 2 \quad \text{and} \quad \frac{\log k}{k} \rightarrow 0 \text{ as } k \rightarrow +\infty$$

moreover

$$\frac{\log k}{k} > \frac{1}{k} \text{ for } k \geq 3 \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{1}{k} \text{ diverges}$$

hence the series does not converge absolutely. Proving algebraically that the sequence $\{a_k\}$ is decreasing is complicated; instead, we pass from the discrete to the continuous setting. We have, for $x \in \mathbb{R}$ and $x \geq 2$:

$$f(x) = \frac{\log x}{x} \quad \text{and} \quad f'(x) = \frac{1 - \log x}{x^2} \leq 0 \text{ for } x \geq e$$

It follows that f is decreasing for $x \geq e$; consequently the sequence $a_k = f(k)$ is decreasing for $k \geq 3$ (the first integer $> e$). Hence the series converges by the Leibniz test.

if $\sum a_k$ converges and $\sum b_k$ converges then $\sum (a_k + b_k)$ converges

if $\sum a_k$ converges and $\sum b_k$ diverges then $\sum (a_k + b_k)$ diverges

This can be checked by viewing the series as the limit of the sequence of partial sums and applying the theorem on the limit of a sum.

Example 5: Leibniz test

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{k+1+(-1)^k k^2}{k^3}$$

We split it into the sum of two series:

$$\sum_{k=1}^{\infty} \frac{k+1+(-1)^k k^2}{k^3} = \sum_{k=1}^{\infty} \frac{k+1}{k^3} + \sum_{k=1}^{\infty} \frac{(-1)^k}{k}$$

For the first series we have:

$$\frac{k+1}{k^3} \geq 0, \forall k \geq 1 \quad \text{and} \quad \frac{k+1}{k^3} \rightarrow 0 \text{ as } k \rightarrow +\infty$$

hence it converges by the limit comparison test, since:

$$\frac{k+1}{k^3} \sim \frac{1}{k^2} \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{1}{k^2} \text{ converges}$$

The second series converges by the Leibniz test; hence the original series converges.

Example 6: Leibniz test

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} (-1)^k \left(\frac{\sqrt{k} + (-1)^k}{k} \right)$$

We split it into the sum of two series:

$$\sum_{k=1}^{\infty} (-1)^k \left(\frac{\sqrt{k} + (-1)^k}{k} \right) = \sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}} + \sum_{k=1}^{\infty} \frac{1}{k}$$

The first series converges by the Leibniz test; the second diverges (harmonic series); hence the original series diverges.

Example 7: Leibniz test

Let us determine the behavior of the series:

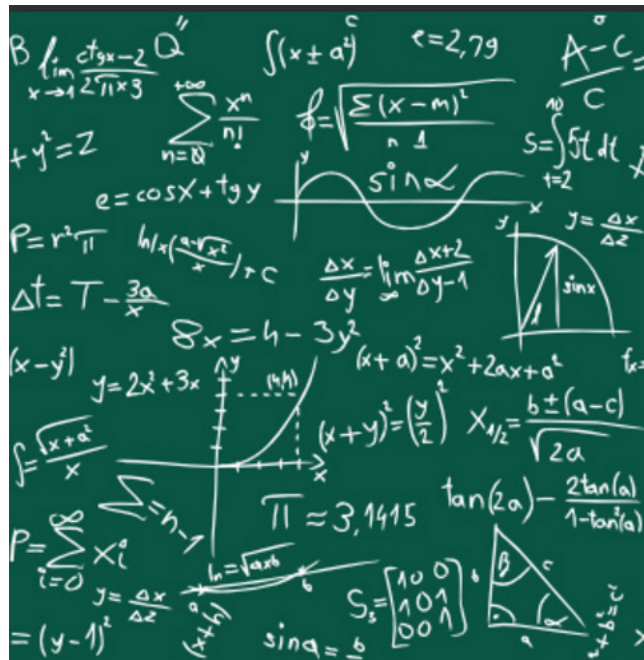
$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

We have:

$$\frac{(-1)^{k+1}}{k} = -\frac{(-1)^k}{k}, \quad \forall k \in \mathbb{N}, k > 1$$

moreover, the sequence $a_k = \frac{1}{k}$ is decreasing and $\frac{1}{k} \rightarrow 0$ as $k \rightarrow +\infty$, hence it satisfies the two conditions of the theorem. Consequently, the series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$ converges.

Series of functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Series of functions

- **Series of functions** are numerical series depending on a parameter x : when these series converge for every value x belonging to a certain interval, they represent **functions of a new kind**
- There are two important classes of series of functions, namely trigonometric series and power series. We will only deal with **power series**, also called **Taylor series**, which can be seen as a natural extension of Taylor expansions.

1.1 Taylor series of the elementary transcendental functions

- Taylor's expansion/formula with Lagrange remainder can be rewritten in the form:

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + E_n(x) \quad (1)$$

where the Lagrange approximation error is

$$E_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1} \quad (2)$$

and c is a suitable point between x_0 and x .

Definition 1: Taylor series

If a function f has derivatives of every order, the series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

is called the **Taylor series** of the function f centered at x_0 .

At a given point $x \in (a, b)$, if the error

$$E_n(x) \rightarrow 0 \text{ as } n \rightarrow +\infty$$

then the Taylor series is convergent and its sum equals $f(x)$. This is equivalent to saying that the n -th partial sum of the series has a finite limit and this limit is precisely $f(x)$. In formulas:

$$\underbrace{T_{n,x_0}(x)}_{\text{Taylor polynomial}} = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \rightarrow f(x) \text{ as } n \rightarrow +\infty$$

Definition 2: function expandable in Taylor series on an interval

Given a function $f : D \rightarrow \mathbb{R}$ with derivatives of every order, if

$$E_n(x) \rightarrow 0 \text{ as } n \rightarrow +\infty, \quad \forall x \in I \subseteq D$$

we say that $f(x)$ is **expandable in Taylor series** on the interval I .

- There are (infinitely differentiable) functions that are expandable in Taylor series on the whole real line, others that are expandable only on a bounded interval, and others for which the interval reduces to a single point (the point x_0).

- We now deal with the three elementary transcendental functions:

$$f(x) = e^x, \quad f(x) = \sin x \quad \text{and} \quad f(x) = \cos x$$

which we will show to be expandable in Taylor series on the whole of $\mathbb{R}$. They are three transcendental (not algebraic) functions.

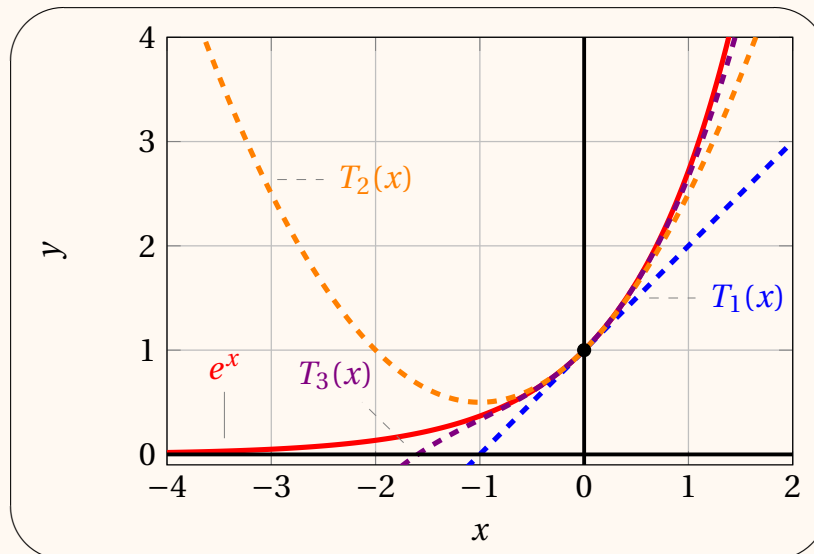
- Real functions of a real variable are classified into **algebraic functions and transcendental functions**. Algebraic functions are built through a finite number of applications of the four arithmetic operations, of raising to a power and of extracting the n -th root. Transcendental functions are all the functions that are not algebraic.

1.1.1 The Taylor series of the exponential function

Example 1: Maclaurin formula/expansion of the exponential

Maclaurin formula/expansion of order n of the exponential with Lagrange remainder:

$$e^x = T_n(x) + \frac{e^c}{(n+1)!} x^{n+1} = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \frac{e^c}{(n+1)!} x^{n+1} \quad \text{with } c \in [0, x]$$



Remark 1: Taylor series of the exponential function

$$\forall x \in \mathbb{R}, \quad e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Proof. From the Maclaurin expansion of order n of e^x with Lagrange remainder, we have that for every integer n and $x \in \mathbb{R}$ there exists a point c , lying between 0 and x , such that:

$$e^x = \sum_{k=0}^n \frac{x^k}{k!} + \frac{e^c}{(n+1)!} x^{n+1}$$

We fix x and let n tend to $+\infty$. The point c may vary with n but, since it always lies between 0 and x , we have:

$$e^c \leq \begin{cases} e^x & \text{if } x > 0 \\ 1 & \text{if } x < 0 \end{cases}$$

and $e^0 = 1$, hence e^c stays bounded. From the hierarchy of infinities theorem we have:

$$\frac{x^{n+1}}{(n+1)!} \rightarrow 0 \text{ as } n \rightarrow +\infty$$

hence the error tends to zero, that is:

$$\frac{x^{n+1}}{(n+1)!} e^c \rightarrow 0 \text{ as } n \rightarrow +\infty$$

since it is the product of an infinitesimal sequence and a bounded one. We have therefore proved that the function e^x can be written as the sum of a power series, its Taylor series, convergent for every $x \in \mathbb{R}$. $\square$

Remark 2

$$e = \sum_{k=0}^{\infty} \frac{1}{k!} \tag{3}$$

Proof. It suffices to set x equal to 1 in the Taylor series of the exponential function. $\square$

- We have derived a second definition of Euler's number e (Napier's constant); the first one was:

$$e = \lim_{k \rightarrow +\infty} \left(1 + \frac{1}{k}\right)^k \tag{4}$$

- Consequently, we have two methods for computing an approximation of the value of e :

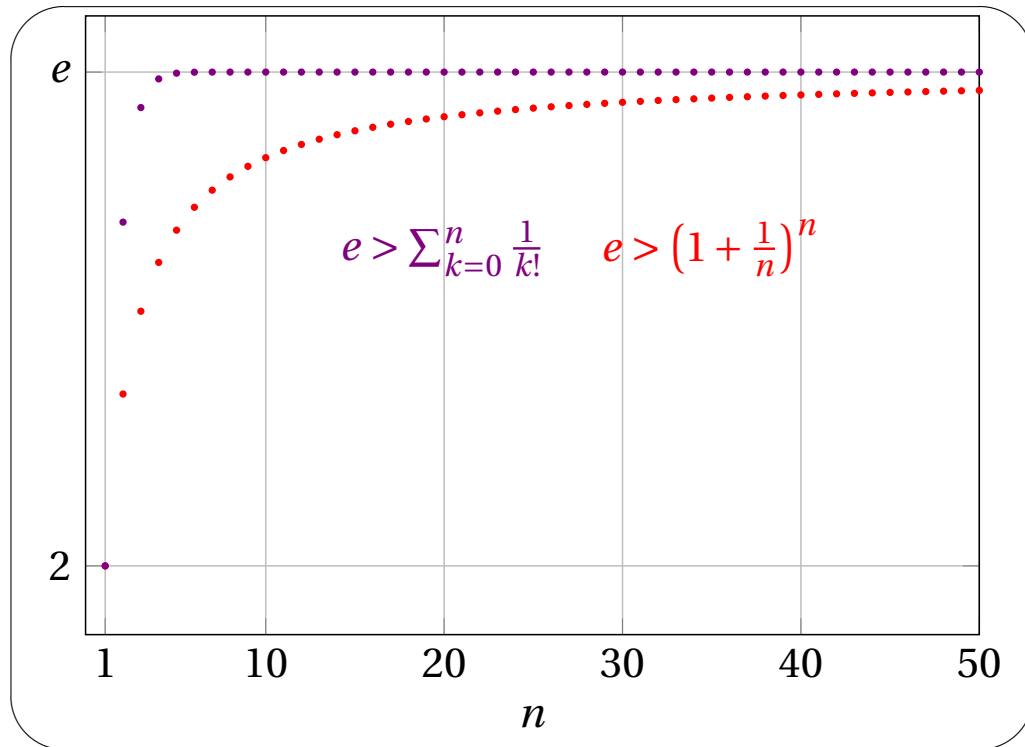
1. The first method is based on formula (4), and by fixing $k = n$ we obtain the approximation:

$$e \approx \left(1 + \frac{1}{n}\right)^n$$

2. The second method is based on formula (3), and by computing the n -th partial sum we obtain the approximation:

$$e \approx \sum_{k=0}^n \frac{1}{k!}$$

	$\left(1 + \frac{1}{n}\right)^n$	$\sum_{k=0}^n \frac{1}{k!}$	e
$n = 1$	2.0000000000...	2.0000000000...	2,7182818284...
$n = 2$	2.2500000000...	2.5000000000...	2,7182818284...
$n = 3$	2.3703703704...	2.6666666666...	2,7182818284...
$n = 4$	2.4414062500...	2.7083333333...	2,7182818284...
$n = 5$	2.4883200000...	2.7166666666...	2,7182818284...



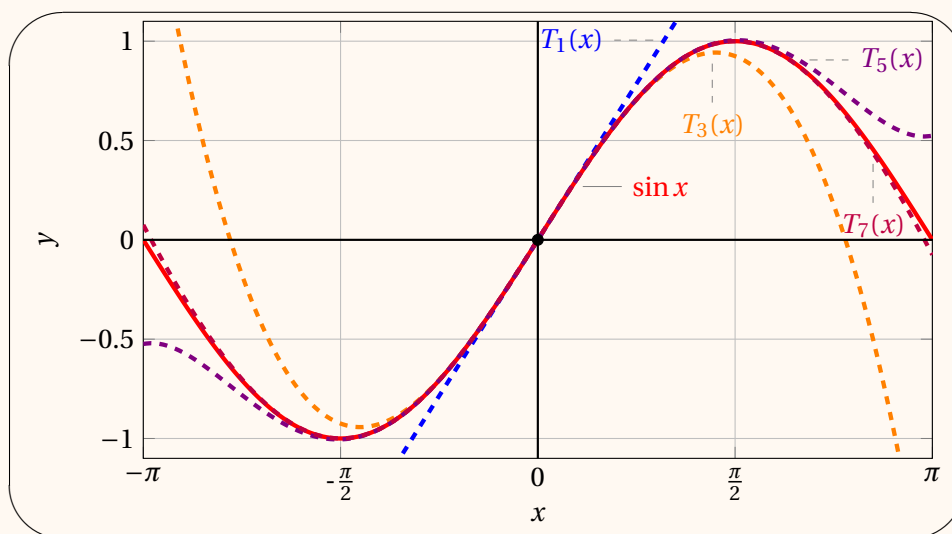
- The second method is therefore much more efficient at approximating the value of e

1.1.2 The Taylor series of the elementary trigonometric functions

Example 2: Maclaurin formula/expansion of sine and cosine

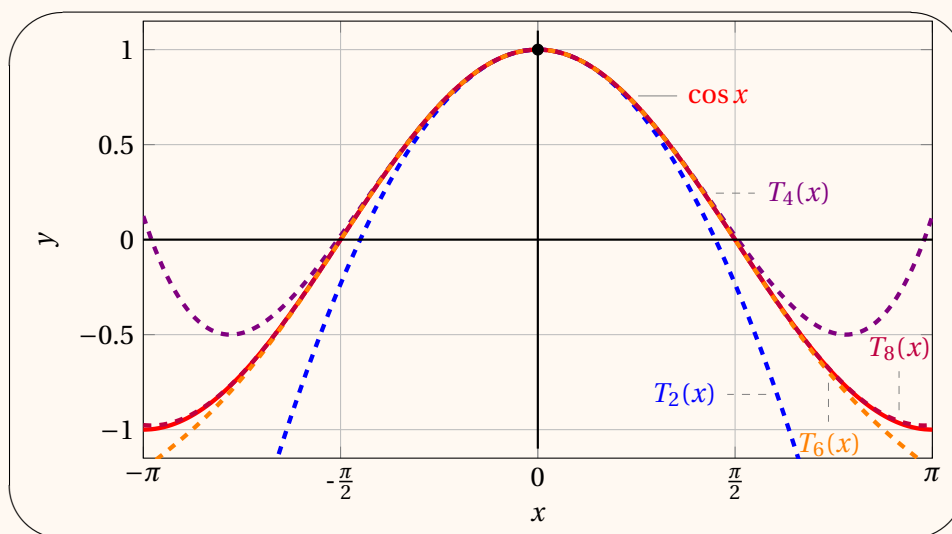
Maclaurin formula/expansion of odd order of the sine with Lagrange remainder:

$$\begin{aligned}\sin x &= T_{2k+1}(x) + \frac{\sin^{(2k+2)}(c)}{(2k+2)!} x^{2k+2} \\ &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^k \frac{x^{2k+1}}{(2k+1)!} + \frac{\sin^{(2k+2)}(c)}{(2k+2)!} x^{2k+2} \text{ with } c \in [0, x]\end{aligned}$$



Maclaurin formula/expansion of even order of the cosine with Lagrange remainder:

$$\begin{aligned}\cos x &= T_{2k}(x) + \frac{\cos^{(2k+1)}(c)}{(2k+1)!} x^{2k+1} \\ &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^k \frac{x^{2k}}{(2k)!} + \frac{\cos^{(2k+1)}(c)}{(2k+1)!} x^{2k+1} \text{ with } c \in [0, x]\end{aligned}$$



Remark 3: Taylor series of the elementary trigonometric functions

$$\forall x \in \mathbb{R}, \quad \sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}, \quad \cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$$

Proof.

□

2 Series in the complex field